

Integrated Medical Foundations

Complete text-only reading edition

Sixty-eight narration-ready chapters integrating physiology, pathology, pharmacology, examination, investigation, treatment logic, safety, and retrieval practice.

Chapter 1: Homeostasis, Cellular Energetics, and Physiological Reserve

Listening guidance

This review explains how the body keeps internal conditions within workable ranges. Pause after each retrieval prompt and answer aloud before continuing.

Learning goals

By the end, you should be able to explain homeostasis, distinguish negative and positive feedback, describe membrane transport, and connect the four tissue classes to function.

The core idea: regulated stability

Physiology asks how living structures work. The body's cells can function only if their surroundings remain reasonably stable. Homeostasis is this dynamic regulation of the internal environment. It is not a perfectly fixed state. Variables such as temperature, blood glucose, blood pressure, acidity, and oxygen concentration fluctuate around useful ranges.

A control system usually has three parts. A receptor detects change. A control centre compares the information with a target or acceptable range. An effector produces a response. In negative feedback, the response opposes the original disturbance. If body temperature rises, heat-loss mechanisms increase. If blood glucose rises, insulin promotes glucose uptake and storage. Negative feedback is the dominant pattern because it stabilises a variable.

Positive feedback amplifies a process until a specific endpoint occurs. During childbirth, stretch of the cervix promotes oxytocin release, which strengthens uterine contraction and causes further stretch. Blood clotting also contains amplifying steps. Positive feedback is useful when the body must rapidly complete an event, but it needs a stopping condition.

Cells and membrane transport

The plasma membrane separates intracellular fluid from extracellular fluid. Its phospholipid bilayer allows some substances to cross more readily than others. Small non-polar molecules, including oxygen and carbon dioxide, can diffuse through the lipid region. Ions and many polar molecules require membrane proteins.

Diffusion is net movement down a concentration gradient. Facilitated diffusion also moves down a gradient, but uses a channel or carrier. Neither process directly requires cellular energy. Osmosis is movement of water across a selectively permeable membrane toward the side with the greater effective concentration of non-penetrating solute.

Tonicity predicts long-term effects on cell volume. In an isotonic solution, cell volume is stable. In a hypotonic solution, water enters and the cell swells. In a hypertonic solution, water leaves and the cell shrinks.

Active transport moves a substance against its electrochemical gradient and therefore requires energy. The sodium-potassium pump directly uses adenosine triphosphate. For each cycle it moves three sodium ions out and two potassium ions in. This maintains gradients that support membrane voltage, cell volume, nerve signalling, and secondary active transport.

Vesicular transport moves larger material. Endocytosis brings material into the cell. Exocytosis releases material, including neurotransmitters and many hormones.

Tissues as functional systems

Epithelial tissue covers surfaces, lines cavities, and forms glands. Closely packed cells and selective junctions make epithelia effective barriers. Their functions include protection, absorption, secretion, filtration, and sensation.

Connective tissue supports, binds, protects, stores, and transports. Its cells are separated by an extracellular matrix made from ground substance and protein fibres. Bone, cartilage, adipose tissue, tendons, ligaments, and blood are all connective tissues, despite their very different appearances.

Muscle tissue converts chemical energy into force. Skeletal muscle produces voluntary movement and contributes to posture and heat. Cardiac muscle

pumps blood. Smooth muscle changes the diameter or contents of hollow organs and vessels.

Nervous tissue detects stimuli, processes information, and communicates rapidly. Neurons generate electrical signals. Glial cells support, nourish, insulate, and regulate the environment around neurons.

Integration

Homeostasis emerges from cooperation across levels. Membrane transport preserves the conditions inside each cell. Tissues specialise. Organs combine tissues. Organ systems exchange information and matter. A failure at one level can propagate. For example, reduced oxygen delivery limits cellular respiration, decreases adenosine triphosphate production, weakens active transport, disturbs ion gradients, and eventually impairs organ function.

Cellular energy and protein traffic

Cells require a continuous energy supply to maintain order. During cellular respiration, chemical energy from nutrients is transferred into adenosine triphosphate. Glycolysis in the cytosol converts glucose into pyruvate and produces a small amount of adenosine triphosphate. When oxygen and mitochondria are available, pyruvate-derived carbon enters the citric acid cycle. Reduced electron carriers then supply the electron transport chain. The resulting proton gradient powers adenosine triphosphate synthase. Oxygen acts as the final electron acceptor and is reduced to water.

This explains why oxygen deprivation rapidly affects excitable tissues. A neuron relies on active transport to maintain sodium and potassium gradients. If oxidative phosphorylation fails, adenosine triphosphate falls, ion pumps weaken, membrane voltage becomes unstable, calcium regulation fails, and damaging enzymes can become active.

Cell structures divide work. Ribosomes translate messenger ribonucleic acid into protein. Proteins destined for secretion, membranes, or lysosomes usually enter the rough endoplasmic reticulum during synthesis. They are folded and modified, sent in vesicles to the Golgi apparatus, then sorted toward their destination. Lysosomes contain acid hydrolases that digest macromolecules and damaged components. Peroxisomes perform oxidative

reactions and help break down very-long-chain fatty acids. The cytoskeleton organises the cell, supports shape, transports cargo, separates chromosomes, and enables movement.

Protein function is controlled at several levels: gene transcription, messenger ribonucleic acid processing and stability, translation, chemical modification, localisation, and degradation. A cell can therefore respond quickly by modifying existing proteins or more slowly by changing gene expression.

Cell communication and adaptation

Cells communicate through direct contact, local chemical signals, synaptic transmitters, and circulating hormones. A signalling molecule binds a receptor, receptor activation changes intracellular pathways, and the cell produces a response. Signal amplification allows a small external event to generate a large response. Signal termination is equally important; without it, a pathway would remain active after the stimulus disappeared.

Receptors can change in number or sensitivity. Persistent high stimulation may cause down-regulation or desensitisation, reducing responsiveness. Low stimulation can promote up-regulation. These adjustments help explain tolerance, altered hormone sensitivity, and adaptation to changing conditions.

Gap junctions directly connect the cytoplasm of neighbouring cells and permit ions or small molecules to pass. They are important where electrical or metabolic coordination is needed, including cardiac and some smooth muscle. Tight junctions limit movement between epithelial cells and preserve distinct compartments. Anchoring junctions distribute mechanical stress.

Tissue repair and physiological limits

Injury triggers haemostasis, inflammation, proliferation, and remodelling. Platelets and clotting proteins reduce blood loss. Inflammatory signals recruit immune cells that remove microbes and damaged material. New epithelial cells, connective tissue, and blood vessels rebuild the region. During remodelling, collagen is reorganised and tensile strength improves, although repaired tissue may not fully regain its original structure.

Regenerative ability differs among tissues. Surface epithelia and blood-forming tissues renew rapidly because stem and progenitor cells

remain active. Liver tissue has substantial regenerative capacity. Cardiac muscle and neurons of the central nervous system generally replace lost functional cells poorly. When regeneration is incomplete, fibrosis can preserve structural integrity at the cost of specialised function.

Physiological reserve is the difference between baseline function and maximum capacity. A person can lose considerable reserve before resting measurements become abnormal. Exercise, illness, ageing, or environmental stress exposes this hidden limitation. Homeostatic failure therefore often appears first under demand rather than at rest.

Ageing changes homeostasis partly by reducing reserve and slowing recovery. Total body water, kidney concentrating ability, vascular flexibility, immune responsiveness, muscle mass, and sensory feedback may decline at different rates. The important principle is not that regulation disappears, but that the workable range narrows. A disturbance that is easily corrected in a young healthy system can exceed the compensatory capacity of an older or chronically ill system. Clinical measurements must therefore be interpreted together with function, context, and response to stress.

TTS module 2: Control systems, energy failure, and hidden reserve

Homeostasis as layered control

Homeostatic variables are rarely governed by one sensor and one effector. Most are controlled by overlapping loops operating over different timescales. Blood pressure, for example, is adjusted within seconds by arterial baroreceptors, over minutes to hours by sympathetic tone and vasoactive hormones, and over days by renal sodium and water handling. The loops interact. A rapid fall in pressure triggers tachycardia and vasoconstriction, while slower renal responses defend circulating volume. This layered design creates resilience because failure of one component can be partly buffered by another.

The regulated quantity is not always identical to the measured quantity. Baroreceptors sense stretch in selected arteries rather than pressure everywhere. Pancreatic beta cells respond to nutrient metabolism and membrane activity rather than to an abstract glucose set point. The brain

integrates temperature signals from both the core and the skin. Clinical measurements are therefore samples from a distributed control system. A normal value can coexist with intense compensation, while an abnormal value may reflect a deliberate adaptive response rather than primary failure.

Set points are also context dependent. Fever is not simply uncontrolled overheating. Inflammatory mediators alter hypothalamic temperature regulation, so the body actively conserves and generates heat until it reaches a higher defended temperature. By contrast, hyperthermia occurs when heat gain exceeds heat loss without a regulated upward shift. This distinction explains why chills can accompany a rising fever and why treatment priorities differ in severe environmental heat illness.

Gain, delay, and oscillation

Control-system gain describes how strongly a response corrects a disturbance. Low gain permits a large residual error. Excessively high gain combined with delay can produce oscillation. Biological control is full of delays: hormones require synthesis and circulation, gene expression takes time, and kidney-mediated volume adjustment is slower than a neural reflex. These delays help explain why correction can overshoot. Glucose may fall after insulin has already been given, carbon dioxide can change rapidly after ventilation is adjusted, and sodium concentration may continue to move after fluid therapy begins.

Feedback can also reset. Persistent hypertension reduces the ability of arterial baroreceptors to signal that the pressure is abnormal. Chronic hypercapnia changes renal bicarbonate retention and alters the acid-base state against which an acute respiratory change must be interpreted. Adaptation may preserve function in the short term but conceal disease and create dependence on the compensated state. Abruptly reversing a long-standing abnormality can therefore be harmful.

Feed-forward control anticipates demand before the regulated variable changes substantially. Heart rate and ventilation rise at the start of exercise partly through central command and sensory input from moving limbs. Salivation and digestive secretion begin in response to sight, smell, and expectation of food. Feed-forward responses reduce the size of the disturbance that feedback must later correct. They are efficient when

prediction is accurate but can be maladaptive when a conditioned response is triggered without the expected physiological need.

Energy supply and cellular triage

Adenosine triphosphate is not stored in quantities sufficient for prolonged demand. It must be continuously regenerated. Cells therefore match energy production to consumption, and tissues differ in their tolerance of interruption. The brain has high, continuous energy requirements and little intrinsic fuel reserve. Cardiac muscle depends heavily on aerobic metabolism. Skeletal muscle can temporarily draw on phosphocreatine and anaerobic glycolysis, accepting lactate accumulation and reduced efficiency to preserve force.

When oxygen delivery falls, the first problem is not simply absence of oxygen in isolation. Oxygen delivery depends on blood flow, haemoglobin concentration, haemoglobin saturation, and the ability of tissue to extract oxygen. Severe anaemia, shock, hypoxaemia, or microvascular failure can each reduce delivery by a different route. Mitochondrial toxins can prevent oxygen use even when measured delivery appears adequate. Clinical reasoning should therefore separate ventilation, gas exchange, circulation, oxygen carriage, microcirculation, and cellular utilisation.

Falling energy supply produces an ordered loss of function. Cells reduce energy-intensive specialised activity before they lose basic integrity. Ion transport becomes harder to sustain, sodium accumulates intracellularly, water follows, and cells swell. Calcium entry activates phospholipases, proteases, and endonucleases. Reactive oxygen species damage membranes, proteins, and nucleic acids. If the disturbance is corrected early, ion gradients and function may recover. Beyond a variable threshold, mitochondrial and membrane damage become self-amplifying and injury becomes irreversible.

Reperfusion is necessary to rescue ischaemic tissue, yet it can add injury through oxidant generation, calcium disturbance, endothelial activation, and inflammation. This does not argue against restoring flow. It explains why speed, controlled resuscitation, temperature, electrolytes, and the condition of the microcirculation influence recovery even after a blocked vessel or low-flow state has been corrected.

Compartments, effective osmoles, and volume

Body water is divided mainly between intracellular and extracellular compartments. The extracellular compartment includes plasma and interstitial fluid. Sodium salts are the major effective osmoles outside cells, while potassium salts and organic phosphates dominate inside. Water crosses many cell membranes readily, so differences in effective osmolality shift water until the osmotic forces approach equilibrium.

Concentration, content, and volume must be distinguished. A low plasma sodium concentration usually indicates excess water relative to exchangeable sodium and potassium, not necessarily a deficit of total-body sodium. A person can be oedematous and still have reduced effective arterial blood volume if cardiac output is poor or fluid is sequestered outside the circulation. The kidneys then retain sodium and water because their sensors detect underfilling, even though total extracellular volume is increased.

Not every solute exerts sustained tonic force. Urea contributes to measured osmolality but crosses many cell membranes and is therefore often less effective than sodium at determining cell volume. Glucose becomes an important effective osmole when markedly elevated, drawing water from cells into extracellular fluid. This movement can dilute measured sodium. Interpreting fluid disorders requires asking which solute changed, whether it crosses the relevant membrane, and how rapidly the disturbance developed.

Rate matters because the brain adapts to persistent osmotic change. With chronic low extracellular tonicity, brain cells shed electrolytes and organic osmolytes to limit swelling. Rapid correction can then dehydrate adapted cells and damage myelin. With chronic hypertonicity, cells accumulate osmolytes; correction that is too rapid can promote cerebral oedema. Exact correction limits are context-sensitive and must follow current local protocols, but the underlying safety principle is universal: chronic adaptation narrows the safe rate of reversal.

Reserve, compensation, and decompensation

Reserve cannot be inferred reliably from a resting measurement. A person with early cardiac dysfunction may maintain normal blood pressure and oxygen saturation at rest by increasing filling pressure and neurohormonal drive. Exertion reveals breathlessness because cardiac output cannot rise appropriately and pulmonary pressures increase. Similarly, serum creatinine

may remain within a reference interval despite substantial loss of filtration in someone with low muscle mass. Function under stress and the trajectory over time can be more informative than a single apparently normal result.

Compensation has costs. Sympathetic activation preserves perfusion but raises myocardial oxygen demand and constricts some vascular beds. Renin-angiotensin-aldosterone signalling retains volume but can worsen congestion. Hyperventilation partly compensates for metabolic acidosis but increases respiratory work. Bone buffering of chronic acid load protects extracellular acidity at the expense of mineral stores. A compensatory response should be recognised, supported when necessary, and not mistaken for correction of the initiating problem.

Decompensation occurs when demand exceeds reserve or when compensation itself becomes damaging. Warning signs include rising work of breathing, altered mental state, falling urine output, increasing lactate, deteriorating mobility, and slower recovery after minor stress. These trends often precede collapse in a headline vital sign. Repeated assessment matters because physiology is a trajectory, not a photograph.

The practical synthesis is to ask five questions. What variable is being defended? Which sensors, integrators, and effectors are involved? What compensation is currently preserving the measured value? What energetic or structural cost is that compensation imposing? Finally, how much reserve remains if demand rises again? Those questions turn homeostasis from a definition into a clinical method.

Retrieval prompts

One. Why is homeostasis described as dynamic rather than static?

Two. Name the receptor, control centre, and effector in a temperature-control example.

Three. What distinguishes facilitated diffusion from active transport?

Four. Predict what happens to a red blood cell in a strongly hypotonic solution.

Five. Match each tissue class with one defining function.

Concise answers

One. Variables fluctuate, but regulation keeps them within viable ranges.

Two. Thermoreceptors detect temperature, the hypothalamus integrates the signal, and effectors such as sweat glands and skin blood vessels change heat loss.

Three. Facilitated diffusion follows an electrochemical gradient without direct energy input. Active transport moves against a gradient and requires energy.

Four. Water enters, the cell swells, and it may rupture.

Five. Epithelium forms barriers and exchange surfaces. Connective tissue supports and binds. Muscle generates force. Nervous tissue communicates.

Source map

Original private-study synthesis. Primary references: OpenStax Anatomy and Physiology 2e, chapters 1, 3, and 4; Guyton and Hall Textbook of Medical Physiology, chapter 1; Robbins Basic Pathology, cellular adaptation and injury. OpenStax content is licensed under Creative Commons Attribution NonCommercial ShareAlike 4.0.

Chapter 2: Membranes, Signalling, and Excitable Cells

Listening guidance

This review compares rapid neural signalling with slower hormonal control. Pause after each prompt and explain the answer aloud.

Learning goals

Explain resting membrane potential, action potentials, synaptic transmission, autonomic control, hormone classes, receptor signalling, and endocrine feedback.

Electrical signalling in neurons

Neurons maintain unequal ion distributions across their membranes. Potassium is concentrated inside. Sodium is concentrated outside. Selective permeability and the sodium-potassium pump help create a resting membrane potential, with the inside electrically negative relative to the outside.

A graded potential is a local change whose size depends on stimulus strength. If depolarisation at the trigger zone reaches threshold, voltage-gated sodium channels open. Sodium enters rapidly and produces the rising phase of an action potential. Sodium channels then inactivate while voltage-gated potassium channels open. Potassium leaves, repolarising the membrane. Continued potassium conductance can briefly cause hyperpolarisation.

An action potential is all or none. A stronger stimulus does not create a taller action potential. It usually increases firing frequency or recruits more neurons. The refractory period limits firing rate and promotes one-way propagation. Myelin increases speed by allowing current to travel efficiently between nodes of Ranvier, a process called saltatory conduction.

Synapses and networks

At a chemical synapse, an action potential opens voltage-gated calcium channels in the presynaptic terminal. Calcium entry triggers neurotransmitter release. The transmitter crosses the cleft and binds receptors on the postsynaptic cell. Ionotropic receptors directly control channels and act quickly. Metabotropic receptors act through signalling pathways and are often slower and longer lasting.

An excitatory postsynaptic potential moves the membrane toward threshold. An inhibitory postsynaptic potential moves it away from threshold or stabilises it. Neurons integrate many inputs through spatial and temporal summation.

The autonomic nervous system regulates cardiac muscle, smooth muscle, and glands. The sympathetic division supports mobilisation during challenge. The parasympathetic division supports rest, digestion, and energy conservation. These are coordinated tendencies, not simple on-off labels, and many organs receive both forms of input.

Hormonal signalling

Hormones travel through body fluids and act only on cells with appropriate receptors. Peptide and catecholamine hormones are generally water-soluble. They bind membrane receptors and use second messengers. Steroid hormones and thyroid hormones are lipid-soluble. They cross membranes or enter cells through transport mechanisms, bind intracellular receptors, and alter gene expression. Their effects often begin more slowly but last longer.

Hormone secretion is commonly controlled by negative feedback. In the hypothalamic-pituitary-thyroid axis, hypothalamic thyrotropin-releasing hormone stimulates pituitary thyroid-stimulating hormone. Thyroid-stimulating hormone promotes thyroid hormone release. Rising thyroid hormone then inhibits the hypothalamus and pituitary.

The endocrine pancreas illustrates antagonistic regulation. When blood glucose rises, beta cells release insulin. Insulin promotes glucose uptake in insulin-sensitive tissues and supports glycogen, fat, and protein synthesis. When blood glucose falls, alpha cells release glucagon. Glucagon promotes hepatic glucose production, especially through glycogen breakdown and gluconeogenesis.

The adrenal response links neural and endocrine control. Sympathetic input stimulates the adrenal medulla to release epinephrine and norepinephrine rapidly. The hypothalamic-pituitary-adrenal axis produces a slower cortisol response. Cortisol supports fuel availability and modifies immune activity, but persistent excess can have harmful effects.

Integration

Neural and endocrine systems solve different timing problems. Neurons deliver targeted signals over milliseconds to seconds. Hormones distribute signals more broadly over seconds to days. They continually interact. The hypothalamus translates neural information into endocrine commands, while hormones alter brain function and behaviour.

Sensory coding and motor control

Sensory receptors convert physical or chemical energy into electrical signals. This process is transduction. Receptors are selectively sensitive rather than perfectly specific. Photoreceptors respond best to light, mechanoreceptors to deformation, chemoreceptors to chemicals, and nociceptors to potentially damaging stimuli. Stimulus intensity can be encoded by action-potential frequency and by recruitment of additional receptors. Location is encoded by the receptive field and pathway carrying the signal.

Adaptation reduces the response to a constant stimulus. Rapidly adapting receptors emphasise change, vibration, or movement. Slowly adapting receptors continue to report magnitude or position. Pain can become more sensitive rather than less sensitive after injury, because inflammatory mediators and central pathways lower response thresholds.

Motor control is hierarchical and distributed. Spinal circuits coordinate reflexes and patterned activity. The brainstem supports posture, balance, eye movement, breathing, and other essential functions. The cerebral cortex contributes voluntary planning and skilled action. The basal nuclei help select and scale actions. The cerebellum compares intended with actual movement and supports error correction and motor learning.

A stretch reflex illustrates feedback control. Muscle stretch activates muscle spindles. Sensory neurons excite motor neurons serving the same muscle,

causing contraction that opposes the stretch. Reciprocal inhibition reduces activity in the antagonist muscle. Descending pathways can adjust reflex gain according to the task.

Endocrine axes in more detail

Hormone concentration depends on secretion, distribution, binding, metabolism, and excretion. Many lipid-soluble hormones circulate bound to carrier proteins. The bound fraction acts as a reservoir, while the free fraction can usually enter tissues and interact with receptors. Hormones are removed mainly by the liver, kidneys, and target tissues. Pulsatile or circadian secretion means that timing can be as important as average concentration.

The hypothalamic-pituitary-adrenal axis begins with corticotropin-releasing hormone. This stimulates adrenocorticotropic hormone, which stimulates cortisol production by the adrenal cortex. Cortisol supports blood glucose, vascular responsiveness, and adaptation to stress. It also restrains inflammatory activity. Excess cortisol can promote muscle protein breakdown, high blood glucose, central fat accumulation, bone loss, and immune suppression. Deficiency can impair blood-pressure support and stress tolerance.

Growth hormone is secreted in pulses, especially during sleep. It directly changes metabolism and stimulates production of insulin-like growth factor one, largely from the liver and local tissues. These signals support growth and protein synthesis. Prolactin supports milk production and is unusual because hypothalamic dopamine tonically inhibits its release.

Calcium regulation demonstrates multi-organ control. Falling ionised calcium stimulates parathyroid hormone. Parathyroid hormone increases renal calcium conservation, reduces renal phosphate reabsorption, and promotes formation of active vitamin D. Active vitamin D increases intestinal calcium and phosphate absorption. Bone participates as a dynamic mineral reservoir, but chronic hormonal imbalance can weaken it.

Receptors, dose-response, and interaction

The magnitude of a response depends on hormone concentration, receptor abundance, receptor affinity, and downstream machinery. At low concentrations, increasing hormone often increases response. As receptors and pathways approach saturation, additional hormone has less effect. A maximal response does not require every receptor to be occupied if spare receptors are present.

Hormones can interact permissively, synergistically, or antagonistically. Thyroid hormone has a permissive effect on catecholamine responsiveness. Glucagon, epinephrine, and cortisol can act synergistically to support blood glucose during stress. Insulin and glucagon are antagonistic in several metabolic pathways, but both are necessary for flexible regulation.

Disease can arise from hormone excess, hormone deficiency, receptor resistance, abnormal transport, or loss of feedback. Measuring one hormone in isolation may be misleading. In primary thyroid failure, thyroid hormone is low while thyroid-stimulating hormone is usually high because feedback inhibition is reduced. In pituitary failure, both may be inappropriately low. The pattern localises the disturbance.

Neural plasticity allows circuits to change with use. Synapses can strengthen or weaken, dendritic structure can change, and networks can reorganise after learning or injury. Short-term changes often modify existing channels or transmitter release. Long-term memory commonly requires altered gene expression and protein synthesis. Plasticity is essential for adaptation, but it can also maintain maladaptive states such as chronic pain, addiction, or persistent fear. The nervous system is therefore neither fixed wiring nor unlimited flexibility; change is constrained by cell type, developmental stage, activity, and environment.

TTS module 2: Electrochemical gradients, synaptic computation, and receptor adaptation

Equilibrium potentials and membrane conductance

Membrane voltage emerges from both ion gradients and selective conductance. An ion's equilibrium potential is the voltage at which its electrical and concentration forces balance. If a membrane were permeable only to

potassium, its voltage would approach the potassium equilibrium potential. A resting neuron is mostly, but not exclusively, permeable to potassium, so its resting potential lies nearer the potassium equilibrium potential than the sodium equilibrium potential. Chloride permeability and electrogenic transport also contribute.

Changing extracellular potassium has a particularly strong effect because it alters the gradient for potassium leaving the cell. Hyperkalaemia makes the resting potential less negative. Mild depolarisation may initially increase excitability, but sustained depolarisation inactivates voltage-gated sodium channels and can ultimately impair conduction. Hypokalaemia generally makes excitation more difficult and also alters cardiac repolarisation. The clinical effect depends on speed, magnitude, acid-base state, medicines, and the condition of excitable tissues.

The sodium-potassium pump maintains gradients rather than directly generating each action potential. During brief activity, existing gradients supply the ion movement. Over longer periods, pump activity restores what repeated signalling dissipates. Pump failure during energy deprivation therefore has a delayed but catastrophic effect: gradients decay, cells swell, calcium regulation fails, and electrical signalling becomes unreliable.

Current depends on conductance and driving force. Opening a channel does not guarantee that its ion will move in one fixed direction; movement depends on membrane voltage relative to that ion's equilibrium potential. This is why opening chloride channels can inhibit a mature neuron by holding voltage near the chloride equilibrium potential, while altered chloride gradients in developing or diseased cells can change the effect.

Action-potential diversity and conduction failure

Action potentials vary among tissues because channel types, timing, and expression differ. A peripheral axon uses a rapid sodium-dependent spike for communication. Ventricular muscle has a prolonged plateau created partly by calcium entry, supporting contraction and a long refractory period that prevents tetanic activation. Pacemaker cells depolarise spontaneously because several inward and outward currents change during diastole. The shared principle is regenerative channel opening, but the waveform is adapted to tissue function.

Refractoriness has two components. During the absolute refractory period, many sodium channels are inactivated and another propagated spike cannot begin. During the relative refractory period, some channels have recovered, but persistent potassium conductance or incomplete recovery means a stronger stimulus is required. Refractory behaviour limits firing frequency, separates impulses, and shapes re-entry in cardiac tissue.

Conduction can fail through several mechanisms. Demyelination increases current leak and may prevent the next node from reaching threshold. Compression impairs membrane function and local blood flow. Ischaemia reduces energy supply. Local anaesthetics bind voltage-gated sodium channels and preferentially affect channels that open repeatedly, producing use-dependent block. Axons with small diameter and high firing activity are often affected before larger motor fibres, although exact clinical patterns depend on the drug, concentration, and anatomy.

Temperature also changes channel kinetics and conduction. Cooling slows nerve activity and can prolong cardiac electrical intervals. Fever may expose channel disorders or increase seizure susceptibility in predisposed patients. These effects illustrate that excitability is a property of proteins operating in a chemical environment, not an abstract electrical circuit isolated from metabolism.

Synapses as computations

Synaptic strength is controlled before and after transmitter release. Presynaptic calcium entry, vesicle availability, autoreceptors, and inhibitory axo-axonic input determine how much transmitter reaches the cleft. Postsynaptic receptor number, receptor type, membrane location, and intracellular signalling determine the response. Uptake transporters and degrading enzymes control duration. A medicine can therefore alter signalling without directly mimicking the natural transmitter.

Spatial summation combines inputs arriving at different locations. Temporal summation combines inputs arriving close together in time. Their effect depends on dendritic cable properties and distance from the trigger zone. An excitatory input on a distal dendrite may weaken as it spreads, while inhibition placed near the cell body can strongly control whether integrated current reaches the axon initial segment.

Inhibition is not merely the opposite of excitation. Hyperpolarising inhibition moves voltage away from threshold. Shunting inhibition increases membrane conductance and dissipates nearby excitatory current even if voltage changes little. Disinhibition removes an inhibitory influence and can activate a pathway without adding direct excitation. These arrangements allow neural networks to select signals, sharpen contrast, coordinate timing, and prevent runaway activity.

Short-term synaptic plasticity reflects recent activity. Repeated impulses may temporarily facilitate release because calcium remains in the terminal, or depress transmission because releasable vesicles are depleted. Long-term strengthening and weakening involve receptor trafficking, kinase activity, gene expression, and structural change. Learning uses these mechanisms, but similar plasticity can amplify nociceptive pathways or reinforce compulsive behaviour.

Receptor families and biased responses

Ligand-gated ion channels translate binding directly into rapid current. G-protein-coupled receptors activate intracellular effectors that can amplify signals, regulate channels, alter enzymes, and change transcription. Enzyme-linked receptors, including many receptor tyrosine kinases, organise phosphorylation networks important in growth and metabolism. Intracellular receptors bind lipid-soluble signals and regulate gene expression, but some also produce faster non-genomic effects.

One receptor can produce different effects in different cells because downstream machinery differs. Activation of a beta adrenergic receptor in cardiac tissue increases rate and force, while related signalling in some smooth muscle promotes relaxation. The ligand does not contain a complete instruction. The cell interprets it through receptor subtype, coupling proteins, enzymes, channels, and gene expression.

Agonism and antagonism are also more nuanced than simple on and off. A partial agonist activates a receptor but produces a lower maximal response than a full agonist in the same system. In the presence of a full agonist, it may reduce overall activation by occupying receptors. An inverse agonist reduces constitutive receptor activity. Allosteric modulators bind away from the main ligand site and change receptor responsiveness.

Receptors can favour different downstream pathways depending on the ligand and cellular context, a phenomenon often called biased signalling. This helps explain why two drugs acting at the same named receptor may differ in therapeutic and adverse effects. Clinical response also depends on receptor reserve: tissues with spare receptors may preserve a maximal response despite partial receptor blockade.

Desensitisation, withdrawal, and diagnostic interpretation

Persistent stimulation commonly triggers phosphorylation, uncoupling, internalisation, or reduced receptor synthesis. The same dose then produces less effect. Downstream adaptation may occur even when receptor number is unchanged. Tolerance can therefore be pharmacokinetic, receptor-level, cellular, behavioural, or a mixture. Escalating dose without identifying the mechanism can increase toxicity while producing diminishing benefit.

Chronic blockade can produce receptor up-regulation or increased pathway sensitivity. Abrupt withdrawal then exposes the adapted system to endogenous transmitter and may cause rebound activity. This principle contributes to rebound tachycardia after sudden withdrawal of some sympathetic antagonists and to other withdrawal syndromes. Medicines that alter neural or endocrine signalling often require planned tapering according to current guidance.

Endocrine tests are snapshots of dynamic axes. A result should be interpreted with time of day, pulsatility, binding proteins, acute illness, medicines, pregnancy, renal or hepatic clearance, and the expected feedback relationship. A hormone within the population reference interval may still be inappropriate for the paired controlling hormone. Dynamic stimulation or suppression tests ask whether an axis can respond, but they also depend on a valid protocol and suitable clinical context.

The unifying method is to move from gradient to channel, channel to cell, cell to network, and network to whole-body feedback. At each level, ask what drives the signal, what terminates it, how repeated exposure changes responsiveness, and what reserve remains. This prevents a receptor name or laboratory value from being mistaken for a complete physiological explanation.

Retrieval prompts

One. Why does opening sodium channels depolarise a typical neuron?

Two. How does myelin increase conduction speed?

Three. What is the functional difference between an excitatory and inhibitory postsynaptic potential?

Four. Why do steroid hormones usually have slower but longer-lasting effects than peptide hormones?

Five. Trace negative feedback in the thyroid axis.

Concise answers

One. Sodium follows its electrochemical gradient into the cell, making the inside less negative.

Two. It reduces current loss and allows action potentials to regenerate mainly at nodes.

Three. Excitation increases the chance of reaching threshold; inhibition decreases or stabilises that chance.

Four. Steroid hormones alter transcription and protein synthesis, while peptide hormones often modify existing proteins through membrane signalling.

Five. Thyrotropin-releasing hormone drives thyroid-stimulating hormone, which drives thyroid hormone; thyroid hormone inhibits the first two levels.

Source map

Original private-study synthesis. Primary references: OpenStax Anatomy and Physiology 2e, chapters 3 and 12 through 17; Guyton and Hall Textbook of Medical Physiology, membrane, neural, and endocrine chapters; Katzung Basic and Clinical Pharmacology, receptor and autonomic pharmacology. OpenStax content is licensed under Creative Commons Attribution NonCommercial ShareAlike 4.0.

Chapter 3: Genes, Proteins, Adaptation, Injury, and Cell Death

Orientation

Disease often begins when a cell cannot maintain its structure, energy supply, information, or environment. The outcome depends on the nature of the stress, its intensity and duration, the affected cell type, and the cell's previous state. A mild stress may produce adaptation. A more severe but brief stress may produce reversible injury. Persistent or overwhelming injury causes death by necrosis, apoptosis, or related pathways. These cellular events scale upward into tissue dysfunction, clinical signs, and organ failure.

Information flow and protein function

Deoxyribonucleic acid stores information in nucleotide sequence. A gene includes coding information and regulatory regions that determine where, when, and how strongly it is expressed. Transcription produces ribonucleic acid. Messenger ribonucleic acid is processed, exported, and translated by ribosomes. Transfer ribonucleic acid matches codons with amino acids, while ribosomal ribonucleic acid contributes to peptide-bond formation.

The simple phrase from gene to protein is useful but incomplete. One gene can produce different transcripts through alternative splicing. Ribonucleic acid can be edited, localised, stabilised, or degraded. A protein can be cleaved, phosphorylated, glycosylated, folded with chaperones, transported into an organelle, assembled with partners, or destroyed. Cellular phenotype therefore depends on regulation at many levels rather than on sequence alone.

Gene expression responds to developmental programs, hormones, nutrients, oxygen, mechanical forces, inflammation, and neural activity. Transcription factors bind regulatory deoxyribonucleic acid. Chromatin accessibility is influenced by histone modification and deoxyribonucleic acid methylation. These epigenetic mechanisms can produce stable changes in expression without changing nucleotide sequence, although they remain biologically regulated rather than acting as a mystical layer above genetics.

Mutation and inheritance

A mutation can substitute one base, insert or delete sequence, alter copy number, rearrange chromosomes, or change chromosome number. Its effect depends on location and consequence. A synonymous coding change may preserve the amino acid. A missense change substitutes one amino acid. A nonsense change introduces a stop codon. A frameshift alters downstream codons. Variants in promoters, splice sites, enhancers, or non-coding ribonucleic acid can change expression without altering the encoded amino-acid sequence.

Dominant means that one altered allele can influence phenotype; it does not mean common, severe, or beneficial. Recessive conditions usually require pathogenic variants in both alleles. X-linked inheritance depends on sex chromosomes and X inactivation. Mitochondrial deoxyribonucleic acid is transmitted mainly through the oocyte, but heteroplasmy can produce variable severity because cells contain mixtures of normal and altered mitochondrial genomes.

Penetrance is the proportion of people with a genotype who express the associated phenotype. Expressivity describes variation in degree or form. Environment, age, modifier genes, mosaicism, and chance events can all alter presentation. Most common diseases are multifactorial: many variants interact with behaviour, exposures, development, and ageing.

Cellular adaptation

Hypertrophy is increased cell size. It occurs when cells face increased workload or hormonal stimulation and have limited capacity to divide. Skeletal muscle hypertrophies with resistance training. The left ventricle hypertrophies in response to pressure overload. Initially this can preserve function; later, increased oxygen demand, fibrosis, altered gene expression, and impaired relaxation may make the adaptation harmful.

Hyperplasia is increased cell number. It may be physiological, as in glandular breast tissue during pregnancy, or pathological, as in excessive endometrial proliferation under unopposed oestrogen. Hyperplasia remains controlled by growth signals and is not itself cancer, although some forms increase malignant risk.

Atrophy is reduced cell size and sometimes cell number. Causes include reduced workload, denervation, diminished blood supply, malnutrition, loss of hormonal stimulation, pressure, and ageing. Protein synthesis falls, protein degradation rises, and autophagy may recycle components. Atrophy conserves resources but reduces reserve.

Metaplasia is a reversible replacement of one mature cell program by another better suited to chronic stress. In smokers, ciliated respiratory epithelium may be replaced by squamous epithelium that tolerates irritation but loses mucociliary function. Persistent stress can progress to dysplasia, in which growth and maturation become disordered. Dysplasia is not synonymous with invasive cancer, but severe dysplasia may be pre-malignant.

Reversible cell injury

The central disturbance in many injuries is impaired adenosine triphosphate production. Ischaemia reduces both oxygen and substrate delivery and prevents waste removal. Hypoxaemia reduces oxygen but may preserve blood flow. Toxins may damage mitochondria directly. Adenosine triphosphate depletion weakens ion pumps, allowing sodium and water to enter cells and potassium to leave. Cells and organelles swell. Calcium accumulates in the cytosol. Anaerobic glycolysis increases, glycogen falls, lactate rises, and intracellular acidity alters proteins and chromatin.

The endoplasmic reticulum can swell and ribosomes detach, reducing protein synthesis. Misfolded proteins activate the unfolded-protein response, which attempts to restore folding, slow translation, and increase degradation. If stress cannot be corrected, death pathways activate. Fatty change can appear in organs handling lipid metabolism, especially the liver and heart.

Reversible injury has a recovery window. Restoring flow can rescue cells before membranes and mitochondria are irreversibly damaged. However, reperfusion can also add injury through reactive oxygen species, calcium loading, inflammation, and abrupt mitochondrial changes. This does not mean reperfusion should be avoided; clinically, timely restoration of perfusion is often essential.

Irreversible injury and necrosis

Irreversibility is associated with failure to restore mitochondrial function and profound membrane damage. Loss of plasma-membrane integrity releases intracellular contents, which activate inflammation. Lysosomal leakage contributes to digestion of cellular components. Nuclear change progresses from chromatin condensation, called pyknosis, to fragmentation, called karyorrhexis, and dissolution, called karyolysis.

Necrosis describes cell death with membrane disruption and tissue reaction. Coagulative necrosis typically preserves tissue outlines temporarily and is common after ischaemia in solid organs other than the brain. Liquefactive necrosis produces digestion into liquid material and is typical of brain infarction and many abscesses. Caseous necrosis has a friable appearance classically associated with tuberculosis. Fat necrosis follows lipase release or direct trauma to fat. Fibrinoid necrosis is seen in certain immune-mediated vascular injuries. Gangrene is a clinical term rather than a distinct microscopic mechanism.

Intracellular proteins released by necrotic cells can serve as biomarkers. Troponin indicates myocardial injury in the appropriate setting but does not by itself specify the cause. Aminotransferases indicate hepatocyte injury but are not direct measurements of liver function. Biomarkers must be interpreted with timing, magnitude, tissue specificity, clearance, and clinical context.

Apoptosis and regulated death

Apoptosis removes cells with minimal leakage. Cells shrink, chromatin condenses, fragments are packaged into membrane-bound bodies, and phagocytes clear them. The intrinsic pathway responds to deoxyribonucleic acid damage, growth-factor withdrawal, and severe organelle stress through mitochondrial control proteins and cytochrome c release. The extrinsic pathway begins with death receptors. Both activate caspases that dismantle the cell.

Apoptosis is physiological during development, immune regulation, hormone-dependent tissue involution, and routine turnover. It is also protective when cells are irreparably damaged or infected. Too little apoptosis can support cancer or autoimmunity. Excess apoptosis can contribute to

neurodegeneration, immune depletion, or tissue loss. Other regulated death pathways, including necroptosis and pyroptosis, combine signalling with membrane disruption and inflammation.

Autophagy delivers cellular components to lysosomes for recycling. It supports survival during nutrient deprivation and removes damaged organelles. Persistent or dysregulated autophagy may also accompany disease. The context, degree, and timing matter more than labelling a pathway simply good or bad.

Clinical integration

Cell injury is not diagnosed from one symptom or laboratory value. The clinician asks what stress is present, which tissue is vulnerable, whether injury is reversible, how much reserve remains, and whether treatment can remove the cause. Acute coronary occlusion threatens myocardium through ischaemia. Sepsis combines inflammatory signalling, microvascular dysfunction, mitochondrial stress, and altered metabolism. A drug overdose may generate a toxic metabolite, deplete protective molecules, and produce zonal liver necrosis. Chronic hypertension drives adaptive hypertrophy before contributing to fibrosis and failure.

Treatment logic follows mechanism: restore oxygen and perfusion, remove toxins, correct temperature or metabolic disturbance, control infection or inflammation when appropriate, and support organ function while recovery occurs. Intervention itself can cause harm, so dose, rate, timing, comorbidity, and monitoring matter.

TTS module 2: Genome maintenance, organelle stress, and tissue consequences

From variant to phenotype

A genetic variant causes disease only through a chain of biological consequences. The first question is whether it changes the amount, location, timing, or function of a gene product. The next is whether other proteins or pathways can compensate. The final clinical phenotype depends on the tissue in which the gene is important, the developmental stage, environmental exposure, and the reserve of the affected organ.

Loss-of-function variants reduce useful activity. A single normal allele may provide enough product, producing recessive inheritance, or the amount may be insufficient, producing haploinsufficiency and a dominant phenotype. A dominant-negative protein interferes with the normal product, often because both must assemble into a larger structure. Gain-of-function variants create excessive, constitutive, or novel activity and are commonly dominant. These mechanisms are more informative than the inheritance label alone because they suggest different therapeutic strategies.

Mosaicism arises when a variant occurs after fertilisation, leaving genetically distinct cell populations. Earlier events affect more tissues. Somatic mosaicism may produce segmental disease or cancer without being present throughout the body. Germline mosaicism can permit recurrence in siblings even when a parental blood test is negative. In mitochondrial disease, heteroplasmy and tissue-specific segregation create threshold effects: symptoms appear when the proportion of dysfunctional mitochondria exceeds what a tissue can tolerate.

Variant interpretation combines population frequency, predicted molecular consequence, segregation, functional evidence, and fit with the phenotype. A variant of uncertain significance is not a diagnosis and should not be treated as pathogenic merely because it was found. Reclassification is possible as evidence accumulates. Testing can also reveal incidental or familial information, so consent and genetic counselling are part of safe investigation.

Protein quality control

Protein production is intrinsically error-prone. Chaperones assist folding, while surveillance systems identify molecules that fail to achieve a stable conformation. Misfolded cytosolic proteins are often tagged with ubiquitin and degraded by proteasomes. Aggregates and damaged organelles can be enclosed within autophagosomes and delivered to lysosomes. These pathways prevent toxic accumulation and recycle amino acids and lipids.

The endoplasmic reticulum monitors proteins entering secretory and membrane pathways. Accumulation of unfolded protein activates a coordinated response that temporarily reduces translation, increases chaperone capacity, and promotes degradation. If balance is restored, the cell survives. Persistent stress activates inflammatory and apoptotic pathways.

This mechanism contributes to disorders involving secretory cells, metabolic overload, neurodegeneration, and some toxic injuries.

Proteostasis declines with ageing. Oxidative modifications accumulate, chaperone and degradation capacity may fall, and damaged proteins can seed further aggregation. Different proteins produce different patterns of injury according to which cells express them and how aggregates interact with membranes, synapses, or intracellular transport. The presence of an aggregate does not by itself prove it is the only toxic species; smaller soluble intermediates may also disrupt function.

Mitochondria, oxidants, and metabolic adaptation

Mitochondria integrate energy production, calcium handling, biosynthesis, and cell-death signalling. They continually divide, fuse, move, and undergo quality control. Damaged regions can be separated by fission and removed through mitophagy. Failure of this process allows poorly functioning mitochondria to generate less adenosine triphosphate and more reactive oxygen species.

Reactive oxygen species are not exclusively harmful. At controlled levels they participate in signalling and host defence. Injury occurs when production exceeds antioxidant and repair capacity. Sources include the respiratory chain, inflammatory cells, radiation, and metabolism of some chemicals. Superoxide dismutase, catalase, glutathione systems, and dietary or endogenous antioxidant molecules limit damage. Excess oxidants modify lipids, proteins, and nucleic acids and can open mitochondrial permeability pathways that commit the cell to death.

Cells adapt to limited oxygen partly through hypoxia-responsive transcription. They increase glycolytic capacity, alter mitochondrial activity, promote angiogenic signalling, and influence red-cell production through kidney-derived erythropoietin. These responses support survival but cannot fully replace oxygen-dependent energy production in highly aerobic tissues. Chronic activation can also contribute to vascular remodelling or tumour adaptation.

Cancer cells frequently reorganise metabolism to support growth rather than simply maximising adenosine triphosphate. Increased glucose uptake and aerobic glycolysis provide intermediates for nucleotides, amino acids, and lipids. This metabolic flexibility is shaped by oncogenic signalling, nutrient

availability, stromal interaction, and oxygen gradients. It is not evidence that oxygen is irrelevant to cancer; tumours contain changing regions of perfusion, hypoxia, and metabolic cooperation.

Membrane injury and intracellular calcium

Cell membranes preserve gradients and compartmentalise enzymes. Lipid peroxidation makes membranes leaky. Cytoskeletal damage weakens the plasma membrane. Reduced phospholipid synthesis and increased breakdown impair repair. Mitochondrial membrane failure collapses energy production, while lysosomal disruption releases hydrolases into an already acidified cell.

Calcium is a powerful messenger because cytosolic concentration is normally kept far below extracellular and storage-compartment concentrations. Injury permits calcium entry and release from intracellular stores. Elevated cytosolic calcium activates phospholipases that damage membranes, proteases that break down structural proteins, endonucleases that fragment nucleic acids, and enzymes that consume energy. Mitochondrial calcium loading worsens permeability and oxidant production. Calcium disturbance therefore links energy failure with structural destruction.

The point of no return is not a single universal event visible at the bedside. It reflects interacting failures of mitochondria, membranes, ion control, and repair. Different cells cross that threshold at different times. This is why early reperfusion can produce dramatic recovery in some tissue while neighbouring regions remain irreversibly damaged.

Patterns of regulated death

Apoptosis is driven by an organised caspase cascade and usually preserves membrane integrity until fragments are cleared. The intrinsic pathway balances pro-survival and pro-death members of the B-cell lymphoma two protein family. Severe deoxyribonucleic acid damage activates checkpoint proteins that pause the cell cycle, promote repair, or trigger apoptosis if repair fails. The extrinsic pathway begins when death receptors recruit intracellular activating complexes.

Other regulated pathways blur the older division between apoptosis and accidental necrosis. Necroptosis produces membrane rupture through a

kinase-controlled pathway when particular receptor signals cannot complete apoptosis. Pyroptosis follows inflammasome activation and pore formation, releasing inflammatory mediators and intracellular contents. Ferroptosis depends on iron-associated lipid peroxidation and failure of protective antioxidant systems. These distinctions matter in research and may create therapeutic targets, but tissue injury commonly contains several pathways at once.

Dead cells influence their neighbours. Necrotic release activates innate recognition and recruits inflammation. Apoptotic cells expose signals that promote quiet phagocytic clearance and can support resolution. If clearance is overwhelmed or defective, apoptotic material may undergo secondary membrane breakdown. The tissue response therefore depends not only on how a cell dies but also on how rapidly debris is removed.

Tissue architecture, biomarkers, and recovery

Organ outcome depends on the extracellular scaffold, microcirculation, stem-cell compartment, and spatial distribution of injury. A small lesion in the brainstem may be devastating because of location, while a larger injury in a redundant tissue may initially produce little dysfunction. Patchy injury can leave viable bridges that support recovery. Diffuse microvascular or toxic injury may impair an entire organ without one obvious focal lesion.

Laboratory biomarkers sample this process indirectly. Release requires cellular injury and access to the circulation. Concentration then reflects the amount released, distribution volume, continuing injury, and clearance. Serial change can distinguish an evolving event from a stable chronic elevation. A highly tissue-associated marker identifies where injury likely occurred, but mechanism still requires history, examination, electrocardiography, imaging, microbiology, toxicology, or other contextual evidence.

Recovery may involve restoration of function in reversibly injured cells, proliferation of surviving cells, activation of progenitors, remodelling of circuits, and scar formation. Apparent clinical improvement can precede complete structural repair, while normalisation of a biomarker can occur despite persistent loss of reserve. Follow-up should therefore assess function, recurrence risk, and the cause of the original stress rather than stopping when one measurement returns to normal.

The practical sequence is to identify the stress, locate the vulnerable cellular process, judge reversibility, remove the cause, support energy and perfusion, and monitor the tissue response over time. Molecular detail becomes clinically useful when it clarifies why injury occurred, what can still be rescued, and what future stress the organ may no longer tolerate.

Retrieval prompts

One. Why can the same stress produce adaptation in one cell and death in another?

Two. Distinguish hypertrophy, hyperplasia, atrophy, metaplasia, and dysplasia.

Three. Trace how adenosine triphosphate depletion causes cellular swelling.

Four. What two broad changes make injury irreversible?

Five. Contrast necrosis with apoptosis.

Six. Why does an elevated tissue biomarker not automatically identify the cause of injury?

Concise answers

One. Outcome depends on stress intensity and duration, cell type, metabolic demand, blood supply, prior state, and adaptive capacity.

Two. Hypertrophy increases size; hyperplasia increases number; atrophy reduces size and sometimes number; metaplasia replaces one mature program with another; dysplasia is disordered growth and maturation.

Three. Reduced adenosine triphosphate weakens sodium pumps, sodium accumulates intracellularly, and water follows osmotically.

Four. Inability to restore mitochondrial function and severe membrane damage.

Five. Necrosis disrupts membranes and provokes inflammation; apoptosis packages controlled cellular dismantling for phagocytic clearance.

Six. Biomarkers show injury and have specific kinetics, but different mechanisms can injure the same tissue.

Source map

Original private-study synthesis informed by OpenStax Biology 2e and Anatomy and Physiology 2e; Robbins Basic Pathology, cellular adaptation, injury, and death; Guyton and Hall Textbook of Medical Physiology, cellular function and homeostasis; and Katzung Basic and Clinical Pharmacology for examples of toxic and therapeutic mechanisms.

Chapter 4: Inflammation, Healing, Immunity, and Infection

Orientation

Inflammation is a protective response to infection, tissue injury, foreign material, and disturbed cells. It recruits fluid, proteins, and leukocytes to contain threats and begin repair. The same response can damage tissue when excessive, misdirected, prolonged, or poorly resolved. Immunity adds specificity and memory, while healing restores continuity through regeneration and scar formation. Infection is one cause of inflammation, but the two terms are not interchangeable.

Recognition and immediate defence

Physical barriers are the first defence. Skin provides keratin, lipids, low surface moisture, resident microbes, and continuous shedding. Respiratory mucus traps particles while cilia move them toward the pharynx. Cough, gastric acid, bile, urine flow, tears, and antimicrobial peptides reduce microbial access. Barrier damage from wounds, devices, burns, impaired perfusion, or inflammation creates entry points.

Innate immune cells recognise recurring microbial structures and signals released by damaged cells. Pattern-recognition receptors activate inflammatory gene expression, antimicrobial pathways, and inflammasomes. Complement proteins circulate as inactive precursors. Activation can mark targets for phagocytosis, recruit leukocytes, and form membrane attack complexes. Complement is powerful and therefore tightly regulated to limit host injury.

Mast cells, macrophages, dendritic cells, and tissue-resident lymphocytes can respond rapidly. Histamine increases arteriolar dilation and venular permeability. Lipid mediators, cytokines, chemokines, complement fragments, and bradykinin coordinate vascular change, pain, fever, leukocyte recruitment, and systemic responses.

Acute inflammation

Increased blood flow produces redness and warmth. Increased microvascular permeability allows protein-rich fluid to enter tissue, producing swelling. Pain reflects mediators and mechanical pressure. Loss of function can follow pain, swelling, and direct tissue damage. These classic features are useful patterns, not requirements for every inflammatory process.

Leukocyte recruitment begins as blood flow slows and cells move toward the vessel wall. Selectins support rolling. Chemokines activate leukocyte integrins, which bind endothelial adhesion molecules firmly. Leukocytes then cross the vessel wall and follow chemical gradients. Neutrophils commonly dominate early bacterial inflammation. Monocytes arrive and differentiate into macrophages, which phagocytose material and organise repair. The timing and dominant cell type vary with the cause.

Phagocytosis requires recognition, engulfment, killing, and digestion. Antibodies and complement can opsonise a target, making recognition more efficient. Reactive oxygen species, nitric oxide, granule enzymes, acidity, and antimicrobial proteins kill microbes. These mechanisms can also injure surrounding tissue when released extracellularly.

Pus contains neutrophils, necrotic debris, protein-rich fluid, and often microbes. An abscess walls off a local collection and may require drainage because antibiotics penetrate poorly and source control is incomplete. Inflammation can resolve, organise into scar, form an abscess, or progress to chronic inflammation.

Adaptive immunity

Dendritic cells link innate recognition with adaptive responses by processing antigen and presenting peptides to T lymphocytes. Major histocompatibility complex class one displays intracellular peptides to CD8 positive T cells. Class two displays extracellularly derived peptides to CD4 positive T cells. Activation also requires co-stimulation and an appropriate cytokine environment; antigen recognition alone does not necessarily produce an effective response.

CD4 positive T cells coordinate immunity through specialised patterns. Some activate macrophages and support responses to intracellular organisms.

Others promote eosinophils and immunoglobulin E responses against parasites and in allergy. Others recruit neutrophils at barriers. Regulatory T cells restrain immune activation and support tolerance. CD8 positive T cells kill infected or abnormal cells through regulated cytotoxic mechanisms.

B cells recognise antigen through surface immunoglobulin. With appropriate signals they become plasma cells that secrete antibody or memory cells that respond rapidly on re-exposure. Immunoglobulin M appears early and efficiently activates complement. Immunoglobulin G is abundant in blood and tissues, supports neutralisation and opsonisation, and crosses the placenta. Immunoglobulin A protects mucosal surfaces and appears in secretions. Immunoglobulin E binds mast cells and contributes to parasite defence and immediate allergy.

Antibodies neutralise toxins and viruses, block attachment, activate complement, and mark targets for phagocytes. Affinity maturation and class switching improve function. Memory permits a faster and stronger response after vaccination or previous infection, although memory durability varies.

Immune injury and immune failure

Hypersensitivity describes immune-mediated tissue injury. Immediate hypersensitivity involves immunoglobulin E, mast-cell activation, and mediators that can cause urticaria, bronchospasm, gastrointestinal symptoms, or anaphylaxis. Antibody-mediated injury can target cell surfaces or extracellular structures. Immune complexes can deposit and activate complement. Delayed hypersensitivity is driven largely by T cells and activated macrophages.

Autoimmunity develops when tolerance fails and genetic susceptibility interacts with environmental triggers. Disease can be organ-specific or systemic. Mechanisms include autoantibodies, autoreactive T cells, immune complexes, complement activation, and cytokine-driven inflammation. A positive autoantibody test is not a diagnosis by itself; pre-test probability, titre, pattern, organ involvement, and alternative explanations matter.

Immunodeficiency may be inherited or acquired. Causes include malignancy, malnutrition, human immunodeficiency virus, diabetes, kidney or liver disease, loss of splenic function, immunosuppressive drugs, and extremes of age. The

pattern of infection can suggest the affected immune component. Neutropenia predisposes to invasive bacterial and fungal infection. T-cell impairment increases vulnerability to intracellular and opportunistic organisms. Antibody deficiency particularly affects responses to encapsulated bacteria.

Fever and systemic inflammation

Microbial products and inflammatory cytokines increase prostaglandin production in the hypothalamus, raising the temperature set point. The person initially feels cold, vasoconstricts, and may shiver until body temperature reaches the new target. When the set point falls, vasodilation and sweating promote heat loss. Fever differs from hyperthermia, in which heat production or environmental load exceeds heat-loss capacity without a regulated set-point rise.

The liver produces acute-phase proteins, including C-reactive protein and fibrinogen, while albumin can fall. Bone marrow output changes, metabolism shifts, and behaviour favours rest. These responses can be adaptive, but severe systemic inflammation causes endothelial dysfunction, vasodilation, capillary leakage, coagulation activation, myocardial depression, and abnormal cellular metabolism.

Sepsis is life-threatening organ dysfunction caused by a dysregulated response to infection. It is not defined by fever alone and can occur with normal or low temperature. Clinical concern rises with altered mental state, hypotension, rapid breathing, hypoxaemia, oliguria, mottled skin, elevated lactate, thrombocytopenia, or deteriorating organ function. Early priorities include recognising the syndrome, obtaining appropriate cultures without dangerous delay, administering suitable antimicrobials, controlling the source, supporting perfusion and oxygenation, and reassessing response.

Healing and fibrosis

Healing overlaps with inflammation. Haemostasis limits blood loss and creates a provisional matrix. Neutrophils and macrophages remove contamination and damaged material. Macrophages then shift signalling toward repair. Growth factors stimulate epithelial proliferation, fibroblast migration, extracellular-matrix production, and angiogenesis.

Regeneration restores specialised cells when surviving cells or stem cells can proliferate and the tissue scaffold remains suitable. Scar formation becomes dominant when damage is extensive, extracellular matrix is destroyed, cells cannot divide effectively, or inflammation persists. Fibroblasts produce collagen, while myofibroblasts contract the wound. Remodelling changes collagen composition and alignment. Tensile strength increases over months but usually does not return fully to that of uninjured tissue.

Healing is delayed by poor perfusion, infection, repeated trauma, foreign material, oedema, malnutrition, glucocorticoid exposure, diabetes, smoking, and advanced age. Local assessment considers wound depth, tissue viability, exudate, odour, surrounding skin, pain, vascular supply, pressure, and evidence of spreading infection. Systemic assessment considers nutrition, glycaemic control, immune function, medications, and capacity for self-care.

Excessive repair causes hypertrophic scars, keloids, adhesions, strictures, or organ fibrosis. Chronic fibrosis replaces functional tissue with extracellular matrix, distorts architecture, and reduces reserve. Examples include cirrhosis, pulmonary fibrosis, chronic kidney scarring, and adverse cardiac remodelling.

Antimicrobial and anti-inflammatory logic

Antimicrobial therapy should match the likely organism, site, severity, host factors, local resistance, and drug penetration. Cultures guide narrowing when reliable samples can be obtained. Dose depends on absorption, distribution, kidney and liver function, and pharmacodynamic target. Source control may require drainage, debridement, removal of an infected device, or relief of obstruction.

Anti-inflammatory drugs modify the host response rather than eliminating most infectious causes. Non-steroidal anti-inflammatory drugs reduce prostaglandin synthesis but can injure the gastrointestinal tract, kidneys, and cardiovascular system. Glucocorticoids broadly alter gene expression and immune function; they can be life-saving but increase infection, hyperglycaemia, bone loss, muscle weakness, and adrenal suppression. Immunomodulation requires a clear indication and monitoring plan.

TTS module 2: Resolution, immune architecture, and repair under stress

Inflammation is an active sequence

Inflammation is best understood as a changing programme rather than a fixed state. Recognition initiates vascular and cellular responses, recruited cells alter the local environment, and resolution must then actively switch the tissue from defence toward repair. Failure can occur at every stage. Weak recognition permits microbial spread. Excess recruitment damages host tissue. Inadequate source control keeps the initiating signal present. Defective resolution converts a useful acute response into chronic inflammation and fibrosis.

Tissue-resident cells determine the first response. Macrophages, mast cells, epithelial cells, endothelial cells, and stromal cells detect injury in different compartments. They release mediators that alter blood flow, permeability, adhesion molecules, pain sensitivity, and leukocyte production. The vascular endothelium is therefore not passive plumbing. It controls which proteins and cells leave blood, regulates tone and coagulation, and changes its behaviour in response to infection, hypoxia, cytokines, and shear stress.

Inflammatory mediators have overlapping effects and short half-lives. Histamine acts rapidly. Prostaglandins influence vasodilation, pain, and temperature regulation. Leukotrienes affect permeability, bronchial smooth muscle, and leukocyte recruitment. Cytokines coordinate local and systemic responses, while chemokines provide directional information. Redundancy makes host defence robust but also means blocking one mediator may not suppress an entire inflammatory syndrome.

Complement and phagocyte function

Complement can be activated through antibody-associated, lectin-associated, or alternative pathways, but the routes converge on cleavage products that amplify the reaction. Some fragments coat a microbial surface and improve phagocytosis. Others recruit and activate leukocytes. Terminal components assemble pores that are especially useful against susceptible Gram-negative organisms. Host cells express regulators that limit activation on their own surfaces.

Complement deficiency does not produce one uniform pattern. Early classical-pathway deficiency can impair immune-complex clearance and favour autoimmune disease. Terminal-pathway deficiency particularly increases susceptibility to invasive *Neisseria* infection. Loss of complement regulation may cause inappropriate host-cell damage, haemolysis, thrombosis, or kidney injury. These patterns show how a defence pathway can become pathogenic through absence or excess.

Neutrophils kill by engulfment, granule release, oxidant production, and formation of extracellular traps. These mechanisms are effective in a confined infection but hazardous when widespread. Activated neutrophils can injure endothelium and epithelium, obstruct small vessels, and release enzymes into tissue. Neutrophil count alone does not describe competence: marrow reserve, migration, phagocytosis, oxidative killing, and tissue access all matter.

Macrophages are more than late-arriving cleaners. They remove microbes and dead cells, present antigen, remodel extracellular matrix, and coordinate repair. Their behaviour exists along a spectrum shaped by local signals rather than in two rigid categories. Clearance of apoptotic cells tends to promote anti-inflammatory and repair signals. Persistent microbial products, crystals, foreign material, or autoimmune stimulation maintain inflammatory programmes.

Antigen presentation and tolerance

Adaptive immunity requires controlled activation. T-cell receptors recognise peptide presented by major histocompatibility molecules, but productive activation also depends on co-stimulation and cytokines. A naive T cell encountering antigen without the correct context may become unresponsive or regulatory. This reduces accidental activation against harmless or self-derived material.

Central tolerance removes or edits many strongly self-reactive lymphocytes during development in the thymus and bone marrow. Peripheral tolerance restrains cells that escape through anergy, deletion, inhibitory receptors, regulatory T cells, and restricted access to some antigens. Autoimmunity can arise when several safeguards fail in a genetically susceptible person, often

after environmental events alter inflammation, tissue exposure, or lymphocyte activation.

B-cell responses also depend on context. T-cell help supports class switching, affinity maturation, and durable memory for many protein antigens. Germinal centres select B cells whose mutated antibody genes bind antigen more effectively. This is powerful but risky because mutation and proliferation can generate autoreactivity or contribute to lymphoid malignancy. Checkpoints normally remove unsuitable clones.

Memory is not absolute protection. Pathogens mutate, immunity wanes, and some infections do not produce strong durable memory. Protection may still reduce severe disease without preventing all infection. Vaccination presents antigen in a controlled context designed to generate memory while avoiding the full risk of natural disease. The relevant schedule, contraindications, and recommendations must be checked against current local guidance.

Chronic inflammation and granulomas

Chronic inflammation combines ongoing injury, mononuclear cell infiltration, tissue destruction, and attempts at repair. Causes include persistent infection, autoimmune disease, prolonged toxic exposure, foreign material, and unresolved metabolic stress. Macrophages and lymphocytes activate each other through cytokines, while fibroblasts and vascular cells reshape tissue. The result may be simultaneous destruction and fibrosis.

A granuloma is an organised collection of activated macrophages, often with surrounding lymphocytes and sometimes multinucleated giant cells. It forms when the immune system attempts to contain material that is difficult to eradicate. Infection, foreign bodies, inflammatory disorders, and some drug reactions can all produce granulomas. Their presence narrows a differential but does not identify one cause. Microbiology, exposure history, distribution, histological features, and immune status remain essential.

Systemic consequences can outlast the local trigger. Cytokines alter appetite, sleep, muscle metabolism, iron handling, and liver protein synthesis. Persistent inflammation can produce anaemia through iron sequestration and reduced erythropoietic response, contribute to muscle wasting, worsen insulin resistance, and increase thrombotic risk. Treating only a laboratory marker without identifying the driver misses the mechanism.

Resolution and tissue repair

Resolution is an active biological process. Neutrophil recruitment stops, existing neutrophils undergo apoptosis, macrophages clear debris, vascular permeability normalises, and specialised lipid mediators help suppress further recruitment. Lymphatic drainage removes fluid and antigen. Failure of clearance leaves intracellular material available to stimulate innate and adaptive immunity.

Repair requires an adequate blood supply, viable cells, a supportive matrix, and controlled mechanical stress. Angiogenesis supplies oxygen and nutrients but newly formed vessels are initially leaky and fragile. Fibroblasts deposit provisional matrix and collagen. Myofibroblasts generate contraction. Matrix metalloproteinases and their inhibitors remodel the scar, so final strength depends on balanced synthesis, cross-linking, degradation, and alignment.

Wounds heal by different proportions of regeneration and contraction. Clean, closely opposed edges generally require less granulation tissue and produce a smaller scar. Large, contaminated, ischaemic, or tissue-deficient wounds heal with more inflammation, granulation, contraction, and scarring. Pressure and shear repeatedly damage tissue, particularly when sensation, mobility, nutrition, or perfusion is impaired.

Wound appearance must be interpreted in context. Redness immediately around a healing edge may reflect normal inflammation, while expanding erythema, increasing pain, purulent discharge, systemic deterioration, or tissue crepitus raises concern for infection. Ischaemic tissue may show little redness because it cannot mount a strong vascular response. Immunosuppressed patients may have severe infection with muted fever and leukocytosis.

Infection, source control, and immune modulation

Antimicrobial selection is only one part of infection treatment. An obstructed infected system, necrotic tissue, abscess, contaminated prosthesis, or leaking viscus can continue to seed inflammation despite an active drug. Source control reduces microbial burden and restores anatomy. Delay may permit ongoing toxin production, tissue destruction, and systemic dysregulation.

Samples are most useful when they represent the infected site and are obtained before antimicrobials without delaying urgent care. Surface colonisers can mislead. A positive molecular test may detect nucleic acid after viable organisms have declined, while a negative culture may follow prior therapy or inadequate sampling. Results must be reconciled with syndrome, host, anatomy, and response.

Immune suppression requires the same mechanistic discipline. Glucocorticoids reduce many inflammatory pathways but also impair glucose control, tissue repair, bone, muscle, and infection defence. Targeted biological therapies may block a cytokine, receptor, cell population, or intracellular enzyme, creating characteristic infection risks. Before and during treatment, clinicians consider vaccination, latent infection, organ function, blood counts, pregnancy, interactions, and signs of toxicity according to current guidance.

The central clinical distinction is between controlling a dangerous host response and preserving enough immunity to clear the trigger. In sterile autoimmune inflammation, suppression may remove the main driver. In infection, suppression without effective antimicrobial treatment and source control can worsen disease. In sepsis, both pathogen burden and dysregulated host physiology require attention. Repeated assessment decides whether the system is moving toward containment and repair or toward persistent injury and organ failure.

Retrieval prompts

One. Why are infection and inflammation not synonyms?

Two. Trace neutrophil recruitment from blood to injured tissue.

Three. How do innate and adaptive immunity connect?

Four. Contrast immunoglobulin M, G, A, and E.

Five. Why can severe systemic inflammation cause hypotension and organ dysfunction?

Six. What determines whether tissue regenerates or scars?

Concise answers

One. Infection is invasion by a pathogen; inflammation is a host response that can be triggered by infection or sterile injury.

Two. Margination and rolling are followed by chemokine activation, firm integrin adhesion, endothelial crossing, and chemotaxis.

Three. Innate cells recognise danger and dendritic cells present antigen with co-stimulation and cytokines to activate lymphocytes.

Four. M is early and activates complement; G dominates blood and supports memory and placental transfer; A protects mucosa; E activates mast cells in allergy and parasite responses.

Five. Vasodilation, capillary leakage, endothelial injury, coagulation, cardiac depression, and metabolic disturbance impair effective perfusion.

Six. Cell proliferative capacity, scaffold integrity, injury extent, perfusion, infection, inflammation, and systemic conditions.

Source map

Original private-study synthesis informed by OpenStax Microbiology, Biology 2e, Anatomy and Physiology 2e, and Medical-Surgical Nursing; Robbins Basic Pathology, inflammation, immune disease, and repair; Katzung and OpenStax Pharmacology for anti-inflammatory and antimicrobial principles; and Talley and O'Connor for clinical recognition and examination context.

Chapter 5: Pharmacokinetics, Pharmacodynamics, and Safe Prescribing

Orientation

Pharmacology connects a dose given to a patient with a concentration at a target and a biological response. Pharmacokinetics asks what the body does to a drug through absorption, distribution, metabolism, and excretion. Pharmacodynamics asks what the drug does to the body through receptors, enzymes, channels, transporters, and non-specific physical or chemical actions. Safe prescribing adds indication, evidence, patient factors, interactions, monitoring, adherence, cost, and a plan to stop or adjust treatment.

Routes and absorption

Intravenous administration delivers drug directly to the systemic circulation and has complete bioavailability. Oral administration is convenient but requires disintegration, dissolution, gastrointestinal stability, membrane passage, and survival through intestinal and hepatic first-pass metabolism. Bioavailability is the fraction of an administered dose reaching systemic circulation unchanged.

Absorption depends on formulation, surface area, blood flow, gastric emptying, intestinal motility, food, pH, transporters, and drug interactions. Lipid-soluble uncharged molecules often cross membranes readily, but many drugs rely on transport proteins. Weak acids and bases exist as charged and uncharged forms according to pH and their dissociation constant. Ionisation influences membrane passage and distribution, although clinical behaviour cannot be predicted from pH alone.

Sublingual, inhaled, transdermal, rectal, subcutaneous, and intramuscular routes solve different problems. Inhalation provides rapid delivery to a large surface and can target the lungs. Transdermal systems provide slow sustained delivery for suitable potent lipid-soluble drugs. Intramuscular and

subcutaneous absorption depends on local perfusion and may be unreliable in shock.

Distribution

After entering blood, drug distributes according to perfusion, capillary permeability, membrane passage, tissue binding, and plasma-protein binding. Albumin commonly binds acidic drugs, while alpha one acid glycoprotein binds several basic drugs. Only unbound drug is immediately available to cross membranes, interact with targets, and undergo filtration, but bound drug can dissociate and act as a reservoir.

Apparent volume of distribution relates the amount of drug in the body to its measured plasma concentration. A small volume suggests confinement mainly to plasma. A large volume suggests extensive tissue distribution or binding. It is a proportional concept, not a literal anatomical space.

Body composition changes distribution. Older adults often have less total body water and lean mass but more fat, altering concentrations and persistence. Oedema, ascites, pregnancy, burns, critical illness, hypoalbuminaemia, and obesity can change distribution. The blood-brain barrier restricts many polar compounds through tight junctions and active efflux. Inflammation can alter barrier permeability.

Clearance, metabolism, and half-life

Clearance is the virtual volume of plasma from which drug is removed per unit time. Total clearance sums removal by kidneys, liver, and other routes. For many drugs, elimination rate equals clearance multiplied by concentration. Half-life is the time for concentration to fall by half during a specified elimination phase. For first-order elimination, half-life depends on volume of distribution divided by clearance.

Hepatic elimination depends on liver blood flow, uptake, enzyme activity, biliary transport, and protein binding. Phase one reactions include oxidation, reduction, and hydrolysis, often involving cytochrome P450 enzymes. Phase two reactions conjugate drugs or metabolites with polar groups. Metabolism can inactivate a drug, activate a prodrug, or create active or toxic metabolites.

Enzyme inhibition can raise substrate concentrations quickly, while enzyme induction usually takes days because new proteins must be produced. Genetic variation, age, nutrition, inflammation, liver disease, smoking, alcohol, and interacting drugs alter metabolism. Grapefruit products inhibit intestinal metabolism of selected drugs, but the effect is drug-specific rather than universal.

Kidneys remove drug by glomerular filtration, active tubular secretion, and reabsorption. Only unbound drug is filtered. Kidney dysfunction can reduce clearance of parent drug or active metabolites. Estimated filtration guides many dose adjustments but is imperfect at extremes of body size, unstable kidney function, low muscle mass, and pregnancy. Dialysis removal depends on molecular size, protein binding, volume of distribution, and membrane properties.

Dosing over time

A loading dose rapidly achieves a target concentration and depends mainly on volume of distribution. A maintenance dose replaces drug removed and depends mainly on clearance. Repeated dosing produces accumulation until average input equals average elimination. With first-order kinetics, near steady state is usually reached after about four to five half-lives, regardless of dose.

Peak-to-trough fluctuation depends on dosing interval, absorption, distribution, and half-life. Short intervals reduce fluctuation but can complicate adherence. Modified-release formulations smooth concentrations but must not always be crushed. In zero-order elimination, a constant amount rather than a constant fraction is removed per unit time; capacity becomes saturated and small dose increases can cause disproportionate concentration rises.

Therapeutic drug monitoring is most useful when concentration relates predictably to response, the therapeutic range is narrow, clinical effect is difficult to measure directly, and substantial pharmacokinetic variability exists. Sampling time matters. A concentration without dose history, timing, adherence, kidney and liver function, and clinical response can mislead.

Receptors and response

An agonist binds and activates a receptor. A full agonist can produce the system's maximal response. A partial agonist produces less maximal effect even with full receptor occupancy and can functionally oppose a full agonist. An antagonist binds without activating and reduces agonist action. Competitive antagonism can often be overcome with increased agonist concentration; irreversible or non-competitive effects reduce achievable response.

Potency is the dose or concentration needed for a given effect. Efficacy is the maximum effect achievable. A more potent drug is not necessarily more effective, safer, or clinically preferable. Receptor occupancy does not always map linearly to response because signalling can amplify and spare receptors may exist.

Repeated exposure can cause tolerance through receptor desensitisation, down-regulation, altered signalling, increased metabolism, or physiological compensation. Tachyphylaxis is rapid loss of effect. Physical dependence means adaptation produces withdrawal when a drug is stopped. It is distinct from substance-use disorder, which involves impaired control, harmful use, and behavioural features.

Variability and interactions

Response varies with genetics, age, pregnancy, organ function, comorbidity, microbiome, diet, adherence, and concurrent substances. Paediatric dosing cannot always be scaled directly from adult weight because maturation changes clearance and sensitivity. Frail older adults are vulnerable to orthostatic hypotension, delirium, falls, bleeding, kidney injury, and anticholinergic effects.

Pharmacokinetic interactions change concentration. Pharmacodynamic interactions change combined effect without necessarily changing concentration. Two sedatives may produce additive respiratory depression. Several drugs that prolong cardiac repolarisation can increase arrhythmia risk. Anticoagulants combined with platelet inhibitors increase bleeding. Drugs that raise potassium can become dangerous together, especially in kidney dysfunction.

Medication reconciliation compares what the patient actually takes with records and intended therapy. Include prescriptions, over-the-counter products, supplements, recreational substances, formulations, dose, route, frequency, indication, and recent changes. Ask how the patient takes each drug rather than assuming the list is accurate.

Adverse drug reactions and safety

Predictable dose-related adverse effects follow known pharmacology. Unpredictable reactions include immune allergy and unusual susceptibility. A side effect is not automatically an allergy. Mislabelled allergy can deny useful treatment, while true anaphylaxis requires prominent documentation and avoidance planning.

Before prescribing, define the problem and treatment goal. Check contraindications, pregnancy potential, allergies, interactions, kidney and liver function, vital signs, and relevant laboratory values. Choose a drug and dose that fit the patient. Explain benefit, important risks, practical administration, expected timing, missed-dose advice, and warning symptoms. Specify monitoring and review. Plan duration and deprescribing.

Medication errors arise from ambiguous names, decimal points, unit confusion, duplicate therapy, transitions of care, interrupted workflows, and failure to monitor. Use leading zeros before decimal fractions and avoid trailing zeros. High-risk medicines deserve independent checking and clear rescue plans. When a patient deteriorates, review new medicines, recent dose changes, renal function, interactions, withdrawal, and whether treatment is masking a diagnosis.

TTS module 2: Exposure, variability, and prescribing as a control process

Concentration over time

A dose is an input, but exposure is the concentration pattern produced after that input. The area under the concentration-time curve describes total systemic exposure over an interval. Peak concentration may predict concentration-related toxicity or the effectiveness of drugs whose action depends on a high peak. Time above a target concentration matters for other

medicines. Some effects correlate with total exposure rather than either peak or trough alone.

These relationships explain why two regimens with the same daily dose are not always equivalent. Dividing a dose lowers peaks and raises troughs. Extended-release formulations prolong absorption. A rapid intravenous bolus creates a high early concentration that may be useful or dangerous depending on the medicine. The relevant pharmacodynamic target must guide regimen design.

Compartment models approximate distribution. After an intravenous dose, plasma concentration can fall rapidly as drug leaves the central circulation and enters tissue, then more slowly as elimination dominates. Sampling during the distribution phase may overstate how quickly the drug is being cleared. Lipid-soluble drugs can accumulate in fat or highly perfused tissue and later redistribute, producing recurrent sedation or a prolonged terminal half-life.

Half-life is not a fixed property independent of the patient. It rises when clearance falls or apparent distribution volume increases. Critical illness can change both at once through capillary leakage, fluid therapy, altered protein binding, organ dysfunction, and extracorporeal circuits. A familiar dose can therefore generate unfamiliar exposure.

Capacity, extraction, and organ disease

Hepatic clearance depends differently on blood flow, protein binding, and enzyme capacity according to the drug. High-extraction medicines are removed efficiently when delivered to the liver, so clearance is strongly influenced by hepatic blood flow and first-pass passage. Low-extraction medicines depend more on intrinsic enzyme or transporter activity and the unbound fraction. Liver disease does not reduce every pathway uniformly, and standard liver tests do not directly quantify drug clearance.

Kidney handling is similarly more complex than filtration alone. Active secretion can remove protein-bound drug after unbound molecules dissociate. Transporters can be inhibited by competing medicines. Urine pH affects reabsorption of some weak acids and bases and can occasionally be manipulated in poisoning. Kidney injury changes over time, so a

creatinine-based estimate that assumes steady state may lag behind true filtration.

Renal replacement therapies differ in membrane, flow, duration, and adsorption. A small, water-soluble, weakly protein-bound drug with a modest distribution volume is more likely to be removed than a highly tissue-bound molecule. If clinically important removal occurs, timing and supplemental dosing may need specialist guidance. Drug levels can help only when the assay, sampling time, and target are meaningful.

Saturable metabolism creates non-linearity. Below capacity, increasing dose may produce a roughly proportional increase in concentration. Near saturation, a small additional dose can produce a much larger increase because elimination cannot rise accordingly. Autoinduction, time-dependent inhibition, and active metabolites add further delay between dose change and observed effect.

Free concentration and binding shifts

Total plasma concentration includes bound and unbound drug. A fall in albumin can increase the unbound fraction of highly albumin-bound medicines. Initially this may increase distribution and clearance as well as effect, so the total concentration can fall even when active unbound exposure is adequate or excessive. Interpreting only the total level can prompt an unsafe dose increase.

Displacement from binding sites is often transient because freed drug distributes and is eliminated, but it becomes important when clearance is limited, distribution is small, or the medicine has a narrow therapeutic margin. Endogenous substances such as bilirubin and uraemic toxins can also alter binding. When available and validated, direct measurement of unbound concentration may clarify selected cases.

Binding in tissue can prolong action without a high plasma concentration. Some medicines accumulate in lysosomes, bone, melanin-containing tissue, or intracellular targets. Toxicity may therefore reflect cumulative exposure and tissue residence rather than the current blood level. A normal concentration after toxicity develops does not necessarily exclude a drug contribution.

Translating receptor pharmacology into dose choice

Dose-response curves describe populations or experimental systems, but an individual patient's curve is rarely known. Therapeutic and adverse effects can have different curves. Increasing dose may add little desired benefit after the therapeutic pathway approaches saturation while continuing to increase off-target or downstream harm. The useful dose is therefore not automatically the maximum tolerated dose.

Competitive antagonists shift an agonist concentration-response relationship when sufficient agonist can overcome blockade. Clinically, that simple model may fail because the body cannot safely generate unlimited agonist, the antagonist has additional actions, or signalling changes over time. Irreversible antagonists reduce available receptor until new receptors are produced. Physiological antagonists act through a different receptor or process to oppose the same endpoint.

Tolerance should trigger a mechanistic review. Faster metabolism may lower exposure. Receptor desensitisation may reduce response at the same concentration. Disease progression may increase the required effect. Poor adherence, formulation changes, or an interaction can imitate tolerance. Increasing the dose without distinguishing these possibilities risks toxicity when the apparent loss of effect has another explanation.

Placebo and nocebo effects are genuine context-dependent changes in symptom experience and behaviour, not proof that symptoms are imaginary. Expectations, previous experience, communication, and conditioning can alter pain, nausea, adherence, and adverse-effect reporting. Clear, balanced explanation can reduce avoidable nocebo burden while preserving informed consent.

Interactions as mechanisms

Interaction checking should move beyond counting alerts. Absorption can fall through chelation, altered gastric acidity, binding in the gut, or changed motility. Metabolism can change through inhibition or induction of a specific enzyme. Transporters can alter intestinal uptake, hepatic entry, biliary secretion, or renal elimination. The same perpetrator affects only medicines that depend materially on that pathway.

Pharmacodynamic interactions may be additive, synergistic, or antagonistic. Sedatives can converge on consciousness, airway tone, and ventilatory drive

through different receptors. Medicines that impair haemostasis can combine effects on coagulation, platelets, and mucosal integrity. Several agents may each slightly reduce blood pressure, sodium, potassium control, or cardiac repolarisation until the combined physiological reserve is exceeded.

Timing matters. Enzyme inhibition can begin when an inhibitor reaches effective concentration and may persist according to its half-life or irreversible binding. Induction develops as gene expression and protein abundance change, then fades gradually after withdrawal. A short course can therefore alter another medicine beyond the final dose. Interaction management may involve avoidance, dose adjustment, level monitoring, clinical surveillance, or temporary substitution.

Prescribing as feedback control

Safe prescribing resembles a control loop. Define a target, choose an intervention, measure response and harm, then adjust or stop. A prescription without a review time or success criterion is an open-loop action. This is especially unsafe when physiology changes, as in acute kidney injury, fluid shifts, pregnancy, critical illness, or recovery after surgery.

The indication should remain attached to each medicine. Without it, clinicians cannot tell whether treatment is effective, duplicated, obsolete, or being used to manage an adverse effect of another drug. Duration should be explicit for antimicrobials, anticoagulation after selected events, acid suppression, analgesia, sedatives, supplements, and many preventive treatments.

Deprescribing is not simply deleting medicines. It prioritises goals, identifies treatments whose harm exceeds likely benefit, anticipates withdrawal or recurrence, changes one variable at a time when practical, and monitors the result. Some drugs require tapering because abrupt cessation exposes physiological adaptation. Others can stop immediately. Current product information and local guidance should direct the method.

Medication safety also depends on the patient's ability to execute the plan. Vision, dexterity, cognition, swallowing, language, cost, access, dosing frequency, packaging, and social support affect real exposure. Simplification, demonstration, written instructions, dose-administration aids, and pharmacist review can be as important as choosing the molecule.

When a patient deteriorates, construct a medication timeline. Link symptom onset to initiation, dose change, organ-function change, missed doses, discontinuation, and new interacting substances. Then ask whether the medicine caused the syndrome, unmasked limited reserve, failed to treat the disease, or obscured an alternative diagnosis. That timeline turns pharmacology into clinical reasoning rather than an isolated list of adverse effects.

Retrieval prompts

One. Distinguish bioavailability, volume of distribution, clearance, and half-life.

Two. What primarily determines a loading dose and a maintenance dose?

Three. Why does kidney dysfunction sometimes require dose reduction or interval extension?

Four. Contrast potency with efficacy.

Five. Distinguish pharmacokinetic and pharmacodynamic interactions.

Six. What information belongs in a safe prescribing and monitoring plan?

Concise answers

One. Bioavailability is the fraction reaching systemic circulation; volume relates body amount to plasma concentration; clearance describes removal capacity; half-life describes concentration decline over time.

Two. Loading dose depends mainly on distribution volume; maintenance dosing depends mainly on clearance and target exposure.

Three. Reduced renal clearance can accumulate parent drug or active metabolites and prolong exposure.

Four. Potency is the amount needed for an effect; efficacy is the maximum effect.

Five. Pharmacokinetic interactions alter concentration; pharmacodynamic interactions alter combined response.

Six. Indication, goal, patient factors, drug and dose, administration, benefits, risks, interactions, baseline data, monitoring, review, duration, and stopping rules.

Source map

Original private-study synthesis informed by OpenStax Pharmacology for Nurses; Katzung Basic and Clinical Pharmacology, pharmacokinetics, pharmacodynamics, and autonomic principles; OpenStax Medical-Surgical Nursing for medication administration and monitoring; Guyton and Hall for organ physiology affecting disposition; and Robbins for organ dysfunction and adverse tissue responses.

Chapter 6: Clinical Reasoning, History, Examination, and Diagnostic Probability

Orientation

Clinical reasoning converts an undifferentiated concern into a working model, tests that model, and guides action under uncertainty. History and examination are not rituals performed before investigations. They define the problem, estimate urgency, select meaningful tests, interpret results, and establish a baseline for change. Good reasoning remains explicit enough to revise when evidence conflicts with the first impression.

Reasoning is iterative rather than linear. New information can change the problem representation, reorder the differential, and make an earlier test more or less meaningful. The aim is not to display certainty, but to choose the safest useful action at each stage while preserving the ability to revise.

Begin with safety and context

Before a detailed history, determine whether the patient is unstable. Assess airway patency, work of breathing, oxygenation, circulation, mental state, temperature, severe pain, active bleeding, and immediate threats. A concise primary survey can occur while monitoring, resuscitation, and help are arranged. Stabilisation and diagnosis proceed together; one should not wait unnecessarily for the other.

Context changes probability. Age, pregnancy, immune status, recent surgery, travel, occupation, medications, substance exposure, implanted devices, and previous disease alter what is likely and dangerous. The same symptom has different implications in a healthy young person and an older patient receiving anticoagulation or chemotherapy.

The history of the presenting problem

Begin with the patient's account in their own words. Open questions reveal priorities and chronology. Focused questions then test mechanisms and red flags. Establish onset, circumstances, location, character, severity, radiation, timing, progression, precipitating and relieving factors, associated symptoms, previous episodes, and functional effect. The structure should serve the symptom rather than force every complaint into one mnemonic.

Chronology is often diagnostic. Sudden maximal symptoms suggest a vascular event, rupture, obstruction, arrhythmia, seizure, or pneumothorax more than a slowly progressive inflammatory or malignant process. Episodic symptoms suggest intermittent obstruction, arrhythmia, seizure, migraine, mediator release, or exposure. Progressive decline raises concern for accumulating structural disease, neurodegeneration, malignancy, organ failure, or unresolved infection.

Ask what the patient thinks is happening, what they fear, and what outcome they seek. Explore how illness affects sleep, mobility, work, relationships, eating, cognition, and self-care. A technically correct diagnosis that ignores the patient's priorities can still produce poor care.

Background and medication history

Past medical and surgical history should include severity, complications, admissions, procedures, and current control rather than a list of labels. Family history can reveal inherited disease, premature vascular events, cancer patterns, or shared exposures. Social history includes living situation, supports, occupation, finances when relevant, smoking, alcohol, other substances, diet, activity, sexual health, travel, animals, and environmental exposure.

Medication history requires name, formulation, dose, route, timing, indication, adherence, benefit, adverse effects, recent changes, and over-the-counter or complementary products. Clarify the reaction behind every stated allergy. Ask about immunisation and preventive care where relevant.

A review of systems can uncover related manifestations but should not become indiscriminate questioning that generates noise. Use it to test the emerging differential and identify overlooked organ involvement.

Examination as hypothesis testing

Obtain consent, preserve dignity, provide a chaperone when appropriate, position the patient safely, and control infection risk. Observe before touching. General appearance, breathing pattern, speech, movement, colour, hydration, distress, body habitus, and interaction provide integrated information.

Measure vital signs accurately and interpret them as a pattern. Heart rate without rhythm, blood pressure without position and cuff context, respiratory rate estimated casually, or oxygen saturation without oxygen delivery information can mislead. Trends often matter more than a single value.

Inspection, palpation, percussion, and auscultation are tools, not mandatory stages for every system. Compare sides when useful. Link each sign to anatomy and mechanism. Elevated jugular venous pressure suggests raised right atrial pressure but depends on technique and context. Crackles can reflect opening of small airways or fluid but are not specific to one disease. Weakness must be separated from pain, poor effort, fatigue, and impaired motor control.

Examination findings have variable reliability. Technique, patient characteristics, disease stage, and observer experience affect accuracy. A sign should change probability rather than be treated as an absolute verdict.

Problem representation and differential diagnosis

After initial assessment, compress the case into a problem representation: relevant demographics, time course, key syndrome, major risk factors, and discriminating findings. For example, an older person with abrupt pleuritic chest pain, hypoxaemia, tachycardia, and recent immobility represents a different problem from a young person with reproducible chest-wall pain after exercise.

Construct the differential by mechanism and anatomy. Categories can include vascular, infectious, inflammatory, neoplastic, degenerative, toxic, metabolic, endocrine, traumatic, obstructive, iatrogenic, congenital, and functional. This reduces availability bias, in which memorable diagnoses dominate thought.

Prioritise with three questions. What is most likely? What is most dangerous if missed? What is most treatable or time-sensitive? A rare catastrophic condition may deserve urgent exclusion even if it is not the leading diagnosis.

Probability and diagnostic tests

Pre-test probability is the estimated chance of disease before a new test. It comes from prevalence, setting, risk factors, history, and examination. Sensitivity is the proportion of people with disease who test positive. Specificity is the proportion without disease who test negative. Likelihood ratios describe how much a result shifts odds.

A negative highly sensitive test can reduce probability when used in the right population. A positive highly specific test can increase probability. These shortcuts fail if disease spectrum, threshold, sampling, or timing differs from the validation setting.

Predictive values depend on prevalence. When disease is rare, even a fairly specific test can produce more false positives than true positives. Testing very-low-risk patients can therefore cause cascades of imaging, invasive procedures, anxiety, and incidental findings. At very high probability, a negative result may be insufficient to exclude disease. Testing is most informative near a decision threshold.

Reference intervals describe distributions in selected populations, not boundaries between health and disease. A result slightly outside range may be normal variation, while a result within range may be dangerous for a particular patient or trend. Interpret units, assay method, timing, biological variation, and pre-analytical error.

Cognitive error and calibration

Anchoring occurs when early information receives excessive weight. Premature closure stops the differential after one plausible answer. Confirmation bias favours supporting evidence. Search satisfaction ends evaluation after one abnormality. Base-rate neglect ignores prevalence. Framing effects allow wording or prior labels to direct conclusions.

Countermeasures include a diagnostic pause: What else could this be? Which finding does not fit? What dangerous alternative remains? What evidence

would change the plan? Seek disconfirming evidence, not only confirmation. Use checklists for high-risk processes but do not substitute them for thought.

Calibration means matching confidence to accuracy. State uncertainty explicitly. Instead of saying a diagnosis is ruled out, describe the residual probability and why it is acceptable or not. Follow-up is a diagnostic tool: define expected course, warning symptoms, and who will review unresolved findings.

Clinical documentation and communication

Document the source and reliability of information, relevant positives and negatives, examination, results, assessment, uncertainty, and plan. Avoid copying outdated statements. A problem-oriented assessment should show reasoning without becoming an unfiltered transcript.

Handover should communicate identity, current situation, relevant background, assessment, immediate risks, completed actions, response, and explicit recommendations. Closed-loop communication confirms that critical instructions were heard and understood. Escalation should state concern directly rather than hiding it inside data.

Shared decision-making combines evidence with the patient's values and circumstances. Explain options, expected benefits, important harms, uncertainty, and the consequence of doing nothing. Capacity is decision-specific and can fluctuate. It requires understanding, retention long enough to decide, use or weighing of information, and communication of a choice, interpreted within applicable law and policy.

TTS module 2: Bayesian updating, decision thresholds, and diagnostic safety

Evidence changes odds rather than declaring truth

Diagnostic information is most useful when it changes the probability of disease enough to alter action. Bayes' principle formalises this: begin with prior odds, multiply by the likelihood ratio of new evidence, and obtain posterior odds. Clinicians need not calculate every case numerically, but they should preserve the direction and scale of the update. A mildly abnormal

finding creates only a small shift. A highly discriminating finding can create a large one.

Sensitivity and specificity describe test performance conditional on disease status. Predictive values answer the inverse clinical question: given this result, how likely is disease? Because predictive values depend on prevalence and patient selection, a result means different things in different settings. A positive test in a high-risk symptomatic group can be persuasive while the same result in mass testing of a low-risk population may largely represent false positives.

Likelihood ratios are not fixed laws of nature. They change with disease spectrum, test threshold, operator skill, timing, and the definition of the reference standard. A physical sign established in severely affected inpatients may perform less strongly in early community disease. A biomarker sampled too soon may be falsely reassuring. Applying published accuracy requires checking whether the patient resembles the population in which the test was evaluated.

Tests are often correlated. Several inflammatory markers do not provide fully independent evidence because they respond to related biology. Repeatedly multiplying their likelihood ratios would exaggerate certainty. History, examination, laboratory tests, and imaging should be integrated according to the distinct information each contributes.

Thresholds for testing and treatment

Decision thresholds connect probability to consequences. Below a testing threshold, disease is sufficiently unlikely that further investigation creates more harm than benefit. Between testing and treatment thresholds, additional information may change management. Above a treatment threshold, action is justified without waiting for a test that would cause dangerous delay or is unlikely to reverse the decision.

Thresholds move with stakes. A dangerous, rapidly treatable disease may justify investigation at a low probability. A toxic or invasive treatment requires stronger evidence. The reliability of follow-up also matters. Watchful waiting is safer when deterioration will be recognised quickly and the patient can return, but unsafe when access is poor or the condition can progress silently.

Empirical treatment can alter subsequent evidence. Antimicrobials may sterilise cultures. Glucocorticoids can suppress inflammatory findings. Anticoagulation changes bleeding risk and some procedural options. If treatment is urgent, collect high-value samples first only when doing so will not create harmful delay, and document the pre-treatment syndrome clearly.

The value of information depends on whether a result can change an action. Ordering a test while intending to ignore either a positive or negative result adds burden without guiding care. Before ordering, state the question, expected result under each leading diagnosis, and what decision each possible result would change.

Building a causal problem representation

A strong problem representation includes semantic qualifiers that separate mechanisms: acute rather than chronic, focal rather than diffuse, painful rather than painless, exertional rather than random, progressive rather than episodic. These contrasts compress many observations into a form that activates useful illness patterns.

However, compression can discard inconvenient facts. After forming the representation, deliberately list findings it does not explain. A patient may have a common disease plus an independent second problem, an atypical presentation, a complication, or an incorrect initial label. Diagnostic parsimony is helpful but not mandatory; multimorbidity is common.

Mechanism-based differentials reduce memory bias. For syncope, consider impaired cerebral perfusion from reflex, orthostatic, cardiac, vascular, metabolic, or medication-related mechanisms, then distinguish events that mimic loss of perfusion such as seizure or psychogenic episodes. For hypoxaemia, separate low inspired oxygen, hypoventilation, diffusion limitation, ventilation-perfusion inequality, and shunt. Mechanisms predict associated findings and useful tests.

Time course supplies causal information. A sudden event suggests a threshold crossing, occlusion, rupture, electrical disturbance, toxin, or seizure. Subacute progression suggests accumulating inflammation, infection, obstruction, or immune injury. Long gradual decline suggests degenerative, neoplastic, chronic organ, or cumulative exposure processes. These are tendencies, not rules, but they structure questioning.

Examination quality and measurement error

An examination finding has technical and biological variance. Cuff size, patient position, ambient noise, lighting, effort, pain, and observer expectation can alter the result. Repeating a surprising measurement with better technique may be more valuable than immediately constructing a disease explanation.

Interobserver agreement and diagnostic accuracy are different. Clinicians may agree consistently on a finding that has weak disease discrimination, or disagree on a useful sign because technique is difficult. Training should therefore include both reliable performance and knowledge of what the sign changes.

Absence of a sign can be uninformative when the disease is early, the patient cannot cooperate, anatomy limits examination, or treatment has modified the finding. Conversely, multiple coherent signs arising from one mechanism can create a stronger pattern than any individual sign. Jugular venous pressure, peripheral oedema, hepatic congestion, and weight change together support a volume and pressure syndrome while each alone has alternatives.

Point-of-care instruments extend examination but require the same discipline. Bedside ultrasound, capillary glucose, pulse oximetry, and electrocardiography produce measurements that can be artefactual or misinterpreted. The operator should know acquisition quality, device limitations, and whether the result fits physiology.

Cognitive forcing and team reasoning

Expert pattern recognition is fast and often accurate when the presentation is familiar and feedback has been reliable. Analytical reasoning is slower and useful when the case is unfamiliar, high stakes, internally inconsistent, or not responding as expected. Safety comes from switching modes deliberately rather than treating intuition or analysis as universally superior.

A diagnostic timeout can be triggered by transfer of care, unexpected deterioration, treatment failure, or a result that does not fit. Restate the problem without inherited labels, rebuild the timeline, review medicines and procedures, and ask which dangerous diagnosis would explain both the original syndrome and the new finding.

Team diversity can uncover blind spots if hierarchy permits challenge. Nurses, pharmacists, therapists, interpreters, patients, and family members often hold different parts of the timeline. Closed-loop communication is especially important when an abnormal result requires action: identify who received it, what action was agreed, and when completion will be checked.

Handover creates vulnerability because uncertainty and pending work can disappear. A safe handover distinguishes established facts, working diagnoses, and unresolved possibilities. It names anticipated deterioration, escalation criteria, outstanding tests, and ownership. "Review bloods" is weaker than specifying which result, the expected time, the action threshold, and the responsible clinician.

Diagnostic harm and safety-netting

Overdiagnosis identifies a condition that would never have caused symptoms or harm during the person's lifetime. It differs from a false positive, which is an incorrect result. Incidental findings can lead to surveillance, procedures, anxiety, and labels even when their clinical importance is uncertain. More detection is not automatically better care.

Underdiagnosis can arise from poor access, atypical presentation, communication barriers, fragmented records, or premature attribution to an existing condition. Diagnostic overshadowing occurs when symptoms are incorrectly explained by psychiatric illness, disability, obesity, age, or another known diagnosis. Rechecking the physical and medication causes of change is a practical countermeasure.

Safety-netting converts uncertainty into a managed plan. Explain the working diagnosis and what remains uncertain, the expected trajectory, specific warning symptoms, where and how urgently to seek help, who will review tests, and what happens if symptoms persist. The advice must be feasible and understood, not merely documented.

Follow-up provides new evidence only when it is designed. Define a time interval appropriate to disease speed and the patient's reserve. Compare symptoms, function, examination, and objective measures with the baseline. Improvement after treatment may support a mechanism but is rarely definitive because many illnesses improve spontaneously and treatments can have non-specific effects.

The final reasoning question is not simply, “What is the diagnosis?” It is, “What action is safest at this probability, what evidence could change that action, and how will failure of the current model be detected?” That formulation preserves diagnostic humility while still producing decisive care.

Retrieval prompts

One. Why should urgency be assessed before a comprehensive history?

Two. What belongs in a concise problem representation?

Three. How should a differential be prioritised?

Four. Why does pre-test probability matter?

Five. Name four cognitive errors and one countermeasure.

Six. What makes follow-up part of diagnosis rather than administration?

Concise answers

One. Immediate threats require parallel stabilisation and focused diagnosis.

Two. Relevant demographics, time course, syndrome, risk factors, and discriminating findings.

Three. Consider the most likely, the dangerous diagnoses not to miss, and time-sensitive treatable causes.

Four. It determines how much a result changes probability and whether testing is likely to help or harm.

Five. Anchoring, premature closure, confirmation bias, and base-rate neglect; counter with a diagnostic pause and search for disconfirming evidence.

Six. Evolution over time can confirm, refute, or reveal a diagnosis, provided expected course and escalation triggers are explicit.

Source map

Original private-study synthesis informed by Talley and O'Connor's Clinical Examination; OpenStax Medical-Surgical Nursing; principles of diagnostic testing and clinical epidemiology; and the physiology, pathology, microbiology, and pharmacology sources listed in the source register.

Chapter 7: Cardiac Electrophysiology and the Cardiac Cycle

Orientation

The heart converts coordinated electrical activation into pressure and flow. Electrical impulses must arise automatically, spread in the correct sequence, trigger calcium-dependent contraction, and then terminate so chambers can relax and refill. Mechanical output depends on preload, contractility, afterload, heart rate, rhythm, valve function, and myocardial oxygen supply. Disturbance at any level can cause syncope, congestion, ischaemia, shock, or sudden death.

Pacemaker activity and conduction

The sinoatrial node normally initiates each beat. Pacemaker cells do not maintain a stable resting voltage. During diastole, inward current through hyperpolarisation-activated channels, reduced potassium efflux, and calcium entry gradually depolarise the membrane toward threshold. Calcium currents generate much of the nodal upstroke. Potassium efflux repolarises the cell. The slope of spontaneous depolarisation helps determine heart rate.

The impulse spreads through atrial myocardium to the atrioventricular node. Slow conduction through the atrioventricular node allows atrial contraction to contribute to ventricular filling before ventricular systole. The His bundle, bundle branches, and Purkinje fibres then distribute activation rapidly through the ventricles. This sequence activates the septum and ventricular apex before much of the free wall, producing efficient ejection.

Working atrial and ventricular myocytes have a fast action potential. Rapid sodium entry produces depolarisation. Brief potassium currents begin early repolarisation. A plateau follows because calcium entry through L-type channels balances potassium efflux. Delayed potassium currents complete repolarisation. The long action potential and refractory period prevent sustained tetanic contraction, allowing rhythmic filling and ejection.

Autonomic control changes pacemaker rate and conduction. Sympathetic beta one receptor activation increases cyclic adenosine monophosphate, steepens pacemaker depolarisation, speeds atrioventricular conduction, and increases calcium handling and contractility. Parasympathetic acetylcholine slows sinoatrial firing and atrioventricular conduction through muscarinic receptors. Resting heart rate reflects intrinsic pacemaker activity modified by autonomic tone.

Excitation-contraction coupling

Depolarisation opens L-type calcium channels in transverse tubules. Incoming calcium triggers larger calcium release from the sarcoplasmic reticulum through ryanodine receptors. Cytosolic calcium binds troponin C, moves tropomyosin, and permits actin-myosin cross-bridge cycling. Relaxation requires calcium removal by sarcoplasmic-reticulum pumps, sodium-calcium exchange, and membrane calcium pumps.

Sympathetic stimulation increases calcium entry, sarcoplasmic-reticulum uptake, release, and cycling speed. Contraction becomes stronger and relaxation faster. This allows greater output without preventing diastolic filling at higher rates. Ischaemia, acidosis, electrolyte disturbance, and many drugs alter ion channels or calcium handling and can impair both rhythm and contraction.

The electrocardiogram

The electrocardiogram records surface voltage differences generated by cardiac depolarisation and repolarisation. The P wave represents atrial depolarisation. The P R interval includes atrial conduction and delay through the atrioventricular node. The Q R S complex represents ventricular depolarisation. The S T segment corresponds broadly to the ventricular plateau, and the T wave represents ventricular repolarisation.

An electrocardiogram does not directly show contraction, cardiac output, or coronary anatomy. Interpretation asks about rate, rhythm, axis, intervals, conduction, chamber patterns, Q R S morphology, repolarisation, and comparison with prior tracings. Lead placement, movement, body habitus, and electrical interference can create artefact.

The Q T interval spans ventricular depolarisation and repolarisation and varies with heart rate. Excessive corrected Q T prolongation can permit early afterdepolarisations and torsades de pointes, especially with interacting drugs, bradycardia, low potassium, low magnesium, congenital channel variants, or acute illness.

Mechanical events of the cardiac cycle

During ventricular diastole, pressure falls below atrial pressure and the atrioventricular valves open. Early filling is largely passive. Reduced ventricular compliance raises filling pressure and makes atrial contraction more important. At the end of diastole, ventricular volume is maximal: the end-diastolic volume.

Ventricular depolarisation triggers contraction. When ventricular pressure exceeds atrial pressure, the mitral and tricuspid valves close, producing the first heart sound. During isovolumetric contraction, all valves are closed, volume is constant, and pressure rises rapidly. When ventricular pressure exceeds aortic or pulmonary-artery pressure, semilunar valves open and ejection begins.

The volume remaining after ejection is end-systolic volume. Stroke volume equals end-diastolic volume minus end-systolic volume. Ejection fraction is stroke volume divided by end-diastolic volume. It describes one aspect of systolic performance but does not equal cardiac output or exclude heart failure when preserved.

As ventricles relax, pressure falls below arterial pressure and the semilunar valves close, producing the second heart sound. Isovolumetric relaxation follows until ventricular pressure falls below atrial pressure and filling resumes. The cycle shortens as heart rate rises, with disproportionate loss of diastolic time. Extreme tachycardia can impair filling and coronary perfusion.

Preload, afterload, and contractility

Preload describes myocardial fibre stretch before contraction and is related to ventricular filling, venous return, compliance, and end-diastolic volume and pressure. Within a physiological range, increased filling increases contraction force through the Frank-Starling mechanism. This helps match right- and left-ventricular output and adjusts stroke volume to venous return.

Afterload is the load against which the ventricle ejects. It reflects arterial pressure, outflow obstruction, vascular impedance, and ventricular geometry. Increased afterload tends to increase end-systolic volume and myocardial work. Chronic pressure load promotes concentric hypertrophy; chronic volume load tends to promote chamber dilation and eccentric remodelling.

Contractility is intrinsic force generation at a given preload and afterload. Sympathetic stimulation and several inotropic drugs increase contractility. Ischaemia, hypoxia, acidosis, and some medicines reduce it. Lusitropy describes relaxation, an energy-dependent process essential for filling.

Cardiac output equals heart rate multiplied by stroke volume. A faster rate can raise output until filling and coronary supply become limiting. A slow rate can be tolerated if stroke volume rises, but severe bradycardia reduces output. Rhythm matters because coordinated atrial activity, atrioventricular timing, and synchronous ventricular activation all support efficient pumping.

Coronary perfusion and oxygen balance

The myocardium extracts a high fraction of delivered oxygen at rest, so increased demand is met mainly by increased coronary flow. Left-ventricular coronary flow occurs predominantly during diastole because systolic contraction compresses intramyocardial vessels. Tachycardia both raises oxygen demand and shortens diastolic supply time.

Demand rises with heart rate, contractility, wall stress, and muscle mass. Supply depends on arterial oxygen content, coronary perfusion pressure, vessel patency, microvascular control, and diastolic duration. Anaemia, hypoxaemia, hypotension, coronary obstruction, spasm, or excessive demand can create ischaemia.

Rhythm disturbances

Arrhythmias arise from abnormal impulse formation, abnormal conduction, or both. Enhanced automaticity can create ectopic beats. Triggered activity follows afterdepolarisations. Re-entry requires a circuit, unidirectional block, and conduction timing that permits tissue to recover and be re-excited.

Bradyarrhythmias include sinus-node dysfunction and atrioventricular block. Tachyarrhythmias may originate above or within the ventricles. Atrial fibrillation produces disorganised atrial activation, loss of effective atrial contraction, irregular ventricular response, and risk of atrial thrombus and embolic stroke. Ventricular tachycardia can severely reduce output and may degenerate into ventricular fibrillation.

Clinical effect depends on ventricular rate, duration, ventricular function, filling, coronary disease, and autonomic state. Palpitations may be benign or dangerous. Syncope during exertion, structural heart disease, family history of sudden death, persistent chest pain, hypotension, or broad-complex tachycardia requires urgent assessment.

Electrolytes are central to rhythm interpretation. Hyperkalaemia can slow conduction, widen the Q R S complex, and progress toward electrical arrest; hypokalaemia increases ectopy and delayed repolarisation. Magnesium deficiency increases susceptibility to ventricular arrhythmia. Calcium concentration changes the action-potential plateau and Q T duration. Electrocardiographic patterns are neither perfectly sensitive nor specific, so suspected severe disturbance requires urgent measurement and treatment based on the whole clinical picture.

TTS module 2: Pressure-volume loops, arrhythmia mechanisms, and pump efficiency

Reading the ventricle as a pressure-volume system

A pressure-volume loop displays one cardiac cycle with ventricular volume on the horizontal axis and pressure on the vertical axis. Filling moves the loop toward greater volume at low pressure. Isovolumetric contraction raises pressure without changing volume. Ejection reduces volume while pressure first rises and then falls. Isovolumetric relaxation lowers pressure at constant end-systolic volume.

The width of the loop is stroke volume. The enclosed area approximates external stroke work. The end-diastolic pressure-volume relationship reflects passive compliance: a stiff ventricle reaches a high pressure at a relatively modest volume. The end-systolic relationship reflects contractile state more closely, although loading and geometry still matter. These relationships explain why the same ejection fraction can arise from very different hearts.

Increasing preload generally shifts end-diastolic volume rightward and widens the loop through the Frank-Starling mechanism, provided the ventricle remains on the useful portion of its length-tension relationship. Increasing afterload makes the ventricle generate more pressure before ejection, raises end-systolic volume, and can narrow stroke volume. Increased contractility lowers end-systolic volume at a given load and often widens the loop.

Valve disease changes loop geometry. In mitral regurgitation, some stroke volume escapes into the low-pressure atrium, so total ejected volume can appear large while effective forward output is reduced. In aortic stenosis, the ventricle must generate pressure above aortic pressure to drive flow through the narrowed valve. Pressure work and wall stress rise, promoting hypertrophy and eventually failure.

Wall stress, hypertrophy, and oxygen demand

Ventricular wall stress rises with intracavitary pressure and chamber radius and falls as wall thickness increases. This relation explains adaptive remodelling. Pressure overload stimulates wall thickening that initially reduces stress for a given pressure. Volume overload increases chamber size and tends to add sarcomeres in series. Both adaptations can preserve output before fibrosis, altered energetics, capillary mismatch, and impaired relaxation reduce reserve.

Hypertrophy is not equivalent to stronger effective pumping. A thick ventricle may contract forcefully but fill poorly because compliance is reduced. Filling pressure then rises into the atrium and pulmonary or systemic veins. Symptoms can appear despite a preserved ejection fraction because ejection fraction describes the proportion ejected from the volume present, not the pressure required to fill the chamber or the output achieved.

Myocardial oxygen demand tracks wall stress, rate, and contractile activity. A dilated ventricle faces greater wall stress, while a hypertrophied ventricle

contains more metabolically active muscle and may have reduced capillary density relative to mass. Tachycardia adds demand and shortens diastole. These factors can produce supply-demand ischaemia without an acute coronary occlusion.

Diastolic filling and ventricular interaction

Filling depends on active relaxation, passive compliance, atrial pressure, pericardial constraint, rhythm, and available diastolic time. Relaxation requires energy to remove calcium from the cytosol and detach cross-bridges. Ischaemia can therefore impair diastole before obvious systolic failure. A stiff ventricle relies more on atrial contraction, so atrial fibrillation may produce disproportionate breathlessness or hypotension.

The right and left ventricles share the septum and are enclosed by the pericardium. Acute right-ventricular dilation can shift the septum toward the left ventricle and reduce left filling. High intrathoracic or pericardial pressure can limit both chambers. This ventricular interdependence helps explain haemodynamic collapse in massive pulmonary vascular obstruction, tamponade, or excessive ventilatory pressure.

Respiration alters filling. Spontaneous inspiration lowers intrathoracic pressure and usually promotes venous return to the right heart. Positive-pressure ventilation reverses this pressure change and can reduce venous return, especially when circulating volume is limited. At the same time, it can reduce left-ventricular transmural afterload. The net effect depends on volume state, ventricular function, pulmonary resistance, and pressure settings.

Arrhythmia initiation and maintenance

Abnormal automaticity occurs when pacemaker-like depolarisation develops in cells outside the sinoatrial node or normal pacemakers accelerate. Ischaemia, catecholamines, stretch, electrolyte disturbance, and drugs can alter the balance of inward and outward currents. Triggered activity differs because it depends on a preceding action potential. Early afterdepolarisations arise during prolonged repolarisation, while delayed afterdepolarisations follow calcium overload after repolarisation.

Re-entry requires more than a circular pathway. Conduction must be blocked in one direction but able to proceed in another, and travel time must allow previously refractory tissue to recover. Scar, fibrosis, unequal refractoriness, accessory pathways, and slowed conduction create substrate. A premature impulse can provide the trigger. Treatment can target the trigger, interrupt the circuit, prolong refractoriness, slow conduction, or remove the anatomical substrate.

Antiarrhythmic actions can also create arrhythmia. Sodium-channel block may slow conduction enough to suppress one circuit but promote another in diseased myocardium. Potassium-channel block can prolong repolarisation and permit early afterdepolarisations. Atrioventricular nodal blockers control many supraventricular rhythms but can worsen selected conduction disease or accessory-pathway situations. Drug choice must fit the rhythm mechanism, structural heart disease, organ function, interactions, and current guidance.

Electrocardiography as spatial sampling

Each lead records the projection of a changing cardiac electrical vector along a particular axis. A positive deflection means net electrical activity is directed toward the lead's positive electrode; a negative deflection means it moves away. Small or biphasic signals can occur when the vector is perpendicular or changes direction during the measured event.

Axis deviation is therefore a clue to the direction of ventricular activation, not a diagnosis by itself. Conduction block, ventricular hypertrophy, prior infarction, pacing, anatomical position, and technical error can alter it. QRS width reflects the duration of ventricular depolarisation. A broad complex may result from bundle-branch block, ventricular origin, pre-excitation, pacing, electrolyte disturbance, or drug effect.

ST and T changes represent altered repolarisation but are not specific to coronary occlusion. Ischaemia, ventricular strain, inflammation, electrolyte imbalance, conduction abnormalities, pacing, and medicines can all change them. Interpretation depends on distribution, morphology, symptoms, timing, and comparison with prior tracings. Serial electrocardiograms can reveal evolution that one snapshot misses.

The corrected QT interval attempts to account for heart rate, but correction formulae become less reliable at rate extremes. Automated values can fail

when T waves are indistinct, U waves are present, rhythm is irregular, or QRS duration is prolonged. Manual confirmation and review of reversible causes are important when the result may alter treatment.

Haemodynamic consequences and emergency priorities

An arrhythmia becomes dangerous through rate, loss of synchrony, or degeneration into a non-perfusing rhythm. Very rapid rates shorten filling and coronary perfusion. Very slow rates reduce cardiac output unless stroke volume can compensate. Loss of atrial contraction matters most when ventricular filling is stiff or preload is limited. Ventricular dyssynchrony reduces mechanical efficiency even when average rate is acceptable.

The first emergency distinction is electrical instability with adverse physiology. Hypotension, shock, ongoing ischaemic chest discomfort, acute pulmonary oedema, or altered consciousness indicates that the rhythm is not merely an electrocardiographic curiosity. Treatment priorities follow current resuscitation protocols and may include synchronised cardioversion, defibrillation, pacing, rate or rhythm medicines, correction of electrolytes, and treatment of the underlying cause.

After stabilisation, identify substrate and trigger. Review onset, previous rhythm, structural disease, ischaemia, infection, endocrine disturbance, alcohol or stimulant exposure, hypoxia, medicines, adherence, and family history. Measure electrolytes and organ function, and use imaging or monitoring according to the syndrome. A transient trigger does not exclude a persistent substrate.

The integrated question is whether electrical timing, filling, force, valve function, vascular load, and oxygen supply are aligned. Cardiac output fails when one component exceeds the reserve of the others. Pressure-volume reasoning and rhythm mechanism together reveal why the same heart rate or ejection fraction can be tolerated by one patient and catastrophic in another. Serial assessment is essential because loading conditions, autonomic tone, ischaemia, and treatment can rapidly transform both the electrical pattern and its mechanical consequence.

Retrieval prompts

One. Why does the atrioventricular node conduct slowly?

- Two. How does calcium couple electrical activation to contraction?
- Three. Trace valve state and pressure during one cardiac cycle.
- Four. Distinguish preload, afterload, contractility, stroke volume, and ejection fraction.
- Five. Why is tachycardia capable of worsening myocardial ischaemia?
- Six. What conditions permit re-entry?

Concise answers

- One. The delay allows atrial emptying before ventricular contraction.
- Two. Membrane calcium entry triggers sarcoplasmic-reticulum release; calcium binds troponin and permits cross-bridge cycling.
- Three. Atrioventricular valves open during filling, close when ventricular pressure exceeds atrial pressure, semilunar valves open when ventricular pressure exceeds arterial pressure, and close during relaxation.
- Four. Preload is pre-contraction filling and stretch; afterload is ejection load; contractility is intrinsic force; stroke volume is ejected volume; ejection fraction is stroke volume divided by end-diastolic volume.
- Five. It increases demand while shortening diastole, when left-coronary perfusion mainly occurs.
- Six. A circuit, unidirectional block, and conduction slow enough for previously activated tissue to recover.

Source map

Original synthesis informed by Guyton and Hall, cardiac muscle, rhythm, electrocardiography, cardiac output, and coronary circulation; OpenStax Anatomy and Physiology 2e; Katzung and OpenStax Pharmacology for autonomic and antiarrhythmic mechanisms; Robbins for myocardial adaptation and injury; and Talley and O'Connor for clinical rhythm assessment.

Chapter 8: Haemodynamics, Vascular Control, and Blood Pressure

Orientation

The circulation delivers oxygen, nutrients, hormones, immune components, and heat while removing carbon dioxide and metabolic products. Flow must be distributed according to tissue need while arterial pressure remains sufficient for perfusion. The heart supplies energy, arteries buffer pulsation, arterioles regulate resistance, capillaries exchange material, veins store volume and return blood, and lymphatics return filtered fluid and macromolecules.

Pressure, flow, and resistance

Flow is driven by a pressure difference and opposed by resistance. In simplified form, flow equals pressure gradient divided by resistance. Absolute pressure at one point is less informative than the gradient between two points. Blood moves from higher to lower total energy, but local flow also depends on vessel geometry, viscosity, pulsatility, and downstream pressure.

For laminar flow in a cylindrical tube, resistance varies directly with length and viscosity and inversely with the fourth power of radius. The circulation is more complex than an ideal tube, but the radius relationship explains why small arteriolar diameter changes have large effects. Vessels in series add resistance. Parallel vascular beds reduce total resistance and permit independent regional control.

Velocity equals flow divided by total cross-sectional area. Although individual capillaries are tiny, their combined area is enormous, so velocity falls in capillary beds. Slow transit supports exchange. Turbulence becomes more likely with high velocity, large diameter, low viscosity, and abrupt narrowing. Turbulence can generate murmurs or bruits but their absence does not exclude disease.

Arterial pressure and pulsation

Systolic pressure is the peak during ventricular ejection. Diastolic pressure is the lowest before the next beat. Pulse pressure is systolic minus diastolic pressure. Mean arterial pressure is closer to diastolic than systolic pressure at normal rates and can be approximated as diastolic pressure plus one third of pulse pressure.

Mean pressure depends mainly on cardiac output and systemic vascular resistance. Cardiac output equals heart rate multiplied by stroke volume. Pulse pressure depends strongly on stroke volume and arterial compliance. Ageing and atherosclerosis stiffen large arteries, raising pulse-wave velocity and often widening pulse pressure.

Elastic arteries store energy during systole and release it during diastole, smoothing intermittent ventricular ejection into more continuous peripheral flow. Reflected pressure waves and arterial stiffness influence central pressure and ventricular workload. Brachial cuff pressure is clinically useful but not identical to pressure at every arterial site.

Arterioles and local control

Arterioles are major resistance vessels. Vascular smooth muscle contracts when cytosolic calcium activates myosin light-chain kinase. Relaxation follows reduced calcium or increased myosin light-chain phosphatase activity. Sympathetic alpha one receptor activation commonly causes constriction. Beta two receptor activation can dilate selected beds, particularly in skeletal muscle, although circulating hormone concentration and receptor distribution matter.

Local metabolic control matches flow to tissue activity. Reduced oxygen and increased carbon dioxide, hydrogen ions, potassium, adenosine, and other metabolites promote dilation in many tissues. The myocardium and active skeletal muscle show strong metabolic regulation. The brain tightly regulates flow in response to carbon dioxide and local activity, although severe pressure extremes can exceed autoregulation.

Myogenic autoregulation occurs when vascular smooth muscle contracts in response to stretch and relaxes when pressure falls. This stabilises flow in organs such as the brain and kidney. Reactive hyperaemia is transient

increased flow after an occlusion is released. Active hyperaemia accompanies increased tissue metabolism.

The endothelium senses shear stress, hormones, platelets, and inflammation. Nitric oxide diffuses to smooth muscle and raises cyclic guanosine monophosphate, causing relaxation. Prostacyclin dilates and inhibits platelets. Endothelin constricts. Endothelial dysfunction reduces vasodilator and antithrombotic properties and contributes to vascular disease.

Capillary exchange and lymph

Most solute exchange occurs by diffusion. Lipid-soluble molecules cross endothelial membranes; water-soluble substances use clefts, pores, or transport pathways depending on tissue. Diffusion rate rises with surface area and concentration gradient and falls with distance.

Bulk fluid movement reflects hydrostatic and oncotic forces across the capillary barrier. Capillary hydrostatic pressure favours filtration. Plasma-protein oncotic pressure favours absorption. Interstitial pressures and protein concentrations also contribute. Modern interpretation emphasises the endothelial glycocalyx and recognises that sustained venous-end reabsorption is less extensive than older diagrams imply. Much filtered fluid returns through lymphatics.

Oedema develops when filtration exceeds lymphatic return. Causes include increased venous or capillary pressure, reduced plasma oncotic pressure, increased permeability, lymphatic obstruction, and sodium retention. Heart failure raises venous pressure. Severe liver disease or protein loss can reduce albumin. Inflammation increases permeability. Malignancy or surgery can obstruct lymphatics.

Lymphatic vessels collect interstitial fluid, proteins, immune cells, and absorbed intestinal lipids. One-way valves, skeletal-muscle movement, respiration, and smooth-muscle activity propel lymph. Lymph nodes filter lymph and organise immune responses.

Veins and venous return

Veins are compliant capacitance vessels containing much of the blood volume. Small changes in venous tone can shift blood centrally. Venous return depends on the pressure gradient toward the right atrium and resistance to return. Sympathetic venoconstriction, skeletal-muscle pumping, respiratory pressure changes, and venous valves support return.

Standing causes gravitational pooling below the heart. Baroreflexes increase heart rate, contractility, and vasoconstriction. Muscle contraction and venous valves limit pooling. Failure of compensation causes orthostatic symptoms. Prolonged immobility promotes venous stasis and thrombosis.

Rapid and long-term pressure control

Baroreceptors in the carotid sinus and aortic arch respond to stretch. When pressure rises, firing increases, promoting parasympathetic activity and reducing sympathetic output. Heart rate, contractility, resistance, and venous tone fall. When pressure drops, the opposite occurs. Baroreflexes act within seconds but reset during sustained pressure change, so they are not the sole mechanism of long-term control.

Chemoreceptors respond to reduced oxygen, elevated carbon dioxide, and acidity, especially when pressure is low. Central nervous system ischaemia can trigger intense sympathetic activity in severe hypotension.

Cardiopulmonary receptors respond to central volume and contribute to hormonal and renal adjustments.

Long-term pressure control depends heavily on sodium balance, extracellular volume, and kidney function. Reduced renal perfusion, sympathetic activation, or reduced sodium delivery to the macula densa promotes renin release. Renin generates angiotensin one, which is converted to angiotensin two. Angiotensin two constricts vessels, stimulates aldosterone, promotes thirst and antidiuretic hormone, and increases renal sodium retention.

Aldosterone increases distal sodium reabsorption and potassium secretion. Antidiuretic hormone increases collecting-duct water permeability and at higher concentrations causes vasoconstriction. Natriuretic peptides are released with cardiac stretch and favour sodium excretion and vasodilation. These systems interact rather than acting as isolated switches.

Hypertension and hypotension

Hypertension often reflects combined genetic, renal, neural, vascular, endocrine, metabolic, and environmental influences. Sustained pressure injures endothelium and small vessels, accelerates atherosclerosis, increases ventricular workload, and damages brain, kidney, retina, and heart. Most patients have no specific symptom, so accurate repeated measurement is essential.

Secondary hypertension should be considered with abrupt onset, young age, resistant pressure, marked hypokalaemia, episodic mediator symptoms, kidney disease, vascular findings, sleep-disordered breathing, or a relevant drug exposure. Treatment combines risk reduction, dietary and activity measures, and medicines targeting volume, vascular tone, neural input, or hormonal pathways.

Hypotension matters when organ perfusion becomes inadequate, not merely when a number is below a population threshold. Causes include reduced volume, impaired pumping, vasodilation, obstruction to flow, and autonomic failure. Clinical effect depends on baseline pressure, speed of change, vascular disease, and compensatory reserve.

Blood-pressure measurement

Measurement requires appropriate cuff size, arm support, patient rest, correct position, and repeated readings. Talking, pain, recent exertion, stimulants, a full bladder, or an unsupported arm can alter results. Compare arms initially when appropriate. Standing measurements help assess orthostatic physiology. Home or ambulatory monitoring can identify white-coat and masked hypertension and reveal nocturnal patterns.

TTS module 2: Vascular impedance, microcirculatory failure, and pressure phenotypes

Beyond steady resistance

The circulation is pulsatile, so ventricular load cannot be described by systemic vascular resistance alone. Resistance captures the relation between mean pressure and mean flow, while impedance also includes arterial compliance, blood inertia, wave reflection, and frequency. A patient may have

a similar mean resistance but a very different systolic load because the aorta is stiff or reflected waves return early.

During systole, compliant proximal arteries expand and store part of the ejected volume. During diastole, recoil maintains forward flow. Loss of compliance raises pulse-wave velocity and causes reflected waves to return during late systole rather than diastole. This augments central systolic pressure, increases left-ventricular work, and may reduce the diastolic support of coronary perfusion.

Pulse pressure is influenced by stroke volume, arterial compliance, and ejection pattern. A wide pulse pressure may reflect stiff arteries or a large, rapid stroke volume. A narrow pulse pressure can occur when stroke volume is reduced, although measurement quality and vascular tone matter. Peripheral pulse pressure can differ from central aortic pulse pressure because waveforms change as they travel and reflect.

Resistance vessels and conduit arteries therefore create distinct problems. Arteriolar constriction raises mean resistance and redistributes flow. Large-artery stiffness increases pulsatile load. Treating blood pressure reduces risk through several mechanisms, but different drug classes may influence volume, resistance, heart rate, and arterial wave behaviour differently.

Autoregulation and perfusion pressure

Organ perfusion pressure is the inflow pressure minus relevant downstream or surrounding pressure. Cerebral perfusion depends on arterial pressure relative to intracranial pressure. Renal filtration depends on pressures across the glomerular circulation and capsule. Coronary perfusion, particularly in the left ventricle, depends strongly on aortic diastolic pressure relative to ventricular pressure.

Autoregulation keeps flow relatively stable across a pressure range by changing arteriolar tone. Chronic hypertension can shift this operating range, so a pressure tolerated by one person may produce cerebral or renal hypoperfusion in another if lowered abruptly. Conversely, severe hypertension can exceed the upper limit, transmitting pressure to fragile microvessels and promoting oedema or injury.

Autoregulation is altered by carbon dioxide, oxygen, inflammation, anaesthesia, trauma, and vascular disease. In the brain, increased carbon dioxide is a potent vasodilator. Hyperventilation lowers carbon dioxide and can temporarily reduce cerebral blood volume, but excessive or prolonged reduction risks ischaemia. Any deliberate manipulation belongs within current specialist protocols.

The kidney uses myogenic responses and tubuloglomerular feedback to stabilise filtration. The afferent arteriole responds to stretch, while sodium chloride delivery at the macula densa modifies tone and renin release. Medicines that alter afferent or efferent arteriolar tone can reduce intraglomerular pressure, particularly when renal perfusion is already dependent on compensatory constriction.

The glycocalyx and transvascular fluid movement

The endothelial glycocalyx is a thin, biologically active surface layer that influences permeability, mechanosensing, coagulation, and leukocyte interaction. Protein concentration immediately beneath it contributes to the effective oncotic gradient. When inflammation, hyperglycaemia, ischaemia-reperfusion, or mechanical stress disrupts the glycocalyx, permeability and microvascular behaviour change.

Traditional descriptions imply filtration at the arterial capillary end and sustained reabsorption at the venous end. In many tissues, net filtration occurs along much of the capillary, with lymphatics returning fluid and proteins. A transient fall in capillary pressure may permit some absorption, but lymphatic function remains central to long-term balance.

Interstitial compliance determines how quickly pressure rises as fluid accumulates. In a low-compliance compartment, a small increase in volume can sharply raise tissue pressure and compromise perfusion, as in compartment syndrome or intracranial swelling. In loose tissue, larger volumes may accumulate before pressure rises substantially. The clinical danger of oedema therefore depends on location as well as amount.

Oedema can coexist with intravascular depletion. Increased permeability allows protein-rich fluid to leave the circulation, reducing effective volume while tissue swelling grows. Low albumin reduces plasma oncotic force but usually interacts with sodium retention, capillary pressure, and lymphatic

capacity. A mechanistic assessment is safer than assuming all oedema requires the same fluid strategy.

Venous function and stressed volume

Total blood volume can be divided conceptually into unstressed volume, which fills vessels without generating much pressure, and stressed volume, which stretches the vascular system and drives venous return. Sympathetic venoconstriction shifts volume from the unstressed toward the stressed compartment, raising mean systemic filling pressure and supporting return without adding fluid.

Venous return falls as right-atrial pressure approaches the upstream filling pressure, but very negative right-atrial pressure cannot increase flow indefinitely because intrathoracic veins collapse. Right-heart failure raises downstream pressure and causes systemic congestion even when total volume is not greatly increased. Elevated venous pressure also impairs renal and hepatic drainage and can reduce organ function independently of arterial hypotension.

The skeletal-muscle pump compresses deep veins, while valves direct flow centrally. Valve incompetence permits reflux and ambulatory venous hypertension. Prolonged elevation damages skin and subcutaneous tissue, causing oedema, pigmentation, inflammation, and ulceration. Compression can improve venous return in appropriate patients, but arterial supply and other contraindications must be assessed first.

Respiratory effects differ between spontaneous and positive-pressure breathing. Spontaneous inspiration lowers right-atrial pressure and often supports venous return. Positive intrathoracic pressure can raise right-atrial pressure and reduce the return gradient, especially with hypovolaemia. It may also reduce left-ventricular afterload and pulmonary congestion, creating a net benefit in selected heart-failure states.

Microcirculatory and distributive failure

A normal or restored arterial pressure does not guarantee normal tissue perfusion. Flow can be heterogeneous at the microvascular level because of endothelial swelling, leukocyte adhesion, microthrombi, altered red-cell deformability, shunting, and impaired vasomotor control. Some capillaries

receive excessive flow while others receive little, limiting extraction despite apparently adequate global delivery.

Distributive shock illustrates the separation between pressure and flow. Widespread vasodilation lowers resistance and effective arterial filling. Cardiac output may initially be high, yet tissue extraction and regional distribution remain abnormal. Later myocardial depression, fluid loss into tissue, and reduced reserve can lower output. Vasopressors may restore pressure, but source control, antimicrobial treatment when infection is present, volume assessment, and organ support address other parts of the syndrome.

Lactate can rise through reduced oxidative metabolism, adrenergic acceleration of glycolysis, impaired clearance, or regional stress. It is a severity and trajectory marker rather than a direct meter of whole-body oxygen absence. Falling concentration after treatment can be reassuring, but interpretation must include liver function, medicines, seizures, muscle activity, and perfusion findings.

Blood-pressure phenotypes and measurement

Clinic, home, daytime, and night-time pressures sample different conditions. White-coat hypertension describes elevated clinic pressure with lower out-of-office values. Masked hypertension is the reverse and can leave risk undetected. Failure of normal nocturnal reduction may accompany sleep-disordered breathing, kidney disease, autonomic dysfunction, or other risk states.

Orthostatic assessment requires standardised timing and symptom correlation. A pressure fall can reflect volume depletion, medicines, autonomic failure, prolonged bed rest, or impaired cardiac response. Heart-rate behaviour may help but is altered by age, rhythm, pacemakers, and rate-limiting drugs. Postprandial or exertional measurements can reveal physiology missed at rest.

Cuff devices infer pressure from arterial oscillations and depend on correct size, position, rhythm, and validated algorithms. Arrhythmia, tremor, severe vascular stiffness, and improper cuff placement can reduce accuracy. An unexpectedly extreme reading should be repeated correctly while urgent symptoms and organ injury are assessed in parallel.

The clinically useful question is not whether pressure is simply high or low. Ask what generates the pressure, whether pulsatile and mean load are appropriate, which organs are autoregulating successfully, whether venous congestion is present, and whether microvascular flow meets metabolic demand. This layered interpretation connects the cuff reading to actual cardiovascular function. Trends during posture, exercise, illness, and treatment often reveal the remaining reserve more clearly than a single resting value.

Retrieval prompts

One. Why does a small arteriolar radius change greatly alter resistance?

Two. Distinguish pressure, flow, velocity, resistance, and compliance.

Three. Name the major causes of oedema.

Four. How does standing challenge venous return?

Five. Contrast rapid baroreflex control with long-term renal control.

Six. Why must blood pressure be interpreted with measurement conditions and organ perfusion?

Concise answers

One. Resistance varies approximately with the inverse fourth power of radius under laminar conditions.

Two. Pressure is force per area; flow is volume per time; velocity is flow per total area; resistance opposes flow; compliance is volume change per pressure change.

Three. Increased hydrostatic pressure, reduced plasma oncotic pressure, increased permeability, lymphatic obstruction, and sodium retention.

Four. Gravity pools blood below the heart, lowering central volume; reflexes, muscle pumps, respiration, and valves compensate.

Five. Baroreflexes alter neural output within seconds but reset; kidneys adjust sodium, water, hormones, and volume over longer periods.

Six. Technique affects the reading, while clinical danger depends on baseline, rate of change, and adequacy of tissue perfusion.

Source map

Original synthesis informed by Guyton and Hall, circulation, microcirculation, vascular control, venous return, and pressure regulation; OpenStax Anatomy and Physiology 2e; Katzung and OpenStax Pharmacology for vascular and antihypertensive mechanisms; Robbins for vascular pathology; and Talley and O'Connor for vascular examination.

Chapter 9: Ischaemia, Heart Failure, Valve Disease, and Arrhythmia

Orientation

Cardiovascular disease often reflects failure of supply, pump, valve, rhythm, or vascular control, with overlap between categories. Coronary ischaemia impairs myocardium and can trigger arrhythmia. Valve disease changes pressure and volume loads and can lead to heart failure. Heart failure activates compensations that initially support perfusion but later worsen congestion and remodelling. Diagnosis requires combining symptoms, examination, electrocardiography, biomarkers, and imaging rather than relying on one finding.

Atherosclerosis and coronary syndromes

Atherosclerosis develops within the arterial intima. Endothelial dysfunction permits entry and modification of apolipoprotein B-containing lipoproteins. Monocytes enter, macrophages take up lipid, and foam cells accumulate. Smooth-muscle migration, extracellular matrix, inflammation, cell death, and calcification form a plaque with a fibrous cap and lipid-rich core.

Plaque burden can narrow an artery and limit flow during exertion. Acute coronary syndromes usually follow plaque disruption or erosion with platelet activation and thrombosis, although spasm, embolism, dissection, microvascular disease, or severe supply-demand imbalance can also cause ischaemia. A plaque need not be the most severely narrowed lesion before disruption.

Stable exertional angina reflects transient demand exceeding supply, commonly producing pressure or discomfort with activity and relief with rest. Acute coronary symptoms may occur at rest, last longer, or accompany sweating, nausea, dyspnoea, or collapse. Presentations vary, especially in older adults, women, and people with diabetes or cognitive impairment.

Myocardial infarction means myocardial injury with evidence of acute ischaemia. Troponin elevation shows myocardial injury but has many causes,

including myocarditis, tachyarrhythmia, heart failure, pulmonary embolism, critical illness, and kidney disease. Diagnosis uses rise and fall, symptoms, electrocardiographic change, imaging, and context.

Early management prioritises recognition, electrocardiography, monitoring, antiplatelet and antithrombotic decisions, relief of ischaemia, and timely reperfusion when an occluded coronary artery is suspected. Oxygen is used for hypoxaemia rather than automatically. Hypotension, bleeding risk, kidney function, drug exposure, and alternative diagnoses influence treatment.

Heart failure

Heart failure is a clinical syndrome in which cardiac abnormality produces symptoms or signs from inadequate output, elevated filling pressure, or both. Ejection fraction may be reduced, mildly reduced, or preserved. A preserved ejection fraction does not mean normal function; impaired relaxation, stiffness, atrial disease, vascular load, and limited reserve can raise filling pressure despite apparently preserved fractional emptying.

Reduced forward output causes fatigue, cool extremities, kidney dysfunction, and neurohormonal activation. Elevated left-sided pressure causes pulmonary venous congestion, exertional dyspnoea, orthopnoea, paroxysmal nocturnal dyspnoea, and pulmonary oedema. Right-sided pressure causes jugular venous elevation, hepatic congestion, ascites, and peripheral oedema.

Compensation includes sympathetic activation, renin-angiotensin-aldosterone signalling, antidiuretic hormone, salt and water retention, tachycardia, vasoconstriction, and structural remodelling. These support pressure in the short term but increase afterload, oxygen demand, congestion, fibrosis, and arrhythmia over time.

Common causes include coronary disease, hypertension, valve disease, cardiomyopathy, toxins, myocarditis, tachyarrhythmia, endocrine disease, and congenital abnormalities. Precipitants of decompensation include infection, ischaemia, arrhythmia, uncontrolled pressure, missed medication, excess salt or fluid, kidney injury, anaemia, pulmonary embolism, and drugs that retain fluid or depress myocardium.

Assessment includes congestion, perfusion, rhythm, precipitant, organ function, and baseline trajectory. Natriuretic peptides support diagnosis but

vary with age, kidney function, obesity, rhythm, and acute versus chronic state. Echocardiography assesses chamber size, ejection fraction, valves, wall motion, pressure estimates, and pericardium.

Treatment logic differs between acute decompensation and long-term disease modification. Acute congestion often requires diuresis and treatment of the trigger; vasodilators or circulatory support depend on pressure and perfusion. Chronic reduced-ejection-fraction therapy combines neurohormonal and renal-metabolic pathways shown to improve outcomes, titrated with monitoring of pressure, pulse, kidney function, potassium, symptoms, and volume. Devices, revascularisation, valve intervention, or transplantation apply to selected patients.

Valve disease

Stenosis obstructs forward flow and creates a pressure load. Regurgitation permits backward flow and creates a volume load. Severity depends on valve area or regurgitant volume, pressure gradient, flow state, chamber response, pulmonary pressure, symptoms, and ventricular function.

Aortic stenosis causes left-ventricular pressure overload and concentric hypertrophy. Severe disease can produce exertional dyspnoea, angina, syncope, and heart failure. The murmur is typically systolic and radiates toward the neck, but intensity does not reliably equal severity when output is low.

Aortic regurgitation returns blood to the left ventricle during diastole, increasing stroke volume and often widening pulse pressure in chronic disease. Acute severe regurgitation is poorly tolerated because the ventricle has not adapted. Mitral regurgitation sends blood into the left atrium during systole and may arise from leaflet, chordal, papillary-muscle, annular, or ventricular disease.

Mitral stenosis obstructs left-atrial emptying, raises pulmonary venous pressure, and favours atrial enlargement and atrial fibrillation. Right-sided valve disease contributes to systemic venous congestion. Echocardiography defines anatomy and haemodynamics. Intervention timing balances symptoms, severity, ventricular response, procedural risk, and likelihood of durable repair.

Arrhythmia in disease

Atrial fibrillation can cause palpitations, dyspnoea, fatigue, heart failure, stroke, or no symptoms. Management addresses haemodynamic stability, ventricular rate, rhythm strategy when appropriate, precipitating conditions, and thromboembolic risk. Anticoagulation decisions balance embolic and bleeding risks and require attention to kidney function, interactions, adherence, and procedures.

Regular narrow-complex tachycardia commonly reflects re-entry above the ventricles. Vagal manoeuvres or atrioventricular-node blocking therapy can terminate selected rhythms when safe. Broad-complex tachycardia should be treated as ventricular tachycardia when uncertain, particularly with structural disease. Unstable tachyarrhythmia requires urgent synchronised cardioversion; pulseless ventricular tachycardia or fibrillation requires defibrillation and resuscitation.

Bradycardia may reflect physiological adaptation, drugs, sinus-node disease, conduction block, ischaemia, or metabolic disturbance. Symptoms result from inadequate output and include presyncope, syncope, dyspnoea, chest discomfort, confusion, or shock. Treatment addresses reversible causes and may require temporary or permanent pacing.

Clinical synthesis

Chest pain assessment considers coronary ischaemia, aortic catastrophe, pulmonary embolism, pneumothorax, pericarditis, oesophageal disease, and musculoskeletal causes. Examine perfusion, heart sounds, murmurs, lungs, jugular venous pressure, pulses, oedema, and signs of systemic disease. A new murmur with shock, acute pulmonary oedema, ongoing ischaemia, or infection suggests urgent structural assessment.

Treatment is not a list of isolated drugs. It is correction of supply-demand imbalance, prevention of thrombosis and remodelling, relief of congestion, control of rhythm and pressure, modification of vascular risk, and timely mechanical intervention when anatomy demands it.

Pericardial and myocardial inflammation

Pericarditis commonly causes sharp pleuritic pain improved by sitting forward, but presentation varies. A pericardial rub is specific when present but transient. Widespread electrocardiographic repolarisation change may occur. Effusion becomes dangerous when intrapericardial pressure impairs filling. Tamponade produces obstructive physiology with tachycardia, raised venous pressure, hypotension, pulsus paradoxus, and chamber collapse, although classic findings may be incomplete. Rapidly accumulating fluid can cause tamponade at a smaller volume than slowly accumulating fluid.

Myocarditis ranges from silent injury to chest pain, arrhythmia, heart failure, or shock. Causes include infection, immune disease, toxins, and drug reactions. Troponin and imaging support assessment but are not individually definitive. Strenuous exercise may increase arrhythmic risk during active inflammation. Selected patients require coronary exclusion, cardiac magnetic resonance imaging, biopsy, cause-specific immunomodulation, or mechanical support.

Prevention and longitudinal care

Cardiovascular prevention addresses smoking, blood pressure, atherogenic lipoproteins, diabetes, kidney disease, activity, diet, sleep, obesity, and psychosocial barriers. Absolute risk guides intensity more usefully than one factor in isolation. Secondary prevention after established atherosclerotic disease commonly includes antiplatelet therapy when indicated, intensive lipid lowering, pressure control, symptom-directed treatment, rehabilitation, and adherence support.

Follow-up tracks symptoms, exercise tolerance, weight, volume status, rhythm, blood pressure, kidney function, electrolytes, adverse effects, and access to care. Patients need explicit advice about chest pain, syncope, rapid weight gain, worsening breathlessness, bleeding, and medication interruption. Rehabilitation combines monitored activity, education, risk modification, and psychological recovery.

TTS module 2: Coronary flow, congestion phenotypes, and structural decision-making

Coronary syndromes as dynamic processes

Coronary flow depends on epicardial arteries, resistance vessels, collateral channels, perfusion pressure, and diastolic time. A large artery can appear open while microvascular dysfunction limits tissue perfusion. Conversely, collateral vessels can partially support myocardium beyond a chronic severe stenosis. Anatomy, flow, and tissue injury therefore provide related but non-identical information.

Atherosclerotic plaques evolve through lipid retention, inflammation, smooth-muscle response, matrix turnover, haemorrhage, and calcification. Plaque vulnerability depends less on percentage narrowing alone than on cap integrity, inflammatory activity, core composition, and mechanical stress. Disruption exposes thrombogenic material. Platelets adhere and activate, thrombin forms fibrin, and the resulting thrombus can obstruct intermittently or persistently.

Ischaemia first alters metabolism and relaxation, then contraction, electrical properties, and finally membrane integrity if severe and prolonged. Regional wall-motion abnormality can therefore appear before necrosis markers rise. Electrical instability reflects altered ion gradients, conduction slowing, catecholamines, and heterogeneity between injured and viable tissue.

Supply-demand imbalance without acute plaque thrombosis occurs when oxygen need rises or delivery falls. Tachyarrhythmia, severe hypertension, hypoxaemia, anaemia, hypotension, and critical illness can produce myocardial injury, especially when fixed coronary disease is present. The label matters because antithrombotic and invasive strategies appropriate for acute coronary thrombosis may not address the primary cause and can add bleeding harm.

Interpreting troponin and ischaemic evidence

Cardiac troponin is highly specific to myocardial tissue, but myocardial injury has many mechanisms. An acute rise or fall establishes dynamic injury. Infarction additionally requires evidence that ischaemia caused the injury. Symptoms, electrocardiographic evolution, new regional imaging changes, or identification of coronary thrombus supply that context.

The sampling interval and assay precision matter. A value below a decision threshold early after symptom onset may require repeat testing. Chronic elevations occur with structural heart disease, kidney dysfunction, and

persistent myocardial stress. Even then, a superimposed rise may indicate new injury. Comparing with previous values and the clinical trajectory is more informative than declaring every elevation an infarction.

Electrocardiographic localisation is probabilistic. Contiguous lead changes suggest a regional process, but conduction abnormalities, ventricular hypertrophy, pacing, pericardial inflammation, stress cardiomyopathy, and electrolyte disturbance can imitate or obscure ischaemia. Persistent high-risk symptoms should not be dismissed because one tracing is non-diagnostic.

Heart failure as pressure, volume, and reserve

Heart failure phenotyping begins with congestion and perfusion rather than ejection fraction alone. A patient may be wet or dry according to filling-pressure excess and warm or cold according to peripheral perfusion. These categories can change during treatment. Cold congestion signals particularly limited reserve, while apparently warm extremities do not exclude substantial pulmonary or venous pressure.

Left-sided congestion raises pulmonary capillary pressure, producing interstitial oedema, reduced lung compliance, and ventilation-perfusion mismatch. Right-sided congestion raises systemic venous pressure. Renal venous hypertension can reduce filtration and promote sodium retention even when arterial pressure appears adequate. Hepatic congestion can impair function and, over time, produce fibrosis.

Natriuretic peptides are released with myocardial wall stress. Low values can make heart failure less likely in the appropriate acute setting, but obesity can suppress concentration, while age, kidney dysfunction, atrial fibrillation, and pulmonary vascular strain can raise it. The value should be integrated with symptoms, examination, imaging, and previous baseline.

Diuretic response depends on delivery of drug to the nephron, secretion into the tubular lumen, filtered sodium load, kidney function, gut absorption, and compensatory sodium reabsorption elsewhere. Apparent resistance may reflect non-adherence, poor absorption, reduced perfusion, low dose, interacting medicines, or advanced nephron adaptation. Escalation should track urine output, weight, symptoms, pressure, kidney function, electrolytes, and residual congestion.

Disease-modifying therapy and haemodynamic tolerance

Long-term treatment for reduced ejection fraction modifies maladaptive pathways rather than merely relieving symptoms. Blocking renin-angiotensin-aldosterone signalling reduces vasoconstriction, sodium retention, and remodelling. Evidence-based beta blockade protects against chronic catecholamine toxicity and arrhythmia while allowing ventricular recovery. Mineralocorticoid receptor antagonism reduces sodium-retaining and fibrotic signalling. Sodium-glucose cotransporter two inhibition improves outcomes through combined renal and cardiovascular effects that are not explained by glucose lowering alone.

Initiation and titration require attention to blood pressure, volume, kidney function, potassium, pulse, and intercurrent illness. A modest early change in filtration after altering intraglomerular haemodynamics may differ from progressive injury due to hypoperfusion or congestion. Decisions depend on trajectory and the whole syndrome rather than one laboratory result.

Preserved-ejection-fraction heart failure is heterogeneous. Ageing, hypertension, obesity, diabetes, kidney disease, atrial fibrillation, vascular stiffness, pulmonary hypertension, and infiltrative disease can converge on impaired filling and reserve. Treatment addresses congestion, comorbidity, pressure, rhythm, ischaemia, and phenotype-specific causes. A normal resting study may miss exertional elevation of filling pressure.

Valve lesions as interacting loads

Valve severity depends on flow. A pressure gradient across a stenotic valve rises with flow, so a low gradient can occur despite severe anatomical narrowing when stroke volume is small. Integrating valve area, gradients, ventricular function, flow state, morphology, symptoms, and sometimes stress testing prevents false reassurance.

Chronic regurgitation permits chamber adaptation. A compliant receiving chamber accepts backward volume at relatively low pressure, delaying symptoms while dilation progresses. Acute regurgitation lacks time for adaptation, causing sharp pressure rises, pulmonary oedema, hypotension, or shock even when chamber size is normal.

Murmur intensity depends on flow and pressure difference, not simply lesion severity. Very low output can soften a severe stenotic murmur. Equalisation of pressures can shorten or reduce a regurgitant sound despite dangerous physiology. The consequences in chambers, pulmonary circulation, and systemic perfusion matter more than loudness.

Intervention timing seeks to act before irreversible ventricular, pulmonary vascular, or end-organ damage while avoiding unnecessary procedural risk. Symptoms may be under-recognised when activity has gradually declined. Exercise testing, serial imaging, biomarkers, and objective functional change can expose reduced reserve. Decisions follow current valve guidelines and multidisciplinary assessment.

Atrial fibrillation and thromboembolic logic

Atrial fibrillation is both an electrical and structural disease. Atrial stretch, fibrosis, inflammation, autonomic triggers, sleep-disordered breathing, obesity, alcohol, valve disease, and ageing create substrate. Once established, rapid activation promotes further electrical and mechanical remodelling, making persistence more likely.

Rate control aims to prevent excessive ventricular response and relieve symptoms. Rhythm control attempts to restore and maintain sinus rhythm through cardioversion, medicines, or ablation in selected patients. Choice depends on duration, symptoms, atrial structure, heart failure, triggers, prior attempts, treatment risk, and patient preference. Neither strategy removes the need to assess stroke risk.

Blood stasis, endothelial change, and a prothrombotic state favour atrial thrombus. Anticoagulation reduces embolic risk but creates bleeding risk, so dose, kidney function, age, weight, interactions, adherence, procedures, and previous bleeding require review. Bleeding-risk assessment should identify modifiable hazards rather than automatically deny effective stroke prevention.

Cardioversion introduces a specific embolic concern because atrial mechanical recovery can lag behind electrical rhythm restoration. Management depends on arrhythmia duration, imaging, anticoagulation status, urgency, and current protocols. Haemodynamic instability still requires prompt treatment, with thromboembolic protection addressed alongside emergency care.

Integrating structural and electrical disease

The same deterioration may contain several mechanisms. Ischaemia can reduce contraction, raise filling pressure, provoke mitral regurgitation, and trigger ventricular arrhythmia. Rapid atrial fibrillation can worsen demand, shorten filling, and reveal previously compensated valve disease. Infection can increase rate and demand while causing vasodilation and myocardial depression.

Serial assessment should ask whether symptoms arise from congestion, low output, ischaemia, rhythm, or mechanical obstruction, and which component is immediately reversible. Echocardiography, electrocardiography, biomarkers, haemodynamics, and response to treatment answer different parts of that question. The objective is not merely to name each abnormality but to identify the dominant mechanism driving present risk. Functional trajectory after intervention then tests whether that causal model was correct or requires revision.

Retrieval prompts

One. How can atherosclerotic plaque cause both stable and acute coronary syndromes?

Two. Why can heart failure occur with preserved ejection fraction?

Three. How do compensatory systems worsen chronic heart failure?

Four. Contrast stenosis with regurgitation.

Five. What are the core decisions in atrial fibrillation?

Six. Why is broad-complex tachycardia treated cautiously?

Concise answers

One. Fixed narrowing limits exertional flow, while disruption or erosion can trigger acute thrombosis.

Two. Stiffness, impaired relaxation, atrial and vascular dysfunction can raise filling pressure despite preserved fractional emptying.

Three. Vasoconstriction, salt retention, tachycardia, and remodelling increase load, congestion, oxygen demand, and fibrosis.

Four. Stenosis obstructs forward flow and causes pressure load; regurgitation permits backward flow and causes volume load.

Five. Stability, rate control, rhythm strategy, stroke prevention, triggers, and comorbid disease.

Six. It may be ventricular tachycardia and can deteriorate rapidly; inappropriate atrioventricular-node blockade can be dangerous in selected rhythms.

Source map

Original synthesis informed by Guyton and Hall, coronary circulation, cardiac failure, valves, and arrhythmia; Robbins, atherosclerosis, ischaemic heart disease, myocardium, and valves; Katzung and OpenStax Pharmacology, cardiovascular drugs; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, cardiovascular assessment.

Chapter 10: Shock, Perfusion Failure, Fluids, and Vasoactive Therapy

Orientation

Shock is acute circulatory failure causing inadequate tissue perfusion and cellular oxygen use. Blood pressure may be low, normal, or temporarily maintained. The key problem is failure to deliver or utilise sufficient oxygen and substrate while removing metabolic products. Shock becomes self-reinforcing through inflammation, endothelial injury, coagulation, myocardial dysfunction, microvascular shunting, and organ failure. Rapid recognition, mechanism-based support, source control, and repeated reassessment are essential.

Oxygen delivery and extraction

Oxygen delivery equals cardiac output multiplied by arterial oxygen content. Cardiac output equals heart rate times stroke volume. Arterial oxygen content depends mainly on haemoglobin concentration and saturation, with a small dissolved component. A patient can therefore have normal oxygen saturation but poor delivery because of anaemia or low output.

Oxygen consumption equals delivery multiplied by tissue extraction. When delivery falls, tissues initially extract more oxygen. Beyond a critical point, consumption becomes supply-dependent, anaerobic metabolism increases, and lactate may rise. Lactate also rises from adrenergic stimulation, reduced clearance, mitochondrial dysfunction, seizures, or drugs, so it is a severity and trend marker rather than a direct meter of tissue hypoxia.

Mean arterial pressure supports organ perfusion, but effective perfusion depends on local resistance, venous pressure, autoregulation, microcirculation, and baseline disease. A congested organ can be poorly perfused despite apparently adequate arterial pressure.

Shock categories

Hypovolaemic shock follows loss of blood, plasma, or extracellular fluid. Causes include haemorrhage, gastrointestinal loss, burns, renal loss, and third spacing. Reduced venous return lowers stroke volume. Sympathetic activity causes tachycardia, vasoconstriction, and cool skin, although medicines, age, pregnancy, or neurogenic states can alter these signs.

Cardiogenic shock reflects pump failure from myocardial infarction, severe ventricular dysfunction, mechanical complication, acute valve disease, myocarditis, or arrhythmia. Filling pressures are often high, causing pulmonary or systemic congestion, but right- and left-sided patterns differ.

Distributive shock features pathological vasodilation and maldistribution. Sepsis is the common example, combining vasodilation, permeability, myocardial depression, and microvascular dysfunction. Anaphylaxis combines mediator-driven vasodilation, leakage, bronchospasm, and upper-airway oedema. Neurogenic shock follows loss of sympathetic tone and may produce bradycardia with warm skin.

Obstructive shock blocks filling or outflow. Causes include massive pulmonary embolism, cardiac tamponade, tension pneumothorax, and dynamic hyperinflation. Treatment requires relieving obstruction, not simply giving escalating fluid or vasopressor doses.

Patients often have mixed shock. Sepsis can include hypovolaemia and cardiomyopathy. Myocardial infarction may cause pump failure and mechanical obstruction. Trauma can combine haemorrhage, tension pneumothorax, and neurogenic physiology.

Recognition and monitoring

Concerning features include altered mental state, cool or mottled skin, delayed capillary refill, weak pulses, oliguria, tachypnoea, increasing lactate, metabolic acidosis, hypotension, narrow pulse pressure, or deteriorating liver and kidney function. Warm vasodilated shock can have bounding pulses early. A normal single blood pressure does not exclude compensated shock.

History and examination search for bleeding, infection, chest pain, allergic exposure, fluid loss, medication effects, pregnancy, trauma, and obstruction.

Compare jugular venous pressure, lung findings, peripheral temperature, oedema, and pulse quality. Bedside ultrasound can assess cardiac activity, ventricular pattern, pericardial fluid, lung signs, aorta, and venous congestion, but findings require skill and context.

Monitoring includes repeated vital signs, mental state, urine output, perfusion examination, electrocardiography, blood gas and lactate trends, haemoglobin, electrolytes, kidney and liver function, coagulation, cultures, and targeted imaging. Invasive monitoring can help selected complex cases but does not replace clinical reassessment.

Fluids

Fluid is a drug with indication, dose, expected effect, adverse effects, and stop criteria. A bolus tests whether increasing stressed intravascular volume improves output. Response should be assessed through pressure, pulse, perfusion, urine output, lung findings, and dynamic measures when available.

Crystalloid solutions distribute through extracellular fluid, so only a fraction remains intravascular. Balanced crystalloids contain electrolyte patterns designed to reduce extreme chloride load, while saline has specific indications and risks. Albumin is a colloid with selected uses but is not universally superior. Synthetic starches can cause harm in critical illness.

Excess fluid causes pulmonary oedema, tissue oedema, impaired oxygen diffusion, abdominal pressure, delayed healing, and venous congestion of kidneys and liver. Lack of response or worsening congestion should stop blind repetition. Dynamic response to passive leg raising, respiratory variation, or a small fluid challenge may be more useful than static filling pressure, but each has limitations.

Haemorrhagic shock requires control of bleeding and appropriate blood-component resuscitation. Large crystalloid volumes can dilute clotting factors, worsen hypothermia, and disrupt clots. Trauma care integrates haemorrhage control, balanced transfusion according to protocol, calcium, warming, coagulation assessment, and selective antifibrinolytic therapy.

Vasoactive and inotropic therapy

Vasopressors raise vascular tone and pressure. Norepinephrine is commonly first-line in septic vasodilatory shock because alpha effects predominate with useful beta support. Vasopressin can supplement catecholamines. Epinephrine has strong alpha and beta effects and is essential in anaphylaxis and cardiac arrest, with selected shock use. Excess vasoconstriction can impair limb, gut, or other regional flow and increase cardiac workload.

Inotropes increase contractility when low output persists despite adequate filling and pressure. Dobutamine stimulates beta receptors and can improve output but may cause tachyarrhythmia or hypotension. Inodilators can reduce resistance while increasing contraction but may accumulate in organ dysfunction. Therapy is chosen from the haemodynamic problem, not from blood pressure alone.

Vasoactive drugs should preferably be delivered through reliable vascular access with close monitoring. In emergencies, selected agents can begin through a well-sited peripheral line while central access is arranged, according to protocol. Extravasation, arrhythmia, ischaemia, and excessive afterload require surveillance.

Mechanism-specific treatment

Septic shock requires prompt antimicrobials, cultures when they do not delay therapy, source control, fluids according to response, vasopressors, oxygenation support, and management of organ failure. Anaphylaxis requires immediate intramuscular epinephrine, airway planning, positioning, fluid, and adjuncts; antihistamines do not replace epinephrine.

Cardiogenic shock may require revascularisation, rhythm correction, careful diuresis or fluid, vasoactive support, and selected mechanical support. Mechanical complications need urgent intervention. Tamponade needs drainage. Tension pneumothorax needs immediate decompression. High-risk pulmonary embolism may require reperfusion. Haemorrhage requires control and blood replacement.

Reassessment and de-resuscitation

Resuscitation is a cycle: identify mechanism, intervene, measure response, and revise. Improvement in pressure without better perfusion is incomplete success. Once shock resolves, accumulated fluid and unnecessary vasoactive support should be reduced. De-resuscitation may use spontaneous diuresis, diuretics, or renal replacement in selected patients while preserving perfusion.

Cellular and organ consequences

Compensated shock preserves central pressure by diverting flow from skin, gut, and kidneys. Prolonged vasoconstriction and low flow then contribute to mucosal injury, acute kidney injury, hepatic dysfunction, and impaired drug clearance. Endothelial activation increases permeability and leukocyte adhesion. Coagulation may shift toward microvascular thrombosis while clotting factors and platelets are consumed, producing simultaneous thrombosis and bleeding.

The lungs develop permeability oedema and acute respiratory distress syndrome. The heart may become depressed by ischaemia, inflammatory mediators, acidosis, and excessive catecholamine exposure. Encephalopathy reflects perfusion, inflammation, metabolic disturbance, sedatives, and organ failure. Kidney injury alters potassium, acidity, volume, and medicine handling, feeding back into cardiovascular instability.

Special physiological contexts

Pregnancy increases plasma volume and cardiac output, so substantial haemorrhage may occur before hypotension appears. The gravid uterus can compress major veins when supine, reducing venous return; left lateral displacement can help. Older adults may not generate marked tachycardia and are more vulnerable to fluid overload. Beta blockers, pacemakers, autonomic neuropathy, and spinal injury alter expected compensatory signs.

Burn shock combines plasma loss through injured tissue, inflammation, evaporation, pain, and heat loss. Formulae are starting estimates, not substitutes for urine output, perfusion, respiratory status, and compartment

assessment. Adrenal insufficiency can cause vasodilatory shock that responds poorly until glucocorticoid deficiency is treated.

Safety traps

A transient pressure response does not prove that perfusion is restored. Repeated fluid can worsen occult cardiogenic or obstructive shock. Vasopressors can conceal uncorrected hypovolaemia. Sedation and positive-pressure ventilation reduce venous return and can precipitate collapse in preload-dependent patients. Transfer for imaging must not delay immediate decompression, haemorrhage control, epinephrine, or another time-critical treatment.

TTS module 2: Haemodynamic phenotyping, oxygen debt, and staged resuscitation

Shock is a mismatch, not a pressure number

The circulation must generate flow, distribute it, and maintain pressure gradients across organs. Shock can arise when any of these functions fails. Arterial hypotension is common but neither necessary nor sufficient. A young person may preserve pressure through intense vasoconstriction despite major blood loss, while a chronically low pressure may be well tolerated when perfusion is stable.

The bedside task is to identify the dominant haemodynamic phenotype. Preload failure reduces venous return. Pump failure reduces forward flow despite filling. Vasoplegia reduces effective arterial tone and stressed volume. Obstruction creates a mechanical barrier. Microcirculatory failure prevents delivered blood from being used evenly. Mixed forms are expected, especially after prolonged illness or treatment.

Compensation can hide severity. Tachycardia and vasoconstriction preserve central pressure at the expense of skin, gut, and kidney flow. Increased oxygen extraction supports consumption while venous oxygen content falls. When extraction reserve is exhausted, oxygen consumption becomes dependent on delivery and cellular dysfunction accelerates.

Oxygen content and extraction

Haemoglobin-bound oxygen dominates arterial oxygen content. Saturation therefore cannot be interpreted without haemoglobin and flow. A profoundly anaemic patient can have a perfect saturation but low content. A patient with low cardiac output can have normal content but inadequate delivery. Carbon monoxide can impair oxygen carriage and release while conventional pulse oximetry appears misleadingly reassuring.

Venous oxygen saturation reflects the balance between delivery and consumption across sampled tissues. A low value can result from low flow, anaemia, hypoxaemia, or high metabolic demand. A high value does not guarantee adequate perfusion; distributive shunting, impaired extraction, or mitochondrial dysfunction can leave oxygen in venous blood while some tissue is starved.

Oxygen debt accumulates when aerobic supply cannot meet demand. Repayment requires restored flow and oxygen content, but reperfusion can expose tissue injury and wash accumulated metabolites into the circulation. Temperature, glucose, calcium, acidity, and coagulation interact with this transition. Restoring one number without stabilising the physiological environment may not restore function.

Lactate and acid-base context

Lactate is produced continuously and can be oxidised as fuel or recycled into glucose. Its concentration rises when production exceeds clearance. Accelerated glycolysis from catecholamines may increase lactate even with oxygen present. Liver dysfunction reduces clearance, while seizures, strenuous muscle activity, selected drugs, toxins, and regional ischaemia provide additional causes.

A rising lactate in a deteriorating patient is concerning, but the mechanism should guide treatment. Blind fluid loading is inappropriate when the driver is cardiogenic congestion, seizure, hepatic failure, or beta-adrenergic stimulation. Trends are useful when interpreted with perfusion examination, urine output, mental state, acid-base status, and the interventions already given.

Metabolic acidosis can impair contractility, alter vascular responsiveness, and increase respiratory demand. The anion gap helps identify unmeasured anions but is affected by albumin and laboratory method. Hyperchloraemic

acidosis can follow chloride-rich fluid administration or gastrointestinal bicarbonate loss. Treating the cause and restoring perfusion are usually more important than chasing pH in isolation.

Fluid responsiveness is not fluid need

Fluid responsiveness means stroke volume is likely to rise after preload increases. It does not prove that fluid is the best treatment or that tissue perfusion will improve. A patient may be responsive but already oedematous, or may need vasopressor support because venous and arterial tone are the main problem. Conversely, a non-responsive patient may still require replacement of ongoing haemorrhage.

Static filling pressures poorly predict response because ventricular compliance varies. Dynamic tests use a reversible preload change. Passive leg raising transfers venous blood centrally and can be assessed with a real-time output measure. Respiratory variation in selected arterial or venous waveforms may help during controlled ventilation and regular rhythm, but fails under many common conditions.

A fluid challenge should specify type, amount, rate, target, and stop condition. The most convincing response is improved stroke volume or perfusion with acceptable filling pressure. A transient blood-pressure rise alone can reflect vascular tone or measurement noise. New crackles, hypoxaemia, rising venous pressure, hepatic congestion, or no output response argues against repetition.

Fluid composition matters. Large chloride loads can alter acid-base balance and renal vascular behaviour. Balanced crystalloids contain buffers and lower chloride but are not identical to plasma. Blood products restore oxygen carriage and haemostatic components in haemorrhage but also carry transfusion risks. The correct fluid follows the deficit being treated.

Vascular tone and vasoactive selection

Mean arterial pressure approximates cardiac output multiplied by systemic resistance, but raising resistance can reduce flow when the ventricle cannot overcome the new load. A vasopressor can improve coronary and cerebral perfusion pressure while simultaneously increasing myocardial work and constricting vulnerable beds. Dose should be titrated to organ perfusion and

the patient's autoregulatory context, not to an unnecessarily high universal target.

Norepinephrine predominantly raises arterial and venous tone with some cardiac beta activity. Venokonstriction recruits unstressed volume and can improve preload as well as pressure. Vasopressin acts through a non-catecholamine pathway and may reduce catecholamine requirements. Epinephrine increases rate, contractility, and tone but can raise lactate and provoke arrhythmia.

An inotrope is appropriate when inadequate contractile output remains an important mechanism after filling and pressure are addressed. Increasing contractility can improve flow but also raises oxygen demand and arrhythmia risk. Inodilators may reduce afterload while improving contraction, yet their vasodilation can worsen hypotension and their effect may persist in organ dysfunction.

Vasoactive support buys time; it does not remove the cause. Escalating doses should trigger a search for ongoing bleeding, unrelieved obstruction, inadequate source control, adrenal failure, severe acidosis, occult pump dysfunction, drug effect, or measurement error.

Congestion and the right ventricle

The right ventricle is sensitive to acute afterload. Pulmonary embolism, hypoxic vasoconstriction, high ventilatory pressure, or severe pulmonary disease can sharply increase pulmonary resistance. The right ventricle dilates, wall stress and oxygen demand rise, tricuspid regurgitation worsens, and the septum shifts leftward, reducing left-ventricular filling.

Large fluid boluses can worsen this cycle by further dilating the right ventricle without increasing forward flow. Management may require cautious preload optimisation, correction of hypoxia and acidosis, reduction of avoidable ventilatory pressure, support of systemic pressure for right-coronary perfusion, and definitive treatment of the pulmonary obstruction or vascular cause.

Venous congestion itself injures organs. High renal venous pressure reduces the gradient across the kidney. Hepatic and gut congestion impair function

and drug handling. Peripheral oedema is visible evidence, but internal congestion can be substantial before it appears.

Phases of resuscitation

Early rescue treats immediate threats with rapid, mechanism-directed action. Optimisation then refines preload, pressure, output, oxygen carriage, ventilation, temperature, electrolytes, and source control. Stabilisation prevents recurrence and reduces unnecessary support. Evacuation or de-resuscitation removes accumulated fluid once shock has resolved and vascular refill can tolerate it.

These phases can overlap, and a patient can regress. Ongoing capillary leak may make fluid removal premature. Persistent congestion may impede organ recovery even after pressure normalises. Daily balance, weight, examination, ultrasound, respiratory status, kidney function, and response to diuretics help define the trajectory.

Mechanical support is not a generic solution. Devices alter preload, afterload, flow, and complications in different ways. Selection depends on cause, reversibility, ventricular involvement, timing, expertise, and goals of care. Support without a recovery, bridge, or destination strategy can prolong harm.

The discipline of shock care is repeated causal testing. State the presumed mechanism, choose an intervention that should change a measurable variable, observe whether it does, and revise quickly when the response contradicts the model. Pressure, flow, congestion, oxygen content, extraction, and cellular response must improve together before resuscitation can be called successful. The final endpoint is durable organ recovery with the least harmful exposure to fluid, catecholamines, ventilation, transfusion, and invasive support.

Retrieval prompts

One. What determines oxygen delivery?

Two. Contrast the four major shock categories.

Three. Why can normal saturation coexist with poor oxygen delivery?

Four. How should a fluid bolus be prescribed and assessed?

Five. Distinguish vasopressors from inotropes.

Six. Why must shock treatment focus on mechanism?

Concise answers

One. Cardiac output multiplied by arterial oxygen content.

Two. Hypovolaemic loses circulating volume; cardiogenic loses pump function; distributive loses vascular tone and distribution; obstructive blocks filling or outflow.

Three. Oxygen content also depends heavily on haemoglobin, and delivery depends on cardiac output.

Four. Give a defined dose for a defined indication, then reassess perfusion, pressure, congestion, output, and adverse effects.

Five. Vasopressors primarily raise vascular tone and pressure; inotropes primarily increase myocardial contraction.

Six. Bleeding, infection, pump failure, anaphylaxis, tamponade, and pulmonary embolism require different definitive actions.

Source map

Original synthesis informed by Guyton and Hall, circulatory shock and cardiac output; Robbins, shock, sepsis, thrombosis, and myocardial injury; Katzung and OpenStax Pharmacology, vasoactive drugs; OpenStax Medical-Surgical Nursing, resuscitation and monitoring; and Talley and O'Connor, acute assessment.

Chapter 11: Red Cells, Anaemia, Haemostasis, Thrombosis, and Transfusion

Orientation

Blood transports respiratory gases, nutrients, hormones, heat, immune cells, and waste. Red cells carry haemoglobin; platelets and coagulation preserve vascular integrity; plasma proteins maintain oncotic pressure and transport molecules. Disorders can reduce oxygen delivery, cause bleeding, promote thrombosis, or create both simultaneously. Interpretation begins with physiology, cell counts, morphology, coagulation tests, clinical context, and time course.

Red-cell production and turnover

Haematopoietic stem cells in marrow generate red cells, white cells, and platelets through regulated progenitors. Erythropoietin is produced mainly by renal interstitial cells when oxygen delivery is inadequate. It supports survival and maturation of erythroid precursors. Effective erythropoiesis requires iron, vitamin B twelve, folate, amino acids, healthy marrow, and suitable hormonal and inflammatory conditions.

Reticulocytes are recently released red cells. A raised appropriately corrected reticulocyte response suggests marrow compensation after blood loss or haemolysis. An inadequate response suggests impaired production, although timing matters because marrow response is not immediate.

Haemoglobin contains globin chains and iron-bearing haem groups that bind oxygen cooperatively. Red cells lack nuclei and mitochondria and depend on glycolysis. Their deformability permits passage through capillaries. Senescent or damaged cells are removed mainly by splenic and hepatic macrophages. Iron is recycled, globin is reused as amino acids, and haem is converted to bilirubin.

Anaemia by mechanism

Anaemia is reduced haemoglobin concentration relative to an appropriate reference and physiological context. It reduces arterial oxygen content. Symptoms depend on severity, speed, age, cardiopulmonary reserve, and demand. Fatigue, dyspnoea, palpitations, dizziness, chest pain, flow murmurs, and heart failure can occur, but mild chronic anaemia may be silent.

Microcytosis suggests impaired haemoglobin synthesis. Iron deficiency is common and should trigger a search for cause: blood loss, increased requirement, reduced intake, or malabsorption. Ferritin reflects iron stores but rises with inflammation, so transferrin saturation, inflammatory markers, and context help. Thalassaemia reduces globin-chain production and can produce marked microcytosis with a relatively preserved red-cell count.

Macrocytosis can reflect vitamin B twelve or folate deficiency, alcohol, liver disease, hypothyroidism, reticulocytosis, medication, or marrow disorder. Vitamin B twelve deficiency can cause neurological injury and must not be treated with folate alone when suspected. Macrocytosis may be absent when mixed deficiencies coexist.

Normocytic anaemia includes acute blood loss, haemolysis, chronic inflammation, kidney disease, endocrine disease, marrow failure, and mixed deficiency. Inflammation raises hepcidin, reducing intestinal iron absorption and trapping iron in storage cells. Kidney disease reduces erythropoietin and shortens red-cell survival.

Haemolysis may be intrinsic to the red cell, as in membrane, enzyme, or haemoglobin disorders, or extrinsic through antibodies, mechanical destruction, infection, toxins, burns, or hypersplenism. Findings can include increased unconjugated bilirubin and lactate dehydrogenase, reduced haptoglobin, reticulocytosis, jaundice, dark urine, and abnormal blood film. Intravascular and extravascular haemolysis produce overlapping but different patterns.

Platelets and primary haemostasis

Vascular injury causes local vasoconstriction and exposes subendothelial matrix. Von Willebrand factor links platelets to collagen through platelet glycoprotein receptors. Platelet activation changes shape, releases granule contents, synthesises thromboxane, and activates glycoprotein two b three a receptors. Fibrinogen bridges these receptors between platelets, forming the primary plug.

Platelet number alone does not determine function. Antiplatelet drugs, uraemia, inherited receptor defects, cardiopulmonary bypass, and systemic illness can impair function. Primary haemostatic disorders often cause petechiae, mucosal bleeding, immediate procedural bleeding, or heavy menstruation, whereas coagulation-factor disorders more often cause deep tissue, joint, or delayed bleeding. Real patients can show mixed patterns.

Coagulation and fibrinolysis

Tissue factor exposed at injury initiates coagulation and generates thrombin. Thrombin converts fibrinogen to fibrin, activates platelets and cofactors, and amplifies its own generation on phospholipid surfaces. Factor thirteen cross-links fibrin. Natural anticoagulant systems include antithrombin, protein C with protein S, tissue-factor pathway inhibitor, and intact endothelium.

The prothrombin time and international normalised ratio assess selected extrinsic and common pathway factors and monitor vitamin K antagonist therapy. The activated partial thromboplastin time assesses selected intrinsic and common factors and monitors some unfractionated heparin use. Normal screening tests do not exclude every bleeding disorder, and prolonged tests do not automatically predict clinical bleeding.

Fibrinolysis removes clot during repair. Tissue plasminogen activator converts plasminogen to plasmin, which degrades fibrin and produces D-dimer fragments. D-dimer is sensitive to recent clot formation and breakdown but rises with age, infection, inflammation, pregnancy, trauma, malignancy, and surgery. It is most useful to exclude venous thromboembolism in appropriately selected low- or intermediate-risk patients.

Thrombosis

Thrombosis is promoted by endothelial injury, abnormal flow, and hypercoagulability: Virchow's triad. Arterial thrombi form in high-flow settings, often on disrupted atherosclerotic plaque, and are platelet-rich. Venous thrombi are promoted by stasis and coagulation and contain abundant fibrin and red cells. This distinction guides the relative roles of antiplatelet and anticoagulant treatment.

Venous risk factors include surgery, immobility, malignancy, pregnancy and postpartum state, oestrogen exposure, previous thrombosis, inherited thrombophilia, inflammation, and central venous devices. Deep-vein thrombosis may cause unilateral swelling, pain, warmth, or no symptoms. Pulmonary embolism may cause sudden dyspnoea, pleuritic pain, tachycardia, syncope, hypoxaemia, or shock.

Disseminated intravascular coagulation is systemic coagulation activation with microvascular fibrin, consumption of platelets and factors, secondary fibrinolysis, organ injury, and bleeding. Triggers include sepsis, trauma, obstetric catastrophe, malignancy, and severe tissue injury. Treat the cause while supporting coagulation and organs according to bleeding, thrombosis, procedures, and laboratory trends.

Antithrombotic drugs

Antiplatelet drugs interfere with platelet activation or aggregation and are central to many arterial syndromes. Anticoagulants reduce thrombin generation or activity and prevent extension and recurrence of venous and cardiac-source thrombi. Fibrinolytic drugs accelerate plasmin generation and can dissolve clot but carry substantial bleeding risk.

Choice depends on indication, onset, reversibility, kidney and liver function, interactions, pregnancy, adherence, body size, cancer, procedures, and bleeding history. Monitoring differs by drug. Reversal may require time, specific antidotes, vitamin K, protamine, factor concentrates, or supportive transfusion. Stopping treatment can also be dangerous, especially after recent thrombosis or coronary stenting.

Transfusion

Red-cell transfusion increases oxygen-carrying capacity but does not treat the cause of anaemia. Decisions use symptoms, haemodynamic state, active bleeding, haemoglobin, trajectory, cardiac disease, and alternatives rather than one universal threshold. One unit should be followed by reassessment when circumstances permit.

Compatibility testing reduces haemolytic reactions. Acute reactions include haemolysis, febrile non-haemolytic reactions, allergy, anaphylaxis, transfusion-associated circulatory overload, transfusion-related acute lung injury, bacterial contamination, and electrolyte or temperature disturbance during massive transfusion. If a reaction is suspected, stop the transfusion, maintain access with appropriate fluid, assess the patient, check identity, notify the blood service, and investigate according to protocol.

Platelets treat selected thrombocytopenic or functional bleeding states. Plasma supplies multiple coagulation factors and is not a general volume expander. Cryoprecipitate or fibrinogen products treat significant hypofibrinogenaemia. Patient blood management reduces avoidable loss, treats deficiency, uses appropriate thresholds, and minimises unnecessary sampling.

Blood-film and marrow logic

The peripheral film can reveal size variation, shape change, fragments, spherocytes, target cells, nucleated cells, parasites, abnormal white cells, and platelet morphology. Schistocytes suggest mechanical fragmentation but require clinical correlation. Spherocytes occur in immune haemolysis and membrane disorders. Pancytopenia raises concern for marrow failure, infiltration, severe deficiency, hypersplenism, drugs, infection, or systemic illness. Bone-marrow examination is considered when peripheral evidence cannot explain cytopenia, abnormal cells suggest malignancy, or direct assessment of production, infiltration, fibrosis, infection, or iron is required.

Splenomegaly can sequester and destroy blood cells and may indicate portal hypertension, infection, haemolysis, malignancy, or infiltrative disease. Lymphadenopathy requires assessment of distribution, size, tenderness, texture, duration, systemic symptoms, and local drainage. Haematology

interpretation therefore joins cell numbers with morphology, production, destruction, distribution, and clinical examination.

TTS module 2: Marrow response, clot architecture, and transfusion physiology

Quantifying red-cell production

The reticulocyte percentage must be interpreted against the degree of anaemia. A seemingly normal percentage can represent inadequate production when the total red-cell mass is low. Correcting for haematocrit and for the prolonged circulation time of prematurely released reticulocytes gives a better estimate of marrow output. The practical distinction is whether the marrow response is proportionate to loss or destruction.

Production failure can occur despite abundant erythropoietin when marrow lacks iron, vitamins, healthy progenitors, or a suitable environment. Inflammation suppresses erythropoietin response and increases hepcidin. Kidney disease reduces hormone production and exposes red cells to uraemic stress. Marrow infiltration physically and chemically disrupts haematopoiesis. Several mechanisms commonly coexist.

Iron moves through a regulated circuit. Enterocytes absorb dietary iron, transferrin carries it in plasma, marrow incorporates it into haem, macrophages recover it from senescent cells, and ferritin stores it. Hepcidin binds the exporter ferroportin and causes its internalisation, reducing intestinal transfer and macrophage iron release. High hepcidin protects against iron availability to pathogens but restricts erythropoiesis.

Ferritin can be low only when stores are depleted, making a low value strongly supportive of deficiency. A normal or high value does not exclude deficiency during inflammation, liver injury, or malignancy. Transferrin saturation estimates circulating iron availability, while soluble transferrin receptor and marrow assessment have selected roles. The cause of deficiency matters more than replacement alone.

Morphology as a mechanism map

Mean cell volume is an average and can conceal mixed populations. Combined iron and vitamin B twelve deficiency may produce a normal

average with increased size variation. Reticulocytosis raises the average because reticulocytes are larger. Blood-film review can therefore reveal information hidden by summary indices.

Target cells form when membrane area is high relative to cell volume and appear in selected haemoglobin disorders, liver disease, and hyposplenic states. Schistocytes reflect mechanical fragmentation in microangiopathic processes, prosthetic devices, or severe vascular injury. Spherocytes have reduced surface area and arise from membrane defects or immune-mediated removal. Bite cells and inclusions can point toward oxidant injury or splenic processing.

Haemolysis should be localised conceptually. Intravascular destruction releases haemoglobin directly into plasma, consuming haptoglobin and potentially causing haemoglobinaemia and haemoglobinuria. Extravascular destruction occurs mainly within macrophages in spleen and liver, producing splenomegaly and bilirubin elevation. Both raise lactate dehydrogenase and reticulocytes when marrow reserve is intact.

The direct antiglobulin test detects immunoglobulin or complement attached to red cells and supports immune haemolysis in the right syndrome. It does not quantify severity or identify every cause, and positive results can occur without active destruction. The film, haemolysis markers, medication history, transfusion history, and clinical trajectory remain necessary.

Haemostasis occurs on cell surfaces

The coagulation cascade is useful for laboratory organisation, but haemostasis in vivo occurs on activated cell and platelet surfaces. Tissue-factor exposure generates a small initial amount of thrombin. Thrombin activates platelets and cofactors, allowing a larger burst on the platelet phospholipid surface. Fibrin then stabilises the platelet plug.

Endothelium normally resists clotting through nitric oxide, prostacyclin, anticoagulant proteins, and a glycocalyx that separates blood from procoagulant structures. Injury or inflammation shifts the surface toward adhesion, tissue factor, reduced anticoagulant activity, and impaired fibrinolysis. The same vessel lining therefore regulates both bleeding prevention and thrombosis.

Von Willebrand factor protects factor eight in circulation and mediates platelet adhesion under shear. Deficiency or dysfunction can cause mucocutaneous bleeding with variable laboratory findings. Stress, pregnancy, inflammation, blood group, and hormones influence concentration, so repeated or specialised testing may be needed when suspicion remains.

Platelet count provides no direct measure of adhesion, secretion, aggregation, or interaction with endothelium. Severe thrombocytopenia increases spontaneous bleeding risk, but procedural risk also depends on platelet function, tissue site, anticoagulants, fibrinogen, vascular integrity, and cause. A stable chronic count may behave differently from a rapidly falling count in systemic illness.

Coagulation tests and mixing logic

The prothrombin time and activated partial thromboplastin time begin clotting under artificial laboratory conditions. They sample selected factor pathways but omit platelet function, vascular integrity, factor thirteen activity, and much of fibrinolysis. A normal result cannot exclude important bleeding disorders.

When a clotting time is prolonged, a mixing study combines patient plasma with normal plasma. Correction suggests a factor deficiency because normal plasma supplies what is missing. Failure to correct suggests an inhibitor, although time-dependent inhibitors and anticoagulant medicines complicate interpretation. Further testing follows the clinical bleeding or thrombosis pattern.

Lupus anticoagulants illustrate the difference between laboratory and physiology. They can prolong phospholipid-dependent clotting tests while associating clinically with thrombosis rather than bleeding. The name reflects an in-vitro effect, not anticoagulation in the patient. Anticoagulant drugs, factor inhibitors, liver disease, vitamin K deficiency, and consumption must be separated mechanistically.

Fibrinogen can fall through consumption, severe liver dysfunction, dilution, or primary fibrinolysis, yet it may initially rise as an acute-phase protein. D-dimer confirms that cross-linked fibrin was formed and degraded but not where or why. Serial platelet count, fibrinogen, clotting times, D-dimer, film, and organ function help identify evolving disseminated intravascular coagulation.

Thrombotic risk and anticoagulant control

Venous thrombosis often begins in valve pockets where flow is slow and local hypoxia activates endothelium. Inflammation, tissue factor, neutrophil products, platelets, and coagulation interact. Cancer can release procoagulant material and compress vessels. Surgery combines injury, immobility, inflammation, and temporary changes in natural anticoagulants.

Thrombophilia testing is most useful when the result will change management or family counselling. Acute thrombosis, pregnancy, anticoagulants, inflammation, and liver disease can distort assays. Testing indiscriminately after every provoked event creates ambiguous labels without improving care. Timing and selection should follow specialist and current local guidance.

Anticoagulants prevent extension and recurrence more reliably than they dissolve established clot. The body's fibrinolytic system remodels thrombus while treatment suppresses new propagation. Duration balances recurrence risk against bleeding and changes with whether the provoking factor was transient, persistent, malignant, or unprovoked.

Renal function, hepatic function, age, weight, interacting drugs, adherence, and procedure timing alter anticoagulant safety. A "therapeutic" prescription can become excessive during acute kidney injury or reduced intake. Bleeding should prompt resuscitation and source control alongside decisions about withholding or reversing anticoagulation.

Transfusion as transplantation of function

Red-cell transfusion increases haemoglobin and therefore potential oxygen content, but benefit depends on cardiac output, saturation, microcirculation, and tissue demand. Stored cells also add volume. In a congested patient, a slower rate, diuretic strategy, or alternative may be needed. In life-threatening haemorrhage, speed and balanced replacement take priority.

Transfusion-associated circulatory overload reflects hydrostatic pulmonary oedema from volume and pressure stress. Transfusion-related acute lung injury reflects inflammatory permeability oedema and can occur without high filling pressure. Both cause respiratory deterioration, but their haemodynamics and future transfusion planning differ.

Haemolytic reactions can be immediate or delayed. Clerical mismatch is a preventable cause of catastrophic acute haemolysis, making bedside identity checks a critical physiological intervention. Delayed alloantibody responses may lower haemoglobin days later and complicate future compatibility. Pregnancy and previous transfusion increase sensitisation opportunities.

Massive transfusion introduces dilution, hypothermia, acidity, citrate-related low ionised calcium, potassium disturbance, and changing coagulation. Warming, calcium assessment, laboratory or viscoelastic guidance, and haemorrhage control accompany component replacement. The endpoint is not merely a normal count but restored perfusion and haemostasis without avoidable overload or thrombosis.

Patient blood management begins before transfusion is needed. It identifies iron or vitamin deficiency, reduces unnecessary phlebotomy, corrects reversible coagulopathy, plans surgery around bleeding risk, and uses cell-salvage or haemostatic techniques when appropriate. After transfusion, document the indication, response, reaction history, and any newly detected antibody. This closes the safety loop and informs future pregnancy, surgery, emergency care, and compatibility testing.

Retrieval prompts

One. How does the reticulocyte response separate production failure from loss or destruction?

Two. Contrast iron deficiency, inflammatory anaemia, vitamin B twelve deficiency, and haemolysis.

Three. Trace primary haemostasis and fibrin formation.

Four. State Virchow's triad.

Five. Why is D-dimer useful only in selected patients?

Six. What is the immediate response to a suspected transfusion reaction?

Concise answers

One. An appropriate rise shows marrow compensation; an inadequate rise suggests impaired production, allowing for delay.

Two. Iron deficiency depletes stores; inflammation traps iron; vitamin B twelve deficiency impairs deoxyribonucleic acid synthesis and may injure nerves; haemolysis shortens red-cell survival.

Three. Platelets adhere through von Willebrand factor, activate and aggregate; tissue factor generates thrombin, which forms and cross-links fibrin.

Four. Endothelial injury, abnormal blood flow, and hypercoagulability.

Five. It is sensitive but non-specific and best excludes thrombosis when pre-test probability is not high.

Six. Stop the transfusion, assess and support the patient, verify identity, maintain access, notify the blood service, and investigate.

Source map

Original synthesis informed by Guyton and Hall, red cells, immunity, blood groups, haemostasis, and coagulation; Robbins, red-cell disease, thrombosis, and shock; Katzung and OpenStax Pharmacology, antithrombotic drugs; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, haematological assessment.

Chapter 12: Cardiovascular History, Examination, and Core Investigations

Orientation

Cardiovascular assessment identifies instability, characterises symptoms, estimates haemodynamics, localises disease, and selects investigations. The same finding can arise from different mechanisms, so history, examination, electrocardiography, biomarkers, and imaging must be integrated. Assessment should establish functional impact and trajectory, not merely name a lesion.

Presenting symptoms

Chest discomfort is described by onset, location, quality, duration, radiation, triggers, relief, associated symptoms, and recurrence. Pressure with exertion suggests myocardial ischaemia, but burning, indigestion, dyspnoea, fatigue, or nausea can be equivalents. Sudden tearing pain to the back raises concern for acute aortic disease. Pleuritic pain suggests pericardial, pulmonary, or chest-wall causes. Reproducibility with palpation supports a musculoskeletal source but does not absolutely exclude coexisting disease.

Dyspnoea requires timing, exertional threshold, orthopnoea, nocturnal episodes, wheeze, cough, oedema, weight change, and cardiopulmonary history. Orthopnoea reflects increased venous return and redistribution when supine in a patient with limited cardiac reserve. Paroxysmal nocturnal dyspnoea wakes the patient after sleep and is more specific for congestion than simple breathlessness.

Palpitations may represent awareness of normal rhythm, ectopy, tachycardia, or irregular rhythm. Ask whether onset and termination are abrupt, whether rhythm feels regular, and whether symptoms include syncope, chest pain, breathlessness, or exertional association. A pulse or wearable recording during symptoms is more useful than a retrospective adjective.

Syncope is transient loss of consciousness from global cerebral hypoperfusion with rapid recovery. Distinguish it from seizure, intoxication, metabolic disturbance, and falls without loss of consciousness. Exertional or supine syncope, structural heart disease, family history of sudden death, abnormal electrocardiogram, bleeding, or persistent instability raises risk. Reflex syncope often has triggers and autonomic prodrome. Orthostatic syncope follows postural pressure failure. Cardiac syncope may be abrupt with little warning.

Peripheral vascular symptoms include exertional limb pain, rest pain, non-healing wounds, colour change, unilateral swelling, and sudden painful cold limb. Venous thrombosis and acute arterial occlusion require timely recognition. Ask about stroke or transient neurological symptoms because cardiac and vascular disease are systemic.

Risk and background

Assess hypertension, lipids, diabetes, kidney disease, smoking, sleep apnoea, inflammatory disease, pregnancy complications, prior vascular events, rheumatic fever, congenital disease, infection, cancer therapy, and family history of premature disease or sudden death. Medication review includes adherence, antithrombotic drugs, anti-inflammatory drugs, stimulants, decongestants, hormones, supplements, and drugs affecting electrolytes or Q T interval.

Functional status can be described by activities achieved, walking distance, stairs, pace, and symptom limitation. Determine baseline and rate of change. Frailty, cognition, falls, support, driving, work, and medication access influence treatment.

General examination and vital signs

Observe distress, work of breathing, colour, sweating, body habitus, movement, and mental state. Measure pulse, blood pressure, respiratory rate, oxygen saturation, and temperature accurately. Compare arms initially when relevant. Measure standing pressure when orthostatic symptoms or autonomic dysfunction are possible.

Examine hands for temperature, perfusion, clubbing, splinter haemorrhages, tendon xanthomata, and nicotine staining, recognising limited specificity. Assess conjunctival pallor, central cyanosis, oral health, and features of endocrine or connective-tissue disease when indicated.

Pulse and venous pressure

Assess pulse rate, rhythm, volume, character, and symmetry. A slow-rising low-volume pulse suggests severe outflow obstruction. A collapsing pulse may accompany high stroke volume and rapid diastolic runoff. Radiofemoral delay or unequal pulses suggest vascular obstruction or aortic disease. Pulse deficit occurs when not every ventricular contraction produces a palpable peripheral wave.

Jugular venous pressure estimates right-atrial pressure. Position the patient, identify internal jugular pulsation, and measure vertical height above the sternal angle. Venous pulsation is usually non-palpable, varies with position and respiration, and has multiple waveforms. Elevated pressure suggests volume overload, right-heart failure, tamponade, constriction, or obstruction. A rise with inspiration can indicate impaired right-sided filling. Technique and body habitus limit accuracy.

Precordial and peripheral examination

Inspect and palpate the chest for scars, devices, apex beat, heaves, and thrills. The apex location and character reflect heart position, chamber enlargement, and chest anatomy. A parasternal heave suggests right-ventricular pressure load. A thrill is a palpable murmur.

Auscultate systematically with diaphragm and bell positions suited to frequency. The first sound reflects atrioventricular valve closure and the second semilunar closure. Physiological splitting of the second sound changes with respiration. Added sounds include a third sound from rapid filling and a fourth from atrial contraction into a stiff ventricle. Context determines whether they are physiological or pathological.

Describe murmurs by timing, site, radiation, intensity, pitch, shape, and response to manoeuvres. Systolic murmurs may be ejection or regurgitant. Diastolic murmurs are generally pathological. Maneuvers alter venous return,

afterload, chamber size, or outflow and can clarify mechanism. Murmur intensity does not reliably equal lesion severity, especially when flow is low.

Examine lungs for congestion or pleural fluid, abdomen for liver enlargement, pulsatile masses or ascites, and legs for oedema, perfusion, ulcers, and venous disease. Oedema is not specific to heart failure; kidney, liver, venous, lymphatic, and medication causes remain possible.

Electrocardiography and monitoring

A twelve-lead electrocardiogram assesses rhythm, rate, conduction, axis, chamber patterns, ischaemia, prior infarction, electrolyte effects, and drug toxicity. Obtain it promptly during ongoing chest pain, arrhythmia, syncope, or instability. Serial tracings can reveal dynamic change. A normal tracing does not exclude intermittent arrhythmia or acute coronary disease.

Ambulatory monitoring is selected by symptom frequency. Continuous short monitoring suits daily symptoms; longer patch, event, or implantable monitoring suits less frequent episodes. Correlating symptoms with rhythm is crucial because incidental ectopy may not explain events.

Biomarkers and imaging

Troponin detects myocardial injury and should be interpreted through assay-specific timing and change. Natriuretic peptides support heart-failure assessment but vary with age, obesity, kidney function, rhythm, and acute strain. Lipids, glycated haemoglobin, blood count, electrolytes, kidney function, thyroid tests, inflammatory markers, and cultures are selected by the clinical problem.

Chest radiography assesses heart size, pulmonary vessels, oedema, pleural fluid, lung disease, and devices but can be normal in important disease. Echocardiography evaluates chamber dimensions, systolic and diastolic function, valves, wall motion, pericardium, congenital anatomy, and haemodynamic estimates. Image quality and operator interpretation matter.

Coronary computed tomography defines coronary anatomy in selected patients. Invasive angiography permits direct visualisation and intervention but carries procedural risk. Stress testing provokes demand through exercise or medication and detects symptoms, electrocardiographic change, perfusion

abnormality, or wall-motion response. Cardiac magnetic resonance characterises tissue, function, inflammation, scar, and infiltrative disease. The best test depends on pre-test probability and the decision it will change.

Integrated bedside synthesis

After assessment, state whether the patient is stable, congested or dry, warm or poorly perfused, in sinus rhythm or arrhythmia, and whether the dominant concern is coronary, pump, valve, pericardial, vascular, or non-cardiac disease. Identify time-critical alternatives and specify the next decision.

Vascular investigations

The ankle-brachial pressure index compares ankle with arm systolic pressure and supports assessment of peripheral arterial disease. Calcified incompressible vessels, especially in diabetes or kidney disease, can produce falsely high values; toe pressure or waveform assessment may help. Duplex ultrasound combines anatomy with flow velocity for carotid, peripheral arterial, and venous disease. Computed-tomography angiography provides rapid vascular mapping but uses radiation and iodinated contrast. Magnetic-resonance angiography avoids ionising radiation but has its own limitations.

Suspected deep-vein thrombosis is assessed through pre-test probability, D-dimer in suitable patients, and compression ultrasonography. Suspected pulmonary embolism uses structured probability, oxygenation and haemodynamic assessment, and selected imaging. Aortic disease may require urgent computed tomography, transoesophageal echocardiography, or magnetic resonance depending on stability and availability. Testing should never delay immediate treatment of a clinically evident limb-threatening occlusion, ruptured aneurysm, or unstable obstruction.

Serial assessment matters after intervention. Recheck symptoms, distal pulses, access sites, kidney function, haemoglobin, rhythm, and perfusion. A technically successful procedure can still be followed by bleeding, embolisation, contrast-associated injury, reperfusion injury, or compartment syndrome.

TTS module 2: Bedside haemodynamics, test selection, and longitudinal cardiovascular inference

Symptoms as physiological stress tests

Cardiovascular symptoms often appear when demand exposes limited reserve. Exertional chest pressure suggests that coronary flow cannot rise sufficiently for myocardial demand. Exertional dyspnoea may reflect rising filling pressure, inadequate output, pulmonary vascular limitation, anaemia, or deconditioning. The activity level that provokes symptoms provides a repeatable measure of functional change.

Ask what the patient has stopped doing. A person may deny exertional symptoms because they have gradually reduced walking, stairs, carrying, or pace. Comparing current ability with previous baseline and observing recovery time can reveal progression more clearly than a generic severity score.

Orthopnoea is influenced by fluid redistribution, diaphragm position, obesity, lung disease, and sleep-related breathing, so the number of pillows is not a diagnosis. Paroxysmal nocturnal dyspnoea describes awakening after a period asleep with marked breathlessness and a need to sit or stand. Its delay and severity distinguish it from immediate positional discomfort.

Palpitation history should reconstruct onset and offset. Abrupt onset suggests an electrical transition, while gradual acceleration may reflect sinus tachycardia responding to fever, pain, hypovolaemia, or anxiety. Regular rapid pounding suggests a different mechanism from irregular fluttering, but patient descriptions are imperfect. Capturing rhythm during symptoms remains the decisive aim.

Perfusion and congestion at the bedside

Peripheral temperature, capillary refill, pulse volume, mental state, and urine output provide a composite view of perfusion. Each is influenced by environment, age, medicines, and local disease. Cold hands may reflect sympathetic vasoconstriction without critical shock; warm skin can coexist with distributive perfusion failure. Trends and coherence matter.

Jugular venous pressure estimates right-atrial pressure but does not directly measure total-body fluid. A high value can arise from right-heart failure, tricuspid disease, tamponade, constriction, pulmonary hypertension, or positive intrathoracic pressure. A low value supports reduced central filling but can be difficult to visualise.

The waveform adds information. The a wave reflects atrial contraction and disappears in atrial fibrillation. A large a wave may occur when the atrium contracts against increased resistance or a closed atrioventricular valve. The v wave reflects atrial filling during ventricular systole and can become prominent in tricuspid regurgitation. Precise interpretation requires a visible waveform and rhythm correlation.

Hepatojugular or abdominojugular pressure tests whether the right heart can accommodate a transient increase in venous return. A sustained rise supports elevated filling pressure. It should be performed with controlled pressure and observed long enough to distinguish a transient normal response from persistent elevation.

Peripheral oedema describes interstitial fluid, not its cause. Bilateral dependent oedema can reflect cardiac, renal, hepatic, venous, lymphatic, medication-related, or immobility mechanisms. Sacral oedema may predominate in bedbound patients. Unilateral swelling raises venous or lymphatic obstruction, infection, and local injury.

Pulses, murmurs, and manoeuvres

Pulse contour expresses ventricular ejection and arterial properties. A delayed low-amplitude carotid upstroke supports significant aortic outflow obstruction. A brisk collapsing pulse suggests rapid systolic ejection with fast diastolic runoff, seen in aortic regurgitation and other high-output states. Severe low output can make every pulse small and obscure distinctive patterns.

Murmur timing localises the pressure gradient. Ejection murmurs rise and fall as forward flow accelerates and decelerates. Holosystolic regurgitant murmurs persist while a systolic pressure difference remains. Diastolic murmurs imply abnormal flow across semilunar or atrioventricular valves. Continuous murmurs require a gradient through systole and diastole.

Dynamic manoeuvres alter loading. Standing or strain reduces venous return and ventricular size. Squatting raises venous return and systemic resistance. Handgrip raises afterload. Inspiration increases right-heart filling and often intensifies right-sided murmurs. These responses help distinguish mechanisms but can be unreliable in tachycardia, low output, obesity, poor cooperation, or multiple lesions.

A murmur with fever, embolic signs, new conduction disturbance, or heart failure raises concern for infection affecting valves or adjacent tissue. A new murmur after myocardial infarction with pulmonary oedema or shock suggests acute mechanical complication. Urgency follows physiological consequence, not auscultatory elegance.

Electrocardiography and rhythm capture

Before interpreting an electrocardiogram, confirm identity, symptoms, time, calibration, speed, and lead placement. Limb-lead reversal and misplaced chest leads can imitate axis change, infarction, or poor progression. Comparing with prior tracings is especially valuable when baseline conduction or repolarisation is abnormal.

Rate and rhythm come before morphology. Determine whether atrial activity is present, whether it relates consistently to QRS complexes, and whether ventricular activation is narrow or broad. Then assess intervals, axis, pathological waves, hypertrophy patterns, ST segments, and T waves. This sequence reduces distraction by one dramatic feature.

Monitoring duration should match event frequency. A short recording has little value for monthly syncope unless it reveals a persistent high-risk substrate. Patient-triggered devices require consciousness and activation, while loop recorders can capture pre-event rhythm automatically. A symptom diary improves correlation with exertion, posture, meals, medicines, and sleep.

Wearables expand access but introduce selection and artefact. A single-lead tracing can identify some rhythms but cannot replace a diagnostic twelve-lead study for morphology or localisation. Automated notifications should be confirmed before major treatment decisions.

Imaging according to the clinical question

Echocardiography can answer whether ventricular function, filling, valves, pericardium, or proximal great vessels explain the syndrome. Measurements depend on acoustic windows, loading conditions, rhythm, assumptions about geometry, and operator interpretation. A reported ejection fraction should not eclipse chamber size, wall motion, right-heart function, valve severity, and pressure estimates.

Computed-tomography coronary angiography is strongest for defining anatomy and excluding important obstructive disease in appropriately selected patients. Calcification, motion, rhythm, kidney function, contrast allergy, and radiation influence suitability. Anatomical stenosis does not always establish whether flow limitation causes symptoms.

Stress tests examine functional reserve. Exercise provides symptoms, workload, blood-pressure response, rhythm, and electrocardiographic data. Imaging can assess perfusion or wall motion when the resting tracing is difficult or greater accuracy is needed. A test may be technically negative because the patient failed to reach adequate stress.

Cardiac magnetic resonance characterises scar, oedema, infiltration, ventricular volumes, and complex anatomy without ionising radiation. It is limited by access, scan duration, implanted-device compatibility, patient tolerance, and selected contrast considerations. Invasive angiography is justified when anatomical definition and potential intervention outweigh procedural risk.

Biomarkers and serial inference

Biomarkers answer focused questions. Troponin indicates myocardial injury. Natriuretic peptides indicate myocardial wall stress. C-reactive protein reflects inflammation without identifying its source. Creatinine and potassium influence both cardiovascular disease and treatment safety. No marker replaces a causal assessment.

Serial change often carries more information than a single result. Dynamic troponin supports acute injury. Falling natriuretic peptide may accompany reduced congestion, although clinical improvement matters more. Worsening creatinine during diuresis can reflect altered haemodynamics, persistent congestion, true injury, or a mixture. The response should be interpreted with volume, pressure, urine output, electrolytes, and symptoms.

Every investigation should end with an action statement. Specify what was found, how confidently it explains the syndrome, which dangerous alternative remains, and what decision follows. Incidental abnormalities require proportionate follow-up rather than automatic escalation.

Exercise capacity is itself a longitudinal vital sign. A standardised walk, stair history, rehabilitation workload, or formal cardiopulmonary exercise test can distinguish circulatory, ventilatory, muscular, and motivational limitations with varying precision. Peak oxygen uptake, ventilatory efficiency, chronotropic response, blood pressure, symptoms, and recovery add prognostic information in selected patients. Repeat testing is useful only when protocol and clinical state are comparable, and specialist interpretation should guide high-stakes decisions.

The final cardiovascular synthesis should describe stability, rhythm, perfusion, congestion, likely anatomical or physiological mechanism, and trajectory. It should also state the next test or intervention and the result that would cause the plan to change. This converts a collection of findings into a monitored clinical model.

Retrieval prompts

One. Which chest-pain and syncope features raise urgency?

Two. How is jugular venous pressure distinguished and interpreted?

Three. How should a murmur be described?

Four. Why can troponin and natriuretic peptides not diagnose disease alone?

Five. How is ambulatory rhythm monitoring selected?

Six. What determines the best cardiac imaging test?

Concise answers

One. Ongoing ischaemic symptoms, instability, tearing pain, exertional or supine syncope, structural disease, abnormal electrocardiogram, and family history of sudden death.

Two. It is non-palpable, positional and respiratory with a venous waveform; height estimates right-atrial pressure.

Three. By timing, site, radiation, intensity, pitch, shape, and manoeuvre response.

Four. They reflect injury or wall stress and are altered by timing, kidney function, rhythm, age, obesity, and other conditions.

Five. Match recording duration and activation method to symptom frequency and severity.

Six. The suspected mechanism, pre-test probability, patient factors, test accuracy, risk, availability, and management decision.

Source map

Original synthesis informed by Talley and O'Connor's Clinical Examination, cardiovascular history and examination; Guyton and Hall, cardiovascular physiology; Robbins, cardiac and vascular pathology; OpenStax Medical-Surgical Nursing; and Katzung and OpenStax Pharmacology for medication and test interpretation context.

Chapter 13: Ventilation, Perfusion, Diffusion, and Gas Transport

Orientation

Respiratory function requires air movement, pulmonary blood flow, gas transfer across a thin interface, carriage in blood, and tissue exchange. Failure at any step can reduce oxygen delivery or retain carbon dioxide. Oxygenation and ventilation are related but distinct: oxygenation describes transfer of oxygen into blood, while alveolar ventilation determines carbon dioxide removal. A patient can fail predominantly in one domain before the other.

Functional anatomy

The conducting airways warm, humidify, and filter air but do not directly exchange gas. They extend from nose and mouth through trachea and branching bronchi to terminal bronchioles. Cartilage supports larger airways; smooth muscle becomes increasingly important in smaller airways. Mucus traps particles and cilia move material toward the pharynx.

Respiratory bronchioles, alveolar ducts, and alveoli form the gas-exchange region. Type one alveolar cells provide a thin diffusion surface. Type two cells produce surfactant and can proliferate after injury. Alveolar macrophages remove particles and microbes. The interface includes alveolar epithelium, interstitium, capillary endothelium, plasma, and red-cell membrane.

Pulmonary arteries carry mixed venous blood from the right ventricle. The pulmonary circulation is normally low pressure and low resistance. Gravity, lung volume, vascular recruitment, hypoxia, and cardiac output influence regional flow. Bronchial arteries from the systemic circulation nourish conducting airways and contribute a small physiological shunt.

Ventilation and dead space

Minute ventilation equals tidal volume multiplied by breathing frequency. Not every inhaled volume reaches functioning alveoli. Anatomical dead space remains in conducting airways. Alveolar dead space reaches ventilated alveoli with little or no perfusion. Physiological dead space is the sum.

Alveolar ventilation equals respiratory rate multiplied by tidal volume minus dead-space volume. Rapid shallow breathing can produce a reasonable minute ventilation but poor alveolar ventilation because dead space consumes a larger fraction of each breath. Carbon dioxide tension varies inversely with effective alveolar ventilation when production is stable.

Regional ventilation is affected by gravity and mechanics. At resting lung volume, dependent alveoli are smaller but often more compliant, so they receive more tidal ventilation. Disease, position, obesity, anaesthesia, pain, and airway closure alter this pattern.

Alveolar gas and partial pressure

Gas molecules exert partial pressures according to their fraction of total gas. Inspired oxygen pressure falls after humidification because water vapour contributes pressure. In alveoli, oxygen is continually removed and carbon dioxide added. Alveolar oxygen therefore depends on inspired oxygen, barometric pressure, and carbon dioxide relative to ventilation.

At altitude, oxygen fraction remains about twenty-one percent but barometric pressure falls, reducing inspired oxygen partial pressure. Supplemental oxygen raises inspired fraction. Hypoventilation raises alveolar carbon dioxide and lowers alveolar oxygen. This coupled change helps distinguish pure hypoventilation from additional gas-exchange defects.

Diffusion

Diffusion rate rises with surface area, diffusion coefficient, and partial-pressure gradient, and falls with membrane thickness. Oxygen moves from alveolar gas to blood; carbon dioxide moves from blood to alveoli. Carbon dioxide is more soluble and usually diffuses more readily despite a smaller pressure gradient.

Emphysema destroys surface area. Interstitial fibrosis and pulmonary oedema increase diffusion distance. Exercise shortens capillary transit time but also recruits and distends vessels and increases diffusion capacity in healthy lungs. Severe diffusion limitation becomes more apparent with exertion or low inspired oxygen.

The diffusing capacity test commonly uses carbon monoxide because its capillary partial pressure remains low. Interpretation depends on alveolar volume, haemoglobin, smoking, pulmonary vascular blood volume, and technical quality. Low values occur in emphysema, fibrosis, anaemia, and pulmonary vascular disease; haemorrhage can raise the measurement.

Ventilation-perfusion matching

Efficient exchange requires regional ventilation matched to perfusion. The average whole-lung ratio hides substantial variation. At the lung apex, perfusion falls more than ventilation, so the ratio is relatively high. At dependent bases, both are greater but perfusion predominates, so the ratio is lower.

Low ventilation relative to perfusion lowers regional oxygen and raises carbon dioxide. Hypoxic pulmonary vasoconstriction diverts blood away from poorly ventilated alveoli. This is useful locally but widespread hypoxia, as at altitude or in diffuse lung disease, raises pulmonary pressure.

Shunt is perfusion without ventilation. Causes include alveolar collapse, filling with fluid or pus, and intracardiac right-to-left flow. True shunt responds incompletely to supplemental oxygen because blood bypasses ventilated gas units. Dead space is ventilation without effective perfusion, as in pulmonary embolism or severe vascular destruction. Most lung disease produces a spectrum rather than pure categories.

The alveolar-arterial oxygen difference compares estimated alveolar oxygen with measured arterial oxygen. It is usually near normal in uncomplicated hypoventilation but increases with ventilation-perfusion mismatch, diffusion limitation, or shunt. It changes with age and inspired oxygen and must be interpreted with assumptions.

Oxygen transport

Most oxygen binds haemoglobin; a small amount dissolves in plasma. Arterial oxygen content depends on haemoglobin concentration, saturation, and dissolved oxygen. Partial pressure determines diffusion and haemoglobin saturation, while content determines how much oxygen is carried.

The oxygen-haemoglobin dissociation curve is sigmoidal because binding is cooperative. The plateau preserves saturation across moderate alveolar pressure changes. The steep tissue region promotes unloading when tissue oxygen falls. Higher carbon dioxide, acidity, temperature, and two-three bisphosphoglycerate shift the curve right, reducing affinity and supporting unloading. Opposite changes shift it left.

Anaemia reduces content despite normal pressure and saturation. Carbon monoxide occupies haemoglobin sites and increases affinity at remaining sites, reducing both carriage and unloading; pulse oximetry can be misleading. Methaemoglobin contains oxidised iron unable to bind oxygen normally. Tissue oxygen delivery further depends on cardiac output and regional flow.

Carbon dioxide transport and acid-base link

Carbon dioxide is carried dissolved, bound to proteins as carbamino compounds, and mainly as bicarbonate. In red cells, carbonic anhydrase accelerates conversion of carbon dioxide and water to carbonic acid, which dissociates into hydrogen and bicarbonate. Haemoglobin buffers hydrogen. Bicarbonate exits in exchange for chloride.

In the lungs, oxygenation of haemoglobin favours release of hydrogen and carbon dioxide, reversing the reactions. Carbon dioxide elimination therefore connects ventilation to acidity. Acute hypoventilation raises carbon dioxide and lowers pH. Hyperventilation lowers carbon dioxide and raises pH.

Applied interpretation

Hypoxaemia can result from low inspired oxygen, hypoventilation, diffusion limitation, ventilation-perfusion mismatch, or shunt. Hypoventilation is the main mechanism of hypercapnia, although increased dead space and carbon dioxide production raise the ventilatory requirement. A normal oxygen saturation on supplemental oxygen does not prove adequate ventilation, and a normal carbon dioxide value can be inappropriate in a patient working hard to breathe.

Exercise, altitude, and fetal circulation

During exercise, oxygen consumption and carbon dioxide production rise. Cardiac output, ventilation, pulmonary perfusion, and diffusing capacity increase together. Mixed venous oxygen falls as tissues extract more, yet arterial oxygen is usually maintained in healthy people. Exercise limitation can arise from ventilation, gas exchange, cardiac output, haemoglobin, muscle conditioning, or symptom perception, so breathlessness alone does not localise failure.

At altitude, hypoxaemia stimulates ventilation, lowering carbon dioxide and causing respiratory alkalosis. Kidneys excrete bicarbonate over days, permitting further ventilation. Erythropoietin increases red-cell production, while two-three bisphosphoglycerate and tissue adaptations change oxygen use. Excessive pulmonary vasoconstriction can cause high-altitude pulmonary oedema; cerebral oedema reflects dangerous brain swelling. Descent and oxygen are priorities in severe altitude illness.

Fetal gas exchange occurs through the placenta rather than lungs. Fetal haemoglobin has higher oxygen affinity, helping transfer oxygen from maternal blood. Fetal shunts direct blood around unventilated lungs. At birth, lung expansion lowers pulmonary resistance, placental separation raises systemic resistance, and shunts begin to close. Failure of this transition contributes to neonatal hypoxaemia.

Limits of pulse oximetry

Pulse oximetry estimates saturation from pulsatile light absorption. Accuracy falls with poor perfusion, movement, nail products, dyshemoglobins, venous pulsation, severe anaemia, and device or skin-related bias. It does not measure carbon dioxide, pH, work of breathing, or oxygen content. A reassuring number must not override clinical deterioration.

TTS module 2: Alveolar gas reasoning, shunt behaviour, and oxygen-delivery traps

Using the alveolar gas relationship

Alveolar oxygen tension depends primarily on inspired oxygen and alveolar carbon dioxide. When inspired oxygen is stable, hypoventilation raises carbon dioxide and lowers alveolar oxygen in a linked way. The alveolar gas equation estimates this relationship and allows comparison between calculated alveolar oxygen and measured arterial oxygen.

The alveolar-arterial oxygen difference is small when hypoxaemia is explained by low inspired oxygen or uncomplicated hypoventilation. It widens when oxygen encounters diffusion limitation, ventilation-perfusion inequality, or shunt. The calculation is an estimate because respiratory exchange ratio, barometric pressure, water vapour pressure, and inspired fraction must be assumed or measured.

Supplemental oxygen changes the interpretation. A high inspired fraction increases alveolar oxygen greatly, so the numerical gradient also grows. Trends across different devices cannot be compared without accounting for inspired oxygen. Variable-performance devices deliver a fraction influenced by the patient's inspiratory flow and breathing pattern.

At altitude, barometric pressure rather than oxygen fraction falls. The resulting low inspired partial pressure stimulates ventilation. Lower carbon dioxide partly raises alveolar oxygen, but respiratory alkalosis develops. Acclimatisation permits greater ventilation and changes oxygen carriage, while severe illness still requires descent and oxygen according to emergency guidance.

Ventilation-perfusion distributions

Real lungs contain a distribution of ventilation-perfusion ratios, not one average value. Units with a low ratio deliver poorly oxygenated blood that mixes with blood from better units. High-ratio units cannot fully compensate because haemoglobin in well-ventilated regions is already near saturation. This asymmetry makes low-ratio disease an efficient cause of hypoxaemia.

Carbon dioxide behaves differently because its dissociation relationship and diffusion properties permit better compensation through increased ventilation of functioning units. Carbon dioxide may remain normal or low while a patient works intensely to maintain elimination. A rising value in severe obstructive or parenchymal disease can therefore signal failing ventilatory reserve.

True shunt represents blood reaching the systemic circulation without contact with ventilated alveoli. Increasing inspired oxygen improves oxygen in ventilated units but cannot directly oxygenate shunted blood. The response is therefore limited, although most clinical disease combines shunt with lower-ratio units and shows some improvement.

Dead-space ventilation wastes respiratory effort. Pulmonary embolism abruptly removes perfusion from ventilated regions. Emphysema destroys capillary bed and alveolar wall. Low cardiac output can increase the proportion of ventilation going to underperfused units. To maintain carbon dioxide removal, total minute ventilation must rise.

Diffusion and capillary transit

Gas transfer requires equilibration during the time red cells spend in pulmonary capillaries. At rest, healthy lungs have reserve: oxygen usually equilibrates before the end of transit. Exercise shortens transit but recruits capillaries and increases diffusing capacity. Fibrosis, emphysema, pulmonary vascular loss, or low alveolar oxygen can consume this reserve and reveal exertional desaturation.

Carbon monoxide diffusing capacity reflects membrane transfer and pulmonary capillary blood volume. Anaemia lowers the measurement because less haemoglobin is available to bind test gas. Polycythaemia and alveolar haemorrhage can raise it. Emphysema lowers surface area, fibrosis thickens the interface, and pulmonary vascular disease reduces participating blood volume.

A reduced diffusing capacity is therefore not a disease label. It should be interpreted with spirometry, lung volumes, haemoglobin, imaging, smoking status, and exercise response. A low value with obstruction suggests a different phenotype from obstruction with preserved transfer.

Oxygen content, delivery, and extraction

Arterial oxygen tension measures dissolved-gas pressure and drives haemoglobin binding. Saturation measures the proportion of available binding sites occupied. Content measures the total oxygen carried per blood volume. Delivery multiplies content by cardiac output. These quantities answer different clinical questions.

A patient with severe anaemia can have normal tension and saturation while content is dangerously low. A patient with carbon monoxide exposure can have a misleading pulse-oximeter value, reduced available sites, and impaired unloading. A patient in shock can have adequate arterial content but inadequate flow. Treating only the saturation misses these mechanisms.

Tissues extract oxygen according to demand, local flow, diffusion distance, capillary recruitment, and mitochondrial function. Mixed venous oxygen reflects the average residual oxygen returning from the body. A low value suggests high extraction relative to delivery. A high value can occur when flow is high, demand is low, or extraction is impaired by shunting or cellular dysfunction.

The oxygen-haemoglobin curve shifts right with acidity, carbon dioxide, heat, and increased two-three biphosphoglycerate, promoting unloading at a given tissue tension. A left shift improves loading but can hinder release. Fetal haemoglobin, hypothermia, alkalosis, and carbon monoxide increase affinity. Clinical impact depends on both loading in lungs and unloading in tissue.

Pulse oximetry and co-oximetry

Pulse oximeters separate pulsatile arterial absorption from surrounding tissue and estimate saturation using selected wavelengths. They assume ordinary oxyhaemoglobin and deoxyhaemoglobin. Dyshemoglobins violate those assumptions. Co-oximetry uses multiple wavelengths to estimate carboxyhaemoglobin and methaemoglobin more directly.

Poor perfusion, vasoconstriction, movement, venous pulsation, nail products, sensor position, ambient light, and device performance can distort readings. Accuracy also varies across saturation range and patient characteristics. The displayed pulse should match the patient's pulse, and an unexpected value should be checked clinically and, when necessary, with arterial sampling.

Oximetry does not assess ventilation. Supplemental oxygen can maintain saturation while carbon dioxide rises from opioid effect, neuromuscular weakness, obesity hypoventilation, or severe airway obstruction. Respiratory rate may also appear normal despite shallow ineffective breaths. Consciousness, effort, tidal movement, and carbon dioxide assessment remain essential.

Carbon dioxide as flow and buffer

Carbon dioxide production depends on metabolism and substrate use. It is transported dissolved, bound to proteins, and as bicarbonate. In tissue capillaries, deoxygenated haemoglobin accepts hydrogen and carbon dioxide more readily. In lungs, oxygenation promotes carbon dioxide unloading, known as the Haldane effect.

This matters during oxygen therapy in chronic hypercapnic disease. Improved oxygenation can release carbon dioxide from haemoglobin, while reversal of hypoxic pulmonary vasoconstriction increases flow to poorly ventilated units. Reduced ventilatory drive may contribute in some patients but is not the sole explanation. Dangerous hypoxaemia should still be treated while oxygen is controlled and gases are reassessed.

Carbon dioxide tension varies inversely with alveolar ventilation when production is stable. A normal value can be inappropriate when metabolic acidosis should provoke compensatory hyperventilation. A patient with severe acidosis and a merely normal carbon dioxide may already have ventilatory failure.

Applying mechanism at the bedside

Begin hypoxaemia analysis with inspired oxygen, ventilation, and the alveolar-arterial difference. Then consider low-ratio units, shunt, diffusion, and circulatory mixing. Pair saturation with haemoglobin and perfusion to judge

delivery. Pair carbon dioxide with pH and expected compensation to judge ventilation.

Response to intervention tests the model. Oxygen-responsive hypoxaemia supports functioning ventilated units but does not identify one disease. Persistent hypoxaemia on high inspired oxygen raises concern for substantial shunt, technical failure, or severe mixed pathology. Improving saturation with worsening consciousness can conceal carbon dioxide retention.

Position can change regional matching within minutes. Upright posture usually directs more perfusion toward dependent basal lung, while supine positioning redistributes both ventilation and blood flow. Unilateral disease may oxygenate better when the healthier lung is dependent because gravity sends it more perfusion, although secretions, mechanics, and haemodynamics can alter the result. Prone positioning in severe diffuse lung injury can recruit dorsal regions, make stress distribution more uniform, and improve matching. A positional response is supportive physiological evidence, not a replacement for treating collapse, fluid, infection, embolism, or another cause.

The final respiratory statement should specify oxygenation mechanism, ventilatory adequacy, oxygen content, work of breathing, and reserve. That synthesis avoids the common error of allowing one attractive saturation number to substitute for complete gas-transport physiology during rapidly changing clinical illness.

Retrieval prompts

One. Distinguish minute ventilation from alveolar ventilation.

Two. What factors determine diffusion rate?

Three. Contrast shunt with dead space.

Four. Why can anaemia cause tissue hypoxia with normal saturation?

Five. What shifts the oxygen-haemoglobin curve right?

Six. List the five major mechanisms of hypoxaemia.

Concise answers

One. Minute ventilation is total tidal volume per minute; alveolar ventilation subtracts dead space from each breath.

Two. Surface area, thickness, diffusion coefficient, and partial-pressure gradient.

Three. Shunt is perfusion without ventilation; dead space is ventilation without effective perfusion.

Four. Saturation is the fraction of available haemoglobin sites occupied; anaemia reduces the number of sites and total oxygen content.

Five. Increased carbon dioxide, hydrogen ions, temperature, and two-three biphosphoglycerate.

Six. Low inspired oxygen, hypoventilation, diffusion limitation, ventilation-perfusion mismatch, and shunt.

Source map

Original synthesis informed by Guyton and Hall, pulmonary ventilation, gas exchange, gas transport, and respiratory regulation; OpenStax Anatomy and Physiology 2e; Robbins, pulmonary pathology; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, respiratory assessment.

Chapter 14: Respiratory Mechanics, Control, and Respiratory Failure

Orientation

Breathing requires a neural drive, a functional pump, patent airways, compliant lungs and chest wall, and sufficient endurance. Respiratory failure occurs when this system cannot maintain appropriate oxygenation, ventilation, or both. A patient can compensate with increased work for hours before fatigue and abrupt deterioration. Assessment must therefore consider effort, trajectory, and reserve as well as gas values.

Pressure and the respiratory pump

Air moves when alveolar pressure differs from atmospheric pressure. During quiet inspiration, the diaphragm contracts and descends, external intercostal activity expands the thorax, pleural pressure becomes more negative, alveoli enlarge, alveolar pressure falls, and air enters. Quiet expiration is largely passive recoil. Forced expiration recruits abdominal and internal intercostal muscles.

Transpulmonary pressure is alveolar minus pleural pressure and keeps lungs expanded. The chest wall tends outward at lower volumes while lungs recoil inward. Their balance defines functional residual capacity, the volume after a normal expiration.

Pneumothorax introduces gas into pleural space, reduces the normal pressure difference, and permits lung recoil. Tension physiology occurs when pressure progressively impairs venous return and compresses cardiopulmonary structures. Pleural fluid separates lung from chest wall and can compress dependent lung.

Compliance and elastic recoil

Compliance is volume change per pressure change. High compliance means a structure expands easily; low compliance means it is stiff. Fibrosis, oedema, acute respiratory distress syndrome, obesity, ascites, and chest-wall deformity reduce respiratory-system compliance. Emphysema increases lung compliance but reduces elastic recoil, making expiration and airway stability difficult.

Alveolar surface tension promotes collapse. Surfactant from type two cells lowers surface tension, increases compliance, stabilises differently sized alveoli, and reduces fluid movement into air spaces. Surfactant deficiency is central to neonatal respiratory distress and can be altered in severe lung injury.

Work of breathing includes elastic work to expand lung and chest wall, resistive work to move air through airways, and tissue resistance. Stiff lungs favour rapid shallow breathing to limit elastic work. Obstructed airways favour slower deeper breathing to reduce resistive losses, until dynamic hyperinflation limits this strategy.

Airway resistance and flow limitation

Airway resistance depends strongly on radius. Parasympathetic activity, inflammatory mediators, mucus, oedema, smooth-muscle contraction, and loss of radial traction narrow airways. Sympathetic beta two receptor stimulation relaxes airway smooth muscle, mainly through circulating catecholamines and medicines rather than dense direct sympathetic innervation.

During forced expiration, pleural pressure can exceed airway pressure beyond an equal-pressure point, dynamically compressing intrathoracic airways. Increased expiratory effort then raises pleural and alveolar pressure together without proportionally increasing flow. Loss of elastic recoil in emphysema moves compression toward smaller collapsible airways.

In obstructive disease, incomplete expiration causes air trapping and raises end-expiratory lung volume. The next breath begins before full exhalation, producing intrinsic positive end-expiratory pressure. Inspiratory muscles must

first overcome this threshold, increasing work and causing dyspnoea. Rapid breathing worsens dynamic hyperinflation.

Lung volumes and spirometry

Tidal volume is a normal breath. Inspiratory and expiratory reserve volumes are additional volumes available. Residual volume remains after maximal expiration and cannot be measured by simple spirometry. Vital capacity is the maximum exhaled after maximal inhalation. Total lung capacity includes residual volume.

Forced expiratory volume in one second and forced vital capacity assess airflow. Obstruction reduces their ratio. Restriction is suggested by reduced vital capacity with preserved or high ratio but requires confirmation of reduced total lung capacity. Bronchodilator response supports variable airflow obstruction but absence does not exclude asthma. Results depend on effort, technique, reference population, and test quality.

Flow-volume loops can suggest obstruction, restriction, or upper-airway lesions. Peak expiratory flow is useful for personal variability and monitoring in selected asthma patients but is less complete than spirometry.

Neural control

Brainstem networks generate respiratory rhythm and coordinate inspiratory and expiratory muscles. The cortex permits voluntary control, while limbic and hypothalamic inputs alter breathing with emotion, pain, temperature, and behaviour. Sleep changes control and upper-airway muscle tone.

Central chemoreceptors respond mainly to carbon dioxide through acidity in brain extracellular fluid. Carbon dioxide crosses the blood-brain barrier, forms acid, and stimulates ventilation. Peripheral carotid and aortic bodies respond to low arterial oxygen, acidity, and carbon dioxide. Their oxygen response becomes strong at substantially reduced arterial oxygen pressure.

Chronic carbon dioxide retention leads kidney bicarbonate retention and partial buffering of central acidity. Oxygen should not be withheld from a dangerously hypoxaemic patient. In susceptible chronic hypercapnic disease, controlled oxygen and gas reassessment are used because worsened

ventilation-perfusion matching, the Haldane effect, and altered ventilatory drive can increase carbon dioxide.

Respiratory failure

Type one respiratory failure is predominantly hypoxaemic with normal or low carbon dioxide. Mechanisms include ventilation-perfusion mismatch, shunt, diffusion limitation, and low inspired oxygen. Type two failure includes hypercapnia from inadequate alveolar ventilation and usually has hypoxaemia as well.

Causes of ventilatory failure include central depression from drugs or neurological disease, spinal-cord or peripheral-nerve dysfunction, neuromuscular-junction disease, respiratory-muscle fatigue, chest-wall restriction, severe obesity, and obstructive lung disease. Increased carbon dioxide production from fever, seizures, overfeeding, or severe work can overwhelm limited reserve.

Acute hypercapnia causes respiratory acidosis. Chronic retention permits renal bicarbonate compensation. An apparently near-normal pH with high carbon dioxide can indicate chronic compensation, while a falling pH suggests acute deterioration. Blood gases must be compared with clinical effort and previous values.

Oxygen and ventilatory support

Supplemental oxygen is titrated to an appropriate target based on clinical context. Delivery devices differ in flow, achievable inspired fraction, patient tolerance, and dependence on breathing pattern. High-flow nasal therapy provides warmed humidified gas, some dead-space washout, and variable positive pressure.

Non-invasive ventilation supports breathing through a mask. Continuous positive airway pressure improves oxygenation and recruits lung but does not directly add inspiratory assistance. Bilevel support adds pressure during inspiration and can increase tidal volume and reduce carbon dioxide. It is useful in selected obstructive exacerbations and cardiogenic pulmonary oedema but requires a cooperative patient able to protect the airway and clear secretions.

Invasive ventilation is indicated when oxygenation or ventilation cannot be maintained, work is unsustainable, airway protection fails, or consciousness and haemodynamics demand control. Ventilation can injure lungs through excessive volume, pressure, repeated opening and closing, oxygen exposure, and diaphragmatic inactivity. Lung-protective strategies use appropriate tidal volume, pressure limitation, positive end-expiratory pressure, and treatment of the cause.

Warning signs and reassessment

Danger signs include exhaustion, reduced consciousness, silent chest with poor airflow, inability to speak, paradoxical movement, cyanosis, worsening acidosis, unstable pressure, or falling respiratory rate despite deterioration. A reduction in wheeze can mean improvement or critically reduced airflow. After intervention, reassess effort, respiratory rate, gas exchange, mental state, auscultation, haemodynamics, and tolerance.

Sleep-disordered breathing

During sleep, ventilatory responses and upper-airway muscle tone fall. Obstructive sleep apnoea causes repeated pharyngeal collapse despite respiratory effort, producing intermittent hypoxaemia, arousal, intrathoracic pressure swings, and sympathetic activation. Risk rises with obesity and craniofacial or upper-airway anatomy, but body size alone neither confirms nor excludes it. Symptoms include snoring, witnessed pauses, unrefreshing sleep, morning headache, and daytime sleepiness.

Central sleep apnoea reflects reduced or unstable ventilatory drive and occurs with heart failure, altitude, opioids, or neurological disease. Obesity hypoventilation includes awake hypercapnia in an obese patient after other causes are considered. Sleep studies measure airflow, effort, oxygenation, heart rate, sleep stage, and events. Treatment may include weight management, positive airway pressure, reduction of respiratory depressants, and cause-specific therapy.

Weaning and respiratory-muscle reserve

Resolution of the precipitating disease does not guarantee immediate freedom from ventilatory support. Readiness includes adequate oxygenation, manageable secretions, haemodynamic stability, sufficient consciousness, and a tolerable respiratory load. A spontaneous breathing trial tests integrated reserve. Failure can arise from cardiac dysfunction, muscle weakness, airway obstruction, excessive load, anxiety, fever, or metabolic demand.

Diaphragmatic weakness develops with critical illness, malnutrition, nerve disease, and prolonged controlled ventilation. Inspiratory pressure, vital capacity, cough strength, and gas trends help in neuromuscular disease, but deterioration can occur before oxygen saturation falls. Early recognition permits airway-clearance support and planned ventilation rather than emergency intubation.

TTS module 2: Time constants, respiratory load, and ventilatory-support reasoning

Compliance, resistance, and the time constant

Each lung unit fills and empties according to its resistance and compliance. Their product is the time constant. A unit with high resistance or high compliance changes volume slowly. A stiff low-compliance unit accepts little volume but may fill quickly. Healthy lungs contain some variation, while disease broadens the distribution and makes ventilation uneven.

In obstructive disease, slowly emptying units retain gas when expiratory time is too short. The next breath begins above the natural resting volume, producing dynamic hyperinflation and intrinsic positive end-expiratory pressure. Faster rate, larger tidal volume, and greater obstruction worsen trapping. Allowing more expiratory time can reduce it.

In acute stiff-lung disease, a given tidal volume requires higher pressure. The same airway pressure can reflect resistance, elastic load, or both. Peak inspiratory pressure includes resistive and elastic components. A pause after flow stops estimates plateau pressure, which more closely reflects alveolar and chest-wall elastic pressure. A high peak with a lower plateau suggests increased airway resistance; both high suggests reduced compliance or excessive volume.

Chest-wall stiffness complicates interpretation. Obesity, ascites, oedema, posture, and abdominal pressure increase airway pressure without necessarily exposing the lung itself to the same transpulmonary stress. Oesophageal-pressure methods can estimate pleural pressure in selected specialist settings, but ordinary bedside interpretation still needs the whole mechanical context.

Work, power, and muscle endurance

Respiratory muscles must overcome elastic recoil, airway resistance, intrinsic positive pressure, and any external apparatus. Work per breath rises with pressure and volume; total mechanical power also depends on respiratory rate and flow. A modestly difficult breath repeated very rapidly can create a large cumulative load.

The diaphragm operates best over a useful length range. Hyperinflation shortens and flattens it, reducing mechanical advantage. Malnutrition, sepsis, electrolyte disturbance, corticosteroid exposure, neuropathy, and inactivity weaken force. Fever, acidosis, agitation, and increased dead space raise ventilatory demand. Failure occurs when load exceeds neuromuscular capacity.

Early compensation includes tachypnoea, accessory-muscle recruitment, and increased neural drive. As fatigue or central depression develops, tidal volume may fall, movement becomes paradoxical, carbon dioxide rises, and consciousness declines. A slowing respiratory rate in an exhausted patient can therefore be ominous rather than reassuring.

Neural respiratory drive can remain high even when mechanical output is poor. Occlusion-pressure measurements and diaphragmatic electrical activity estimate drive in specialised contexts. At the bedside, severe effort with small tidal movement suggests a large load or weak pump and should prompt urgent support.

Flow limitation and bronchodilator logic

Airway resistance is greatest in medium-sized bronchi under many normal conditions, but small airways become important when inflammation, mucus, wall remodelling, or loss of radial traction narrows many of them in parallel. Because they contribute little resistance until widespread disease is present,

significant small-airway pathology can develop before routine spirometry changes dramatically.

Bronchodilators reduce smooth-muscle tone but do not directly remove mucus, reverse fixed fibrosis, restore destroyed elastic recoil, or treat infection. A limited immediate spirometric response does not prove there is no symptomatic benefit or no asthma, because baseline calibre, technique, timing, and airway inflammation influence the test.

Inhaled delivery depends on particle size, inspiratory flow, device resistance, coordination, breath hold, and airway anatomy. A correctly chosen drug can fail when technique is poor. Spacers reduce coordination demands for metered-dose devices, while dry-powder devices require sufficient inspiratory flow. Severe distress may require nebulised or assisted delivery according to protocol.

Control of breathing and compensation

Central chemoreceptors respond to acidity generated when carbon dioxide crosses into brain extracellular fluid. Metabolic hydrogen ions cross the blood-brain barrier less readily, so peripheral chemoreceptors provide much of the rapid ventilatory response to metabolic acidosis. Renal and respiratory systems then coordinate over different timescales.

Ventilation normally rises when carbon dioxide rises. The relationship can be blunted by opioids, sedatives, neurological injury, sleep, severe obesity, or chronic adaptation. It can be exaggerated by pain, anxiety, fever, pregnancy, hypoxaemia, and acidosis. Observed carbon dioxide must be judged against the drive expected from the clinical state.

Chronic hypercapnia permits kidney bicarbonate retention, buffering pH over days. An acute additional rise then produces a mixed acute-on-chronic respiratory acidosis. A bicarbonate concentration that seems high may be appropriate chronic compensation or may include a separate metabolic alkalosis from diuretics, vomiting, or volume contraction.

Non-invasive support as a physiological intervention

Continuous positive airway pressure raises pressure throughout the cycle. It splints collapsible upper airways, recruits selected alveoli, increases functional residual capacity, and can reduce left-ventricular afterload. It does

not add a separate inspiratory pressure boost, so ventilation improves mainly through recruitment and reduced work.

Bilevel support provides a higher inspiratory than expiratory pressure. The difference supports tidal volume and carbon dioxide clearance, while expiratory pressure recruits lung and offsets intrinsic positive pressure. Excess expiratory pressure can worsen hyperinflation or reduce venous return, while insufficient support leaves respiratory muscles overloaded.

Success requires a treatable syndrome, suitable consciousness, manageable secretions, mask tolerance, and close reassessment. Failure signs include worsening acidosis, deteriorating consciousness, refractory hypoxaemia, haemodynamic instability, inability to protect the airway, or exhaustion. Persisting too long with ineffective non-invasive support can delay safer invasive control.

Invasive ventilation and lung stress

Mechanical ventilation applies pressure to create flow. Volume-controlled modes target volume while pressure varies; pressure-controlled modes limit pressure while volume varies. Neither mode is inherently protective unless tidal volume, driving pressure, positive end-expiratory pressure, rate, flow, and patient interaction are appropriate.

Overdistension produces volutrauma and barotrauma. Repeated collapse and reopening produces atelectrauma. Inflammatory signalling caused by mechanical stress is biotrauma. High inspired oxygen can add toxicity. Patient effort can also generate large transpulmonary swings, creating self-inflicted injury when respiratory drive is extreme.

Positive end-expiratory pressure prevents selected units from collapsing and can improve oxygenation, but excessive pressure overdistends recruitable lung, raises dead space, impairs venous return, and increases right-ventricular afterload. Recruitment depends on the underlying morphology; not every opaque region is safely recruitable.

Ventilator dyssynchrony occurs when machine timing or flow does not match the patient's drive. The response should identify pain, anxiety, mode settings, trigger sensitivity, flow demand, intrinsic pressure, and disease progression

before simply deepening sedation. Sedation can facilitate safety but prolong ventilation, delirium, and weakness.

Weaning as a whole-system test

A spontaneous breathing trial removes substantial assistance and tests respiratory, cardiac, neurological, and metabolic reserve together. Failure may reflect excessive respiratory load, diaphragm weakness, cardiac filling-pressure rise, ischaemia, secretion burden, fever, anxiety, or electrolyte disturbance.

During unsupported breathing, negative intrathoracic pressure increases venous return and left-ventricular afterload. A patient with limited cardiac reserve can develop pulmonary oedema despite improving lung infection. Recognising this phenotype changes management from repeated respiratory escalation toward cardiac and volume assessment.

Extubation readiness also requires upper-airway patency, cough, secretion clearance, alertness, and a plan for post-extubation support. Passing a breathing trial does not guarantee airway protection. Conversely, prolonged intubation creates pneumonia, laryngeal injury, weakness, and sedation exposure.

Airway clearance can determine success independently of gas exchange. Humidification, hydration, mobilisation, cough augmentation, suction, physiotherapy, bronchoscopy in selected obstruction, and treatment of pain help move secretions. Neuromuscular patients may generate an adequate tidal volume yet lack effective expiratory force. Measuring cough strength and observing bulbar function can therefore identify extubation risk that oxygenation and carbon dioxide values miss.

The final respiratory-failure assessment should name the failing component: drive, neuromuscular pump, chest wall, airway, parenchyma, pulmonary circulation, or cardiovascular interaction. It should then state whether the present support reduces load, improves gas exchange, and protects tissue while the cause is corrected. Device escalation without that causal statement risks treating numbers while respiratory reserve continues to disappear.

Retrieval prompts

One. What creates inspiratory airflow?

Two. Contrast low and high lung compliance.

Three. Explain dynamic airway compression and hyperinflation.

Four. How does spirometry distinguish obstruction from suspected restriction?

Five. Contrast type one and type two respiratory failure.

Six. What is the difference between continuous and bilevel positive pressure?

Concise answers

One. Inspiratory muscle contraction lowers alveolar pressure below atmospheric pressure.

Two. Low compliance means stiffness and high elastic work; high compliance means easy expansion but potentially poor recoil.

Three. Positive pleural pressure compresses intrathoracic airways during forced expiration; incomplete emptying raises end-expiratory volume and intrinsic pressure.

Four. Obstruction lowers the forced expiratory ratio; restriction requires confirmation of reduced total lung capacity.

Five. Type one is predominantly hypoxaemic; type two includes hypercapnia from inadequate alveolar ventilation.

Six. Continuous pressure mainly recruits and splints; bilevel support adds inspiratory assistance and augments ventilation.

Source map

Original synthesis informed by Guyton and Hall, respiratory mechanics, regulation, and insufficiency; OpenStax Anatomy and Physiology 2e; Robbins, obstructive, restrictive, and acute lung injury; Katzung and OpenStax Pharmacology, respiratory drugs; OpenStax Medical-Surgical Nursing; and Talley and O'Connor.

Chapter 15: Obstructive, Restrictive, Vascular, Infectious, and Neoplastic Lung Disease

Orientation

Respiratory diseases alter airways, alveoli, interstitium, vessels, pleura, respiratory muscles, or control. Similar symptoms can arise from different compartments, while several mechanisms often coexist. Diagnosis uses exposure history, tempo, physiology, imaging, microbiology, and pathology. Treatment combines removal of causes, disease-specific therapy, vaccination and prevention, rehabilitation, oxygen or ventilation when indicated, and management of systemic effects.

Asthma

Asthma is variable airflow obstruction with airway hyperresponsiveness and inflammation. Triggers include allergens, viral infection, exercise, cold air, smoke, occupational exposure, and selected drugs. Mast cells, eosinophils, lymphocytes, epithelial signals, mucus, oedema, and smooth-muscle constriction contribute, with different inflammatory patterns between patients.

Symptoms include episodic wheeze, cough, chest tightness, and dyspnoea, often varying by time and trigger. Normal examination and spirometry between episodes do not exclude asthma. Evidence of variable airflow can come from bronchodilator response, peak-flow variability, challenge testing, or change after anti-inflammatory therapy.

Inhaled corticosteroids reduce exacerbation risk by treating inflammation. Bronchodilators relax smooth muscle; short- and long-acting beta two agonists differ in onset and duration. Long-acting beta agonists should be paired with inhaled corticosteroid in asthma. Antimuscarinic drugs, leukotriene modifiers, and targeted biological therapies serve selected patients. Technique, spacer use, adherence, trigger control, and a written action plan are essential.

Severe exacerbation produces increased work, tachypnoea, inability to speak, reduced peak flow, hypoxaemia, and possibly silent airflow. A normal or rising carbon dioxide in a distressed patient can indicate fatigue. Treatment includes oxygen to target, repeated inhaled bronchodilator, systemic corticosteroid, and selected adjuncts; exhaustion, altered consciousness, or worsening gas exchange may require ventilatory support.

Chronic obstructive pulmonary disease

Chronic obstructive pulmonary disease combines persistent airflow limitation with airway inflammation, small-airway remodelling, mucus, and emphysema. Tobacco smoke is a major cause, but biomass exposure, occupation, early-life factors, and alpha one antitrypsin deficiency also contribute. Chronic bronchitis is a clinical mucus-hypersecretion phenotype. Emphysema is permanent enlargement and destruction distal to terminal bronchioles.

Loss of elastic recoil, expiratory collapse, and mucus obstruction cause air trapping and hyperinflation. Ventilation-perfusion mismatch causes hypoxaemia; advanced disease can cause hypercapnia, pulmonary hypertension, and right-heart failure. Systemic effects include muscle loss, osteoporosis, anxiety, cardiovascular risk, and frailty.

Diagnosis requires post-bronchodilator airflow obstruction in the appropriate context. Management includes smoking cessation, vaccination, activity and pulmonary rehabilitation, inhaled bronchodilators, selected inhaled corticosteroid use based on exacerbations and inflammatory markers, nutrition, and comorbidity care. Long-term oxygen improves survival in selected severe chronic resting hypoxaemia, not simply breathlessness.

Exacerbations are commonly triggered by infection or pollution but alternatives include heart failure, embolism, pneumothorax, and medication effects. Treatment uses bronchodilators, systemic corticosteroids, antibiotics when bacterial features or ventilatory support indicate, controlled oxygen, and non-invasive ventilation for selected hypercapnic acidosis.

Bronchiectasis and cystic fibrosis

Bronchiectasis is irreversible bronchial dilation caused by cycles of impaired clearance, infection, inflammation, and wall damage. Causes include prior infection, immune deficiency, allergic bronchopulmonary fungal disease, aspiration, ciliary disorders, obstruction, and cystic fibrosis. Features include chronic productive cough, recurrent infection, haemoptysis, and coarse crackles.

Management includes airway-clearance techniques, microbiological sampling, targeted antibiotics, vaccination, treatment of cause, and selected long-term antimicrobial or anti-inflammatory strategies. Massive haemoptysis requires airway and bleeding control. Cystic fibrosis results from dysfunctional chloride transport, producing dehydrated secretions in lungs and other organs; modulator therapy can improve function for eligible variants.

Pneumonia and tuberculosis

Pneumonia is infection of lung parenchyma. Pathogens vary with community or healthcare exposure, aspiration, immune status, age, structural lung disease, geography, and recent antibiotics. Fever, cough, sputum, pleuritic pain, breathlessness, confusion, and hypoxaemia can occur; older or immunocompromised patients may have subtle signs.

Diagnosis combines syndrome and imaging, with cultures and antigen or molecular tests in selected severity. Treatment considers likely pathogen, resistance, allergy, organ function, aspiration, and severity. Complications include sepsis, respiratory failure, abscess, necrosis, and parapneumonic effusion or empyema. Empyema usually requires drainage as well as antibiotics.

Tuberculosis spreads through airborne particles and can remain latent or cause active disease. Pulmonary disease may produce prolonged cough, weight loss, fever, night sweats, haemoptysis, and upper-lobe or disseminated patterns. Diagnosis uses microbiology and imaging; immune tests show infection but do not alone prove active disease. Multidrug therapy and public-health measures are necessary because resistance emerges rapidly with inadequate treatment.

Interstitial and occupational lung disease

Interstitial lung disease includes many disorders causing inflammation, fibrosis, or both in alveolar walls and supporting tissue. Patients often develop progressive exertional dyspnoea, dry cough, fine crackles, reduced volumes, impaired diffusion, and exertional desaturation. High-resolution computed tomography patterns narrow the differential.

Causes include idiopathic pulmonary fibrosis, connective-tissue disease, hypersensitivity pneumonitis, sarcoidosis, drugs, radiation, and occupation. Ask about birds, mould, dust, silica, asbestos, metals, farming, hobbies, and temporal relation to environments. Management removes exposure, treats inflammation when responsive, uses antifibrotic therapy in selected progressive fibrosis, and includes rehabilitation, oxygen, transplantation assessment, and palliative symptom care.

Pulmonary vascular disease

Pulmonary embolism obstructs pulmonary arteries and increases dead space and right-ventricular afterload. Presentation ranges from asymptomatic clot to pleuritic pain, dyspnoea, syncope, right-heart failure, or shock. Use clinical probability to guide D-dimer and imaging. Treatment is anticoagulation when appropriate; high-risk haemodynamic compromise may require reperfusion or intervention.

Pulmonary hypertension is elevated pulmonary arterial pressure from pulmonary arterial disease, left-heart disease, lung disease or hypoxia, chronic thromboembolism, or mixed mechanisms. Symptoms include exertional dyspnoea, fatigue, chest discomfort, syncope, and oedema. Echocardiography estimates probability, but right-heart catheterisation defines haemodynamics. Therapy depends on classification; pulmonary vasodilator drugs beneficial in one group can be ineffective or harmful in another.

Pleural disease and pneumothorax

Pleural effusion results from altered hydrostatic or oncotic forces, inflammation, infection, malignancy, lymphatic disruption, or blood. Thoracentesis can distinguish transudative from exudative patterns and assess cells, organisms, chemistry, and malignancy. Drainage depends on size, symptoms, cause, and complexity.

Pneumothorax may be spontaneous, traumatic, or iatrogenic. Sudden pleuritic pain and dyspnoea are common. Tension physiology requires immediate decompression based on clinical instability rather than waiting for imaging. Persistent air leak can reflect a fistula or ongoing parenchymal disease.

Lung cancer

Lung cancers are broadly small-cell and non-small-cell, with molecular subtypes affecting therapy. Tobacco is the major risk, but radon, occupational exposure, air pollution, and genetic factors contribute. Symptoms include cough, haemoptysis, recurrent infection, pain, weight loss, hoarseness, neurological signs, or paraneoplastic syndromes.

Diagnosis requires imaging, tissue, histology, molecular testing, and staging. Treatment can include surgery, radiation, cytotoxic chemotherapy, targeted therapy, and immune checkpoint inhibition. Fitness, pulmonary reserve, goals, and symptom burden matter. Prevention through smoking reduction and screening of eligible high-risk people can reduce mortality.

Acute lung injury and aspiration

Acute respiratory distress syndrome is diffuse inflammatory lung injury with increased permeability, bilateral opacities, reduced compliance, severe ventilation-perfusion mismatch and shunt, and hypoxaemia not fully explained by cardiac failure. Triggers include sepsis, pneumonia, aspiration, trauma, pancreatitis, transfusion, and inhalational injury. Neutrophils, endothelial and epithelial damage, protein-rich oedema, surfactant dysfunction, and microvascular thrombosis contribute.

Treatment addresses the trigger and uses lung-protective ventilation, appropriate positive end-expiratory pressure, conservative fluid strategy after shock is controlled, prone positioning in selected severe disease, thrombosis

prevention, nutrition, and avoidance of secondary injury. Prolonged disease can organise and fibrose. Survivors may have weakness, cognitive symptoms, psychological injury, and reduced exercise capacity.

Aspiration ranges from chemical pneumonitis after sterile gastric contents to bacterial pneumonia after colonised secretions. Risk rises with impaired consciousness, swallowing dysfunction, reflux, vomiting, tubes, poor dentition, and neurological disease. Immediate care prioritises airway and oxygenation. Antibiotics are used for established or strongly suspected infection rather than automatically after every witnessed aspiration. Recurrent aspiration requires swallowing assessment, positioning, oral care, feeding review, and treatment of causes.

Infection in immune compromise

Immune status changes the differential. Neutropenia, T-cell impairment, transplantation, glucocorticoids, biological therapy, and structural disease predispose to different organisms. Fungi can cause allergy, colonisation, chronic cavities, invasive vascular disease, or dissemination. Parasites can migrate through lungs or cause eosinophilic syndromes. Pneumocystis causes diffuse hypoxaemic pneumonia in susceptible hosts. Diagnosis may require computed tomography, antigen or molecular testing, bronchoalveolar sampling, and early specialist treatment.

TTS module 2: Respiratory phenotypes, progression mechanisms, and treatment selection

Asthma beyond episodic bronchospasm

Asthma contains several inflammatory and clinical phenotypes. Type two inflammation is commonly associated with eosinophils, immunoglobulin E, and cytokines that promote mucus and airway hyperresponsiveness. Other patients have neutrophilic, mixed, or paucigranulocytic patterns. Phenotype influences response to corticosteroids and eligibility for targeted biological therapy.

Airway remodelling develops through repeated epithelial injury, smooth-muscle enlargement, mucus-gland change, and subepithelial matrix deposition. Variable obstruction can become partly fixed. Preventing

exacerbations therefore protects future function as well as relieving present symptoms.

Symptoms and inflammation do not always move together. A patient can feel well while eosinophilic risk remains high, or remain breathless because of dysfunctional breathing, obesity, reflux, inducible laryngeal obstruction, or deconditioning despite controlled airway inflammation. Before escalating treatment, confirm diagnosis, inhaler technique, adherence, exposure, and competing mechanisms.

Severe exacerbation narrows airways unevenly, creating low ventilation-perfusion units and dynamic hyperinflation. Increasing effort may initially lower carbon dioxide. A normalising or rising value during persistent distress indicates reduced effective ventilation. Intubation is high risk because induction removes muscle tone and positive-pressure breaths can worsen trapping; specialist preparation prioritises haemodynamics and long expiratory time.

Chronic obstructive disease as multiple traits

Chronic obstructive pulmonary disease is not one uniform entity. Emphysema, small-airway disease, mucus hypersecretion, frequent exacerbation, eosinophilic inflammation, pulmonary vascular disease, and systemic frailty vary independently. Spirometric obstruction confirms physiology but does not capture symptom burden, exacerbation risk, gas transfer, hyperinflation, or comorbidity.

Hyperinflation can be more disabling than the forced expiratory measurement suggests. End-expiratory volume rises during exertion as breathing accelerates, reducing inspiratory capacity and creating intolerable dyspnoea. Long-acting bronchodilation can improve exercise by reducing trapping even when the absolute spirometric change appears modest.

Inhaled corticosteroids reduce exacerbations in selected patients but increase pneumonia risk and are not universal treatment. Previous exacerbation pattern, eosinophil count, asthma overlap, and adverse effects guide use. Long-term oral corticosteroids create substantial muscle, bone, metabolic, infection, and adrenal harm and are generally reserved for specific circumstances.

Pulmonary rehabilitation interrupts a cycle of dyspnoea, inactivity, deconditioning, and greater dyspnoea. Exercise training improves peripheral efficiency even when lung structure cannot recover. Nutrition, anxiety management, breathing strategies, smoking cessation, and advance care planning address the systemic disease.

Bronchiectasis and microbial ecology

Dilated airways retain secretions, but disease activity also depends on mucus properties, ciliary function, immune response, aspiration, and microbial community. Repeated broad antibiotics can select resistance and alter flora. Sputum obtained during stability establishes a baseline; exacerbation samples help distinguish new pathogens from chronic colonisation.

Pseudomonas infection is associated with greater disease burden in many patients, but eradication, suppression, or acute treatment should follow specialist protocols. Long-term macrolide therapy may reduce exacerbations through antimicrobial and immunomodulatory effects, yet requires assessment of cardiac repolarisation, hearing, resistance, and non-tuberculous mycobacteria.

Airway clearance is individualised to anatomy, sputum load, haemoptysis, reflux, musculoskeletal ability, and preference. Hydration and inhaled mucoactive therapy can support selected patients. Treating immune deficiency, allergic fungal disease, obstruction, reflux, or aspiration may reduce the driver rather than only managing infection.

Interstitial disease and progressive fibrosis

Interstitial lung diseases differ in whether inflammation, fibrosis, granulomas, exposure, or vascular injury predominates. High-resolution imaging patterns reflect spatial distribution, reticulation, ground-glass change, traction bronchiectasis, honeycombing, nodules, and air trapping. A pattern can strongly suggest a diagnosis but should be reconciled in multidisciplinary discussion.

Ground-glass opacity is non-specific and can represent partial air-space filling, interstitial thickening below resolution, inflammation, haemorrhage, oedema, or collapse. Fibrotic signs such as architectural distortion and traction indicate more established structural change. Serial imaging must

balance information against radiation and is interpreted with symptoms and pulmonary function.

Forced vital capacity tracks volume but can remain deceptively stable when emphysema counterbalances restriction. Diffusing capacity and exertional oxygenation add information. Progressive fibrosis is defined by worsening across symptoms, physiology, or imaging despite management, not by one noisy measurement.

Exposure removal can be decisive in hypersensitivity pneumonitis and occupational disease. A generic occupational title is insufficient; ask about tasks, materials, ventilation, protective equipment, co-workers, hobbies, home dampness, birds, and temporal improvement away from exposure. Workplace control can protect others as well as the patient.

Pulmonary vascular classification

Pulmonary hypertension is a haemodynamic finding with several causal groups. Pre-capillary disease raises pulmonary resistance without elevated left-sided filling pressure. Post-capillary disease follows left-heart pressure. Lung disease and hypoxia reduce vascular bed and cause constriction. Chronic thromboembolic disease creates organised obstruction with secondary microvascular remodelling.

Echocardiography estimates probability through tricuspid velocity, right-heart size and function, septal position, pulmonary-artery features, and venous pressure. Right-heart catheterisation measures pressures, output, and resistance and distinguishes pre- from post-capillary physiology. Measurements are sensitive to volume state, respiratory swings, zero reference, and technique.

Pulmonary arterial vasodilators can improve selected pre-capillary disease but worsen pulmonary oedema in left-heart disease or impair gas matching in lung disease. Chronic thromboembolic disease may be surgically or catheter treatable. Classification is therefore a therapeutic requirement, not academic taxonomy.

Acute pulmonary embolism strains the right ventricle according to clot burden, baseline pulmonary reserve, and haemodynamic response. Risk assessment includes pressure, right-heart dysfunction, biomarkers, oxygenation, and

clinical trajectory. Reperfusion decisions balance death from obstruction against major bleeding and follow current protocols.

Infection, pleura, and source control

Pneumonia is a syndrome of infection plus parenchymal response. Imaging may lag, especially early or with dehydration. Microbiology is most valuable in severe disease, immune compromise, unusual exposure, treatment failure, or when resistance will change therapy. Detection of an organism does not always prove causation because colonisation and prolonged nucleic-acid shedding occur.

Aspiration location depends on position and anatomy. Chemical pneumonitis can improve with support alone, whereas bacterial pneumonia develops from inoculated organisms and impaired clearance. Repeated aspiration requires assessment of swallowing, consciousness, reflux, oral health, feeding method, and medicines rather than indefinite antibiotics.

Pleural infection progresses from free-flowing reactive fluid to bacterial invasion, acidity, fibrin loculation, and organisation. Antibiotic penetration and drainage become increasingly important. Ultrasound reveals septations and guides access. Failure to drain a complex infected space allows persistent sepsis despite appropriate antimicrobial susceptibility.

Tuberculosis management joins individual and public health. Microbiological confirmation supports susceptibility testing. Airborne precautions, contact assessment, adherence support, toxicity monitoring, and combination therapy prevent spread and resistance. Immune restoration can temporarily intensify inflammation even while microbial treatment works.

Lung cancer and treatment reserve

Lung-cancer staging describes local tumour, nodal spread, and metastasis, while molecular profiling identifies actionable drivers and immune markers. Tissue acquisition should obtain enough material safely for both histology and biomarkers. The easiest lesion to sample may also establish the highest stage.

Operability depends on anatomy, but physiological resectability depends on predicted postoperative lung function, exercise capacity, cardiac risk, frailty, and patient goals. Radiation, systemic therapy, and local ablative options may

offer control when surgery is unsuitable. Treatment toxicity must be separated from infection, embolism, progression, and pre-existing lung disease.

Immune checkpoint therapy can produce inflammatory injury in lung and other organs. Targeted treatments create pathway-specific toxicities and resistance. New dyspnoea in a treated patient therefore requires a broad differential and rapid coordination rather than automatic attribution to cancer.

The final disease synthesis should name the affected compartment, dominant physiological defect, cause or exposure, activity, structural reversibility, exacerbation risk, and systemic consequence. Treatment becomes precise only when those dimensions are separated instead of labelling every chronic cough or breathless patient with one broad diagnosis. Longitudinal review should then test whether symptoms, exacerbations, function, gas exchange, imaging, and treatment burden move in the expected direction, while preserving rapid access for haemoptysis, pneumothorax, infection, embolism, or acute respiratory failure during any unexpected or accelerating deterioration.

Retrieval prompts

One. What defines asthma physiologically?

Two. How do emphysema and small-airway disease produce obstruction?

Three. Why does bronchiectasis perpetuate infection?

Four. What distinguishes pneumonia treatment from empyema treatment?

Five. How should interstitial lung disease be approached?

Six. Why must pulmonary hypertension be classified before therapy?

Concise answers

One. Variable airflow obstruction with airway hyperresponsiveness and inflammation.

Two. Loss of recoil promotes collapse and air trapping, while narrowed inflamed airways increase resistance.

Three. Damaged dilated airways clear mucus poorly, allowing infection and further inflammation.

Four. Pneumonia may respond to antimicrobials; organised infected pleural fluid usually also needs drainage.

Five. Use tempo, exposure, autoimmune features, physiology, high-resolution imaging, and selected tissue or laboratory tests.

Six. Causes have different mechanisms, and drugs useful for pulmonary arterial disease may harm other groups.

Source map

Original synthesis informed by Robbins, pulmonary pathology; Guyton and Hall, respiratory pathophysiology; OpenStax Microbiology and Medical-Surgical Nursing; Katzung and OpenStax Pharmacology, respiratory and antimicrobial therapy; and Talley and O'Connor, respiratory disease correlation.

Chapter 16: Respiratory History, Examination, Blood Gases, and Investigations

Orientation

Respiratory assessment determines urgency, anatomical compartment, physiological defect, cause, and functional effect. Symptoms overlap with cardiac, haematological, metabolic, neuromuscular, and psychological disease. A structured history and examination establish pre-test probability and make gas, imaging, and pulmonary-function results interpretable.

Symptom history

Dyspnoea is characterised by onset, tempo, triggers, position, exertional threshold, variability, and associated chest pain, cough, wheeze, fever, oedema, palpitations, and weakness. Sudden dyspnoea suggests embolism, pneumothorax, acute airway obstruction, oedema, arrhythmia, or metabolic disturbance. Progressive exertional dyspnoea suggests obstructive, interstitial, vascular, cardiac, anaemic, or neuromuscular disease.

Cough is acute, subacute, or chronic and may be dry or productive. Ask about timing, triggers, sputum volume and colour, blood, reflux, nasal symptoms, medicines, smoking, infection, and aspiration. Purulent colour reflects leukocytes but does not alone prove bacterial infection. Haemoptysis must be distinguished from nasopharyngeal or gastrointestinal bleeding. Estimate volume and assess airway risk.

Wheeze suggests narrowed airways but can be focal from obstruction or diffuse from asthma, chronic obstructive disease, oedema, or anaphylaxis. Stridor is loud upper-airway obstruction and demands urgency. Pleuritic pain worsens with breathing and occurs with pleural inflammation, embolism, pneumothorax, infection, or chest-wall disease.

Ask about smoking and vaping in pack-year and temporal detail; occupational dust, fumes, asbestos, silica, animals, mould, and metal exposure; tuberculosis contact; travel; immune suppression; aspiration; drugs causing

cough, bronchospasm, bleeding, or interstitial disease; and family disease. Determine vaccination, previous ventilation, exacerbation frequency, home oxygen, inhaler technique, and baseline function.

Initial observation and vital signs

Observe position, speech, respiratory rate and pattern, accessory muscles, chest movement, audible sounds, cough, colour, sweating, and consciousness. Count respiratory rate rather than estimating it. Record oxygen device and flow with saturation. Fever suggests infection or inflammation but may be absent. Tachycardia can reflect hypoxaemia, work, fever, embolism, or medication.

In severe distress, inability to speak, exhaustion, reduced consciousness, paradoxical movement, poor air entry, cyanosis, hypotension, or worsening gas values requires immediate support. Do not complete a ceremonial examination before treating instability.

Hands, face, and neck

Hands may show nicotine staining, peripheral cyanosis, clubbing, tremor, asterixis, or inflammatory-joint disease. Clubbing occurs with lung cancer, bronchiectasis, fibrosis, abscess, cyanotic heart disease, and gastrointestinal disorders, but not uncomplicated asthma or chronic obstructive disease.

Assess conjunctival pallor, central cyanosis at tongue and mucosa, nasal disease, oral infection, dentition, and upper-airway anatomy. Palpate cervical and supraclavicular lymph nodes. Assess tracheal position; marked deviation can reflect volume loss, mass, large effusion, or tension pneumothorax.

Chest examination

Inspect shape, scars, deformity, symmetry, and respiratory expansion. Hyperinflation increases anteroposterior diameter but visual assessment is imprecise. Unilateral lag suggests pain, collapse, pleural disease, or obstruction.

Palpate expansion and tactile fremitus when useful. Fremitus increases over consolidated lung that transmits sound and decreases with pleural fluid, pneumothorax, obstruction, or thick chest wall. Percussion is resonant over

aerated lung, dull over consolidation or fluid, and hyperresonant with excess gas. Technique and comparison matter.

Auscultate systematically. Vesicular breath sounds are soft with longer inspiration. Bronchial breathing over peripheral lung suggests transmission through consolidation or a cavity. Crackles arise from airway opening or secretions; fine late inspiratory crackles support fibrosis or oedema, while coarse sounds suggest larger-airway secretions, but overlap is substantial. Wheeze is musical airflow through narrowed airways. A pleural rub is grating and localised.

Assess vocal resonance when consolidation or effusion is suspected. Examine cardiovascular status, jugular venous pressure, oedema, calf asymmetry, skin, joints, muscle strength, and abdomen because respiratory disease can be systemic or secondary.

Arterial and venous blood gases

An arterial gas measures pH, carbon dioxide, oxygen, and derived bicarbonate. A venous gas often estimates pH and carbon dioxide sufficiently for initial metabolic assessment but cannot replace arterial oxygen measurement. Sampling errors include air bubbles, delay, venous contamination, and failure to document oxygen therapy.

Interpret systematically. First assess pH. Second identify whether carbon dioxide or bicarbonate changes explain the primary direction. Third judge expected compensation. Fourth calculate the anion gap when metabolic acidosis exists. Fifth assess oxygenation with inspired oxygen and clinical context. Mixed disorders are common.

Acute respiratory acidosis has limited bicarbonate rise; chronic retention has greater renal compensation. Respiratory alkalosis occurs with hypoxaemia, pain, anxiety, sepsis, pregnancy, liver disease, pulmonary embolism, or central stimulation. A normal pH does not exclude a mixed or compensated disorder.

The alveolar-arterial oxygen difference separates uncomplicated hypoventilation from gas-exchange impairment, but estimation depends on barometric pressure, inspired fraction, and respiratory quotient. Co-oximetry

directly measures dyshemoglobins when carbon monoxide or methaemoglobin is suspected.

Imaging

Chest radiography assesses lung volume, focal opacity, interstitial pattern, pleural fluid, pneumothorax, heart size, vessels, bones, and devices. Portable films, rotation, inspiration, exposure, and supine position alter appearance. A normal radiograph does not exclude early pneumonia, embolism, small pneumothorax, asthma, or airway disease.

Computed tomography provides detailed parenchymal, airway, pleural, and mediastinal anatomy. High-resolution technique characterises interstitial disease and bronchiectasis. Computed-tomography pulmonary angiography assesses emboli. Contrast, radiation, motion, and incidental findings require consideration.

Ultrasound detects pleural fluid, peripheral consolidation, interstitial artefacts, pneumothorax signs, and diaphragmatic movement at bedside. It cannot see through aerated lung to all central lesions. Ventilation-perfusion scanning is useful for embolism in selected patients. Positron emission tomography supports cancer staging but inflammation can also be avid.

Pulmonary-function and exercise tests

Spirometry measures forced expiratory volume and vital capacity. Confirm obstruction with a reduced ratio and restriction with total lung capacity. Lung volumes quantify hyperinflation and gas trapping. Diffusing capacity assesses transfer and pulmonary capillary blood volume with haemoglobin correction.

Bronchial challenge tests hyperresponsiveness when asthma remains suspected. Six-minute walk testing measures functional capacity and desaturation. Cardiopulmonary exercise testing integrates ventilation, gas exchange, circulation, and metabolism when routine testing does not explain limitation.

Microbiology, bronchoscopy, and pleural testing

Sputum quality matters; saliva contamination reduces usefulness. Gram stain, culture, acid-fast testing, fungal studies, and molecular assays are selected by syndrome. Blood cultures are appropriate in severe infection. Bronchoscopy visualises airways, samples distal lung, removes plugs or foreign bodies, and obtains biopsy, with sedation, bleeding, and respiratory risks.

Pleural fluid is assessed for protein and enzyme patterns, pH, glucose, cells, microbiology, cytology, and appearance. Very low pH in parapneumonic effusion supports drainage. Tissue diagnosis may require image-guided biopsy, bronchoscopy, thoracoscopy, or surgery.

Severity, longitudinal monitoring, and procedures

Severity tools can standardise assessment in pneumonia, embolism, and chronic lung disease, but they supplement rather than replace judgement. A low score may not capture rapid trajectory, frailty, immune suppression, social barriers, or inability to maintain hydration and treatment. Reassessment should document oxygen requirement, respiratory rate, work, mental state, mobility, intake, and response.

Longitudinal monitoring uses symptoms, exacerbations, activity, inhaler use, spirometry, oxygenation, imaging, and adverse effects according to disease. A small test change can be important with functional decline, while apparent decline may reflect technique. Pulmonary rehabilitation assesses exercise, muscle strength, breathing strategies, nutrition, education, and psychological support.

Thoracentesis, chest-drain insertion, bronchoscopy, lung biopsy, and pleural procedures require a clear indication, imaging review, consent, anticoagulation and platelet assessment, monitoring, and complication planning. Complications include pneumothorax, bleeding, infection, re-expansion oedema, hypoxaemia, arrhythmia, and sedation-related failure. Ultrasound improves pleural site selection.

Preoperative respiratory assessment identifies active infection or exacerbation, smoking, functional limitation, sleep apnoea, aspiration risk, airway difficulty, and surgery-specific risk. Routine testing without an actionable question is less useful than optimisation through smoking

cessation, bronchodilator control, activity, secretion management, and postoperative lung expansion and mobilisation.

TTS module 2: Pattern localisation, blood-gas compensation, and investigation quality

Localising the respiratory problem

Respiratory assessment becomes clearer when symptoms are assigned to a probable compartment. Upper-airway disease produces stridor, voice change, swallowing difficulty, or positional obstruction. Conducting-airway disease produces wheeze, cough, mucus, and variable flow. Alveolar disease produces hypoxaemia, crackles, and opacities. Interstitial disease reduces compliance and diffusion. Vascular disease increases dead space and right-heart load. Pleural disease causes pain and mechanical restriction.

The respiratory pump may fail with relatively normal lung parenchyma. Central depression, spinal injury, neuropathy, neuromuscular-junction disease, myopathy, chest-wall restriction, and obesity reduce effective ventilation. Weak cough and bulbar dysfunction increase secretion and aspiration risk. Normal early saturation does not exclude impending pump failure.

Tempo refines localisation. Abrupt pleuritic pain and dyspnoea suggest pneumothorax, embolism, or acute pleural irritation. Hours to days favour infection, oedema, aspiration, exacerbation, or inflammatory injury. Months to years suggest chronic airway, interstitial, vascular, neoplastic, neuromuscular, or cardiac disease. Acute illness can be superimposed on any chronic substrate.

Examination as a set of transmission tests

Percussion asks how underlying tissue transmits sound. Air produces resonance, fluid or dense tissue produces dullness, and excess pleural or lung gas can produce hyperresonance. Obesity, muscle, technique, and lesion depth limit sensitivity. Comparison across symmetric sites is more useful than an isolated note.

Tactile fremitus and vocal resonance ask how voice vibration travels. Consolidated airless lung can transmit vibration strongly when the airway remains open. Pleural fluid or gas separates lung from chest wall and reduces

transmission. Obstruction can also reduce distal signal. A large effusion may produce bronchial breathing just above its upper border through compressed lung.

Crackles are brief discontinuous sounds. Fine late-inspiratory crackles often reflect sudden opening of small stiff airways in fibrosis or oedema. Coarse sounds may arise from larger-airway secretions and can change after coughing. Wheeze requires airflow; a severely obstructed patient with minimal flow may become quiet. Sound intensity must be paired with effort and air movement.

Asymmetry has high localising value. Unilateral reduced expansion, breath sounds, and fremitus with dull percussion suggests pleural fluid; hyperresonance suggests pleural gas. Local bronchial breathing supports consolidation. Findings are imperfect, so urgent ultrasound or radiography should confirm when management depends on anatomy.

Blood-gas compensation step by step

Begin with pH but remember that a value within the reference interval can conceal two opposing disorders. Next decide whether carbon dioxide and bicarbonate move in the direction that explains acid or alkali change. Then compare compensation with the expected physiological response. Compensation reduces pH disturbance but does not overshoot into the opposite primary state.

In respiratory acidosis, carbon dioxide is the primary retained acid. Intracellular and extracellular buffers produce a small immediate bicarbonate increase. Over days, kidneys increase acid excretion and bicarbonate retention. In respiratory alkalosis, acute buffering lowers bicarbonate modestly and renal adaptation lowers it further over time.

Metabolic acidosis should trigger increased ventilation and reduced carbon dioxide. If measured carbon dioxide is higher than expected, a superimposed respiratory acidosis exists. If it is lower, an additional respiratory alkalosis exists. Metabolic alkalosis normally produces limited hypoventilatory compensation because hypoxaemia constrains how far ventilation can fall.

The anion gap estimates unmeasured anions. Albumin is a major unmeasured anion, so hypoalbuminaemia can conceal a clinically important

gap. Comparing the rise in gap with the fall in bicarbonate can suggest a concurrent normal-gap acidosis or metabolic alkalosis, but baseline values and measurement error limit precision.

Oxygenation and sampling reliability

Arterial oxygen must be interpreted with inspired oxygen, barometric pressure, ventilation, haemoglobin, and timing. A specimen collected after changing the oxygen device may not represent equilibrium. The recorded device and flow are part of the result, not optional administrative detail.

Air bubbles allow equilibration with room air, tending to raise a low oxygen tension and lower a high carbon dioxide tension. Delayed analysis permits ongoing cellular metabolism to consume oxygen and generate carbon dioxide. Venous contamination lowers measured oxygen. A result inconsistent with the patient should prompt a sample-quality review before elaborate explanation.

Venous gases are useful for pH and broad carbon dioxide assessment in many contexts but do not provide reliable arterial oxygenation. Capillary samples vary with perfusion and collection. Pulse oximetry supplies continuous trend but not pH or carbon dioxide. Choosing the least invasive adequate method reduces harm without losing the needed information.

Pulmonary-function quality

Spirometry requires a maximal inhalation, rapid forceful start, sustained expiration, and reproducible effort. A poor start lowers the first-second volume. Early termination lowers forced vital capacity and can falsely raise the ratio. Cough, leak, glottic closure, pain, and submaximal effort create characteristic errors that should be reviewed before interpretation.

Lower-limit-of-normal values account for age, sex, height, and reference population more accurately than one fixed ratio in many settings. A statistically low value is not automatically clinically important, and a value above threshold does not exclude early disease. Symptoms, exposure, trend, and test quality matter.

Restriction cannot be confirmed from low forced vital capacity alone because air trapping, weak effort, obesity, or neuromuscular weakness can reduce it.

Total lung capacity establishes restriction. Residual volume and the ratio of residual volume to total capacity quantify trapping and hyperinflation.

Diffusing capacity should be corrected or interpreted for haemoglobin. Transfer coefficient relates diffusing capacity to measured alveolar volume but is not a simple correction that reveals a hidden normal lung. Low alveolar volume from incomplete inspiration, prior resection, or uneven ventilation changes its meaning.

Imaging patterns and pre-test probability

Image interpretation begins with technical adequacy. Rotation changes mediastinal appearance. Low inspiration mimics basal opacity and magnifies the heart. Supine radiographs redistribute pleural fluid and can hide pneumothorax. Portable anteroposterior technique enlarges the cardiac silhouette.

Air-space opacity may represent infection, oedema, haemorrhage, aspiration, or tumour. Interstitial patterns can arise from oedema, fibrosis, inflammation, lymphatic tumour spread, or technical factors. Distribution, volume change, air bronchograms, pleural findings, and time course narrow the cause.

Computed tomography detects more detail but also more incidental abnormalities. Small nodules require management according to size, morphology, risk, and prior imaging rather than immediate biopsy. Contrast-enhanced studies answer vascular and mediastinal questions; high-resolution non-contrast technique answers many interstitial questions. Protocol should match the clinical decision.

Ultrasound artefacts become useful signs. Pleural sliding argues against pneumothorax at the scanned point, while its absence has several causes. Vertical artefacts increase when subpleural interstitial fluid or density changes, but they do not distinguish cardiac oedema from all inflammatory disease alone. Consolidation near pleura can show tissue-like appearance and dynamic air bronchograms.

Microbiology and invasive sampling

Specimen quality determines microbiological value. Expecterated sputum with many squamous cells suggests oral contamination. An endotracheal aspirate samples proximal secretions and can detect colonisers. Bronchoalveolar

lavage samples a defined distal region but may worsen hypoxaemia or provoke bleeding. Tissue is needed when architecture, invasion, granuloma, malignancy, or treatment resistance must be established.

Molecular assays can remain positive after viable organism declines and can detect colonisation. Culture takes longer but provides susceptibility and viable isolate. Antigen tests have organism-specific strengths and limitations. Results must be interpreted with host immunity, pre-treatment, specimen site, and imaging.

Pleural fluid classification begins with whether systemic pressure forces or local pleural disease predominate. Protein and enzyme ratios support transudative or exudative categorisation but can be altered by diuresis. Cell differential, pH, glucose, microbiology, cytology, triglycerides, haematocrit, and selected specialised tests answer further questions.

Closing the diagnostic loop

Every test result should be reconciled with the original physiological model. If spirometry is normal despite episodic wheeze, consider variable disease and alternative upper-airway or cardiac mechanisms. If imaging shows severe abnormality with mild symptoms, assess adaptation and reserve rather than dismissing either. If treatment fails, revisit diagnosis, adherence, delivery technique, resistance, complication, and second pathology.

Document oxygen requirement, respiratory rate, work, speech, mental state, gas trajectory, and functional baseline so later clinicians can recognise change. Specify pending cultures, imaging follow-up, tissue results, and ownership. Respiratory diagnosis is safest when anatomy, physiology, cause, severity, and trajectory converge rather than when one impressive image or isolated gas value dominates.

Retrieval prompts

One. Which dyspnoea and haemoptysis features demand urgency?

Two. What do fremitus, percussion, and bronchial breathing suggest?

Three. Give a systematic blood-gas sequence.

Four. Why can a normal chest radiograph be falsely reassuring?

Five. What does each pulmonary-function component contribute?

Six. How should pleural fluid and microbiological samples be chosen?

Concise answers

One. Sudden onset, instability, hypoxaemia, exhaustion, altered consciousness, large bleeding, or airway compromise.

Two. Increased fremitus and bronchial breathing support consolidation; dullness suggests fluid or solid tissue; reduced fremitus suggests fluid, gas, or obstruction.

Three. Assess pH, primary carbon dioxide or bicarbonate change, compensation, anion gap, and oxygenation.

Four. Early, vascular, airway, or small pleural disease may not be visible, and technique limits sensitivity.

Five. Spirometry measures flow, volumes confirm restriction or trapping, diffusion assesses transfer, and exercise tests integrated reserve.

Six. Match specimen and method to syndrome, site, severity, contamination risk, and the decision results will change.

Source map

Original synthesis informed by Talley and O'Connor's Clinical Examination, respiratory history, examination, and correlations; Guyton and Hall, respiratory physiology and blood gases; Robbins, pulmonary pathology; OpenStax Microbiology and Medical-Surgical Nursing; and Katzung and OpenStax Pharmacology.

Chapter 17: Glomerular Filtration, Tubular Transport, Concentration, and Clearance

Orientation

The kidneys filter plasma, selectively reclaim useful substances, secrete others, regulate water and electrolytes, control acidity, participate in blood-pressure regulation, produce erythropoietin, and activate vitamin D. Renal function depends on perfusion, glomerular barrier integrity, tubular transport, interstitial architecture, drainage, and hormonal control. A normal urine volume does not guarantee normal filtration, and a reduced filtration estimate does not reveal mechanism by itself.

Renal circulation and nephron organisation

Renal arteries divide into progressively smaller vessels ending in afferent arterioles. Each afferent arteriole supplies a glomerular capillary tuft. Blood leaves through an efferent arteriole rather than a venule. Efferent vessels form peritubular capillaries around cortical tubules or vasa recta that descend into the medulla.

The nephron begins at Bowman's capsule, followed by proximal tubule, loop of Henle, distal tubule, connecting segment, and collecting duct. Cortical nephrons have short loops; juxtamedullary nephrons have long loops essential for urine concentration. Collecting ducts receive fluid from several nephrons and pass through the medullary gradient.

Renal blood flow is high relative to organ mass because filtration requires delivery. The cortex receives most flow. The medulla has lower flow, preserving its osmotic gradient but making it vulnerable to hypoxic injury. Oxygen demand is driven strongly by active sodium transport.

Glomerular filtration barrier

Filtration crosses fenestrated endothelium, glomerular basement membrane, and podocyte slit structures. Size, shape, and charge influence passage. Water and small solutes filter freely, while cells and most large proteins remain in blood. Albumin restriction reflects multiple barrier properties and proximal reclamation of small filtered amounts.

Glomerular injury can produce haematuria, proteinuria, reduced filtration, or combinations. Dysmorphic red cells and red-cell casts support glomerular bleeding. Heavy albumin loss produces nephrotic physiology with oedema, hypoalbuminaemia, lipid abnormalities, and thrombosis risk. Inflammatory glomerular syndromes can produce haematuria, hypertension, reduced filtration, and variable protein loss.

Filtration pressures and rate

Glomerular hydrostatic pressure favours filtration. Hydrostatic pressure in Bowman's space and plasma-protein oncotic pressure oppose it. Filtration rate equals net filtration pressure multiplied by the filtration coefficient, which reflects permeability and surface area.

Afferent dilation raises glomerular flow and pressure. Afferent constriction lowers both. Moderate efferent constriction can raise glomerular pressure while reducing renal blood flow; severe constriction eventually reduces filtration as oncotic pressure rises and flow falls. Systemic pressure, volume, neurohormonal state, and drugs alter these relationships.

Autoregulation stabilises renal blood flow and filtration across a pressure range. The myogenic response constricts afferent muscle when stretched. Tubuloglomerular feedback uses the macula densa to sense sodium chloride delivery. High delivery promotes afferent constriction and reduced renin; low delivery promotes renin and adjustments that support filtration and sodium retention.

Proximal tubule

The proximal tubule reabsorbs most filtered sodium and water, nearly all glucose and amino acids under normal conditions, most bicarbonate, and substantial phosphate and other solutes. Sodium gradients maintained by the basolateral sodium-potassium pump drive apical cotransport and exchange. Water follows solute, so reabsorption is approximately iso-osmotic.

Filtered bicarbonate combines with secreted hydrogen in the lumen, forms carbon dioxide through carbonic anhydrase, enters cells, and is regenerated into bicarbonate returned to blood. This process reclaims bicarbonate but does not by itself create new bicarbonate.

Glucose reabsorption has a transport maximum. When filtered load exceeds capacity, glucose appears in urine and causes osmotic diuresis. Drugs that inhibit sodium-glucose cotransport deliberately increase glucose and sodium excretion and alter glomerular haemodynamics.

The proximal tubule secretes organic acids and bases, including many drugs and metabolites. Competition at transporters can change drug elimination. Injury produces loss of reabsorptive capacity, bicarbonate wasting, phosphate loss, glucose in urine despite normal blood glucose, and aminoaciduria.

Loop of Henle and countercurrent multiplication

The thin descending limb is highly permeable to water and less permeable to solute, so tubular fluid becomes concentrated as it descends into hyperosmotic medulla. The thick ascending limb reabsorbs sodium, potassium, and chloride but is relatively impermeable to water, diluting tubular fluid and adding solute to the medullary interstitium.

Repeated flow through opposing limbs multiplies a small transverse gradient into a large corticomedullary gradient. Urea recycling contributes, especially in inner medulla. Vasa recta act as countercurrent exchangers, supplying tissue while limiting washout. Excess medullary blood flow, loop-diuretic action, or structural disease reduces concentrating ability.

Distal nephron and collecting duct

The early distal tubule reabsorbs sodium and chloride and remains relatively water-impermeable. Later segments contain principal cells and intercalated cells. Principal cells reabsorb sodium through epithelial sodium channels and secrete potassium. Aldosterone increases sodium transport and potassium secretion by changing channel and pump expression and activity.

Antidiuretic hormone inserts aquaporin two channels into collecting-duct apical membranes, increasing water permeability. Water then follows the medullary gradient, concentrating urine. Without antidiuretic hormone effect, collecting ducts remain less permeable and dilute urine is excreted.

Alpha intercalated cells secrete hydrogen and return bicarbonate to blood, while beta intercalated cells can secrete bicarbonate. Distal acid secretion and ammonium handling generate new bicarbonate and permit excretion of non-volatile acid.

Clearance and renal measurement

Clearance is the virtual plasma volume cleared of a substance per unit time. For a freely filtered marker neither reabsorbed nor secreted, clearance equals glomerular filtration rate. Inulin approximates this ideal experimentally. Creatinine is produced from muscle and filtered, with some secretion; measured or estimated creatinine clearance therefore has limitations.

Serum creatinine depends on production, distribution, filtration, and time. A small rise can represent substantial filtration loss in a person with low baseline creatinine. During acute change, steady-state estimating equations are inaccurate. Cystatin C offers a filtration marker with different non-renal influences and can refine estimates.

Renal plasma flow can be estimated using a substance filtered and strongly secreted so most delivered amount is excreted. Filtration fraction is filtration rate divided by renal plasma flow. Fractional excretion calculations relate excreted to filtered amounts but are altered by diuretics, chronic disease, sepsis, and timing.

Urinalysis

Dipstick assesses concentration, pH, blood, protein, glucose, ketones, leukocyte esterase, and nitrite, with false positives and negatives. Microscopy identifies cells, casts, crystals, organisms, and contamination. Hyaline casts can occur normally. Red-cell casts support glomerulonephritis; white-cell casts suggest renal inflammation; granular muddy casts support tubular injury.

Albumin-to-creatinine ratio estimates albumin excretion while accounting approximately for urine concentration. Protein-to-creatinine ratio includes non-albumin proteins. Persistent abnormality should be confirmed and interpreted with exercise, fever, infection, blood, menstruation, and systemic disease.

Endocrine and metabolic renal functions

Renal interstitial cells produce erythropoietin in response to hypoxia-inducible signalling. Chronic kidney disease can therefore cause hypoproliferative anaemia, compounded by inflammation, iron restriction, blood loss, and shortened red-cell survival. Erythropoiesis-stimulating treatment requires iron assessment and careful haemoglobin and pressure monitoring because excessive correction increases vascular risk.

Proximal tubular cells convert vitamin D to its active form under stimulation by parathyroid hormone and low phosphate. Active vitamin D increases intestinal calcium and phosphate absorption. Kidney disease reduces activation and phosphate excretion, contributing to hypocalcaemia, secondary hyperparathyroidism, altered bone turnover, vascular calcification, and renal osteodystrophy.

The kidneys perform gluconeogenesis during fasting, metabolise peptide hormones, and clear many endogenous molecules. Loss of renal function changes insulin requirement, drug exposure, and endocrine measurements. Conversely, diabetes, paraproteins, urate, oxalate, and metabolic disease can damage glomeruli and tubules.

Diuretic sites as physiology probes

Carbonic-anhydrase inhibitors reduce proximal bicarbonate reclamation. Loop diuretics block sodium-potassium-chloride transport in the thick ascending limb, weakening the medullary gradient and increasing calcium and magnesium excretion. Thiazide-type drugs block distal sodium-chloride transport and reduce calcium excretion. Potassium-sparing drugs inhibit aldosterone signalling or epithelial sodium channels. Their effects reveal segment function but also produce predictable volume, electrolyte, acid-base, and kidney adverse effects.

TTS module 2: Glomerular haemodynamics, tubular energetics, and renal diagnostic patterns

Filtration is pressure constrained

Glomerular filtration requires adequate renal plasma flow, a favourable transcapillary pressure, and an intact filtration surface. Afferent tone controls inflow. Efferent tone controls outflow resistance. Moderate efferent constriction can preserve glomerular pressure when systemic pressure falls, but it reduces renal blood flow and raises filtration fraction. Excessive constriction eventually limits flow enough to reduce filtration and increase peritubular oncotic pressure.

Angiotensin two supports efferent tone during low effective arterial volume. Blocking this pathway lowers intraglomerular pressure and can reduce filtration, particularly with bilateral renal-artery disease, severe volume depletion, or dependence on compensatory tone. The same haemodynamic effect can be beneficial over time by reducing damaging glomerular pressure in proteinuric disease.

Prostaglandins help maintain afferent dilation during stress. Inhibiting their synthesis can constrict afferent flow, especially in hypovolaemia, heart failure, cirrhosis, older age, or existing kidney disease. Combining reduced afferent support, reduced efferent support, and diuretic-associated volume loss creates a characteristic risk for haemodynamic acute kidney injury.

Autoregulation has limits. Below its range, filtration becomes pressure dependent. Above it, transmitted pressure can injure glomerular capillaries. Diabetes, chronic hypertension, vascular disease, inflammation, and ageing

alter the autoregulatory response, making a pressure tolerated by one kidney harmful to another.

Filtration barrier selectivity

The glomerular endothelium restricts cells and interacts with the glycocalyx. The basement membrane provides structural and charge-related selectivity. Podocyte foot processes and slit proteins create a final specialised barrier. Injury to any layer can produce a different combination of haematuria, albuminuria, reduced filtration, and thrombosis.

Nephritic patterns reflect inflammatory capillary injury. Red cells enter urine, filtration falls, and sodium retention produces hypertension and oedema. Nephrotic patterns reflect severe protein permeability, leading to hypoalbuminaemia, oedema, lipid change, infection susceptibility, and thrombosis. Mixed patterns are common, and syndromes describe physiology rather than one histological diagnosis.

Protein in urine can be glomerular, tubular, overflow, or post-renal. Albumin predominance supports glomerular leakage. Tubular injury prevents reclamation of low-molecular-weight proteins. Overflow occurs when circulating proteins exceed reabsorptive capacity, as with selected immunoglobulin light chains. Inflammation or bleeding beyond the kidney can add protein after filtration.

Tubular transport consumes oxygen

Filtered sodium load creates most renal transport work. Basolateral sodium-potassium pumps maintain gradients that power apical cotransport and exchange. Oxygen demand is therefore highest where active sodium reclamation is greatest. The medulla receives relatively low oxygen delivery while performing substantial transport, creating vulnerability to hypoperfusion and toxins.

Proximal reabsorption normally keeps solute and water together. When transport fails, bicarbonate, phosphate, glucose, amino acids, urate, and small proteins can be lost in varying combinations. Generalised proximal dysfunction produces a Fanconi pattern. Causes include inherited disease, toxins, medicines, paraproteins, and severe tubular injury.

The thick ascending limb both dilutes tubular fluid and constructs the medullary gradient. Loop-diuretic blockade increases distal sodium delivery, weakens concentrating ability, and increases potassium, calcium, and magnesium loss. Distal segments can compensate partially, but chronic high delivery changes their transport capacity.

The distal nephron makes fine adjustments under aldosterone and antidiuretic hormone. Sodium entry through epithelial channels creates a lumen-negative voltage that supports potassium and hydrogen secretion. Potassium loss therefore depends not only on aldosterone but also on distal sodium delivery, tubular flow, dietary load, acid-base state, and magnesium.

Concentrating and diluting urine

Urine concentration requires three intact elements: a medullary osmotic gradient, collecting-duct water permeability, and delivery of fluid to that system. Antidiuretic hormone supplies permeability through aquaporin insertion. The loop and urea recycling supply the gradient. Vasa recta preserve it through countercurrent exchange.

Polyuria can be water diuresis or solute diuresis. Water diuresis produces dilute urine through excess intake, reduced antidiuretic hormone, or renal resistance. Solute diuresis produces greater urine osmoles through glucose, urea, sodium, mannitol, or other particles. Measuring urine volume and osmolality, then estimating osmole excretion, separates these mechanisms.

Concentrating failure can precede a large filtration decline in tubulointerstitial disease. Nocturia may appear because the kidney cannot produce a small concentrated overnight volume. Conversely, severe low filtration limits total water excretion even if antidiuretic hormone is suppressed because insufficient filtrate reaches diluting segments.

Clearance and non-steady-state traps

Clearance links urinary excretion to plasma concentration. A substance filtered and reabsorbed has clearance below filtration rate. A filtered and secreted substance has higher clearance. Fractional excretion expresses clearance relative to filtration and can suggest how the nephron handles a solute during a specific physiological state.

Serum creatinine changes slowly after an abrupt filtration loss because it must accumulate through its distribution volume. Early values therefore underestimate severity. During recovery, concentration can remain elevated after filtration has improved. Fluid accumulation dilutes it, while low muscle mass reduces production. A stable creatinine is reassuring only when production, volume, and kidney function are truly stable.

Estimated filtration equations are population tools, not direct measurements. They are less reliable during acute change, pregnancy, extremes of muscle mass, unusual diet, amputation, and selected drug effects on creatinine secretion. Cystatin C has different biases, including inflammation, corticosteroids, thyroid state, and body composition. Concordance strengthens inference; discordance requires explanation.

Urine output is an immediate but non-specific marker. Oliguria can result from low perfusion, obstruction, glomerular failure, tubular injury, or intense sodium and water retention. Non-oliguric injury remains possible when damaged tubules cannot concentrate urine or when diuretics preserve volume without restoring filtration.

Urine sediment as a renal biopsy in miniature

Red-cell casts form when glomerular bleeding is moulded within tubules and strongly support glomerular inflammation. White-cell casts point toward inflammation within kidney interstitium or tubules rather than uncomplicated lower urinary infection. Pigmented granular casts and tubular epithelial cells support acute tubular injury.

Crystals may be incidental or reveal supersaturation, toxin, medicine, metabolic disease, or stone risk. Their meaning depends on urine pH, shape, abundance, symptoms, and timing. Automated urinalysis can miss morphology, so manual microscopy remains valuable when glomerular, tubular, crystalline, or haematological disease is suspected.

Dipstick blood detects haem pigment, not intact red cells specifically. A positive result with few red cells suggests haemoglobin or myoglobin, although lysed cells and technical effects occur. Dipstick protein is most sensitive to albumin and can miss light chains. Concentrated or alkaline urine can distort results.

Integrating kidney function

Renal diagnosis should state filtration, urine output, sediment, protein pattern, tubular handling, volume status, and obstruction risk. Ultrasound assesses kidney size, structure, and drainage but normal imaging does not exclude functional or microscopic disease. Bladder scanning can rapidly identify retention.

A kidney result becomes actionable only when linked to trend and physiology. Ask whether perfusion is inadequate, venous pressure is excessive, the filtration barrier is inflamed, tubules are injured, a toxin is present, or drainage is blocked. Review all medicines and adjust doses according to current function and trajectory.

Renal reserve matters. Recovery is not simply creatinine normalisation. Residual albuminuria, impaired concentrating ability, reduced reserve, hypertension, and recurrent susceptibility can persist. Follow-up after acute injury should reassess kidney function, urine protein, pressure, medicines, and the cause that created vulnerability.

Kidney endocrine failure also deserves careful longitudinal attention. Reduced erythropoietin contributes to anaemia, but iron restriction, blood loss, inflammation, haemolysis, and vitamin deficiency should be assessed before attributing every low haemoglobin to kidney disease. Phosphate retention, reduced active vitamin D, altered calcium, and rising parathyroid hormone progressively remodel bone and vessels. Treatment depends on stage, diet, biochemical pattern, symptoms, and dialysis context. Excessive erythropoiesis stimulation or indiscriminate calcium and vitamin therapy can create vascular harm, so targets and monitoring should follow current specialist guidance rather than attempting to normalise every isolated laboratory value without considering symptoms, trajectory, calcification risk, and treatment burden.

Retrieval prompts

One. Name the filtration-barrier layers and major driving forces.

Two. How do afferent and efferent tone alter filtration?

Three. What are the proximal tubule's major functions?

Four. How does the loop create a medullary gradient?

Five. Contrast aldosterone with antidiuretic hormone.

Six. Why is creatinine an imperfect filtration marker?

Concise answers

One. Fenestrated endothelium, basement membrane, and podocyte slits; glomerular hydrostatic pressure favours, while capsular pressure and plasma oncotic pressure oppose.

Two. Afferent dilation raises flow and pressure; afferent constriction lowers them; moderate efferent constriction raises glomerular pressure but severe constriction can reduce filtration.

Three. Bulk sodium and water reclamation, nutrient and bicarbonate reabsorption, and organic-solute secretion.

Four. Water leaves the descending limb while salt leaves the water-impermeable ascending limb; repeated countercurrent flow multiplies the gradient.

Five. Aldosterone increases distal sodium reabsorption and potassium secretion; antidiuretic hormone increases collecting-duct water permeability.

Six. Production varies with muscle and diet, secretion contributes, and concentration lags during acute change.

Source map

Original synthesis informed by Guyton and Hall, renal circulation, filtration, tubular transport, concentration, and clearance; OpenStax Anatomy and Physiology 2e; Robbins, glomerular and tubular pathology; Katzung and OpenStax Pharmacology, diuretics and renal drug handling; and OpenStax Medical-Surgical Nursing.

Chapter 18: Sodium, Water, Potassium, and Volume Regulation

Orientation

Sodium content is the major determinant of extracellular fluid volume, while water relative to body solute largely determines plasma sodium concentration and osmolality. These are related but distinct variables. A patient can be volume depleted with high, normal, or low sodium, and oedematous with low sodium. Potassium is mostly intracellular, so dangerous plasma changes can arise from redistribution without equivalent change in total body content.

Body-fluid compartments

Total body water varies with age, sex, and body composition. Intracellular fluid contains most body water. Extracellular fluid includes plasma and interstitial fluid, with smaller transcellular compartments. Cell membranes are highly water-permeable, so effective osmoles determine water distribution between intracellular and extracellular spaces.

Sodium and accompanying anions dominate extracellular osmolality. Potassium salts dominate intracellular fluid. The sodium-potassium pump maintains these gradients. Water moves until effective osmotic pressure equilibrates, so a change in extracellular tonicity alters cell volume.

Osmolality counts particles per mass of water; osmolarity uses solution volume. Clinically they are often similar. Measured osmolality can be compared with calculated values to identify an osmolar gap from unmeasured solutes, but formula, laboratory method, and timing matter.

Tonicity refers to effective osmoles that do not freely cross cell membranes and therefore alter cell volume. Urea contributes to measured osmolality but crosses many membranes and is less effective for sustained tonicity. Hyperglycaemia raises extracellular tonicity, drawing water out of cells and lowering measured sodium through dilution.

Water balance and antidiuretic hormone

Osmoreceptors respond to increased plasma tonicity by stimulating thirst and antidiuretic hormone. Antidiuretic hormone increases collecting-duct water permeability, producing concentrated urine. Reduced effective circulating volume also stimulates antidiuretic hormone through baroreceptor pathways, even when tonicity is low. Volume defence can therefore override osmotic regulation.

Dilute urine requires suppression of antidiuretic hormone, adequate filtration, intact diluting segments, and sufficient distal delivery. Concentrated urine requires antidiuretic hormone, a medullary gradient, and responsive collecting ducts. Water excretion is limited by the amount of solute available for excretion; very low solute intake can impair the ability to excrete free water.

Hyponatraemia

Hyponatraemia usually means excess water relative to exchangeable sodium and potassium. First determine whether it is hypotonic. Hyperglycaemia causes hypertonic hyponatraemia; severe lipid or protein effects can create laboratory pseudohyponatraemia with selected methods.

Hypotonic hyponatraemia is assessed by symptoms, duration, volume state, urine osmolality, urine sodium, kidney function, endocrine context, drugs, and fluid intake. Hypovolaemic causes include gastrointestinal or renal sodium loss. Euvolaemic patterns include inappropriate antidiuretic hormone effect, glucocorticoid deficiency, hypothyroidism, low solute intake, and excess water. Hypervolaemic patterns occur in heart failure, cirrhosis, kidney failure, and nephrotic states.

Acute severe hyponatraemia can cause cerebral oedema, headache, vomiting, confusion, seizures, and coma because water enters brain cells. Chronic hyponatraemia permits brain osmolyte adaptation, reducing swelling but creating vulnerability to osmotic demyelination if corrected too rapidly. Treatment is guided by neurological severity, duration, mechanism, and trajectory. Hypertonic saline is used for severe symptoms with closely controlled correction; cause-specific fluid, solute, hormone, or drug management follows.

Hypernatraemia

Hypernatraemia means water deficit relative to body sodium and is nearly always hypertonic. It results from inadequate access or thirst, impaired concentration, gastrointestinal or skin water loss, osmotic diuresis, or rarely excessive sodium gain. Neurological impairment, infancy, frailty, and dependence increase risk.

Diabetes insipidus causes excessive dilute urine through deficient antidiuretic hormone production or renal resistance. Water deprivation and response testing require specialist supervision because severe hypernatraemia can develop. Hypernatraemia is corrected with free water while addressing volume, ongoing losses, and cause. Chronic cases require gradual correction to reduce cerebral-oedema risk.

Effective circulating volume and oedema

The body responds to effective arterial filling rather than total extracellular volume alone. Heart failure and cirrhosis can produce oedema while kidneys sense underfilling, activating sympathetic tone, renin-angiotensin-aldosterone, antidiuretic hormone, and thirst. Additional sodium and water then worsen oedema and dilutional hyponatraemia.

Clinical volume assessment integrates history, weight change, postural pressure, pulse, jugular venous pressure, oedema, lung findings, mucosa, urine output, laboratory trends, and ultrasound when useful. No single sign is definitive. Venous congestion itself can reduce kidney function.

Potassium distribution

Most potassium resides inside cells. The sodium-potassium pump moves potassium inward. Insulin and beta two receptor stimulation promote cellular uptake. Acidosis can shift potassium outward depending on acid type and transport. Cell lysis, intense exercise, hyperosmolality, and low insulin raise extracellular potassium. These shifts can change plasma potassium rapidly without changing total content.

The kidneys regulate long-term balance. Potassium is filtered, largely reabsorbed proximally and in the loop, then variably secreted distally. Distal secretion rises with aldosterone, plasma potassium, distal sodium delivery,

flow, and a negative luminal voltage. Low distal delivery, kidney failure, low aldosterone, or epithelial sodium-channel blockade reduces secretion.

Hypokalaemia

Hypokalaemia results from low intake with other factors, gastrointestinal loss, renal loss, or cellular shift. Diuretics, mineralocorticoid excess, vomiting with secondary renal loss, diarrhoea, magnesium deficiency, insulin, and beta agonists are common mechanisms. Urine potassium and acid-base status help distinguish causes.

Effects include weakness, ileus, cramps, impaired concentration ability, glucose intolerance, and arrhythmia. Electrocardiographic changes can include flattened T waves, S T depression, U waves, and prolonged repolarisation. Risk is greater with heart disease, digoxin, magnesium deficiency, and rapid change.

Replacement route and rate depend on severity, symptoms, electrocardiogram, ongoing loss, kidney function, and ability to take oral therapy. Magnesium deficiency makes potassium difficult to correct and increases arrhythmic risk.

Hyperkalaemia

Hyperkalaemia can result from sample haemolysis, cellular release, shift, reduced renal excretion, or excessive intake in a susceptible patient. Repeat testing may be appropriate for an unexpected result without risk, but electrocardiographic or clinical danger requires immediate action.

Electrocardiographic progression can include peaked T waves, P-wave loss, conduction slowing, Q R S widening, sine-wave pattern, ventricular arrhythmia, and arrest, but severe hyperkalaemia can occur without classic changes. Immediate therapy stabilises cardiac membranes with intravenous calcium when indicated, shifts potassium into cells with insulin and glucose and selected adjuncts, and removes potassium through kidneys, gastrointestinal binders, or dialysis. Stop contributing drugs and treat the cause.

Calcium, magnesium, and phosphate

Although sodium, water, and potassium dominate acute fluid management, calcium, magnesium, and phosphate strongly affect neuromuscular, cardiac, skeletal, and metabolic function. Ionised calcium is biologically active; total calcium changes with albumin and pH. Alkalosis increases protein binding and can lower ionised calcium enough to cause paraesthesia, tetany, or arrhythmia despite a less dramatic total value.

Hypocalcaemia follows vitamin D deficiency, hypoparathyroidism, kidney disease, pancreatitis, massive transfusion, and severe magnesium deficiency. Hypercalcaemia commonly reflects primary hyperparathyroidism or malignancy, with other endocrine, granulomatous, drug, and immobilisation causes. Severe hypercalcaemia causes dehydration, kidney injury, constipation, confusion, and arrhythmia; treatment combines volume restoration when appropriate, cause-specific therapy, and monitoring.

Magnesium depletion occurs with gastrointestinal loss, diuretics, alcohol use, malnutrition, and tubular injury. It causes weakness, tremor, seizures, refractory hypokalaemia, hypocalcaemia, and torsades risk. Excess magnesium occurs mainly with kidney failure and magnesium administration, causing reduced reflexes, hypotension, bradycardia, and respiratory weakness.

Phosphate depletion can impair adenosine triphosphate, muscle function, red-cell oxygen handling, and bone. Refeeding drives phosphate into cells and can cause respiratory or cardiac failure. Hyperphosphataemia occurs with kidney failure, cell breakdown, and phosphate load and promotes hypocalcaemia and tissue calcification.

Fluid prescription and monitoring

Maintenance fluid replaces expected water and electrolyte needs; replacement fluid matches abnormal ongoing losses; resuscitation fluid addresses impaired perfusion. Prescription accounts for oral intake, fever, drains, urine, gastrointestinal loss, heart and kidney function, body size, and laboratory trends. Daily weight, cumulative balance, examination, sodium, potassium, creatinine, and urine output guide revision. A positive charted balance can be inaccurate, but should prompt examination rather than dismissal.

TTS module 2: Effective osmoles, renal responses, and safe electrolyte correction

Separate concentration from content

Plasma sodium is a ratio between exchangeable body sodium and potassium and total body water. It does not directly report total sodium stores. Extracellular volume, by contrast, is strongly influenced by total sodium content. This distinction explains why saline can improve hypovolaemia without predictably correcting every hyponatraemia, and why an oedematous patient can have low plasma sodium despite excess body sodium.

Water moves between extracellular and intracellular spaces according to effective osmoles. Adding isotonic sodium expands extracellular volume with little sustained cell-volume change. Adding free water distributes across both compartments and lowers tonicity. Losing hypotonic fluid can reduce volume while raising sodium. The composition of input and loss determines the resulting pattern.

Translocational hyponatraemia occurs when an extracellular effective osmole, commonly glucose, draws water from cells and dilutes sodium. Correcting glucose reverses the water shift and raises measured sodium.

Pseudohyponatraemia is a measurement artefact in selected indirect laboratory methods when plasma water fraction is reduced by extreme lipid or protein concentration; tonicity remains normal.

Reading urine osmolality and sodium

Urine osmolality reveals whether antidiuretic hormone effect is appropriately suppressed. In hypotonic hyponatraemia, maximally dilute urine suggests excess intake relative to excretory capacity or very low solute. Concentrated urine shows that water is being retained, whether from appropriate volume defence, nausea, pain, cortisol deficiency, drugs, or inappropriate hormone activity.

Urine sodium helps infer renal sodium avidity, but diuretics, kidney disease, recent saline, adrenal disorders, and timing complicate interpretation. Low urine sodium can reflect effective arterial underfilling from true volume loss, heart failure, or cirrhosis. Higher values can reflect renal salt loss, diuretics, adrenal deficiency, or a euvolaemic antidiuretic-hormone syndrome.

Urine studies should be paired with serum samples and collected before major treatment when feasible. A later sample may describe the treatment response rather than the original disease. Serial changes can nevertheless be useful: falling urine osmolality after volume restoration supports removal of a non-osmotic hormone stimulus.

Hyponatraemia and brain adaptation

Acute hypotonicity drives water into brain cells. The rigid skull limits expansion, creating headache, vomiting, confusion, seizures, respiratory arrest, and herniation in severe cases. Brain cells first lose electrolytes and then organic osmolytes to reduce swelling. This adaptation develops over time and lowers immediate oedema risk.

The adapted brain becomes vulnerable when extracellular tonicity rises rapidly. Water leaves cells before osmolytes can be restored, and oligodendrocytes or myelin can be injured. Risk is especially concerning with very low chronic sodium, malnutrition, alcohol-use disorder, liver disease, hypokalaemia, and other severe illness.

Severe neurological symptoms require controlled hypertonic therapy aimed at a modest prompt rise, followed by careful limitation of total correction. Sodium can accelerate unexpectedly when antidiuretic hormone switches off and a water diuresis begins. Frequent measurements, urine monitoring, and strategies to slow or relower correction belong in protocol-driven care.

Potassium replacement can also raise plasma sodium because potassium is an exchangeable body cation and intracellular uptake changes water distribution. Correction calculations are estimates, while ongoing urine and gastrointestinal losses change the result. Treatment must follow observed trajectory rather than formula alone.

Hypernatraemia and water loss

Hypernatraemia usually indicates inadequate water relative to solute. Thirst and access normally prevent it, so severe cases often signal impaired consciousness, dependence, infancy, frailty, hypothalamic disease, or inability to communicate. The water deficit estimate is a starting point and does not include ongoing loss.

Urine concentration distinguishes appropriate renal conservation from diabetes insipidus or osmotic diuresis. Very concentrated urine points toward extrarenal water loss or prior sodium gain. Inappropriately dilute urine suggests reduced antidiuretic hormone effect or renal resistance. Intermediate values occur with partial disease, kidney dysfunction, diuretics, or solute diuresis.

Central diabetes insipidus lacks adequate hormone production, while nephrogenic disease lacks renal response. Causes include neurosurgery, trauma, tumours, infiltrative disease, genetic disorders, lithium, high calcium, low potassium, and tubulointerstitial injury. Treatment must address cause, volume, electrolyte state, and safe access to water.

Chronic hypertonicity drives brain cells to accumulate osmolytes. Rapid free-water administration can then cause cerebral oedema. Correct volume-threatening hypoperfusion with suitable isotonic resuscitation first when necessary, then replace free water according to duration, loss, and monitored sodium trajectory.

Potassium is governed by distribution and excretion

Only a small fraction of body potassium is extracellular, so shifts can rapidly alter plasma concentration. Insulin activates cellular uptake. Beta two stimulation increases pump activity. Hypertonicity draws water and potassium from cells. Tissue breakdown releases intracellular stores. Acidosis effects

depend on the accompanying anion and transport response rather than one universal hydrogen-potassium exchange.

Kidney secretion occurs mainly through principal cells and increases when potassium is high, aldosterone is active, distal sodium reaches epithelial channels, and tubular flow is adequate. Non-reabsorbable anions can increase potassium loss. Low effective volume may reduce distal delivery despite high aldosterone, limiting secretion.

Acid-base pattern and blood pressure help organise renal potassium disorders. Hypokalaemic metabolic alkalosis with hypertension suggests mineralocorticoid or channel-driven sodium retention. The same acid-base pattern without hypertension suggests vomiting, diuretics, or inherited salt-wasting physiology. Hyperkalaemic normal-gap acidosis suggests reduced aldosterone effect or distal acid-secretory dysfunction.

Urine potassium should be interpreted with urine concentration and intake. A low absolute concentration in very high urine volume can still represent substantial loss. A spot ratio or timed excretion may help, but no index replaces clinical context and treatment history.

Emergency potassium treatment

Hyperkalaemia kills through impaired cardiac conduction and excitability. Intravenous calcium stabilises the myocardial membrane but does not lower potassium. Insulin with glucose shifts potassium into cells, while beta agonists provide an additional shift in selected patients. Bicarbonate is most useful when significant metabolic acidosis contributes and should not delay proven emergency measures.

Definitive removal occurs through urinary excretion, gastrointestinal binding, or dialysis. Diuretics require kidney function and adequate flow. Binders differ in onset and suitability and do not replace urgent stabilisation. Dialysis provides reliable removal when kidney failure, severe refractory elevation, or ongoing release overwhelms other methods.

Glucose must be monitored after insulin because delayed hypoglycaemia can occur, especially with kidney dysfunction. Potassium can rebound as shifting treatments wear off if removal has not occurred. Repeat electrocardiography and laboratory measurement close the loop.

Hypokalaemia replacement is limited by route, concentration, rate, kidney function, and redistribution. Oral replacement is safer when feasible. Intravenous delivery requires controlled dilution and monitoring. Correct magnesium because deficiency increases renal potassium loss and membrane instability. Ongoing diarrhoea, diuretics, insulin, or alkalosis can consume replacement.

Calcium, magnesium, phosphate, and refeeding

Ionised calcium changes with pH because hydrogen and calcium compete for albumin binding. Acute alkalosis increases binding and lowers ionised concentration, producing paraesthesia or tetany without a major total-calcium change. Direct ionised measurement is useful in critical illness, massive transfusion, and rapid pH shifts.

Magnesium regulates potassium channels, parathyroid function, and cardiac repolarisation. Severe deficiency can make hypokalaemia and hypocalcaemia refractory. Replacement requires caution in kidney failure because magnesium accumulates and suppresses reflexes, pressure, conduction, and respiration.

Refeeding after prolonged undernutrition raises insulin and drives phosphate, potassium, and magnesium into cells while increasing thiamine demand and sodium-water retention. Consequences include respiratory weakness, heart failure, arrhythmia, neurological injury, and haemolysis. Prevention uses risk recognition, cautious energy introduction, thiamine, electrolyte replacement, and close monitoring.

Prescribing fluid as composition and trajectory

Resuscitation treats perfusion, maintenance supplies expected needs, and replacement matches abnormal losses. A single bag cannot serve all three purposes by default. Gastrointestinal fluid, high-output stoma, urine, drains, fever, and ventilation each remove different proportions of water and electrolytes.

Daily weight often detects net fluid change more reliably than an incomplete balance chart. Examination should look for both arterial underfilling and venous congestion. Rising creatinine can follow either. Cumulative sodium

exposure from fluids and medicines matters even when charted water balance appears acceptable.

The safe electrolyte plan states the mechanism, symptom urgency, target direction, maximum correction pace, chosen fluid or drug, monitoring interval, and response to overshoot. Concentration values must be managed as dynamic consequences of total content, distribution, kidney function, and ongoing loss.

Retrieval prompts

One. Distinguish sodium content, plasma sodium, osmolality, and tonicity.

Two. Why can heart failure cause oedema and hyponatraemia simultaneously?

Three. What determines dilute versus concentrated urine?

Four. Why is chronic hyponatraemia corrected cautiously?

Five. What shifts potassium into cells?

Six. State the three immediate goals in dangerous hyperkalaemia.

Concise answers

One. Sodium content governs extracellular volume; plasma sodium reflects water relative to exchangeable cations; osmolality counts particles; tonicity reflects effective osmoles and cell volume.

Two. Reduced effective arterial filling activates sodium and water retention, while antidiuretic hormone retains proportionally more water.

Three. Antidiuretic-hormone level, distal delivery, intact tubular segments, medullary gradient, and solute excretion.

Four. Brain adaptation reduces swelling but rapid correction can cause osmotic demyelination.

Five. Insulin, beta two stimulation, and sodium-potassium pump activity.

Six. Stabilise the myocardium, shift potassium intracellularly, and remove it from the body.

Source map

Original synthesis informed by Guyton and Hall, body fluids, sodium, potassium, and volume regulation; OpenStax Anatomy and Physiology 2e; Robbins, oedema and kidney disease; Katzung and OpenStax Pharmacology, diuretics and electrolyte-altering drugs; and OpenStax Medical-Surgical Nursing.

Chapter 19: Acid-Base Physiology and Clinical Interpretation

Orientation

Hydrogen ion concentration influences protein structure, enzyme activity, membrane excitability, vascular tone, potassium distribution, and oxygen binding. The body controls pH through immediate chemical buffers, respiratory carbon dioxide elimination, and renal handling of bicarbonate and non-volatile acid. Acid-base disorders are processes, not merely abnormal numbers; several can coexist and partially conceal one another.

The bicarbonate-carbon dioxide system

Carbon dioxide combines with water to form carbonic acid, which dissociates into hydrogen and bicarbonate. Carbonic anhydrase accelerates this reaction in red cells, kidney, and other tissues. The Henderson-Hasselbalch relationship expresses pH as a function of bicarbonate relative to dissolved carbon dioxide. Lungs regulate the volatile acid component within minutes; kidneys regulate bicarbonate and fixed acid over hours to days.

Buffers minimise pH change by accepting or donating hydrogen. Bicarbonate is important in extracellular fluid because it is abundant and open to regulation by lungs and kidneys. Haemoglobin, plasma proteins, phosphate, and intracellular proteins also buffer. Bone contributes during chronic acid loads at the cost of mineral balance.

Metabolism produces large quantities of carbon dioxide and smaller quantities of non-volatile acids from proteins, phosphates, and incomplete organic-acid metabolism. Healthy lungs excrete carbon dioxide continuously. Kidneys must excrete daily fixed acid while reclaiming virtually all filtered bicarbonate and generating replacement bicarbonate.

Renal acid handling

The proximal tubule reclaims filtered bicarbonate through hydrogen secretion and carbonic-anhydrase reactions. This recovers existing buffer. New bicarbonate is generated when hydrogen is excreted with urinary buffers rather than returned to blood with bicarbonate loss.

Phosphate accepts secreted hydrogen as titratable acid. Ammonium is quantitatively adaptable. Proximal cells metabolise glutamine to ammonium and bicarbonate. Ammonia diffuses into the collecting system, binds secreted hydrogen, and becomes trapped as ammonium. Acidosis increases ammoniogenesis. Kidney failure reduces acid excretion and bicarbonate regeneration.

The minimum urine pH limits free hydrogen excretion, so buffers are essential. Urine pH alone does not measure total acid excretion: a patient can have acidic urine yet inadequate ammonium production.

A systematic interpretation

First assess pH: acidemia is low blood pH and alkalemia high blood pH. Second identify whether carbon dioxide or bicarbonate changes in the direction explaining the pH. Third test whether compensation is appropriate. Fourth calculate anion gap when metabolic acidosis exists. Fifth compare delta changes for mixed metabolic disorders. Sixth integrate electrolytes, albumin, lactate, ketones, renal function, chloride, history, and treatment.

Compensation limits pH change but does not overcorrect beyond normal. If measured compensation differs substantially from expected, another primary disorder is present. Approximate rules are clinical tools, not physical laws, and depend on acute versus chronic timing.

Metabolic acidosis

Metabolic acidosis is primary bicarbonate reduction. The lungs compensate by increasing ventilation and lowering carbon dioxide. In severe acidosis, deep rapid Kussmaul breathing reflects respiratory compensation rather than primary lung disease.

The anion gap equals sodium minus chloride and bicarbonate, with potassium commonly omitted. It estimates unmeasured anions. Albumin is a major unmeasured anion, so low albumin lowers the expected gap and can conceal pathology. A high gap results from lactate, ketones, kidney-retained acids, toxic alcohol metabolites, salicylate-related processes, and other organic acids.

Normal-gap acidosis usually reflects bicarbonate loss or impaired renal acid excretion with chloride retention. Causes include diarrhoea, pancreatic or intestinal loss, renal tubular acidosis, early kidney disease, and chloride-rich fluid administration.

Lactic acidosis arises when production exceeds clearance. Causes include tissue hypoperfusion, severe hypoxaemia, seizures, adrenergic stimulation, liver dysfunction, mitochondrial toxins, drugs, malignancy, and thiamine deficiency. Treating the lactate number without identifying mechanism is inadequate.

Ketoacidosis occurs in diabetes, alcohol-associated states, starvation, and selected drug contexts. Insulin deficiency and counter-regulatory hormones drive lipolysis and hepatic ketone production. Treatment differs by cause but often addresses volume, insulin when deficient, electrolytes, carbohydrate, thiamine, and precipitating illness.

Metabolic alkalosis

Metabolic alkalosis is primary bicarbonate elevation. Generation can follow hydrogen loss from vomiting or gastric drainage, mineralocorticoid activity, diuretics, alkali load, or intracellular hydrogen shift. Persistence requires impaired renal bicarbonate excretion, commonly from volume and chloride depletion, reduced filtration, potassium deficiency, or ongoing mineralocorticoid effect.

Urine chloride helps separate chloride-responsive states, such as remote vomiting or diuretic effect, from chloride-resistant states with ongoing mineralocorticoid activity, though active diuretics and kidney disease complicate interpretation. Severe alkalemia reduces cerebral and coronary flow, lowers ionised calcium, shifts potassium intracellularly, and promotes arrhythmia.

Treatment replaces chloride and volume when depleted, corrects potassium and magnesium, stops the cause, reduces mineralocorticoid effect when present, and occasionally uses carbonic-anhydrase inhibition or dialysis. Giving saline indiscriminately is harmful in volume-overloaded alkalosis.

Respiratory acidosis

Respiratory acidosis is primary carbon dioxide retention from inadequate alveolar ventilation. Causes include sedatives, central neurological disease, neuromuscular weakness, chest-wall restriction, severe obesity, airway obstruction, and ventilatory fatigue. Acute retention causes marked pH fall with limited bicarbonate rise. Over days, kidneys retain bicarbonate and excrete acid, producing chronic compensation.

An acute rise above a chronic baseline creates acute-on-chronic respiratory acidosis. A near-normal pH can therefore hide severe chronic hypercapnia. Treatment restores ventilation and addresses cause while avoiding abrupt inappropriate correction in selected chronically adapted patients.

Respiratory alkalosis

Respiratory alkalosis is primary carbon dioxide reduction from ventilation exceeding metabolic production. Causes include hypoxaemia, pulmonary embolism, sepsis, fever, pain, anxiety, pregnancy, liver disease, central lesions, and salicylate toxicity. Acute reduction causes modest bicarbonate fall through buffering; chronic reduction produces greater renal bicarbonate loss.

Do not label unexplained hyperventilation as anxiety before excluding dangerous causes. In metabolic acidosis, low carbon dioxide is expected compensation, not a separate respiratory alkalosis unless it falls more than predicted.

Mixed disorders and delta analysis

A patient with sepsis can have lactic acidosis and respiratory alkalosis. Vomiting can add metabolic alkalosis to ketoacidosis. Chronic lung disease can combine respiratory acidosis with diuretic-related metabolic alkalosis. Kidney failure can combine high-gap and normal-gap acidosis.

In high-gap metabolic acidosis, compare the rise in gap with the fall in bicarbonate. A larger bicarbonate fall suggests an additional normal-gap acidosis. A smaller fall suggests concurrent metabolic alkalosis or pre-existing high bicarbonate. This delta approach is approximate and should be corrected for albumin and baseline when known.

Toxicological patterns

Salicylate toxicity classically produces respiratory alkalosis from central stimulation together with high-gap metabolic acidosis; a near-normal pH can therefore conceal severe poisoning. Clinical features include nausea, tinnitus, tachypnoea, fever, confusion, pulmonary oedema, hypoglycaemia in the central nervous system, and deterioration after acidemia increases tissue entry. Management uses repeated concentrations, glucose and electrolytes, serum and urine alkalinisation, and dialysis for severe features.

Methanol and ethylene glycol generate toxic organic acids after metabolism. Early osmolar gap can fall as anion gap rises, so neither excludes late poisoning. Visual toxicity suggests methanol; kidney injury and calcium oxalate can accompany ethylene glycol. Prompt alcohol-dehydrogenase inhibition, cofactor therapy, supportive care, and dialysis are time-critical. Propylene glycol, pyroglutamic acidosis, toluene, and other exposures produce distinct mixed patterns.

Physiological contexts

Pregnancy normally lowers carbon dioxide through progesterone-stimulated ventilation, with renal bicarbonate reduction and a slightly higher pH. Values normal for a non-pregnant adult can therefore indicate respiratory impairment in pregnancy. Fever and sepsis increase acid production and ventilatory demand. After cardiac arrest, mixed lactic and respiratory acidosis is common; improving circulation and ventilation matters more than chasing pH alone.

Albumin, phosphate, chloride, and unmeasured ions influence acid-base interpretation. The physicochemical approach describes pH through carbon dioxide, strong-ion difference, and weak acids. It can clarify complex critical illness, but should complement rather than obscure bedside recognition of perfusion, ventilation, kidney function, fluid composition, and toxins.

Potassium and treatment safety

Acid-base and potassium interact, but simple rules have exceptions. Mineral acidosis tends to shift potassium out of cells; organic acids may behave differently. Insulin treatment of ketoacidosis shifts potassium into cells even when total body potassium is depleted. Alkalosis promotes hypokalaemia and increases renal potassium loss.

Bicarbonate therapy is indicated in selected severe or specific disorders, not every acidosis. It adds sodium, generates carbon dioxide, can lower ionised calcium, shift potassium, expand volume, and overshoot. Definitive treatment corrects perfusion, ventilation, insulin deficiency, toxin, kidney failure, gastrointestinal loss, or hormonal disturbance.

TTS module 2: Quantitative compensation, hidden mixtures, and acid-base treatment traps

Start with process, then use equations

An acid-base result is a snapshot of several processes acting at different speeds. Carbon dioxide changes within minutes through ventilation. Extracellular buffers respond immediately. Renal bicarbonate and ammonium handling changes over hours to days. Fluid, chloride, potassium, albumin, and treatment modify the pattern. Timing is therefore part of diagnosis.

The Henderson-Hasselbalch relationship shows that pH follows the ratio of bicarbonate to dissolved carbon dioxide. A fall in either component alone cannot be labelled without examining the other. Low bicarbonate can reflect metabolic acidosis or renal compensation for respiratory alkalosis. High bicarbonate can reflect metabolic alkalosis or compensation for respiratory acidosis.

Compensation rules estimate what the other component should be after one primary disturbance. They are ranges, not exact targets. A measured value outside the expected range indicates another primary process. A value within range supports appropriate compensation but does not identify the underlying cause.

Respiratory compensation in metabolic disease

In metabolic acidosis, ventilation rises rapidly. A useful expected-carbon-dioxide estimate is approximately one and a half times bicarbonate plus eight, with a small acceptable range. A higher measured carbon dioxide indicates concurrent ventilatory failure. A lower value indicates an additional respiratory alkalosis.

This distinction is clinically urgent. A patient with severe metabolic acidosis must sustain high minute ventilation. Sedation, fatigue, chest disease, or neuromuscular weakness can remove that compensation and cause a sudden pH fall. Intubation is hazardous if post-intubation ventilation fails to match the previous compensatory demand.

In metabolic alkalosis, hypoventilation raises carbon dioxide, but compensation is limited by the need to maintain oxygenation. A very high carbon dioxide should not be assumed to be appropriate without considering chronic lung disease, sedation, obesity hypoventilation, or muscle weakness.

Acute and chronic respiratory change

Acute carbon dioxide retention produces only a small bicarbonate rise from buffering. Chronic retention produces a larger rise through renal acid excretion and bicarbonate generation. A carbon dioxide increase with less bicarbonate than expected suggests a concurrent metabolic acidosis. More bicarbonate suggests metabolic alkalosis or a higher previous baseline.

The same logic applies to respiratory alkalosis. Acute reduction lowers bicarbonate modestly. Sustained reduction allows kidney bicarbonate loss. Pregnancy provides a physiological example of chronic mild respiratory alkalosis, while sepsis can add acute hyperventilation and lactic acidosis.

After correcting chronic hypercapnia rapidly with ventilation, retained bicarbonate may remain temporarily high, producing post-hypercapnic metabolic alkalosis. Volume depletion, chloride deficiency, potassium loss, and diuretics can maintain it. Abruptly treating the bicarbonate number without understanding this sequence can destabilise ventilation and electrolytes.

Correcting and dissecting the anion gap

The anion gap represents unmeasured negatively charged molecules. Albumin contributes substantially, so the expected normal gap falls when albumin is low. Correcting approximately for albumin can reveal hidden

organic acid accumulation in critical illness. Laboratory reference ranges differ, and local methods should be used.

The delta gap compares the increase in anion gap with the decrease in bicarbonate from baseline. If bicarbonate has fallen much more than the gap rose, a concurrent normal-gap acidosis is likely. If it has fallen less, a concurrent metabolic alkalosis or pre-existing high bicarbonate is likely. Baseline uncertainty and fluid treatment make this an interpretive aid rather than proof.

The gap can change while illness continues. Chloride-rich resuscitation may replace an organic-gap acidosis with hyperchloraemic acidosis as lactate or ketones clear. The pH may remain low even though perfusion improves. Serial lactate, ketones, chloride, bicarbonate, kidney function, and clinical state prevent misreading this transition as treatment failure.

Normal-gap acidosis and urinary response

Normal-gap metabolic acidosis usually reflects bicarbonate loss or impaired renal acid excretion with chloride replacement. Diarrhoea loses gastrointestinal bicarbonate. Proximal renal tubular acidosis loses filtered bicarbonate until plasma concentration falls enough that the reduced load can be reclaimed. Distal disease fails to acidify urine adequately. Reduced aldosterone impairs distal hydrogen and potassium handling.

Urinary ammonium is the major adaptive acid excretion, but it is not routinely measured everywhere. The urine anion gap uses sodium plus potassium minus chloride as an indirect estimate. A negative value suggests high ammonium chloride excretion and an appropriate renal response, as in diarrhoea. A positive value suggests reduced ammonium, but unusual urinary anions, low sodium delivery, kidney disease, and measurement conditions limit the inference.

Urine pH alone is insufficient. It may be low even when total ammonium excretion is inadequate. Bacterial urease, diet, specimen delay, volume, and alkali therapy alter it. Interpret pH alongside plasma potassium, kidney function, urine electrolytes, stones, medicines, and autoimmune disease.

Metabolic alkalosis as generation plus maintenance

Vomiting removes hydrogen and chloride, but the kidney would normally excrete excess bicarbonate. Alkalosis persists when volume depletion activates sodium reclamation, chloride is unavailable for bicarbonate exchange, filtration falls, potassium is low, or mineralocorticoid signalling drives distal hydrogen secretion.

Urine chloride is low when the kidney avidly conserves chloride after remote gastric or diuretic loss. It remains high with ongoing diuretics, mineralocorticoid excess, and selected tubulopathies. A single value can be misleading soon after medication, and blood pressure helps separate volume-depleted from sodium-retaining states.

Potassium depletion sustains alkalosis through intracellular shifts, increased proximal bicarbonate reclamation, ammonium production, and distal hydrogen secretion. Correcting chloride without potassium may be incomplete. Magnesium depletion can make potassium replacement ineffective.

Lactate, ketones, and toxins

Lactate is an endpoint shared by several mechanisms. Low oxygen delivery, regional ischaemia, seizures, beta-adrenergic stimulation, mitochondrial inhibition, impaired hepatic clearance, malignancy, and thiamine deficiency can all raise it. A falling value can indicate improved balance, but normalisation is not required before every other clinical sign improves.

Ketoacidosis contains multiple acid-base changes. Osmotic diuresis causes volume, sodium, potassium, phosphate, and chloride loss. Vomiting adds alkalosis. Hyperventilation adds expected compensation or an additional respiratory process. Plasma potassium may be high while total body stores are depleted, then fall rapidly with insulin and volume.

Toxic alcohols evolve from an early osmolar gap toward a later anion gap as parent alcohol is metabolised to acids. A normal osmolar gap late does not exclude poisoning. Salicylate stimulates ventilation while generating organic acids, so pH may appear near normal. Acidification increases tissue penetration and neurological toxicity.

Physicochemical interpretation

The physicochemical approach describes pH through carbon dioxide, strong-ion difference, and weak acids such as albumin and phosphate. Giving

chloride narrows the strong-ion difference and promotes acidosis. Losing chloride or adding sodium without accompanying chloride widens it and promotes alkalosis. Low albumin exerts an alkalinising effect that can offset other acidosis.

This framework is particularly helpful in critical illness with saline, hypoalbuminaemia, lactate, phosphate, and kidney dysfunction. It complements bicarbonate-centred analysis because both describe the same electroneutral system from different perspectives.

Treatment should follow the cause

Bicarbonate can be appropriate for selected severe acid states, bicarbonate loss, toxin management, or kidney failure, but it adds sodium and carbon dioxide, lowers ionised calcium, shifts potassium, and can overshoot. If ventilation cannot eliminate generated carbon dioxide, intracellular or central acidity may worsen despite a higher blood bicarbonate.

Ventilation should restore adequate carbon dioxide removal without erasing chronic adaptation unnecessarily fast. Chloride-responsive alkalosis needs volume and chloride when safe. Mineralocorticoid states need cause-specific treatment. Dialysis removes selected acids or toxins and corrects electrolytes when endogenous kidney capacity is insufficient.

The final interpretation should state every primary process, expected compensation, gap and albumin context, likely cause, and immediate physiological danger. A normal pH should never end the analysis when carbon dioxide and bicarbonate are both markedly abnormal. Serial measurements should confirm that ventilation, perfusion, renal response, and treatment are moving each process toward recovery without creating another disorder.

Retrieval prompts

One. How do lungs and kidneys divide acid-base regulation?

Two. Give the six-step interpretation sequence.

Three. Contrast high-gap and normal-gap metabolic acidosis.

Four. Why does metabolic alkalosis persist after its initial generation?

Five. Contrast acute with chronic respiratory acidosis.

Six. Why can a normal pH conceal severe disease?

Concise answers

One. Lungs regulate carbon dioxide rapidly; kidneys reclaim and generate bicarbonate and excrete fixed acid slowly.

Two. Assess pH, primary process, compensation, anion gap, delta pattern, and clinical context.

Three. High-gap acidosis adds unmeasured anions; normal-gap acidosis loses bicarbonate or impairs acid excretion with chloride retention.

Four. Reduced filtration, chloride or volume depletion, potassium deficiency, or mineralocorticoid activity prevents bicarbonate excretion.

Five. Acute retention has little bicarbonate compensation and greater pH fall; chronic retention has renal bicarbonate elevation.

Six. Opposing or compensated disorders can bring pH toward normal while carbon dioxide, bicarbonate, and underlying illness remain severely abnormal.

Source map

Original synthesis informed by Guyton and Hall, acid-base regulation and respiratory control; OpenStax Anatomy and Physiology 2e; Robbins, shock and kidney disease; Katzung and OpenStax Pharmacology, diuretics, toxins, and bicarbonate; and OpenStax Medical-Surgical Nursing.

Chapter 20: Acute Kidney Injury, Chronic Kidney Disease, and Renal Pharmacology

Orientation

Kidney disease presents through reduced filtration, abnormal urine, disordered volume or electrolytes, endocrine failure, structural findings, or systemic complications. Acute kidney injury develops over hours to days; chronic kidney disease persists for at least months through reduced filtration or markers of damage. Acute injury can occur on chronic disease, and repeated episodes accelerate long-term loss.

Acute kidney injury definition and recognition

Acute kidney injury is identified by rising creatinine, reduced urine output, or both. Creatinine rises after filtration falls and is diluted or concentrated by fluid balance, so early injury may be underestimated. Oliguria is sensitive to haemodynamics but can occur without structural damage; non-oliguric injury can still be severe.

Classify mechanism as reduced perfusion, intrinsic renal injury, or obstruction, while recognising overlap. Review baseline function, blood pressure, volume, sepsis, surgery, contrast, medications, toxins, muscle injury, urinary symptoms, and systemic disease. Examine perfusion, congestion, bladder, rash, joints, and signs of obstruction or immune disease.

Reduced perfusion and haemodynamic injury

Reduced effective renal perfusion follows haemorrhage, dehydration, vasodilation, low cardiac output, or severe congestion. Autoregulation maintains filtration through afferent and efferent tone, but fails when pressure falls beyond range or drugs block compensation.

Non-steroidal anti-inflammatory drugs reduce afferent prostaglandin-mediated dilation. Renin-angiotensin system blockers reduce efferent constriction. Diuretics lower volume. Each can be beneficial chronically yet contribute

during acute hypoperfusion. A modest creatinine change after haemodynamic therapy may reflect altered filtration rather than structural toxicity, but large or progressive change requires evaluation.

Prolonged ischaemia causes tubular injury. Sepsis produces complex microvascular, inflammatory, and metabolic dysfunction rather than simple whole-kidney underperfusion. Venous congestion raises interstitial pressure and reduces filtration in heart failure.

Intrinsic renal disease

Acute tubular injury follows ischaemia, sepsis, pigments, or nephrotoxins. Tubular cells lose polarity and transport, detach, obstruct lumens, and trigger inflammation. Urine may show granular casts. Recovery requires survival or regeneration and can pass through a high-output phase with electrolyte loss.

Acute interstitial nephritis is often drug-related but also follows infection or immune disease. Fever, rash, and eosinophilia are classic but frequently absent. Urine can show white cells and casts. Diagnosis sometimes requires biopsy, and treatment includes stopping the cause with selected immunosuppression.

Glomerulonephritis produces haematuria, proteinuria, hypertension, oedema, and reduced filtration in varying combinations. Rapidly progressive disease can destroy function over days to weeks and requires urgent serology, microscopy, biopsy, and cause-specific immune treatment. Thrombotic microangiopathy causes endothelial injury, platelet consumption, microangiopathic haemolysis, and organ damage.

Pigment nephropathy follows myoglobin from muscle breakdown or haemoglobin from intravascular haemolysis. Rhabdomyolysis can cause high creatine kinase, hyperkalaemia, hypocalcaemia, acidosis, and compartment syndrome. Early volume and cause control are important while avoiding overload.

Obstruction

Obstruction can occur at urethra, bladder outlet, ureters, or renal pelvis. Causes include prostate disease, stones, tumours, clots, strictures, neurogenic bladder, retroperitoneal fibrosis, and blocked devices. Bilateral obstruction or obstruction of a solitary functioning kidney reduces overall filtration, but unilateral obstruction may preserve creatinine.

Bladder scanning and renal imaging support detection, but early obstruction may lack dilation and chronic dilation may persist after relief. Decompression can cause post-obstructive diuresis with water, sodium, potassium, and magnesium loss, requiring monitoring and partial replacement.

Immediate management and dialysis

Stop avoidable nephrotoxins, adjust medicine doses, restore appropriate perfusion, treat infection, relieve obstruction, manage volume and electrolytes, and monitor urine, weight, creatinine, potassium, bicarbonate, and symptoms. Do not use diuretics to convert kidney injury into a better laboratory category; use them for volume management when responsive.

Kidney replacement therapy is considered for refractory hyperkalaemia, severe acidemia, pulmonary oedema or volume overload, selected toxins, and uraemic complications such as pericarditis or encephalopathy. Creatinine alone does not determine timing. Intermittent haemodialysis removes solute rapidly; continuous therapy provides slower fluid and solute control for selected unstable patients. Peritoneal dialysis uses the peritoneal membrane.

Chronic kidney disease

Chronic kidney disease is staged by filtration category and albuminuria because both predict progression and cardiovascular risk. Causes include diabetes, hypertension, glomerular disease, inherited disorders, reflux, obstruction, interstitial disease, and repeated acute injury. Determine chronicity from previous results, imaging, anaemia, mineral-bone changes, and clinical history.

Remaining nephrons hyperfilter and hypertrophy, maintaining total filtration initially but increasing intraglomerular stress. Proteinuria and inflammation promote tubulointerstitial fibrosis. Progression varies and can be slowed

through pressure control, reduction of albuminuria, diabetes management, renin-angiotensin blockade when indicated, sodium-glucose cotransporter inhibition in eligible patients, avoidance of injury, smoking cessation, and treatment of cause.

Systemic complications

Sodium retention causes hypertension and oedema. Potassium and acid accumulate. Reduced erythropoietin and iron restriction cause anaemia. Phosphate retention, low active vitamin D, hypocalcaemia, and secondary hyperparathyroidism alter bone and vessels. Uraemic toxins contribute to anorexia, pruritus, neuropathy, platelet dysfunction, pericarditis, cognitive change, and immune impairment.

Cardiovascular risk is high from shared risk factors, vascular calcification, inflammation, anaemia, pressure and volume load, and myocardial remodelling. Nutrition requires balancing protein-energy needs with sodium, potassium, phosphate, and fluid issues without causing malnutrition.

Renal pharmacology and prescribing

Drug dosing depends on indication, filtration estimate, acute stability, dialysis, protein binding, volume of distribution, active metabolites, and therapeutic window. A filtration-adjusted dose may reduce amount, extend interval, or both. Loading doses often still depend on distribution and may need adjustment in oedema or critical illness.

Nephrotoxicity mechanisms include haemodynamic change, tubular toxicity, crystal precipitation, interstitial inflammation, microangiopathy, and obstruction. Risk rises with cumulative exposure, combinations, dehydration, age, and pre-existing disease. Monitor the specific expected injury rather than ordering undirected tests.

Dialysis removes small water-soluble, poorly protein-bound drugs with low distribution volume more effectively. Timing and supplemental post-dialysis doses vary. Prescribing references and pharmacy support are essential because modality and residual function change.

Kidney history, examination, and investigation

Ask about urine volume, colour, blood, foam, pain, dysuria, obstruction, nocturia, swelling, weight, breathlessness, systemic symptoms, pregnancy, family disease, stones, infections, and medicines. Examine pressure, volume, bladder, skin, joints, eyes, neuropathy, and vascular access.

Investigation integrates blood trends, urinalysis and microscopy, albumin or protein quantification, ultrasound, immune and paraprotein tests, microbiology, and biopsy when histology will change treatment. Referral urgency rises with rapid loss, active sediment, heavy protein, resistant pressure, severe electrolyte disturbance, suspected inherited disease, or preparation for replacement therapy.

Dialysis access and complications

Haemodialysis requires vascular access. An arteriovenous fistula is durable but needs time to mature. Grafts mature faster but have more thrombosis and infection. Central venous catheters provide immediate access but carry bloodstream infection, thrombosis, and central stenosis risk. Examine access for a continuous thrill and bruit, infection, distal ischaemia, aneurysm, or heart-failure burden; avoid unnecessary pressure and venepuncture in a protected fistula arm.

Haemodialysis complications include hypotension, cramps, arrhythmia, access bleeding, electrolyte shifts, disequilibrium, and air or haemolysis emergencies. Peritoneal dialysis complications include peritonitis, catheter infection, hernia, protein loss, glucose exposure, and membrane failure. Cloudy effluent with abdominal pain requires urgent assessment and sampling.

Kidney transplantation and conservative care

Transplantation can improve survival and quality of life but requires donor and recipient assessment, surgery, immunosuppression, infection prevention, malignancy surveillance, and management of cardiovascular and metabolic risk. Rejection can be antibody- or T-cell-mediated and may present through rising creatinine without symptoms. Drug levels, adherence, interactions, and opportunistic infection remain central.

Not every patient chooses or benefits from dialysis or transplantation. Comprehensive conservative kidney care treats symptoms, anaemia, volume, acidity, pruritus, nausea, restless legs, pain, and psychosocial needs while planning future deterioration. Shared decisions consider prognosis, frailty, treatment burden, cognition, support, and patient goals. Advance care planning should occur before crisis.

Prevention across transitions

After acute kidney injury, document the episode, reconcile medicines, recheck filtration and albuminuria, and reduce future exposure risk. Hospital discharge is not proof of renal recovery. Patients need advice about hydration during illness, temporary medicine review according to clinician guidance, avoidance of non-prescribed nephrotoxins, and when to seek care. Pregnancy, contrast procedures, surgery, and infection create predictable high-risk transitions requiring proactive planning.

TTS module 2: Kidney injury phenotypes, recovery trajectories, and replacement decisions

Acute kidney injury is a syndrome

Acute kidney injury describes loss of filtration or urine output, not one pathological lesion. Haemodynamic filtration change, tubular injury, interstitial inflammation, glomerular disease, vascular injury, and obstruction can produce the same creatinine rise. Several mechanisms may coexist and evolve during treatment.

The traditional pre-renal label suggests intact tissue responding to low perfusion, but prolonged or severe haemodynamic stress can become structural injury. Conversely, tubular injury still contains haemodynamic components that can improve. Rather than forcing a binary distinction, assess perfusion, congestion, exposure, sediment, protein, obstruction, and trajectory.

Creatinine-based staging estimates severity but not cause. Oliguria provides earlier evidence and carries prognostic information, yet antidiuretic hormone, catheter obstruction, diuretics, and intake affect it. A patient with preserved

urine can have major filtration loss, while transient oliguria can reverse without lasting damage.

Haemodynamic and congestive phenotypes

Reduced arterial filling lowers afferent pressure and activates sympathetic, renin-angiotensin, and antidiuretic responses. The kidney conserves sodium and water, increasing urine concentration. When autoregulation fails, filtration falls. Restoring suitable pressure and volume can reverse the process before structural injury develops.

Congestion creates a different haemodynamic problem. High renal venous and interstitial pressure reduces the filtration gradient, impairs lymphatic drainage, and activates sodium retention. Giving repeated fluid because creatinine rose can worsen this phenotype. Jugular venous pressure, oedema, liver congestion, ultrasound, weight, and diuretic response help identify it.

Abdominal pressure can compress renal veins and parenchyma in severe oedema, ascites, burns, or surgical illness. Positive-pressure ventilation alters venous return and right-heart load. Renal perfusion is therefore determined by inflow, outflow, and surrounding pressure rather than mean arterial pressure alone.

Tubular injury and repair

Acute tubular injury affects especially metabolically active and relatively hypoxic segments. Cells lose polarity and brush border, transporters redistribute, and detached cells combine with proteins to obstruct lumens. Back-leak, inflammation, vasoconstriction, and impaired microcirculation sustain reduced filtration.

Surviving epithelial cells dedifferentiate, migrate, proliferate, and restore tubular structure. Recovery can include a polyuric phase because filtration improves before concentrating and reabsorptive capacity. Sodium, potassium, magnesium, phosphate, and water losses may require replacement while avoiding recurrence of overload.

Maladaptive repair produces persistent inflammation, capillary loss, fibroblast activation, and fibrosis, linking acute injury to chronic disease. Apparent biochemical recovery may leave fewer functioning nephrons and reduced

reserve. Recurrent dehydration, infection, surgery, or nephrotoxin exposure then produces a larger effect.

Pigment injury combines direct toxicity, tubular obstruction, and vasoconstriction. Myoglobin release in rhabdomyolysis often follows crush, ischaemia, seizure, heat, drugs, or metabolic muscle disease. Early potassium and calcium hazards may precede creatinine rise. Fluid therapy requires individualisation when heart or kidney reserve limits volume.

Interstitial, glomerular, and vascular urgency

Drug-associated interstitial nephritis may develop after variable exposure and lacks the classic fever-rash-eosinophilia triad in many patients. Pyuria, white-cell casts, modest protein, and systemic clues support the possibility, but biopsy may be needed when diagnosis changes immunosuppressive treatment.

An active sediment with red-cell casts, dysmorphic cells, substantial albumin, or rapid filtration loss raises glomerular urgency. Serological testing searches for immune-complex, complement, antibody, infection, and systemic causes, but tissue often defines the lesion and guides risk. Delay can convert reversible inflammation into crescent formation and scar.

Thrombotic microangiopathy presents with endothelial injury, thrombocytopenia, mechanical haemolysis, and variable kidney or neurological dysfunction. Severe hypertension, pregnancy syndromes, infection, medicines, transplantation, complement dysregulation, and enzyme deficiency are among causes. Time-critical plasma exchange or complement-directed treatment applies to selected mechanisms, so specialist classification is urgent.

Renal vascular disease includes arterial occlusion, cholesterol emboli, venous thrombosis, vasculitis, and malignant hypertension. A relatively bland urine does not exclude vascular injury. Timing after angiography, livedo, eosinophilia, asymmetric kidneys, flank pain, infarction, or abrupt pressure change can provide clues.

Obstruction and decompression

Obstruction must be considered even when urine continues, because unilateral disease, partial blockage, or intermittent flow can preserve output.

Pelvic tumour, retroperitoneal fibrosis, clots, papillary debris, stones, prostate enlargement, neurogenic bladder, and device failure produce different levels of blockage.

Bladder volume identifies lower retention but not upper obstruction.

Ultrasound detects dilation and structural clues, yet early acute blockage may show little hydronephrosis and chronic dilation can persist without active pressure. Computed tomography better identifies stones and masses in selected cases.

Decompression can cause haematuria, pressure change, and post-obstructive diuresis. The diuresis may be appropriate excretion of retained salt and water or reflect impaired concentrating ability. Replace a proportion of losses according to pressure, electrolytes, thirst, and volume rather than automatically matching every millilitre.

Chronic progression and risk

Chronic kidney disease risk is two-dimensional: filtration category and albuminuria. Albumin signals glomerular or systemic vascular injury and predicts cardiovascular events even when filtration is relatively preserved. Rate of change adds a third dimension. A stable low estimate differs from rapid decline.

Adaptive hyperfiltration by surviving nephrons maintains total clearance but raises single-nephron pressure and protein traffic. Filtered proteins activate tubular inflammation and fibrosis. Renin-angiotensin blockade and sodium-glucose cotransporter inhibition can reduce intraglomerular load, producing an early haemodynamic filtration dip while improving long-term trajectory in eligible patients.

Diabetes and hypertension are common labels but should not end investigation when the pattern is atypical. Abrupt decline, active sediment, short diabetes duration, absent expected microvascular disease, heavy protein, systemic symptoms, paraprotein clues, or family history may indicate another treatable cause.

Dialysis as controlled exchange

Haemodialysis moves small solutes across a membrane by diffusion and removes water through transmembrane pressure. Clearance rises with blood

flow, dialysate flow, membrane efficiency, and treatment time. Larger or protein-bound toxins are removed less efficiently. Rapid urea decline in severe chronic uraemia can create osmotic brain swelling, so initial treatment may need gentler delivery.

Continuous therapies trade speed for prolonged steady control and are useful when rapid fluid or solute shifts are poorly tolerated. They still require anticoagulation decisions, access, dose delivery, temperature management, and drug adjustment. Filter clotting, interruptions, and access dysfunction reduce actual treatment below the prescribed dose.

Peritoneal dialysis uses peritoneal capillaries and dialysate. Glucose provides an osmotic gradient for ultrafiltration. Membrane transport characteristics, dwell time, residual kidney function, catheter performance, nutrition, and infection determine effectiveness. Cloudy effluent should be sampled and treated urgently because delay risks membrane and systemic injury.

The decision to start replacement follows refractory physiology rather than a creatinine threshold. Potassium, acidity, volume, uraemic organ effects, toxin, trajectory, and feasibility of medical treatment matter. The modality should fit haemodynamic stability, neurological risk, urgency, access, and long-term plan.

Renal prescribing and transitions

Dose adjustment should distinguish loading from maintenance. Loading depends mainly on distribution volume and target concentration, while maintenance depends on clearance. Critical illness, oedema, obesity, hypoalbuminaemia, and extracorporeal circuits change distribution. Dialysis adds intermittent or continuous clearance.

Medicines can alter creatinine without injuring tissue by inhibiting tubular secretion. Others lower filtration haemodynamically but provide long-term protection. True nephrotoxicity can affect tubules, interstitium, glomeruli, vessels, or urinary crystallisation. Review timing, dose, combinations, levels, hydration, and characteristic urine findings.

Transitions are high risk. After acute injury, previously appropriate antihypertensives, diuretics, diabetes medicines, anticoagulants, analgesics, and antimicrobials may require staged reintroduction or continued adjustment.

Failure to restart beneficial therapy can be harmful, while automatic resumption before stability can cause recurrence.

The final kidney plan should state mechanism, baseline, current filtration and urine trajectory, volume and electrolyte phenotype, offending exposures, obstruction status, dose changes, monitoring interval, and recovery follow-up. Kidney disease is safest when managed as changing physiology rather than a static creatinine label. Patient goals, frailty, cardiovascular reserve, nutrition, access to monitoring, and treatment burden should remain clearly visible throughout every acute and chronic decision.

Retrieval prompts

- One. Why can creatinine lag acute filtration loss?
- Two. Contrast reduced perfusion, intrinsic injury, and obstruction.
- Three. What are the major urgent indications for kidney replacement therapy?
- Four. Why are filtration and albuminuria both used in chronic staging?
- Five. List major systemic complications of chronic kidney disease.
- Six. What determines renal drug adjustment and dialysis removal?

Concise answers

- One. It must accumulate in body water before concentration rises and is influenced by production and fluid balance.
- Two. Reduced perfusion alters haemodynamics; intrinsic injury damages glomeruli, tubules, interstitium, or vessels; obstruction blocks urine flow.
- Three. Refractory hyperkalaemia, severe acidaemia, pulmonary oedema or overload, selected toxins, and uraemic organ complications.
- Four. Each independently predicts progression, cardiovascular risk, and management need.
- Five. Volume and pressure disorders, hyperkalaemia, acidosis, anaemia, mineral-bone disease, uraemia, malnutrition, and cardiovascular disease.

Six. Filtration, acute stability, drug distribution, binding, metabolites, therapeutic window, dialysis modality, and residual function.

Source map

Original synthesis informed by Robbins, glomerular, tubular, vascular, and chronic renal disease; Guyton and Hall, renal physiology and diuretics; Katzung and OpenStax Pharmacology, renal dosing and nephrotoxicity; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, genitourinary assessment.

Chapter 21: Digestion, Absorption, Liver Function, and Nutrition

Orientation

The gastrointestinal system converts food into absorbable molecules, moves water and electrolytes between lumen and blood, excludes pathogens, and communicates with neural, endocrine, immune, and metabolic networks. The liver receives portal blood, processes nutrients and toxins, synthesises essential proteins, produces bile, and regulates whole-body fuel availability. Clinical disease follows when motility, secretion, mucosal integrity, perfusion, digestion, absorption, or hepatic processing fails.

Organisation and control

The gut wall contains mucosa, submucosa, muscular layers, and serosa or adventitia. Surface area is amplified by folds, villi, and microvilli. Tight junctions regulate paracellular movement, while epithelial transporters mediate selective uptake. Rapid epithelial renewal helps repair injury but also makes the mucosa vulnerable to antimetabolites, radiation, inflammation, and ischaemia.

The enteric nervous system coordinates local secretion, blood flow, and motility through myenteric and submucosal plexuses. Parasympathetic activity generally promotes digestion; sympathetic activity generally inhibits motility and secretion while constricting vessels. Long and short reflexes integrate distension, nutrient composition, emotion, and danger signals. Interstitial cells of Cajal generate electrical slow waves that organise smooth-muscle contraction.

Gastrointestinal hormones coordinate regional responses. Gastrin promotes gastric acid secretion and mucosal growth. Secretin responds to duodenal acid by stimulating pancreatic and biliary bicarbonate. Cholecystokinin responds particularly to fat and protein, stimulating pancreatic enzymes, contracting the gallbladder, and slowing gastric emptying. Glucose-dependent insulinotropic peptide and glucagon-like peptide one contribute to the incretin response. Motilin supports fasting migrating motor complexes.

Motility from mouth to colon

Chewing reduces particle size and mixes food with saliva. Swallowing begins voluntarily, then becomes a coordinated pharyngeal and oesophageal reflex that protects the airway and propels the bolus. Lower oesophageal sphincter relaxation permits entry to the stomach; impaired relaxation causes achalasia, whereas inappropriate reflux exposes squamous epithelium to acid.

The proximal stomach accommodates a meal without a large pressure rise. The distal stomach grinds solids and meters chyme into the duodenum. Emptying slows when duodenal acid, fat, hypertonicity, or distension signals that processing capacity is exceeded. Vomiting coordinates gastric and intestinal movement with diaphragm, abdominal muscles, glottic closure, and autonomic responses; persistent vomiting causes chloride, hydrogen, potassium, and volume loss.

Small-intestinal segmentation mixes chyme, while peristalsis advances it. After absorption, the fasting migrating motor complex clears residue. The ileocaecal region limits reflux from the colon. Colonic haustral contractions support water absorption; mass movements propel stool. Defecation combines rectal distension, internal sphincter relaxation, voluntary external sphincter control, pelvic-floor coordination, and increased abdominal pressure.

Secretion and digestion

Saliva lubricates, buffers, protects teeth, and begins starch and lipid digestion. Gastric parietal cells secrete hydrochloric acid and intrinsic factor. Acid denatures protein, activates pepsin, and limits ingested microbes. Histamine, acetylcholine, and gastrin stimulate parietal cells through converging pathways; somatostatin and prostaglandins restrain secretion and support mucosal defence. Mucus, bicarbonate, epithelial blood flow, tight junctions, and rapid restitution protect the stomach from itself.

The exocrine pancreas delivers bicarbonate-rich fluid and digestive enzymes. Proteases are secreted as inactive precursors and activated in the intestinal lumen, reducing pancreatic autodigestion risk. Amylase digests starch; lipase with colipase digests triglyceride; nucleases digest nucleic acids. Pancreatic

insufficiency therefore causes maldigestion, particularly steatorrhoea and fat-soluble vitamin deficiency.

Bile acids are synthesised from cholesterol, stored and concentrated in the gallbladder, and released after meals. Their amphipathic structure emulsifies fat and forms micelles that carry lipids toward the brush border. Most bile acids are reclaimed in the terminal ileum and returned through the portal circulation. Interruption of this enterohepatic cycle can cause diarrhoea, fat malabsorption, or both.

Absorption

Carbohydrates are absorbed mainly as glucose, galactose, and fructose. Sodium-coupled glucose transport explains why oral rehydration solution remains effective in many secretory diarrhoeas: glucose uptake pulls sodium and water with it. Proteins are reduced to amino acids and small peptides and absorbed through several transport systems.

Long-chain fatty acids and monoglycerides enter enterocytes from micelles, are re-esterified, packaged into chylomicrons, and delivered through lymph before reaching blood. Shorter fatty acids enter portal blood more directly. Fat-soluble vitamins accompany lipid absorption. Vitamin B twelve binds intrinsic factor and is absorbed in terminal ileum. Iron is absorbed mainly in proximal small bowel and regulated by hepatic hepcidin; calcium absorption is promoted by active vitamin D.

The small intestine absorbs most fluid presented to it, including large endogenous secretions. The colon salvages water and sodium and secretes potassium and bicarbonate. Diarrhoea results from excessive secretion, unabsorbed osmoles, inflammation, impaired absorption, or rapid transit. Fasting tends to improve osmotic but not purely secretory diarrhoea. High stool volume can rapidly cause hypovolaemia, potassium loss, acidosis, and kidney injury.

Barrier, microbiome, and splanchnic circulation

The intestinal barrier combines mucus, epithelium, antimicrobial molecules, secretory immunoglobulin A, immune cells, and resident microorganisms. The microbiome metabolises otherwise indigestible substrates, produces short-chain fatty acids, modifies bile acids and drugs, and influences immune development. Antibiotics, diet, infection, inflammation, and altered transit can disrupt this ecology, but association does not prove that a microbial pattern causes disease.

Splanchnic blood flow rises after meals and drains largely through the portal vein. Villi are efficient exchange structures but their countercurrent circulation leaves their tips vulnerable to hypoxia. Severe hypotension, arterial occlusion, venous thrombosis, or low-flow states can produce intestinal ischaemia. Pain disproportionate to early examination findings is an important warning, although later infarction causes peritonism and systemic collapse.

Liver architecture and blood flow

The liver receives nutrient-rich portal venous blood and oxygenated hepatic arterial blood. Blood traverses sinusoids toward central veins while hepatocytes exchange substances across a fenestrated endothelium. Bile flows in the opposite direction through canaliculi toward ducts. This arrangement exposes hepatocytes to absorbed nutrients, microbial products, and orally administered drugs before systemic distribution.

Functional zones differ in oxygenation and enzyme expression. Periportal hepatocytes encounter more oxygen and support oxidative metabolism and urea formation. Centrilobular cells have lower oxygen exposure and prominent drug-metabolising activity, making them vulnerable to hypoxia and selected toxins. Stellate-cell activation after chronic injury drives collagen deposition and fibrosis.

Metabolic, synthetic, and detoxifying functions

The liver buffers blood glucose through glycogen synthesis and breakdown, gluconeogenesis, and substrate exchange. It performs fatty-acid oxidation and synthesis, produces ketones and lipoproteins, converts nitrogen to urea, and interconverts amino acids. It synthesises albumin, most clotting factors, transport proteins, complement components, and regulatory molecules. Albumin supports oncotic pressure and binds many drugs and endogenous compounds.

Biotransformation commonly introduces or exposes functional groups through phase-one reactions and conjugates compounds through phase-two reactions, facilitating elimination. Metabolism can inactivate drugs, activate prodrugs, or create toxic intermediates. Enzyme induction, inhibition, genetics, nutrition, liver disease, and portal-systemic shunting alter exposure. The liver also clears bacteria and debris through resident macrophages.

Bilirubin derives largely from haem breakdown. Unconjugated bilirubin travels bound to albumin, enters hepatocytes, is conjugated, and is excreted in bile. Gut metabolism generates pigments eliminated in stool and urine. Predominantly unconjugated elevation suggests overproduction or impaired conjugation; conjugated elevation suggests impaired excretion or obstruction, although mixed patterns are common.

Nutrition and clinical assessment

Energy balance reflects intake, absorption, storage, expenditure, and losses. Carbohydrate, fat, and protein provide fuel and substrates; vitamins and minerals enable enzymatic, structural, haematological, neurological, and endocrine functions. Protein-energy malnutrition reduces muscle, immunity, healing, respiratory strength, and physiological reserve. Obesity can coexist with micronutrient deficiency and sarcopenia.

Nutritional assessment integrates weight trajectory, intake, gastrointestinal symptoms, function, muscle and fat stores, oedema, disease severity, and relevant laboratory evidence. Albumin falls with inflammation and fluid shifts and is not a standalone nutrition marker. Refeeding after prolonged undernutrition can drive phosphate, potassium, and magnesium into cells, causing arrhythmia, weakness, respiratory failure, and neurological injury.

Identify risk, replace thiamine and electrolytes, begin cautiously, and monitor closely.

When the gut works, enteral feeding generally preserves mucosal function and is preferred to parenteral nutrition. Tube position, aspiration risk, tolerance, hydration, electrolytes, and medication formulation require review. Parenteral nutrition can be lifesaving when intestinal use is impossible but carries catheter infection, thrombosis, metabolic disturbance, and hepatobiliary complications.

TTS module 2: Gut transport, portal integration, and failure of digestive reserve

Motility is coordinated propulsion

Gastrointestinal movement depends on smooth-muscle electrical activity, enteric circuits, autonomic input, hormones, luminal contents, and mechanical anatomy. Slow waves set windows in which contraction can occur, while neural and chemical signals determine whether threshold is reached. Contraction must be spatially coordinated; stronger activity without coordination can worsen pain or obstruction.

Peristalsis moves a bolus through contraction behind and relaxation ahead. Segmentation repeatedly divides contents and improves contact with mucosa. The fasting migrating motor complex clears residual material between meals and is interrupted by feeding. Loss of this clearance can favour small-intestinal bacterial overgrowth in selected dysmotility states.

Gastric emptying depends on particle size and duodenal feedback. Fat, acid, hyperosmolality, and nutrient load slow delivery to protect intestinal processing. Autonomic neuropathy, surgery, medicines, inflammation, and metabolic disturbance can delay or accelerate emptying. Symptoms correlate imperfectly with measured delay because accommodation and sensory processing also matter.

Mechanical obstruction must be distinguished from ileus or pseudo-obstruction. Obstruction creates a transition point and proximal distension. Ileus reflects diffuse failure of propulsion after surgery, inflammation, electrolyte disturbance, medicines, or critical illness. Both can

cause vomiting, sequestration, aspiration, and perforation, but definitive treatment differs.

Secretion and mucosal defence

Gastric acid secretion depends on a final proton pump stimulated by histamine, acetylcholine, and gastrin pathways. Acid suppression is effective because blocking the final pump overcomes several upstream signals. However, long-term suppression can alter mineral or vitamin handling, infection risk, microbiology, and drug absorption in selected patients, so indication and duration should be reviewed.

The mucosal barrier uses mucus, bicarbonate, epithelial restitution, tight junctions, prostaglandins, and blood flow. Non-steroidal anti-inflammatory drugs reduce prostaglandin support and can cause injury throughout the gastrointestinal tract. *Helicobacter pylori* adds chronic inflammation and ulcer risk. Smoking, anticoagulation, physiological stress, and previous ulcer modify the likelihood of bleeding.

Pancreatic enzymes are activated through a controlled cascade in the intestinal lumen. Premature intrapancreatic activation contributes to pancreatitis, while duct obstruction, alcohol, triglycerides, medicines, trauma, metabolic factors, and genetic susceptibility can initiate injury. Severe inflammation causes capillary leak, systemic inflammatory response, necrosis, infection, and organ failure.

Absorption requires digestion and surface

Malabsorption can arise from failure to digest nutrients, insufficient bile, reduced mucosal surface, transporter disease, lymphatic obstruction, or rapid transit. Pancreatic insufficiency leaves macronutrients incompletely digested. Bile-acid deficiency impairs micelle formation. Coeliac disease damages proximal small-bowel surface. Ileal resection interrupts bile-acid and vitamin B twelve recovery.

Bile-acid loss has two phases. Modest ileal dysfunction sends excess bile acids into colon, stimulating secretion and diarrhoea. Extensive loss depletes the total bile-acid pool, causing fat malabsorption and steatorrhoea. Unabsorbed fatty acids bind calcium, leaving oxalate available for colonic absorption and increasing kidney-stone risk.

Short-bowel physiology depends on which segments remain, whether colon is in continuity, adaptation, and intake. The colon can salvage water, sodium, and energy from fermentation products. Gastric hypersecretion and rapid transit can worsen early losses. Treatment combines diet, oral rehydration, antimotility or antisecretory therapy, micronutrients, and parenteral support when intestinal autonomy is insufficient.

Carbohydrate malabsorption leaves osmoles in the lumen and supplies colonic bacteria, producing gas and acidic stool. Fat malabsorption carries fat-soluble vitamins and can cause weight loss. Protein loss can occur through inflamed or congested mucosa even when digestion is intact.

Diarrhoea as a transport phenotype

Osmotic diarrhoea improves with fasting because the unabsorbed substance is removed. Secretory diarrhoea persists because active ion secretion or impaired absorption continues. Inflammatory diarrhoea adds mucosal damage, protein, blood, leukocytes, and reduced absorptive area. Rapid transit reduces contact time and may coexist with each mechanism.

Stool electrolyte measurements can estimate the osmotic gap when mechanism remains unclear. A low gap suggests measured electrolytes account for most stool osmolality and supports secretion. A high gap suggests unmeasured osmoles. Sample contamination, fermentation, and collection difficulty limit use.

High-volume diarrhoea causes more than dehydration. Bicarbonate and potassium loss produce normal-gap acidosis and weakness. Magnesium loss, malnutrition, kidney injury, and medication malabsorption can follow. Oral rehydration succeeds because sodium-glucose cotransport remains functional in many secretory illnesses, but correct composition matters.

Portal circulation and hepatic zonation

Portal blood carries nutrients, microbial products, and oral drugs directly to liver sinusoids. The hepatic artery supplies additional oxygen. These inflows mix, while bile moves oppositely toward ducts. Disturbance of portal flow changes both delivery and pressure.

Periportal hepatocytes receive more oxygen and favour oxidative metabolism, gluconeogenesis, and urea synthesis. Centrilobular hepatocytes receive less

oxygen and express many drug-metabolising enzymes. Low-flow states therefore preferentially injure centrilobular regions, while some toxins damage zones according to where reactive metabolites form.

Portal hypertension follows increased resistance within liver, portal vein, or downstream circulation. Splanchnic vasodilation then reduces effective arterial filling and activates sodium retention. Collaterals create oesophageal or gastric varices and shunt ammonia and drugs around hepatic clearance. Splenic congestion causes thrombocytopenia.

Ascites reflects portal hydrostatic pressure, renal sodium retention, lymph formation, and reduced effective filling more than low albumin alone. Infection can develop within ascitic fluid without an obvious perforation. New pain, fever, encephalopathy, kidney deterioration, or decompensation should prompt urgent sampling according to current protocols.

Liver tests are compartment signals

Aminotransferases indicate hepatocyte membrane injury but do not measure synthetic function. Alkaline phosphatase and gamma-glutamyl transferase support a cholestatic or ductal pattern, although bone and other sources require consideration. Bilirubin reflects production, uptake, conjugation, and excretion. Albumin and clotting time reflect synthesis but change with nutrition, inflammation, loss, vitamin K, anticoagulants, and half-life.

Massive aminotransferase elevation can follow ischaemia, toxin, or acute viral injury, but falling values may mean recovery or loss of viable hepatocytes. Rising bilirubin, clotting impairment, hypoglycaemia, acidosis, encephalopathy, and kidney dysfunction better describe acute failure. Time course and medication or exposure history are critical.

Hepatic encephalopathy reflects interacting ammonia, inflammation, altered neurotransmission, muscle loss, shunting, and precipitating illness. Gastrointestinal bleeding, infection, constipation, dehydration, kidney failure, sedatives, and electrolyte disturbance can trigger deterioration. Treatment identifies precipitants and reduces gut-derived nitrogen while protecting airway and nutrition.

Nutrition as functional physiology

Nutrition assessment should identify intake, absorption, inflammatory demand, losses, body composition, and function. Weight can be concealed by oedema or ascites. Hand strength, mobility, muscle area, dietary history, and trajectory often provide more actionable information than serum proteins.

Enteral feeding uses the gut but does not eliminate aspiration. Tube location, gastric emptying, consciousness, reflux, positioning, oral hygiene, and airway protection determine risk. Post-pyloric feeding helps selected patients but does not prevent aspiration of oral secretions.

Parenteral nutrition bypasses absorption and requires precise glucose, amino acid, lipid, electrolyte, vitamin, trace-element, and fluid delivery. Excess energy drives hyperglycaemia, carbon dioxide production, fatty liver, and infection risk. Underfeeding perpetuates catabolism. Catheter and metabolic monitoring are integral parts of the prescription.

The microbiome participates in metabolism without acting as an isolated organ. Colonic organisms ferment fibre into short-chain fatty acids that nourish colonocytes and influence immune and metabolic signalling. They modify bile acids, medicines, and dietary compounds. Antibiotics, transit, acidity, inflammation, surgery, and diet reshape the community. Dysbiosis associations are common, but a measured composition rarely proves causation or defines treatment. Faecal microbiota therapy has strong use only in selected conditions under regulated protocols; indiscriminate supplements can add infection, interaction, cost, and false reassurance.

The final gastrointestinal synthesis should state the failing process: propulsion, digestion, absorption, barrier, perfusion, portal flow, hepatic processing, or nutritional delivery. It should then identify the anatomical cause, systemic consequence, reversible trigger, and measure that will demonstrate recovery. Hydration, electrolytes, weight trajectory, muscle function, stool or stoma output, and medication absorption provide practical longitudinal checks across transitions of care.

Retrieval prompts

One. How do neural and hormonal signals coordinate digestion?

Two. Why does terminal ileal disease impair fat and vitamin B twelve absorption?

Three. How does oral rehydration solution exploit normal transport physiology?

Four. What major functions make liver failure a multisystem disorder?

Five. Why is albumin an unreliable isolated nutrition marker?

Six. What makes refeeding syndrome dangerous?

Concise answers

One. Enteric circuits, autonomic input, reflexes, and hormones match motility, secretion, blood flow, and emptying to luminal conditions.

Two. The terminal ileum reclaims bile acids and absorbs intrinsic-factor-bound vitamin B twelve.

Three. Coupled glucose and sodium uptake remains active and promotes water absorption despite secretory loss.

Four. The liver regulates fuel metabolism, synthesis, detoxification, immune clearance, bile formation, and bilirubin handling.

Five. Inflammation, capillary leak, distribution, hydration, and liver synthesis alter albumin independently of nutrient stores.

Six. Insulin-driven intracellular shifts can rapidly deplete circulating phosphate, potassium, and magnesium and cause organ failure.

Source map

Original synthesis informed by Guyton and Hall, gastrointestinal physiology, hepatic metabolism, and nutrition; Robbins, gastrointestinal and liver pathology; Katzung and OpenStax Pharmacology, acid suppression, motility, and hepatic drug handling; OpenStax Anatomy and Physiology, Biology, Microbiology, and Medical-Surgical Nursing; and Talley and O'Connor, abdominal and nutritional assessment.

Chapter 22: Diabetes, Energy Metabolism, Obesity, and Metabolic Emergencies

Orientation

Energy metabolism matches nutrient supply to cellular demand. Insulin dominates the fed state, promoting storage and restraining fuel production. Glucagon and stress hormones defend fuel during fasting and illness. Diabetes develops when insulin supply, action, or both become inadequate. Obesity and ectopic fat intensify resistance, but size alone does not define metabolic health. Acute dysregulation can cause dehydration, electrolyte disturbance, acidosis, cerebral dysfunction, and death.

Fed, fasting, and stress states

After a carbohydrate-containing meal, glucose enters portal blood and stimulates pancreatic beta cells. Rising adenosine triphosphate closes potassium channels, depolarises the membrane, opens calcium channels, and triggers insulin release. Incretin hormones amplify glucose-dependent secretion. Insulin acts through a receptor tyrosine kinase to promote glucose uptake in muscle and adipose tissue, glycogen formation, lipid synthesis, protein synthesis, and potassium entry into cells. It suppresses hepatic glucose output, lipolysis, proteolysis, and ketogenesis.

Glucagon from alpha cells rises as glucose and insulin fall. It promotes hepatic glycogen breakdown, gluconeogenesis, and ketogenesis. During an overnight fast, liver glycogen is an early glucose source; as fasting continues, gluconeogenesis from lactate, glycerol, and amino acids becomes dominant. Adipose lipolysis releases fatty acids for oxidation. The liver converts fatty acids into ketone bodies, which increasingly fuel brain and muscle and reduce obligatory protein breakdown.

Catecholamines, cortisol, and growth hormone support fuel mobilisation during stress. They oppose aspects of insulin action and can expose limited beta-cell reserve. Severe infection, infarction, trauma, glucocorticoid

treatment, pregnancy, and surgery therefore precipitate hyperglycaemia. Counter-regulation protects against hypoglycaemia, but recurrent episodes can blunt autonomic warning and create dangerous unawareness.

Diabetes classification and pathogenesis

Type one diabetes results from immune-mediated destruction of beta cells in susceptible people, producing absolute insulin deficiency. Autoantibodies can support classification but are not themselves the direct cause of tissue injury. Presentation may be gradual or occur as ketoacidosis. Exogenous insulin is essential for survival.

Type two diabetes combines insulin resistance with progressive beta-cell dysfunction. Genetic susceptibility interacts with age, visceral and ectopic fat, inactivity, sleep disturbance, medications, and social environment. Beta cells initially compensate with increased secretion, but glucotoxicity, lipotoxicity, islet stress, and limited reserve eventually produce hyperglycaemia. The process is heterogeneous: some people have marked resistance, others predominant secretory failure.

Other forms include monogenic diabetes, pancreatic destruction, endocrinopathies, drug-induced disease, and gestational diabetes. Classification matters because treatment, inheritance, and ketosis risk differ; age or body size alone does not establish type.

Diagnosis and monitoring

Diabetes is diagnosed through validated thresholds using fasting plasma glucose, an oral glucose tolerance test, glycated haemoglobin, or unequivocal symptomatic hyperglycaemia. Unless the presentation is clear, an abnormal result requires confirmation. Glycated haemoglobin estimates recent average exposure but can mislead when red-cell lifespan, haemoglobin structure, pregnancy, kidney disease, transfusion, or assay interference changes.

Monitoring should connect measurements to decisions. Continuous sensors reveal direction, variability, nocturnal patterns, and time in range, but still require clinical interpretation. Measure ketones during insulin deficiency, acute illness, pregnancy, or sodium-glucose cotransporter inhibitor use.

Chronic tissue injury

Persistent hyperglycaemia alters proteins, signalling, oxidation, and inflammation, damaging microvessels. Retinopathy progresses from leakage to ischaemia and neovascularisation. Kidney disease causes albuminuria, filtration loss, or both. Peripheral neuropathy causes sensory loss, pain, weakness, and foot injury; autonomic neuropathy disrupts cardiovascular, gastrointestinal, bladder, sweating, and sexual function.

Diabetes also accelerates atherosclerotic cardiovascular disease through clustering of hypertension, dyslipidaemia, inflammation, endothelial dysfunction, kidney disease, and smoking exposure. Foot ulceration arises from neuropathy, pressure, deformity, trauma, infection, and ischaemia. Prevention therefore requires retinal surveillance, kidney assessment, foot examination, pressure and lipid management, vaccination, dental care, smoking cessation, and attention to mental health—not glucose treatment alone.

Principles of glucose-lowering therapy

Nutrition, activity, sleep, weight management, education, and reduction of treatment burden are foundations. Therapy should be individualised to diabetes type, cardiovascular and kidney disease, hypoglycaemia risk, weight effects, pregnancy, frailty, cost, preferences, and ability to use the regimen safely.

Metformin reduces hepatic glucose production with little hypoglycaemia; gastrointestinal effects and kidney function influence use. Sulfonylureas stimulate insulin but can cause hypoglycaemia and weight gain. Thiazolidinediones improve sensitivity but may cause oedema, fractures, and heart-failure worsening. Dipeptidyl peptidase four inhibitors modestly prolong incretin action.

Glucagon-like peptide one receptor agonists increase glucose-dependent insulin, suppress glucagon, slow gastric emptying, and reduce appetite; selected agents improve cardiovascular outcomes and weight. Sodium-glucose cotransporter two inhibitors cause glycosuria and natriuresis with heart and kidney benefits, but can cause genital infection, volume depletion, and ketoacidosis without extreme hyperglycaemia.

Insulins differ in onset, peak, and duration. Basal insulin restrains fasting hepatic output; prandial insulin covers meals and correction. Needs vary with food, activity, stress, kidney clearance, steroids, and residual secretion. Safety requires correct technique, monitoring, hypoglycaemia treatment, and sick-day planning.

Hypoglycaemia

Hypoglycaemia most often reflects insulin or insulin-secretagogue exposure but can occur with alcohol, organ failure, sepsis, endocrine deficiency, malnutrition, or tumours. Adrenergic features include tremor, sweating, palpitations, hunger, and anxiety. Neuroglycopenia causes confusion, behavioural change, visual disturbance, seizure, coma, or focal deficits.

An awake person able to swallow should receive rapid carbohydrate followed by reassessment and longer-acting intake when appropriate. Severe impairment requires intravenous glucose or glucagon, airway protection, and investigation of cause. Sulfonylurea-associated recurrence may require prolonged observation and suppression of insulin release. Prevention requires regimen review, education, and avoidance of recurrent lows.

Diabetic ketoacidosis

Diabetic ketoacidosis results from severe effective insulin deficiency combined with excess counter-regulatory hormones. Hepatic glucose production and impaired uptake cause hyperglycaemia and osmotic diuresis. Lipolysis drives ketone production, consuming bicarbonate and producing high-gap acidosis. Total-body potassium is depleted despite normal or high initial serum potassium because insulin deficiency, hypertonicity, and acidaemia shift potassium outward.

Diagnosis integrates ketonaemia, acidosis, and the clinical context; glucose may be only moderately elevated. Treatment restores circulation with carefully selected fluid, gives insulin to stop ketogenesis, replaces potassium according to level and urine function, and adds glucose when needed so insulin can continue until ketosis resolves. Search for infection, omitted insulin, infarction, pregnancy, pancreatitis, or drugs. Bicarbonate is rarely required. Closing the glucose value alone does not prove resolution.

Hyperosmolar hyperglycaemic state

Hyperosmolar hyperglycaemic state usually develops through relative insulin deficiency sufficient to restrain major ketogenesis but insufficient to control glucose. Profound osmotic diuresis causes water loss, hypertonicity, kidney impairment, and neurological dysfunction. Evolution is often slower than ketoacidosis and occurs in older or vulnerable people with limited access to water.

Treatment prioritises gradual restoration of circulation and water, cautious correction of tonicity, electrolyte monitoring, insulin after initial fluid response, thrombosis prevention when indicated, and treatment of the precipitant. Rapid shifts can worsen neurological injury. Mixed ketoacidotic and hyperosmolar presentations are common and should be managed according to the abnormalities present.

Obesity and metabolic dysfunction

Body weight is regulated through interacting neural, endocrine, gastrointestinal, adipose, behavioural, environmental, and social signals. Leptin communicates energy stores, while hypothalamic pathways integrate leptin, insulin, gut peptides, reward, stress, and sensory cues. Weight loss triggers hunger and reduced expenditure, which explains biological resistance to maintenance without implying personal failure.

Visceral and ectopic fat promote insulin resistance, fatty liver, dyslipidaemia, hypertension, sleep apnoea, osteoarthritis, reproductive dysfunction, and cardiovascular disease. Assessment integrates complications, medications, eating, sleep, mobility, mental health, stigma, goals, and fat distribution rather than body-mass index alone.

Treatment combines sustainable dietary energy reduction, adequate protein and micronutrients, physical activity, sleep and behavioural support, management of secondary causes, anti-obesity pharmacotherapy when appropriate, and metabolic surgery for selected patients. Effective treatment can improve diabetes, blood pressure, fatty liver, sleep apnoea, function, and quality of life even when an arbitrary ideal weight is not reached.

Dyslipidaemia and fatty liver

Insulin resistance raises triglyceride-rich and small dense atherogenic particles while lowering high-density lipoprotein cholesterol. Treatment follows total cardiovascular risk. Statins reduce cholesterol synthesis and increase low-density lipoprotein receptor clearance; additional agents target absorption, proprotein convertase, or triglycerides.

Metabolic dysfunction-associated steatotic liver disease ranges from fat accumulation to inflammation, fibrosis, cirrhosis, and cancer. Aminotransferases can be normal despite significant fibrosis. Risk assessment therefore combines metabolic context with non-invasive fibrosis estimates, imaging, and specialist evaluation when risk is elevated. Weight reduction, activity, diabetes and lipid treatment, alcohol review, and cardiovascular prevention are central.

TTS module 2: Fuel partitioning, insulin reserve, and metabolic crisis trajectories

Fuel selection across time

Metabolic regulation prioritises continuous fuel for the brain and red cells while matching muscle and storage tissues to feeding and activity. Insulin signals nutrient availability. It promotes glycogen and lipid storage, protein synthesis, and glucose uptake while restraining hepatic glucose production and adipose lipolysis.

During fasting, falling insulin and rising glucagon permit glycogen breakdown, gluconeogenesis, and fat mobilisation. Liver glycogen supplies early glucose but is finite. Lactate returns through the Cori cycle, alanine carries carbon and nitrogen from muscle, and glycerol enters gluconeogenesis. Fatty acids fuel liver and muscle, while ketones gradually spare muscle protein.

Stress hormones make substrate available for immediate survival. Catecholamines stimulate glycogenolysis and lipolysis. Cortisol supports gluconeogenesis and protein catabolism. Growth hormone reduces glucose uptake in selected tissues. In infection or trauma, these responses combine with inflammatory insulin resistance and can produce hyperglycaemia even without previous diabetes.

Metabolic flexibility is the ability to switch appropriately between carbohydrate and fat oxidation. Physical inactivity, ectopic fat, mitochondrial stress, sleep disruption, and chronic overnutrition reduce flexibility. Exercise improves insulin-independent muscle glucose uptake acutely and insulin sensitivity over time.

Beta-cell sensing and failure

Beta cells sense glucose through metabolism rather than a surface glucose receptor alone. Increased adenosine triphosphate closes potassium channels, depolarises the membrane, opens calcium channels, and triggers granule release. A rapid first phase uses ready granules; a sustained second phase recruits additional granules and synthesis.

Incretins amplify insulin only when glucose is present and also affect glucagon, gastric emptying, and satiety. This glucose dependence explains the relatively low intrinsic hypoglycaemia risk of incretin-based therapies when not combined with insulin or secretagogues.

Type two diabetes develops when insulin secretion cannot match resistance. Early hyperinsulinaemia may preserve glucose but imposes secretory demand. Ectopic lipid, islet amyloid, oxidative and endoplasmic-reticulum stress, inflammation, genetics, and ageing reduce beta-cell capacity. Glucose toxicity then worsens both secretion and sensitivity, creating a self-reinforcing cycle.

Some patients with apparent type two diabetes have autoimmune, monogenic, pancreatic, medication-induced, or endocrine disease. Rapid insulin need, ketosis, low body mass, strong multigenerational pattern, pancreatic history, or unusual treatment response should prompt reconsideration.

Interpreting glucose markers

Fasting glucose reflects hepatic output and basal insulin action. Post-meal glucose reflects meal composition, absorption, incretin effect, insulin timing, muscle uptake, and gastric emptying. Glycated haemoglobin averages exposure but gives limited information about variability and hypoglycaemia.

Red-cell survival changes glycated haemoglobin. Iron deficiency may raise it, while haemolysis, blood loss, treatment of deficiency, or recent transfusion

can lower or distort it. Haemoglobin variants affect some assays. Kidney disease creates competing effects through anaemia, altered lifespan, and therapy.

Fructosamine or glycated albumin reflects a shorter period and can help in selected red-cell disorders, but albumin turnover, protein loss, liver disease, and thyroid state influence results. Continuous sensors measure interstitial glucose with a lag during rapid change and require confirmation when symptoms conflict with the reading.

Insulin therapy as dynamic replacement

Basal insulin suppresses hepatic glucose release between meals. Prandial insulin covers absorbed carbohydrate and protein-related glucose effects. Correction insulin addresses unexpected elevation but cannot replace an inadequate basal or meal strategy. Repeated correction without identifying the pattern creates oscillation and hypoglycaemia.

Insulin absorption varies with injection site, depth, temperature, exercise, lipohypertrophy, dose volume, and formulation. Repeated injection into lipohypertrophic tissue produces unpredictable exposure. Technique review includes storage, needle use, priming, site rotation, timing, and understanding of the specific preparation.

Kidney failure reduces insulin clearance and gluconeogenic reserve, increasing hypoglycaemia risk. Infection and glucocorticoids increase need. Exercise can lower glucose during and after activity through increased uptake. Alcohol suppresses hepatic gluconeogenesis and can cause delayed hypoglycaemia after glycogen is depleted.

Automated delivery systems reduce burden but still depend on sensor function, infusion-site integrity, carbohydrate estimates, algorithm settings, and sick-day response. Pump interruption can produce rapid insulin deficiency because no long-acting depot remains.

Ketoacidosis as staged physiology

Insulin deficiency removes restraint from lipolysis and hepatic ketogenesis. Counter-regulatory hormones increase substrate flow. Acetoacetate and beta-hydroxybutyrate consume bicarbonate, while hyperglycaemia drives

osmotic diuresis. Vomiting and reduced intake worsen sodium, chloride, potassium, phosphate, and water loss.

Measured sodium can be low from glucose-driven extracellular water shift, yet total body sodium is depleted. Effective osmolality reflects sodium and glucose and helps track neurological risk. Potassium may begin high from insulin deficiency, hypertonicity, acidosis, and cell stress while total stores are low.

Fluid restores perfusion and kidney clearance. Insulin stops ketone production and moves potassium intracellularly. Glucose is added before ketosis resolves so insulin can continue safely. The anion gap and beta-hydroxybutyrate describe resolution better than glucose alone.

Euglycaemic ketoacidosis occurs with reduced carbohydrate intake, pregnancy, prolonged fasting, partial insulin treatment, and sodium-glucose cotransporter inhibition. A near-normal glucose must not override ketones, acidosis, and clinical context. Sick-day plans address hydration, carbohydrate, ketone testing, and when to pause selected medicines or seek care according to current guidance.

Hyperosmolar crisis

Hyperosmolar hyperglycaemic state evolves when enough insulin restrains major ketogenesis but not hyperglycaemia. Glucose rises over days, causing profound osmotic diuresis. Frailty, cognitive impairment, infection, diuretics, glucocorticoids, and poor water access increase risk.

Neurological dysfunction relates to hypertonicity, dehydration, vascular disease, seizure, infection, and sometimes focal pathology. Fluid treatment changes glucose and sodium together. Tonicity must fall in a controlled way because rapid water entry into adapted brain can be harmful.

Insulin is often delayed until initial volume replacement because fluids alone lower glucose and improve perfusion. Potassium, phosphate, kidney function, thrombosis risk, heart failure, and the precipitating cause require parallel management. Mixed hyperosmolar and ketoacidotic states are common.

Hypoglycaemia and impaired awareness

Falling glucose normally suppresses endogenous insulin, increases glucagon and catecholamines, then produces autonomic warning. Recurrent hypoglycaemia shifts warning thresholds lower, creating impaired awareness. Sleep, alcohol, exercise, kidney disease, older age, and autonomic neuropathy increase vulnerability.

Treatment should be measured rather than followed by uncontrolled eating that causes rebound hyperglycaemia. Severe episodes require airway protection and parenteral glucose or glucagon, but depleted glycogen, alcohol, liver disease, or prolonged fasting can reduce glucagon effectiveness.

After recovery, identify excess basal insulin, unmatched meal dose, delayed absorption, exercise, alcohol, kidney change, or medication error. Sensor alarms, higher temporary targets, structured avoidance, and regimen simplification can help restore awareness and prevent recurrence.

Adipose tissue and ectopic fat

Adipose tissue stores energy safely until expansion exceeds local vascular, extracellular-matrix, and cellular capacity. Hypertrophic adipocytes release more fatty acids and inflammatory signals. Lipid accumulates in liver, muscle, pancreas, and around organs, interfering with insulin signalling and function.

Visceral fat drains toward portal circulation and is strongly associated with metabolic risk, but genetics, fitness, sleep, medications, socioeconomic environment, and fat-storage capacity modify outcomes. Body-mass index estimates size, not composition, distribution, or individual disease burden.

Weight loss lowers energy expenditure and raises hunger signals, creating biological pressure for regain. Effective long-term care treats obesity as chronic relapsing physiology. Nutrition, resistance and aerobic activity, sleep, behavioural support, medicines, and surgery can be combined according to complications, preference, safety, access, and response.

Loss of muscle during weight reduction reduces function and metabolic capacity. Adequate protein, resistance exercise, and gradual monitoring are particularly important in older or frail patients. Improvement in liver fat, glucose, pressure, mobility, fertility, or sleep may be clinically meaningful before a large weight change.

The final metabolic plan should separate glycaemic exposure, variability, hypoglycaemia, insulin reserve, cardiovascular and kidney risk, adipose complications, nutrition, and treatment burden. Success is durable reduction in organ risk and improved function, not pursuit of one glucose or weight number in isolation during lifelong metabolic care in clinical practice.

Retrieval prompts

One. How do insulin and glucagon divide control between fed and fasting states?

Two. Why can serum potassium be high in ketoacidosis despite total-body depletion?

Three. How does hyperosmolar hyperglycaemic state differ mechanistically from ketoacidosis?

Four. Which chronic complications require systematic surveillance?

Five. Why does weight regain commonly follow weight loss?

Six. Why can normal aminotransferases not exclude important fatty liver disease?

Concise answers

One. Insulin promotes uptake and storage after feeding; glucagon supports hepatic fuel release and ketogenesis during fasting.

Two. Insulin deficiency, hypertonicity, and acidaemia shift potassium out of cells while osmotic diuresis removes it from the body.

Three. Relative insulin activity limits major ketogenesis, but extreme glucose-driven water loss produces profound hypertonicity and dehydration.

Four. Eye, kidney, nerve, foot, cardiovascular, dental, and psychosocial complications require planned review.

Five. Neuroendocrine adaptation increases hunger and lowers energy expenditure after weight loss.

Six. Enzyme concentrations reflect injury imperfectly and do not directly measure fibrosis.

Source map

Original synthesis informed by Guyton and Hall, insulin, glucagon, fuel metabolism, and obesity physiology; Robbins, diabetes, vascular injury, pancreatic disease, and fatty liver; Katzung and OpenStax Pharmacology, glucose-lowering and lipid therapies; OpenStax Anatomy and Physiology, Biology, Chemistry, and Medical-Surgical Nursing; and Talley and O'Connor, metabolic history and examination.

Chapter 23: Hypothalamic-Pituitary Control and Endocrine Feedback

Orientation

Endocrine systems communicate through hormones carried in blood to receptors in distant or local tissues. Their concentrations reflect secretion, distribution, binding, metabolism, and clearance rather than gland activity alone. The hypothalamus and pituitary link neural information to thyroid, adrenal, gonadal, growth, lactation, and water-balance systems. Most axes use negative feedback, pulsatility, and circadian timing. Clinical interpretation therefore depends on paired hormones, physiological context, and dynamic response.

Hormone classes and receptors

Peptide hormones are synthesised as precursors, stored in vesicles, and released by exocytosis. They circulate mainly free and act through membrane receptors, second messengers, kinases, and ion channels. Their effects can begin rapidly but may also alter gene expression. Steroid hormones derive from cholesterol, are usually synthesised on demand, circulate substantially protein-bound, cross membranes, and regulate transcription through intracellular receptors. Thyroid hormones are amino-acid derivatives but act through nuclear receptors; catecholamines act through surface receptors.

Only unbound hormone is readily available to tissues and clearance. Changes in binding proteins can alter total concentration without changing free biological activity. Pregnancy, oestrogen, liver disease, protein loss, drugs, and inherited variants can therefore distort total-hormone interpretation. Receptor number, downstream signalling, local activation, and tissue sensitivity also determine response.

Feedback, rhythms, and dynamic testing

In a typical axis, hypothalamic releasing hormone stimulates a pituitary trophic hormone, which stimulates a peripheral gland. Peripheral hormone feeds back at pituitary and hypothalamus. Primary gland failure produces low peripheral hormone with high trophic drive; central failure produces low peripheral hormone with an inappropriately low or normal trophic hormone. Autonomous secretion suppresses upstream signals.

Hormones are often pulsatile. Gonadotropin-releasing hormone must be intermittent to sustain gonadotropins; continuous exposure suppresses the axis. Growth hormone pulses strongly during sleep. Cortisol normally peaks near waking and falls around midnight. Meals, exercise, stress, sleep, posture, menstrual stage, and acute illness alter results. A single value can therefore be misleading.

Dynamic tests challenge physiology. Suppression tests ask whether autonomous secretion can be restrained; stimulation tests ask whether reserve can respond. Test selection, preparation, timing, medication interference, assay limitations, and safety matter. Dynamic testing should answer a defined question rather than compensate for indiscriminate screening.

Hypothalamus and anterior pituitary

Hypothalamic neurons release regulatory peptides into the portal circulation. Corticotropin-releasing hormone drives adrenocorticotrophic hormone. Thyrotropin-releasing hormone drives thyroid-stimulating hormone and can stimulate prolactin. Gonadotropin-releasing hormone drives luteinising and follicle-stimulating hormones. Growth-hormone-releasing hormone promotes growth hormone, while somatostatin inhibits growth hormone and thyroid-stimulating hormone. Dopamine tonically inhibits prolactin.

Anterior pituitary cell types include corticotrophs, thyrotrophs, gonadotrophs, somatotrophs, and lactotrophs. Pituitary masses can secrete hormone, compress normal tissue, or both. Upward extension may compress the optic chiasm, classically impairing temporal visual fields. Lateral invasion can affect ocular motor nerves and sympathetic fibres. Sudden haemorrhage or

infarction within a tumour can cause severe headache, visual loss, ophthalmoplegia, altered consciousness, and acute adrenal insufficiency.

Posterior pituitary and water balance

Arginine vasopressin is synthesised in hypothalamic nuclei, transported down axons, and released from the posterior pituitary. Osmoreceptors respond sensitively to rising effective osmolality; substantial volume depletion provides a powerful non-osmotic stimulus. Vasopressin acts at renal V-two receptors to insert aquaporin water channels in collecting ducts. V-one effects include vasoconstriction.

Deficient secretion or renal resistance causes diabetes insipidus: excretion of large volumes of dilute urine, thirst, and risk of hypernatraemic dehydration when water access is limited. Central disease responds to desmopressin; nephrogenic disease requires correction of cause and reduction of solute and urine load. Primary polydipsia can mimic the pattern. Diagnosis compares plasma and urine concentration under carefully supervised conditions.

In inappropriate antidiuresis, water retention lowers sodium and osmolality while urine remains inappropriately concentrated. Diagnosis requires assessment of volume, kidney function, adrenal and thyroid status, drugs, pulmonary or neurological disease, pain, nausea, and malignancy. Treatment depends on symptom severity and duration because overly rapid correction of chronic hyponatraemia can cause osmotic demyelination.

Growth hormone physiology and disease

Growth hormone supports linear growth through insulin-like growth factor one and has direct metabolic effects including lipolysis and reduced insulin sensitivity. Secretion depends on sleep, nutrition, exercise, stress, age, and feedback. Insulin-like growth factor one integrates secretion over time and is useful for screening, but nutrition, liver disease, kidney disease, pregnancy, and age influence it.

Excess before epiphyseal closure causes gigantism; after closure it causes acromegaly. Features include enlargement of hands, feet, jaw, and soft tissue; sweating; headaches; sleep apnoea; hypertension; cardiomyopathy; arthritis; neuropathy; glucose intolerance; and increased neoplasia risk in

selected tissues. Diagnosis uses insulin-like growth factor one and failure of glucose to suppress growth hormone, followed by pituitary imaging.

Growth hormone deficiency in children impairs growth; in adults it can reduce lean mass, bone density, exercise capacity, and quality of life. Because symptoms are non-specific and random growth hormone is unhelpful, diagnosis usually requires assessment of other axes and stimulation testing. Replacement is individualised and avoided or used cautiously in relevant active disease.

Prolactin physiology and hyperprolactinaemia

Prolactin promotes milk production. Suckling reduces hypothalamic dopamine and increases secretion; pregnancy prepares the breast, while falling sex steroids after delivery permit lactation. High prolactin suppresses gonadotropin-releasing hormone, causing menstrual disturbance, infertility, sexual dysfunction, and reduced bone protection. Galactorrhoea may occur but is neither required nor specific.

Causes include pregnancy, lactation, prolactinoma, pituitary-stalk interruption, hypothyroidism, kidney disease, chest-wall stimulation, and dopamine-blocking or dopamine-depleting drugs. Mild elevation should be repeated under appropriate conditions and assessed for macroprolactin and assay effects. Very large tumours can rarely produce falsely modest results unless the sample is diluted.

Dopamine agonists suppress secretion and usually shrink prolactinomas. Surgery is considered for intolerance, resistance, selected compressive disease, or urgent circumstances. Management must distinguish a secreting tumour from medication effect or stalk compression because the same laboratory abnormality has different implications.

Hypopituitarism

Pituitary hormone loss can follow tumours, surgery, radiation, trauma, vascular injury, inflammation, infection, infiltrative disease, genetic disorders, or postpartum haemorrhage. Deficits may emerge gradually and in variable order. Symptoms include fatigue, weakness, weight or temperature change, amenorrhoea, infertility, sexual dysfunction, reduced body hair, impaired growth, polyuria, and inability to respond to stress.

Adrenocorticotrophic deficiency is immediately dangerous because cortisol support is lost, although aldosterone is usually preserved by the renin-angiotensin system. Thyroid replacement before recognising severe cortisol deficiency can precipitate crisis. Evaluate paired target and trophic hormones, electrolytes, osmolality, visual fields, and pituitary imaging. Replace deficient cortisol before thyroid hormone when both may be lacking, then address gonadal, growth, and water-balance needs.

Functional pituitary tumours

Adrenocorticotrophic tumours cause pituitary-dependent cortisol excess. Thyroid-stimulating tumours cause central hyperthyroidism with non-suppressed thyroid-stimulating hormone. Gonadotroph tumours are often clinically non-functioning and present through mass effect. Most pituitary adenomas are benign, but size, location, hormonal activity, growth, and treatment effects determine morbidity.

Management may use transsphenoidal surgery, receptor-targeted or synthesis-inhibiting drugs, radiotherapy, and hormone replacement. Long-term surveillance is required because recurrence, evolving deficiencies, visual change, and treatment complications can appear years later.

Endocrine investigation in practice

Begin with the phenotype and medication list. Decide whether hormone excess, deficiency, resistance, or incidental structural disease is suspected. Measure the peripheral hormone with its controlling hormone whenever possible. Repeat surprising results and consider timing, acute illness, pregnancy, binding proteins, supplements, heterophile antibodies, and cross-reactivity.

Imaging should usually follow biochemical reasoning. Incidental pituitary abnormalities are common and do not prove secretion. Conversely, very small functional lesions may not be obvious. Urgent assessment is needed for visual compromise, pituitary apoplexy, severe hyponatraemia, symptomatic hyponatraemia, or suspected adrenal crisis.

TTS module 2: Endocrine pattern recognition, assay interference, and pituitary emergencies

Hormone concentration is an integrated signal

A measured hormone concentration reflects secretion, pulsatility, binding, distribution, conversion, degradation, and clearance. The biological effect also depends on receptor abundance and downstream sensitivity. An abnormal number can therefore arise without excessive gland secretion, and normal concentration can coexist with resistance.

Free hormones are generally available to receptors and clearance, while bound hormone forms a circulating reservoir. Increased binding protein raises total concentration until free equilibrium is restored. Pregnancy and oestrogen can therefore raise total thyroid or cortisol measures without equivalent tissue excess. Protein loss or liver disease can lower totals.

Local conversion adds another layer. Deiodinases activate or inactivate thyroid hormone within tissues. Eleven-beta hydroxysteroid dehydrogenase controls local cortisol access to mineralocorticoid receptors. Aromatase converts androgens to oestrogens. Endocrine physiology is consequently both systemic and tissue-specific.

Paired tests localise the axis

The target hormone and its trophic hormone should be interpreted together. Low free thyroid hormone with high thyroid-stimulating hormone suggests primary gland failure. Low target hormone with a low or inappropriately normal trophic value suggests central failure. A normal-range trophic hormone can be pathological when the target is clearly deficient.

Hormone excess suppresses upstream drive when feedback is intact. High thyroid hormone with suppressed thyroid-stimulating hormone suggests primary thyrotoxicosis. High target hormone with non-suppressed trophic

hormone raises assay interference, resistance, or autonomous central secretion and requires confirmation before imaging.

Feedback patterns become unreliable during severe acute illness because secretion, binding, conversion, and clearance all change. Testing without a strong indication can identify transient non-thyroidal or stress patterns that provoke unnecessary treatment. Repeat assessment after recovery is often safer unless true endocrine failure is clinically urgent.

Pulses, cycles, and sample timing

Growth hormone is secreted in pulses and may be nearly undetectable between them, so a random value cannot diagnose deficiency. Insulin-like growth factor one integrates exposure but is modified by age, nutrition, liver function, kidney disease, and pregnancy.

Cortisol follows a circadian rhythm linked to sleep and waking. Shift work, critical illness, glucocorticoid exposure, oestrogen-related binding changes, and assay method alter interpretation. Midnight or waking samples answer different questions, and suppression or stimulation protocols require precise timing.

Gonadotropins vary across the menstrual cycle, puberty, menopause, and pulsatile gonadotropin-releasing hormone. Prolactin rises with sleep, stress, nipple stimulation, pregnancy, and selected medicines. Mild unexpected abnormalities should be repeated under controlled conditions before structural conclusions.

Assay interference

Immunoassays depend on antibody binding and can be distorted by heterophile antibodies, human anti-animal antibodies, autoantibodies, biotin, cross-reacting metabolites, and extreme analyte concentrations. Interference should be suspected when results conflict with physiology or with other tests.

High-dose biotin can produce falsely high results in some competitive assays and falsely low results in some sandwich assays. This can imitate thyrotoxicosis or obscure trophic-hormone elevation. Supplement history matters because patients may not consider vitamins to be medicines.

The hook effect occurs when an extremely high analyte saturates assay antibodies and generates a falsely modest result. A very large pituitary mass with only mild prolactin elevation should prompt sample dilution.

Macroprolactin contains immunoreactive prolactin complexes with limited biological activity and can cause laboratory elevation without the expected phenotype.

Confirmation strategies include repeating with a different platform, serial dilution, blocking reagents, measuring free rather than total hormone, or using mass spectrometry for selected steroids. Imaging should not precede resolution of a biochemically implausible pattern unless compression itself is urgent.

Dynamic tests ask a directional question

Stimulation testing assesses reserve. Synthetic adrenocorticotrophic hormone asks whether adrenal cortex can produce cortisol, although early central deficiency may retain adrenal responsiveness temporarily. Insulin-induced hypoglycaemia stresses growth and cortisol axes but carries neurological and cardiac risk and is reserved for supervised selected use.

Suppression testing assesses autonomy. Dexamethasone should suppress normal adrenocorticotrophic drive, while autonomous cortisol secretion escapes. Oral glucose suppresses growth hormone in healthy physiology, while acromegaly may fail to suppress. Preparation, absorption, interacting medicines, and assay thresholds affect validity.

Water-deprivation testing separates excessive intake from impaired vasopressin secretion or response, but dehydration and hypernatraemia can develop. Copeptin-based strategies provide a more stable surrogate of vasopressin in selected protocols. These investigations require specialist supervision.

Pituitary mass effects

The pituitary sits beneath the optic chiasm and beside cavernous sinus structures. Upward expansion classically compresses crossing nasal retinal fibres, reducing temporal visual fields. Lateral extension can impair ocular movements or facial sensation. Headache is common but non-specific.

Pituitary lesions may secrete one hormone while compressing other cell lines. The stalk can interrupt dopamine delivery and raise prolactin modestly. Growth, gonadal, thyroid, and adrenal deficits occur in variable order. Posterior involvement can produce diabetes insipidus, which is less typical of an uncomplicated adenoma and may suggest another lesion.

Formal visual-field testing detects deficits not obvious to confrontation. Imaging should describe size, suprasellar and cavernous extension, optic relationship, haemorrhage, and comparison over time. Incidental small lesions require hormonal assessment and proportionate surveillance rather than automatic surgery.

Pituitary apoplexy and hormone-order safety

Pituitary apoplexy is haemorrhage or infarction within the gland, often in a tumour. Sudden severe headache, vomiting, visual loss, ophthalmoplegia, reduced consciousness, meningism, hypotension, or hyponatraemia can occur. It can mimic subarachnoid haemorrhage or meningitis.

Acute adrenocorticotrophic loss makes cortisol deficiency immediately dangerous. Empirical parenteral glucocorticoid is given when haemodynamic or neurological features suggest crisis, after obtaining samples only if this causes no delay. Neurosurgical, endocrine, ophthalmological, electrolyte, and imaging assessment proceed together.

Thyroid replacement increases metabolic demand and cortisol clearance. In combined pituitary failure, giving thyroid hormone before securing cortisol can precipitate adrenal crisis. This ordering principle applies beyond apoplexy whenever severe central deficiency is possible.

Growth, prolactin, and gonadal consequences

Growth-hormone excess causes acral and soft-tissue growth, sleep apnoea, insulin resistance, cardiovascular remodelling, neuropathy, arthropathy, and tumour-related symptoms. Normalisation of insulin-like growth factor one does not instantly reverse established structural disease, so comorbidity surveillance continues.

Hyperprolactinaemia suppresses pulsatile gonadotropin-releasing hormone, lowering gonadal steroids. Consequences include infertility, menstrual change, sexual dysfunction, and bone loss. Dopamine-blocking medicines are

common causes; abrupt psychiatric-medication changes without coordination can be harmful.

Pituitary gonadal deficiency must be distinguished from functional suppression due to low energy availability, severe illness, stress, obesity, hyperprolactinaemia, menopause, or primary gonadal failure. Fertility goals influence treatment because sex-steroid replacement does not reproduce gonadotropin-driven gamete production.

Water balance and masked diabetes insipidus

Vasopressin release responds to tonicity and, at stronger thresholds, effective arterial volume. Cortisol deficiency can increase vasopressin and impair water excretion, producing hyponatraemia. Treating cortisol deficiency can unmask previously hidden diabetes insipidus and cause sudden polyuria and rising sodium.

Central diabetes insipidus responds to desmopressin, but excessive dosing with unrestricted water can cause hyponatraemia. Nephrogenic disease requires cause removal, solute reduction, and selected diuretic or prostaglandin approaches. Intake, urine volume, sodium, osmolality, and thirst must be reviewed together.

Incidental endocrine imaging should be managed from function outward. A structural lesion may be non-functioning, while a very small lesion can produce substantial hormone excess. Begin with symptoms, paired biochemical testing, medication and supplement review, then characterise anatomy. Repeat imaging is justified by growth risk and consequence rather than anxiety alone. Surveillance also watches for new compression or hormone loss, because a stable diameter does not guarantee stable pituitary function. Shared decisions should include fertility, pregnancy plans, vision, replacement burden, and the long-term effects of surgery or radiation.

The final pituitary interpretation should state which axis is excessive or deficient, whether the lesion is primary or central, how timing and binding affect the sample, whether assay interference is plausible, and whether vision, cortisol, sodium, or consciousness creates immediate urgency.

Retrieval prompts

- One. How do primary and central endocrine failure differ on paired testing?
- Two. Why can total and free hormone concentrations disagree?
- Three. What controls vasopressin release, and what does it do in the kidney?
- Four. Why is random growth hormone unsuitable for diagnosis?
- Five. How does hyperprolactinaemia impair reproduction?
- Six. Why must cortisol deficiency be considered before thyroid replacement?

Concise answers

- One. Primary failure lowers target hormone and raises trophic drive; central failure has inappropriately low trophic drive.
- Two. Binding-protein changes alter total hormone while free hormone can remain physiologically regulated.
- Three. Osmolality and effective volume regulate release; renal V-two signalling inserts collecting-duct aquaporins.
- Four. Secretion is strongly pulsatile and varies with sleep, stress, nutrition, and age.
- Five. Prolactin suppresses gonadotropin-releasing hormone and consequently gonadal steroid production and fertility.
- Six. Thyroid hormone increases metabolic demand and cortisol clearance, potentially precipitating adrenal crisis.

Source map

Original synthesis informed by Guyton and Hall, hypothalamic-pituitary physiology, growth, prolactin, and vasopressin; Robbins, pituitary pathology; Katzung and OpenStax Pharmacology, endocrine diagnostics and therapy; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and Talley and O'Connor, endocrine examination and visual-field assessment.

Chapter 24: Thyroid, Adrenal, Calcium, and Bone Physiology and Disease

Orientation

The thyroid controls metabolism, growth, and neurological development. The adrenal cortex regulates stress, vascular tone, electrolytes, and androgens. Parathyroid hormone and vitamin D coordinate calcium and phosphate with bone, kidney, and intestine. These systems interact: glucocorticoids alter bone, thyroid excess accelerates turnover, kidney disease disrupts mineral regulation, and endocrine failure can cause cardiovascular or neurological collapse.

Thyroid hormone synthesis and action

Thyroid follicles concentrate iodide through a sodium-iodide symporter. Thyroid peroxidase oxidises iodide and attaches it to tyrosines within thyroglobulin, then couples iodinated residues to form thyroxine and triiodothyronine. Colloid stores hormone extracellularly until endocytosis and proteolysis release it. Most secretion is thyroxine; peripheral deiodinases produce active triiodothyronine or inactive metabolites.

Thyroid hormones circulate strongly bound to proteins. Free hormone enters cells, binds nuclear receptors, and alters transcription affecting heat production, oxygen use, carbohydrate and lipid metabolism, cardiac responsiveness, gut motility, bone turnover, growth, and brain development. Thyroid-stimulating hormone supports synthesis and gland growth under negative feedback from free hormone.

Hypothyroidism

Primary hypothyroidism commonly follows autoimmune destruction, thyroid treatment, iodine imbalance, or drugs. Central hypothyroidism reflects pituitary or hypothalamic disease. Features include fatigue, cold intolerance, weight gain, constipation, dry skin, hair loss, bradycardia, slow relaxation of reflexes, menstrual disturbance, cognitive slowing, and hyponatraemia. Symptoms are non-specific; diagnosis relies on paired thyroid-stimulating hormone and free thyroxine interpreted in context.

Primary disease produces high thyroid-stimulating hormone and low free thyroxine. In central disease, thyroid-stimulating hormone is inappropriately low or normal. Acute illness, pregnancy, binding proteins, biotin, amiodarone, lithium, iodine, and assay interference can distort testing.

Levothyroxine replaces hormone, with dose determined by body composition, age, cardiac disease, pregnancy, absorption, interactions, and residual function. Iron, calcium, food, and several gastrointestinal conditions reduce absorption. Severe decompensation with hypothermia, hypoventilation, bradycardia, hypotension, altered consciousness, and metabolic disturbance constitutes myxoedema coma and requires urgent thyroid hormone, glucocorticoid cover until adrenal status is safe, supportive care, and treatment of triggers.

Thyrotoxicosis and structural thyroid disease

Thyrotoxicosis means excessive tissue exposure to thyroid hormone. Graves disease stimulates the thyroid-stimulating hormone receptor; toxic nodules secrete autonomously; thyroiditis releases stored hormone; exogenous hormone bypasses synthesis. Features include heat intolerance, sweating, weight loss, tremor, anxiety, proximal weakness, tachycardia, atrial fibrillation, increased stool frequency, and bone loss. Graves disease may add diffuse goitre, eye disease, and dermatopathy.

Low thyroid-stimulating hormone with high free hormones establishes overt primary thyrotoxicosis. Receptor antibodies and radionuclide uptake help identify cause: increased uptake suggests active synthesis, whereas low uptake suggests destructive thyroiditis or exogenous hormone. Ultrasound answers structural questions but does not replace biochemical classification.

Beta blockade controls adrenergic symptoms. Thionamides inhibit hormone synthesis; serious adverse effects include agranulocytosis and hepatic injury, so fever or sore throat requires urgent blood assessment. Radioiodine and surgery provide definitive treatment in selected disease. Thyroid storm combines severe thyrotoxicosis with fever, cardiovascular dysfunction, gastrointestinal or hepatic disturbance, and altered consciousness. Treatment rapidly blocks adrenergic effects, synthesis, release, peripheral conversion, and the precipitant.

Thyroid nodules require thyroid-stimulating hormone, ultrasound risk assessment, and selective needle sampling. Compression, suspicious nodes, radiation history, rapid growth, voice change, or high-risk imaging increase concern. Differentiated cancers retain follicular origin; calcitonin-producing medullary carcinoma can be hereditary.

Adrenal cortical organisation

The zona glomerulosa produces aldosterone under angiotensin two and potassium control. The zona fasciculata produces cortisol under adrenocorticotrophic hormone. The zona reticularis produces adrenal androgens. Cortisol supports vascular responsiveness, gluconeogenesis, protein and fat mobilisation, and modulation of immune and inflammatory activity. Secretion follows a circadian rhythm and rises during physiological stress.

Aldosterone acts in distal nephron principal cells to increase sodium reabsorption and potassium secretion, while promoting hydrogen secretion indirectly and in intercalated cells. Renin release responds to reduced renal perfusion, reduced macula-densa sodium chloride, and sympathetic stimulation. Volume expansion suppresses the system.

Adrenal insufficiency and crisis

Primary adrenal failure destroys the cortex, causing cortisol and aldosterone deficiency with high adrenocorticotrophic hormone and renin. Autoimmunity is common, but infection, haemorrhage, infiltration, genetic disease, and drugs occur. Central failure reduces adrenocorticotrophic drive; aldosterone is usually preserved. Chronic glucocorticoid exposure suppresses the axis and abrupt withdrawal can expose inadequate reserve.

Symptoms include fatigue, weight loss, nausea, abdominal pain, postural dizziness, weakness, and hypoglycaemia. Primary disease may cause hyperpigmentation, salt craving, hyperkalaemia, and marked volume depletion. Hyponatraemia occurs in both forms through impaired water excretion. Morning cortisol and adrenocorticotrophic hormone guide evaluation; stimulation testing assesses reserve when needed.

Adrenal crisis presents with hypotension or shock, vomiting, abdominal pain, fever, confusion, hypoglycaemia, and electrolyte disturbance. Do not delay treatment for testing: obtain samples if immediately possible, then give parenteral hydrocortisone, isotonic fluid, glucose when required, and treatment for the precipitant. People with adrenal insufficiency need sick-day dose education, emergency identification, and injectable rescue therapy.

Cortisol excess

Cushing syndrome is chronic pathological glucocorticoid excess, most often iatrogenic. Endogenous disease may be adrenocorticotrophic-dependent from pituitary or ectopic secretion, or independent from an adrenal lesion. Features include proximal weakness, easy bruising, broad violaceous striae, facial plethora, central fat gain, hypertension, diabetes, osteoporosis, infection, mood change, menstrual disturbance, and thrombosis risk.

Screening tests assess loss of normal suppression, excess free cortisol, or disrupted midnight nadir. Abnormal screening must be confirmed and exogenous exposure excluded before localisation with adrenocorticotrophic hormone and imaging. Random cortisol is unsuitable. Treatment targets the source, sometimes supported by steroidogenesis inhibitors or receptor blockade. After cure, suppressed normal tissue may require temporary glucocorticoid replacement.

Aldosterone disorders and adrenal masses

Primary aldosteronism causes hypertension, renin suppression, potassium wasting, and alkalosis, although potassium may remain normal. Screening uses the aldosterone-renin relationship under controlled conditions, followed by confirmation and localisation. Unilateral disease may be surgical; bilateral disease usually receives mineralocorticoid-receptor blockade.

Adrenal catecholamine tumours cause headache, sweating, palpitations, and hypertension. Biochemical testing precedes imaging; alpha blockade precedes beta blockade. Adrenal incidentalomas require assessment of imaging phenotype, growth, cortisol autonomy, catecholamines, and aldosterone when relevant.

Calcium regulation

Ionised calcium supports membrane excitability, contraction, secretion, coagulation, and intracellular signalling. Albumin binds a substantial fraction, so total calcium changes with albumin and pH; direct ionised measurement is preferable in critical uncertainty. Parathyroid hormone rises as ionised calcium falls. It increases renal calcium reabsorption, phosphate excretion, active vitamin D production, and bone resorption when persistently elevated.

Vitamin D is produced in skin or obtained from diet, modified in liver, and activated in kidney. It increases intestinal calcium and phosphate absorption and supports mineralisation. Fibroblast growth factor twenty-three promotes phosphate excretion and reduces active vitamin D. Magnesium is required for parathyroid secretion and action; severe deficiency can cause treatment-resistant hypocalcaemia.

Hypercalcaemia and hypocalcaemia

Primary hyperparathyroidism and malignancy cause most significant hypercalcaemia. Other causes include vitamin D excess, granulomatous activation, drugs, immobilisation, and endocrine disease. Manifestations include polyuria, dehydration, stones, constipation, weakness, confusion, shortened QT interval, and kidney injury. Measure parathyroid hormone early: an elevated or inappropriately normal value suggests parathyroid dependence; suppression redirects investigation.

Severe symptomatic hypercalcaemia requires volume restoration, rapid antiresorptive or calcium-lowering therapy according to cause, and monitoring of kidney and cardiac status. Long-term management treats the source. Hypocalcaemia causes perioral tingling, cramps, tetany, seizures, laryngospasm, and QT prolongation. Causes include hypoparathyroidism, vitamin D deficiency, kidney disease, pancreatitis, massive transfusion, and magnesium deficiency. Severe symptoms require intravenous calcium with

cardiac monitoring; chronic treatment corrects the underlying mineral and hormonal disorder.

Bone remodelling and osteoporosis

Osteoclasts resorb bone, osteoblasts form osteoid and support mineralisation, and osteocytes sense mechanical load. Remodelling repairs microdamage and releases mineral but excessive resorption weakens architecture. Peak bone mass, ageing, menopause, inactivity, low weight, nutrition, smoking, alcohol, endocrine disease, inflammation, kidney disease, and glucocorticoids shape fracture risk.

Osteoporosis is reduced bone strength predisposing to fragility fracture. Dual-energy X-ray absorptiometry estimates density, while clinical risk tools incorporate age and other factors. Vertebral fractures may be silent; hip fracture carries major mortality and loss of independence. Evaluate secondary causes when suggested by age, severity, history, or laboratory abnormalities.

Management combines resistance and weight-bearing activity, falls prevention, adequate calcium and vitamin D, risk-factor reduction, and pharmacotherapy. Antiresorptive drugs reduce breakdown; anabolic agents stimulate formation. Choice depends on fractures, kidney function, contraindications, adherence, and treatment sequence. Some agents cause rebound after abrupt withdrawal and require an exit strategy.

TTS module 2: Endocrine synthesis pathways, crisis physiology, and skeletal treatment sequencing

Thyroid synthesis and iodine effects

Thyroid hormone production requires iodide uptake, transport into follicular lumen, oxidation, organification onto thyroglobulin, coupling, storage, endocytosis, and proteolytic release. Thyroid peroxidase supports organification and coupling, while hydrogen peroxide supplies oxidative power. Disruption at different steps produces distinct drug and genetic effects.

Iodine has a biphasic relationship with the gland. Deficiency limits synthesis and can cause goitre through increased thyroid-stimulating hormone. A large acute iodine load temporarily suppresses organification and release in normal physiology. Autonomous tissue may instead use the substrate to produce

excess hormone, while susceptible autoimmune glands may become hypothyroid.

Most circulating triiodothyronine is generated outside the gland by deiodination of thyroxine. Severe illness, fasting, medicines, and organ disease alter conversion. Reverse triiodothyronine rises in some non-thyroidal illness, but measuring it rarely solves routine diagnostic uncertainty.

Thyroid tests in context

Thyroid-stimulating hormone is highly sensitive to small free-thyroxine changes when the pituitary axis is intact, making it an effective primary screen. It becomes unreliable for central disease, very recent treatment change, severe illness, assay interference, or selected drug effects.

After adjusting levothyroxine, thyroid-stimulating hormone takes weeks to reach a new steady state because hormone and feedback turnover are slow. Testing too early can provoke repeated dose changes and oscillation. Pregnancy increases hormone requirement in many treated patients and uses gestation-specific interpretation.

Absorption varies with fasting state, gastric acidity, coeliac disease, bowel surgery, iron, calcium, bile-acid binders, and some foods. Apparent treatment resistance should prompt administration and interaction review before major escalation. Inconsistent use followed by intense dosing just before testing can produce discordant free hormone and trophic hormone.

Thyrotoxic mechanisms and emergencies

Thyrotoxicosis describes excessive hormone action regardless of source. Hyperthyroidism specifically means increased gland synthesis. Thionamides help synthesis-driven disease but cannot stop release from destructive thyroiditis. Radionuclide uptake, antibodies, tenderness, exposure, and imaging separate mechanisms.

Thyroid hormone increases beta-adrenergic responsiveness, heat production, gut motility, bone turnover, and cardiac workload. Atrial fibrillation, high-output failure, angina, muscle wasting, and osteoporosis reflect systemic exposure. Older patients may present with weight loss, apathy, or cardiac deterioration rather than obvious tremor.

Thyroid storm is a clinical syndrome of severe decompensation, not a hormone threshold. Fever, tachyarrhythmia, heart failure, gastrointestinal or hepatic dysfunction, and neurological change occur after infection, surgery, medication interruption, iodine exposure, or other stress. Treatment sequence blocks adrenergic effects, new synthesis, hormone release after synthesis blockade, peripheral conversion, and the trigger.

Myxoedema coma similarly reflects decompensated hypothyroid physiology, often without actual coma. Hypothermia, hypoventilation, bradycardia, hypotension, hyponatraemia, hypoglycaemia, and reduced consciousness can worsen after sedatives, cold, infection, or cardiac events. Empirical glucocorticoid protects against unrecognised adrenal failure while thyroid and supportive therapy begin.

Adrenal steroid pathways

All adrenal steroids derive from cholesterol, but enzyme expression differs by zone. The zona glomerulosa lacks key cortisol-producing capacity and responds primarily to angiotensin and potassium. The zona fasciculata produces cortisol under adrenocorticotrophic hormone. The reticularis contributes adrenal androgens.

Inherited enzyme defects redirect precursors into alternative pathways. Cortisol deficiency raises adrenocorticotrophic hormone and causes adrenal hyperplasia. Depending on the block, mineralocorticoid activity and androgen production may rise or fall. Salt loss, hypertension, virilisation, or undervirilisation provide mechanistic clues.

Cortisol circulates bound mainly to corticosteroid-binding globulin and albumin. Oestrogen raises binding protein and total cortisol. Critical illness, liver disease, and protein loss lower it. Free exposure and tissue metabolism may diverge from total measurements.

Adrenal insufficiency and stress dosing

Primary adrenal failure removes cortisol and often aldosterone. Central failure removes cortisol but usually preserves aldosterone because renin-angiotensin regulation remains. Primary disease therefore more often produces hyperkalaemia, salt craving, pronounced volume loss, and high renin.

Chronic exogenous glucocorticoids suppress hypothalamic and pituitary drive. Risk depends on dose, duration, timing, potency, route, and individual response; inhaled, topical, injected, and interacting medicines can contribute. Recovery is variable, so abrupt cessation after meaningful exposure can be dangerous.

During illness, normal physiology increases cortisol. A patient with adrenal failure cannot mount that response and needs planned stress dosing. Vomiting can prevent oral absorption, requiring parenteral rescue. Education, emergency identification, spare medication, and clear perioperative planning are as important as the maintenance dose.

Adrenal crisis causes vasodilation, reduced catecholamine responsiveness, volume loss, hypoglycaemia, and water-retaining hormone excess. Treatment should not await definitive testing. Draw cortisol and adrenocorticotropic hormone if immediately feasible, then give hydrocortisone, suitable fluid, glucose, and cause-specific care.

Cortisol and aldosterone excess

Glucocorticoid excess redistributes substrate and connective tissue. Protein catabolism causes proximal weakness and thin skin. Bone formation falls and resorption rises. Glucose production, appetite, vascular responsiveness, coagulation, and infection susceptibility increase. The pattern and progression distinguish pathological exposure from common non-specific obesity.

Testing for cortisol excess first establishes autonomous exposure, then measures adrenocorticotropic hormone to localise dependence. Imaging before biochemical confirmation finds incidental lesions and can misdirect care. Cyclical disease may require repeated testing when phenotype is strong.

Primary aldosteronism increases distal sodium retention, potassium and hydrogen loss, and cardiovascular injury beyond pressure alone. Potassium can remain normal because intake and kidney adaptation compensate. Renin suppression is central to screening, but posture, sodium, potassium, medicines, kidney function, and assay conditions affect the ratio.

Catecholamine-secreting tumours create episodic or sustained receptor stimulation. Biochemical confirmation precedes imaging because adrenal

nodules are common. Alpha blockade is established before beta blockade to avoid unopposed vasoconstriction. Volume restoration is often needed before surgery because chronic vasoconstriction reduces circulating volume.

Calcium sensing and phosphate balance

The parathyroid calcium-sensing receptor detects ionised calcium. Falling calcium increases parathyroid hormone secretion. The kidney reabsorbs calcium, excretes phosphate, and activates vitamin D. Active vitamin D increases intestinal mineral absorption. Bone supplies calcium and phosphate through regulated remodelling.

Phosphate and calcium must be considered together. Kidney failure retains phosphate and reduces vitamin D activation, promoting secondary hyperparathyroidism. Long-standing stimulation can become partly autonomous. Fibroblast growth factor twenty-three rises early to increase phosphate excretion but reduces active vitamin D.

Magnesium is needed for parathyroid release and action. Mild reduction can stimulate secretion, while severe deficiency paradoxically suppresses it and causes resistance. Calcium replacement may fail until magnesium is corrected.

Hypercalcaemia and hypocalcaemia logic

The first hypercalcaemia branch point is parathyroid hormone. If it is elevated or inappropriately normal, primary hyperparathyroidism, tertiary disease, lithium effect, and familial calcium-sensing variants are considered. If suppressed, malignancy, vitamin D pathways, granulomatous disease, medicines, endocrine disease, and immobilisation become more likely.

Volume depletion worsens hypercalcaemia by reducing filtration and calcium excretion. Isotonic fluid can restore renal handling when appropriate, but heart and kidney reserve limit rate. Antiresorptive agents act over different timescales, while calcitonin is faster but transient. Dialysis is reserved for selected severe or refractory states.

Hypocalcaemia should be confirmed with ionised calcium when albumin or pH makes total concentration unreliable. Acute symptoms reflect membrane excitability and cardiac repolarisation. Determine magnesium, phosphate,

parathyroid hormone, kidney function, vitamin D, pancreatitis, transfusion, and recent neck surgery.

Bone strength and treatment sequence

Bone density measures mineral quantity projected through an image, while strength also depends on geometry, cortical porosity, trabecular architecture, turnover, microdamage, and falls. A fragility fracture can establish high risk even when density is not in the lowest range.

Antiresorptive therapy reduces osteoclast activity. Bisphosphonates bind bone and have persistent effects; denosumab suppresses resorption through a reversible antibody pathway. Abrupt denosumab discontinuation can cause rebound turnover and vertebral fractures, so a planned subsequent antiresorptive strategy is essential.

Anabolic therapy stimulates formation and is used for selected very-high-risk patients. Sequence matters because previous and subsequent agents influence gains and preservation. Kidney function, calcium, dental issues, cancer history, adherence, dosing route, and cost shape selection.

Fracture prevention includes muscle strength, balance, vision, footwear, home hazards, sedative reduction, nutrition, and treatment of endocrine causes. Monitoring should assess fractures and adherence, not density alone. The final endocrine-bone plan links hormone mechanism to present organ risk, treatment sequence, and a safe transition whenever therapy stops.

Retrieval prompts

One. How does thyroid hormone synthesis depend on iodide and thyroid peroxidase?

Two. What distinguishes primary from central adrenal failure?

Three. Why should thyroid storm and adrenal crisis be treated before complete confirmation?

Four. How does parathyroid hormone raise ionised calcium?

Five. Why can total calcium misrepresent biological calcium status?

Six. What determines fracture risk beyond bone-density measurement?

Concise answers

One. Iodide is concentrated, oxidised, organified onto thyroglobulin, and coupled into thyroid hormones by thyroid peroxidase.

Two. Primary failure loses cortisol and aldosterone with high trophic drive; central failure usually preserves aldosterone.

Three. Both can deteriorate rapidly and treatment delay is more dangerous than appropriately managed empirical therapy.

Four. It increases renal calcium retention, phosphate loss, vitamin D activation, and sustained bone resorption.

Five. Albumin binding and pH alter total concentration without equivalent change in ionised calcium.

Six. Age, prior fractures, falls, architecture, secondary disease, medication exposure, and clinical context modify risk.

Source map

Original synthesis informed by Guyton and Hall, thyroid, adrenal, calcium, and bone physiology; Robbins, endocrine and skeletal pathology; Katzung and OpenStax Pharmacology, endocrine replacement, suppression, and bone therapies; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and Talley and O'Connor, endocrine and musculoskeletal examination.

Chapter 25: Reproductive Physiology, Pregnancy, and Lactation

Orientation

Reproduction coordinates hypothalamic, pituitary, gonadal, anatomical, and developmental processes. Meiosis produces gametes, fertilisation restores diploidy, and implantation connects embryo with maternal circulation. Pregnancy requires cardiovascular, respiratory, renal, haematological, and metabolic adaptation. Reproductive care includes puberty, fertility, contraception, sexual function, pregnancy complications, menopause, and medication safety.

Sexual differentiation and puberty

Chromosomal sex is established at fertilisation, but gonadal and anatomical differentiation depend on gene expression and hormone action. Testicular development produces anti-Müllerian hormone and androgens that direct internal and external differentiation. In their absence, ovarian and female tract development proceeds through alternative pathways. Variations can occur at chromosomal, gonadal, hormonal, receptor, or anatomical levels and require precise, respectful assessment rather than assumptions based on appearance.

During childhood, hypothalamic gonadotropin-releasing hormone activity is restrained. Puberty begins when pulsatile activity increases, driving luteinising and follicle-stimulating hormones. Gonadal steroids produce secondary sexual characteristics, growth acceleration, bone maturation, and reproductive capacity. Adrenal androgen production contributes to pubic and axillary hair but is partly independent. Early or delayed puberty may reflect normal variation, central activation, gonadal failure, chronic disease, nutrition, exercise, genetic conditions, or structural pathology.

Ovarian cycle

Follicle-stimulating hormone recruits ovarian follicles. Granulosa cells convert theca-derived androgens to oestradiol. Rising oestradiol initially restrains gonadotropins, but sustained high levels generate positive feedback and a luteinising-hormone surge. This triggers ovulation and luteinisation. The corpus luteum produces progesterone and oestradiol, transforming the endometrium into a secretory state suitable for implantation.

Without pregnancy, luteal regression lowers progesterone and oestradiol, spiral arteries constrict, and the functional endometrium sheds. Cycle length varies mainly through the follicular phase; the luteal phase is more consistent. Cervical mucus becomes thin and penetrable near ovulation, then thickens under progesterone. Basal temperature rises slightly after ovulation.

Amenorrhoea can result from pregnancy, hypothalamic suppression, pituitary disease, ovarian failure, endocrine disturbance, uterine outflow problems, or medications. Heavy or irregular bleeding may reflect ovulatory dysfunction, structural uterine disease, coagulopathy, endometrial pathology, pregnancy complications, or treatment. Pregnancy testing is an early step whenever biologically possible.

Testicular function and sexual response

Luteinising hormone stimulates Leydig-cell testosterone production. Follicle-stimulating hormone and intratesticular testosterone support Sertoli cells and spermatogenesis. Inhibin contributes feedback. Sperm mature in the epididymis and combine with secretions from seminal vesicles, prostate, and other glands. Production takes weeks and is vulnerable to heat, toxins, systemic illness, hormonal suppression, genetic defects, obstruction, and testicular injury.

Erection depends mainly on parasympathetic nitric-oxide signalling, smooth-muscle relaxation, arterial inflow, and venous occlusion. Ejaculation uses sympathetic and somatic pathways. Sexual dysfunction may arise from vascular, neurological, endocrine, medication, pain, psychological, relational, or structural factors. Assessment should separate desire, arousal, orgasm, ejaculation, pain, fertility, and satisfaction rather than using one label.

Fertilisation, implantation, and placenta

Fertilisation usually occurs in the uterine tube after capacitated sperm penetrate the oocyte coverings. The conceptus divides while travelling toward the uterus, forming a blastocyst that implants into receptive endometrium. Trophoblast invades maternal tissue and contributes to the placenta. Human chorionic gonadotropin preserves the corpus luteum until placental steroid production is established.

The placenta exchanges gases, nutrients, water, antibodies, wastes, hormones, and drugs without normally mixing the circulations. Exchange depends on flow, surface area, membrane properties, metabolism, and gradients. It produces chorionic gonadotropin, progesterone, oestrogens, placental lactogen, and other signals. It is an active endocrine and metabolic organ, not a passive filter.

Abnormal implantation outside the uterine cavity can rupture and haemorrhage. Defective placental invasion and vascular remodelling contribute to fetal growth restriction and pre-eclampsia. Placental separation before birth causes painful bleeding and fetal compromise; implantation over the cervical opening causes placenta praevia and typically painless bleeding. Antepartum bleeding requires urgent location and stability assessment.

Maternal physiological adaptation

Plasma volume, red-cell mass, cardiac output, and heart rate rise during pregnancy. Plasma expansion exceeds red-cell expansion, producing physiological haemodilution. Systemic vascular resistance falls, and blood pressure often decreases mid-pregnancy. Venous stasis and coagulation changes increase thromboembolic risk. Supine uterine compression can reduce venous return late in pregnancy.

Ventilation rises under progesterone influence, lowering carbon dioxide and bicarbonate with mild respiratory alkalosis. Oxygen consumption increases and diaphragmatic elevation reduces reserve. Glomerular filtration and renal blood flow rise, lowering usual creatinine. Sodium and water are retained to support expansion. Urinary stasis increases infection risk.

Insulin resistance rises later in pregnancy to preserve nutrients for the fetus; pancreatic secretion normally compensates. Failure produces gestational

diabetes. Thyroid-binding proteins rise, altering total hormone levels. Gastrointestinal motility slows, reflux and constipation are common, and gallbladder emptying decreases. These normal adaptations change laboratory reference ranges, pharmacokinetics, and tolerance of acute illness.

Fetal circulation and transition at birth

Oxygenated placental blood enters through the umbilical vein. The ductus venosus partly bypasses the liver, the foramen ovale directs blood from right to left atrium, and the ductus arteriosus connects pulmonary artery to aorta. High pulmonary resistance and low placental resistance maintain these shunts.

At birth, lung expansion lowers pulmonary resistance and cord clamping raises systemic resistance. Increased left atrial pressure functionally closes the foramen ovale; oxygen and reduced prostaglandins promote ductal closure. Failure or premature closure of fetal channels can cause major disease. Some congenital heart lesions depend on a patent ductus until definitive treatment.

Labour and postpartum physiology

Labour emerges from increasing uterine excitability, cervical ripening, prostaglandins, oxytocin responsiveness, fetal and placental signals, and mechanical stretch. Positive feedback between cervical stretch and oxytocin strengthens contractions. The stages encompass cervical dilation, birth, and placental delivery. Effective contractions, fetal position, pelvic dimensions, and maternal effort determine progression.

After placental separation, uterine contraction compresses vessels and limits bleeding. Uterine atony is a major cause of postpartum haemorrhage; retained tissue, trauma, and coagulation failure are other causes. Pregnancy-related hypertensive disease can worsen after delivery, and thrombotic, infectious, psychiatric, and cardiomyopathic complications remain possible postpartum.

Lactation

Oestrogen and progesterone promote breast development during pregnancy while restraining full milk secretion. Their fall after placental delivery permits prolactin-driven synthesis. Suckling reduces dopamine, increases prolactin, and triggers oxytocin-mediated milk ejection. Frequent effective milk removal maintains supply through neural and local mechanisms.

Early milk contains concentrated immune and nutritional components. Mature milk changes during feeds and over time to match developmental needs. Difficult feeding may reflect positioning, attachment, pain, infection, infant anatomy, neurological function, inadequate transfer, endocrine disease, retained placental tissue, medications, or insufficient stimulation. Support should assess both parent and infant and respect informed feeding choices.

Contraception and fertility

Contraception may prevent ovulation, thicken cervical mucus, alter endometrium, impair sperm transport, or provide a physical barrier. Long-acting reversible methods are highly effective because they reduce user dependence. Combined hormonal methods increase thrombotic risk and are unsuitable in selected vascular, migraine, smoking, postpartum, and other contexts. Progestogen-only, intrauterine, barrier, fertility-awareness, emergency, and permanent methods have distinct benefits and limitations.

Infertility is failure to achieve pregnancy over a context-dependent interval and warrants earlier assessment with age, cycle abnormality, known disease, or male risk factors. Evaluate both partners: ovulation, ovarian reserve when relevant, uterine and tubal anatomy, semen, sexual timing, medications, infection, and systemic health. Treatment ranges from education and cause-specific therapy to ovulation induction, surgery, insemination, and assisted reproduction.

Menopause and reproductive ageing

Loss of ovarian follicular function lowers oestradiol and inhibin, raising gonadotropins. Menopause is diagnosed clinically after sustained cessation of menstruation when no competing cause exists. Vasomotor symptoms, sleep disturbance, genitourinary atrophy, sexual symptoms, mood effects, and accelerated bone loss vary widely. Cardiovascular risk changes with age and metabolic factors rather than hormone concentration alone.

Menopausal hormone therapy is the most effective treatment for vasomotor symptoms and prevents bone loss while used, but benefit-risk depends on age, time since menopause, uterus, thrombosis, cancer history, cardiovascular disease, route, dose, and goals. Oestrogen without endometrial protection increases cancer risk in a person with a uterus. Non-hormonal and local treatments provide alternatives.

Pregnancy prescribing and teratology

Medication decisions balance maternal disease, fetal development, placental transfer, dose, timing, and alternatives. Untreated illness can be more dangerous than treatment. Early exposure may disrupt organ formation; later exposure can affect growth, physiology, labour, or neonatal adaptation. Population data are often incomplete, so categorical labels oversimplify risk.

Review medicines, supplements, alcohol, nicotine, and other exposures before conception and during pregnancy. Folate reduces neural-tube risk; selected medicines require higher-dose planning. Avoid abrupt withdrawal of essential treatment. Lactation decisions consider milk transfer, infant absorption, prematurity, toxicity, and feed timing.

TTS module 2: Reproductive timing, placental exchange, and adaptation across pregnancy

Pulsatility and reproductive competence

Gonadotropin-releasing hormone must be pulsatile to sustain luteinising and follicle-stimulating hormone secretion. Pulse frequency and amplitude change across development and the ovarian cycle. Continuous exposure desensitises pituitary receptors, a property used therapeutically after an initial stimulatory phase.

Energy availability, stress, sleep, exercise, leptin, prolactin, and illness influence hypothalamic drive. Functional hypothalamic amenorrhoea reflects adaptive suppression when energy or stress signals are unsuitable for reproduction. Gonadotropins and oestradiol fall, with consequences for fertility, bone, cardiovascular physiology, and wellbeing.

Puberty requires reactivation of pulsatile signalling. Central precocious puberty follows early hypothalamic activation, while peripheral sex-steroid exposure bypasses central drive. Delayed puberty can reflect constitutional timing, chronic disease, undernutrition, primary gonadal failure, or central deficiency. Growth pattern and bone age help separate mechanisms.

Follicle selection and luteal physiology

Early-cycle follicle-stimulating hormone recruits a cohort, but one follicle usually becomes dominant through greater sensitivity and local signalling. Theca cells produce androgen under luteinising hormone, and granulosa cells convert it to oestradiol under follicle-stimulating hormone. This two-cell cooperation explains why both gonadotropins matter.

Sustained oestradiol from the dominant follicle switches feedback and triggers the luteinising-hormone surge. Ovulation releases the oocyte and transforms the follicle into a progesterone-producing corpus luteum. Progesterone organises secretory endometrium, thickens cervical mucus, and raises basal temperature.

Implantation requires synchrony between embryo development and a receptive endometrial window. Human chorionic gonadotropin rescues the corpus luteum until placental steroid production is sufficient. Inadequate timing, uterine pathology, chromosomal abnormality, inflammation, or anatomical transport problems can disrupt the sequence.

Ovarian reserve estimates remaining follicular pool and response to stimulation rather than guaranteeing natural fertility or oocyte quality. Age strongly influences chromosomal competence. Results must be explained without converting a probability marker into a deterministic prognosis.

Spermatogenesis and androgen regulation

Spermatogenesis proceeds from stem cells through meiosis and maturation within seminiferous tubules. Sertoli cells create a specialised environment and

blood-testis barrier. Intratesticular testosterone concentration is much higher than blood concentration and is required even when circulating testosterone seems adequate.

Exogenous androgen suppresses gonadotropins and intratesticular testosterone, reducing sperm production and sometimes causing azoospermia. It cannot be assumed to improve fertility. Recovery after cessation varies with duration, dose, baseline function, age, and additional agents.

Semen analysis samples concentration, motility, morphology, volume, and related features but varies with illness, abstinence interval, collection, fever, and laboratory technique. One abnormal result should be repeated when appropriate. Severe abnormalities prompt endocrine, genetic, anatomical, and exposure assessment.

Erectile function integrates vascular inflow, endothelial nitric oxide, autonomic nerves, smooth muscle, venous occlusion, hormones, sensation, and psychological context. Erectile dysfunction can precede overt vascular disease. Medication and cardiovascular review should accompany sexual assessment.

Placental exchange is flow limited and selective

The placenta maintains separate maternal and fetal circulations while bringing them into close exchange. Oxygen transfer depends on maternal arterial content, uterine flow, placental surface, membrane distance, fetal haemoglobin, and umbilical flow. Carbon dioxide diffuses readily in the opposite direction.

Glucose crosses by facilitated transport and generally follows maternal concentration. Amino acids and ions use active or facilitated systems. Maternal immunoglobulin G crosses through receptor-mediated transport, especially later in pregnancy. Many drugs cross according to size, ionisation, lipid solubility, binding, transporters, and placental metabolism.

The placenta is not an impermeable shield. Maternal hypoxaemia, anaemia, hypotension, vascular disease, smoking, infection, inflammation, and placental lesions can reduce fetal delivery. Conversely, treating maternal disease often benefits the fetus more than withholding necessary therapy.

Spiral-artery remodelling normally converts high-resistance maternal vessels into high-flow channels. Inadequate remodelling contributes to placental ischaemia and release of factors that produce maternal endothelial dysfunction. Hypertension, proteinuria, liver injury, thrombocytopenia, neurological symptoms, fetal growth restriction, and placental separation can follow.

Maternal cardiovascular and respiratory reserve

Cardiac output rises through greater stroke volume and heart rate, while systemic resistance falls. Plasma volume expands substantially. This supports uteroplacental flow and provides reserve for delivery blood loss, but it can expose stenotic valves, pulmonary hypertension, cardiomyopathy, or aortopathy.

Supine vena-caval compression reduces venous return late in pregnancy. Left lateral displacement can improve circulation during symptoms or resuscitation. Labour adds autotransfusion from uterine contraction, pain-related sympathetic activity, and abrupt postpartum volume shifts. Cardiovascular complications can emerge after birth rather than ending with delivery.

Progesterone increases ventilation and lowers carbon dioxide. Bicarbonate falls through renal compensation. A carbon dioxide value normal for a non-pregnant adult can indicate inadequate ventilation in pregnancy. Reduced functional residual capacity and increased oxygen use make apnoea and respiratory disease less well tolerated.

Renal, haematological, and metabolic adaptation

Renal blood flow and filtration rise early, lowering creatinine and urea. Glycosuria can occur at lower blood glucose because filtered load and tubular handling change. Dilatation and urinary stasis increase infection risk. Protein excretion changes modestly, but substantial new protein requires assessment in context.

Plasma expansion exceeds red-cell expansion, creating physiological haemodilution. Iron demand rises, and true deficiency remains common. Coagulation shifts toward thrombosis while fibrinolysis and natural

anticoagulants change. Venous stasis and vascular injury complete a pregnancy-specific thrombotic risk.

Insulin resistance rises later through placental and maternal signals. This supplies fetal nutrients but can exceed beta-cell reserve. Gestational diabetes predicts future metabolic risk for parent and child and warrants postpartum reassessment even when glucose normalises after delivery.

Medication distribution and clearance change through plasma volume, albumin, body water, fat, filtration, hepatic enzymes, and placental transfer. A dose appropriate outside pregnancy may under- or overexpose. Pregnancy-specific evidence and specialist guidance should direct high-risk medicines.

Fetal circulation and birth transition

Umbilical venous blood has the highest fetal oxygen content. Preferential streaming directs relatively oxygenated blood through the foramen ovale toward left heart and brain. Less oxygenated systemic venous blood enters the right ventricle and largely bypasses high-resistance lungs through the ductus arteriosus.

At birth, aeration and oxygen lower pulmonary resistance, increasing pulmonary blood flow and left-atrial pressure. Cord separation raises systemic resistance and removes placental prostaglandin influence. Pressure closes the foramen ovale functionally, while the ductus arteriosus constricts.

Prematurity, hypoxia, acidosis, lung disease, congenital anatomy, and prostaglandin exposure alter transition. Some cardiac lesions require the ductus to remain open for pulmonary or systemic flow. Prostaglandin therapy can preserve it while definitive assessment occurs.

Labour, haemorrhage, and postpartum change

Labour progress depends on uterine power, fetal position and size, pelvic anatomy, cervical response, and time. Excessive stimulation can impair placental perfusion or rupture a scarred uterus, while inadequate contraction prolongs labour and raises infection and haemorrhage risk.

Postpartum haemorrhage is organised by tone, tissue, trauma, and thrombin. Uterine atony is common; retained placenta prevents contraction, genital

trauma bleeds despite a firm uterus, and coagulation failure worsens all sources. Resuscitation, uterotonic treatment, examination, blood products, procedural control, and escalation proceed together.

Placental delivery abruptly lowers progesterone and oestrogen, enabling prolactin-driven milk secretion. Oxytocin ejects milk but stress and pain can inhibit the reflex. Milk supply follows effective removal, so transfer, latch, infant feeding strength, breast anatomy, endocrine function, and frequency should be assessed together.

Reproductive ageing gradually reduces follicle number and changes cycle predictability before final menstruation. Oestradiol can fluctuate substantially during transition, so one hormone value may not define status. Loss of ovarian function accelerates bone turnover and changes genitourinary tissue, while vascular risk remains governed by age and broader factors. Symptom treatment should separate systemic vasomotor needs from local genitourinary needs and account for uterus, thrombosis, cancer, migraine, cardiovascular disease, and informed preference.

The final reproductive synthesis should state hypothalamic drive, gonadal function, anatomy, fertility goal, pregnancy possibility, vascular and thrombotic context, and medication exposure. Pregnancy assessment must always consider both normal adaptation and the reduced reserve it creates during acute illness.

Retrieval prompts

One. What creates the mid-cycle luteinising-hormone surge?

Two. How do gonadotropins divide control of testicular function?

Three. Why is the placenta considered an active organ?

Four. Which pregnancy adaptations alter laboratory interpretation?

Five. What circulatory changes occur at birth?

Six. Which mechanisms maintain lactation?

Concise answers

One. Sustained high oestradiol switches to positive feedback and triggers the surge.

Two. Luteinising hormone supports Leydig testosterone; follicle-stimulating hormone and intratesticular testosterone support Sertoli cells and sperm production.

Three. It performs selective exchange, metabolism, immune transfer, and extensive endocrine signalling.

Four. Haemodilution, increased filtration, respiratory alkalosis, binding-protein changes, insulin resistance, and altered pharmacokinetics.

Five. Lung expansion lowers pulmonary resistance, cord clamping raises systemic resistance, and fetal shunts begin functional closure.

Six. Prolactin drives synthesis, oxytocin causes ejection, and frequent effective milk removal preserves supply.

Source map

Original synthesis informed by Guyton and Hall, gonadal physiology, pregnancy, fetal circulation, and lactation; Robbins, reproductive and placental pathology; Katzung and OpenStax Pharmacology, contraception, fertility, and pregnancy prescribing; OpenStax Anatomy and Physiology, Biology, and Medical-Surgical Nursing; and Talley and O'Connor, reproductive and pregnancy assessment.

Chapter 26: Endocrine and Reproductive Examination and Investigation

Orientation

Endocrine disorders often evolve slowly and affect several systems. Diagnosis requires recognising physiology, measuring paired hormones, and separating gland disease from medication, illness, pregnancy, binding proteins, and assay interference. Reproductive assessment additionally requires privacy, consent, trauma-informed communication, and attention to fertility, sexual function, pregnancy possibility, and safeguarding.

Endocrine history

Begin with time course and function. Ask about weight trajectory, appetite, heat or cold tolerance, sweating, tremor, thirst, urine volume, bowel pattern, energy, sleep, muscle strength, cramps, fractures, skin and hair change, headaches, vision, mood, cognition, and episodic symptoms. Clarify whether weight change reflects fat, fluid, muscle, intake, malabsorption, or catabolism.

Explore pressure changes, palpitations, dizziness, collapse, pigmentation, salt craving, nausea, abdominal pain, recurrent infection, bruising, wound healing, kidney stones, and altered shoe or ring size. Document menstrual pattern, libido, erections, ejaculation, vaginal symptoms, galactorrhoea, fertility, pregnancy history, and pubertal development where relevant. Symptoms such as fatigue or weight gain are weak in isolation but meaningful in coherent clusters.

Record all prescribed, injected, inhaled, topical, non-prescribed, and complementary agents. Glucocorticoids, thyroid hormone, amiodarone, lithium, antipsychotics, opioids, sex steroids, supplements containing biotin, anabolic agents, and immune therapies can create or conceal endocrine syndromes. Ask about recent contrast, surgery, radiation, head injury, postpartum haemorrhage, autoimmune disease, cancer, infection, shift work, eating patterns, exercise intensity, and family syndromes.

General endocrine examination

Observe body habitus, distribution of fat and muscle, movement, behaviour, voice, and apparent distress. Measure temperature, pulse rhythm, pressure lying and standing when appropriate, respiratory pattern, weight, height, and waist contextually. Look for pallor, jaundice, dehydration, oedema, pigmentation, bruises, striae, acne, hirsutism, hair loss, vitiligo, and altered skin texture.

Proximal weakness is tested by rising from a chair or lifting arms against resistance. Tremor, reflex relaxation, visual fields, eye movements, peripheral sensation, and gait can reveal thyroid, pituitary, calcium, diabetes, or cortisol disease. Examine the cardiovascular system for rate, rhythm, flow murmurs, failure, and pressure effects; examine abdomen for organ enlargement, masses, striae, injections, and tenderness.

No single feature proves an endocrine diagnosis. Rounded face, obesity, tiredness, or sparse hair are common outside hormone disease. Greater weight belongs to discriminatory findings such as broad violaceous striae with proximal weakness and bruising, true acral enlargement, persistent hyperpigmentation with postural hypotension, or objective visual-field loss.

Thyroid and pituitary examination

Inspect the neck at rest and during swallowing. Note diffuse or nodular enlargement, asymmetry, scars, venous congestion, and movement. Palpate from a comfortable position, identifying gland size, consistency, nodules, tenderness, mobility, and cervical nodes. Listen for a bruit only when increased flow is suspected. Retrosternal extension may cause pressure symptoms even when little is palpable.

Assess thyroid effects beyond the neck: pulse, tremor, warm or dry skin, eye signs, lid movement, reflexes, proximal power, and heart failure. Graves orbitopathy requires visual acuity, fields, colour vision, pupils, ocular movement, corneal exposure, and urgent review for optic neuropathy.

Pituitary examination includes visual fields by confrontation, acuity, pupils, ocular movements, cranial nerves, and evidence of hormone excess or loss. Headache with sudden visual or ocular-motor change, vomiting, hypotension, or altered consciousness suggests apoplexy. Acromegaly assessment notes

acral and facial change, jaw, tongue, voice, sweating, neuropathy, pressure, cardiac disease, visual fields, and sleep-apnoea features.

Diabetes and foot assessment

Diabetes review should examine glucose patterns, hypoglycaemia, ketosis risk, medications, injection or device technique, eating, activity, driving, illness plans, and psychosocial burden. Assess pressure, weight trajectory, injection sites, hydration, vascular disease, eyes, kidney surveillance, and autonomic symptoms.

Foot assessment inspects skin, nails, callus, deformity, footwear, infection, ulceration, and prior amputation. Palpate pulses and assess temperature, capillary return, and signs of ischaemia. Test protective sensation with a calibrated monofilament at recommended sites and add vibration or other modalities when useful. Risk classification should lead to education, footwear, podiatry, vascular review, and follow-up rather than documentation alone.

Reproductive and sexual history

Use neutral language and establish the patient's terms for body parts, partners, and identity. Explain why questions matter. Ask about cycle timing and volume, pain, intermenstrual or postcoital bleeding, discharge, pelvic pressure, menopausal symptoms, sexual practices, contraception, pregnancy intention, infection risk, function, coercion, and safety. Do not assume anatomy, fertility, sexual activity, or pregnancy risk from identity or relationship status.

Obstetric history records every pregnancy and outcome, gestation, mode of birth, complications, birth weights, haemorrhage, hypertensive disease, diabetes, thrombosis, infection, neonatal outcomes, and postpartum mental health. In a current pregnancy ask about pain, bleeding, fluid loss, fetal movement, headache, visual symptoms, epigastric pain, swelling, breathlessness, fever, urinary symptoms, and contractions.

Male reproductive history includes testicular development, trauma, torsion, infection, surgery, undescended testes, genital symptoms, erections, ejaculation, fertility, systemic disease, heat or toxin exposure, and androgen

use. Semen varies substantially; abnormal results require appropriate repetition rather than premature conclusions.

Intimate examination

Perform only when clinically indicated, with explicit consent, privacy, explanation, appropriate exposure, and a chaperone according to preference and policy. The patient may pause or stop at any time. Offer self-collection where validated. Document the indication, consent, chaperone, findings, specimens, and tolerance.

Abdominal and groin examination may precede genital examination. External inspection can identify lesions, inflammation, discharge, trauma, atrophy, prolapse, masses, or anatomical variation. Speculum examination visualises vagina and cervix and permits sampling. Bimanual examination assesses tenderness, uterine characteristics, adnexal masses, and cervical excitation but has limited sensitivity for many disorders. Rectal or rectovaginal examination is selective, not routine.

Testicular examination assesses position, size, consistency, tenderness, masses, epididymis, cord, hernia, and varicocele. A hard intratesticular mass requires urgent ultrasound. Sudden severe pain with a high or abnormal testis suggests torsion and requires immediate surgical assessment without delaying for low-value testing.

Laboratory reasoning

Start by defining the suspected axis. Measure a target hormone with its regulator: free thyroxine with thyroid-stimulating hormone, cortisol with adrenocorticotrophic hormone when appropriate, calcium with parathyroid hormone, and gonadal steroid with gonadotropins. A normal regulator can be abnormal when physiology demands elevation or suppression.

Confirm unexpected abnormalities. Check sampling time, fasting or posture requirements, cycle stage, pregnancy, acute illness, kidney and liver function, binding proteins, and medications. Assays can be distorted by biotin, heterophile antibodies, macrohormones, cross-reactivity, and extreme-concentration hook effects. Contact the laboratory when results conflict with the phenotype.

Dynamic testing evaluates reserve or suppressibility. Cortisol stimulation tests, dexamethasone suppression, glucose suppression of growth hormone, water-balance testing, and confirmatory aldosterone procedures have protocol-specific preparation and hazards. Interpretation requires pretest probability and validated thresholds for the assay used.

Reproductive and pregnancy investigations

Pregnancy testing detects chorionic gonadotropin in urine or blood. Serial concentrations and transvaginal ultrasound may help evaluate early pregnancy, but neither should be interpreted from one number without gestational context. Pain, bleeding, haemodynamic instability, or peritoneal signs raise urgency for ectopic pregnancy or haemorrhage.

Amenorrhoea assessment commonly begins with pregnancy exclusion, then thyroid, prolactin, gonadotropin, and androgen evaluation guided by history. Abnormal bleeding investigation depends on age, pregnancy possibility, haemodynamic state, anaemia, structural risk, and endometrial cancer risk. Pelvic ultrasound describes anatomy; hysteroscopy and tissue sampling answer different questions.

Infertility evaluation assesses ovulation, semen, and tubal or uterine factors in parallel. Ovarian-reserve tests estimate response to stimulation better than natural conception. Infection testing, cervical screening, genetic testing, and prenatal screening should follow consent and local evidence-based pathways. Screening estimates risk; diagnostic testing establishes or excludes defined conditions with different certainty and procedural consequences.

Imaging and tissue diagnosis

Use imaging after biochemical and clinical formulation. Ultrasound is central for thyroid nodules, pelvic anatomy, testes, and pregnancy. Magnetic resonance imaging defines pituitary and selected pelvic disease. Computed tomography characterises adrenal anatomy and stages malignancy but radiation matters in pregnancy. Nuclear imaging assesses thyroid function and selected tumours but is contraindicated or restricted in pregnancy and lactation.

An incidental lesion may be unrelated to symptoms. Decide whether it secretes, compresses, appears malignant, or is growing before attributing disease. Tissue diagnosis is valuable for selected thyroid, endometrial, cervical, gonadal, and other lesions, but pituitary and adrenal masses are generally not sampled without specialist biochemical reasoning.

Emergency recognition

Treat suspected adrenal crisis, thyroid storm, myxoedema coma, severe calcium disturbance, diabetic ketoacidosis, hyperosmolar state, pituitary apoplexy, testicular torsion, ruptured ectopic pregnancy, severe pre-eclampsia, and major obstetric haemorrhage as time-critical syndromes. Stabilisation and targeted empirical treatment may precede definitive confirmation. Obtain critical samples when this causes no meaningful delay.

TTS module 2: Phenotype-led endocrine testing and reproductive diagnostic safety

Build an endocrine phenotype before ordering

Endocrine symptoms are common because hormones influence almost every organ. Fatigue, weight change, low mood, hair change, and altered sleep have low specificity. Diagnostic value rises when symptoms form a mechanistically coherent cluster with objective signs and a compatible time course.

Before testing, define the proposed axis and whether excess, deficiency, resistance, or structural compression is suspected. Then identify the target hormone, controlling hormone, relevant timing, and factor that could distort measurement. Broad panels without a physiological question create incidental abnormalities and contradictory repeats.

Medication history must include route. Inhaled, injected, topical, ocular, and intra-articular glucocorticoids can suppress the adrenal axis. Supplements can contain biotin, iodine, glandular extracts, or undeclared hormones. Opioids suppress gonadal drive; dopamine blockers raise prolactin; immune therapies can inflame pituitary, thyroid, adrenal, or pancreas.

Examination should test the proposed mechanism

True proximal weakness is demonstrated by impaired shoulder or hip power, not fatigue alone. It supports cortisol excess, thyroid disease, inflammatory myopathy, steroid myopathy, or neuromuscular disease and requires localisation. Muscle bulk, reflexes, sensation, pain, and gait separate alternatives.

Skin findings gain specificity through morphology. Broad violaceous striae with bruising and weakness support cortisol excess more than pale narrow stretch marks. Diffuse hyperpigmentation in palmar creases, scars, pressure points, and mucosa supports high adrenocorticotrophic drive. Velvety hyperpigmentation at flexures suggests insulin resistance but can have other causes.

Eye assessment in thyroid disease includes more than prominence. Pain, colour desaturation, reduced acuity, relative pupillary defect, field loss, exposure, corneal injury, and impaired movement identify sight-threatening orbitopathy. Urgent specialist review is required when optic nerve or cornea is endangered.

Thyroid palpation describes diffuse versus nodular enlargement, texture, tenderness, fixation, retrosternal signs, and nodes. A normal-feeling gland does not exclude functional disease. A large gland can be euthyroid. Biochemical and structural questions remain separate.

Pair the hormone and regulator

Thyroid-stimulating hormone with free thyroxine distinguishes primary from central patterns. Calcium with parathyroid hormone identifies whether the response is appropriate. Cortisol with adrenocorticotrophic hormone localises confirmed deficiency or excess. Gonadal steroid with luteinising and follicle-stimulating hormone distinguishes primary gonadal failure from central suppression.

The controlling hormone must be judged against what physiology demands, not only its printed reference range. A normal parathyroid hormone is inappropriate during hypercalcaemia because it should be suppressed. A normal thyroid-stimulating hormone is inappropriate when free thyroxine is low in central disease.

Paired samples should be taken under the same physiological conditions where possible. Acute illness, fasting, posture, cycle phase, sleep, stress, and exogenous hormones alter both components. Discordance should prompt repetition and laboratory discussion before rare diagnoses.

Dynamic tests and pre-test probability

A stimulation test is useful when baseline results are indeterminate and reserve matters. A suppression test is useful when autonomous secretion is suspected. Test performance deteriorates when used in populations unlike those validated or when preparation is inadequate.

Cortisol testing requires review of glucocorticoids, oestrogen, binding proteins, sleep schedule, and acute stress. Dexamethasone suppression can fail through poor absorption, enzyme induction, non-adherence, or assay cross-reactivity. An abnormal screen establishes neither source nor imaging target.

Aldosterone-renin testing depends on posture, time, sodium, potassium, kidney function, pressure medicines, and assay. Hypokalaemia can suppress aldosterone and create a false-negative screen. Potentially interfering medicines are adjusted only when clinically safe under protocol.

Water-balance testing can provoke dangerous sodium change and should not be improvised. Baseline plasma and urine osmolality, thirst, urine output, medicine exposure, and kidney function often narrow the diagnosis before supervised testing.

Reproductive history as temporal physiology

Menstrual history should record cycle interval, predictability, duration, volume, pain, intermenstrual bleeding, and change from personal baseline. Irregular cycles suggest inconsistent ovulation but do not identify the cause.

Pregnancy, energy deficit, polycystic ovarian physiology, prolactin, thyroid disease, perimenopause, medicines, and primary ovarian failure require separation.

Heavy bleeding is quantified through flooding, clots, product changes, nocturnal protection, missed activity, and anaemia symptoms rather than subjective labels alone. Structural lesions, ovulatory dysfunction, endometrial

disorder, coagulation disease, medicines, and pregnancy complications can overlap.

Sexual history should separate desire, arousal, lubrication or erection, orgasm, ejaculation, pain, relationship context, trauma, and reproductive intent. Confidentiality boundaries and safeguarding are explained. Neutral questions reduce assumptions but do not remove the need to ask directly about coercion and safety when indicated.

Intimate examination as a consent process

Consent is ongoing rather than a one-time permission. Explain the indication, steps, expected sensations, alternatives, sampling, chaperone options, and right to stop. Reconfirm before progressing from external to internal examination. Avoid unnecessary repetition by coordinating examinations and reviewing existing findings.

Trauma-informed practice offers control over positioning, support person, examiner characteristics when feasible, and self-insertion of instruments where appropriate. Silence is not consent. Distress, pain, guarding, or withdrawal should prompt pause and reassessment.

Bimanual examination has limited sensitivity for many ovarian and uterine lesions. A normal result does not exclude ectopic pregnancy, endometriosis, torsion, or malignancy. Its value lies in selected tenderness, mobility, uterine size, and mass questions combined with imaging and laboratory context.

Sudden testicular pain is a time-critical vascular syndrome. Ultrasound can help when diagnosis is uncertain without causing delay, but a strong torsion presentation requires urgent surgical assessment. A hard painless intratesticular mass is malignant until assessed and requires rapid ultrasound and referral.

Early pregnancy and uncertainty

Early pregnancy location cannot always be established at one visit. Human chorionic gonadotropin trends, transvaginal ultrasound, dates, symptoms, and haemodynamic state are integrated. A value above a locally used discriminatory range without an intrauterine sac raises concern but does not alone prove ectopic pregnancy.

A pregnancy of unknown location is a temporary classification requiring explicit follow-up. It may represent a very early intrauterine pregnancy, failing pregnancy, or ectopic pregnancy. The patient needs clear return advice for pain, bleeding, dizziness, shoulder pain, or collapse and ownership of repeat testing.

Rh-negative status, haemoglobin, infection risk, blood group, and treatment implications are assessed according to presentation and local protocols. Haemodynamic instability or peritoneal signs demand resuscitation and urgent surgical or obstetric care rather than prolonged serial testing.

Infertility as a paired assessment

Ovulation, sperm, anatomy, timing, and age-related probability are assessed in parallel. Testing only one partner delays care and can create misplaced blame. Semen results vary and should be repeated when abnormal unless severity requires immediate action.

Ovulation can be inferred from regular cycles, luteal progesterone timing, or monitoring. A single progesterone must be timed relative to ovulation rather than a fixed calendar day. Ovarian-reserve tests estimate response to stimulation and remaining pool, not natural conception certainty.

Tubal patency tests show passage but not all function. Ultrasound describes uterus and ovaries but cannot exclude every endometrial or tubal lesion. Hysteroscopy sees the cavity, while laparoscopy assesses selected pelvic disease and carries procedural risk.

Diabetes and endocrine surveillance

Diabetes examination should close actionable loops. A foot-risk category determines education, follow-up, footwear, podiatry, or vascular referral. Injection-site findings prompt technique change. Orthostatic pressure, heart-rate response, gastrointestinal, bladder, sweating, and sexual symptoms can reveal autonomic neuropathy.

Retinal, kidney, cardiovascular, dental, vaccination, pregnancy, and mental-health surveillance should be scheduled rather than left to symptoms. Glucose technology data should be reviewed for time below range, variability, patterns, device burden, and whether alerts produce useful action.

The final endocrine-reproductive report should distinguish observed phenotype, biochemical localisation, assay uncertainty, anatomical finding, fertility or pregnancy goal, immediate danger, and ownership of follow-up. Sensitive assessment is clinically strongest when consent and physiological precision reinforce each other. Results should be communicated with calibrated uncertainty, explicit next steps, and a clear route for urgent change in clinical practice.

Retrieval prompts

One. Why are symptom clusters more useful than isolated endocrine symptoms?

Two. What makes a paired-hormone result physiologically appropriate or inappropriate?

Three. Which elements belong in a diabetes foot assessment?

Four. What principles govern intimate examination?

Five. Why should imaging usually follow biochemical formulation?

Six. Which endocrine and reproductive presentations require immediate action?

Concise answers

One. Common symptoms gain specificity when objective findings align within one physiological axis.

Two. The regulator must respond in the direction and magnitude demanded by the target hormone.

Three. Skin, ulcer, infection, deformity, footwear, pulses, ischaemia, protective sensation, and actionable risk classification.

Four. Indication, explicit consent, privacy, explanation, minimal exposure, chaperone choice, and the right to stop.

Five. Incidental lesions are common and structural appearance alone does not establish hormonal function.

Six. Hormonal crises, severe metabolic disturbance, pituitary apoplexy, torsion, ectopic rupture, pre-eclampsia, and obstetric haemorrhage.

Source map

Original synthesis informed by Talley and O'Connor, endocrine, diabetic, reproductive, obstetric, and intimate examination; Guyton and Hall, axis physiology; Robbins, endocrine and reproductive pathology; Katzung and OpenStax Pharmacology, medication and dynamic-test effects; and OpenStax Anatomy and Physiology and Medical-Surgical Nursing.

Chapter 27: Neural Coding, Synapses, Sensory Systems, and Pain

Orientation

The nervous system represents the body and environment through patterns of electrical activity. Receptors transduce energy into graded potentials; neurons convert those signals into action-potential timing; synapses select, amplify, inhibit, and reshape information. Sensation is not a direct recording of reality but an inference constructed from receptor properties, pathways, context, attention, and prior experience. Pain adds protective meaning, autonomic response, emotion, and learning to nociceptive input.

Electrical coding

Ion gradients and selective membrane permeability create resting potential. Depolarisation reaching threshold opens voltage-gated sodium channels, producing the action-potential upstroke. Sodium-channel inactivation and potassium efflux repolarise the membrane. Absolute and relative refractory periods constrain firing frequency and direction. Myelin increases membrane resistance and allows saltatory conduction between nodes; axon diameter also increases conduction velocity.

Action-potential amplitude is all-or-none, so stimulus intensity is encoded mainly by firing frequency, recruited neurons, temporal pattern, and population distribution. Graded receptor and postsynaptic potentials vary in amplitude and can summate. Adaptation changes discharge during sustained stimulation: rapidly adapting receptors emphasise change, whereas slowly adapting receptors preserve information about duration.

Synaptic transmission and integration

At chemical synapses, presynaptic depolarisation opens calcium channels. Calcium triggers vesicle fusion and transmitter release. Transmitter binds ionotropic receptors for rapid conductance change or metabotropic receptors for slower, amplified signalling. Reuptake, enzymatic degradation, diffusion, and glial uptake terminate action. Electrical synapses pass current through gap junctions and synchronise selected networks.

Excitatory postsynaptic potentials move membrane potential toward firing; inhibitory potentials stabilise or hyperpolarise it, often through chloride or potassium conductance. Their location and timing matter.

Axon-initial-segment integration determines output from thousands of inputs. Presynaptic inhibition alters transmitter release before it reaches the target. Feedforward, feedback, reciprocal, and lateral inhibition sharpen timing and contrast.

Glutamate is the principal excitatory transmitter in much of the central nervous system. Gamma-aminobutyric acid is the principal inhibitory transmitter in the brain, while glycine is prominent in spinal cord and brainstem. Acetylcholine, dopamine, noradrenaline, serotonin, histamine, peptides, endocannabinoids, and gases regulate networks through diverse receptors. A transmitter's effect depends on receptor and circuit, not its name alone.

Plasticity and glia

Synapses strengthen or weaken with activity. Short-term facilitation and depression reflect presynaptic dynamics. Long-term potentiation and depression involve receptor trafficking, kinase signalling, gene expression, and structural change. Plasticity supports learning and adaptation but also contributes to addiction, chronic pain, epilepsy, and maladaptive recovery.

Astrocytes regulate ions, transmitter clearance, metabolism, blood-flow coupling, and synaptic environment. Oligodendrocytes myelinate central axons; Schwann cells myelinate peripheral axons. Microglia survey tissue and participate in inflammation and pruning. The blood-brain barrier uses specialised endothelium, pericytes, and astrocytic support to regulate entry; inflammation, trauma, and tumours can disrupt it.

Sensory organisation

A sensory pathway includes receptor, afferent neuron, ascending relay, thalamic or brainstem integration, cortex, and descending modulation. Receptive fields define the region influencing a neuron. Small fields and lateral inhibition improve spatial discrimination. Overlapping population codes allow finer localisation than any one receptor. Cortical maps allocate disproportionate area to behaviourally important regions and can reorganise after injury or learning.

The dorsal-column-medial-lemniscal system carries discriminative touch, vibration, and conscious proprioception. Fibres ascend on the same side of the spinal cord, cross in the medulla, relay through thalamus, and reach somatosensory cortex. The anterolateral system carries pain, temperature, itch, and crude touch; fibres cross near spinal entry and ascend contralaterally. Lesions therefore create characteristic modality and side patterns.

Proprioceptors include muscle spindles sensing length and change, Golgi tendon organs sensing tension, and joint and skin receptors. Their input supports body position, reflexes, posture, and movement calibration. Sensory ataxia worsens when vision is removed because visual compensation is lost.

Vision

The cornea supplies most optical refraction; the lens changes shape for accommodation. Pupillary constriction increases depth of field and regulates retinal illumination. Photoreceptors hyperpolarise in light. Rods are highly sensitive and support dim vision; cones support acuity and colour. Retinal circuits process contrast before information leaves through ganglion-cell axons.

Nasal retinal fibres cross at the optic chiasm while temporal fibres remain uncrossed, so each optic tract represents the opposite visual field. Fibres relay mainly in the lateral geniculate nucleus and project through optic radiations to visual cortex. Lesion location predicts monocular loss, bitemporal loss, homonymous field defects, or cortical patterns. Pupillary reflex pathways diverge before visual cortex, so light responses can be preserved despite cortical blindness.

Accommodation requires parasympathetic ciliary contraction, lens rounding, and convergence. Sympathetic and parasympathetic pathways control pupil size. Eye movements use vestibular, brainstem, cerebellar, and cortical systems to hold images on the fovea. Misalignment produces diplopia unless suppression occurs.

Hearing and balance

The outer ear directs sound to the tympanic membrane. Ossicles transmit and impedance-match vibration to cochlear fluid. Basilar-membrane mechanics separate frequency by place: high frequencies peak near the base and low frequencies near the apex. Hair-cell stereocilia convert mechanical displacement into receptor potentials; inner hair cells provide most afferent output, while outer hair cells amplify and sharpen tuning.

Auditory pathways relay bilaterally through brainstem, thalamus, and cortex, so unilateral central lesions rarely cause complete deafness in one ear. Timing and intensity differences support sound localisation. Conductive loss impairs sound delivery; sensorineural loss affects cochlea, nerve, or central processing. Loud sound, ageing, infection, drugs, genetics, and vascular injury can damage hair cells.

Semicircular canals detect angular acceleration through endolymph movement and cupular deflection. Utricle and saccule detect linear acceleration and head tilt through weighted otolith membranes. Vestibular nuclei integrate these signals with vision and proprioception to drive vestibulo-ocular reflexes, posture, and spatial orientation. Mismatch between systems can produce vertigo, nystagmus, nausea, and imbalance.

Taste, smell, and visceral sensation

Taste receptors detect sweet, salty, sour, bitter, and umami through ion channels or G-protein pathways. Smell receptors in olfactory epithelium project through the cribriform plate to bulb and limbic-associated cortex, linking odour strongly to memory and emotion. Flavour combines taste, retronasal smell, texture, temperature, and trigeminal sensation.

Visceral afferents monitor stretch, chemistry, ischaemia, and inflammation. Their sparse representation and convergence with somatic pathways make

visceral pain diffuse and referred. Autonomic responses often accompany it. Referred pain follows shared spinal segments rather than actual injury at the perceived skin site.

Nociception and pain

Nociceptors are free nerve endings activated by mechanical, thermal, and chemical danger. Tissue injury releases potassium, protons, prostaglandins, bradykinin, cytokines, and growth factors that lower thresholds. Fast myelinated A-delta fibres carry sharp, localised pain; slower unmyelinated C fibres carry burning, aching, and autonomic-affective components.

Primary afferents enter the dorsal horn, where glutamate and peptides activate projection neurons and interneurons. Ascending pathways reach thalamus, cortex, insula, limbic structures, and brainstem. Pain experience integrates sensory discrimination, salience, fear, expectation, attention, culture, and perceived control. Pain can be real and severe without proportionate visible tissue damage.

Descending pathways from cortex, hypothalamus, periaqueductal grey, and medulla can inhibit or facilitate dorsal-horn transmission through endogenous opioids, noradrenaline, serotonin, and local interneurons. Placebo and nocebo effects recruit genuine modulatory physiology; they do not imply fabrication.

Acute, inflammatory, neuropathic, and nociplastic pain

Acute nociceptive pain warns of injury and guides protection. Inflammatory pain increases sensitivity around damaged tissue. Peripheral sensitisation lowers nociceptor thresholds; central sensitisation amplifies spinal and supraspinal responses, enlarges receptive fields, and makes normally innocuous input painful. Persistent activation can outlast the initiating lesion.

Neuropathic pain results from disease or lesion of the somatosensory system and may cause burning, electric shocks, allodynia, sensory loss, and autonomic change. Nociplastic pain reflects altered nociceptive processing without clear tissue or nerve damage sufficient to explain severity. Categories can overlap. Treatment follows mechanism and function rather than pain intensity alone.

Analgesic principles

Paracetamol provides analgesia with limited peripheral anti-inflammatory action; overdose causes hepatic injury. Non-steroidal anti-inflammatory drugs reduce prostaglandin synthesis but can cause gastrointestinal bleeding, kidney injury, fluid retention, and cardiovascular harm. Local anaesthetics block voltage-gated sodium channels. Opioids activate inhibitory receptors but cause sedation, respiratory depression, constipation, tolerance, dependence, and overdose risk.

Neuropathic pain may respond to selected antidepressants, calcium-channel ligands, sodium-channel agents, or topical treatments. Non-pharmacological care includes explanation, graded activity, sleep support, psychological strategies, rehabilitation, and treatment of the underlying disease. Multimodal treatment seeks improved function and tolerable symptoms while minimising cumulative harm.

TTS module 2: Population coding, predictive perception, and persistent pain networks

Neural information is distributed

No single action potential carries the complete identity of a stimulus. Information emerges from which neurons fire, their rate, timing, synchrony, and relationship to ongoing network state. Population coding allows overlapping neurons to represent a continuous range of direction, intensity, colour, pitch, or position.

Rate coding is limited by refractory periods and metabolic cost. Temporal coding uses precise spike timing relative to stimulus or network rhythms. Recruitment extends dynamic range as progressively less sensitive receptors or neurons become active. Adaptation permits the system to devote activity to change rather than constant background.

The nervous system also codes uncertainty. Sensory noise, ambiguous input, and incomplete sampling mean perception combines current evidence with prior expectation. This predictive process is usually efficient but can generate illusions, perceptual bias, or persistent threat interpretation when priors become excessively strong.

Receptive fields and contrast

A receptor's physical transduction zone and a central neuron's receptive field are not identical. Convergent inputs can enlarge a central field, while inhibition sharpens its functional boundaries. Small fields and dense innervation support high acuity in fingertips and fovea.

Lateral inhibition suppresses neighbouring pathways around a strong signal, increasing contrast at edges. In the retina, centre-surround organisation highlights spatial change rather than absolute illumination. In touch, neighbouring inhibition improves two-point discrimination. Similar inhibitory motifs shape sound frequency and movement direction.

Convergence increases sensitivity but reduces localisation. Visceral afferents converge onto spinal neurons also receiving somatic input, helping explain referred pain. Divergence sends one input into sensory, autonomic, motor, affective, and arousal pathways, explaining why a noxious event produces more than a location report.

Synaptic gain and network state

Synaptic effect depends on release probability, receptor number, driving force, dendritic location, and concurrent conductance. An inhibitory synapse near the axon initial segment can exert strong control. Shunting inhibition reduces the impact of excitatory current without a large visible hyperpolarisation.

Neuromodulators change network gain rather than simply turning neurons on or off. Noradrenaline alters arousal and signal-to-noise. Acetylcholine changes attention and plasticity. Dopamine influences salience, learning, and action value. Serotonin has receptor-specific effects across mood, pain, sleep, appetite, and autonomic systems.

Astrocytes take up glutamate and potassium, supply metabolic substrates, and influence local blood flow. Failure of uptake increases extracellular excitation and can promote seizures or excitotoxicity. Microglial inflammatory signalling can change synapses during injury and chronic pain. Glia are active regulators of information processing.

Sensory adaptation and amplification

Rapidly adapting receptors respond strongly to onset and offset, making them effective detectors of vibration or movement. Slowly adapting receptors encode sustained pressure, stretch, and position. Adaptation can occur at receptor, synapse, and central network levels.

Sensitisation is the opposite directional change. Inflammatory mediators phosphorylate nociceptor channels and reduce activation thresholds. Repeated central input removes magnesium block from selected glutamate receptors, increases calcium signalling, changes transcription, and strengthens dorsal-horn responses.

Sensory loss can paradoxically coexist with pain. Damaged peripheral nerves generate ectopic impulses, neighbouring fibres cross-excite, and central circuits increase gain after reduced input. The result can be numbness plus burning, shocks, allodynia, or pain referred beyond the original lesion.

Visual computation

Photoreceptors signal light by reducing transmitter release when they hyperpolarise. Bipolar-cell pathways preserve or invert this change, creating on and off channels. Ganglion cells encode contrast and send action potentials along the optic nerve. Horizontal and amacrine cells shape spatial and temporal interactions.

Colour perception compares cone classes rather than measuring wavelength with one receptor. Opponent channels organise red-green and blue-yellow contrast. Colour constancy uses context to infer surface reflectance under changing illumination, another example of perception as interpretation.

Depth is inferred from binocular disparity, eye convergence, motion parallax, perspective, shading, and familiar size. Loss of one cue does not eliminate all depth perception. Visual cortex contains specialised but interacting streams for object identity, spatial relationships, and visually guided action.

Field defects localise pathways because retinal geometry remains ordered. Optic-nerve lesions cause monocular loss. Chiasmal compression affects crossing nasal fibres and temporal fields. Post-chiasmal lesions produce contralateral homonymous defects, with pattern reflecting optic radiation or cortex.

Auditory and vestibular inference

The travelling wave along the basilar membrane converts frequency into place. Inner hair cells transmit sound information, while outer hair cells actively sharpen and amplify mechanical tuning. Damage to outer cells raises thresholds and broadens frequency discrimination.

Conductive hearing loss reduces sound delivery but preserves cochlear neural machinery. Sensorineural loss reduces transduction or neural processing and may distort speech disproportionately to loudness. Bedside tuning-fork patterns are screening tools; audiometry quantifies air, bone, frequency, speech, and masking relationships.

The brain localises low-frequency sound mainly from timing differences and high-frequency sound from intensity shadows. Bilateral brainstem pathways create redundancy. Sudden unilateral sensorineural loss remains an emergency despite that central redundancy because cochlear or nerve function is acutely threatened.

Vestibular hair cells encode head acceleration. The vestibulo-ocular reflex drives eyes opposite head movement to stabilise vision. A unilateral imbalance produces predictable nystagmus and vertigo. Central lesions alter ocular alignment, pursuit, coordination, or neurological function and may not suppress with fixation.

Pain is a protective inference

Nociception is neural encoding of potential tissue threat; pain is the conscious protective experience. They usually correlate but can diverge. Anaesthesia can block pain despite injury. Central sensitisation or neuropathy can produce severe pain after tissue danger has reduced.

Peripheral sensitisation makes injured tissue tender through prostaglandins, bradykinin, hydrogen ions, cytokines, and growth factors. Central sensitisation amplifies spinal transmission, enlarges fields, reduces inhibition, and recruits normally innocuous input. Descending systems can inhibit or facilitate according to context.

Attention, fear, expectation, memory, sleep, mood, and perceived control modify pain through real networks. Catastrophic interpretation increases threat and avoidance; safe graded experience can update predictions. This

does not deny pathology. It expands the treatment target beyond one injured structure.

Neuropathic and nociplastic patterns

Neuropathic pain follows a lesion or disease of the somatosensory system. Distribution should fit nerve, root, plexus, spinal, or brain anatomy. Positive signs include burning, shocks, allodynia, and hyperalgesia; negative signs include reduced sensation. Autonomic and motor changes can coexist.

Nociplastic pain reflects altered processing when tissue or nerve pathology does not sufficiently explain extent or persistence. Widespread sensitivity, fatigue, sleep disturbance, cognitive symptoms, and variable triggers can occur. Nociceptive, neuropathic, and nociplastic mechanisms often overlap within one patient.

Treatment should match dominant mechanisms and goals. Anti-inflammatory treatment helps inflammatory sensitisation. Sodium-channel, calcium-channel, or monoamine approaches help selected neuropathic states. Rehabilitation, pacing, graded exposure, sleep care, psychological treatment, and social support help restore function and update threat networks.

Opioids can reduce acute nociception but also produce tolerance, dependence, hyperalgesia, endocrine effects, constipation, sedation, and respiratory depression. Long-term benefit should be demonstrated through function and harm review rather than assumed from short-term relief.

Cortical maps remain plastic after amputation, nerve injury, immobilisation, and intensive practice. Adjacent representations can expand into deprived territory, while remembered body models persist. Phantom sensation therefore reflects continuing central representation rather than an imaginary complaint. Mirror therapy, sensory discrimination, prosthetic use, and graded motor imagery can provide coherent visual and motor feedback in selected patients. Plasticity is neither automatically beneficial nor limitless; training must be meaningful, repeated, and matched to intact pathways, while pain, sleep, attention, mood, and fear influence whether new patterns consolidate.

The final sensory-pain formulation should name pathway, modality, distribution, positive and negative signs, tissue state, central amplification,

and functional consequence. It should distinguish urgent new neurological injury from persistent network change and specify how treatment will test the proposed mechanism through measurable changes in sensation, pain, participation, and daily function.

Retrieval prompts

One. How is stimulus intensity encoded when action-potential amplitude is fixed?

Two. How do the dorsal-column and anterolateral systems differ?

Three. Why do optic lesions produce predictable field defects?

Four. How do cochlear mechanics encode frequency?

Five. What is central sensitisation?

Six. Why should analgesia be mechanism-based and multimodal?

Concise answers

One. Through firing frequency, timing, recruited neurons, and population pattern.

Two. Dorsal columns carry fine touch, vibration, and proprioception and cross in medulla; anterolateral fibres carry pain and temperature and cross near entry.

Three. Nasal fibres cross while temporal fibres remain uncrossed, organising each post-chiasmal pathway by contralateral field.

Four. Different frequencies maximally displace different positions along the basilar membrane.

Five. Activity-dependent amplification in central pathways produces exaggerated or persistent responses and pain from innocuous input.

Six. Different mechanisms respond differently, while combining approaches improves function and limits toxicity from any one treatment.

Source map

Original synthesis informed by Guyton and Hall, membrane physiology, synapses, sensory pathways, special senses, and pain; Robbins, neural injury and degeneration; Katzung and OpenStax Pharmacology, neurotransmission and analgesia; OpenStax Anatomy and Physiology and Biology; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, sensory and pain assessment.

Chapter 28: Motor Control, Reflexes, Cerebellum, and Basal Nuclei

Orientation

Movement emerges from distributed control. Spinal circuits organise reflexes and patterned output. Brainstem systems regulate posture, balance, gaze, and orientation. Motor cortex specifies skilled action. Basal nuclei select and scale actions, while cerebellum predicts consequences, compares intended with actual performance, and drives adaptation. Sensory feedback updates every level. Lesions are localised by weakness, tone, reflexes, coordination, involuntary movement, and gait.

Motor units and force

An alpha motor neuron and all muscle fibres it innervates form a motor unit. Small units permit fine control; large units generate power. Force rises by increasing motor-neuron firing and recruiting progressively larger units. Asynchronous activation smooths contraction. The final common pathway integrates descending commands, spinal interneurons, sensory afferents, and intrinsic excitability.

Neuromuscular transmission begins when motor-neuron calcium entry releases acetylcholine. Nicotinic receptors generate an end-plate potential that activates muscle sodium channels. Acetylcholinesterase terminates the signal. Safety factor normally makes transmission reliable, but receptor loss, impaired release, toxins, drugs, or repeated activity can expose failure.

Lower motor-neuron loss produces weakness, reduced tone, depressed reflexes, denervation atrophy, and fasciculations. Disease at neuromuscular junction produces fatigable weakness without sensory loss; reflexes vary by mechanism. Primary muscle disease often causes symmetric proximal weakness with preserved sensation and relatively preserved reflexes until advanced loss.

Muscle spindles and tendon organs

Muscle spindles lie in parallel with ordinary fibres and detect length and rate of stretch. Ia afferents excite alpha motor neurons supplying the same muscle and inhibit antagonists through interneurons. Gamma motor neurons adjust spindle tension during contraction, preserving sensitivity through alpha-gamma coactivation. This system contributes to tone, posture, and rapid correction.

Golgi tendon organs lie in series with muscle and sense tension through Ib afferents. Their spinal actions distribute load and can inhibit the same muscle under selected conditions, but they also participate flexibly in locomotion and force control. Joint, skin, vestibular, and visual input combine with spindle and tendon information; no receptor acts in isolation.

Spinal reflexes

The stretch reflex is short-latency resistance to sudden lengthening. The tendon tap is a clinical probe of afferent nerve, spinal segment, motor neuron, neuromuscular junction, muscle, and descending modulation. Reflex amplitude varies with attention, temperature, age, technique, anxiety, and reinforcement. Symmetry and relation to other signs matter more than an isolated grade.

Painful stimulation activates flexor withdrawal, while crossed extension supports the opposite limb. Interneurons coordinate several segments and muscles. Recurrent inhibition from Renshaw cells shapes motor-neuron output. Reciprocal inhibition permits agonist contraction with antagonist relaxation. After spinal cord injury, reflexes may initially disappear during spinal shock, then become exaggerated as descending control reorganises.

Central pattern generators can produce rhythmic locomotor output, but human walking requires descending initiation and sensory adjustment. Spinal circuits adapt stepping to load and phase; brainstem and cerebellar systems maintain equilibrium; cortex handles obstacles and deliberate changes.

Descending pathways

Corticospinal neurons arise mainly from motor and premotor regions and descend through corona radiata, internal capsule, brainstem, and pyramids. Most fibres cross in the lower medulla and form the lateral corticospinal tract controlling contralateral distal skilled movement. Uncrossed or bilaterally projecting fibres influence axial muscles.

Corticobulbar projections control cranial motor nuclei. Many nuclei receive bilateral input, but lower facial muscles and tongue have important contralateral predominance. A hemispheric lesion therefore often weakens the opposite lower face while sparing forehead movement; a peripheral facial lesion weakens the entire ipsilateral face.

Reticulospinal tracts influence tone, automatic posture, locomotion, and anticipatory adjustments. Vestibulospinal pathways stabilise head and body relative to gravity. Tectospinal circuits orient head and neck toward salient stimuli. Rubrospinal influence is modest in humans but participates in upper-limb control.

An upper motor-neuron lesion causes impaired fractionated movement, weakness in characteristic groups, increased tone, hyperreflexia, clonus, and an extensor plantar response after acute shock resolves. Spasticity is velocity-dependent and reflects exaggerated stretch responses; rigidity is not velocity-dependent and affects agonists and antagonists more evenly. Weakness distribution and associated cortical, cranial, sensory, or sphincter signs localise the lesion.

Motor cortex and planning

Primary motor cortex contributes strongly to force direction and skilled execution. Premotor cortex uses external cues and integrates posture with action; supplementary motor areas contribute internally generated sequences and bimanual coordination. Posterior parietal cortex transforms sensory spatial information into action plans. Prefrontal and limbic systems provide goals, motivation, and rules.

Motor maps overlap and change with learning or injury; they are not a keyboard of individual muscles. Population activity specifies movement through distributed patterns. An efference copy of motor command allows

predicted sensory consequences to be compared with actual feedback. Practice improves performance through synaptic and network adaptation, then reduces attentional demand as skills become more automatic.

Cerebellar architecture and function

The cerebellum receives intended motor plans from cortex and actual performance signals from spinal, vestibular, visual, and other systems. Its cortex uses a repeated circuit: mossy fibres influence granule cells and parallel fibres, climbing fibres provide powerful teaching signals, and Purkinje cells inhibit deep nuclei. Deep nuclei provide major cerebellar output.

Midline regions emphasise axial control, stance, and gait. Intermediate regions influence limb execution. Lateral hemispheres participate in planning, timing, and complex cognitive-motor operations. Vestibulocerebellar regions regulate balance and eye movements. These divisions overlap functionally rather than forming absolute compartments.

The cerebellum builds internal models that predict how commands will change the body. It adjusts timing, scaling, and coordination and supports error-based learning such as adapting the vestibulo-ocular reflex. Because cerebellar influence effectively returns to the same side of the body after pathway crossings, hemispheric lesions usually cause ipsilateral limb signs.

Cerebellar dysfunction causes dysmetria, decomposition, intention tremor, dysdiadochokinesia, rebound, hypotonia, nystagmus, scanning dysarthria, and broad unsteady gait. Sensory ataxia can mimic limb incoordination but worsens markedly with eye closure and accompanies impaired proprioception. Weakness, pain, drugs, alcohol, and vestibular disease must also be separated from true cerebellar signs.

Basal nuclei circuits

The basal nuclei include striatum, globus pallidus, subthalamic nucleus, and substantia nigra, connected with cortex and thalamus in parallel motor, cognitive, and limbic loops. Their output is largely inhibitory. Rather than directly driving motor neurons, they regulate which cortical action patterns are facilitated or suppressed.

In a simplified model, the direct pathway reduces inhibitory output and facilitates selected action; the indirect pathway increases inhibition and suppresses competing action. Dopamine from substantia nigra promotes movement through D-one effects on direct-pathway neurons and D-two effects reducing indirect-pathway activity. Real circuits include hyperdirect, collateral, oscillatory, and learning-related interactions beyond this model.

Parkinsonism reflects reduced dopamine and produces bradykinesia, rigidity, rest tremor, impaired postural responses, reduced automatic movement, and gait freezing. Dopamine replacement improves many features but can cause nausea, hypotension, hallucinations, dyskinesia, and fluctuations. Dopamine agonists and enzyme inhibitors extend dopaminergic effects; deep-brain stimulation modifies selected circuits in appropriate patients.

Excess or poorly constrained movement produces chorea, dystonia, tics, ballism, myoclonus, or dyskinesia according to pattern and network. Huntington disease causes progressive chorea, cognitive change, and psychiatric symptoms through inherited neurodegeneration. Subthalamic lesions can cause contralateral ballism. Dopamine-blocking drugs can cause acute dystonia, akathisia, parkinsonism, tardive syndromes, or malignant hyperthermic rigidity.

Posture, gait, and balance

Quiet stance is active. Vestibular, visual, and proprioceptive signals estimate body motion; ankle, hip, and stepping strategies restore the centre of mass. Anticipatory postural adjustments occur before voluntary movement. Ageing, neuropathy, vestibular loss, weakness, sedatives, impaired vision, cognitive load, and environmental hazards reduce reserve.

Gait analysis considers initiation, base, stride, arm swing, foot clearance, turning, tandem walking, and response to perturbation. Spastic gait is stiff and may circumduct; parkinsonian gait is short-stepped with reduced swing and freezing; cerebellar gait is broad and irregular; sensory gait relies heavily on vision; myopathic gait reflects pelvic weakness; frontal gait disorders impair initiation despite adequate power.

Motor learning and rehabilitation

Recovery after injury uses restitution, compensation, practice, and environmental adaptation. Repetition must be task-specific, sufficiently challenging, and safe. Feedback helps early learning but excessive external correction can impair independent control. Sleep, motivation, cardiovascular fitness, pain, mood, cognition, and treatment timing influence plasticity.

Spasticity treatment is justified when it causes pain, contracture, hygiene difficulty, or impaired function; reducing tone can sometimes worsen transfers when patients use stiffness for support. Management combines positioning, stretching, strengthening, splinting, focal chemodenervation, systemic drugs, and selected procedures around explicit goals.

TTS module 2: Motor-unit physiology, lesion localisation, and adaptive movement control

Force is recruited hierarchically

Motor units differ in fibre number, contraction speed, fatigue resistance, and recruitment threshold. Small fatigue-resistant units are activated first for posture and fine control. Larger faster units join as force demand rises. This size principle provides smooth economical output and preserves high-force capacity.

Rate coding increases force by temporal summation of twitches. At higher discharge, individual contractions fuse. Recruitment and rate interact according to muscle and task. During fatigue, additional units and higher central drive may preserve external force while effort rises.

Surface electromyography records summed activity near electrodes rather than direct muscle force. Needle examination samples individual units and spontaneous activity. Denervation produces fibrillation potentials, while reinnervation creates larger longer-duration units as surviving axons sprout to abandoned fibres. Interpretation depends on timing and distribution.

Neuromuscular transmission has a safety margin

Each nerve impulse normally releases more acetylcholine than required to reach muscle threshold. Repeated activity changes transmitter availability and calcium. Postsynaptic receptor loss reduces the end-plate response,

producing fatigable weakness. Presynaptic release failure may improve briefly with repeated activation as calcium accumulates.

Myasthenic weakness often affects ocular, bulbar, neck, or proximal muscles and fluctuates with use. Sensation is preserved and pupils are generally spared. Respiratory and swallowing weakness can deteriorate rapidly. Infection, surgery, pregnancy, heat, electrolyte disturbance, and selected medicines can worsen transmission.

Botulinum toxin blocks acetylcholine release and is dangerous systemically but useful when injected locally for dystonia, spasticity, autonomic hyperactivity, or pain-related conditions. Dose and placement must balance targeted weakening against swallowing, respiratory, or unwanted regional effects.

Localising weakness

Upper motor-neuron weakness affects movements in characteristic pyramidal patterns and is accompanied later by hyperreflexia, spasticity, clonus, and plantar extension. Acute lesions may initially produce low tone and reduced reflexes. Associated cortical, visual, cranial, sensory, or bladder signs locate the level.

Lower motor-neuron lesions produce weakness within root, plexus, nerve, or motor-neuron distributions, with atrophy, fasciculation, reduced tone, and reduced reflexes. Root lesions often include radicular pain and myotomal weakness. Peripheral nerve lesions fit named sensory and motor territories.

Neuromuscular-junction disease produces fatigability without sensory loss and usually without early atrophy. Myopathy commonly produces symmetric proximal weakness, difficulty rising or lifting, and preserved sensation. Reflexes remain until force becomes severe, though specific myopathies differ.

Functional neurological weakness is diagnosed through positive inconsistency and preserved automatic capacity, not merely normal tests. It is genuine and potentially treatable. Pain, fear, attention, and learned prediction alter motor output; explanation and rehabilitation use preserved pathways rather than accusing the patient of fabrication.

Tone and reflex physiology

Tone is resistance to passive movement produced by tissue mechanics, background activation, reflex gain, and state. Spasticity increases with movement velocity and direction after upper motor-neuron injury. Rigidity is more uniform and can be smooth or ratcheting. Paratonia varies with attention and effort, often in frontal or cognitive disease.

Hyperreflexia reflects increased gain after loss of descending regulation, not simply stronger muscle. Clonus is a self-sustaining stretch-activation cycle. Reduced reflexes arise from afferent, motor-neuron, peripheral nerve, junctional, or severe muscle dysfunction.

Reflex examination must control position, relaxation, tendon strike, and comparison. Reinforcement increases central excitability and can reveal a present reflex. A brisk symmetric reflex may be normal; asymmetry combined with weakness or plantar change carries greater localising value.

Spinal networks and pattern generation

Withdrawal reflexes recruit several muscles and segments according to stimulus location. Crossed extension stabilises the opposite limb. Reflex organisation is task dependent; the same sensory input can have different effects during standing, walking, or sleep.

Central pattern generators can produce alternating locomotor activity, but human gait requires supraspinal initiation and continuous sensory adaptation. Load receptors extend stance when weight is present, hip position helps trigger phase transition, and cutaneous input clears obstacles.

Spinal cord lesions create a level with long-tract findings below and segmental findings at the lesion. Hemicord injury combines ipsilateral motor and proprioceptive loss with contralateral pain-temperature loss below different levels. Central lesions preferentially affect crossing pain-temperature fibres and can involve arms more than legs when cervical.

Cerebellar prediction and error

The cerebellum receives a copy of intended commands and detailed sensory feedback. It predicts expected consequences, compares outcome, and modifies future commands. This feed-forward learning permits rapid smooth movement before slower feedback could correct every deviation.

Climbing-fibre activity provides a powerful teaching signal, while mossy and parallel-fibre patterns represent context. Purkinje cells shape deep-nuclear output through inhibition. Plasticity at several sites updates internal models for limb dynamics, eye movement, speech, and cognitive sequencing.

Cerebellar signs worsen near a target because predictive braking and scaling fail. Dysmetria overshoots or undershoots. Decomposition separates a normally blended movement. Intention tremor grows during approach. Dysdiadochokinesia impairs rapid alternating timing.

Midline disease disrupts stance, trunk, and gait. Hemispheric disease affects ipsilateral limbs. Vestibular regions affect balance and eye stability. Alcohol, sedatives, anticonvulsants, demyelination, stroke, tumour, degeneration, immune disease, and nutritional deficiency can produce overlapping syndromes.

Basal-ganglia selection and reinforcement

Basal-ganglia output tonically inhibits thalamic and brainstem targets. Selecting an action involves reducing inhibition for desired programmes while maintaining suppression of competitors. The hyperdirect pathway can rapidly increase global stopping before more selective pathways resolve the next action.

Dopamine carries movement and reinforcement information through receptor-specific effects. Loss reduces initiation, speed, amplitude, and automaticity. The person may retain strength yet struggle to start, sequence, or scale movement. External visual or rhythmic cues can bypass some impaired automatic control.

Parkinsonian bradykinesia includes decrement with repetition. Rigidity, rest tremor, postural impairment, reduced facial expression, soft voice, and autonomic or cognitive features vary. Levodopa restores dopamine precursor but produces fluctuations as disease advances and buffering capacity declines.

Dyskinesia can result from pulsatile dopaminergic exposure and altered network plasticity. Deep-brain stimulation changes pathological circuit activity but does not cure degeneration and can affect speech, mood, cognition, or

balance. Selection depends on levodopa response, phenotype, cognition, goals, and support.

Dystonia produces sustained or intermittent patterned contractions and abnormal postures. Chorea is irregular flowing movement. Myoclonus is shock-like. Tics are suppressible urges and movements or sounds. Akathisia is inner restlessness. Precise phenomenology determines cause and treatment.

Posture, gait, and dual-task reserve

Balance combines vestibular, visual, somatosensory, strength, cerebellar, basal-ganglia, and cognitive systems. A person can compensate for one impaired input until darkness, uneven ground, distraction, or illness removes reserve. Falls assessment should recreate relevant conditions safely.

Gait begins before the first step with weight shift and anticipatory postural adjustment. Turning requires braking, reorientation, and reacceleration. Freezing often appears at doorways, turns, or competing demands. Arm swing, base, stride, foot clearance, and variability reveal different control failures.

Dual-task walking tests cognitive-motor allocation. Marked slowing or instability while speaking or calculating predicts reduced reserve but is influenced by education, anxiety, hearing, and familiarity. Assistive devices help only when fitted, learned, and used in the environment where falls occur.

Rehabilitation and adaptive control

Motor recovery combines restitution of impaired networks, compensation by alternative strategies, prevention of secondary weakness or contracture, and environmental adaptation. High-repetition practice alone is insufficient if movements are careless or irrelevant. Salient task-specific feedback drives learning.

Error size must be tolerable. Too little error provides no teaching signal; too much causes failure, fear, and compensatory habits. Variable practice improves transfer, while blocked repetition can establish an early pattern. Rest and sleep consolidate learning.

Spasticity treatment should target pain, hygiene, sleep, contracture, positioning, or function. Tone may assist standing or transfers, so indiscriminate reduction can worsen independence. Strength, selective control, sensation, equipment, caregiver capacity, and goals determine the plan.

The final motor formulation should state lesion level, motor phenotype, sensory contribution, movement disorder, balance reserve, and functional task. Serial examination should separate recovery, compensation, medication effect, fatigue, and new disease rather than reducing movement to a single strength grade.

Retrieval prompts

One. How do motor-unit recruitment and firing rate increase force?

Two. What does a tendon reflex test physiologically?

Three. How do upper and lower motor-neuron syndromes differ?

Four. What computational role does the cerebellum perform?

Five. How do basal nuclei influence action selection?

Six. Which systems must cooperate for stable gait?

Concise answers

One. Progressively larger units are recruited and active neurons discharge more frequently.

Two. It probes afferent, segmental, efferent, junctional, muscular, and descending components of a stretch circuit.

Three. Upper lesions impair skilled patterns and later increase tone and reflexes; lower lesions cause flaccidity, areflexia, atrophy, and fasciculation.

Four. It predicts consequences, compares intended with actual movement, corrects timing and scale, and supports error learning.

Five. Inhibitory loops facilitate selected cortical programmes while suppressing competitors.

Six. Spinal patterning, descending command, vestibular and visual orientation, proprioception, cerebellar correction, basal-nuclei scaling, strength, and cognition.

Source map

Original synthesis informed by Guyton and Hall, spinal reflexes, descending pathways, cerebellum, basal nuclei, and gait; Robbins, motor-system pathology; Katzung and OpenStax Pharmacology, neuromuscular and movement-disorder drugs; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and Talley and O'Connor, motor and gait examination.

Chapter 29: Higher Function, Sleep, Consciousness, and Autonomic Control

Orientation

Higher function emerges from interacting networks. Attention selects information; memory preserves and reconstructs it; language maps concepts to symbols; executive systems maintain goals. Consciousness requires arousal and organised awareness. Sleep cycles through states supporting memory and physiology. Autonomic networks coordinate viscera with emotion, posture, temperature, and threat.

Cortical organisation

Primary sensory and motor cortices handle relatively specific signals, while association regions integrate across modalities and time. White-matter tracts connect frontal, parietal, temporal, occipital, limbic, thalamic, basal-nuclear, and cerebellar networks. Function depends on connectivity as much as local tissue. A strategic lesion can disconnect a network and cause disproportionate disability.

The two hemispheres are specialised but cooperative. Language is left-dominant in most people, while right-hemisphere networks are important for spatial attention, prosody, and global context. Corpus-callosal fibres share information. Dominance is probabilistic, so lesion effects require examination rather than assumption.

Attention and executive control

Attention includes alertness, sustained focus, selection, shifting, and divided processing. Frontal and parietal networks interact with thalamus, brainstem arousal systems, and salience networks. Attention failure degrades every cognitive test; poor recall may reflect failure to encode rather than storage loss.

Prefrontal systems hold information in working memory, inhibit inappropriate responses, switch rules, sequence behaviour, estimate consequences, and monitor errors. Orbitofrontal injury can cause disinhibition and poor social judgement; medial frontal injury can cause apathy and reduced initiation; dorsolateral injury impairs planning and mental flexibility. Real syndromes overlap and are shaped by premorbid personality and environment.

Neglect is failure to attend to one side of space or body, usually after right hemispheric injury affecting left-sided attention. It is not explained by primary sensory loss. Patients may ignore food, grooming, or threats on the affected side and can lack awareness of deficit. Extinction appears when a contralesional stimulus is perceived alone but missed during bilateral stimulation.

Language and related symbolic functions

Language requires comprehension, word access, grammar, repetition, speech, reading, and writing. Dominant perisylvian networks interact with semantic and executive systems. Non-fluent aphasia produces effortful reduced output with relatively better comprehension. Fluent receptive aphasia produces effortless but empty speech with impaired comprehension. Conduction syndromes disproportionately impair repetition; anomia occurs widely.

Dysarthria is impaired motor execution of speech with language intact. Apraxia of speech disrupts planning of articulatory sequences. Dysphonia reflects voice production. These distinctions determine localisation and therapy. Reading and writing can dissociate from spoken language according to network damage.

Apraxia is impaired execution of a learned skilled action not explained by weakness, sensory loss, poor comprehension, or incoordination. Agnosia is failure to recognise despite adequate primary sensation. Both expose how association networks transform perception into meaning and action.

Memory systems

Working memory temporarily holds and manipulates information. Episodic memory records personally situated events; semantic memory stores facts and concepts; procedural memory supports skills and habits. These systems use overlapping but different networks. Medial temporal structures, especially hippocampal circuits, are essential for consolidating new declarative memories but are not a permanent warehouse for all detail.

Encoding requires attention and organisation. Consolidation stabilises traces over time and sleep. Retrieval reconstructs information and is influenced by cues, emotion, expectation, and interference. Memory is therefore fallible even in health. Repeated retrieval strengthens access but can also update the trace.

Anterograde amnesia impairs formation of new memories; retrograde amnesia affects information from before injury, often with a temporal gradient. Frontal dysfunction impairs strategies and source monitoring. Basal-nuclear and cerebellar disease can disrupt procedural learning. Confabulation fills gaps without deliberate deception.

Emotion, motivation, and behaviour

Emotion coordinates appraisal, autonomic response, action, memory, and feeling. Amygdala networks detect salience and learn threat; hippocampal systems add context; prefrontal regions regulate response. Dopamine reward signals encode differences between expected and obtained outcomes rather than pleasure alone.

Stress responses mobilise autonomic and endocrine resources. Acute responses can be adaptive; persistent uncontrollable stress can disturb sleep, mood, immunity, metabolism, pain, and cognition. Psychiatric symptoms can arise from primary mental illness, neurological disease, endocrine or metabolic disturbance, drugs, infection, or delirium. Brain and mind are not competing explanations.

Sleep architecture

Sleep is regulated by circadian timing and homeostatic pressure accumulated during wakefulness. The suprachiasmatic nucleus aligns rhythms to light through retinal input. Melatonin signals biological night but does not simply switch consciousness off. Behaviour, meals, activity, social timing, and medications also entrain rhythms.

Non-rapid-eye-movement sleep progresses from light stages to slow-wave sleep, with reduced sympathetic activity and characteristic electroencephalographic patterns. Rapid-eye-movement sleep includes vivid dreaming, cortical activation, rapid eye movements, autonomic variability, and skeletal-muscle atonia except for diaphragm and selected muscles. Cycles recur through the night, with more slow-wave sleep earlier and more rapid-eye-movement sleep later.

Sleep supports memory, emotional regulation, synaptic recalibration, metabolism, immunity, and clearance. Deprivation impairs vigilance, reaction time, judgement, glucose regulation, mood, and pain tolerance. Subjective adaptation often exceeds objective recovery, so people underestimate impairment.

Sleep disorders

Insomnia is persistent difficulty initiating or maintaining sleep, or early waking, with daytime consequence despite opportunity. Perpetuating behaviours and conditioned arousal can outlast the original trigger. Cognitive behavioural therapy for insomnia targets timing, stimulus control, sleep pressure, beliefs, and arousal; sedatives may help selected short-term situations but can cause dependence, falls, cognitive effects, and respiratory worsening.

Obstructive sleep apnoea causes recurrent upper-airway collapse, intermittent hypoxaemia, pressure swings, and arousal. Features include snoring, witnessed pauses, unrefreshing sleep, morning headache, nocturia, and daytime sleepiness, though absence of sleepiness does not exclude disease. Treatment includes positive airway pressure, weight management, positional or oral devices, and selected surgery.

Central sleep apnoea reflects unstable or absent ventilatory drive. Narcolepsy causes sleepiness with rapid-eye-movement intrusion and may include

cataplexy. Parasomnias arise from incomplete state transitions. Restless legs produces an urge to move at rest, relieved by movement, and may relate to iron deficiency or drugs. Diagnosis uses a targeted history, sleep diary, actigraphy, or polysomnography according to the question.

Consciousness and arousal

Consciousness has arousal and content. Arousal depends on distributed ascending systems in upper brainstem, hypothalamus, thalamus, and basal forebrain. Awareness requires organised bilateral cortical and thalamocortical function. Coma generally results from bilateral hemispheric dysfunction or injury to the ascending arousal network, not a small unilateral hemispheric lesion alone.

Assess airway, breathing, circulation, glucose, temperature, toxins, seizures, trauma, infection, and metabolic disturbance immediately. Examine response to voice and pain, pupils, eye position and movements, motor pattern, tone, reflexes, and respiration. Sedatives, paralysis, hypothermia, and organ failure confound assessment. Structural lesions often produce focal or rostrocaudal signs; metabolic disorders more often cause symmetric findings, but exceptions are common.

Delirium is acute fluctuating failure of attention and cognition caused by physiological disturbance. Hypoactive delirium is easily missed. Dementia is acquired chronic decline interfering with independence; delirium can coexist with it. A minimally conscious state shows reproducible but limited awareness, whereas unresponsive wakefulness has eye opening without behavioural evidence of awareness. Locked-in syndrome preserves consciousness but removes most motor output.

Autonomic organisation

The autonomic system regulates pressure, heart rate, temperature, pupils, airways, gut, bladder, sexual function, and glands. Thoracolumbar sympathetic preganglionic neurons use acetylcholine; most postganglionic neurons use noradrenaline except sweat fibres. Craniosacral parasympathetic neurons use acetylcholine at ganglia and targets.

Sympathetic activity supports pressure during standing, redistributes flow, mobilises fuel, dilates pupils, and coordinates threat responses. Parasympathetic activity supports digestion, storage, and selected restorative functions, but the systems are not simple opposites. Most organs receive patterned region-specific control. The enteric system can coordinate locally while receiving autonomic modulation.

Baroreceptors sense arterial stretch and rapidly adjust sympathetic and vagal output. Standing shifts blood downward; vasoconstriction, increased heart rate, muscle pumping, and hormonal support preserve cerebral perfusion. Orthostatic hypotension can reflect volume depletion, drugs, autonomic neuropathy, neurodegeneration, or deconditioning. Heart-rate response and clinical context help distinguish mechanisms.

Autonomic failure and emergencies

Autonomic dysfunction can cause postural dizziness, syncope, abnormal sweating, heat intolerance, bowel dysmotility, bladder dysfunction, erectile failure, pupillary abnormalities, and impaired hypoglycaemia awareness. Evaluation includes lying and standing pressure and pulse, medication review, hydration, neuropathy assessment, and specialised autonomic tests when needed.

High spinal cord injury can permit uncontrolled sympathetic reflexes below the lesion. Bladder or bowel distension may trigger severe hypertension, headache, sweating, flushing, and bradycardia: autonomic dysreflexia. Sit the patient upright, remove triggers, monitor pressure, and use rapid treatment when needed. Failure to recognise it risks stroke, seizure, or cardiac injury.

TTS module 2: Network cognition, sleep-state control, and disorders of consciousness

Higher function is best understood as coordinated activity across distributed networks rather than as a catalogue of isolated cortical centres. A patient can have intact primary sensation yet fail to use sensory information because attention, recognition, spatial representation, or connection to language has failed. Similarly, apparently poor memory may arise because information was never attended to, because it was not consolidated, because retrieval strategies are ineffective, or because the response cannot be expressed. The clinical task is therefore to identify the failed operation, its time course, and the network capable of producing it.

Attention depends on interacting alerting, orienting, and executive systems. Brainstem and hypothalamic arousal nuclei establish wakefulness. Dorsal frontoparietal networks direct goal-driven attention, while ventral and salience networks interrupt current activity when an unexpected or important stimulus appears. The thalamus regulates access and synchronisation rather than merely relaying signals. Diffuse metabolic encephalopathy degrades these networks globally, whereas a strategic right parietal lesion may produce profound left neglect. Bedside testing should begin with arousal and sustained attention, using observation, digit span, months backward, or response consistency before interpreting memory or language scores.

Working memory maintains a limited representation long enough to manipulate it. The prefrontal cortex does not act alone; it coordinates sensory association cortex, basal ganglia, thalamus, and cerebellum according to the current rule. Executive failure appears as perseveration, impulsivity, poor sequencing, reduced error awareness, or inability to shift set. A patient may describe what each item in a task requires but still fail to organise the sequence. Frontal syndromes are influenced by lesion site, network disconnection, fatigue, medication, mood, and the structure supplied by the environment, so behaviour in a quiet examination can underestimate disability in daily life.

Declarative memory formation requires encoding, consolidation, storage, and retrieval. Hippocampal circuits bind the elements of an episode and gradually support their integration into distributed cortical representations. Bilateral medial temporal injury therefore causes severe anterograde amnesia while

immediate span and learned motor skills may remain relatively intact. Remote memories are not uniformly protected because vivid detail and repeated reconstruction may continue to depend on hippocampal interaction. Frontal disease more often impairs organisation, free recall, source monitoring, and verification; cueing or recognition may improve performance. Depression, pain, sleep loss, and delirium impair attention and encoding and can imitate a primary memory disorder.

Language assessment separates spontaneous output, comprehension, naming, repetition, reading, writing, and motor speech. Fluent speech is not necessarily meaningful, and apparent comprehension must be tested with commands that cannot be solved from context. Repetition depends on auditory analysis, phonological working memory, and speech production, making it vulnerable to several network lesions. Naming failure may reflect degraded semantic knowledge, impaired access to the word form, or a motor-output problem. Acute aphasia is a stroke warning even when weakness is absent. A patient who cannot understand language may also seem confused, while a severely dysarthric patient may be cognitively intact; establishing a reliable yes-or-no channel prevents dangerous misclassification.

Consciousness combines wakeful arousal with the contents of awareness. Ascending cholinergic, noradrenergic, serotonergic, dopaminergic, histaminergic, and orexinergic systems project through the hypothalamus, thalamus, basal forebrain, and cortex. No single transmitter is a consciousness switch. Wakefulness is a stabilised network state produced by mutual reinforcement among arousal systems and inhibition of sleep-promoting neurons. Bilateral cortical or thalamocortical dysfunction can abolish awareness despite an anatomically intact brainstem, while a lesion of the upper brainstem arousal system can abolish wakefulness despite relatively preserved cortex.

The sleep-wake switch includes sleep-promoting neurons in the ventrolateral preoptic region that inhibit arousal systems. During wakefulness, orexin neurons help stabilise arousal and prevent inappropriate transitions. Loss of orexin signalling underlies narcolepsy type one and helps explain sleep attacks, cataplexy, sleep paralysis, and rapid-eye-movement phenomena intruding into wakefulness. Adenosine accumulation contributes to

homeostatic sleep pressure; caffeine promotes alertness mainly by antagonising adenosine receptors. The circadian system, synchronised principally by retinal light input to the suprachiasmatic nucleus, determines when sleep is biologically favoured. Homeostatic pressure and circadian alerting can oppose one another, explaining the temporary evening second wind despite prolonged wakefulness.

Non-rapid-eye-movement sleep becomes progressively synchronised, culminating in slow-wave activity associated with high arousal thresholds, memory processing, and substantial parasympathetic predominance. Rapid-eye-movement sleep resembles activated wakefulness on the electroencephalogram but includes pontine mechanisms that suppress spinal motor output. Failure of this atonia permits dream enactment and may precede synuclein neurodegenerative disease. Normal cycling is altered by age, illness, alcohol, antidepressants, sedatives, pain, and environmental interruption. Sedation is not physiological sleep: a sedated patient may be unconscious without receiving the normal architecture or restorative functions of sleep.

Obstructive sleep apnoea repeatedly couples airway collapse to hypoxaemia, hypercapnia, sympathetic surges, and cortical arousal. The patient may not remember waking, yet sleep is fragmented and cardiovascular load rises throughout the night. Anatomical airway susceptibility interacts with sleep-dependent muscle relaxation, ventilatory control instability, arousal threshold, obesity, alcohol, and posture. Severity is not captured by event count alone; oxygen burden, sleep fragmentation, symptoms, comorbidity, and driving risk matter. Positive airway pressure splints the airway but success depends on mask fit, pressure tolerance, humidification, education, and follow-up. Persistent sleepiness despite treatment warrants confirmation of adherence, adequate sleep opportunity, medication review, and consideration of another sleep disorder.

Insomnia is maintained by more than insufficient sedation. Time spent awake in bed conditions the bed as a cue for vigilance, while irregular rising times weaken circadian stability and excessive time in bed reduces sleep pressure. Cognitive behavioural therapy for insomnia consolidates sleep opportunity, strengthens bed-sleep association, regularises timing, and addresses catastrophic interpretation of wakefulness. Long-term sedative prescribing

can worsen falls, cognition, tolerance, dependence, and sleep-disordered breathing. Abrupt withdrawal from some sedatives can cause severe rebound insomnia, autonomic activation, delirium, or seizures, so deprescribing may require a planned taper.

Disorders of consciousness demand parallel stabilisation and localisation. First correct immediately reversible threats: hypoxaemia, hypoglycaemia, hypotension, hyperthermia or hypothermia, opioid toxicity, seizure, and dangerous electrolyte disturbance. Then establish onset, medication and toxin exposure, trauma, infection risk, organ failure, and preceding focal symptoms. Pupillary size and reactivity, spontaneous eye position, vestibulo-ocular responses when safe, corneal reflexes, motor asymmetry, posturing, and respiratory pattern help test brainstem pathways. A unilateral dilated poorly reactive pupil with declining consciousness suggests compression of the third nerve and requires emergency action, although ocular disease and drugs can mimic it.

The Glasgow Coma Scale communicates eye, verbal, and motor responsiveness but is not a complete neurological examination. Intubation, aphasia, deafness, language difference, paralysis, sedation, and facial trauma alter scores. Trend and component scores are more informative than an isolated total. Subtle non-convulsive status epilepticus may present as unexplained coma or fluctuating responsiveness and requires electroencephalography. Structural and metabolic causes overlap: sepsis can produce focal-appearing signs, and bilateral strokes can appear symmetric. Imaging, laboratory testing, toxicology, lumbar puncture, and electroencephalography should be selected from the physiological question rather than performed as a ritual sequence.

Delirium is a network failure characterised by acute change, fluctuating attention, altered arousal, and disorganised cognition. It is often multifactorial: infection, dehydration, pain, urinary retention, constipation, hypoxia, drug toxicity or withdrawal, sleep disruption, sensory deprivation, and unfamiliar surroundings may combine in a vulnerable brain. Hypoactive delirium carries substantial risk because quiet withdrawal is mistaken for cooperation or fatigue. Management treats causes while protecting orientation, hydration, mobility, hearing, vision, sleep, and family contact. Antipsychotics do not cure

delirium and are reserved for selected severe distress or immediate safety risk after reversible triggers and non-drug measures are addressed.

Autonomic control integrates central state with organ demand. The insula, anterior cingulate, amygdala, hypothalamus, brainstem nuclei, spinal cord, peripheral ganglia, and afferent receptors form a continuous control network. Orthostatic symptoms occur when cerebral perfusion falls during standing. In volume depletion, heart rate usually rises as pressure falls; in neurogenic orthostatic hypotension, the compensatory heart-rate increase may be disproportionately small because sympathetic efferent failure is part of the disease. Measurements should follow adequate supine rest and continue after standing because delayed falls may be missed. Meals, heat, exercise, alcohol, antihypertensives, diuretics, and dopaminergic drugs can expose limited reserve.

Treatment of orthostatic intolerance begins with mechanism and safety. Review medications, correct volume loss, rise gradually, use physical counter-pressure manoeuvres, and consider compression or abdominal support. Increased salt or fluid is inappropriate in some patients with heart or kidney failure. Drugs that expand volume or increase vascular tone can cause supine hypertension, electrolyte disturbance, urinary retention, or cardiac stress. The goal is improved function and prevention of syncope rather than a perfect standing pressure. Taken together, cognition, sleep, consciousness, and autonomic function reveal a common principle: clinical states emerge from interacting networks, and careful bedside perturbation often identifies the failed control loop before any single test does.

Retrieval prompts

One. Why can attention failure mimic memory loss?

Two. How do aphasia, dysarthria, and apraxia of speech differ?

Three. Which networks support new declarative memory?

Four. How do circadian and homeostatic processes regulate sleep?

Five. What anatomy must fail to produce coma?

Six. How does the body preserve pressure on standing?

Concise answers

One. Information not attended to is poorly encoded and therefore unavailable for later recall.

Two. Aphasia impairs language, dysarthria impairs motor execution, and apraxia disrupts planning of speech movements.

Three. Medial temporal and hippocampal circuits are central to consolidation, interacting with distributed cortex.

Four. Circadian timing defines biological phase while accumulated wakefulness creates sleep pressure.

Five. Bilateral hemispheric function or the ascending brainstem-diencephalic arousal network must be substantially disrupted.

Six. Baroreflex vasoconstriction and tachycardia combine with muscle pumping, volume, and hormonal support.

Source map

Original synthesis informed by Guyton and Hall, cortical function, memory, sleep, consciousness, and autonomic control; Robbins, cognitive and neurodegenerative pathology; Katzung and OpenStax Pharmacology, autonomic and sedative drugs; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and Talley and O'Connor, cognitive, coma, sleep, and autonomic assessment.

Chapter 30: Stroke, Seizure, Neuropathy, Neurodegeneration, and Raised Intracranial Pressure

Orientation

Neurological disease is interpreted through anatomy, mechanism, and time course. Sudden focal deficit suggests vascular injury until proven otherwise; episodic stereotyped dysfunction suggests seizure or transient disturbance; progressive decline suggests degeneration, tumour, inflammation, or compression. Immediate priorities are airway, breathing, circulation, glucose, temperature, seizures, and rapidly reversible causes. Definitive treatment often depends on minutes, so localisation and stabilisation proceed together.

Cerebral ischaemia

Brain tissue requires continuous oxygen and glucose but stores little substrate. Arterial occlusion creates an irreversibly injured core surrounded by hypoperfused penumbra that may recover if flow returns. Excitotoxic glutamate, calcium entry, free radicals, mitochondrial failure, inflammation, oedema, and microvascular dysfunction expand injury. Collateral circulation, occlusion site, pressure, temperature, glucose, and time determine survival.

Thrombosis can arise from large-artery atherosclerosis, small penetrating-vessel disease, cardiac embolism, arterial dissection, hypercoagulability, or less common vasculopathies. A transient ischaemic attack is transient focal dysfunction without established infarction, but early stroke risk can be high. Rapidly resolved symptoms still require urgent evaluation.

Anterior-circulation stroke may cause contralateral face-arm or leg weakness and sensory loss, aphasia in the dominant hemisphere, neglect in the non-dominant hemisphere, visual-field loss, or gaze deviation. Posterior-circulation disease causes combinations of diplopia, dysarthria, dysphagia, vertigo, ataxia, crossed cranial and body signs, visual loss, and

impaired consciousness. Isolated dizziness is usually not stroke, but abrupt severe persistent vestibular syndrome with central signs demands caution.

Acute stroke treatment

Record last known well, deficit severity, medications, anticoagulants, baseline function, and contraindications. Non-contrast computed tomography rapidly distinguishes haemorrhage and major structural mimics; vascular imaging identifies treatable occlusion. Normal early computed tomography does not exclude ischaemia. Magnetic resonance diffusion is sensitive but should not delay eligible treatment.

Selected patients receive intravenous thrombolysis within a validated time window after haemorrhage and major contraindications are excluded. Mechanical thrombectomy can restore flow in selected large-vessel occlusion, including some patients presenting later when imaging shows salvageable tissue. Pressure targets depend on reperfusion plan and comorbidity; indiscriminate rapid reduction can reduce penumbral flow.

Give antiplatelet therapy when appropriate after haemorrhage and thrombolysis timing are considered. Manage oxygen only when needed, glucose extremes, fever, hydration, swallowing, aspiration, venous thrombosis, pressure injury, bladder, nutrition, and early mobilisation. Decompressive surgery can save life in selected malignant hemispheric or cerebellar infarction. Secondary prevention targets mechanism through antithrombotic therapy, rhythm detection, vascular risk control, carotid intervention when indicated, and rehabilitation.

Intracranial haemorrhage

Intracerebral haemorrhage causes direct tissue disruption, mass effect, oedema, and sometimes ventricular obstruction. Hypertension, cerebral amyloid angiopathy, anticoagulation, vascular malformation, tumour, and drugs are important causes. Treatment includes rapid imaging, airway and pressure management, reversal of anticoagulation, neurosurgical assessment, and control of complications. Routine seizure prophylaxis is not universally beneficial.

Subarachnoid haemorrhage classically causes sudden maximal headache, often with vomiting, neck stiffness, collapse, or focal signs. Aneurysmal rupture can rebleed and later cause vasospasm, hydrocephalus, seizures, and sodium disturbance. If initial imaging is non-diagnostic but suspicion remains, further validated testing is required. Secure the aneurysm early, control pressure, administer nimodipine, and monitor neurological and systemic complications.

Seizures and epilepsy

A seizure is transient excessive or synchronous neuronal activity causing motor, sensory, autonomic, cognitive, emotional, or behavioural phenomena. Focal seizures begin in one hemisphere and may remain aware, impair awareness, or spread bilaterally. Generalised seizures engage bilateral networks from onset. Syncope, migraine, sleep disorders, movement disorders, metabolic events, and functional attacks can mimic epilepsy.

Epilepsy is an enduring predisposition to unprovoked seizures, not every provoked event. Causes include genetic networks, developmental abnormalities, scars, stroke, tumour, infection, immune disease, and degeneration. Electroencephalography supports classification and recurrence risk but a normal recording does not exclude epilepsy. Magnetic resonance imaging seeks structural cause.

During a convulsive seizure, protect from injury, support airway and oxygenation, check glucose, and do not restrain or place objects in the mouth. A seizure lasting about five minutes or recurrent without recovery is status epilepticus: give a benzodiazepine promptly, followed by an effective longer-acting antiseizure medicine and investigation of cause. Persistent coma may represent non-convulsive status and requires electroencephalography.

Long-term drug choice depends on seizure type, age, pregnancy potential, comorbidity, interactions, organ function, and adverse effects. Adherence, sleep, alcohol, driving rules, water safety, bathing, work hazards, contraception, and pregnancy planning require discussion. Some focal epilepsies benefit from surgery, stimulation, or dietary therapy when medication fails.

Raised intracranial pressure

The rigid skull contains brain, blood, and cerebrospinal fluid. Compensation initially shifts venous blood and cerebrospinal fluid, but once reserve is exhausted, small volume increments sharply raise pressure. Causes include mass, haemorrhage, infarct oedema, infection, hydrocephalus, impaired venous drainage, and diffuse injury. Cerebral perfusion pressure approximates mean arterial pressure minus intracranial pressure.

Features include headache, vomiting, declining consciousness, sixth-nerve palsy, and papilloedema; the late combination of hypertension, bradycardia, and irregular respiration signals dangerous brainstem distortion. Herniation syndromes compress specific structures, producing pupil, motor, respiratory, and consciousness changes. Papilloedema can be absent in acute deterioration.

Elevate the head, maintain oxygenation and perfusion, treat fever and seizures, avoid hypotonic fluid, and obtain urgent imaging and specialist help. Hypertonic saline or mannitol can temporarily reduce swelling. Brief controlled hyperventilation is a rescue bridge in imminent herniation, not routine therapy. Definitive treatment may require lesion evacuation, cerebrospinal-fluid drainage, decompression, antimicrobial therapy, or tumour treatment. Lumbar puncture can be dangerous when pressure gradients or obstructive lesions are suspected.

Central nervous system infection and inflammation

Meningitis presents through fever, headache, neck stiffness, photophobia, vomiting, altered consciousness, or rash, but classic combinations may be incomplete. Encephalitis adds brain dysfunction such as confusion, seizures, focal signs, or behavioural change. Obtain cultures and begin empirical antimicrobials promptly; imaging or lumbar puncture must not create harmful delay. Cerebrospinal-fluid pattern, molecular tests, epidemiology, immune status, and exposure refine diagnosis.

Autoimmune encephalitis can produce subacute memory loss, psychiatric change, seizures, movement disorder, dysautonomia, and reduced consciousness. Multiple sclerosis causes inflammatory demyelinating lesions disseminated in time and central nervous system space. Optic neuritis,

sensory or motor episodes, brainstem syndromes, and myelitis are common patterns. Magnetic resonance imaging and cerebrospinal fluid support diagnosis while mimics are excluded. Acute relapses may receive corticosteroids; disease-modifying therapy reduces future inflammatory activity.

Peripheral neuropathy

Peripheral neuropathy is classified by distribution, fibre type, pathology, and tempo. Length-dependent polyneuropathy begins distally and symmetrically, often affecting feet before hands. Small-fibre injury causes burning pain and autonomic symptoms with relatively preserved strength and routine nerve-conduction studies. Large-fibre injury impairs vibration, proprioception, reflexes, and balance. Motor-predominant disease raises a different differential.

Causes include diabetes, alcohol, kidney disease, nutritional deficiency, toxins, medications, paraproteins, infection, autoimmunity, hereditary disorders, and entrapment. Nerve-conduction studies distinguish axonal loss from demyelination and localise large-fibre lesions. Investigation should be targeted by phenotype rather than exhaustive screening.

Guillain-Barre syndrome causes rapidly progressive weakness and areflexia, often after infection. Facial, bulbar, respiratory, sensory, pain, and autonomic features vary. Monitor vital capacity, swallowing, rhythm, and pressure; deterioration can be abrupt. Intravenous immunoglobulin or plasma exchange modifies disease, whereas corticosteroids alone do not. Chronic inflammatory demyelinating polyneuropathy evolves over a longer course and uses different criteria and maintenance treatment.

Neurodegeneration

Alzheimer disease causes progressive impairment of episodic memory followed by broader cognitive decline, associated with amyloid and tau pathology. Vascular cognitive impairment varies with infarcts, small-vessel disease, and network disconnection. Lewy body disease combines cognitive fluctuation, visual hallucinations, parkinsonism, sleep behaviour disorder, and autonomic dysfunction. Frontotemporal degeneration often begins with behaviour, executive, or language change.

Parkinson disease reflects degeneration of dopaminergic and wider networks, producing motor and non-motor disease. Motor-neuron disease causes progressive upper and lower motor-neuron loss while sensation is usually preserved; respiratory and bulbar failure determine risk. Huntington disease is inherited and combines movement, cognitive, and psychiatric change.

Diagnosis remains clinical, supported by imaging, laboratory exclusion, biomarkers in selected contexts, and longitudinal change. Reversible contributors such as drugs, depression, sleep apnoea, sensory loss, thyroid disturbance, vitamin deficiency, hydrocephalus, infection, and subdural blood should be considered. Management combines symptom therapy, rehabilitation, safety, caregiver support, advance planning, and treatment of complications. Disease labels should not eclipse capacity, autonomy, and personal goals.

TTS module 2: Time-critical neurology, localisation, and longitudinal decline

Neurological diagnosis begins with a compact question: where is the lesion, what process can act there, and how quickly did it develop? Sudden maximal deficit points toward vascular interruption, haemorrhage, seizure, or trauma. Evolution over hours to days suggests inflammation, infection, metabolic injury, or an expanding lesion. Weeks to months favour tumour, immune disease, compression, or degeneration, although every rule has exceptions. A precise timeline includes last known normal, first symptom, maximal severity, fluctuation, recovery, recurrence, and associated headache, fever, trauma, drug exposure, or systemic illness.

Acute focal deficit is stroke until proved otherwise because reperfusion benefit decays with time. The examination should identify a disabling syndrome while glucose, oxygenation, pressure, temperature, and airway are assessed. Facial asymmetry, arm drift, aphasia, neglect, gaze preference, visual-field loss, limb ataxia, and sensory asymmetry quickly sample major networks. Posterior circulation stroke may instead cause diplopia, dysphagia, dysarthria, severe gait or truncal ataxia, crossed findings, or reduced consciousness. A low screening score does not make these deficits benign. Stroke mimics include hypoglycaemia, seizure with postictal weakness, migraine aura,

functional neurological disorder, intoxication, tumour, and infection, but uncertainty should accelerate imaging rather than justify delay.

Ischaemic injury reflects both arterial occlusion and failure of collateral supply. Within the core, energy failure causes membrane depolarisation, cytotoxic oedema, calcium-mediated enzyme activation, and cell death. In the penumbra, electrical function fails before membrane integrity, leaving tissue potentially salvageable. Systemic hypotension, hypoxaemia, fever, and marked glucose disturbance can recruit penumbra into the core. Conversely, rapid recanalisation and effective collateral flow may preserve tissue beyond conventional clock windows. This is why modern selection combines time, clinical deficit, vessel imaging, infarct burden, perfusion pattern, bleeding risk, and premorbid context.

Non-contrast computed tomography rapidly detects most acute haemorrhage and major mass effect but may be normal early in ischaemic stroke. Computed-tomographic or magnetic-resonance angiography identifies large-vessel occlusion and arterial dissection. Perfusion imaging estimates severely injured and hypoperfused tissue in selected patients. Intravenous thrombolysis treats eligible patients after exclusion of haemorrhage and important contraindications; mechanical thrombectomy physically removes selected large-artery thrombi. These are complementary rather than competing therapies. Treatment should not await complete diagnostic certainty when validated eligibility is met, yet uncontrolled pressure, anticoagulant effect, recent surgery, bleeding history, and imaging findings require disciplined assessment.

After reperfusion decisions, stroke care remains physiological. Swallow screening precedes oral intake because silent aspiration is common. Fever, urinary infection, venous thrombosis, immobility, dehydration, malnutrition, pressure injury, depression, and delirium worsen recovery. Early rehabilitation uses repetitive task-specific practice while avoiding forced mobilisation in an unstable patient. Secondary prevention follows mechanism: antiplatelet therapy for most non-cardioembolic disease, anticoagulation for appropriate atrial fibrillation, lipid and pressure treatment, smoking cessation, diabetes care, carotid intervention in selected symptomatic stenosis, and investigation for dissection or unusual thrombophilia when the phenotype warrants it.

Intracerebral haemorrhage may enlarge during the first hours. Rapid reversal of anticoagulation, controlled pressure reduction, neurosurgical review, and repeated assessment target preventable expansion and mass effect. The location suggests mechanism: deep basal ganglia, thalamic, pontine, and cerebellar haemorrhages commonly reflect small-vessel hypertensive damage, while lobar bleeding in an older adult raises cerebral amyloid angiopathy among other causes. Vascular malformation, venous thrombosis, tumour, sympathomimetic drugs, and coagulopathy remain important. Cerebellar haemorrhage can rapidly compress the brainstem or obstruct cerebrospinal fluid and may require urgent evacuation.

Subarachnoid haemorrhage should be considered when headache reaches maximal intensity within seconds or minutes, especially with collapse, neck pain, vomiting, exertional onset, meningism, or neurological deficit. A normal early examination does not exclude it. Diagnostic strategy depends on timing and local validation of computed tomography, lumbar puncture, and vascular imaging. Once aneurysmal haemorrhage is established, early securing of the aneurysm limits rebleeding. Nimodipine reduces poor outcome related to delayed cerebral ischaemia. Hydrocephalus, sodium disturbance, seizures, cardiopulmonary complications, fever, and vasospasm require surveillance.

Seizure description should reconstruct the event before assigning a label. An aura may reveal focal onset through a rising epigastric sensation, sudden fear, an unusual smell, visual distortion, unilateral tingling, or forced head turn. Impaired awareness, automatisms, asymmetric tonic posture, rhythmic jerking, eye deviation, duration, cyanosis, injury, and recovery add localisation. Lateral tongue injury and prolonged postictal confusion support a convulsive seizure but are not mandatory. Brief myoclonic jerks after syncope and urinary incontinence are not specific. Eyewitness video, when safely obtained, may be more informative than a retrospective adjective such as “fit.”

Status epilepticus is a failure of seizure termination and becomes progressively harder to stop. At five minutes of continuous convulsion, give an adequate benzodiazepine dose promptly rather than repeated small doses that delay definitive therapy. Follow with a longer-acting antiseizure drug, support ventilation, check glucose and electrolytes, treat fever, and search for stroke, haemorrhage, infection, toxin, withdrawal, non-adherence, pregnancy-related disease, or metabolic failure. Persistent unresponsiveness

after motor activity stops may be postictal, drug-related, structurally caused, or non-convulsive status; electroencephalography resolves an otherwise invisible emergency. Refractory status requires anaesthetic-level therapy and intensive monitoring.

Raised intracranial pressure is dangerous because the skull cannot expand. Compensation initially displaces cerebrospinal fluid and venous blood, so pressure may remain deceptively normal while reserve disappears. Further volume then produces a steep pressure rise, reduced cerebral perfusion, tissue shifts, and herniation. Headache and vomiting are nonspecific, and papilloedema takes time to develop. Declining arousal, new anisocoria, abnormal eye movements, asymmetric motor response, extensor posturing, or irregular respiration are late physiological warnings. Hypotension is especially harmful because cerebral perfusion pressure falls from both sides of the equation.

Emergency management protects oxygenation, venous drainage, and perfusion while definitive treatment is arranged. Elevate the head with the neck neutral, avoid fever and seizures, prevent hypotonic fluid, and correct severe metabolic disturbances. Hypertonic saline or mannitol creates an osmotic gradient, but each has volume, sodium, renal, and haemodynamic consequences. Brief hyperventilation lowers carbon dioxide and constricts cerebral vessels, buying time during imminent herniation; prolonged use risks cerebral ischaemia. Surgical evacuation, decompression, ventricular drainage, antimicrobial treatment, or removal of an obstructing mass addresses the cause. Lumbar puncture is deferred when an intracranial pressure gradient could drive tissue shift.

Peripheral weakness is localised by pattern. Upper motor-neuron disease produces weakness with increased tone, brisk reflexes, and pathological plantar responses after the acute phase. Lower motor-neuron disease produces reduced tone, wasting, fasciculation, and depressed reflexes. Neuromuscular-junction failure is often fluctuating and fatigable with preserved sensation, while muscle disease tends to cause proximal weakness without sensory loss. Polyneuropathy commonly gives distal symmetric sensory loss and reduced ankle reflexes. Nerve-conduction studies test large myelinated fibres; normal results do not exclude small-fibre

neuropathy. Cerebrospinal fluid, antibody testing, imaging, biopsy, or genetics should answer a focused phenotype-based question.

Guillain–Barré syndrome is a neurological and autonomic emergency because limb strength can underestimate respiratory decline. Serial forced vital capacity, inspiratory force where available, cough, swallowing, voice, neck flexion, oxygenation, rhythm, and pressure are monitored. Pulse oximetry can remain normal until ventilatory failure is advanced. Bulbar weakness and inability to clear secretions may mandate intubation before carbon dioxide rises. Dysautonomia produces extreme pressure fluctuation and arrhythmia, making indiscriminate vasoactive treatment hazardous. Intravenous immunoglobulin and plasma exchange are effective; combining them routinely adds no established advantage.

Neurodegenerative diagnosis relies on the first and dominant network to fail. Alzheimer disease commonly begins with progressive episodic-memory impairment. Dementia with Lewy bodies is suggested by fluctuation, recurrent visual hallucinations, rapid-eye-movement sleep behaviour disorder, and spontaneous parkinsonism; marked antipsychotic sensitivity is clinically important. Behavioural-variant frontotemporal degeneration begins with disinhibition, apathy, loss of empathy, compulsive behaviour, dietary change, or executive dysfunction. Parkinson disease includes bradykinesia plus rigidity or rest tremor, but constipation, anosmia, depression, sleep disturbance, orthostatic hypotension, and cognitive change may precede or dominate later disability.

Motor-neuron disease combines progressive upper and lower motor-neuron findings without a matching sensory syndrome. Bulbar function, nutrition, communication, cough strength, and ventilation require proactive review because respiratory failure can develop without dramatic limb decline. Across all degenerative diseases, a label is only the beginning. Medication burden, hearing and vision, sleep, mood, pain, infection, metabolic disturbance, caregiver strain, driving, falls, swallowing, capacity, and goals of care alter function and safety. Longitudinal observation often clarifies phenotype better than a single biomarker. Acute neurology rewards speed; chronic neurology rewards pattern recognition, repeated measurement, and anticipation of the next loss of physiological reserve.

Retrieval prompts

- One. What is the ischaemic penumbra?
- Two. Which factors determine eligibility for acute reperfusion?
- Three. When does a convulsive seizure become status epilepticus?
- Four. Why can lumbar puncture be dangerous with raised pressure?
- Five. How are peripheral neuropathies classified?
- Six. What distinguishes common neurodegenerative syndromes clinically?

Concise answers

- One. Hypoperfused, dysfunctional but potentially salvageable tissue surrounding an infarct core.
- Two. Time, imaging, deficit, occlusion, salvageable tissue, bleeding risk, contraindications, and baseline context.
- Three. At about five minutes, or with recurrent seizures without recovery.
- Four. Removing spinal fluid can worsen a pressure gradient and precipitate herniation.
- Five. By distribution, sensory-motor-autonomic fibre involvement, axonal versus demyelinating pathology, and tempo.
- Six. The initial pattern of memory, behaviour, language, movement, fluctuation, autonomic change, and motor-neuron signs.

Source map

Original synthesis informed by Guyton and Hall, cerebral blood flow, seizures, pressure, and neural systems; Robbins, vascular, infectious, demyelinating, peripheral-nerve, and neurodegenerative pathology; Katzung and OpenStax Pharmacology, reperfusion, antiseizure, and neurological therapy; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, acute neurological assessment.

Chapter 31: Neurological History, Examination, Localisation, and Investigations

Orientation

Neurological diagnosis asks where the lesion is, then what process produces it at that speed. History defines evolution; examination samples cognition, cranial nerves, motor, reflex, sensory, coordination, gait, and autonomic function. Coherent localisation guides testing. Normal findings require adequate testing, and incidental imaging never replaces anatomy.

Time course and symptom definition

Establish last-known-normal time for sudden symptoms. Maximal deficit at onset suggests embolism, haemorrhage, seizure, migraine, or trauma. Evolution over hours to days suggests inflammation, infection, infarct progression, metabolic disturbance, or compression. Weeks to months suggest tumour, immune disease, or degeneration; recurrent stereotyped episodes suggest seizure, migraine, vascular events, or channel disorders.

Ask the patient to describe what failed rather than accepting labels such as dizziness, weakness, numbness, blackout, or confusion. Dizziness may mean vertigo, presyncope, imbalance, or non-specific disequilibrium. Weakness may be loss of power, pain inhibition, fatigue, incoordination, or reduced initiation. Numbness may mean absent sensation, tingling, burning, heaviness, or altered body perception.

Clarify distribution, triggers, duration, recovery, associated headache, fever, neck pain, vision, speech, swallowing, hearing, bladder, bowel, sexual function, pain, cognition, and consciousness. Obtain witness descriptions of episodic events, including posture, prodrome, colour, eye and head movement, motor sequence, responsiveness, duration, injury, tongue biting, recovery, and provoking circumstances.

Background and safety

Record vascular risks, migraine, epilepsy, cancer, immune disease, infection, trauma, pregnancy, surgery, toxins, alcohol, nutrition, exposures, and family disease. Review anticoagulants, sedatives, dopamine blockers, antiseizure medicines, immunotherapy, supplements, and recreational drugs. Establish baseline mobility, cognition, communication, handedness, driving, supports, and wishes.

Immediate red flags include sudden focal deficit, thunderclap headache, meningism, new seizure, rapidly progressive weakness, respiratory or bulbar symptoms, acute visual loss, spinal pain with sphincter disturbance, trauma, fever with altered cognition, and declining consciousness. Stabilisation and urgent treatment precede a leisurely complete examination.

Mental status

First assess arousal and attention because impairment invalidates downstream cognitive testing. Observe behaviour, spontaneous speech, orientation, and ability to follow commands. Test sustained attention with a suitable task, then memory registration and delayed recall, language, visuospatial processing, praxis, calculation, abstraction, and executive control according to the clinical question.

Screening scores can document impairment but are influenced by education, language, culture, sensory loss, distress, fatigue, and disability. Delirium is suggested by acute fluctuation and inattention. Aphasia must be distinguished from dysarthria, deafness, low arousal, and language unfamiliarity. Assess capacity for the specific decision rather than inferring it from diagnosis or a score.

Cranial nerves

Test visual acuity in each eye, fields, pupils, fundi when indicated, colour, and eye movements. A relative afferent pupillary defect indicates asymmetric retinal or optic-nerve dysfunction. Papilloedema suggests raised pressure but is neither immediate nor universal. Eye position, diplopia direction, nystagmus, and vestibulo-ocular responses help localise brainstem, nerve, neuromuscular, orbital, or vestibular disease.

Assess facial sensation and corneal pathways. Inspect forehead elevation, eye closure, smile, and cheek inflation. Upper motor-neuron lesions usually spare the forehead; lower motor-neuron lesions affect the whole ipsilateral face. Distinguish conductive from sensorineural hearing loss before formal audiology.

Evaluate palate movement, voice, cough, articulation, secretion control, and swallowing safety. Test shoulder elevation, head turn, and tongue movement. Lower cranial dysfunction can threaten airway and nutrition. Tongue deviation, atrophy, fasciculation, palate asymmetry, and vocal quality contribute to brainstem, nerve, junction, and muscle localisation.

Motor examination

Inspect bulk, fasciculations, involuntary movement, posture, and asymmetry. Assess tone by moving relaxed joints at different speeds. Spasticity is velocity-dependent; rigidity is more uniform; paratonia varies with cooperation or frontal dysfunction. Test power in anatomically informative groups and grade consistently, distinguishing true weakness from pain, poor comprehension, fatigue, and give-way performance.

Look for drift and loss of fine fractionated movement. Pyramidal weakness often affects upper-limb extensors and lower-limb flexors more than their opposites. Root and peripheral-nerve lesions follow myotomal or named-nerve patterns. Neuropathy is often distal; myopathy is often proximal; neuromuscular-junction weakness fluctuates and fatigues.

Reflexes include biceps, supinator, triceps, knee, ankle, plantar responses, and selected superficial reflexes. Compare sides and relate results to power and tone. Hyperreflexia, clonus, and extensor plantar responses support upper motor-neuron dysfunction. Reduced reflexes suggest lower motor neuron, nerve, root, junction, or acute spinal shock but can be normal variants.

Sensory examination

Map symptoms before testing. With eyes closed, compare light touch, pinprick, temperature when needed, vibration, and joint position. Avoid predictable rhythmic testing and ask what is felt rather than leading the answer. Cortical sensory tests such as extinction, graphesthesia, stereognosis, and point localisation require intact primary sensation and attention.

A sensory level suggests spinal cord disease. Saddle loss suggests cauda equina or conus involvement. Dermatomal loss suggests root disease; named-nerve loss suggests mononeuropathy; stocking distribution suggests polyneuropathy. Hemibody loss can localise above the brainstem, while crossed face-body patterns suggest brainstem. Functional sensory findings may be internally inconsistent, but positive diagnostic signs are preferable to diagnosis by exclusion.

Coordination, stance, and gait

Test finger-nose, heel-shin, rapid alternating movement, rebound, and rhythm. Dysmetria and intention tremor suggest cerebellar dysfunction but can result from weakness, sensory loss, or severe tremor. Romberg testing assesses ability to maintain stance when visual input is removed; falling only with eye closure suggests impaired proprioceptive or vestibular compensation, not a general cerebellar test.

Observe rising, initiation, base, stride, foot clearance, arm swing, turning, tandem gait, heel and toe walking, and safe response to perturbation. Describe the pattern rather than merely calling it abnormal. Do not test an unsafe patient without assistance.

Localisation framework

A cerebral-hemisphere lesion may combine contralateral weakness or sensation with aphasia, neglect, field loss, seizure, or cortical sensory dysfunction. Brainstem lesions often cause cranial-nerve signs with contralateral body findings, gaze abnormalities, ataxia, or long-tract signs. Cerebellar lesions produce ipsilateral coordination deficits, eye signs, dysarthria, and gait disturbance without primary weakness.

Spinal-cord lesions produce bilateral signs below a level, upper motor-neuron findings beneath the lesion, segmental lower motor-neuron findings at the lesion, and possible sphincter dysfunction. Hemicord injury can combine ipsilateral motor and proprioceptive loss with contralateral pain-temperature loss below. Cauda equina lesions produce asymmetric lower motor-neuron and root signs with saddle and sphincter involvement.

Root disease causes radiating pain, myotomal weakness, dermatomal sensory change, and reduced corresponding reflex. Plexus lesions cross several roots and nerves. Mononeuropathy follows one named nerve. Polyneuropathy is usually length-dependent and symmetric. Neuromuscular-junction disease causes fatigable motor dysfunction without sensory loss. Myopathy produces proximal weakness without sensory loss, often with preserved reflexes until late.

Imaging

Non-contrast computed tomography rapidly detects blood, mass effect, hydrocephalus, and fractures. Angiography assesses arteries; venography assesses venous thrombosis. Magnetic resonance provides superior contrast for infarction, demyelination, tumour, infection, posterior fossa, spinal cord, nerve, and muscle. Sequences answer different mechanisms.

Imaging urgency follows clinical risk, not convenience. Contrast choice considers kidney function, pregnancy, allergy, and the diagnostic question. Incidental white-matter lesions, vascular variants, degenerative spine change, and benign masses are common. Correlate lesion side, level, timing, and pathway with the examination before assigning causality.

Cerebrospinal fluid and electrophysiology

Lumbar puncture measures opening pressure and samples cells, protein, glucose, microbiology, cytology, immune markers, and selected biomarkers. Consider imaging first when focal mass effect, obstructive hydrocephalus, markedly reduced consciousness, papilloedema with concerning features, or other herniation risk is suspected. Check bleeding risk and do not delay essential antimicrobial therapy.

Electroencephalography records cortical electrical activity and supports seizure classification, encephalopathy assessment, and detection of non-convulsive status. A normal short recording does not exclude epilepsy. Nerve-conduction studies and electromyography localise large-fibre nerve, root, motor-neuron, junctional, and muscle dysfunction and distinguish axonal from demyelinating patterns. Results depend on timing, temperature, sampling, and operator interpretation.

Evoked potentials test conduction through visual, auditory, or somatosensory pathways. Formal autonomic testing may assess heart-rate variability, pressure responses, sweat function, and small fibres. Neuropsychological assessment characterises cognition beyond screening and helps separate encoding, retrieval, language, attention, executive, and behavioural deficits.

Integration and communication

Summarise in one sentence: syndrome, localisation, likely mechanism, tempo, severity, and urgent alternatives. For example, abrupt dominant hemispheric cortical syndrome suggests acute arterial ischaemia until haemorrhage and mimics are excluded. State uncertainty explicitly and repeat examination because neurological signs evolve.

Communicate driving restrictions, seizure and water safety, falls, swallowing, medication changes, return precautions, and follow-up. Diagnose functional neurological disorder through positive signs and explain it without implying fabrication. Rehabilitation and support begin alongside diagnosis.

TTS module 2: Bedside localisation, examination reliability, and test selection

A neurological consultation should transform a narrative into a testable anatomical model. The most useful opening summary specifies the failed function, distribution, onset, evolution, associated features, and baseline. “The left side suddenly became weak and numb while speech remained normal” is more localising than “possible stroke.” Before refining anatomy, identify instability: impaired airway protection, respiratory muscle weakness, rapidly declining consciousness, status epilepticus, meningism with sepsis, acute spinal compression, or an evolving vascular syndrome changes the order and speed of assessment.

History often localises more precisely than a hurried examination because symptoms fluctuate and signs may recover. Ask what the patient was doing at onset and whether the deficit was maximal immediately. Positive phenomena such as flashing lights, marching tingling, jerking, or formed hallucinations suggest excessive or spreading neural activity; negative phenomena such as visual loss, numbness, weakness, or aphasia suggest loss of function. This distinction is helpful but not absolute. Migraine aura typically spreads over minutes and may move between modalities, whereas embolic ischaemia commonly starts abruptly. Focal seizure may be followed by temporary weakness, and transient ischaemia can occasionally produce positive symptoms.

Episodic loss of consciousness requires reconstruction from witnesses. Presyncope often includes warmth, nausea, sweating, visual dimming, pallor, and rapid recovery when supine. Arrhythmic syncope may be abrupt and unheralded. Convulsive movements can occur in syncope because cerebral perfusion is briefly inadequate; their presence alone does not establish epilepsy. Prolonged post-event confusion, stereotyped focal onset, lateral tongue injury, and sustained rhythmic motor activity increase the likelihood of seizure. Posture, exertion, emotion, pain, urination, cough, medication, family history of sudden death, and recovery time direct cardiovascular and neurological testing.

Examination reliability begins with conditions. Ensure the patient can hear, see, understand the language, and remain awake. Explain each manoeuvre and compare like with like. Record whether pain, fear, fatigue, sedation, splinting, or cognitive impairment limited performance. Repeating a finding after repositioning or distraction distinguishes a stable deficit from measurement noise. Serial examinations are especially valuable in evolving stroke, intracranial pressure, spinal compression, neuromuscular respiratory failure, and delirium. A documented change is often more informative than a technically elaborate single assessment.

Mental-status examination starts with spontaneous behaviour. Observe whether the patient initiates conversation, attends to both sides, maintains a goal, repairs errors, and understands social context. Orientation is relatively insensitive: an attentive patient with aphasia may fail questions, while a delirious patient may recite date and place. Test attention before delayed

recall. If free recall is poor, use category cues and recognition to distinguish retrieval inefficiency from failed encoding, while remembering that no bedside pattern is perfectly disease-specific. Language testing samples fluency, comprehension, naming, repetition, reading, and writing; use objects and commands that minimise educational bias.

Visual assessment separates eye, optic nerve, chiasm, tract, radiations, and cortex. Monocular loss lies anterior to the chiasm. Bitemporal field loss suggests chiasmal disease. A homonymous defect localises behind the chiasm on the contralateral side. Visual acuity should be measured with correction and each eye separately. Red desaturation and a relative afferent pupillary defect support asymmetric optic-nerve or severe retinal dysfunction. Pupillary constriction tests afferent input through the optic nerve and efferent parasympathetic output through the third nerve; anisocoria that differs in light and dark helps identify which pathway is failing.

Diplopia is characterised by direction, separation, torsion, gaze dependence, ptosis, pupil involvement, and fatigability. Binocular diplopia disappears when either eye is covered and reflects misalignment; monocular diplopia usually arises within the eye. A third-nerve palsy can impair adduction, elevation, depression, and eyelid elevation, with pupillary involvement raising concern for compression. Sixth-nerve weakness limits abduction but may be a non-localising consequence of raised intracranial pressure. Internuclear ophthalmoplegia indicates disruption of the medial longitudinal fasciculus. Complex gaze palsies and skew deviation point toward central pathways rather than a single ocular motor nerve.

Facial weakness distinguishes corticobulbar from peripheral patterns, but forehead sparing is not infallible. A lower motor-neuron seventh-nerve lesion weakens eye closure and the lower and upper face on one side; taste, lacrimation, hyperacusis, and other cranial findings refine the level. Bulbar examination deserves physiological attention. Wet voice, repeated throat clearing, weak cough, nasal speech, drooling, fatigable chewing, and inability to manage secretions predict aspiration or ventilatory risk. Tongue wasting and fasciculation support lower motor-neuron disease; slow spastic tongue movement and brisk jaw reflex support bilateral corticobulbar involvement.

Motor testing separates activation, force, tone, pattern, and endurance. Pronator drift is sensitive to subtle pyramidal weakness because the affected

arm pronates and descends; an arm that drifts without pronation or moves inconsistently requires broader interpretation. Grade power only after placing the joint correctly and stabilising adjacent segments. Compare movements supplied by different roots and nerves rather than relying on a single muscle. Proximal symmetric weakness suggests muscle disease, but endocrine myopathy, steroid exposure, critical illness, pain, and deconditioning are common alternatives. Fatigable ptosis, diplopia, dysarthria, or neck weakness suggests neuromuscular-junction failure, where reflexes and sensation are usually preserved.

Reflexes are meaningful as a pattern. Reinforcement can reveal a genuinely reduced lower-limb reflex, and asymmetry is often more useful than absolute briskness. An absent ankle reflex with distal sensory loss supports length-dependent neuropathy; an absent knee reflex with weak knee extension suggests femoral or relevant root dysfunction. Brisk legs with extensor plantar responses and a sensory level indicate spinal cord disease. Acute cord injury may initially produce flaccidity and absent reflexes before spasticity appears. A brisk jaw reflex with limb upper motor-neuron signs suggests disease above the cervical cord or widespread corticobulbar involvement.

Sensory testing should answer an anatomical question rather than cover the body indiscriminately. Begin where sensation is expected to be normal and move toward the abnormal region. Pinprick samples small-fibre pathways; vibration and joint position sample large-fibre dorsal-column pathways. A spinal sensory level can lie below the structural lesion and should be checked from both directions. In thalamic or cortical disease, primary modalities may be altered across the contralateral body, while neglect, extinction, or impaired graphesthesia indicates higher processing failure. Dissociated loss of pain and temperature with preserved proprioception narrows the involved pathways.

Gait integrates nearly every neurological system. A hemiparetic gait circumducts a stiff extended leg; a spastic paraparesis may scissor; foot drop produces a high-stepping gait; proximal weakness produces pelvic instability and difficulty rising; cerebellar disease broadens the base with irregular timing; sensory ataxia worsens without visual input; parkinsonism reduces stride and arm swing and impairs turning. Frontal gait disorders cause

initiation failure and short, poorly lifted steps despite preserved strength when seated. Observe turning and recovery from perturbation only when safe, because these reveal postural reserve but can provoke a fall.

Localisation is built from concordant findings. Contralateral face, arm, and leg weakness with aphasia indicates a dominant cerebral hemisphere. Ipsilateral cranial-nerve dysfunction with contralateral limb signs suggests brainstem. Bilateral upper motor-neuron signs below a sensory level indicate spinal cord. Dermatomal pain with myotomal weakness and a corresponding reflex change indicates root disease. Weakness and sensory loss spanning multiple named nerves but not an entire limb length-dependently suggest plexus disease. A single named-nerve pattern suggests mononeuropathy, although anatomical variants and overlapping territories require electrodiagnostic confirmation when management depends on precision.

Investigations test the model. Computed tomography is fast for haemorrhage, fracture, hydrocephalus, and gross mass effect. Magnetic resonance diffusion detects acute infarction; fluid-sensitive sequences reveal oedema, demyelination, tumour, and gliosis; susceptibility sequences detect blood products; contrast patterns assess barrier disruption and vascularity. Spinal imaging must include the suspected level and sometimes adjacent regions because clinical levels are approximate. Imaging abnormalities are causal only when side, tract, level, age, and syndrome align. Common degenerative disc disease should not explain symptoms it cannot anatomically produce.

Lumbar puncture can measure pressure and identify infection, inflammation, haemorrhage, malignant cells, and selected biomarkers. Opening pressure requires appropriate positioning and interpretation in context. Cell count, differential, protein, glucose paired with blood glucose, culture, and molecular tests are selected before the procedure so limited fluid is used intelligently. Antibiotics and antiviral therapy must not wait when meningitis or encephalitis is strongly suspected. Electroencephalography answers whether abnormal cortical electrical activity or encephalopathy is present during the recording; prolonged monitoring increases yield for intermittent events. Nerve conduction and electromyography distinguish axonal loss, demyelination, root disease, motor-neuron dysfunction, junctional failure, and myopathy, but timing matters because denervation changes may not yet be detectable.

The final synthesis should state both the best localisation and evidence against dangerous alternatives. For example: an acute right hemispheric cortical syndrome with left neglect and weakness is most consistent with arterial ischaemia, while haemorrhage and seizure remain urgent exclusions. This formulation directs the next action and exposes uncertainty. Neurological expertise is not the performance of every manoeuvre; it is the disciplined selection of observations that change localisation, urgency, or management, followed by reassessment when the patient or evidence changes.

Retrieval prompts

One. Why must neurological symptoms be translated from labels into failed functions?

Two. What distinguishes cerebral, brainstem, and spinal localisations?

Three. How do root, nerve, junction, and muscle patterns differ?

Four. What does Romberg testing assess?

Five. When can lumbar puncture require preceding imaging?

Six. What belongs in a useful neurological summary?

Concise answers

One. Common labels combine physiologically different experiences requiring different localisations.

Two. Cortex adds higher-function signs; brainstem combines cranial and long-tract signs; cord produces bilateral findings below a level.

Three. Roots follow myotomes and dermatomes, nerves follow named territories, junctions cause fatigable pure motor weakness, and muscle disease is usually proximal.

Four. Dependence on vision for stance when proprioceptive or vestibular input is impaired.

Five. When mass effect, obstruction, concerning papilloedema, or reduced consciousness creates herniation concern.

Six. Syndrome, localisation, mechanism, tempo, severity, urgent alternatives, and degree of certainty.

Source map

Original synthesis informed principally by Talley and O'Connor, neurological history, examination, and localisation; Guyton and Hall, pathway anatomy and reflex physiology; Robbins, neurological pathology; Katzung and OpenStax Pharmacology, medication effects; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and contemporary diagnostic principles synthesised in original prose.

Chapter 32: Skeletal Muscle, Bone, Joint, and Connective Tissue Physiology

Orientation

The musculoskeletal system converts chemical energy into force, transmits that force through tendons and joints, supports and protects organs, stores mineral, houses marrow, and adapts to load. Function depends on muscle fibres, motor control, extracellular matrix, bone architecture, cartilage, synovium, ligaments, tendons, fascia, vessels, and nerves. Disease can arise from failure of contractile machinery, connective tissue, mineralisation, perfusion, innervation, immune tolerance, or mechanical adaptation.

Skeletal-muscle organisation

Muscle is organised from whole muscle into fascicles, fibres, myofibrils, and sarcomeres. Connective-tissue layers transmit force both longitudinally and laterally. A fibre is a multinucleated cell containing repeating sarcomeres bounded by Z discs. Thin filaments contain actin, tropomyosin, and troponin; thick filaments contain myosin. Titin centres thick filaments and contributes passive elasticity.

Sarcomeres shorten as actin slides past myosin; filaments do not themselves shorten. Myosin binds actin, performs a power stroke, releases when adenosine triphosphate binds, and recocks after hydrolysis. Without adenosine triphosphate, cross-bridges remain attached. Force depends on overlap: excessive shortening or stretch reduces productive cross-bridge formation.

Muscle architecture determines function. Parallel fibres permit greater excursion and speed; pennate fibres pack more contractile material into a given volume and generate greater force. Physiological cross-sectional area predicts force better than visible bulk. Tendon compliance allows temporary elastic-energy storage and alters the relation between fibre shortening and joint movement.

Excitation-contraction coupling

A motor-neuron action potential releases acetylcholine at the neuromuscular junction. The muscle action potential spreads along sarcolemma and transverse tubules. Voltage-sensitive proteins activate ryanodine receptors in sarcoplasmic reticulum, releasing calcium. Calcium binds troponin C, moves tropomyosin from actin sites, and permits cross-bridge cycling.

Calcium pumps return calcium to sarcoplasmic reticulum and allow relaxation. Repeated stimuli arrive before complete removal, summing force; sufficiently rapid stimulation produces tetanus. Neural recruitment and firing rate regulate whole-muscle force. Length, shortening velocity, fatigue, temperature, metabolic state, and prior activity modify output.

Fibre types and metabolism

Slow oxidative fibres have abundant mitochondria, myoglobin, and capillaries, produce lower force more economically, and resist fatigue. Fast oxidative-glycolytic fibres combine power with moderate endurance. Fast glycolytic fibres generate high power but fatigue rapidly. Human muscles contain mixtures, and training changes metabolic and contractile properties within biological limits.

Immediate adenosine triphosphate stores sustain only brief activity. Phosphocreatine rapidly buffers adenosine triphosphate. Anaerobic glycolysis provides fast substrate-level phosphorylation and lactate, while oxidative phosphorylation supplies most energy during sustained activity. Lactate is a transferable fuel and gluconeogenic substrate, not merely waste. Fat oxidation contributes substantially at lower intensities and longer durations.

Fatigue can arise from metabolite accumulation, substrate limitation, excitation-contraction changes, impaired transmission, reduced central drive, pain, heat, or cardiorespiratory limitation. Delayed soreness after unfamiliar eccentric exercise reflects microinjury and inflammation rather than retained lactate.

Mechanical actions and adaptation

Concentric contraction shortens active muscle; eccentric contraction lengthens it under load and can generate high force economically; isometric contraction develops tension without meaningful length change. Most movement combines phases. Moment arm and joint angle change mechanical advantage, so the same muscle force produces different joint torque through range.

Resistance training increases neural activation and later fibre hypertrophy through increased protein synthesis and satellite-cell contribution. Endurance training increases mitochondrial density, oxidative enzymes, capillary support, and fuel handling. Adaptation is specific to load, velocity, range, and metabolic demand. Recovery requires sleep, energy, protein, and time; excessive load without recovery causes declining performance and injury risk.

Disuse rapidly reduces neural drive, fibre size, strength, oxidative capacity, tendon tolerance, and bone loading. Bed rest also impairs cardiovascular and metabolic function. Rehabilitation therefore uses progressive loading early when safe. Ageing reduces motor units, type-two fibre area, anabolic responsiveness, and reserve, but resistance and balance training remain effective.

Bone composition and architecture

Bone contains collagen-rich organic matrix mineralised with hydroxyapatite. Collagen provides tensile toughness; mineral provides compressive stiffness. Cortical bone forms dense shafts and shells; trabecular bone forms a porous load-aligned network with high surface area. Osteons organise cortical vessels and matrix. Periosteum supports growth, repair, innervation, and tendon attachment.

Osteoblast-lineage cells produce osteoid and coordinate mineralisation. Osteocytes embedded in matrix sense strain and communicate through canaliculi. Osteoclasts derive from monocyte-lineage precursors and resorb mineral and matrix. Remodelling couples resorption to formation, repairs microdamage, adjusts architecture, and participates in mineral homeostasis.

Parathyroid hormone, vitamin D, calcium, phosphate, sex steroids, glucocorticoids, growth factors, inflammatory signals, and mechanical load

regulate turnover. Intermittent and sustained signals can have different effects. Bone mass alone does not fully determine strength; geometry, microarchitecture, material quality, damage, and falls matter.

Growth and fracture healing

Long bones grow through endochondral ossification at growth plates, where cartilage proliferates, hypertrophies, calcifies, and is replaced by bone. Flat bones form mainly through intramembranous ossification. Sex steroids accelerate growth and ultimately close plates. Childhood nutrition, endocrine disease, inflammation, and mechanical use influence peak mass and shape.

Fracture healing begins with haematoma and inflammation. Granulation tissue and a soft callus stabilise the site; woven bone forms a hard callus; remodelling restores lamellar architecture along stress lines. Stability and blood supply are essential. Excess motion, infection, smoking, malnutrition, vascular disease, drugs, and severe tissue damage impair union. Primary healing can occur across rigidly compressed, anatomically reduced surfaces with little callus.

Cartilage and synovial joints

Articular cartilage contains chondrocytes within a matrix rich in type-two collagen, proteoglycans, and water. Collagen resists shear and maintains shape; negatively charged proteoglycans attract water and resist compression. Loading expresses fluid and unloading permits recovery, distributing force with low friction. Cartilage is avascular and receives nutrients mainly from synovial fluid, which limits repair.

Synovial joints contain cartilage-covered surfaces, a fibrous capsule, synovial lining, and lubricating fluid. Synoviocytes produce hyaluronan and lubricating proteins and clear debris. Ligaments guide and restrain movement; menisci and labra improve congruence, distribute load, and contribute stability. Bursae reduce friction where tissues move over one another.

Joint stability combines shape, capsule, ligaments, muscle control, proprioception, and pressure. Excess laxity permits injury, while excessive stiffness concentrates load and limits function. Movement circulates synovial

fluid and nourishes cartilage; prolonged immobilisation weakens multiple joint tissues.

Tendon, ligament, fascia, and matrix

Tendons transmit muscle force to bone and store elastic energy. Ligaments connect bone to bone and constrain motion. Both contain aligned type-one collagen, cells, ground substance, vessels, and nerves, but architecture varies by function. Their stress-strain curve includes a low-stiffness toe region as collagen crimp straightens, followed by a stiffer linear region and then microscopic and macroscopic failure.

Matrix adapts more slowly than muscle. Progressive loading increases tendon stiffness and capacity, while abrupt spikes, compression, age, systemic disease, and some drugs increase injury risk. Tendinopathy reflects disordered matrix and load response rather than simple inflammation alone. Rest without graded reloading often fails to restore capacity.

Fascia links compartments and permits force transmission and tissue glide. Extracellular matrix contains collagens, elastin, proteoglycans, adhesive proteins, water, and bound growth factors. Fibroblasts respond to strain and injury by altering synthesis and organisation. Excess repair produces fibrosis; inadequate synthesis or structurally abnormal proteins cause fragility.

Connective-tissue diversity

Collagen types have specialised distributions: type one predominates in bone, tendon, and skin; type two in cartilage; type three in reticular networks and early repair; type four in basement membranes. Elastin permits recoil in arteries, lungs, and selected ligaments. Proteoglycans regulate hydration and compression.

Inherited defects can affect collagen synthesis, fibril structure, elastin-associated microfibrils, basement membrane, or matrix enzymes, producing combinations of joint hypermobility, skin fragility, vascular risk, ocular disease, bone weakness, and organ dysfunction. Phenotype and family history guide genetic evaluation; flexibility alone does not establish a syndromic diagnosis.

Integrated movement and clinical reserve

Efficient movement requires adequate motor command, sensory feedback, muscle power, tendon transfer, joint range, stable connective tissue, pain control, perfusion, and energy supply. Failure at different levels can feel like weakness. Examination must separate loss of force from pain inhibition, stiffness, instability, fatigue, and poor coordination.

Physical capacity reflects strength, power, endurance, balance, mobility, and confidence. Power declines faster than maximal strength with ageing and is critical for preventing falls and rising from a chair. Training should match goals while accounting for disease, tissue healing, medications, nutrition, and recovery. Load is both a potential injury and the principal signal for adaptation.

TTS module 2: Load, tissue adaptation, movement economy, and rehabilitation physiology

Movement is an integrated transfer of force from neural command through muscle, tendon, joint, and bone into the environment. A loss at any link can be experienced as weakness. Pain inhibits recruitment, joint stiffness shortens the usable range, tendon dysfunction wastes force transfer, sensory loss destabilises posture, and cardiopulmonary limitation reduces sustainable work. Musculoskeletal assessment therefore asks whether the limiting variable is activation, force, power, endurance, range, stability, tissue tolerance, coordination, or confidence.

The motor unit is one alpha motor neuron and all muscle fibres it supplies. Small motor units permit fine control; large units generate substantial force. During ordinary voluntary contraction, units are recruited broadly from low-threshold fatigue-resistant units toward larger fast units as demand rises. Increasing discharge frequency then summates force. This size principle permits economical activity while preserving high-force reserve. Motor-unit synchronisation and rate coding adapt with training, explaining why strength can improve before muscle hypertrophy becomes substantial. After denervation, surviving axons may sprout into abandoned fibres, producing larger units but reducing fine control and reserve.

At the sarcomere, force depends on calcium availability, cross-bridge number, fibre length, and shortening velocity. Isometric force is greatest near an

optimal actin–myosin overlap. At very short length, filament interference reduces force; at excessive length, fewer cross-bridges can form. During concentric shortening, force falls as velocity rises because less time is available for attachment. During eccentric lengthening, active muscle can resist greater force at lower metabolic cost, with contributions from cross-bridges and elastic proteins. These properties explain why lowering a load may be possible when lifting it is not, and why unfamiliar eccentric exercise produces marked delayed soreness.

Whole-limb output differs from fibre force because leverage changes through joint range. Joint torque equals muscle force multiplied by its moment arm. A changing moment arm, tendon angle, antagonistic co-contraction, and passive tissue tension alter observable strength. Co-contraction increases stability but raises energy cost and joint compression. Biarticular muscles transfer energy between joints and may shorten less than joint movement suggests. Clinical testing at one angle samples only one mechanical state; rehabilitation should restore force through the functional range and at the speed demanded by the task.

Muscle power is force multiplied by shortening velocity. Daily independence often requires power rather than maximal static strength: rising before momentum is lost, stepping rapidly after a perturbation, or climbing a stair within balance limits. Ageing preferentially reduces fast motor units and type-two fibre area, so power can decline earlier than manual strength grades reveal. Timed chair rise, stair climbing, gait speed, and loaded movement provide functional information. Training power safely uses moderate resistance moved with deliberate rapid intent after adequate strength, technique, and tissue tolerance are established.

Energy systems overlap continuously. Phosphocreatine buffers adenosine triphosphate during brief high-power work, glycolysis responds rapidly as demand rises, and mitochondrial oxidation dominates sustained production. Oxygen consumption may remain elevated after exercise while phosphocreatine, temperature, ventilation, hormones, and fuel stores recover. Lactate carries carbon and reducing equivalents between tissues; its concentration reflects production, oxidation, and clearance rather than oxygen absence alone. Endurance adaptation increases mitochondrial content,

capillary exchange, fat oxidation, and the power sustainable below disruptive metabolite accumulation.

Fatigue is task-specific. Central fatigue reduces motor drive; peripheral fatigue arises at the neuromuscular junction, membrane, calcium handling, cross-bridge machinery, or metabolism. Heat increases cardiovascular and central strain, while dehydration reduces circulating reserve. Glycogen depletion limits prolonged or repeated high-intensity work. In disease, anaemia, inflammation, endocrine disturbance, medication, sleep loss, depression, and malnutrition can reduce performance without primary muscle pathology. Recovery intervals should be matched to the limiting system rather than prescribed as a universal number.

Hypertrophy occurs when repeated mechanical tension and associated signals produce cumulative protein synthesis exceeding breakdown. Satellite cells can donate nuclei and support remodelling, especially with substantial growth or injury. Adequate protein supplies amino acids, but total energy, training stimulus, distribution across meals, age, illness, and anabolic resistance influence response. Unloading reverses signalling rapidly. During bed rest, antigravity muscles lose size and force, insulin sensitivity declines, bone resorption rises, and orthostatic tolerance deteriorates. Even brief frequent loading and mobilisation can preserve function when intensive exercise is impossible.

Bone is a living composite whose toughness depends on collagen and mineral together. Increasing mineral raises stiffness but excessive mineralisation can reduce toughness; defective collagen weakens the framework even when density appears adequate. Cortical geometry strongly affects bending resistance because material distributed farther from the central axis contributes disproportionately. Trabecular orientation reflects habitual loading, and perforation of connecting plates can weaken structure beyond the loss of mineral quantity alone. Bone-density measurement predicts population fracture risk but does not directly measure every component of an individual bone's strength.

Osteocytes act as a mechanosensory network. Fluid movement through canaliculi during strain alters signals including sclerostin, which normally restrains bone formation. Appropriate dynamic loading suppresses inhibitory signals and favours formation, whereas unloading promotes loss. The

response is site-specific, saturates after repeated cycles, and is greatest when strain is novel enough to challenge the existing architecture. Extremely low loading fails to stimulate; excessive loading accumulates microdamage faster than repair. Rest between bouts can restore mechanosensitivity and permit repair.

Remodelling replaces microscopic regions of old or damaged bone through coordinated osteoclast resorption and osteoblast formation. Because resorption precedes refilling, an increase in remodelling temporarily creates spaces and can reduce strength if formation does not catch up. Oestrogen deficiency increases turnover and uncoupling. Glucocorticoids impair osteoblast survival, alter calcium handling, and weaken muscle, combining skeletal fragility with fall risk. Parathyroid hormone given intermittently can stimulate formation, whereas persistent excess favours resorption. Vitamin D permits mineral homeostasis and muscle function, but supplementation cannot substitute for treating severe undernutrition, endocrine disease, immobility, or fall hazards.

Fracture repair is both biological and mechanical. Haematoma provides inflammatory signals and a temporary matrix. Cells from periosteum, marrow, vessels, and surrounding tissues form cartilage and woven bone according to oxygen supply and stability. Relative stability permits callus that bridges the fracture; absolute stability across precisely reduced surfaces permits direct remodelling with little visible callus. Too much motion disrupts vascular ingrowth and mineralisation, while complete removal of useful strain can also limit adaptation later. Rehabilitation must respect fixation, tissue injury, pain, and radiographic progress while restoring load before systemic deconditioning becomes dominant.

Articular cartilage behaves as a fibre-reinforced hydrated material. Proteoglycans attract water and create swelling pressure, while the collagen network restrains expansion and resists shear. On initial loading, interstitial fluid supports much of the compression and limits friction. With sustained loading, fluid slowly leaves and the solid matrix bears more load; unloading permits rehydration. Chondrocytes experience mechanical and chemical signals but have limited capacity to replace major structural loss. Regular moderate movement supports nutrient exchange, whereas abnormal focal

stress, instability, malalignment, inflammation, and prior injury accelerate degeneration.

Synovial joints rely on congruence, labra or menisci where present, capsule, ligaments, muscle, and sensorimotor control. Ligaments provide passive restraint near the limits of movement and sensory information throughout range. Muscle provides adaptable dynamic stability. After ligament injury, mechanical laxity and altered proprioception may persist even after pain settles, so rehabilitation includes strength, perturbation, landing, and task-specific control. Joint swelling itself inhibits surrounding muscle activation, particularly the quadriceps, creating weakness disproportionate to structural muscle loss.

Tendon transmits force and stores energy through aligned collagen fascicles. Cyclic loading initially produces reversible deformation; sufficient recovery allows matrix adaptation, but repeated demand beyond capacity produces accumulating disorganisation. Tendon pain is not a direct gauge of structural damage, and imaging abnormalities occur in asymptomatic people. Management uses symptom-guided progressive loading to restore force tolerance, then speed and energy-storage tasks. Complete rest reduces capacity, while sudden return to previous volume recreates overload. Fluoroquinolone exposure, systemic glucocorticoids, diabetes, inflammatory disease, age, and lipid disorders may increase vulnerability.

Connective tissue heals more slowly than muscle because cell density, vascularity, and turnover differ. Early inflammation removes damaged material and recruits repair cells. Proliferation produces collagen and ground substance, initially organised for speed rather than strength. Remodelling aligns and cross-links fibres according to load over months. Smoking, poor perfusion, infection, hyperglycaemia, malnutrition, and repeated mechanical disruption impair this progression. Scar can restore continuity without reproducing native architecture, so successful recovery may depend on compensatory strength and control as well as tissue repair.

Exercise prescription is therefore a controlled perturbation. Dose includes load, repetitions, range, speed, frequency, density, impact, novelty, and recovery. Progress one or several variables according to response, and distinguish expected short-lived discomfort from worsening night pain, swelling, neurological symptoms, loss of function, or systemic illness. The aim

is not merely to make tissue tolerate the clinic exercise. It is to build enough reserve that ordinary life occupies a smaller fraction of maximum capacity, leaving room for perturbation, fatigue, and ageing without crossing the threshold into failure.

Retrieval prompts

One. How does calcium couple muscle excitation to contraction?

Two. Why do muscle architecture and joint angle alter force output?

Three. How do resistance and endurance training adaptations differ?

Four. What determines bone strength beyond mineral density?

Five. How does articular cartilage bear compressive load?

Six. Why must tendon rehabilitation use progressive loading?

Concise answers

One. Sarcoplasmic calcium binds troponin, displaces tropomyosin, and permits actin-myosin cross-bridge cycling.

Two. Fibre orientation controls physiological cross-section and excursion, while moment arm and overlap vary through range.

Three. Resistance training increases neural drive and fibre size; endurance training increases mitochondrial, capillary, and oxidative capacity.

Four. Geometry, architecture, material quality, turnover, microdamage, loading, and falls contribute.

Five. Collagen restrains a water-rich proteoglycan matrix, distributing fluid pressure and resisting shear.

Six. Tendon adapts slowly and needs graded strain to restore stiffness and load capacity without renewed overload.

Source map

Original synthesis informed by Guyton and Hall, skeletal-muscle contraction, energetics, and movement; Robbins, bone, joint, muscle, and connective-tissue pathology; Katzung and OpenStax Pharmacology, neuromuscular and bone influences; OpenStax Anatomy and Physiology, Biology, Chemistry, and Medical-Surgical Nursing; and Talley and O'Connor, musculoskeletal assessment.

Chapter 33: Injury, Degeneration, Inflammatory Arthritis, and Autoimmune Disease

Orientation

Musculoskeletal symptoms arise from trauma, mechanical overload, degeneration, infection, crystal deposition, immune inflammation, vascular disease, malignancy, or referred pain. The first distinction is often urgent versus non-urgent: threatened limb, open fracture, compartment syndrome, septic joint, spinal compression, or systemic vasculitis cannot wait. The second is inflammatory versus predominantly mechanical disease. Pattern, tempo, distribution, extra-articular features, and targeted tests are more informative than indiscriminate antibody panels.

Acute soft-tissue injury

Injury disrupts cells and matrix, causing bleeding, inflammatory signalling, pain, swelling, and temporary loss of function. Inflammation clears debris and recruits repair cells; proliferation creates new matrix; remodelling aligns tissue with load. Too little protection permits repeated disruption, but prolonged immobilisation causes stiffness, atrophy, reduced matrix capacity, thrombosis, and delayed return.

Initial assessment establishes mechanism, deformity, skin integrity, neurovascular status, active and passive movement, focal tenderness, and ability to bear load. Reassess after splinting or reduction. Cold, compression, elevation, and analgesia can reduce symptoms, but definitive care depends on tissue and severity. Early protected movement and progressive loading are appropriate for many stable injuries.

Sprains injure ligaments; strains injure muscle-tendon units. Tendon rupture may produce a palpable gap, altered contour, weakness, and loss of a normal mechanical test despite surprisingly modest pain. Dislocation can damage vessels, nerves, cartilage, and skin and usually requires urgent reduction with appropriate imaging and analgesia.

Fractures and limb emergencies

Fractures are described by bone, location, displacement, angulation, rotation, shortening, comminution, articular involvement, and whether open.

Mechanism and bone quality matter: low-energy fragility fractures imply reduced strength, while pathological fractures occur through focal disease. In children, growth-plate injury can disturb future growth.

Open fracture communicates with the environment and requires urgent antibiotics, tetanus assessment, sterile coverage, surgical debridement, and stabilisation. Do not repeatedly probe the wound. Neurovascular compromise requires immediate reduction or surgical action. Some apparently normal early radiographs miss occult fractures; persistent focal signs warrant immobilisation and repeat or advanced imaging.

Compartment syndrome occurs when pressure within a closed fascial space impairs perfusion. Severe pain out of proportion, pain with passive stretch, tense swelling, and neurological change are warnings; pulses may remain present. Remove constriction and obtain emergency fasciotomy assessment. Delay causes muscle necrosis, nerve injury, kidney damage, contracture, and limb loss.

Osteoarthritis

Osteoarthritis is whole-joint failure involving cartilage, subchondral bone, synovium, ligaments, capsule, and muscle. Mechanical stress interacts with ageing, previous injury, anatomy, obesity, genetics, and metabolic factors. Cartilage loss is accompanied by attempted repair, osteophytes, bone remodelling, and intermittent synovitis. Radiographic severity correlates imperfectly with pain.

Pain is typically activity-related with brief stiffness after rest; later disease may hurt at rest. Common sites include knees, hips, spine, first metatarsophalangeal joint, thumb base, and selected finger joints. Diagnosis is usually clinical with targeted radiography when it changes decisions. Marked inflammation, prolonged morning stiffness, systemic features, or unusual distribution suggests another process.

Management combines education, graded strengthening and aerobic activity, weight reduction when relevant, pacing, footwear or aids, topical therapy, and

cautious systemic analgesia. Intra-articular corticosteroid may provide short-term relief in selected joints. Joint replacement can restore function in severe disease after non-operative care fails. Arthroscopy does not reverse diffuse degeneration.

Crystal arthritis

Monosodium urate crystals cause gout when tissue urate supersaturation permits deposition. Flares produce abrupt severe pain, swelling, warmth, and erythema, often in one joint. Serum urate can be normal during a flare. Needle-shaped negatively birefringent crystals support diagnosis, but crystals do not exclude concurrent infection.

Treat flares early with a non-steroidal anti-inflammatory drug, colchicine, or corticosteroid according to comorbidity. Long-term urate lowering is indicated by recurrent or severe disease, tophi, selected kidney disease, or stones. Start low, titrate to a target urate, provide flare prophylaxis during initiation, and address adherence and interacting medicines. Urate lowering can initially trigger flares but should usually continue during one.

Calcium pyrophosphate deposition can mimic gout, osteoarthritis, or chronic inflammatory arthritis. Rhomboid weakly positive crystals support diagnosis. Associations include age, previous joint injury, hyperparathyroidism, haemochromatosis, hypomagnesaemia, and hypothyroidism. Treatment controls inflammation; no established therapy removes the crystal burden.

Septic arthritis and bone infection

A hot swollen joint with severe pain and reduced movement is septic arthritis until adequately assessed. Risk rises with age, prosthetic joints, immune suppression, bacteraemia, skin infection, injection, and recent surgery, but healthy people are not exempt. Fever may be absent. Obtain blood cultures and urgent synovial fluid for cell count, Gram stain, culture, and crystals, then begin antibiotics without harmful delay and arrange drainage.

Osteomyelitis can spread through blood, contiguous tissue, surgery, trauma, or diabetic foot infection. Acute disease causes pain and systemic inflammation; chronic disease may form necrotic bone, sinus tracts, and biofilm. Magnetic resonance imaging is sensitive, but microbiological

diagnosis from deep tissue or bone is often needed. Treatment combines source control, targeted antimicrobials, perfusion and pressure management, and stabilisation.

Rheumatoid arthritis

Rheumatoid arthritis is immune-mediated synovitis that can destroy cartilage, bone, tendon, and ligament. It commonly causes symmetric small-joint pain, swelling, prolonged morning stiffness, and reduced function. Cervical spine, lungs, heart, eyes, nerves, skin, blood, and vessels may be involved. Smoking and genetic susceptibility contribute.

Rheumatoid factor is sensitive but not specific; anti-citrullinated-peptide antibodies are more specific and may predict erosive disease. Normal markers or negative antibodies do not exclude clinical synovitis. Ultrasound or magnetic resonance can detect inflammation, while radiographs document structural damage.

Early disease-modifying treatment improves long-term outcome. Methotrexate is a common anchor drug with folate supplementation and monitoring. Other conventional, biological, and targeted synthetic agents inhibit specific immune pathways. Screen for infection and vaccination needs, monitor toxicity, and minimise prolonged glucocorticoid exposure. Treatment aims for remission or low activity, not analgesia alone.

Spondyloarthritis

Spondyloarthritis includes axial disease, psoriatic arthritis, reactive arthritis, and bowel-disease-associated arthritis. Features include inflammatory back pain, sacroiliitis, enthesitis, dactylitis, asymmetric lower-limb arthritis, psoriasis, uveitis, and intestinal inflammation. Human leukocyte antigen B twenty-seven supports risk in context but is not diagnostic.

Inflammatory back pain often improves with movement, disturbs the second half of the night, and begins younger than typical degeneration. Magnetic resonance can show active sacroiliac inflammation before radiographic change, but incidental abnormalities require clinical correlation. Exercise and physiotherapy are foundational; non-steroidal drugs and pathway-targeted biological therapy treat active disease.

Systemic lupus and connective-tissue disease

Systemic lupus erythematosus results from loss of tolerance, autoantibody formation, immune complexes, complement activation, and tissue-specific injury. Manifestations include inflammatory joints, photosensitive skin disease, oral ulcers, cytopenias, serositis, kidney disease, neurological syndromes, thrombosis, and constitutional symptoms. Disease is heterogeneous and fluctuating.

Antinuclear antibody is sensitive but poorly specific. More specific antibodies, complement, blood counts, urine sediment, protein excretion, kidney function, and phenotype determine likelihood and activity. Kidney biopsy guides treatment when lupus nephritis is suspected. Hydroxychloroquine benefits most suitable patients; glucocorticoids control acute inflammation but cumulative toxicity is substantial. Additional immunosuppression follows organ threat and severity.

Systemic sclerosis combines vasculopathy, immune disturbance, and fibrosis, causing Raynaud phenomenon, skin thickening, digital ischaemia, reflux, lung disease, pulmonary hypertension, kidney crisis, and cardiac involvement. Sjogren disease causes dry eyes and mouth with systemic neurological, pulmonary, renal, and lymphoma risks. Mixed and overlap syndromes cross conventional categories.

Vasculitis and inflammatory muscle disease

Vasculitis injures vessel walls and causes ischaemia, haemorrhage, aneurysm, or organ dysfunction. Classification by vessel size narrows patterns. Warning features include headache with visual symptoms, jaw claudication, pulmonary-kidney syndrome, palpable purpura, mononeuritis multiplex, unexplained organ infarction, and severe systemic inflammation. Biopsy or vascular imaging often establishes diagnosis; autoantibodies support but do not replace phenotype.

Inflammatory myopathies cause subacute proximal weakness, sometimes with dysphagia, lung disease, rash, or malignancy association. Creatine kinase can be high but is not universally so. Electromyography, imaging, antibodies, and biopsy help classify disease. Treatment uses immunotherapy plus rehabilitation and complication screening.

Immunomodulatory safety

Before immunosuppression, define diagnosis and severity, screen relevant latent infections, update non-live vaccines, assess pregnancy plans, organ function, malignancy risk, and drug interactions. Fever in an immunosuppressed patient may reflect infection, disease flare, or both. Do not reflexively escalate immune therapy without excluding dangerous infection.

Monitor blood counts, liver and kidney function, drug-specific toxicities, bone protection, cardiovascular risk, and adherence. Live vaccines may be unsafe during selected therapy. Shared decisions should include time to benefit, route, monitoring, cost, fertility, infection precautions, and plans for surgery or acute illness.

TTS module 2: Pattern-based rheumatology, limb emergencies, and immune-treatment strategy

Musculoskeletal diagnosis is accelerated by defining the structure, inflammatory state, distribution, and tempo before naming a disease. Joint pain may originate from synovium, cartilage, subchondral bone, capsule, tendon, bursa, muscle, nerve, or referred viscera. True synovitis produces swelling, warmth, and restriction of both active and passive movement, although deep joints can conceal swelling. Tendon pain is often reproduced by resisted contraction or stretch, while bursitis localises around a predictable friction surface. Pain on active but not passive movement suggests contractile tissue; pain through passive range suggests joint or capsule, but guarding and severe pain blur these distinctions.

Inflammatory disease tends to cause prolonged morning stiffness, rest or night pain, swelling, warmth, and improvement with gentle movement. Mechanical disease more often worsens with use and settles with rest, with brief stiffness after inactivity. Neither pattern is absolute. Osteoarthritis can produce inflammatory flares, and active inflammatory arthritis becomes painful with loading. Distribution adds discriminatory power: symmetric metacarpophalangeal and proximal interphalangeal synovitis suggests rheumatoid arthritis; distal interphalangeal disease with nail change suggests psoriatic arthritis; abrupt first metatarsophalangeal inflammation suggests gout; axial and enthesal disease suggests spondyloarthritis.

Trauma assessment begins before imaging. Mechanism predicts hidden injury: twisting under load injures ligaments and menisci, a fall on an outstretched hand transmits force through wrist, elbow, and shoulder, and high-energy axial load may injure several spinal or limb levels. Inspect skin, deformity, swelling, and compartment tension. Document motor and sensory function and palpable pulses both before and after reduction or splinting. A warm pink limb may still have important arterial injury through collateral flow, so asymmetry, expanding haematoma, bruit, neurological deficit, and mechanism can mandate vascular imaging or exploration.

Compartment syndrome is a perfusion emergency, not a diagnosis based on absent pulses. Tissue pressure rises within a non-compliant fascial space after fracture, crush, bleeding, reperfusion, burns, or tight dressings. Venous outflow fails first, raising interstitial pressure until capillary flow and nerve conduction decline. Escalating analgesic requirement, pain disproportionate to injury, pain with passive stretch, tense swelling, paraesthesia, and evolving weakness are warning signs. Sedation, regional anaesthesia, altered consciousness, and young age can conceal pain. Remove external constriction and obtain immediate surgical assessment; pressure measurement assists when examination is unreliable but should not delay fasciotomy in a convincing syndrome.

A hot swollen joint is approached as infection until adequate evidence supports another cause. Synovial-fluid appearance and leukocyte count overlap among bacterial, crystal, and immune arthritis. Gram stain lacks sufficient sensitivity to exclude infection, and crystals can coexist with bacteria. Blood cultures may be positive even when joint culture is not. Antibiotics follow prompt aspiration when this causes no dangerous delay, and drainage reduces organism and inflammatory burden. Prosthetic-joint infection often presents more subtly and requires coordinated orthopaedic and microbiological sampling before antibiotics when the patient is stable.

Osteomyelitis differs by route and host. Haematogenous infection commonly seeds metaphyseal bone in children and vertebrae in adults. Contiguous infection follows trauma, surgery, ulcers, or adjacent soft-tissue disease. Ischaemic and neuropathic feet combine repetitive trauma, impaired immunity, poor antibiotic delivery, and reduced healing. Plain radiographs may be initially normal; magnetic resonance detects marrow change but can

be nonspecific after surgery or in neuropathic arthropathy. Deep tissue or bone culture is more reliable than superficial swabs. Cure may require removal of necrotic bone or hardware, restoration of perfusion, off-loading, glycaemic management, and prolonged targeted therapy.

Osteoarthritis reflects maladaptation of the entire joint. Cartilage matrix loses mechanical integrity, subchondral bone remodels, osteophytes enlarge the articular margin, synovium becomes intermittently inflamed, and periarticular muscle weakens. Pain can arise from innervated bone, synovium, capsule, tendon, and sensitised neural pathways; cartilage itself lacks nerves. This explains discordance between radiographs and symptoms. Management targets function and modifiable load: strength, aerobic capacity, weight where relevant, sleep, mood, footwear, occupational demand, and confidence. Analgesics create an opportunity for movement but do not replace rehabilitation, and systemic non-steroidal drugs require gastrointestinal, kidney, pressure, and cardiovascular risk assessment.

Crystal deposition illustrates how chemistry becomes episodic inflammation. Urate concentration depends on production and predominantly kidney and intestinal excretion. Hyperuricaemia is necessary for monosodium urate deposition but does not guarantee gout. Crystals may persist between flares, while innate immune activation produces abrupt neutrophilic inflammation when deposits are disturbed. Effective long-term urate lowering gradually dissolves the burden; serum concentration is therefore treated to target and adherence followed over years. Starting therapy can mobilise deposits and provoke early flares, so prophylaxis and explanation prevent premature abandonment. During an acute flare, existing urate-lowering therapy usually continues.

Rheumatoid arthritis develops when loss of immune tolerance sustains synovial inflammation. Activated immune cells, cytokines, fibroblast-like synoviocytes, and osteoclast pathways create invasive pannus that damages cartilage, bone, tendon, and ligament. The window for preventing irreversible damage is why disease-modifying therapy begins early. Disease activity is judged from swollen and tender joints, patient impact, inflammatory markers, imaging where needed, and function, rather than antibody titre alone. Methotrexate requires attention to pregnancy, liver disease, alcohol, blood counts, lung symptoms, renal clearance, interactions, and folate. Biological

and targeted drugs improve control but increase selected infection and other pathway-specific risks.

Spondyloarthritis centres on entheses, axial joints, synovium, digits, skin, gut, and eye. Inflammatory back pain is a pattern, not a diagnosis: onset at younger age, gradual development, improvement with activity, night waking, and alternating buttock pain increase suspicion. Human leukocyte antigen B twenty-seven changes probability according to ancestry and phenotype but cannot screen indiscriminately. Acute painful red photophobic eye suggests anterior uveitis and needs urgent ophthalmic assessment. In psoriatic disease, skin extent may be small or hidden in scalp, umbilicus, natal cleft, or behind ears; nail pitting and onycholysis add evidence.

Systemic lupus erythematosus is classified by coherent clinical and immunological evidence, not a positive antinuclear antibody in isolation. Antinuclear antibodies occur in healthy people, infection, drugs, and other autoimmune disease. More specific antibodies, low complement, cytopenias, proteinuria, active urine sediment, serositis, inflammatory skin disease, and thrombosis define organ risk. Urine examination is essential because nephritis may be clinically quiet. Hydroxychloroquine reduces flares and long-term damage in suitable patients but requires dosing and retinal surveillance. Glucocorticoids act quickly yet accumulate infection, metabolic, bone, cardiovascular, ocular, and psychiatric toxicity, making steroid-sparing control a major objective.

Systemic sclerosis combines small-vessel dysfunction with immune activation and fibrosis. Raynaud phenomenon may precede skin thickening, but new severe Raynaud, digital ulcers, abnormal nailfold capillaries, reflux, dyspnoea, or puffy fingers raises concern for systemic disease. Interstitial lung disease and pulmonary arterial hypertension require different physiological assessment and treatment. Abrupt hypertension with kidney injury suggests scleroderma renal crisis, where rapid angiotensin-converting-enzyme inhibition is central despite rising creatinine. High-dose glucocorticoid exposure may increase crisis risk.

Vasculitis should be suspected when apparently unrelated organs share an ischaemic or inflammatory mechanism. Palpable purpura, haematuria with red-cell casts, pulmonary haemorrhage, mononeuritis multiplex, jaw claudication, visual symptoms, mesenteric ischaemia, and unexplained

constitutional inflammation are high-value clues. Vessel size predicts manifestations, but infection, emboli, atherosclerosis, drugs, and malignancy mimic vasculitis. Tissue biopsy from an active accessible site or vascular imaging should be obtained when feasible before prolonged immunosuppression. Suspected giant-cell arteritis with visual threat is treated immediately because waiting for biopsy can cost sight.

Inflammatory myopathy produces weakness more reliably than pain. Difficulty rising, climbing, lifting arms, swallowing, or holding the head suggests proximal or axial involvement. Creatine kinase reflects membrane leakage and can be normal in selected subtypes or advanced atrophy. Rash, interstitial lung disease, arthritis, Raynaud phenomenon, mechanic-like hand changes, and autoantibody pattern refine phenotype and malignancy risk. Drug toxicity, endocrine disease, inherited myopathy, motor-neuron disease, and inclusion-body myositis require separation because immunosuppression may not help and can harm.

Before immune treatment, estimate both untreated disease risk and treatment risk. Screen for tuberculosis, hepatitis, vaccination needs, pregnancy, malignancy history, heart failure, demyelination, renal and liver function according to the intended drug. Establish baseline blood counts and infection education. Fever or organ dysfunction during therapy may represent infection, immune flare, drug toxicity, or all three, and stopping or escalating treatment reflexively can be dangerous. The best strategy uses the least cumulative toxicity that achieves control, measures response explicitly, preserves exercise and bone health, and treats the patient's functional goals rather than a laboratory marker alone.

Retrieval prompts

One. Which findings warn of compartment syndrome?

Two. Why do osteoarthritis symptoms correlate imperfectly with radiographs?

Three. What must be tested in an acutely hot swollen joint?

Four. Why is early disease-modifying treatment central in rheumatoid arthritis?

Five. Which features suggest spondyloarthritis rather than mechanical back pain?

Six. Why must infection be considered before escalating immunosuppression?

Concise answers

One. Disproportionate pain, passive-stretch pain, tense swelling, and evolving neurological deficit.

Two. Pain reflects synovium, bone, capsule, muscle, sensitisation, and context as well as cartilage loss.

Three. Synovial cell count, Gram stain, culture, and crystals, alongside blood cultures and clinical assessment.

Four. It suppresses synovitis before irreversible cartilage, bone, tendon, and functional damage accumulates.

Five. Young onset, night disturbance, improvement with movement, sacroiliitis, enthesitis, dactylitis, psoriasis, uveitis, or bowel inflammation.

Six. Infection can mimic inflammatory flare and may worsen catastrophically under additional immune suppression.

Source map

Original synthesis informed by Robbins, bone, joint, muscle, autoimmune, and vascular pathology; Guyton and Hall, tissue and immune physiology; Katzung and OpenStax Pharmacology, analgesic, anti-inflammatory, urate-lowering, and immunomodulatory therapy; OpenStax Medical-Surgical Nursing; OpenStax Microbiology; and Talley and O'Connor, musculoskeletal and rheumatological assessment.

Chapter 34: Skin Function, Wounds, Infection, Tumours, and Clinical Examination

Orientation

Skin is a barrier, immune organ, sensory surface, thermoregulatory system, and record of internal disease. Diagnosis relies on morphology and distribution: describe before naming. Urgent threats include spreading infection, necrosis, mucosal drug reactions, blistering, burns, vascular compromise, and melanoma. Erythema, pallor, cyanosis, and inflammation appear differently across pigmentation.

Structure and barrier function

The epidermis is stratified squamous epithelium dominated by keratinocytes. Basal cells proliferate, migrate outward, produce keratin and lipids, flatten, and form the stratum corneum. Corneocytes embedded in extracellular lipid layers create a water-resistant barrier. Continuous shedding limits microbial persistence. Barrier disruption increases water loss and permits irritants, allergens, and pathogens to enter.

Melanocytes produce melanin and transfer pigment to keratinocytes, where it absorbs ultraviolet radiation and limits DNA damage. Differences in skin colour mainly reflect melanosome activity and distribution rather than melanocyte number. Langerhans cells participate in immune surveillance. Merkel cells contribute to touch.

The dermis contains matrix, vessels, nerves, follicles, and glands. Its papillary layer interfaces with epidermis; reticular tissue supplies strength and elasticity. Subcutaneous fat cushions, insulates, stores energy, permits mobility, and contains larger vessels and nerves. Ageing, ultraviolet exposure, glucocorticoids, malnutrition, oedema, and connective-tissue disease alter these layers.

Thermoregulation, sensation, and secretion

Cutaneous vasodilation transfers core heat toward the environment; vasoconstriction conserves it. Eccrine sweat cools through evaporation and is controlled mainly by sympathetic cholinergic fibres. Heat loss depends on humidity, airflow, clothing, hydration, acclimatisation, and cardiovascular reserve. Impaired sweating or sensation increases heat-injury and wound risk.

Cutaneous receptors detect touch, vibration, stretch, temperature, itch, and pain. Itch uses overlapping but distinct neural pathways from pain and can arise from skin inflammation, infestation, kidney or liver disease, haematological disease, drugs, or neuropathy. Scratching briefly inhibits itch but damages barrier and sustains an itch-scratch cycle.

Sebaceous glands release lipid-rich sebum into follicles. Apocrine glands contribute scent-associated secretion in selected regions. Skin initiates vitamin D synthesis under ultraviolet B exposure, although exposure recommendations must balance deficiency against cancer and photoageing risk.

Describing a rash

Begin with site, number, arrangement, symmetry, and distribution: localised or generalised; flexural or extensor; dermatomal, acral, facial, intertriginous, sun-exposed, or pressure-related. Note scalp, nails, hair, palms, soles, and mucosa. Ask where it began and how it spread.

Primary lesions include macules, patches, papules, plaques, vesicles, bullae, pustules, nodules, wheals, and cysts. Secondary changes include scale, crust, erosion, ulcer, fissure, excoriation, lichenification, atrophy, scar, and pigment change. Describe colour, surface, border, shape, blanching, tenderness, temperature, and mobility.

Diascopy tests whether colour blanches under pressure. Dermoscopy reveals subsurface pigment and vascular structures but requires training. Photograph with consent and a scale when useful. Palpate regional nodes and examine systemic features. A precise description allows another clinician to visualise the eruption and narrows mechanism.

Inflammatory dermatoses

Atopic dermatitis combines barrier dysfunction, inflammation, itch, and microbial interaction. Distribution changes with age and commonly affects flexures later. Management uses emollients, irritant avoidance, topical anti-inflammatory therapy, infection treatment when present, and escalation for severe disease. Food avoidance without evidence can cause nutritional harm.

Contact dermatitis is irritant from direct barrier injury or allergic through delayed hypersensitivity. Distribution provides exposure clues. Patch testing identifies selected allergens; it is not the same as immediate allergy testing. Removal of the cause is central.

Psoriasis produces sharply demarcated scaly plaques, often on extensor surfaces and scalp, with nail change and possible arthritis. It is systemic inflammatory disease associated with cardiometabolic and psychological burden. Topical therapy, phototherapy, conventional systemic drugs, and targeted biologics are chosen by extent, site, comorbidity, and impact.

Urticaria consists of transient itchy wheals caused by mast-cell mediators. Individual lesions usually resolve within a day without scale or bruising. Angioedema affects deeper tissue. Airway, breathing, circulation, and anaphylaxis features require immediate adrenaline treatment. Chronic spontaneous urticaria is rarely explained by broad allergy panels.

Skin infection

Impetigo causes superficial crusted lesions; folliculitis affects follicles; abscesses form purulent cavities requiring drainage when suitable. Cellulitis is spreading infection of dermis and subcutaneous tissue, usually unilateral with warmth, tenderness, erythema, and swelling. Bilateral red legs more often reflect non-infectious inflammation or venous disease.

Pain out of proportion, rapid progression, bullae, skin anaesthesia, crepitus, shock, or organ dysfunction suggests necrotising infection. Immediate surgical exploration and broad antimicrobial treatment are essential; imaging must not delay action. Infection risk and organisms vary with wounds, water, bites, injection, diabetes, immune suppression, and surgery.

Herpes simplex causes grouped painful vesicles or erosions and can disseminate in vulnerable patients. Varicella-zoster reactivation follows a sensory distribution and can cause persistent neuralgia, eye disease, ear-facial syndromes, or dissemination. Early antiviral treatment is important in high-risk sites and hosts.

Dermatophytes infect keratinised skin, hair, and nails, often with an active scaly border. Candida favours moist folds and mucosa. Sampling prevents inappropriate treatment. Scabies causes nocturnal itch and burrows or papules; contacts and environment require coordinated management.

Wounds and chronic ulcers

Wound healing proceeds through haemostasis, inflammation, proliferation, and remodelling. Keratinocytes close the surface; fibroblasts deposit matrix; angiogenesis supplies granulation tissue; contraction reduces area. Scar strength increases for months but never fully matches uninjured skin. Infection, ischaemia, oedema, pressure, diabetes, smoking, glucocorticoids, malnutrition, and foreign material delay healing.

Assess cause, depth, dimensions, edge, base, exudate, surrounding skin, pain, perfusion, sensation, pressure, and infection. Debridement removes non-viable tissue when perfusion and goals permit. Dressing choice follows moisture, tissue, location, comfort, cost, and frequency rather than brand.

Venous ulcers commonly occur near the gaiter region with oedema, pigmentation, and lipodermatosclerosis; compression treats venous hypertension after arterial supply is assessed. Arterial ulcers are painful, distal, and associated with poor perfusion; revascularisation may be essential. Neuropathic ulcers occur at pressure points with sensory loss. Atypical, undermined, inflamed, or non-healing ulcers may require biopsy for vasculitis, inflammation, infection, or cancer.

Pressure injury

Pressure and shear deform tissue between body and support surface, compromising microcirculation. Risk rises with immobility, reduced sensation, moisture, poor perfusion, malnutrition, friction, devices, and critical illness. Injury can begin in deep tissue before surface breakdown appears.

Prevention requires risk assessment, repositioning, pressure redistribution, heel off-loading, device checks, moisture management, nutrition, mobility, and inspection. Do not massage damaged areas. Staging describes visible depth, not a simple linear progression; healing lesions are not reverse-staged.

Burns

Burn severity depends on depth, total body surface area, site, inhalation injury, electrical or chemical mechanism, age, and comorbidity. Superficial burns are painful and blanching; deeper burns may appear pale, mottled, leathery, or insensate. Circumferential burns can compromise circulation or ventilation. Facial burns, soot, voice change, enclosed-space exposure, or respiratory distress suggest inhalation injury.

Stop burning, remove constricting items, cool thermal burns with running water without causing hypothermia, cover cleanly, provide analgesia, and assess tetanus. Avoid ice. Major burns require response-guided fluid, temperature control, wound care, nutrition, surveillance, and specialist transfer. Do not give routine systemic antibiotics without infection.

Drug eruptions and blistering emergencies

Morbilloform drug eruptions are common, but timing and systemic findings determine concern. Drug reaction with eosinophilia and systemic symptoms can cause fever, facial oedema, organ injury, and delayed progression. Stevens-Johnson syndrome and toxic epidermal necrolysis cause painful skin, mucosal erosions, blistering, and epidermal detachment. Stop suspected drugs and obtain urgent specialist supportive care.

Autoimmune blistering diseases arise from antibodies against adhesion structures. Blister level affects tension and mucosal involvement. Diagnosis uses biopsy for routine histology and perilesional immunofluorescence. Extensive blistering causes fluid loss, infection risk, pain, temperature instability, and nutritional burden.

Skin tumours

Basal-cell carcinoma commonly appears as a slowly growing pearly, ulcerated, or scar-like lesion and rarely metastasises but can invade locally. Squamous-cell carcinoma may be scaly, keratotic, tender, ulcerated, or rapidly growing and has greater metastatic potential in high-risk contexts. Chronic ultraviolet exposure, immune suppression, scars, and selected infections increase risk.

Melanoma arises from melanocytes and is dangerous because of metastatic potential. Warning features include asymmetry, border irregularity, colour variation, diameter or growth, and especially evolution or an outlier lesion. Nodular melanoma may be symmetric but rapidly enlarging. Suspicious pigmented lesions generally require complete excision biopsy with appropriate margins for diagnosis and depth measurement, rather than destructive treatment.

Prevention includes shade, clothing, sunscreen, avoidance of tanning devices, and high-risk surveillance. Examine scalp, nails, soles, and mucosa when indicated. Appearance alone cannot exclude malignancy.

TTS module 2: Barrier failure, lesion morphology, wound physiology, and dermatological emergencies

Skin disease becomes manageable when observation is separated from interpretation. Before calling an eruption eczema, infection, allergy, or cancer, describe its primary lesion, border, surface, colour, arrangement, distribution, and evolution. A plaque is a raised broad lesion; scale implies altered keratinisation; a vesicle contains clear fluid; a pustule contains purulent material but is not necessarily infected; a wheal is transient dermal oedema. Crust, erosion, ulceration, excoriation, lichenification, and scar are secondary changes that may obscure the initiating lesion. Photographs from early disease can recover lost morphology.

Distribution frequently identifies mechanism. A sharply exposed pattern suggests contact or light; flexural disease supports atopic dermatitis; extensor plaques with scalp or nail involvement support psoriasis; grouped vesicles in a dermatome support zoster; annular scale advancing at the edge supports dermatophyte infection. Palms, soles, scalp, nails, folds, genital skin, and oral

and ocular mucosa should be examined when relevant. Drug eruptions and viral exanthems often begin on the trunk and become symmetric, but timing, pain, mucosal involvement, systemic features, and laboratory abnormalities determine urgency.

The epidermal barrier is created by differentiating keratinocytes, structural proteins, natural moisturising factors, and organised lipids between corneocytes. The result limits water loss while resisting chemical, mechanical, and microbial invasion. Barrier damage increases transepidermal water loss and allows irritants and antigens deeper access, promoting inflammation that further damages the barrier. This feedback loop explains why bland emollients and avoidance of harsh cleansers are mechanistic treatments rather than cosmetic additions. Occlusion improves hydration and drug penetration but may also increase maceration, folliculitis, and systemic absorption.

Topical treatment depends on molecule, vehicle, site, and surface area. Ointments are occlusive and useful for dry thick lesions; creams are less greasy and often acceptable in moist or flexural sites; lotions and solutions spread through hairy regions. Thin eyelid, facial, genital, and intertriginous skin absorbs more drug and is more vulnerable to corticosteroid atrophy than palms or soles. Potency, quantity, duration, and intermittent maintenance should be specified. Incorrect fear can produce undertreatment, while unrestricted potent steroid use can cause striae, telangiectasia, infection masking, ocular complications, and systemic effects.

Atopic dermatitis arises from interacting barrier susceptibility, immune signalling, microbiome change, itch, and environment. Scratching mechanically disrupts the stratum corneum and releases inflammatory signals, perpetuating itch. Treatment interrupts several links: regular emollient restores water retention, topical anti-inflammatory therapy suppresses active disease, trigger reduction limits injury, and sleep and behavioural strategies reduce scratching. Weeping or crusting does not automatically prove bacterial infection because inflamed eczema can exude; spreading pain, pustules, fever, or systemic illness increase concern. Monomorphic painful punched-out erosions suggest eczema herpeticum and require urgent antiviral treatment.

Psoriasis reflects immune-driven acceleration and altered differentiation of keratinocytes. Plaque extent alone underestimates burden when hands, feet, face, scalp, genitals, or nails impair function or identity. Obesity, smoking,

alcohol, metabolic disease, depression, and inflammatory arthritis commonly coexist. Sudden widespread pustular or erythrodermic psoriasis can disturb temperature, fluid, protein, and cardiovascular homeostasis and may be precipitated by infection, drugs, or corticosteroid withdrawal. Systemic treatment selection considers arthritis, bowel disease, infection risk, pregnancy, malignancy, organ function, and the immune pathway targeted.

Urticaria results from superficial dermal mast-cell mediator release, producing itchy transient wheals that migrate and leave normal skin. Individual lesions persisting beyond about a day, bruising, pain, fever, or systemic inflammation suggest urticarial vasculitis or another mimic. Angioedema affects deeper dermis and submucosa. Histamine-mediated angioedema often accompanies wheals and responds to anaphylaxis treatment. Bradykinin-mediated forms, including selected drug-related and hereditary disease, usually lack urticaria and respond poorly to antihistamines or corticosteroids. Any tongue, laryngeal, breathing, pressure, or multisystem involvement requires immediate airway and anaphylaxis assessment.

Cellulitis is a clinical syndrome of spreading dermal and subcutaneous infection, commonly unilateral, tender, warm, and poorly demarcated. Venous stasis dermatitis, lymphoedema, contact dermatitis, gout, and deep-vein thrombosis can imitate it. Bilateral symmetric lower-leg erythema without fever should prompt reconsideration before antibiotics. Purulence suggests a drainable collection and changes likely organisms and management. Marking the margin and recording systemic state help assess response, although redness can transiently expand after treatment because inflammation does not cease immediately when bacteria are controlled.

Necrotising soft-tissue infection progresses along fascia and causes thrombosis, ischaemia, and systemic toxicity. Early skin findings may be modest because destruction is deep. Pain beyond visible change, rapid progression, wooden induration, oedema extending past erythema, bullae, dusky colour, anaesthesia, crepitus, hypotension, confusion, or severe metabolic disturbance demand surgical review. Laboratory scores and imaging cannot safely rule out disease when clinical suspicion is high. Exploration and debridement are diagnostic and therapeutic; antibiotics alone cannot penetrate and control devitalised tissue adequately.

Wound healing proceeds through overlapping phases rather than a rigid sequence. Platelets establish haemostasis and release signals. Neutrophils and macrophages clear contaminants and coordinate repair. Fibroblasts deposit provisional matrix, endothelial cells form new vessels, keratinocytes migrate across a viable moist surface, and myofibroblasts contract the wound. Collagen is reorganised and cross-linked for months, but scar never regains the exact architecture or strength of intact skin. Persistent inflammation from infection, foreign body, pressure, oedema, or ischaemia arrests progression.

Chronic-wound evaluation treats cause before dressing brand. Palpate pulses, assess temperature and capillary refill, measure ankle pressure or toe perfusion where appropriate, test protective sensation, identify oedema, inspect footwear and pressure, and record depth and undermining. Compression is central for venous hypertension but can worsen severe arterial insufficiency. Neuropathic plantar ulcers require off-loading even when painless. Dry stable heel eschar in an ischaemic limb may protect tissue and should not be automatically debrided. Increasing pain, friable edge, unusual location, excessive granulation, or failure despite appropriate care raises the need for biopsy.

Pressure injury arises from prolonged deformation, shear, and impaired reperfusion, not simply surface pressure. Deep muscle may be injured before epidermis because tissue strains differ over bone. Repositioning schedules must reflect mobility, support surface, perfusion, skin tolerance, comfort, and goals rather than a universal interval. Devices such as oxygen tubing, masks, splints, and catheters create focal risk and require direct inspection. Nutrition supports repair when deficiency exists, but supplements cannot overcome ongoing pressure or critical ischaemia.

Burn depth evolves over time because marginal tissue may recover or progress with oedema and hypoperfusion. Total body surface estimation generally excludes simple erythema and guides major-burn fluid planning. Electrical injury can cause deep muscle damage, arrhythmia, compartment syndrome, and kidney injury despite a small skin wound. Chemical burns require removal of contaminated clothing and prolonged appropriate irrigation, with special handling for selected agents. Inhalation injury may cause early airway oedema or later toxic and lower-airway injury. Escalating hoarseness, stridor, soot, facial burns, enclosed-space exposure, or reduced

consciousness lowers the threshold for securing the airway before swelling makes intubation difficult.

Severe drug reactions are recognised by systemic context. A simple morbilliform eruption is usually symmetric and mildly itchy without mucosal injury or organ dysfunction. Drug reaction with eosinophilia and systemic symptoms often develops after a latency and may include facial oedema, lymphadenopathy, hepatitis, kidney or lung injury, eosinophilia, and prolonged relapse. Stevens–Johnson syndrome and toxic epidermal necrolysis cause skin pain, dusky lesions, epidermal detachment, and prominent mucosal erosions. Immediate withdrawal of likely drugs, burn-level supportive care, ophthalmic and mucosal management, fluid and temperature control, and specialist treatment are more important than assigning a culprit from appearance alone.

Pigmented-lesion assessment prioritises evolution and difference from the patient's background pattern. Melanoma may be asymmetric and multicoloured, but nodular lesions can be smooth, uniform, and rapidly growing. Amelanotic melanoma may be pink or red. Dermoscopy improves discrimination in trained hands but does not replace biopsy of a suspicious lesion. Excision biopsy preserves architecture and permits measurement of Breslow thickness, which guides staging and margins. When anatomical constraints prevent routine excision, biopsy planning should involve clinicians who can avoid compromising definitive management.

Skin examination must work across pigmentation. Erythema may appear violaceous, grey, or primarily as warmth and swelling rather than bright red. Cyanosis, pallor, and pressure injury can likewise be missed if assessment relies on colour alone. Palpation, blanching, tenderness, temperature, contour, mucosa, palms, soles, and comparison with unaffected skin increase reliability. Dermatology is visual, but its emergencies are physiological: barrier loss, infection, airway oedema, ischaemia, thermal failure, and malignancy require the same disciplined attention to time, mechanism, and systemic reserve as disease in any internal organ.

Retrieval prompts

One. What structures create the epidermal water barrier?

- Two. Which descriptors belong in a rash examination?
- Three. What suggests necrotising rather than uncomplicated skin infection?
- Four. How do venous, arterial, and neuropathic ulcers differ?
- Five. Which burn features require specialist assessment?
- Six. What is the safest diagnostic approach to suspected melanoma?

Concise answers

- One. Cornified keratinocytes embedded in organised extracellular lipid layers.
- Two. Site, distribution, primary lesion, colour, border, surface, arrangement, secondary change, mucosa, hair, and nails.
- Three. Disproportionate pain, rapid spread, bullae, anaesthesia, crepitus, shock, or organ dysfunction.
- Four. Venous ulcers accompany oedema and gaiter changes; arterial ulcers accompany poor perfusion; neuropathic ulcers occur at pressure sites with sensory loss.
- Five. Major area or depth, critical sites, circumferential injury, inhalation, electrical or chemical cause, extremes of age, or major comorbidity.
- Six. Complete excision biopsy where feasible, preserving architecture and allowing measurement of invasion depth.

Source map

Original synthesis informed by Robbins, skin inflammation, infection, blistering, and tumours; Guyton and Hall, barrier, sensation, thermoregulation, and repair physiology; Katzung and OpenStax Pharmacology, dermatological and antimicrobial therapy; OpenStax Anatomy and Physiology, Microbiology, and Medical-Surgical Nursing; and Talley and O'Connor, skin morphology and examination.

Chapter 35: Bacteria, Viruses, Fungi, Parasites, and Microbial Genetics

Orientation

Microorganisms differ fundamentally in cellular organisation, replication, metabolism, and dependence on hosts. These differences determine diagnosis, transmission, immune response, and therapy. Bacteria are prokaryotic cells; fungi and parasites are eukaryotic; viruses are genetic programmes packaged for transfer between cells. Normal microbiota support colonisation resistance and metabolism, while pathogens exploit particular niches. Detection of microbial material does not always prove that an organism is causing disease.

Bacterial cell structure

Bacteria lack a membrane-bound nucleus. Their usually circular chromosome occupies a nucleoid, and plasmids may carry accessory genes. Seventy-S ribosomes translate protein. The cytoplasmic membrane controls transport, energy generation, and signalling. Many bacteria possess a peptidoglycan wall that prevents osmotic rupture and determines staining and antibiotic susceptibility.

Gram-positive bacteria have thick exposed peptidoglycan containing teichoic acids. Gram-negative bacteria have a thin wall between inner and outer membranes; the outer membrane contains lipopolysaccharide and porins. Lipid A can provoke powerful inflammation. The periplasm contains enzymes and transport proteins. Mycobacteria have lipid-rich, acid-fast envelopes; mycoplasmas lack a cell wall and are intrinsically resistant to wall-active drugs.

Capsules impede phagocytosis and support adherence. Pili and fimbriae attach to host surfaces; specialised pili transfer DNA. Flagella enable motility and chemotaxis. Some Gram-positive genera form metabolically dormant endospores resistant to heat, drying, and chemicals. Biofilms embed communities in matrix, altering growth, immune exposure, and antimicrobial penetration.

Bacterial growth and metabolism

Binary fission produces clonal descendants, but mutation and gene exchange create diversity. Growth proceeds through lag, exponential, stationary, and death phases under closed conditions. Nutrient supply, oxygen, temperature, pH, osmolarity, and competing organisms shape growth in vivo. Slow, dormant, intracellular, and biofilm-associated populations respond differently from rapidly dividing laboratory cultures.

Obligate aerobes require oxygen; obligate anaerobes are harmed by it; facultative anaerobes switch metabolism; microaerophiles prefer lower oxygen. Respiration uses an electron-transport chain and terminal acceptor, while fermentation regenerates cofactors without an external acceptor. Metabolic products aid identification but vary with conditions.

Bacterial taxonomy and clinical patterns

Morphology describes cocci, rods, curved forms, spirals, and branching filaments, arranged in pairs, chains, clusters, or other patterns. Stain, oxygen tolerance, spores, motility, enzymes, antigens, mass spectrometry, and nucleic-acid sequence refine identification. Taxonomy changes as genomic relationships become clearer; clinical reasoning should not rely on names alone.

Extracellular pyogenic bacteria often cause neutrophilic inflammation, pus, and abscess. Intracellular bacteria require entry into host cells and stronger cell-mediated immunity. Toxin-mediated disease may occur with limited invasion. Fastidious organisms need specialised nutrients; obligate intracellular organisms cannot reproduce on routine media. Spirochaetes have distinctive motility and can disseminate widely.

Viruses

Viruses contain DNA or RNA enclosed by protein capsid, sometimes surrounded by a host-derived lipid envelope. They lack independent ribosomes and energy metabolism and must use host machinery. Genome may be single- or double-stranded, linear, circular, segmented, positive-sense, negative-sense, or reverse transcribed. These choices dictate replication strategy and mutation opportunities.

Attachment to specific receptors influences tissue and species tropism. Entry occurs through fusion, endocytosis, or genome injection. The virus uncoats, expresses genes, copies genome, assembles particles, and exits by lysis or budding. Enveloped viruses are generally more vulnerable to drying, detergents, and environmental disruption; non-enveloped viruses often persist longer on surfaces and through the gastrointestinal tract.

Cytopathic effects include host shutoff, membrane damage, syncytia, inclusion bodies, apoptosis, and immune-mediated injury. Some infections are acute and cleared; others persist through chronic replication, latency, genomic integration, or infection of immune-privileged or long-lived cells. Reactivation follows altered immunity or stress. Oncogenic viruses promote cancer through chronic inflammation, interference with cell-cycle control, or genome integration.

RNA polymerases often lack proofreading, allowing rapid variation, while segmented viruses can reassort genomes during coinfection. DNA viruses generally mutate more slowly, although exceptions exist. Viral evolution reflects mutation, recombination, selection, population bottlenecks, and host immunity rather than a predetermined march toward greater or lesser virulence.

Fungi

Fungi are eukaryotes with nuclei, organelles, chitin-containing walls, and membranes containing ergosterol. Yeasts are single cells reproducing by budding or fission. Moulds form multicellular hyphae; septate or non-septate morphology and branching patterns assist identification. Dimorphic fungi switch form according to temperature or environment.

Superficial fungi inhabit keratinised tissue. Subcutaneous fungi enter through trauma. Systemic endemic fungi are often inhaled and may disseminate. Opportunistic fungi exploit neutropenia, T-cell impairment, transplantation, diabetes, broad antibiotics, devices, or barrier failure. *Candida* can colonise mucosa yet invade blood and organs in vulnerable hosts; *Aspergillus* can cause allergy, chronic cavities, or angioinvasive disease depending on immunity.

Fungal diagnosis uses microscopy, culture, histology, antigen, antibody, mass spectrometry, and molecular tests. Culture may be slow and contamination difficult to distinguish from disease. Antifungal selectivity is challenging because fungal and human cells are both eukaryotic; therapeutic targets include ergosterol, its synthesis, cell wall, nucleic-acid processes, and mitosis.

Protozoa and helminths

Protozoa are single-celled eukaryotes with varied life cycles. Some multiply in intestine, blood, tissue, or vectors; cyst forms may survive outside hosts. Disease can result from invasion, cell destruction, inflammation, malabsorption, or immune evasion. Malaria parasites cycle between mosquito and human liver and red cells; species and parasite burden influence severity and relapse.

Helminths are multicellular worms classified broadly as nematodes, cestodes, and trematodes. Adults, larvae, and eggs may occupy different tissues or hosts. Many do not multiply substantially within humans, so burden reflects exposure, but strongyloides can autoinfect and become fatal during immunosuppression. Tissue migration causes eosinophilic inflammation; chronic infection can obstruct organs, consume nutrients, bleed, or promote cancer.

Diagnosis depends on life cycle and timing: stool ova and parasite examination, blood films, antigen, serology, molecular tests, imaging, or tissue sampling. One negative specimen may be insufficient when shedding is intermittent. Eosinophilia supports selected tissue-invasive helminths but is neither sensitive nor specific for all parasites.

Prions and other infectious entities

Prions template abnormal folding of host protein, producing transmissible spongiform neurodegeneration without conventional nucleic acid. They resist standard inactivation and have long incubation. Disease may be sporadic, inherited, or acquired; diagnosis combines clinical syndrome with specialised neurological tests. Viroids infect plants, not humans.

Microbial genomes and mutation

Mutations arise through replication error, chemical damage, radiation, mobile elements, and imperfect repair. They may be silent, harmful, beneficial, or conditionally advantageous. Selection increases variants that reproduce better in the current environment. Antibiotic exposure does not purposefully create resistance but selects resistant organisms already generated or newly acquired.

Horizontal gene transfer accelerates bacterial adaptation. Transformation takes up free DNA. Transduction transfers DNA through bacteriophages. Conjugation transfers plasmids or chromosomal segments through cell contact. Transposons and integrons mobilise and collect genes. Resistance, toxins, adhesion factors, and metabolic pathways can therefore move between strains and species.

Viruses recombine or reassort; fungi undergo sexual, parasexual, and asexual variation; parasites alter surface antigens and life-cycle expression. Host-pathogen interaction is an evolutionary contest shaped by immune pressure, treatment, vectors, reservoirs, and transmission opportunity.

Regulation and virulence expression

Microbes regulate genes according to nutrients, population density, temperature, iron, oxygen, and host signals. Operons coordinate bacterial pathways. Quorum sensing uses secreted signals to estimate local population and can regulate biofilm and virulence. Two-component systems couple environmental sensors to transcriptional responses.

Bacteriophages can lyse bacteria or integrate as prophages. Lysogenic conversion may add toxin or virulence genes. CRISPR-Cas systems provide adaptive defence against foreign nucleic acid and have become laboratory gene-editing tools. Restriction enzymes and other defences also limit incoming DNA.

Microbiota and dysbiosis

Resident microorganisms occupy niche-specific communities, competing with pathogens, modifying nutrients, producing metabolites, training immunity, and maintaining barriers. Composition varies with age, diet, geography, drugs, illness, and sampling. Antibiotics can reduce colonisation resistance.

Dysbiosis describes alteration, not a proven mechanism; associations require control for confounding, and interventions require demonstrated benefit.

Laboratory cultivation and identification

Specimen quality precedes technology. Collect from the infected site before antimicrobials when safe, avoid contamination, use correct transport, and provide clinical context. Superficial swabs may misrepresent deep infection.

Microscopy gives rapid morphology and host-cell information. Culture recovers viable organisms and susceptibility but misses non-cultivable or treated pathogens. Antigen tests target defined organisms. Nucleic-acid amplification detects organisms, resistance genes, or residual non-viable material. Sequencing supports outbreaks and difficult diagnosis but does not establish causality automatically.

TTS module 2: Microbial strategy, ecological niches, and evolution under selection

Microbial classification matters clinically because cellular architecture constrains what an organism can do and what a drug can target. Bacteria synthesize their own proteins and energy but lack a nucleus. Fungi share eukaryotic machinery with humans, narrowing selective toxicity. Parasites may change tissue and metabolic state during complex life cycles. Viruses depend on host cells and expose different targets during entry, genome replication, protein processing, assembly, and release. A useful diagnosis therefore identifies not only a name but the organism's state, anatomical niche, burden, and relationship to the host.

The bacterial envelope controls shape, osmotic survival, permeability, immune recognition, and antibiotic access. Peptidoglycan consists of glycan chains cross-linked by short peptides. Wall-active drugs weaken this expanding structure, making active growth important for killing. Gram-positive

organisms expose a thick wall but lack an outer membrane. Gram-negative organisms place a thin wall in the periplasm behind an asymmetric outer membrane. Porins restrict entry by size and chemistry, while efflux pumps and periplasmic enzymes reduce intracellular drug exposure. Loss or alteration of porins can therefore combine with beta-lactamases to create substantial resistance.

Mycobacterial envelopes contain mycolic acids and other lipids that slow permeability and contribute to acid-fast staining, intracellular persistence, and prolonged treatment. Mycoplasmas cannot be treated with beta-lactams because they possess no peptidoglycan wall. Bacterial membranes also generate electrochemical gradients, making their integrity and metabolic state relevant to aminoglycoside uptake and membrane-active therapy. Capsules and surface polysaccharides inhibit complement deposition or phagocytosis, while opsonising antibody can restore efficient clearance. This is why absent splenic function makes encapsulated bloodstream infection particularly dangerous.

Bacteria are metabolically flexible populations rather than identical cells. Nutrient-rich planktonic growth differs from slow growth inside an abscess, an acidic phagolysosome, necrotic bone, or a device biofilm. Oxygen gradients, iron restriction, pH, host metabolites, and immune pressure alter gene expression and drug susceptibility. Persister cells are transient phenotypic survivors rather than necessarily genetically resistant organisms; when drug pressure ends, their descendants may again be susceptible. This helps explain relapse from protected sites despite a laboratory result obtained from rapidly growing cells.

Biofilms begin with surface attachment, then matrix production and community maturation. The matrix contains polysaccharides, proteins, lipids, and extracellular nucleic acids and limits immune access and diffusion. Cells communicate, exchange genes, and occupy metabolic zones. Prosthetic material, catheters, valves, chronic wounds, and damaged airways provide durable surfaces. Antibiotics may suppress planktonic shedding without sterilising the adherent reservoir, so source control or device removal can be decisive. A positive culture from a device must still be distinguished from contamination and harmless colonisation.

Viruses solve a sequence of compulsory tasks: reach a susceptible cell, bind a receptor, enter, uncoat, make messenger RNA and protein, copy the genome, assemble, and escape. Positive-sense RNA can often function directly as messenger RNA, whereas negative-sense RNA viruses must carry or produce an RNA-dependent RNA polymerase. Retroviruses reverse-transcribe RNA into DNA and integrate it, creating persistent cellular reservoirs. Many DNA viruses use host nuclear machinery, while important exceptions encode specialised enzymes or replicate partly in cytoplasm. Antivirals exploit these required steps, but intracellular dependence makes toxicity and resistance challenging.

Tropism is more than receptor presence. A receptor may permit attachment, yet intracellular factors, temperature, polarity, innate responses, and access route determine productive infection. Respiratory viruses favour airway environments; hepatotropic viruses exploit liver-cell entry and replication factors; neurotropic viruses may travel through nerves or blood. Tissue injury can result from direct cytopathic effect, immune killing of infected cells, vascular injury, or systemic inflammation. Therefore viral load, symptoms, and tissue damage do not always peak together.

Enveloped viruses acquire lipid membrane from host cells. Detergent, drying, heat, and solvents usually disrupt that membrane, making many enveloped viruses less stable outside the body. Non-enveloped capsids often withstand environmental and gastrointestinal stress and may spread efficiently by contaminated surfaces or faecal-oral routes. These are tendencies, not sufficient infection-control rules. Particle shedding, infectious dose, ventilation, humidity, host behaviour, and route all matter.

Viral variation occurs through mutation, recombination, and, in segmented genomes, reassortment. An RNA virus can generate a diverse swarm within one host, but most mutations are neutral or harmful. Selection enriches variants that reproduce and transmit in the current environment. A transmission bottleneck may pass only a small subset to the next host. Immune escape can improve growth in immune populations while compromising another property. Evolution has no intention and does not guarantee reduced virulence; disease severity is an outcome of replication, transmission route, host immunity, and ecological opportunity.

Fungal structure creates both diagnostic clues and therapeutic targets. Chitin and glucans support the wall, while ergosterol stabilises the membrane. Azoles inhibit ergosterol synthesis, polyenes bind membrane sterol, and echinocandins inhibit glucan synthesis, but activity varies by species and site. Yeast detected in a non-sterile specimen may represent colonisation, whereas recovery from blood is usually significant. Hyphae invading tissue or vessels establish disease more convincingly than respiratory culture alone. In neutropenia, angioinvasion can cause infarction and haemorrhage with surprisingly little inflammatory response.

Dimorphic fungi adopt different forms according to environmental conditions, often growing as mould outside the host and a yeast or specialised tissue form after inhalation. Geography and exposure are therefore part of microbiology. Disturbance of cell-mediated immunity may permit dissemination long after initial exposure. Serology, antigen, microscopy, culture, and histology each sample different aspects of infection, and immunosuppression may blunt antibody or inflammatory findings.

Parasite life cycles determine symptoms and test timing. Malaria sporozoites first infect liver cells; later blood-stage replication produces fever and haemolysis. Some species leave dormant liver forms capable of relapse, so blood-stage treatment alone is incomplete. Parasite density and species influence emergency risk, while partial immunity and prior treatment modify presentation. Thick blood films increase detection sensitivity and thin films aid species and burden assessment; repeated sampling may be necessary when initial burden is low.

Helminths can produce disease through adult obstruction, larval migration, blood loss, nutrient competition, cyst formation, or chronic immune stimulation. Eosinophilia is most associated with tissue migration and may disappear when worms remain confined to a lumen. Strongyloides can maintain infection through autoinfection for decades; corticosteroid or other immunosuppression can trigger accelerated dissemination and secondary bacterial sepsis. Exposure history must therefore precede immune therapy in people from endemic settings. Schistosome eggs, rather than adult worms alone, drive granulomatous and fibrotic injury in urinary or portal tissues.

Resistance emerges when heritable variation meets antimicrobial selection and transmission. A spontaneous target mutation may reduce drug binding;

enzymes may destroy or modify drugs; pumps export them; permeability falls; bypass pathways replace inhibited metabolism; and target production increases. Horizontal transfer allows several mechanisms to travel together on plasmids, transposons, or integrons. Antibiotic exposure suppresses susceptible competitors, enriching resistant populations even outside the treated infection. Heavy use across hospitals, communities, agriculture, and environments therefore shapes a shared microbial ecology.

Resistance genotype and phenotype are related but not identical. A detected gene may be poorly expressed, while an untested mechanism may confer resistance. Minimum inhibitory concentration measures growth inhibition under standard laboratory conditions, not eradication in biofilm, abscess, cerebrospinal fluid, urine, or intracellular tissue. Inoculum, protein binding, immune function, dose, and source control alter outcome. Conversely, an organism labelled resistant at standard exposure may be treatable with a validated higher exposure in a suitable site. Interpretation requires the organism, breakpoint system, dosing strategy, and infection compartment.

The microbiome offers colonisation resistance through competition for nutrients and attachment, production of inhibitory metabolites, maintenance of mucus and epithelium, and immune education. Antibiotics, diet, inflammation, hospital exposure, and host physiology can disrupt these communities. Yet “dysbiosis” is descriptive and does not establish causality. Stool composition may not represent mucosa or another body site, and relative abundance can change when one taxon falls without another increasing in absolute number. Microbiome interventions should be judged by clinical outcomes, not by a promise to restore an undefined ideal community.

The central microbiological habit is to distinguish presence from pathogenic action. Nucleic acid can persist after death, antibodies can reflect remote exposure, culture can recover colonisers, and microscopy can miss sparse organisms. Causation becomes stronger when the organism is found at the diseased site, burden fits the syndrome, tissue shows invasion or characteristic injury, host risk and timing align, and alternative causes are weaker. Microbes obey cellular and evolutionary constraints; clinical microbiology succeeds by connecting those constraints to the patient's anatomy and physiology.

Retrieval prompts

- One. How do Gram-positive and Gram-negative envelopes differ?
- Two. Why do viral genome types require different replication strategies?
- Three. Which features distinguish yeasts, moulds, and dimorphic fungi?
- Four. Why must parasite testing follow life-cycle biology?
- Five. How does horizontal gene transfer spread bacterial traits?
- Six. Why can a positive molecular test fail to prove active disease?

Concise answers

- One. Gram-positive organisms expose thick peptidoglycan; Gram-negative organisms have thin peptidoglycan plus an outer lipopolysaccharide membrane.
- Two. Host cells cannot directly copy or translate every genome orientation, so viruses supply or encode specialised enzymes and intermediates.
- Three. Yeasts are unicellular, moulds form hyphae, and dimorphic fungi switch form with environment.
- Four. Organisms, eggs, antigens, and antibodies appear in different sites and times across the cycle.
- Five. Transformation, transduction, conjugation, transposons, and integrons move DNA between organisms.
- Six. It may detect colonisation, contamination, latent infection, or nucleic acid remaining after viability is lost.

Source map

Original synthesis informed principally by OpenStax Microbiology, bacterial, viral, fungal, parasitic, and genetic foundations; OpenStax Biology and Chemistry, cell and molecular mechanisms; Robbins, infectious pathology; Katzung and OpenStax Pharmacology, microbial targets and resistance; OpenStax Medical-Surgical Nursing; and Guyton and Hall, host physiological context.

Chapter 36: Transmission, Virulence, Host Defence, and Diagnostic Microbiology

Orientation

Infection occurs when an organism reaches a susceptible host, enters through an appropriate portal, overcomes local and systemic defences, and damages tissue directly or through the immune response. Exposure does not guarantee colonisation; colonisation does not guarantee disease; detection does not guarantee causation. Clinical interpretation connects organism, site, burden, host, timing, syndrome, and test performance.

Reservoirs and transmission

Reservoirs include humans, animals, soil, water, food, equipment, and built environments. A human reservoir may be symptomatic, presymptomatic, asymptomatic, chronic, or latent. Zoonoses cross from animals directly, through food, or through vectors. Environmental organisms often exploit disrupted barriers or immune impairment rather than routine person-to-person spread.

Direct contact transfers organisms through skin, mucosa, sex, bites, or body fluids. Indirect contact uses contaminated objects or hands. Droplets travel over short distances; aerosols remain suspended and can move with airflow. Faecal-oral transmission follows contaminated hands, food, or water. Blood-borne transmission occurs through needles, transfusion, procedures, or mucosal exposure. Vertical transmission occurs before, during, or after birth.

Vectors may passively carry organisms or support replication. Transmission depends on dose, environmental survival, portals, contact, immunity, and behaviour. Incubation ends with symptoms; latency ends with infectiousness. Communicability may precede symptoms or persist after recovery. Reproduction and generation measures vary with immunity, interventions, biology, and contact opportunity.

Colonisation and invasion

Microbes first adhere using pili, surface proteins, capsules, or receptor-binding molecules. Tissue tropism reflects receptor distribution, temperature, nutrients, oxygen, local immunity, and route. Organisms compete with resident microbiota and resist mechanical clearance by mucus, cilia, flow, shedding, coughing, and peristalsis.

Some pathogens remain superficial and release toxins. Others invade between or through cells, enter phagocytes, spread through lymph or blood, or travel along nerves. Enzymes can degrade matrix, clot or dissolve fibrin, modify antibodies, or damage membranes. Intracellular survival may require blocking phagolysosome fusion, escaping into cytosol, resisting oxidative killing, or moving cell to cell.

A pathogen's infectious dose and invasion strategy shape presentation. A small inoculum of an acid-resistant organism may cause enteric disease, while other organisms require large exposure. Breached skin, devices, obstruction, ischaemia, aspiration, and altered microbiota create new portals and niches.

Toxins and tissue injury

Exotoxins are secreted proteins with specific targets. A-B toxins use one component for binding and entry and another for enzymatic injury. Cytolysins disrupt membranes. Superantigens activate many T cells without ordinary antigen specificity, producing massive cytokine release. Neurotoxins alter neurotransmission; enterotoxins alter intestinal transport.

Endotoxin refers to lipopolysaccharide from Gram-negative outer membranes. Its lipid A component activates innate sensors and can drive fever, vasodilation, endothelial activation, coagulation, capillary leak, and shock. Similar inflammatory structures exist in other microbes. Tissue damage also arises from replication, cell lysis, vascular invasion, immune complexes, granulomas, cytotoxic lymphocytes, and prolonged inflammation.

Virulence is relative and context-dependent. A factor advantageous in one tissue may be irrelevant elsewhere. Opportunists cause severe disease when neutrophils, T cells, antibodies, complement, spleen, barriers, or devices alter

the host environment. Host damage can reflect too little defence or an excessive, misdirected response.

Innate barriers and recognition

Intact skin, mucus, cilia, tears, saliva, stomach acid, bile, urine flow, vaginal acidity, antimicrobial peptides, and resident microbiota provide immediate defence. Epithelial cells are active sensors that release cytokines and recruit leukocytes. Fever and acute-phase proteins coordinate systemic response.

Pattern-recognition receptors detect conserved microbial structures and damage signals. Toll-like and cytosolic receptors activate transcription, inflammasomes, interferons, and cell death pathways. Complement is triggered through classical, lectin, or alternative routes, then opsonises microbes, recruits inflammation, and forms membrane attack complexes. Regulatory proteins protect host surfaces.

Neutrophils migrate rapidly, phagocytose, degranulate, generate reactive oxygen species, and form extracellular traps. Macrophages ingest organisms and debris, produce cytokines, present antigen, and organise repair. Natural killer cells detect stressed or inadequately labelled cells and kill through granules while producing interferon gamma. Dendritic cells link innate sensing to adaptive activation.

Adaptive defence

B cells become antibody-secreting plasma or memory cells. Immunoglobulin M appears early; immunoglobulin G dominates systemic memory and crosses placenta; immunoglobulin A protects mucosa and milk; immunoglobulin E supports mast-cell and antiparasite responses. Antibodies neutralise, opsonise, activate complement, and recruit cellular killing. T-cell help enables affinity maturation and class switching; conjugate vaccines create memory against poorly immunogenic capsules.

CD-four T cells coordinate macrophage, eosinophil, mast-cell, neutrophil, and regulatory responses according to subtype. CD-eight T cells kill infected cells displaying peptide on major histocompatibility complex class one. Memory responds faster than primary immunity. Vaccines may prevent infection,

disease, severe disease, or transmission to different degrees, and protection can wane.

Immune evasion

Capsules resist phagocytosis; antigenic variation alters recognised targets; latency hides gene expression; intracellular residence reduces antibody access. Microbes inhibit complement, degrade antibodies, suppress interferon, block antigen presentation, mimic host molecules, or induce immunoregulatory pathways. Biofilms reduce penetration and expose slow-growing cells.

Immunodeficiency patterns

Defect type predicts infection. Neutropenia predisposes to invasive bacterial and fungal disease with muted pus. T-cell deficiency permits opportunistic viral, fungal, protozoal, and intracellular infections. Antibody deficiency causes recurrent sinopulmonary infection, especially with encapsulated bacteria and selected enteric organisms. Complement terminal-pathway defects predispose particularly to *Neisseria*; absent splenic function increases risk from encapsulated organisms.

Secondary immune impairment from diabetes, malnutrition, kidney or liver disease, cancer, human immunodeficiency virus, immunosuppressive drugs, burns, devices, and critical illness is more common than primary deficiency. Exposure history and prophylaxis modify actual risk. Fever may be absent, so subtle deterioration warrants action.

Choosing a specimen

Sample the involved site, not a convenient colonised surface. Blood cultures are appropriate for suspected bloodstream infection, endocarditis, severe sepsis, and selected focal infections. Cerebrospinal fluid, deep tissue, joint fluid, pleural fluid, urine, respiratory material, stool, genital samples, and aspirated pus answer different questions. Tissue architecture may be as important as organism recovery.

Collect before antimicrobial therapy when feasible without delaying urgent treatment. Use aseptic technique, sufficient volume, correct container, and

rapid transport under suitable temperature and atmosphere. Label site and time precisely and provide syndrome, travel, immune status, exposure, and treatment. Rejection of a poor specimen can be safer than a misleading result.

Microscopy, culture, and identification

Microscopy reveals organisms, inflammatory cells, contamination, and inclusions. Gram stain rapidly shows wall class and morphology but sensitivity depends on burden and specimen. Specialised stains answer acid-fast, fungal, parasitic, and tissue questions.

Culture demonstrates viability, enables identification and susceptibility, and can reveal mixed infection. Selective, differential, and enrichment media serve different organisms. Atmosphere and incubation must match biology. Negative culture may follow treatment, poor sampling, slow growth, intracellular dependence, or wrong conditions.

Identification uses biochemistry, antigen, mass spectrometry, or sequence. Quantitation sometimes separates contamination from infection, but thresholds are contextual. Flora from a sterile site may be significant; on skin it may be expected.

Antigen, nucleic acid, and serology

Antigen tests rapidly detect defined targets. Nucleic-acid amplification can multiplex pathogens, quantify burden, and detect resistance markers, but may detect colonisation or non-viable material. A gene does not guarantee expression, and one absent marker does not prove susceptibility.

Serology detects antibody or antigen. Immunoglobulin M can persist or cross-react; seroconversion, rising titre, avidity, or antigen pattern may be stronger evidence. Antibody may be absent early or with immune deficiency, and vaccination complicates interpretation.

Susceptibility and result interpretation

Susceptibility testing measures inhibition or killing and applies exposure-informed breakpoints. Minimum inhibitory concentrations cannot rank different drugs directly. Penetration, dose, binding, biofilm, burden, source control, immunity, and organ function still determine success.

Interpret through pretest probability: sensitivity affects false negatives, specificity affects false positives, and predictive value changes with prevalence. A low-pathogenicity skin organism in one blood-culture set may be contamination, but devices or immune impairment can make it real.

A result must fit the syndrome, site sterility, viability, treatment timing, and alternatives. Repeated low-value testing creates incidental positives and unnecessary therapy.

TTS module 2: Exposure chains, immune phenotypes, and microbiological inference

Infection is a causal chain: a viable organism leaves a reservoir, survives transmission, reaches an appropriate portal, establishes itself in a susceptible host, and produces damage. Breaking any link can prevent disease. The same organism may cause harmless colonisation in one compartment, local infection in another, and bloodstream invasion in a vulnerable host. Clinical reasoning therefore integrates exposure, anatomical site, barrier integrity, immune phenotype, organism burden, inflammatory response, and test timing rather than treating a positive assay as the diagnosis.

Respiratory spread lies on a continuum of particle sizes rather than two perfectly separated categories. Large wet particles settle more rapidly, while smaller aerosols remain suspended and accumulate in poorly ventilated indoor air. Distance, time, airflow, respiratory activity, masks, filtration, and ultraviolet or environmental inactivation affect dose. Contact transmission can occur when contaminated hands bridge a surface to mucosa, but detection of microbial material on a surface does not prove that route contributed meaningfully. Infection-control measures should target the organism's demonstrated pathways and the procedure's aerosol-generating potential.

Entry requires compatibility between portal and niche. Gastric acidity destroys many swallowed organisms, ciliary clearance removes inhaled particles, urine

flow limits ascending infection, and intact keratinised skin resists invasion. Acid suppression, aspiration, obstruction, catheterisation, wounds, burns, oedema, ischaemia, and antibiotics change these barriers. An infectious dose is therefore conditional: fewer organisms may establish disease when delivered directly into tissue or when local defences are absent. Repeated low-dose exposure can differ from one large inoculum, and pre-existing immunity can reduce progression without preventing entry.

Virulence factors are advantageous traits, not moral properties. Adhesins retain organisms at a surface. Capsules and complement-binding proteins prevent opsonisation. Secretion systems inject effectors into host cells. Siderophores acquire scarce iron. Toxins alter signalling, translation, membranes, or neurotransmission. Intracellular organisms manipulate phagosome maturation or escape into cytoplasm. These mechanisms produce disease only in context. A capsule is especially valuable in blood, while acid tolerance matters during gastric passage; a factor can increase persistence without increasing acute tissue destruction.

Host injury may be microbe-driven, immune-driven, or both. Toxin can cause severe physiology after a small local infection. Cytotoxic lymphocytes eliminate virus-infected cells but damage the organ they protect. Immune complexes deposit in vessels or glomeruli after infection. Granulomas contain persistent organisms yet distort tissue over time. Sepsis reflects a dysregulated host response with endothelial, coagulation, metabolic, and organ effects disproportionate to a local lesion. Clinical improvement can lag behind microbial killing because inflammatory mediators and damaged tissue require time to resolve.

The innate response buys time and defines the adaptive response. Epithelial sensors, complement, tissue macrophages, and dendritic cells detect microbial patterns and damaged cells. Chemokines recruit neutrophils and monocytes, while cytokines alter vessels, temperature, liver protein synthesis, marrow output, and behaviour. Complement fragments opsonise, attract leukocytes, and assemble membrane attack complexes. Excess activation can injure endothelium, while inherited or acquired deficiency reveals pathway-specific susceptibility. Regulatory proteins normally prevent complement from damaging host surfaces.

Adaptive immunity is organised around antigen recognition and context. B cells produce neutralising and opsonising antibody, but high-affinity class-switched responses usually require T-cell help. CD-four subsets activate macrophages, recruit neutrophils or eosinophils, support B cells, and constrain inflammation. CD-eight cells kill infected host cells presenting intracellular peptide. Memory cells respond faster and at lower antigen burden than naive cells. Vaccines exploit this memory, but correlates of protection differ: circulating antibody may prevent invasion, mucosal antibody may reduce entry, and T-cell memory may chiefly prevent severe disease.

Immune defects create recognisable patterns. Neutropenia impairs control of invasive extracellular bacteria and moulds and may produce little pus or radiographic inflammation. T-cell deficiency permits opportunistic viruses, fungi, mycobacteria, and protozoa and weakens granuloma formation. Antibody deficiency produces recurrent infection with encapsulated respiratory bacteria and selected gastrointestinal pathogens. Terminal complement deficiency strongly predisposes to invasive meningococcal disease. Asplenia impairs clearance of opsonised encapsulated organisms and infected erythrocytes. These are probability shifts rather than exclusive organism lists.

Secondary immunodeficiency is shaped by dose and duration. A brief low glucocorticoid course differs from prolonged high-dose therapy combined with another agent. Transplant timing predicts changing risks from surgery, latent reactivation, and environmental opportunists. Diabetes impairs several immune and vascular functions, especially when control is poor. Kidney and liver failure, malnutrition, cancer, burns, critical illness, and advanced age alter barriers and cellular responses. Prophylaxis, vaccination, geography, prior colonisation, pets, occupation, and travel modify the final phenotype.

A specimen is a physical sample of a diagnostic hypothesis. If pneumonia is suspected, a saliva-contaminated expectorate may describe oral flora rather than alveolar infection. If osteomyelitis is suspected, a surface ulcer swab may not represent organisms in bone. Aspirated pus, deep tissue, fluid from a normally sterile space, and multiple adequately filled blood-culture bottles generally provide stronger evidence. The request should state exact site, syndrome, immune status, exposure, and current antimicrobials so the laboratory can select media, incubation, stains, and safety procedures.

Timing and technique alter yield. Antimicrobials reduce culture viability, so cultures are collected first when this does not delay urgent therapy. Blood cultures require adequate volume because bloodstream organisms may be sparse; increasing volume often improves yield more than simply adding repeated low-volume sets. Separate venepunctures help distinguish contamination and assess persistence. Catheter-related infection may be supported when paired catheter and peripheral cultures become positive at meaningfully different times, interpreted with organism and context. Skin antisepsis matters because false positives expose patients to antibiotics and device removal.

Microscopy is rapid but burden-dependent. A Gram stain can reveal morphology, inflammatory cells, epithelial contamination, and mixed flora within minutes, yet a negative result cannot exclude low-density disease. Culture establishes viability and permits susceptibility testing, but fastidious, intracellular, slow-growing, previously treated, or mishandled organisms may not grow. Histology shows tissue invasion, necrosis, granulomas, viral cytopathic change, fungi, or parasites and may establish causality when culture is contaminated or negative. Special stains improve visibility but vary in sensitivity.

Nucleic-acid amplification is analytically powerful. It can detect an organism below culture thresholds, identify difficult pathogens rapidly, quantify selected genomes, and reveal resistance determinants. Its very sensitivity creates interpretive problems: colonisation, latent carriage, transient passage, contamination, and dead organisms can all be detected. Multiplex panels increase the chance of incidental positives, especially when ordered in low-probability syndromes. A positive result is most useful when the target, sample site, and clinical phenotype correspond; a negative result is useful only if sampling occurred at the right site and time and the assay covers the relevant variant.

Serology measures the host response and is therefore delayed and host-dependent. A single immunoglobulin M result can be false-positive, cross-reactive, or persistent. Demonstrated seroconversion, a significant titre rise in paired samples, antigen detection, or antibody avidity may better establish timing. Immunosuppressed patients may never generate expected antibody. Previous vaccination or infection complicates interpretation. For

infections where antibodies remain for life, serology proves exposure but not active symptoms; for others, antigen or direct detection is necessary.

Sensitivity and specificity describe test behaviour under defined conditions, while predictive values depend on pretest probability. In a low-risk patient, even a highly specific test can generate more false than true positives when used broadly. In a high-risk syndrome, a moderately sensitive negative test may leave substantial residual probability. Likelihood ratios provide a formal bridge from prior to posterior odds, but specimen quality and spectrum effects must still be considered. Laboratory validation in obvious disease may overestimate performance in early, partially treated, or immunocompromised patients.

Susceptibility testing places an organism's measured inhibitory concentration against breakpoints integrating achievable drug exposure and outcome evidence. "Susceptible" assumes an appropriate regimen; some categories explicitly require increased exposure. Site matters because urine, blood, cerebrospinal fluid, bone, lung, and intracellular compartments achieve different free concentrations. Inoculum, abscess acidity, foreign material, biofilm, immunity, and source control are not reproduced by a standard plate. Repeat susceptibility may be needed during persistent infection because resistant subpopulations can emerge under therapy.

Diagnostic stewardship asks whether a result will change management and whether the test can answer the question. Repeating cultures after expected clearance is useful in selected bloodstream infections but creates noise in others. Testing formed stool for enteric pathogens or urine without compatible symptoms commonly finds carriage and promotes unnecessary antibiotics. Conversely, failing to collect cultures in severe infection loses the opportunity to narrow treatment. The aim is neither maximal nor minimal testing: it is high-quality sampling at the decision point, interpreted through anatomy, host defence, and probability.

The final microbiology statement should distinguish observation from inference. For example, growth of a typical skin organism in one low-volume bottle is compatible with contamination, but persistent growth in several sets from a patient with an intravascular device may indicate true bacteraemia. This explicit reasoning prevents both dismissal and overtreatment. Infection

diagnosis is strongest when organism, route, host, tissue response, and test limitations form one coherent causal account.

Retrieval prompts

- One. How do incubation and latent periods differ?
- Two. Which mechanisms allow organisms to invade or persist?
- Three. How do innate and adaptive immunity cooperate?
- Four. Which infection patterns suggest specific immune defects?
- Five. What makes a microbiological specimen high quality?
- Six. Why must molecular and susceptibility results be interpreted clinically?

Concise answers

One. Incubation ends with symptoms; latency ends when infectiousness begins.

Two. Adhesion, barrier disruption, toxins, enzymes, intracellular survival, antigenic variation, latency, biofilm, and immune interference.

Three. Innate sensing contains infection and presents antigen; adaptive cells add specificity, antibodies, cytotoxicity, regulation, and memory.

Four. Neutrophil, T-cell, antibody, complement, and splenic defects each create characteristic organism and site risks.

Five. Correct diseased site, sufficient volume, asepsis, proper container and transport, useful timing, and clinical context.

Six. Detection may represent colonisation or dead material, while laboratory susceptibility does not capture site, exposure, source control, or host defence.

Source map

Original synthesis informed principally by OpenStax Microbiology, transmission, pathogenesis, immunity, and laboratory diagnosis; OpenStax Biology, adaptive and innate mechanisms; Robbins, host injury and immunodeficiency; Katzung and OpenStax Pharmacology, susceptibility and resistance; OpenStax Medical-Surgical Nursing; Guyton and Hall, integrative host physiology; and Talley and O'Connor, infectious history and examination.

Chapter 37: Antibacterial, Antiviral, Antifungal, and Antiparasitic Therapy

Orientation

Antimicrobial therapy succeeds when an active drug reaches the site at sufficient exposure, the pathogen is vulnerable, source control occurs, and host defence completes clearance. Choice integrates syndrome, likely organisms, resistance, specimens, allergy, organ function, pregnancy, interactions, toxicity, route, and urgency. Empirical breadth can save life but also selects resistance and causes harm.

Selective toxicity and exposure

Antimicrobials exploit microbial features absent from or sufficiently different from human cells: bacterial walls and ribosomes, viral enzymes, fungal ergosterol and walls, or parasite metabolism. Selectivity is relative. Mitochondria resemble bacterial ancestors, fungi are eukaryotic, and parasites share many human pathways, so toxicity remains substantial.

Pharmacokinetics determines concentration over time; pharmacodynamics relates exposure to killing. Some antibacterials work best when free concentration remains above the minimum inhibitory concentration. Others depend on peak relative to minimum inhibitory concentration or total area under the concentration-time curve. Dose, interval, infusion duration, protein binding, tissue penetration, kidney or liver clearance, dialysis, and critical illness alter target attainment.

Bactericidal drugs kill under defined conditions; bacteriostatic drugs inhibit growth, leaving immune clearance. The distinction varies by organism, exposure, and method and does not automatically rank clinical effectiveness. Difficult sites such as central nervous system, bone, prostate, abscess, vegetation, biofilm, and poorly perfused tissue require specific penetration and source-control reasoning.

Cell-wall-active antibacterials

Beta-lactams bind penicillin-binding proteins and disrupt peptidoglycan cross-linking. Penicillins, cephalosporins, carbapenems, and monobactams differ in spectrum, stability, penetration, and adverse effects. They generally show time-dependent activity. Allergy labels are often inaccurate; careful history and testing can restore safer narrow options. Immediate anaphylaxis differs from delayed benign rash and severe immune-mediated reactions.

Resistance arises through beta-lactamases, altered binding proteins, reduced permeability, and efflux. Beta-lactamase inhibitors protect selected partners but do not inhibit every enzyme. Extended-spectrum enzymes, AmpC, and carbapenemases have different implications. Susceptibility results and infection site should guide definitive therapy.

Glycopeptides bind cell-wall precursors and act mainly against Gram-positive organisms. Vancomycin exposure is monitored in serious infection to balance efficacy and kidney toxicity. Oral vancomycin acts within the gut and does not treat systemic infection. Other wall agents include fosfomycin and bacitracin in selected roles.

Protein-synthesis inhibitors

Aminoglycosides bind the bacterial thirty-S subunit, cause mistranslation, and kill concentration-dependently. They act best against aerobic Gram-negative organisms and may synergise with wall-active drugs. Kidney and vestibular or auditory toxicity limit use; dosing and monitoring account for weight, clearance, and cumulative exposure. Anaerobes cannot provide the oxygen-dependent uptake needed.

Tetracyclines block aminoacyl transfer-RNA binding and cover many intracellular, vector-borne, zoonotic, and atypical organisms. Adverse effects include gastrointestinal irritation, photosensitivity, oesophageal injury, and effects on developing teeth and bone. Macrolides inhibit translocation on the fifty-S subunit and cover respiratory and atypical pathogens but can prolong QT interval and cause interactions.

Clindamycin inhibits the fifty-S subunit and suppresses selected toxin production but strongly predisposes to *Clostridioides difficile* disease. Linezolid blocks initiation and treats resistant Gram-positive infection;

prolonged use risks cytopenias, neuropathy, lactic acidosis, and serotonergic interaction. Chloramphenicol has broad activity but serious marrow toxicity and limited indications.

Nucleic-acid and metabolic inhibitors

Fluoroquinolones inhibit DNA gyrase and topoisomerase. They penetrate many tissues but can cause tendon injury, neuropathy, dysglycaemia, central effects, QT prolongation, vascular concerns, and *Clostridioides difficile* infection. Resistance develops readily through target mutation, efflux, and protective genes, so convenience should not drive use.

Rifamycins inhibit RNA polymerase and are central to mycobacterial therapy. Resistance emerges quickly during monotherapy for active tuberculosis. Potent enzyme induction creates many interactions and changes hormone, anticoagulant, antiviral, and immunosuppressant exposure. Red-orange body fluids are expected.

Sulfonamides and trimethoprim block sequential folate metabolism. The combination treats selected urinary, skin, respiratory, and opportunistic infections. Risks include allergy, marrow suppression, hyperkalaemia, kidney effects, haemolysis in glucose-six-phosphate dehydrogenase deficiency, and interactions. Metronidazole generates toxic intermediates in anaerobes and selected protozoa and is effective in low-oxygen sites.

Membrane-active and specialised agents

Daptomycin disrupts Gram-positive membranes but is inactivated by pulmonary surfactant and cannot treat pneumonia; monitor muscle toxicity. Polymyxins disrupt Gram-negative outer membranes but cause kidney and neurological toxicity and are reserved for selected resistant infections. Nitrofurantoin concentrates in urine for lower urinary infection but does not achieve useful renal-tissue or systemic concentrations.

Tuberculosis treatment uses multiple drugs to prevent selection across large, heterogeneous bacterial populations. Standard first-line agents target mycolic-acid synthesis, transcription, membrane energetics, and other processes, with distinct hepatic, neurological, ocular, and interaction risks.

Adherence, susceptibility, site, disease burden, immune status, and public-health coordination determine regimen and duration.

Resistance mechanisms

Microbes resist by destroying or modifying drugs, altering targets, bypassing pathways, reducing entry, increasing efflux, protecting targets, and changing growth state. Resistance can be intrinsic or acquired through mutation and horizontal transfer. Biofilm and persister states create tolerance without standard genetic resistance.

Combination therapy can broaden empirical coverage, prevent resistance in infections such as tuberculosis and human immunodeficiency virus, provide synergy, or treat polymicrobial disease. It can also increase toxicity, antagonism, cost, and ecological pressure. De-escalate when cultures, clinical response, and source clarify the need.

Antiviral therapy

Viruses use host machinery, so therapy targets virus-specific entry, uncoating, polymerases, proteases, integrases, neuraminidase, or assembly. Timing matters because replication often peaks early. Drugs may suppress replication without eradicating latent genomes, requiring prolonged or episodic therapy.

Nucleoside and nucleotide analogues mimic substrates and terminate or corrupt viral genome synthesis after selective activation. Herpesvirus polymerase inhibitors shorten disease and suppress recurrence; resistance rises with profound immune impairment. Influenza neuraminidase inhibitors reduce release and work best early, with greatest value in severe or high-risk disease.

Human immunodeficiency virus therapy combines drugs from different classes to suppress replication, restore immunity, prevent resistance, and eliminate sexual transmission when sustained viral suppression is achieved. Classes target reverse transcriptase, integrase, protease, entry, or capsid. Adherence, interactions, resistance testing, organ function, pregnancy, and coinfection guide selection.

Hepatitis B therapy usually suppresses polymerase and reduces progression but may not eliminate covalently closed nuclear DNA. Direct-acting hepatitis C combinations target viral protease, replication complex, and polymerase and can cure most treated infection. Exposure prophylaxis and treatment for emerging viral disease require current evidence and resistance surveillance.

Antifungal therapy

Polyenes bind ergosterol and disrupt membranes. Amphotericin has broad serious-infection activity but causes infusion reactions, kidney injury, potassium and magnesium loss, and anaemia; lipid formulations reduce some toxicity. Azoles inhibit ergosterol synthesis and differ in spectrum, absorption, QT effect, hepatic toxicity, and cytochrome interactions. Therapeutic monitoring is useful for selected agents and invasive disease.

Echinocandins inhibit beta-glucan cell-wall synthesis and treat invasive *Candida* and selected *Aspergillus* disease with limited oral availability. Flucytosine disrupts fungal nucleic-acid metabolism and is combined in selected infections because resistance emerges rapidly; monitor marrow and liver toxicity. Terbinafine inhibits squalene epoxidase and is useful for dermatophytes, including nail disease.

Invasive fungal treatment depends strongly on immune recovery, species, site, surgical control, and removal of infected devices. Neutropenia, transplantation, and prolonged immunosuppression alter spectrum and prophylaxis. Colonisation of non-sterile sites should not trigger toxic systemic therapy without evidence of invasion.

Antiparasitic therapy

Antiprotozoal drugs target folate pathways, redox metabolism, haem detoxification, electron transport, membranes, or parasite-specific enzymes. Malaria treatment depends on species, geography, resistance, severity, pregnancy, and relapse biology. Severe malaria requires urgent parenteral treatment; selected species require therapy against dormant liver forms after checking glucose-six-phosphate dehydrogenase status.

Helminth drugs disrupt microtubules, neuromuscular transmission, membrane permeability, or energy metabolism. Choice follows worm and life stage.

Killing parasites can provoke inflammation, especially in high tissue burden, and corticosteroid or supportive management may be required. *Strongyloides* should be identified or treated before major immunosuppression in at-risk patients because hyperinfection can be fatal.

Ectoparasites require treatment of patient, close contacts when indicated, clothing or bedding, and sometimes repeat dosing according to life cycle. Apparent failure may reflect reinfestation, incorrect application, resistance, or post-treatment itch rather than living organisms.

Safe prescribing and monitoring

Obtain cultures before therapy when safe, then treat severe syndromes promptly. Dose for site and physiology. Review organ function, blood counts, electrolytes, electrocardiogram, pregnancy, allergy, interactions, weight, immunity, and prior resistance.

Monitor response and toxicity. Failure may reflect wrong diagnosis, resistance, inadequate exposure, adherence, undrained pus, obstruction, infected device, poor perfusion, immune deficit, or complication. Longer is not automatically better. Convert intravenous to oral therapy when absorption, agent, site, and stability permit.

TTS module 2: Exposure-targeted antimicrobial treatment and resistance-aware prescribing

An antimicrobial prescription is a quantitative hypothesis: a defined regimen will create sufficient free-drug exposure at the infected site to inhibit or kill the suspected organism without unacceptable harm. Success also depends on viable perfusion, drainage, removal of foreign material where needed, and host immunity. A report that says “susceptible” cannot compensate for an undrained abscess, obstructed kidney, necrotic limb, infected prosthesis, or dose that never reaches the compartment. Every review should therefore revisit diagnosis, organism, site, exposure, source control, and host response.

Pharmacodynamic targets differ among drug classes. For many beta-lactams, efficacy relates to the fraction of the dosing interval during which unbound concentration exceeds the minimum inhibitory concentration. Shortening the interval or extending the infusion can improve target attainment without

necessarily increasing total daily dose. Aminoglycosides rely substantially on a high peak relative to the inhibitory concentration and have a post-antibiotic effect, supporting large separated doses in suitable patients. Vancomycin and several fluoroquinolones are better related to area under the concentration-time curve relative to inhibition. These indices are population models, not bedside guarantees.

Critical illness profoundly alters exposure. Capillary leak and fluid resuscitation expand the apparent volume of distribution of hydrophilic drugs, reducing initial concentration. Hypoalbuminaemia changes free fractions. Kidney clearance may be sharply reduced or paradoxically augmented in a hyperdynamic patient. Dialysis and extracorporeal circuits remove or sequester selected agents. A loading dose is determined mainly by distribution and desired concentration and may remain necessary despite renal failure; maintenance dosing then accounts for clearance. Under-dosing early severe infection risks failure and selection, while accumulation later causes toxicity.

Tissue penetration depends on free concentration, lipid solubility, ionisation, transport, inflammation, perfusion, and anatomical barriers. Meningeal inflammation can increase entry of selected beta-lactams, but cerebrospinal-fluid targets remain stricter than serum targets. Urine may achieve very high concentrations even when tissue levels are modest, explaining why nitrofurantoin treats lower urinary infection but not pyelonephritis or sepsis. Daptomycin is neutralised by pulmonary surfactant. Aminoglycosides function poorly in acidic, anaerobic abscess conditions. Drug choice must match the infected compartment rather than the culture source alone.

Beta-lactams are frequently first-line because of strong efficacy and generally favourable toxicity, but allergy labels unnecessarily exclude many patients. A history should identify the drug, timing, symptoms, treatment, severity, and later tolerance. Isolated gastrointestinal upset is intolerance, not allergy. A remote mild rash differs from immediate urticaria, bronchospasm, hypotension, or a severe delayed syndrome such as epidermal necrolysis, hepatitis, nephritis, haemolysis, or drug reaction with eosinophilia. Risk-based direct challenge or formal testing can remove inaccurate labels; severe immune reactions are not challenged casually.

Beta-lactamases vary in substrate and inhibitor susceptibility. Some hydrolyse extended-spectrum cephalosporins, inducible AmpC enzymes may emerge during therapy with selected agents, and carbapenemases compromise last-line drugs through several molecular families. Altered penicillin-binding proteins produce methicillin resistance and reduced pneumococcal susceptibility. Gram-negative outer-membrane permeability and efflux can amplify enzyme effects. A molecular resistance marker gives rapid warning but does not describe every expressed mechanism; phenotypic susceptibility and clinical context remain necessary.

Aminoglycosides provide rapid concentration-dependent Gram-negative activity but distribute poorly into some tissues and cause kidney, vestibular, and cochlear toxicity. Monitoring uses renal function, interval, levels according to regimen, and cumulative exposure. Neuromuscular blockade can worsen weakness. Vancomycin dosing for serious systemic infection increasingly targets calculated exposure rather than a high trough alone, reducing kidney injury while maintaining efficacy. Infusion-related flushing reflects rate-dependent mediator release and differs from immunoglobulin-E-mediated anaphylaxis.

Protein-synthesis inhibitors have class-specific advantages. Macrolides and tetracyclines cover atypical intracellular respiratory organisms because they penetrate cells. Doxycycline also treats many tick-borne and zoonotic infections; taking it with water while upright reduces oesophageal injury. Linezolid has excellent oral bioavailability and penetrates lung, but prolonged exposure can suppress marrow, injure optic or peripheral nerves, and impair mitochondrial metabolism. Interaction with serotonergic drugs is a risk assessment rather than an automatic prohibition. Clindamycin can suppress toxin production in selected invasive infections but carries substantial *Clostridioides difficile* risk.

Fluoroquinolones are highly bioavailable and penetrate many compartments, which makes them convenient but ecologically powerful. Tendon injury, neuropathy, central effects, dysglycaemia, corrected-QT prolongation, aortic concerns in selected patients, and *C. difficile* disease narrow their optimal role. Chelation by iron, calcium, magnesium, or aluminium reduces oral absorption. Rifampicin similarly penetrates well and acts on biofilm-associated staphylococci in selected combinations, but resistance emerges rapidly if it is

used alone against a large active population. Its enzyme induction can cause contraceptive, anticoagulant, antiretroviral, antifungal, opioid, and transplant-drug failure.

Combination therapy has four defensible purposes: ensure broad empirical activity in life-threatening uncertainty, treat polymicrobial infection, create synergy, or suppress resistance when single-step escape is likely. Once microbiology and anatomy clarify the syndrome, unnecessary overlap should stop. Double Gram-negative coverage seldom remains beneficial after susceptibility is known. Tuberculosis, human immunodeficiency virus, and selected endocardial, prosthetic, or difficult resistant infections are distinct because population heterogeneity, protected compartments, or validated synergy justify combinations.

Antiviral treatment is most effective when active replication drives disease. Herpes antivirals require viral or cellular phosphorylation before inhibiting polymerase, while resistance may alter activating enzymes or polymerase. Dose and route depend on mucocutaneous disease versus encephalitis, neonatal infection, or disseminated disease. Kidney adjustment and hydration reduce crystal-related injury for selected agents. Influenza therapy has greatest benefit when started early, yet severe hospitalised disease can warrant treatment later because replication may persist. Viral pneumonia can continue to worsen from inflammation after replication falls, so clinical trajectory does not directly measure antiviral failure.

Combination antiretroviral therapy suppresses human immunodeficiency virus below effective transmission thresholds and permits immune recovery, but latent integrated provirus prevents routine eradication. Regimen design considers resistance, hepatitis coinfection, kidney and bone health, pregnancy, cardiovascular and metabolic effects, and drug interactions. Adherence interruptions expose partially active regimens as drugs decay at different rates, selecting resistance. Long-acting therapy reduces dosing frequency for selected stable patients but makes missed injections and prolonged pharmacokinetic tails important.

Antifungal dosing is constrained by shared eukaryotic biology and variable absorption. Azoles differ substantially: food, gastric acidity, formulation, hepatic metabolism, and interactions can determine exposure, while some prolong and others shorten cardiac repolarisation. Therapeutic drug

monitoring is valuable when absorption is unpredictable, the therapeutic window is narrow, or invasive disease demands reliable levels. Amphotericin rapidly covers many severe mycoses but causes infusion reactions, renal vasoconstriction, tubular electrolyte wasting, and anaemia. Echinocandins are useful for invasive *Candida* but achieve poor concentrations in urine, eye, and central nervous system.

Invasive fungal disease requires reversal of the ecological advantage where possible. Recover neutrophils, reduce immunosuppression cautiously, remove an infected vascular catheter, debride devitalised tissue, or correct obstruction. Species-level identification matters because *Candida*, *Aspergillus*, *Cryptococcus*, *Mucorales*, and endemic fungi have different intrinsic susceptibility and tissue behaviour. Respiratory *Candida* usually represents colonisation; treating it exposes toxicity without benefit. By contrast, *Candida* in blood should trigger systemic therapy, repeat clearance cultures, source review, and assessment for metastatic complications according to phenotype.

Antiparasitic therapy follows species, life stage, geography, severity, and host. Severe malaria is treated urgently with parenteral therapy because parasite burden, sequestration, haemolysis, acidosis, brain injury, and organ failure can advance quickly. Eradication of dormant liver hypnozoites for relevant species requires a separate drug and prior glucose-six-phosphate dehydrogenase assessment to avoid haemolysis. Killing tissue parasites may release antigens and provoke dangerous inflammation, so corticosteroids or supportive treatment accompany therapy in selected neurological, ocular, or high-burden syndromes.

Treatment failure should trigger structured review rather than reflexive broadening. Was the original syndrome infectious? Was the organism sampled from the true site? Is susceptibility reliable and is the drug active at the achieved exposure? Has the patient absorbed and received every dose? Is there pus, obstruction, necrosis, foreign material, thrombosis, or a metastatic focus? Is immune recovery possible? Fever can persist from tissue inflammation, drug reaction, thrombosis, or another infection even after microbial control. Broader therapy may add toxicity while delaying the needed procedure.

The end of antimicrobial therapy is an active decision. Define duration from syndrome, source control, response, immune state, and trial evidence rather than from habit. Convert to oral treatment when a bioavailable active drug reaches the site and the patient can absorb it; intravenous access is not inherently stronger. Document indication, agent, dose, route, review point, and stop date. The best antimicrobial course is not the narrowest or shortest imaginable; it is the least harmful regimen that reliably attains the biological target and completes cure.

Retrieval prompts

One. How do pharmacokinetic-pharmacodynamic targets differ among antibacterials?

Two. Which mechanisms produce beta-lactam resistance?

Three. Why are combinations essential for some infections but harmful when unnecessary?

Four. Why can antivirals suppress without curing infection?

Five. What limits antifungal selective toxicity?

Six. Which factors should be checked before and during antimicrobial therapy?

Concise answers

One. Activity may track time above inhibitory concentration, peak-to-inhibitory ratio, or exposure area relative to inhibition.

Two. Drug-destroying enzymes, altered binding proteins, reduced entry, and efflux.

Three. They can prevent resistance, broaden or synergise, but also add toxicity, antagonism, and ecological pressure.

Four. Latent, integrated, or stable nuclear viral genomes may persist outside active replication.

Five. Fungi and humans are both eukaryotic and share many cellular processes.

Six. Site, organism, susceptibility, dose exposure, source control, organ function, pregnancy, allergy, interactions, response, and toxicity.

Source map

Original synthesis informed principally by Katzung and OpenStax Pharmacology, antimicrobial mechanisms, kinetics, toxicity, and resistance; OpenStax Microbiology, microbial targets and resistance; Robbins, infectious pathology; OpenStax Medical-Surgical Nursing; OpenStax Biology and Chemistry; Guyton and Hall, organ function and drug handling; and Talley and O'Connor, clinical monitoring.

Chapter 38: Sepsis, Antimicrobial Stewardship, and Infection Prevention

Orientation

Sepsis is life-threatening organ dysfunction from a dysregulated response to infection. It is a syndrome, not a culture result, and may occur without fever or hypotension. Care combines recognition, resuscitation, cultures, antimicrobials, source control, and reassessment. Stewardship is disciplined selection, optimisation, narrowing, and stopping—not weak treatment. Prevention reduces infection and antimicrobial use.

Pathophysiology of sepsis

Microbial structures and injured cells activate innate receptors, complement, coagulation, endothelium, leukocytes, and neuroendocrine responses. Cytokines, nitric oxide, capillary leak, altered vascular tone, microthrombi, mitochondrial dysfunction, and impaired cellular oxygen use produce heterogeneous organ injury. Simultaneous pro- and anti-inflammatory signals can cause both tissue damage and immune exhaustion.

Vasodilation and leakage reduce effective circulating volume. Myocardial depression and arrhythmia can reduce flow. Microcirculatory shunting means apparently adequate global pressure may coexist with impaired tissue perfusion. Lung permeability causes hypoxaemia; kidney filtration falls; liver and coagulation fail; gut barrier changes; brain dysfunction causes delirium or coma.

Serum lactate can rise from hypoperfusion, adrenergic glycolysis, reduced clearance, drugs, seizures, and other causes. It is a risk marker and trend, not a direct meter of oxygen debt. Normal lactate does not exclude serious sepsis, and persistent elevation demands renewed assessment rather than automatic fluid.

Recognition and differential diagnosis

Suspect sepsis when infection accompanies acute deterioration in consciousness, pressure, breathing, oxygenation, urine output, perfusion, coagulation, liver function, or platelets. Older, pregnant, neonatal, immunosuppressed, and medicated patients may present atypically. Neutropenia can blunt local inflammation; beta blockade can blunt tachycardia.

Scores can identify risk but do not replace judgement. Compare observations with baseline and trajectory. Examine likely sources: lung, urinary tract, abdomen, biliary system, skin and soft tissue, bone and joint, central nervous system, heart valves, teeth, reproductive tract, wounds, and devices. Search for non-infectious mimics including haemorrhage, pancreatitis, pulmonary embolism, adrenal crisis, anaphylaxis, toxic ingestion, drug reaction, infarction, and inflammatory disease.

Septic shock is sepsis with persistent circulatory and metabolic abnormality despite adequate initial volume assessment, commonly operationalised by vasopressor need to maintain pressure with elevated lactate. Do not wait for formal criteria when perfusion is failing.

Immediate assessment and cultures

Assess airway, breathing, circulation, disability, exposure, glucose, temperature, perfusion, and urine output. Establish access, monitor rhythm and pressure, obtain blood count, electrolytes, kidney and liver function, coagulation, blood gas, lactate when useful, and targeted imaging. Repeat clinical assessment after each intervention.

Obtain blood cultures from appropriately prepared separate sites before antibiotics when this causes no meaningful delay. Culture suspected focal sites with good-quality specimens. Device cultures and removal decisions depend on device and syndrome. Molecular tests can accelerate diagnosis but may detect colonisation; negative panels do not exclude untargeted organisms.

Antimicrobial timing and selection

In shock or high-likelihood severe infection, administer active intravenous antimicrobials urgently after rapid cultures. In stable patients with uncertain infection, focused investigation can reduce unnecessary exposure without compromising safety. The correct principle is proportional urgency, not one identical clock for every possible syndrome.

Empirical selection uses likely source and organisms, community versus healthcare exposure, previous cultures, local resistance, recent antibiotics, immune status, devices, allergies, organ function, and tissue penetration. Dosing in sepsis must account for expanded distribution, altered protein binding, augmented or reduced renal clearance, and organ support. A loading dose may be required despite kidney dysfunction.

Reassess at least daily and whenever results return. Confirm whether infection remains likely, narrow spectrum, optimise dose and route, define duration, and stop duplicate or unnecessary agents. Clinical improvement does not prove the original diagnosis; deterioration may reflect resistance, inadequate exposure, wrong source, immune injury, or non-infectious disease.

Fluids, vasoactive therapy, and organ support

Give isotonic crystalloid in indicated boluses while assessing response, congestion, cardiac function, perfusion, and losses. Fixed large volumes can harm limited cardiac or renal reserve. Passive-leg-raise response, capillary refill, urine, mentation, and echocardiography guide further fluid better than static pressure alone.

Norepinephrine is the usual first vasopressor for distributive shock. Add or change agents according to pressure, rhythm, cardiac output, and adverse effects. Inotropes may help selected myocardial dysfunction with persistent hypoperfusion. Corticosteroids are considered in vasopressor-dependent shock, not as routine treatment for every infection.

Support oxygenation and ventilation, avoid excessive tidal stretch, manage glucose without severe hypoglycaemia, provide kidney replacement when indicated, prevent thrombosis and pressure injury, support nutrition, and treat

pain and delirium. Transfusion follows overall oxygen-delivery and bleeding context rather than sepsis alone.

Source control

Antibiotics cannot drain pus, remove necrosis, relieve obstruction, or sterilise many foreign bodies. Source control includes drainage, debridement, surgery for perforation or ischaemia, relief of urinary or biliary obstruction, device removal, and management of endocardial or prosthetic infection.

Act as soon as safely feasible. Imaging defines anatomy but must not delay exploration in necrotising infection or perforation. Obtain deep cultures. Failure to improve requires another search for collections, devitalised tissue, device infection, or a second source.

Antimicrobial stewardship

Stewardship seeks the best outcome with the least toxicity, resistance, ecological disruption, and cost. Core actions include diagnostic stewardship, guideline-concordant empirical therapy, preauthorisation or review of selected agents, prospective audit and feedback, pharmacy support, allergy clarification, dose optimisation, intravenous-to-oral conversion, and documented indication and stop date.

Do not treat asymptomatic bacteriuria except in defined situations such as pregnancy and selected urological procedures. Do not treat colonised wounds, airways, urine catheters, or skin solely because cultures are positive. Avoid antibiotics for uncomplicated viral syndromes. Biomarkers can support but cannot independently start or stop therapy.

Duration follows syndrome, source control, immune status, response, and evidence. Many infections need shorter courses than traditional practice; endocarditis, bone infection, tuberculosis, and selected deep or immune-compromised infections may require longer. Automatic extension for persistent radiographic change or nonspecific inflammation can be harmful.

Standard and transmission-based precautions

Standard precautions apply to every patient: hand hygiene, risk-based protective equipment, respiratory and sharps safety, aseptic technique, cleaning, and safe injection. Gloves do not replace hand hygiene. Use alcohol rub routinely; use soap and water for visible soil and selected spore or outbreak contexts.

Contact precautions address touch and contaminated environments; droplet precautions address short-range particles; airborne precautions require respiratory protection and ventilation. Some infections require combinations. Start and stop isolation according to organism, symptoms, treatment, and guidance.

Select protective equipment for anticipated exposure and remove it without self-contamination. Training, access, observation, and safety culture matter. Isolation can cause loneliness and missed care, so deliberate contact remains essential.

Device- and procedure-associated infection

Every invasive device breaches defence and needs continuing indication. Vascular-catheter prevention uses hand hygiene, sterile central insertion, skin antisepsis, appropriate site and dressing, aseptic access, and daily review. Suspected bloodstream infection requires paired cultures and removal when indicated.

Insert urinary catheters aseptically only for valid indications, maintain a closed unobstructed system, and remove early. Ventilator prevention includes minimising intubation and sedation, appropriate positioning, oral care, aspiration reduction, mobilisation, and weaning assessment.

Surgical prevention combines optimisation, timely prophylaxis, clipping rather than shaving, skin preparation, temperature and glucose management, sterile technique, and wound care. Drains alone do not justify continued prophylaxis.

Environmental control and outbreaks

Cleaning removes soil; disinfection inactivates many pathogens; sterilisation destroys all microbial life including spores. Processing depends on whether equipment enters sterile tissue, contacts mucosa, or touches skin. Organic material impairs disinfection, so clean first.

Outbreak investigation verifies diagnosis, defines cases, maps person-place-time, identifies exposures and routes, and implements early control. Line lists and epidemic curves reveal clustering; genomic links require epidemiological context.

Ventilation, water, food, laundry, dust, and equipment can sustain outbreaks. Report promptly, preserve specimens, communicate, and audit whether control interrupts transmission.

Vaccination and occupational protection

Vaccines reduce susceptibility, severe disease, and sometimes transmission. Live vaccines can be unsafe in severe immune suppression or pregnancy. Inactivated, subunit, conjugate, toxoid, vector, and nucleic-acid platforms differ in schedule and response.

Healthcare workers need vaccination, fit-tested respiratory protection, sharps prevention, and rapid exposure pathways. After needlestick or mucosal exposure, provide first aid, document, test confidentially with consent, and offer time-sensitive prophylaxis when indicated.

Recovery after sepsis

Survivors may have weakness, cognitive and psychological change, organ impairment, and recurrent infection. Reconcile medicines, document resistance and allergies, assess function and nutrition, plan rehabilitation, and explain red flags. Prevention includes vaccination, device removal, chronic-disease, dental, and wound care.

TTS module 2: Dynamic sepsis care, diagnostic stewardship, and interruption of transmission

Sepsis is recognised when a suspected infection produces new organ dysfunction, but neither suspicion nor dysfunction is specific. Pneumonia may cause hypoxaemia without systemic sepsis, while pancreatitis, haemorrhage, pulmonary embolism, adrenal crisis, anaphylaxis, toxic exposure, and inflammatory disease can closely mimic it. The safest approach runs two processes in parallel: treat immediately reversible physiological threats and build evidence for infection, source, severity, and alternatives. Repeated examination is central because response to intervention changes probability.

The septic circulation is heterogeneous. Inflammatory mediators reduce vascular tone, increase endothelial permeability, alter glycocalyx and coagulation, and redistribute microvascular flow. Effective arterial volume falls even when total body water is increased. Myocardial depression, right-heart strain, arrhythmia, mechanical ventilation, and pre-existing cardiac disease may further limit output. Cells can experience regional oxygen-delivery failure while a global pressure appears acceptable. Conversely, low pressure in an alert, warm patient with preserved urine and improving perfusion may require a different response from the same number accompanied by mottling and confusion.

Organ dysfunction arises through interacting macro- and microcirculatory, inflammatory, metabolic, and iatrogenic mechanisms. Acute respiratory distress syndrome increases alveolar-capillary permeability and decreases aerated lung. Kidney injury reflects haemodynamics, inflammation, microvascular change, congestion, drugs, and underlying reserve rather than simple low flow alone. Encephalopathy may precede hypotension. Cholestasis, thrombocytopenia, coagulopathy, ileus, hyperglycaemia, and myocardial injury are common. Failure of one organ increases demand on others, so trajectory and reserve matter more than a single score.

Serum lactate is a useful risk signal but an ambiguous mechanism. Adrenergic stimulation accelerates glycolysis and lactate production even with oxygen present. Regional hypoperfusion, impaired liver clearance, seizures, beta agonists, mitochondrial dysfunction, and thiamine deficiency can contribute. A falling lactate may accompany improved perfusion, but forcing normalisation with repeated fluid can cause pulmonary oedema and

abdominal or venous congestion. Interpret lactate alongside capillary refill, skin temperature, mentation, urine output, pressure, ultrasound, acid-base state, and the intervention just given.

Initial fluid should be a therapeutic trial, not an irreversible commitment to a fixed volume. Balanced isotonic crystalloid is commonly used. Give an amount appropriate to severity, then ask whether stroke volume or tissue perfusion improved and whether congestion developed. Passive leg raising creates a reversible internal volume challenge when measured with a responsive output marker. Pulse-pressure or stroke-volume variation can help only under suitable rhythm, ventilation, and tidal conditions. Static central venous pressure poorly predicts responsiveness. A patient can be fluid responsive yet not need fluid if perfusion is already adequate or pulmonary risk outweighs benefit.

Norepinephrine restores vascular tone and can begin early when hypotension is profound or fluid response is limited. Peripheral administration through a well-sited monitored vein can avoid harmful delay while definitive access is considered. The pressure target is individualised from perfusion, chronic hypertension, bleeding, and cardiac state. Vasopressin can reduce norepinephrine requirement in selected shock. Epinephrine adds inotropic and vasoconstrictor effects but can raise lactate and provoke arrhythmia. Inotropes are reserved for evidence of low output with persistent hypoperfusion, because increasing contractility raises oxygen demand and rhythm risk.

Antibiotic urgency is proportional to the likelihood and consequence of untreated infection. Shock, meningitis, neutropenic sepsis, necrotising infection, and severe bacterial syndromes warrant immediate active therapy after rapid cultures. In a stable patient with weak evidence, a focused diagnostic interval may prevent unnecessary broad treatment. Empirical selection combines source, severity, prior organisms, recent drugs, healthcare contact, devices, immune status, local resistance, allergy phenotype, and organ function. The first dose must be adequate; loading is often preserved even when later maintenance requires renal adjustment.

Blood cultures should be obtained from separate properly prepared sites with sufficient volume. Poorly filled bottles reduce sensitivity; contamination increases unnecessary therapy. Culture urine only when symptoms or a

compatible systemic syndrome make urinary infection plausible, with defined exceptions. Respiratory samples must be interpreted through collection quality and colonisation. Molecular panels can accelerate targeted action but cannot detect organisms outside the panel and may identify irrelevant carriage. Diagnostic stewardship means collecting the right specimen before treatment when safe, not withholding tests indiscriminately.

Source control converts an antimicrobial problem into an anatomical one. Drain abscesses, relieve urinary or biliary obstruction, debride dead tissue, repair perforation or ischaemic bowel, and remove infected devices when indicated. Delay permits ongoing microbial growth, toxin production, and inflammatory signalling even when serum drug concentrations are adequate. Imaging clarifies anatomy in stable patients, but necrotising infection, abdominal catastrophe, or rapidly compressive collections may require exploration before perfect imaging. Deep operative samples should be collected because superficial swabs can misdirect definitive therapy.

Reassessment begins within hours, not at the end of a customary course. Review physiology, source, cultures, imaging, dose exposure, organ function, and alternative diagnoses. Narrow when an organism and susceptibility are reliable, stop redundant coverage, and define a review date and expected duration. Culture-negative sepsis may reflect prior antibiotics, inaccessible focus, non-cultivable organism, inadequate sampling, or a non-infectious mimic. Clinical improvement does not retrospectively prove bacterial disease, and persistent fever alone does not prove inadequate breadth.

Stewardship protects the treated patient first. Avoiding unnecessary antimicrobials reduces allergy, kidney and liver injury, cytopenia, cardiac effects, interactions, *C. difficile* disease, and disruption of microbiota. It also preserves future options by reducing selection and transmission of resistant organisms. Allergy assessment, dose optimisation, therapeutic drug monitoring, intravenous-to-oral conversion, and evidence-based duration are clinical quality interventions. Restriction and approval systems work best when paired with rapid advice and feedback rather than functioning as barriers during emergencies.

Asymptomatic bacteriuria illustrates diagnostic harm. Bacteria commonly inhabit catheterised or abnormal urinary tracts without causing disease. Pyuria reflects inflammation and is also common; cloudy or odorous urine is

nonspecific. Treating a positive result without compatible symptoms rarely improves outcomes outside defined pregnancy and urological contexts, but it causes adverse effects and resistance. Likewise, chronic wounds, tracheostomies, and respiratory devices become colonised. A culture should explain new tissue invasion or systemic dysfunction before it drives systemic therapy.

Infection prevention follows the hierarchy of eliminating unnecessary exposure, engineering safer environments, changing work systems, and using personal protection. Hand hygiene interrupts transfer but fails when access, staffing, skin tolerance, or workflow make adherence difficult. Gloves protect against anticipated contamination yet become another vehicle if not changed and followed by hand hygiene. Standard precautions apply to all patients because carriage is frequently unknown. Added contact, droplet, or airborne measures depend on route, setting, procedure, and current guidance.

Ventilation is a population-level control for airborne risk. Outdoor air exchange, filtration, pressure-controlled rooms, and well-fitting respiratory protection reduce inhaled dose. Surgical masks chiefly provide source control and droplet protection, while respirators are designed to reduce inhalation when correctly selected and fit-tested. Eye protection blocks mucosal splash and particle exposure in appropriate contexts. Personal protective equipment fails when donning, fit, removal, disposal, or supply is poor, so observation and training matter as much as written policy.

Every invasive device should have an indication, insertion bundle, maintenance plan, and removal trigger. Central-line infection prevention includes maximal sterile insertion, appropriate skin antisepsis, dressing and hub care, and daily necessity review. Urinary catheter prevention begins by not inserting one for convenience, maintaining closed dependent drainage, and removing it promptly. Ventilator prevention combines avoidance of unnecessary intubation, sedation minimisation, head positioning, oral care, secretion management, mobility, and readiness-to-wean assessment. A checklist succeeds only when staff can stop unsafe practice.

Cleaning physically removes soil and microbial burden. Disinfection inactivates organisms to a level appropriate for environmental or device use, while sterilisation destroys all viable microbial forms including spores. Critical devices entering sterile tissue require sterilisation; semicritical devices

contacting mucosa require validated high-level processing; noncritical equipment touching intact skin uses lower-level disinfection. Organic matter shields organisms and consumes disinfectant, so cleaning precedes chemical processing. Contact time, concentration, compatibility, and recontamination determine actual effectiveness.

Outbreak investigation starts by confirming that cases exceed expectation and share a valid definition. A line list maps person, place, time, exposure, procedure, organism, and outcome. The epidemic curve and common links suggest point source, continuing exposure, or propagation. Immediate controls need not wait for final proof when consequences are high. Environmental or genomic similarity supports a link but cannot replace epidemiology: common organisms may match by chance, while sampled intermediates may be missing.

Sepsis recovery is an active phase of care. Survivors may have muscle loss, neuropathy, cognitive change, post-traumatic symptoms, sleep disturbance, organ impairment, and medication confusion. Document the infection, source control, culture results, resistant organisms, adverse reactions, and true allergy status. Remove obsolete devices, restore mobility and nutrition, update vaccination, and arrange organ and functional follow-up. Acute survival is only the first outcome; good sepsis care preserves long-term capacity while preventing the next avoidable infection.

Retrieval prompts

One. Why can lactate not be interpreted as a pure measure of oxygen debt?

Two. How should fluid response be assessed in sepsis?

Three. What forms can source control take?

Four. What are the core actions of antimicrobial stewardship?

Five. How do standard, contact, droplet, and airborne precautions differ?

Six. Which strategies prevent device-associated infection?

Concise answers

One. Perfusion, adrenergic glycolysis, clearance, drugs, and seizures can all change lactate.

Two. Use small indicated boluses and repeated dynamic assessment of perfusion, congestion, cardiac function, and physiological response.

Three. Drainage, debridement, surgery, relief of obstruction, and removal of infected foreign material.

Four. Better diagnostics, appropriate empirical choice, review, narrowing, dose optimisation, oral conversion, duration control, and stopping unnecessary therapy.

Five. Standard precautions always apply; additional categories interrupt touch, short-range respiratory, or suspended-air transmission.

Six. Valid indication, aseptic insertion, correct maintenance, minimal manipulation, daily review, and early removal.

Source map

Original synthesis informed by OpenStax Microbiology, epidemiology, infection control, and host-pathogen interaction; Katzung and OpenStax Pharmacology, antimicrobial selection and stewardship; Guyton and Hall, shock and organ physiology; Robbins, sepsis pathology; OpenStax Medical-Surgical Nursing, infection prevention and acute care; and Talley and O'Connor, sepsis recognition and clinical assessment.

Chapter 39: White-Cell Disorders, Marrow Failure, and Haematological Malignancy

Orientation

Bone marrow continually produces red cells, platelets, and diverse leukocytes from self-renewing stem cells. Abnormal counts may reflect altered production, maturation, release, distribution, consumption, destruction, or clonal expansion. A count is not a diagnosis: morphology, lineage, tempo, symptoms, organ enlargement, and molecular findings establish mechanism. Immediate threats include febrile neutropenia, hyperleukocytosis, tumour lysis, severe cytopenia, disseminated coagulation, and spinal or airway compression.

Haematopoiesis and marrow organisation

Haematopoietic stem cells balance self-renewal with differentiation into myeloid and lymphoid lineages. Stromal cells, extracellular matrix, vessels, osteolineage cells, macrophages, cytokines, and growth factors form specialised niches. Erythropoietin promotes red-cell precursors; thrombopoietin supports megakaryocytes; colony-stimulating factors influence granulocyte and monocyte production.

Granulocytes mature through recognisable stages before release. Neutrophils circulate briefly, marginate along vessels, enter tissues, phagocytose, and die. Monocytes become macrophages or dendritic cells. B lymphocytes mature in marrow and secondary lymphoid tissue; T lymphocytes mature in thymus; natural killer cells provide innate cytotoxicity.

Clonal haematopoiesis becomes more common with age when a stem-cell mutation expands without diagnostic malignancy. It can increase future blood-cancer and cardiovascular risk but does not mean leukaemia is inevitable. Additional genetic, epigenetic, and environmental changes shape progression.

Interpreting white-cell counts

Use absolute cell counts rather than percentages alone. Neutrophilia accompanies bacterial infection, inflammation, tissue necrosis, glucocorticoids, smoking, stress, and myeloid neoplasia. A left shift means increased immature granulocytes; toxic granulation and vacuoles support reactive activation. Demargination can rapidly raise count without increased production.

Lymphocytosis may accompany viral infection, pertussis, smoking, stress, or lymphoid neoplasia. Atypical reactive lymphocytes differ from a monomorphic malignant population. Eosinophilia suggests allergy, drug reaction, tissue-invasive parasites, selected immune disease, endocrine deficiency, or clonal disease. Monocytosis occurs in infection, inflammation, recovery, and myeloid neoplasia.

Leukopenia can arise from marrow suppression, immune destruction, drugs, infection, nutritional deficiency, hypersplenism, or inherited disease. Trends and accompanying anaemia or thrombocytopenia reveal whether one lineage or the whole marrow is affected. A blood film can show blasts, dysplasia, parasites, fragmentation, inclusions, or abnormal lymphocytes.

Neutropenia and febrile neutropenia

Infection risk rises with the depth and duration of neutropenia and defects in skin, mucosa, devices, and cellular immunity. Profound neutropenia can produce serious infection without pus, focal signs, or fever. Causes include chemotherapy, drugs, autoimmune destruction, viral infection, marrow infiltration, aplasia, nutritional deficiency, and inherited syndromes.

Fever during significant neutropenia is an emergency. Obtain cultures and assess skin, mouth, lungs, abdomen, perineum, devices, and neurological state, then give broad antipseudomonal antibiotics promptly. Avoid rectal procedures. Add coverage according to instability, source, resistance, devices, and local guidance. Persistent fever triggers reassessment for occult bacterial, fungal, viral, drug, or inflammatory causes.

Granulocyte colony-stimulating factor prevents or shortens selected treatment-related neutropenia but can cause bone pain and rarely splenic or

pulmonary complications. It is not a substitute for antimicrobial treatment in established sepsis.

Bone-marrow failure

Aplastic anaemia is hypocellular marrow failure causing pancytopenia without malignant infiltration. Immune destruction of stem cells is common; drugs, toxins, radiation, viruses, pregnancy, inherited repair disorders, and paroxysmal nocturnal haemoglobinuria associations must be considered. Symptoms reflect anaemia, infection, and bleeding, usually without major lymph-node or spleen enlargement.

Management depends on severity, age, donor availability, and cause. Stem-cell transplantation can cure selected patients; immunosuppression with thrombopoietic support treats others. Transfusion requires careful matching and planning because alloimmunisation can complicate transplantation.

Marrow failure also results from myelodysplasia, fibrosis, metastatic cancer, leukaemia, infection, haemophagocytic syndromes, and severe nutritional deficiency. Bone-marrow aspirate assesses cellular detail and flow cytometry; trephine biopsy shows architecture, fibrosis, cellularity, granulomas, and infiltration. Cytogenetic and molecular tests classify clonal disease and prognosis.

Myelodysplastic and myeloproliferative neoplasms

Myelodysplastic neoplasms are clonal stem-cell disorders with ineffective, dysplastic haematopoiesis, cytopenias, and risk of acute myeloid leukaemia. Macrocytosis, abnormal neutrophils, and platelets may appear. Exclude reversible nutritional, toxic, drug, and infectious mimics. Classification integrates blood, marrow, chromosomes, and mutations.

Myeloproliferative neoplasms produce excessive mature myeloid cells. Polycythaemia vera expands red cells and often leukocytes and platelets, causing thrombosis, microvascular symptoms, itch, and splenomegaly. Essential thrombocythaemia elevates platelets with thrombotic or bleeding risk. Primary myelofibrosis causes marrow fibrosis, extramedullary haematopoiesis, splenomegaly, constitutional symptoms, and a leukoerythroblastic film.

Driver mutations activate growth signalling. Treatment balances thrombosis, bleeding, symptoms, progression, and therapy toxicity using venesection, antiplatelet treatment, cytoreduction, pathway inhibitors, or transplantation. A high count alone does not determine risk.

Acute leukaemia

Acute leukaemias are expansions of immature blasts that suppress normal marrow and infiltrate tissues. Acute myeloid leukaemia predominates in adults; acute lymphoblastic leukaemia is common in children but occurs at all ages. Patients present with fatigue, infection, bleeding, bone pain, organ enlargement, gum or skin infiltration, neurological signs, or incidental cytopenia.

Diagnosis uses film, marrow, flow cytometry, cytogenetics, and molecular testing. Lineage and genotype determine therapy and prognosis. Acute promyelocytic leukaemia is a distinct emergency associated with severe coagulopathy; start differentiation therapy immediately when suspected while confirming diagnosis.

Treatment combines induction, consolidation, targeted drugs, immune therapy, central nervous system prophylaxis where relevant, and transplantation according to risk. Supportive care includes antimicrobial prophylaxis, irradiated or matched blood products when indicated, fertility discussion, and management of mucositis and organ toxicity.

Hyperleukocytosis can impair microvascular flow, particularly in brain and lungs. Symptoms include headache, confusion, visual change, breathlessness, and hypoxaemia. Urgent specialist cytoreduction is required; routine red-cell transfusion can increase viscosity and needs careful judgement.

Chronic leukaemias

Chronic myeloid leukaemia arises from a fusion tyrosine kinase created by a chromosome translocation. It causes granulocytic proliferation with left-shifted cells and splenomegaly and can progress from chronic phase to blast crisis. Tyrosine-kinase inhibitors transform prognosis, but adherence, interactions, molecular response, resistance, and cardiovascular or metabolic toxicities require monitoring.

Chronic lymphocytic leukaemia is a clonal mature B-cell disorder ranging from incidental lymphocytosis to lymph-node, spleen, marrow, immune, and constitutional disease. Hypogammaglobulinaemia, autoimmune cytopenias, infection, and transformation can occur. Early asymptomatic disease is observed; treatment begins for active clinical indications, not lymphocyte count alone.

Lymphoma

Lymphomas are clonal lymphoid tumours presenting in nodes or extranodal tissue. Hodgkin lymphoma has characteristic malignant cells within a rich reactive background and often spreads contiguously. Non-Hodgkin lymphomas encompass indolent and aggressive B-, T-, and natural-killer-cell neoplasms with varied biology.

Persistent enlarging, firm, or unexplained lymphadenopathy, constitutional fever, drenching sweats, weight loss, itch, cytopenia, organ enlargement, or extranodal mass warrants assessment. Infection, immune disease, drugs, and metastatic solid cancer are important alternatives. Excision or adequate core biopsy preserves architecture; fine-needle aspiration alone often cannot classify lymphoma.

Staging uses clinical assessment, blood tests, metabolic imaging or computed tomography, and selective marrow or fluid testing. Treatment may use observation, chemotherapy, antibodies, small-molecule inhibitors, radiation, cellular therapy, or transplantation. Histological subtype and molecular risk matter more than stage alone in many entities.

Plasma-cell disorders

Plasma-cell neoplasms produce a monoclonal immunoglobulin or light chain. Monoclonal gammopathy of undetermined significance lacks defining organ injury but can progress and needs risk-based follow-up. Multiple myeloma causes marrow replacement, bone destruction, anaemia, kidney injury, hypercalcaemia, infection, and abnormal protein effects.

Investigate with blood and urine protein studies, free light chains, counts, calcium, kidney function, marrow, and whole-body imaging. A small clone can still damage kidney, nerve, or other organs. Therapy combines proteasome inhibitors, immunomodulatory and targeted agents, antibodies, corticosteroids, transplantation in eligible patients, bone protection, infection prevention, and supportive care.

Tumour lysis and other emergencies

Rapid malignant-cell breakdown releases potassium, phosphate, and nucleic acids, producing hyperkalaemia, hypocalcaemia, urate kidney injury, arrhythmia, seizure, and death. Risk depends on tumour burden, proliferation, treatment sensitivity, kidney function, and hydration. Prevent with risk assessment, fluid, frequent monitoring, urate-lowering therapy, and planned treatment setting. Established severe disturbance may require rasburicase and dialysis.

Other emergencies include spinal-cord compression, superior vena-cava obstruction, airway compromise, severe haemolysis, hypercalcaemia, thrombosis, bleeding, and haemophagocytic lymphohistiocytosis. Treat physiological threat while establishing the malignancy.

Diagnostic and care principles

History asks about infection, bleeding, fatigue, bone pain, nodes, abdominal fullness, weight, sweats, itch, drugs, toxins, immune disease, travel, family disease, and prior cancer treatment. Examine pallor, bruising, infection, nodes, liver, spleen, bone tenderness, neurological signs, and perfusion. Never dismiss persistent abnormalities without trend review.

Treatment decisions incorporate disease biology, curability, age, frailty, organ function, fertility, infection, patient goals, and treatment burden. Survivorship

includes vaccination, secondary-cancer and cardiovascular risk, endocrine and fertility effects, neuropathy, cognition, fatigue, psychological health, and return to function.

TTS module 2: Marrow pattern recognition, clonal disease, and haematological emergencies

Abnormal blood counts are interpreted as dynamic outputs of marrow production, maturation, release, distribution, tissue consumption, and clearance. A low circulating neutrophil count may reflect absent production, immune destruction, splenic sequestration, migration into infected tissue, or drug effect. A high count may reflect clonal proliferation, cytokine-driven production, marrow release, or demargination. The diagnostic unit is therefore the absolute count plus trend, film morphology, other lineages, clinical tempo, organ enlargement, and physiological consequences.

The marrow niche regulates stem-cell dormancy, self-renewal, and lineage commitment through stromal contact, oxygen gradients, growth factors, extracellular matrix, and vascular access. Mature cells enter sinusoidal blood only after ordered differentiation. Injury to an early multipotent stem cell can reduce all lineages, whereas a lineage-specific immune process may produce isolated cytopenia. Malignant clones can fail to produce useful mature cells despite a cellular marrow, so peripheral pancytopenia does not necessarily mean an empty marrow.

When reviewing leukocytosis, absolute differential and blood film come first. Reactive neutrophilia often includes bands, toxic granulation, vacuoles, and Döhle-like inclusions, while clonal myeloid disease may show a broad maturation spectrum, basophilia, dysplasia, or blasts. Glucocorticoids rapidly increase circulating neutrophils through demargination and reduced tissue migration while often lowering lymphocytes and eosinophils. Stress, smoking, pregnancy, inflammation, and tissue necrosis can produce substantial counts. Persistent unexplained abnormality requires comparison with baseline and molecular or marrow assessment rather than a single numerical threshold.

Lymphocytosis is interpreted by age and cell appearance. Reactive populations are often morphologically varied and arise with viral infection or immune stimulation. A monomorphic population suggests a clone but does not define whether it requires treatment. Flow cytometry identifies lineage and

light-chain or antigen restriction. Chronic lymphocytic leukaemia can be discovered incidentally and observed for years; therapy begins for progressive marrow failure, symptomatic nodes or spleen, constitutional disease, rapid progression, or selected complications, not because the lymphocyte count looks dramatic.

Neutropenic infection is dangerous because inflammation becomes diagnostically silent. Mucositis and devices breach barriers while neutrophils cannot form expected pus, infiltrates, or local tenderness. Risk rises sharply with profound and prolonged neutropenia but also depends on T-cell function, corticosteroids, prior colonisation, prophylaxis, and organ reserve. Fever may be the sole sign; hypothermia, confusion, hypotension, abdominal discomfort, or subtle catheter change can be equally important. Blood cultures and focused specimens are obtained promptly, followed by an antipseudomonal regimen appropriate to local resistance and previous isolates.

Persistent fever during neutropenia is not automatically bacterial resistance. Possibilities include an occult abscess, invasive mould, viral reactivation, drug fever, thrombosis, mucosal inflammation, tumour fever, or marrow recovery. Reassessment includes repeated examination, imaging based on symptoms and duration, drug exposure, microbiology, and immune trajectory. Rectal temperature, examination, enemas, and suppositories are avoided because mucosal injury can introduce organisms. Granulocyte colony-stimulating factor is used according to expected risk and regimen; it does not replace prompt antibiotics or source control.

Pancytopenia is organised by marrow cellularity and peripheral destruction. Nutritional deficiencies can produce ineffective haematopoiesis, macrocytosis, and hypersegmented neutrophils. Aplastic anaemia produces a hypocellular marrow without malignant replacement. Myelodysplastic neoplasia produces clonal ineffective blood formation with dysplasia and mutation-defined risk. Marrow infiltration by leukaemia, lymphoma, fibrosis, granuloma, or metastatic cancer can release nucleated red cells and immature myeloid cells into blood, creating a leukoerythroblastic film. Hypersplenism sequesters and destroys several lineages in an enlarged spleen.

Marrow aspirate and trephine biopsy answer complementary questions. Aspirate spreads individual cells for morphology, differential count, flow cytometry, cytogenetics, and molecular testing. A dry tap can occur with

fibrosis or densely packed disease. Trephine preserves architecture, cellularity, fibrosis, bone relationship, focal infiltration, necrosis, and granulomas. Samples should be planned before collection so tissue is allocated correctly. Molecular findings are interpreted with morphology because age-related clonal haematopoiesis can produce mutations without diagnostic cancer.

Myelodysplastic neoplasms arise from clonal stem cells that produce abnormal and ineffective descendants. Cytopenias cause fatigue, infection, and bleeding, while mutation and chromosomal patterns predict progression to acute myeloid leukaemia. Alcohol, copper or vitamin deficiency, drugs, infection, and toxins can mimic dysplasia and must be addressed. Treatment ranges from observation and transfusion support to growth factors, epigenetic therapy, targeted drugs, or allogeneic transplantation, the principal potentially curative approach. Risk combines blast proportion, cytopenias, chromosomes, mutations, age, frailty, and goals.

Myeloproliferative neoplasms produce mature cells but also abnormal vascular and inflammatory physiology. In polycythaemia vera, increased red-cell mass raises viscosity, while activated leukocytes and platelets contribute thrombosis. Aquagenic itch, erythromelalgia, headache, visual disturbance, splenomegaly, and unusual-site thrombosis are clues. Essential thrombocythaemia can paradoxically bleed when extreme platelet numbers consume large von Willebrand multimers. Myelofibrosis replaces useful marrow with fibrosis and drives blood production into spleen and liver. Driver mutation, age, thrombosis history, symptoms, and count guide therapy rather than count alone.

Acute leukaemia is a failure of differentiation with expansion of blasts that displace normal blood production and infiltrate tissues. Presentation may be pancytopenic rather than leukocytic. Flow cytometry establishes lineage; chromosomes and molecular alterations define entities, treatment targets, measurable residual disease strategy, and prognosis. Acute promyelocytic leukaemia is uniquely urgent because abnormal promyelocytes activate coagulation and fibrinolysis, causing catastrophic bleeding. Differentiation therapy starts on suspicion while diagnostic confirmation proceeds, alongside aggressive blood-product support.

Hyperleukocytosis becomes leukostasis when poorly deformable cells obstruct and activate microvasculature, most often affecting lung and brain. Breathlessness, hypoxaemia, headache, confusion, focal deficit, or visual change requires emergency cytoreduction and specialist care. The absolute count associated with risk differs by lineage. Dehydration, tumour lysis, and coagulation abnormalities may coexist. Red-cell transfusion can further increase viscosity and is timed cautiously unless immediately necessary. Leukapheresis has selected roles but cannot substitute for definitive antileukaemic treatment.

Lymphoma diagnosis depends on architecture. Excisional biopsy is preferred when feasible because relationships among malignant cells, follicles, sinuses, stroma, and reactive background distinguish subtypes. Corticosteroids can rapidly shrink lymphoid tissue and obscure diagnosis, so they are avoided before biopsy unless airway, cord, brain, or other organ threat demands treatment. Metabolic imaging maps active disease and guides biopsy toward a representative site, but inflammation also takes up tracer. Stage describes extent; histological and molecular biology frequently determines urgency and curability more strongly.

Plasma-cell clones injure through mass and protein. Marrow replacement causes anaemia; osteoclast activation creates lytic bone disease and hypercalcaemia; filtered free light chains damage kidney tubules; impaired normal immunoglobulin increases infection. Monoclonal protein concentration alone does not define myeloma. A small clone can produce amyloid, neuropathy, glomerular disease, or other organ injury and become clinically significant. Serum and urine electrophoresis, immunofixation, free-light-chain ratio, marrow, calcium, renal function, counts, and whole-body imaging build the diagnosis.

Tumour lysis syndrome occurs when rapid cellular breakdown releases potassium, phosphate, and nucleic acids faster than homeostasis can compensate. Hyperkalaemia causes arrhythmia, phosphate complexes calcium and promotes kidney deposition, hypocalcaemia causes neuromuscular irritability, and uric acid obstructs tubules. Prevention begins before cytotoxic treatment by estimating tumour burden, proliferation, treatment sensitivity, kidney function, and baseline chemistry. Hydration, frequent laboratory monitoring, allopurinol to prevent new urate, or

rasburicase to degrade existing urate are selected by risk. Dialysis treats refractory electrolyte, fluid, or kidney failure.

Haematological treatment repeatedly trades disease control against marrow, immune, organ, and functional reserve. Support includes compatible and appropriately modified blood products, infection prophylaxis, vaccination planning, fertility preservation, mucositis care, thrombosis and bleeding management, and early palliative input when useful. Measurable residual disease can detect a malignant population below morphological thresholds and refine relapse risk, but it is assay- and disease-specific. The final treatment decision incorporates biological curability, expected toxicity, frailty, donor options, patient priorities, and the consequences of both action and delay.

Retrieval prompts

One. Why must absolute leukocyte counts be used rather than percentages alone?

Two. What makes febrile neutropenia an emergency?

Three. How do myelodysplastic and myeloproliferative neoplasms differ?

Four. Which tests classify acute leukaemia?

Five. Why is excisional or core biopsy preferred for suspected lymphoma?

Six. What abnormalities define tumour lysis syndrome?

Concise answers

One. Percentages can appear normal despite severe absolute excess or deficiency.

Two. Infection can progress rapidly with muted local signs and inadequate neutrophil defence.

Three. Myelodysplasia produces ineffective dysplastic cytopenias; myeloproliferation produces excess mature cells, though overlap exists.

Four. Blood and marrow morphology, flow cytometry, cytogenetics, and molecular analysis.

Five. Architecture and adequate tissue are needed to determine lineage and subtype.

Six. Hyperkalaemia, hyperphosphataemia, secondary hypocalcaemia, hyperuricaemia, kidney injury, and their cardiac or neurological effects.

Source map

Original synthesis informed principally by Robbins, marrow failure, leukaemia, lymphoma, plasma-cell, and myeloid pathology; Guyton and Hall, blood-cell production and immune physiology; Katzung and OpenStax Pharmacology, growth factors and antineoplastic therapy; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; OpenStax Microbiology, infection risk; and Talley and O'Connor, haematological examination.

Chapter 40: Cancer Biology, Invasion, Metastasis, and Treatment Principles

Orientation

Cancer is clonal evolution within tissue. Genetic and epigenetic alterations favour proliferation, survival, resource acquisition, immune escape, and spread. Malignancy is not simply rapid growth: tumours vary in differentiation, invasion, metastasis, sensitivity, and host interaction. Diagnosis requires tissue and molecular context; treatment combines local and systemic control with toxicity prevention and patient goals.

Drivers, passengers, and clonal evolution

Proto-oncogenes normally promote growth or survival; activating mutation, amplification, or fusion can convert them into oncogenic drivers.

Tumour-suppressor genes restrain proliferation, repair damage, or trigger senescence and death; both copies often need functional loss at the cellular level. DNA-repair defects increase mutation rate and characteristic genomic signatures.

Not every tumour mutation drives disease. Passenger changes accumulate without providing major advantage. A driver can be essential in one clone but bypassed by another. Tumours therefore contain subclones with different vulnerabilities. Treatment removes sensitive cells and selects resistant populations, while new mutations and non-genetic plasticity continue evolution.

Inherited pathogenic variants can confer predisposition, but most cancers arise from acquired alterations. A germline defect exists throughout the body and has implications for relatives, surveillance, and therapy. Tumour-only sequencing can reveal a possible inherited variant that requires confirmatory counselling and testing.

Hallmarks of malignancy

Cancer cells sustain proliferative signals, evade growth suppression, resist death, maintain replicative capacity, reprogramme metabolism, induce vessels, invade, metastasise, and escape immunity. Genomic instability and tumour-promoting inflammation enable these traits. No one hallmark is unique to cancer; malignancy emerges from their combination and persistence.

Growth signalling may become autonomous through receptor, kinase, transcription-factor, or cell-cycle alterations. Loss of checkpoints allows damaged cells to divide. Telomere maintenance prevents crisis. Apoptotic pathways are disabled, but other death and stress pathways remain therapeutic targets. Metabolism shifts to support biomass, redox balance, and survival in hypoxia rather than merely maximising energy efficiency.

Carcinogens and prevention

Carcinogens directly damage DNA or require metabolic activation. Tobacco causes many cancers through mutagens and inflammation. Ultraviolet radiation creates characteristic lesions; ionising radiation causes breaks. Dose, tissue, repair, age, and latency influence risk.

Oncogenic infections act through microbial proteins, inflammation, immune suppression, or regeneration. Examples include human papillomavirus, hepatitis viruses, Epstein-Barr virus, *Helicobacter pylori*, human T-cell leukaemia virus, and parasites. Vaccination, antimicrobial treatment, tobacco cessation, alcohol reduction, healthy weight, activity, and occupational control prevent cancer.

Risk is probabilistic. Exposure does not guarantee cancer, and absence of known exposure does not imply fault. Prevention messages should avoid blame and account for social, commercial, and environmental determinants.

Tumour microenvironment and angiogenesis

Cancer cells interact with fibroblasts, immune cells, vessels, nerves, extracellular matrix, adipocytes, and microbes. Stromal cells provide growth factors, remodel matrix, suppress immunity, or restrain tumour depending on context. Hypoxia stabilises transcriptional programmes that support angiogenesis, glycolysis, invasion, and treatment resistance.

New vessels are structurally abnormal, leaky, heterogeneous, and poorly organised. They deliver resources yet create uneven perfusion. Anti-angiogenic therapy can inhibit growth but also cause hypertension, thrombosis, bleeding, poor wound healing, proteinuria, and rare perforation. Tumours can recruit alternative vascular pathways or invade along existing vessels.

Invasion and metastasis

Carcinomas breach basement membrane and invade stroma by altering adhesion, polarity, cytoskeleton, and matrix degradation. Epithelial-mesenchymal programmes can increase movement and plasticity, but metastasis is not one irreversible switch. Tumour cells enter lymphatic or blood vessels, survive shear and immunity, arrest in distant capillaries, exit, and adapt to a foreign niche.

Most disseminated cells fail. Metastasis depends on blood flow, adhesion, chemokines, niche, and tumour-tissue compatibility. Cells can remain dormant for years before outgrowth. Lymph-node involvement reflects spread and prognosis but does not mean every distant metastasis passed sequentially through each node.

Metastatic patterns are tissue-specific: portal drainage favours liver, systemic venous drainage favours lung, and some cancers favour bone, brain, or peritoneum. Metastases can cause organ failure, fracture, compression, bleeding, thrombosis, and paraneoplastic syndromes.

Immune surveillance and escape

Mutated or aberrantly expressed proteins can generate tumour antigens. Dendritic cells present antigen, T cells kill recognised cells, and natural killer cells target stressed cells. Tumours evade through antigen loss, impaired presentation, checkpoint ligands, suppressive cytokines, regulatory cells, metabolic competition, and exclusion of lymphocytes.

Checkpoint inhibitors release inhibitory brakes on T cells and can produce durable control in selected cancers. Biomarkers such as ligand expression, mismatch-repair deficiency, mutation burden, or specific viral association enrich probability but are imperfect. Immune activation can inflame almost any

organ, causing dermatitis, colitis, hepatitis, pneumonitis, endocrinopathy, myocarditis, nephritis, or neurological disease. Early recognition and graded immunosuppression are crucial.

Naming, grading, and staging

Benign tumours remain local but can still secrete hormones, bleed, obstruct, or compress. Malignant epithelial tumours are carcinomas; mesenchymal tumours are sarcomas; haematological tumours use lineage-specific names. Dysplasia is disordered pre-invasive growth; carcinoma in situ has malignant cytology without basement-membrane invasion.

Grade describes differentiation, mitoses, architecture, and sometimes necrosis, approximating biological aggressiveness. Stage describes anatomical extent through primary tumour, regional nodes, and metastasis or disease-specific systems. Stage often dominates prognosis, but molecular subtype, response, host factors, and treatment access also matter.

Diagnosis and biomarkers

Imaging identifies extent, but tissue usually establishes diagnosis. Biopsy must preserve surgical planes, yield enough tissue, and minimise complication. Histology, immunohistochemistry, flow cytometry, cytogenetics, and sequencing determine lineage and targets. Liquid biopsy detects circulating tumour material but has limitations.

Predictive biomarkers estimate likelihood of response; prognostic markers estimate outcome independent of a particular treatment; monitoring markers track burden. A target may be present without being biologically essential, and absence in one sample may miss heterogeneity. Tumour markers rarely diagnose cancer alone because benign disease can raise them and some cancers do not secrete them.

Surgery and radiation

Surgery can diagnose, stage, cure local disease, reduce bulk, prevent complications, reconstruct, or palliate. Oncological surgery aims for appropriate margins and nodal assessment while preserving function. Neoadjuvant treatment precedes surgery to shrink or test biology; adjuvant treatment follows to eradicate microscopic disease.

Radiation damages DNA directly and through radicals. Fractionation allows normal tissue repair while repeatedly injuring tumour. External-beam, stereotactic, brachytherapy, and systemic radionuclide techniques differ in geometry and indication. Toxicity depends on dose, fraction, volume, organ, and concurrent drugs; effects can be acute inflammatory or late fibrotic, vascular, endocrine, cognitive, or malignant.

Cytotoxic and targeted systemic therapy

Cytotoxic drugs damage DNA, inhibit nucleotide synthesis, disrupt mitosis, or inhibit topoisomerases. They preferentially affect dividing cells but also injure marrow, mucosa, follicles, gonads, nerves, heart, kidney, lung, or bladder according to agent. Combination regimens attack different pathways and resistance mechanisms but accumulate toxicity.

Targeted agents inhibit specific receptors, kinases, repair pathways, epigenetic regulators, or tumour dependencies. Monoclonal antibodies can block signals, recruit immunity, deliver toxins, or redirect T cells. Antibody-drug conjugates link targeting to a cytotoxic payload. Endocrine therapy deprives hormone-dependent breast or prostate cancers of growth signals.

Targeted does not mean harmless or universally effective. Tumours resist through target mutation, amplification, bypass signalling, lineage change, altered drug transport, or protective niches. Repeat tissue or circulating DNA can reveal resistance and guide subsequent therapy.

Cellular and transplant therapies

Stem-cell transplantation permits marrow replacement after intensive therapy and can provide donor immune attack. Risks include infection, organ toxicity, graft failure, graft-versus-host disease, infertility, and secondary cancer. Engineered immune cells target tumour antigens but can cause cytokine-release syndrome, neurological toxicity, cytopenia, infection, and target-cell depletion, requiring specialised monitoring.

Supportive care and emergencies

Supportive oncology prevents and treats nausea, pain, mucositis, constipation, diarrhoea, infection, thrombosis, anaemia, bone loss, neuropathy, cachexia, fatigue, and psychological distress. Fertility preservation and pregnancy discussion should occur before gonadotoxic therapy when time permits. Palliative care improves symptom control and decision quality alongside disease-directed treatment.

Oncological emergencies include febrile neutropenia, tumour lysis, hypercalcaemia, spinal-cord compression, superior vena-cava obstruction, malignant pericardial effusion, airway compromise, raised intracranial pressure, pathological fracture, haemorrhage, and immune-therapy toxicity. Treat physiological threat immediately while confirming cause.

Screening, surveillance, and survivorship

Screening tests asymptomatic people and is justified when benefit exceeds false positives, overdiagnosis, harm, and cost. Diagnostic testing investigates symptoms. Eligibility depends on cancer, age, anatomy, exposure, family history, and evidence.

Surveillance seeks recurrence, second cancers, and late effects while supporting function. Survivors may face cardiac, pulmonary, endocrine, neurological, sexual, fertility, bone, cognitive, and psychological consequences. Plans document treatment, risks, warning symptoms, prevention, and follow-up responsibility.

TTS module 2: Tumour evolution, biomarker reasoning, treatment resistance, and oncological emergencies

Cancer is an evolving ecosystem rather than a uniform mass. A founding altered cell expands, but its descendants acquire genetic mutations, chromosomal changes, epigenetic states, and reversible phenotypic adaptations. Nutrient limitation, hypoxia, immunity, anatomy, and treatment select among these variants. A biopsy therefore samples one region at one time. The dominant clone in a primary tumour may differ from a metastasis or later relapse, explaining why repeated tissue or circulating tumour analysis can change treatment.

Oncogenic drivers create dependencies by activating growth receptors, intracellular kinases, transcription factors, or cell-cycle machinery. Tumour suppressors normally restrain division, repair damage, regulate contact and differentiation, or trigger senescence and death. Loss of DNA-repair pathways does more than add mutations: it produces characteristic patterns such as mismatch-repair instability or defective homologous recombination that may predict therapy. Yet a mutation labelled pathogenic is not automatically the active driver in that tumour. Allele fraction, copy number, expression, clonality, and biological context determine significance.

Cancer cells tolerate stresses that stop normal proliferation. They maintain telomeres, evade checkpoints, resist apoptosis, and remodel metabolism to supply nucleotides, lipids, amino acids, reducing power, and energy. Aerobic glycolysis allows carbon intermediates to be diverted into biomass even when mitochondria remain functional. Oncogenic signalling can create dependence on a nutrient or repair pathway, providing a therapeutic window. Normal proliferating tissues share many requirements, however, which explains marrow, mucosal, hair, gonadal, and immune toxicity from cytotoxic treatment.

The microenvironment includes fibroblasts, endothelial cells, pericytes, extracellular matrix, nerves, adipocytes, macrophages, lymphocytes, and microbes. Cancer-associated fibroblasts can stiffen matrix, release growth factors, and restrict drug or lymphocyte access. Tumour-associated macrophages may kill malignant cells or support angiogenesis and immune suppression according to signals. Hypoxia selects cells capable of surviving acidosis and low nutrient supply and activates angiogenic programmes.

Abnormal vessels leak, shunt, collapse under pressure, and deliver drugs unevenly, leaving sanctuaries for resistant cells.

Invasion requires loss or reorganisation of epithelial polarity and adhesion, motility, matrix interaction, and survival outside the native compartment. Tumour cells cross basement membrane, enter lymphatic or blood vessels, withstand shear and immune attack, lodge in distant microvessels, exit, and adapt to a foreign organ. Each step is inefficient; many disseminated cells die or remain dormant. A supportive pre-metastatic niche may be prepared by tumour-derived factors and bone-marrow cells before arrival. Organ tropism reflects circulation and molecular compatibility, explaining recurrent patterns beyond simple blood flow.

Metastatic dormancy makes cure and surveillance biologically difficult. Disseminated cells can remain non-dividing or balanced by immune and vascular restraint for years before reactivation. Surgery can remove all visible disease without proving that microscopic cells are absent, which is the rationale for adjuvant systemic treatment when recurrence risk justifies toxicity. Neoadjuvant treatment can shrink disease, improve operability, and reveal sensitivity through pathological response, but progression during therapy exposes aggressive resistance early.

The immune system both constrains and shapes cancer. T cells recognise peptides presented by major histocompatibility molecules; natural killer cells respond to stress and deficient self-recognition; antibodies and phagocytes contribute. Selection favours tumour cells that lose antigen, reduce presentation, express inhibitory ligands, exclude lymphocytes, recruit regulatory cells, or consume critical metabolites. Checkpoint blockade restores activity across many existing tumour-reactive T cells rather than directly poisoning malignant cells. Durable responses are possible, but only where antigenicity, presentation, trafficking, and effector function remain sufficient.

Immune-related adverse events arise when released immunity attacks normal organs. They may occur during treatment or after it stops and can affect skin, bowel, liver, lung, pituitary, thyroid, adrenal, heart, kidney, nerve, muscle, blood, and eye. New diarrhoea, dyspnoea, weakness, chest pain, headache, visual change, or endocrine collapse requires prompt grading and exclusion of infection, progression, and other drugs. Corticosteroids and additional

immunosuppression are matched to severity. Hormone deficiency may be permanent even after inflammation resolves.

Diagnosis begins with obtaining representative tissue safely. Biopsy route should not contaminate future surgical planes, fracture a vulnerable bone, seed a compartment, or expose a vascular tumour. Core tissue preserves architecture and permits histology, immunohistochemistry, and molecular testing; cytology can be adequate for selected sites but may exhaust material. Fixation and decalcification can impair nucleic acid or antigen assays. A multidisciplinary plan is valuable when tissue is scarce and several predictive tests compete for it.

Grade estimates microscopic aggressiveness from differentiation, mitotic activity, architecture, and necrosis. Stage estimates anatomical extent. Neither is interchangeable with molecular subtype. A small early-stage tumour may have high-risk biology, while extensive indolent disease may evolve slowly. Prognostic markers estimate outcome under a defined context; predictive markers estimate differential benefit from a treatment. A marker can be both. Apparent association does not prove that targeting the marker improves outcome, so validated companion tests and clinical evidence matter.

Imaging measures anatomy and, in selected modalities, metabolism or receptor expression. Treatment can produce necrosis, inflammation, fibrosis, oedema, or immune-cell infiltration that temporarily enlarges a lesion without viable progression. Conversely, stable size may hide biological response. Response criteria standardise measurement but do not replace symptoms and organ function. Tumour markers can follow a known secreting cancer, yet changes may reflect benign inflammation, impaired clearance, or assay variation. Screening a low-risk population with nonspecific markers creates false positives and overdiagnosis.

Surgery cures when all clonogenic disease can be removed with acceptable function. Margin requirements depend on tumour biology and anatomy rather than a universal distance. Regional-node assessment provides staging and sometimes control but can cause lymphoedema and nerve injury. Radiation deposits energy that creates DNA damage; fractionation exploits differences in repair, redistribution, reoxygenation, and repopulation. Highly conformal techniques spare surrounding tissue but cannot eliminate toxicity when critical

organs overlap the target. Late fibrosis, vascular injury, endocrine failure, cognitive effects, and second malignancy may appear years later.

Cytotoxic combinations attack different cellular processes and reduce the probability that one resistant clone survives, but overlapping marrow, nerve, heart, kidney, lung, and mucosal injury constrains dose. Dose intensity can influence cure in sensitive cancers; delays or reductions should be deliberate rather than automatic. Growth-factor support and antiemetics preserve delivery but also carry harms. Extravasation of a vesicant can destroy tissue and requires immediate drug-specific management. Cumulative anthracycline cardiac injury, platinum kidney and nerve injury, and alkylator infertility or secondary malignancy shape long-term planning.

Targeted therapy works when the inhibited alteration is both present and necessary. Resistance may exist before treatment in a minor subclone or arise through target mutation, amplification, pathway bypass, histological transformation, altered transport, or protective stroma. Combination can suppress several escape routes but increases toxicity. A later biopsy can reveal that the original target was lost or that a new actionable dependency emerged. “Targeted” describes selection, not harmlessness: kinase inhibitors can cause pressure, vascular, cardiac, pulmonary, skin, endocrine, liver, and metabolic injury.

Endocrine treatment withdraws proliferative signals from hormone-dependent breast, prostate, and selected other tumours. Oestrogen blockade can cause vasomotor symptoms, bone loss, thrombosis, uterine effects, sexual dysfunction, and cardiovascular or metabolic change depending on class. Androgen deprivation causes hot flushes, muscle and bone loss, insulin resistance, fatigue, mood and sexual effects. Exercise, bone assessment, cardiovascular prevention, and symptom care are part of anticancer treatment because long survival magnifies chronic toxicity.

Oncological emergencies demand physiological treatment before complete histology. Spinal-cord compression causes back pain, weakness, sensory change, and sphincter dysfunction; urgent magnetic resonance, corticosteroid in appropriate malignant compression, and radiation or surgery preserve function. Hypercalcaemia causes dehydration, kidney injury, constipation, confusion, and arrhythmia; restore volume carefully, inhibit bone resorption, and treat cancer. Superior vena-cava obstruction produces facial and

upper-limb swelling, venous distension, dyspnoea, and cerebral or airway symptoms; urgency depends on physiological compromise, and tissue diagnosis should be secured when stable.

Other emergencies include febrile neutropenia, tumour lysis, malignant pericardial tamponade, airway obstruction, raised intracranial pressure, pathological fracture, thrombosis, haemorrhage, and immune toxicity. Palliative care participates from diagnosis when symptom or decision burden is high; it is not synonymous with stopping treatment. Pain control, nutrition, rehabilitation, fertility, sexuality, sleep, cognition, family support, and advance planning preserve the person receiving therapy.

Screening is justified only when earlier detection improves meaningful outcomes enough to offset false positives, invasive follow-up, treatment toxicity, anxiety, inequity, and overdiagnosis of disease that would never become symptomatic. Survivorship likewise requires more than recurrence scans. A treatment summary should identify cardiac, endocrine, bone, neurological, fertility, infection, and second-cancer risks and assign follow-up responsibility. Cancer medicine is longitudinal evolutionary management: repeatedly match tumour biology, anatomical burden, host reserve, and the patient's goals as all four change.

Retrieval prompts

One. How do driver and passenger mutations differ?

Two. Which steps must a cell complete to form a metastasis?

Three. How do grade and stage differ?

Four. What distinguishes predictive from prognostic biomarkers?

Five. Why do targeted and immune therapies still cause major toxicity?

Six. What trade-offs determine whether screening is beneficial?

Concise answers

One. Drivers confer selected biological advantage; passengers accumulate without materially driving the clone.

Two. Invasion, entry into vessels, survival, arrest, exit, niche adaptation, dormancy escape, and outgrowth.

Three. Grade estimates microscopic aggressiveness; stage measures anatomical extent.

Four. Predictive markers estimate treatment response, while prognostic markers estimate outcome independent of that treatment.

Five. Targets exist in normal physiology, immune activation damages healthy organs, and delivery or class effects create off-target injury.

Six. Mortality or morbidity reduction must outweigh false positives, overdiagnosis, treatment harm, anxiety, and cost.

Source map

Original synthesis informed principally by Robbins, neoplasia, invasion, metastasis, tumour immunity, and organ cancers; Katzung and OpenStax Pharmacology, cytotoxic, targeted, endocrine, and immune therapy; Guyton and Hall, tissue and organ physiology; OpenStax Biology and Chemistry, genetics and cell signalling; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, cancer examination and systemic assessment.

Chapter 41: Multisystem Autoimmune Disease and Vasculitis

Orientation

Autoimmune disease develops when immune tolerance fails and adaptive responses injure self tissues. The same diagnosis can affect different organs, and similar syndromes can arise from infection, malignancy, drugs, or inherited immune dysregulation. Diagnosis rests on phenotype and objective organ involvement; autoantibodies modify probability but do not replace clinical reasoning. Treatment balances suppression of damaging immunity against infection, malignancy, metabolic toxicity, and loss of vaccine response.

Tolerance and loss of self-control

Central tolerance deletes or edits strongly self-reactive lymphocytes during development. Peripheral tolerance restrains escaped cells through anergy, regulatory T cells, inhibitory receptors, antigen sequestration, and absence of costimulation. Clearance of apoptotic material limits exposure of nuclear antigens. Genetic variants in antigen presentation, signalling, complement, interferon, and regulation alter susceptibility.

Environmental triggers include infection, ultraviolet radiation, smoking, silica, drugs, hormones, and tissue injury. Molecular mimicry, epitope spreading, bystander activation, and modified self proteins can expand responses. Most exposed individuals do not develop disease; autoimmunity reflects interaction among inherited risk, immune history, sex and age, environment, and chance.

Damage mechanisms include autoantibodies that block or stimulate receptors, antibodies against cells or matrix, immune-complex deposition, complement activation, cytotoxic T cells, macrophage activation, granulomas, thrombosis, and fibrosis. A patient may have several mechanisms simultaneously.

Interpreting autoantibodies

Test only when pretest probability justifies it. Antinuclear antibodies occur in many connective-tissue diseases but also in healthy people, infection, drugs, cancer, and relatives. Titre and staining pattern can guide follow-up, yet neither establishes a diagnosis. Broad repeated panels generate incidental positives.

Disease-associated antibodies differ in sensitivity, specificity, organ correlation, and relationship to activity. Anti-double-stranded-DNA titres and complement may track lupus kidney activity in some patients; anti-Ro antibodies have pregnancy implications; antiphospholipid antibodies require persistent confirmation and a compatible clinical event. Antineutrophil-cytoplasmic antibodies support selected small-vessel vasculitis but occur in mimics.

Inflammatory markers are nonspecific and can be normal during serious organ disease. Complement consumption, blood counts, urine sediment, protein excretion, kidney function, muscle enzymes, imaging, pulmonary testing, nailfold vessels, and biopsy often provide stronger evidence of damage.

Systemic lupus erythematosus

Lupus features autoantibodies, immune complexes, complement activation, interferon signalling, and fluctuating inflammation. Manifestations include photosensitive rash, oral or nasal ulcers, non-erosive inflammatory arthritis, cytopenias, serositis, kidney disease, neurological syndromes, thrombosis, fever, fatigue, and vascular risk. Hair loss and Raynaud phenomenon are common but nonspecific.

Kidney involvement ranges from mild urinary abnormalities to proliferative or membranous glomerular disease. Regular pressure, creatinine, urine sediment, and protein measurement detect silent nephritis. Biopsy classifies mechanism and activity when treatment decisions require it. Neurological symptoms demand careful exclusion of infection, thrombosis, metabolic disturbance, and primary psychiatric disease.

Hydroxychloroquine reduces flares and improves outcomes for most suitable patients but requires weight-based dosing and retinal surveillance.

Glucocorticoids control acute inflammation but cause infection, diabetes, osteoporosis, cardiovascular disease, cataracts, and adrenal suppression. Immunosuppressive selection follows organ threat, fertility, pregnancy, comorbidity, and patient preference. Sun protection, vaccination, cardiovascular prevention, bone health, and pregnancy planning are core care.

Antiphospholipid syndrome

Antiphospholipid syndrome combines persistently positive defined antibodies with venous, arterial, or microvascular thrombosis or characteristic pregnancy morbidity. Antibodies alone do not establish the syndrome. Tests are affected by anticoagulants and acute illness, so timing and repetition matter.

Clinical patterns include deep-vein thrombosis, stroke, placental disease, thrombocytopenia, livedo, valve lesions, and kidney microangiopathy. Catastrophic disease causes rapid multiorgan small-vessel thrombosis and requires urgent anticoagulation, immunomodulation, and trigger treatment. Long-term antithrombotic strategy depends on event type, antibody profile, bleeding risk, pregnancy, and recurrence.

Systemic sclerosis

Systemic sclerosis combines vasculopathy, immune activation, and progressive fibrosis. Raynaud phenomenon and swollen fingers may precede skin thickening. Limited and diffuse cutaneous patterns carry different probabilities but both can affect oesophagus, lung, pulmonary vessels, heart, kidney, gut, joints, and digital circulation.

Interstitial lung disease and pulmonary arterial hypertension are major causes of death and require systematic screening. Scleroderma renal crisis causes abrupt hypertension and kidney injury and is treated urgently with angiotensin-converting-enzyme inhibition. High-dose glucocorticoids increase crisis risk. Digital ischaemia requires warmth, smoking cessation, vasodilator therapy, wound care, and urgent action for threatened tissue.

Treatment targets organ mechanisms rather than fibrosis as one entity: acid suppression and motility care, vasodilators, immune therapy for inflammatory lung or skin disease, pulmonary-hypertension treatment, rehabilitation, and

nutrition. Blood-pressure self-monitoring can detect renal crisis early in high-risk patients.

Sjögren disease

Sjögren disease targets exocrine glands but can affect nerves, lung, kidney, vessels, joints, skin, and blood. Dry eyes cause grittiness and corneal injury; dry mouth causes swallowing difficulty, dental decay, candidiasis, and gland enlargement. Medicines, dehydration, diabetes, radiation, hepatitis, human immunodeficiency virus, and ageing can mimic dryness.

Assessment combines symptoms with tear or salivary function, ocular staining, antibodies, ultrasound, or minor salivary-gland biopsy when needed. Management uses tear and saliva substitutes, stimulation where appropriate, meticulous dental prevention, topical therapy, and systemic immunomodulation for organ disease. Persistent gland enlargement, nodes, purpura, low complement, or cryoglobulins raise lymphoma concern.

Inflammatory myopathy

Immune-mediated myopathies cause progressive proximal weakness, difficulty rising or lifting, neck weakness, dysphagia, respiratory failure, or cardiac involvement. Dermatomyositis adds characteristic skin findings. Antisynthetase syndromes combine myositis with interstitial lung disease, arthritis, fever, Raynaud phenomenon, and roughened hands. Inclusion-body myositis often affects finger flexors and quadriceps asymmetrically and responds poorly to conventional immunosuppression.

Creatine kinase, electromyography, magnetic resonance, antibodies, and biopsy help classify disease, but enzyme levels can be modest. Drugs, endocrine disease, inherited myopathy, infection, motor-neuron disease, and deconditioning are alternatives. Treatment combines immune therapy where responsive, malignancy screening in appropriate phenotypes, swallowing and respiratory assessment, and graded rehabilitation.

Vasculitis framework

Vasculitis inflames vessel walls, causing stenosis, occlusion, aneurysm, rupture, and downstream ischaemia. Classify by predominant vessel size, organ pattern, histology, antibodies, and cause. Infection, drugs, malignancy, emboli, cholesterol, thrombosis, and vasospasm can mimic vasculitis. Immunosuppression before excluding infection can be catastrophic.

Large-vessel vasculitis includes giant-cell arteritis and Takayasu arteritis. Giant-cell arteritis affects older adults and can cause new headache, scalp tenderness, jaw or tongue claudication, visual symptoms, polymyalgia, constitutional illness, and aortic disease. Threatened vision requires immediate glucocorticoids before definitive imaging or biopsy. Takayasu disease affects the aorta and branches, causing limb claudication, pulse or pressure asymmetry, bruits, hypertension, and organ ischaemia.

Medium-vessel disease causes organ infarction, aneurysm, skin nodules, livedo, neuropathy, abdominal pain, and kidney ischaemia without primary glomerulonephritis. Polyarteritis nodosa may relate to hepatitis B. Kawasaki disease in children causes mucocutaneous inflammation and coronary aneurysm risk.

Small-vessel vasculitis produces palpable purpura, glomerulonephritis, pulmonary haemorrhage, neuropathy, gut disease, and skin injury. Antineutrophil-cytoplasmic-antibody-associated syndromes differ in granulomatous airway disease, eosinophilia, asthma, and organ preference. Immune-complex vasculitis includes immunoglobulin-A, cryoglobulinaemic, lupus, infection-related, and drug-related forms.

Organ-threatening presentations

Pulmonary-kidney syndrome combines alveolar haemorrhage and rapidly progressive glomerulonephritis. Patients may have breathlessness, falling haemoglobin, diffuse lung opacities, blood and protein in urine, and rapid kidney loss; haemoptysis can be absent. Urgent serology, microbiology, imaging, bronchoscopy or biopsy, and specialist treatment are required.

Mononeuritis multiplex causes painful asymmetric deficits from nerve-vessel ischaemia. Mesenteric vasculitis causes severe abdominal pain, bleeding,

perforation, or infarction. Retinal or cerebral ischaemia, myocarditis, digital gangrene, and rapidly progressive kidney disease demand immediate action.

Treatment strategy and safety

Separate induction of remission from maintenance. Organ-threatening disease may require high-dose glucocorticoids with cyclophosphamide, B-cell depletion, or other targeted therapy. Plasma exchange has selected indications rather than universal benefit. Less severe disease can use lower-toxicity conventional or biological agents.

Before treatment, screen relevant hepatitis, tuberculosis, human immunodeficiency virus, pregnancy, vaccination, blood counts, organ function, malignancy risk, and drug interactions. Provide prophylaxis against opportunistic infection and bone loss when indicated. Monitor disease activity and treatment toxicity separately; normal inflammatory markers do not prove remission.

Fever, cytopenia, lung opacity, or kidney injury during therapy may be infection, drug toxicity, thrombotic microangiopathy, or disease flare. Obtain cultures and targeted diagnostics before reflexively escalating immunosuppression when the patient is stable enough. Shared decisions address fertility, contraception, vaccination, surgery, travel, cost, monitoring, and long-term cumulative toxicity.

TTS module 2: Autoimmune probability, vascular injury patterns, and safe immune suppression

Autoimmune medicine is a problem of causal coherence. Fatigue, pain, rash, a positive antibody, and an elevated inflammatory marker are common individually and do not necessarily share one mechanism. A convincing syndrome links specific clinical features to objective tissue injury, a plausible immune pathway, and a compatible time course while actively testing infection, malignancy, drugs, vascular disease, and inherited disorders. The purpose of an autoantibody is to modify probability or define phenotype, not to replace this synthesis.

Self-tolerance is multilayered because no single checkpoint is perfect. Developing lymphocytes with strong self-reactivity are deleted, edited, or

redirected, but some escape. Peripheral restraint then depends on absent costimulation, anergy, regulatory T cells, inhibitory receptors, controlled antigen presentation, and removal of apoptotic debris. Genetic susceptibility may alter several weak checkpoints rather than one deterministic gene. Infection or tissue injury can supply inflammatory costimulation, expose hidden antigens, modify proteins, or activate cross-reactive cells. Disease emerges only when these influences converge in a permissive host.

Autoimmune injury occurs through different effectors. Antibodies may stimulate a receptor, block it, destroy a cell, or bind structural tissue. Soluble antigen–antibody complexes deposit in vessels, kidney, skin, and joints, activating complement and leukocytes. T cells directly kill cells or organise macrophage and granulomatous inflammation. Cytokines sustain synovitis or fibrosis, while antiphospholipid antibodies promote thrombosis rather than classic vessel-wall inflammation. Naming the mechanism predicts which laboratory, biopsy, and treatment signals are meaningful.

Testing should follow a defined clinical question. Antinuclear antibody is sensitive for lupus but has limited positive predictive value in people without compatible features. Repeating it rarely measures disease activity. Antineutrophil cytoplasmic antibodies help classify selected small-vessel vasculitis, yet infection, inflammatory bowel disease, drugs, and other autoimmune conditions can produce positivity. Rheumatoid factor occurs in rheumatoid arthritis, chronic infection, age, and other immune disease. A negative test cannot exclude every seronegative phenotype, and laboratory reference ranges do not replace pretest probability.

Objective organ surveillance detects silent threat. In lupus, serial blood pressure, creatinine, urine protein, and sediment can reveal nephritis before symptoms. Haematuria with dysmorphic red cells or casts suggests glomerular inflammation, while protein quantity helps assess severity. Falling complement and rising double-stranded-DNA antibody can support activity in some patients but should not overrule a stable kidney phenotype. Kidney biopsy identifies class, activity, chronic scar, and alternative pathology, allowing treatment intensity to match salvageable inflammation rather than serum markers alone.

Neuropsychiatric symptoms in systemic disease require especially sceptical attribution. Headache, depression, cognitive change, seizure, psychosis,

neuropathy, or stroke may arise from inflammation, thrombosis, infection, hypertension, kidney failure, corticosteroids, primary psychiatric illness, or unrelated neurological disease. Cerebrospinal fluid, imaging, electroencephalography, vascular studies, antibody context, and treatment timing are selected from the syndrome. Labelling every symptom "lupus cerebritis" risks both missing infection and exposing the patient to unnecessary immunosuppression.

Antiphospholipid syndrome is defined by events plus persistent laboratory evidence. Lupus anticoagulant paradoxically prolongs phospholipid-dependent clotting assays while increasing thrombosis risk in vivo. Acute thrombosis, inflammation, pregnancy, and anticoagulants can distort tests, so confirmation timing matters. Arterial, venous, placental, and microvascular events have different recurrence and treatment implications. Catastrophic antiphospholipid syndrome causes rapidly progressive multiorgan thrombosis and is treated with urgent anticoagulation, suppression of the inflammatory trigger, plasma exchange or immunoglobulin in selected protocols, and treatment of infection or other precipitant.

Systemic sclerosis should be conceptualised as interacting vasculopathy, autoimmunity, and fibrosis. Raynaud phenomenon is episodic digital vasospasm, but persistent ulcers, pitting scars, capillary loss, or gangrene signal structural vascular disease. Nailfold capillaroscopy can distinguish a scleroderma pattern from uncomplicated primary Raynaud. Breathlessness may arise from interstitial lung disease, pulmonary arterial hypertension, cardiac dysfunction, anaemia, deconditioning, aspiration, or infection. Pulmonary function trends, echocardiography, high-resolution imaging, biomarkers, and right-heart catheterisation answer different parts of this differential.

Scleroderma renal crisis often presents with abrupt pressure rise, acute kidney injury, headache, visual change, encephalopathy, heart failure, or microangiopathic haemolysis. Angiotensin-converting-enzyme inhibition is started promptly and continued despite an early creatinine rise when the diagnosis fits, because renin-driven vascular injury otherwise progresses. Normal pressure does not entirely exclude crisis in a person whose usual pressure is low. High-dose glucocorticoids are avoided where possible

because they increase risk. This emergency differs from immune-complex nephritis and is not treated by simply escalating steroids.

Sjögren disease demonstrates how chronic local dysfunction becomes systemic. Loss of tears damages corneal epithelium; loss of saliva impairs buffering, lubrication, swallowing, antimicrobial defence, and dental remineralisation. Objective tear and salivary tests separate true gland failure from subjective dryness caused by anticholinergic drugs, dehydration, mouth breathing, anxiety, diabetes, or menopause. Peripheral neuropathy, interstitial lung disease, renal tubular acidosis, vasculitic purpura, cryoglobulinaemia, low complement, persistent salivary enlargement, and lymphadenopathy suggest systemic activity or lymphoma risk.

Inflammatory myopathy is recognised through loss of power, not creatine kinase alone. Weak neck flexion, shoulder and hip movement, finger flexion, swallowing, cough, and respiration should be measured serially. Dermatomyositis skin signs may be subtle across pigmentation and can precede muscle disease. Antisynthetase syndromes frequently make lung disease the dominant threat. Inclusion-body myositis progresses slowly with disproportionate finger-flexor and quadriceps weakness and usually resists conventional immune therapy, making correct classification essential. Magnetic resonance guides biopsy toward active, not end-stage, muscle.

Vasculitis is localised through vessel calibre and downstream organ pattern. Large-vessel inflammation produces pulse deficits, bruits, limb claudication, aortic pain or dilatation, renovascular hypertension, and branch-organ ischaemia. Medium-vessel injury causes aneurysm, infarction, skin nodules, livedo, abdominal catastrophe, and mononeuritis without primary glomerulonephritis. Small-vessel disease reaches capillaries, venules, and arterioles, producing palpable purpura, glomerular haematuria, pulmonary capillaritis, and peripheral-nerve ischaemia. Overlap occurs, but this framework directs imaging versus tissue biopsy.

Giant-cell arteritis is treated immediately when visual ischaemia is threatened. New headache, scalp tenderness, jaw or tongue claudication, transient visual symptoms, polymyalgia, constitutional inflammation, thrombocytosis, or an abnormal temporal artery increase probability. Normal inflammatory markers reduce but do not eliminate risk. Ultrasound, arterial imaging, and temporal-artery biopsy retain diagnostic value after treatment for a period and

should be arranged without delaying glucocorticoids. Long-term care monitors aortic complications and cumulative steroid toxicity.

Pulmonary–kidney syndrome is a time-critical pattern. Alveolar haemorrhage can present with dyspnoea, falling haemoglobin, diffuse opacities, and hypoxaemia without haemoptysis. Glomerulonephritis produces blood, protein, casts, and rapidly declining filtration. Antibody-associated vasculitis, anti-basement-membrane disease, lupus, infection, cardiac disease, drugs, and coagulopathy are alternatives. Cultures, serology, imaging, bronchoscopy, and kidney biopsy are coordinated while respiratory and renal threats are supported. Plasma exchange benefits selected mechanisms and severities, not all antibody-associated disease automatically.

Induction therapy aims to stop organ-threatening inflammation quickly; maintenance prevents recurrence with less cumulative toxicity. High-dose glucocorticoids are effective but begin causing metabolic, psychiatric, infectious, bone, eye, muscle, and adrenal harm immediately. Cyclophosphamide threatens marrow, bladder, fertility, and later malignancy. B-cell depletion, antimetabolites, calcineurin inhibitors, and pathway-targeted agents carry distinct infection and organ risks. Choice should be tied to the endangered organ, expected speed of response, previous therapy, fertility, pregnancy, comorbidity, and patient preference.

Before major immune suppression, document baseline infection symptoms, vaccination, tuberculosis and hepatitis risk, blood counts, kidney and liver function, immunoglobulins where relevant, pregnancy plans, and prior malignancy. Provide prophylaxis against *Pneumocystis*, viral reactivation, osteoporosis, gastric complications, or thrombosis when the regimen and host justify it. Live vaccines may become unsafe after treatment begins, while inactive vaccines may produce weaker responses. Exposure history for *Strongyloides* matters before high-dose corticosteroids because hyperinfection can be fatal.

During treatment, disease activity and toxicity are two separate axes. Fever, lung opacity, cytopenia, liver injury, confusion, or kidney decline may be flare, infection, thrombosis, malignancy, or drug toxicity. Escalating immune suppression without renewed specimens and anatomy can worsen the true cause. Conversely, waiting for perfect certainty during visual loss, alveolar haemorrhage, renal crisis, or mesenteric ischaemia can cause irreversible

damage. Expert care makes this trade-off explicit, acts proportionately to organ threat, and revises the diagnosis as objective response and new evidence arrive.

Retrieval prompts

One. Which mechanisms maintain immune tolerance?

Two. Why can a positive autoantibody fail to establish disease?

Three. Which tests detect silent lupus nephritis?

Four. What distinguishes systemic sclerosis renal crisis?

Five. How does vessel size guide vasculitis presentation?

Six. Why must infection be excluded during apparent autoimmune flare?

Concise answers

One. Central deletion or editing, anergy, regulatory cells, inhibitory receptors, antigen sequestration, and lack of costimulation.

Two. Autoantibodies occur in healthy people and many unrelated conditions; probability depends on phenotype, titre, specificity, and objective injury.

Three. Blood pressure, creatinine, urine sediment, and quantified protein excretion, followed by biopsy when indicated.

Four. Abrupt severe hypertension with kidney injury in a susceptible systemic-sclerosis patient.

Five. Large vessels cause pulse and branch-organ syndromes; medium vessels cause infarcts and aneurysms; small vessels cause purpura, glomerulonephritis, and alveolar haemorrhage.

Six. Infection can mimic flare and deteriorate rapidly under additional immune suppression.

Source map

Original synthesis informed principally by Robbins, autoimmunity, immune-complex disease, connective-tissue disease, and vasculitis; Katzung and OpenStax Pharmacology, glucocorticoid and immunomodulatory therapy; Guyton and Hall, immune and organ physiology; OpenStax Biology, Microbiology, and Medical-Surgical Nursing; and Talley and O'Connor, multisystem rheumatological and vascular assessment.

Chapter 42: Paediatric and Neonatal Physiology and Assessment

Orientation

Children are not small adults. Anatomy, physiology, drug handling, disease expression, communication, and measurements change through development. Illness may progress rapidly because airway diameter, fluid, glucose, and cardiorespiratory reserve are limited. Assessment begins with appearance, breathing, circulation, interaction, and trajectory. Care is family-centred while preserving the child's voice, privacy, assent, and safety.

Transition at birth

Fetal gas exchange occurs through placenta. Pulmonary resistance is high, placental resistance is low, and blood bypasses lungs through the foramen ovale and ductus arteriosus. At birth, aeration lowers pulmonary resistance; cord clamping raises systemic resistance; pulmonary venous return raises left-atrial pressure; fetal shunts begin functional closure.

Failure of lung expansion, persistent pulmonary hypertension, congenital heart disease, prematurity, infection, hypothermia, or metabolic disturbance can disrupt transition. Some cardiac lesions depend on ductal flow and deteriorate as the duct closes. Prostaglandin infusion can preserve patency while anatomy is defined, but may cause apnoea and requires specialist support.

Newborn resuscitation prioritises warmth, positioning, drying and stimulation when appropriate, then effective ventilation if breathing or heart rate is inadequate. Heart-rate response indicates ventilation success. Compressions and medicines are secondary to lung inflation in most neonatal collapse.

Neonatal physiological reserve

Neonates lose heat rapidly through high surface-area-to-mass ratio, thin skin, limited insulation, and poor behavioural control. Brown-fat thermogenesis consumes oxygen and glucose. Hypothermia worsens respiratory distress, hypoglycaemia, and acidosis. Maintain neutral temperature without overheating.

Glucose shifts from placental transfer to feeding and endogenous production. Prematurity, growth restriction, maternal diabetes, infection, endocrine disease, and feeding difficulty increase hypoglycaemia risk. It may be silent or cause jitteriness, poor feeding, apnoea, lethargy, hypothermia, or seizure. Treatment depends on age, symptoms, and risk.

Renal filtration, concentration, water, sodium, drug, and acid handling are immature, especially in prematurity. Hepatic conjugation and metabolism are also immature. Fetal haemoglobin is initially high, followed by a physiological haemoglobin fall as erythropoietin decreases.

Neonatal jaundice and feeding

Bilirubin production is high because red-cell turnover is increased, while hepatic conjugation and enterohepatic handling are immature. Physiological jaundice appears after the first day and follows an expected pattern. Jaundice in the first day, rapid rise, prolonged elevation, pallor, illness, dark urine, pale stool, or high concentration suggests haemolysis, infection, liver disease, obstruction, enzyme deficiency, or another pathological process.

Unconjugated bilirubin can enter brain and cause acute encephalopathy or kernicterus, particularly with prematurity, haemolysis, acidosis, sepsis, or low albumin binding. Interpret concentration by postnatal age and risk. Phototherapy changes bilirubin into excretable forms; exchange transfusion is reserved for severe risk. Conjugated jaundice is never physiological.

Assess feeding by frequency, attachment, suck-swallow-breathe coordination, transfer, maternal comfort, urine, stool, weight, hydration, and infant alertness. Weight loss after birth is expected within limits, but excessive loss can cause hypernatraemic dehydration. Feeding support should identify tongue, palate, cardiac, respiratory, neurological, endocrine, infectious, and social barriers without blame.

Growth and development

Growth charts describe population trajectories for weight, length or height, head circumference, and body mass. A single centile is less informative than serial velocity and relation among measurements. Correct age for prematurity when appropriate. Faltering growth can reflect inadequate intake, feeding dysfunction, malabsorption, excess loss, increased metabolic demand, endocrine disease, chronic inflammation, neglect, or normal familial pattern.

Development spans gross motor, fine motor and vision, language and hearing, social, cognitive, and adaptive domains. Milestones are ranges, not deadlines. Regression is more concerning than isolated mild delay and may signal neurodegenerative, metabolic, epileptic, structural, or psychosocial disease. Hearing and vision impairment can mimic global delay.

Assessment combines caregiver report, observation, validated screening, school information, examination, growth, and family context. Early intervention can improve function before a final cause is known.

Age-specific physiology

Infants have compliant chest walls, fewer fatigue-resistant respiratory fibres, smaller airways, and higher oxygen consumption. Small reductions in airway radius greatly increase resistance. Respiratory distress appears as tachypnoea, recession, nasal flaring, grunting, head bobbing, poor feeding, apnoea, or exhaustion. Cyanosis is late and can be difficult to identify across pigmentation.

Heart rate and respiratory rate normally decrease with age while pressure rises. Interpret observations against age, fever, pain, crying, sleep, medication, and baseline. Cardiac output in infants depends strongly on heart rate because stroke-volume reserve is limited. Bradycardia often reflects hypoxia and is ominous.

Children have higher total-body water and proportionally larger extracellular volume when young. Vomiting, diarrhoea, fever, tachypnoea, and poor intake therefore cause rapid dehydration. They can preserve pressure through vasoconstriction until late shock; tachycardia, prolonged capillary refill, cool peripheries, altered interaction, reduced urine, and weak pulses precede hypotension.

Nutrition and immunisation

Nutrition supports brain, bone, muscle, immune, and organ development. Needs vary with gestation, growth, activity, and disease. Breast milk provides nutrition and immune factors; prepared formula is a safe alternative. Complementary foods provide iron, energy, protein, texture progression, and safe allergen exposure while avoiding choking.

Micronutrient deficiency can impair development before physical signs. Iron affects cognition and haemoglobin; vitamin D supports mineralisation; vitamin B twelve deficiency can injure the nervous system. Obesity can coexist with deficiency and requires non-stigmatising family and environmental assessment.

Vaccination creates memory before exposure. Schedule and contraindications are age- and risk-specific; mild illness rarely requires delay. Live vaccines require caution in severe immune deficiency and pregnancy. Use registries and catch-up guidance when records are uncertain.

Paediatric history and observation

First observe from a distance: colour, tone, alertness, eye contact, cry or speech, posture, spontaneous movement, respiratory effort, and interaction with caregivers. The child's ability to play, feed, smile, or resist examination provides physiological information. A quiet motionless child may be more concerning than a crying one.

History includes pregnancy, birth, gestation, neonatal care, feeding, growth, development, immunisation, exposures, medicines, family disease, school, sleep, continence, behaviour, and social environment. Ask adolescents privately about mood, safety, substances, sexual health, eating, bullying, and confidentiality.

Caregivers often detect deviation before objective signs. Clarify onset, intake, urine, breathing, pain, responsiveness, rash, fever, and trajectory. Measure medicines in milligrams and verify concentration; volume errors can be dangerous.

Examination

Sequence examination from least intrusive to most intrusive and adapt through play, caregiver holding, and explanation. Record weight for dosing. Measure head circumference in infants when relevant. Assess hydration, perfusion, fontanelle, skin, lymph nodes, ears, throat, chest, heart, abdomen, hips, spine, genital or perianal region only when indicated, and neurological development.

Use age-appropriate pain tools. Examine growth and pubertal stage respectfully with consent and chaperoning. Blood pressure cuff must fit. Oxygen saturation can be affected by movement, perfusion, probe placement, and skin pigmentation; reconcile monitor values with the child.

Common acute patterns

Bronchiolitis causes viral small-airway obstruction; judge severity by breathing, oxygenation, hydration, apnoea, and risk. Croup causes barking cough and inspiratory stridor; stridor at rest and fatigue indicate severity. Sudden cough or unilateral signs suggest aspiration. Drooling, tripod posture, toxicity, or severe obstruction requires minimal disturbance and expert airway help.

Gastroenteritis requires dehydration assessment and oral rehydration when possible. Bilious vomiting suggests obstruction. Projectile infant vomiting, currant-jelly stool, severe abdominal pain, guarding, or testicular pain requires urgent surgical assessment.

Fever risk is age-dependent. Young infants, immune-compromised children, and those with poor perfusion, altered responsiveness, breathing difficulty, non-blanching rash, neck stiffness, or focal signs need urgent care. Antipyretics improve comfort but neither prevent febrile seizures nor treat infection.

Paediatric resuscitation and medicines

Use a structured airway-breathing-circulation approach. Open airway in a neutral age-appropriate position, provide oxygen and effective ventilation, and recognise that hypoxia commonly drives arrest. Vascular or intraosseous access permits urgent treatment. Fluid boluses are weight-based and reassessed because cardiac, renal, malarial, or malnourished states alter safety.

Medication dosing uses current measured weight, indication, age, organ function, maximum adult limits, and an independently checked calculation. Decimal and unit errors are major hazards. Pharmacokinetics changes with total-body water, protein binding, maturation of metabolism and filtration, and puberty. Avoid unnecessary cough and cold products and consider excipients.

Safeguarding and communication

Consider abuse or neglect with inconsistent history, developmentally implausible injury, delay, multiple healing stages, or concerning interaction. Bruising in non-mobile infants, patterned injury, burns, fractures, intracranial or genital injury, poisoning, and faltering growth require evaluation. Consider medical mimics without delaying protection.

Document exact words, observations, images according to policy, people present, and discrepancies without accusation. Follow reporting and multidisciplinary pathways. Speak developmentally, explain confidentiality limits, seek assent, and avoid promises.

TTS module 2: Developmental physiology, neonatal transition, and recognition of paediatric deterioration

Paediatric physiology changes continuously with gestational age, postnatal age, growth, and puberty. Normal heart rate, respiratory rate, pressure, kidney clearance, body-water distribution, airway anatomy, communication, and developmental behaviour cannot be inferred from adult values. Illness also removes reserve quickly: an infant has high oxygen demand, a narrow airway, limited stroke-volume adaptation, and small glycogen and fluid stores. Assessment therefore emphasises trajectory, age-specific ranges, appearance, work of breathing, perfusion, intake, urine, and interaction before a child becomes hypotensive.

At birth, the first effective breaths replace lung liquid with air and greatly lower pulmonary vascular resistance. Increased pulmonary blood flow raises left-atrial pressure, while removal of the low-resistance placenta raises systemic resistance and right-atrial pressure falls. These changes functionally close the foramen ovale and reverse ductal flow before later anatomical closure. Oxygen tension and reduced prostaglandin exposure constrict the ductus arteriosus. Failure of aeration, acidosis, hypoxia, or pulmonary vascular disease preserves fetal shunting and can create severe cyanosis.

Neonatal resuscitation is principally respiratory because bradycardia usually follows inadequate lung aeration and hypoxia. Warmth, positioning, clearing only obstructing secretions, drying and stimulation are followed by positive-pressure ventilation when breathing or heart rate is inadequate. Visible chest movement and rising heart rate are the critical feedback that ventilation is effective. Mask seal, head position, airway patency, pressure, and alternative airway are corrected before escalating. Chest compressions are reserved for persistent severe bradycardia despite effective ventilation.

Some congenital cardiac lesions become evident only as the ductus closes. Duct-dependent pulmonary flow causes worsening cyanosis; duct-dependent systemic flow can cause shock, weak femoral pulses, acidosis, and organ injury; mixing lesions produce variable cyanosis. A newborn in shock without an obvious cause needs simultaneous consideration of sepsis, hypovolaemia, metabolic disease, arrhythmia, and congenital heart disease. Prostaglandin E one can reopen or maintain ductal flow while echocardiography defines anatomy, but apnoea and hypotension require monitored airway capability.

Temperature is a metabolic intervention. Neonates lose heat by evaporation, conduction, convection, and radiation because of large surface area, thin skin, limited fat, and inability to alter clothing or environment. Brown adipose tissue generates heat without shivering but consumes oxygen and glucose, worsening hypoxaemia, acidosis, and hypoglycaemia when cold stress persists. Preterm infants have even greater evaporative loss. Drying, warm surfaces, hats, plastic wrapping for very preterm infants, warmed gases, and skin-to-skin care are selected while preventing overheating.

Neonatal glucose concentration reflects interrupted placental supply, glycogenolysis, gluconeogenesis, feeding, insulin, and metabolic demand. Infants of diabetic mothers may remain hyperinsulinaemic after cord separation. Preterm or growth-restricted infants have limited glycogen and fat, while sepsis and cold raise consumption. Because operational treatment thresholds vary with age, risk, symptoms, and guideline, a number is interpreted in context and confirmed appropriately. Symptomatic hypoglycaemia requires prompt glucose; recurrent or high-infusion requirement raises endocrine or metabolic causes.

Bilirubin handling illustrates developmental physiology. High red-cell mass and shorter neonatal red-cell lifespan increase bilirubin production. Hepatic uptake and conjugation are immature, and intestinal deconjugation promotes reabsorption. Unconjugated bilirubin bound to albumin is less available to brain, but prematurity, acidosis, sepsis, displacement by drugs, and low albumin increase vulnerability. Treatment thresholds therefore depend on postnatal age, gestation, and neurotoxicity risks. Conjugated hyperbilirubinaemia signals hepatobiliary disease and demands early investigation, particularly when stool is pale or urine dark.

Feeding is a coordinated cardiorespiratory and neurological task. Effective transfer requires latch or seal, tongue and jaw movement, suck–swallow–breathe coordination, alertness, endurance, and safe airway protection. Tachypnoea, congenital heart disease, hypotonia, cleft palate, infection, pain, and maternal factors can reduce intake. Observe a feed rather than relying only on reported duration. Serial weight, urine, stool, mucous membranes, fontanelle, sodium, and behaviour identify inadequate transfer and hypernatraemic dehydration. Support should preserve parental confidence while treating the physiological barrier.

Growth is assessed as trajectory. Weight responds quickly to intake and fluid; length reflects longer-term skeletal growth; head circumference samples brain and skull growth in early life. Crossing centiles can be normal after birth as genetic potential declares itself, but sustained deceleration, disproportion, or loss after prior growth warrants evaluation. Corrected age avoids misclassifying preterm developmental progress. Causes of faltering growth are organised as inadequate intake, poor absorption, excessive loss, increased energy requirement, impaired utilisation, endocrine disease, or psychosocial constraint, often in combination.

Developmental surveillance considers gross motor, fine motor and vision, language and hearing, social reciprocity, cognition, and adaptive skills. A milestone is a probability distribution, not a pass–fail deadline. The pattern matters: isolated expressive-language delay differs from delay across domains, and loss of acquired skill is more concerning than slow acquisition. Hearing impairment can masquerade as language or behavioural disorder; visual impairment alters motor development. Examination includes head growth, tone, symmetry, reflexes, movement quality, skin markers, dysmorphism, and interaction, followed by early supports while aetiology is investigated.

Respiratory deterioration appears first as compensation. Infants raise respiratory rate and recruit compliant chest-wall muscles, producing recession, nasal flaring, grunting, and head bobbing. Feeding declines because sucking competes with breathing. As fatigue develops, respiratory rate and effort may fall despite worsening gas exchange; a quiet child with reduced air movement, apnoea, poor tone, or altered alertness is more dangerous than one vigorously compensating. Oxygen saturation is interpreted with probe quality, perfusion, motion, haemoglobin, and skin-pigmentation limitations. Ventilation failure requires carbon-dioxide and clinical assessment, not saturation alone.

The paediatric airway is narrow, so small mucosal swelling greatly increases resistance. Croup produces inspiratory stridor from upper-airway narrowing; stridor at rest, fatigue, cyanosis, or reduced consciousness indicates severity. Drooling, tripod position, muffled voice, or toxic appearance suggests a supraglottic or deep-neck emergency and unnecessary throat examination can precipitate obstruction. Sudden cough, unilateral wheeze, or focal

reduced air entry suggests foreign-body aspiration even with a normal radiograph. Skilled airway help and minimal agitation take priority.

Circulatory compensation preserves blood pressure through tachycardia and vasoconstriction. Early shock appears as altered interaction, cool or mottled skin, prolonged capillary refill, weak peripheral pulses, reduced urine, and increasing respiratory effort. Hypotension is late. Fever, pain, anxiety, and crying also raise heart rate, so trends after comfort and treatment matter. Fluid is weight-based and given as reassessed aliquots rather than an automatic large total. Myocarditis, congenital heart disease, severe anaemia, malnutrition, renal disease, and some infections can deteriorate with excessive fluid.

Dehydration assessment integrates weight change, intake, losses, urine, mental state, pulse quality, breathing, mucosa, tears, eyes, skin, and perfusion. No single sign precisely measures deficit. Oral rehydration uses coupled sodium–glucose absorption and is preferred for most mild to moderate gastroenteritis, delivered in frequent small amounts with replacement of ongoing losses. Shock, severe lethargy, ileus, or persistent inability to drink may require intravenous or enteral supervised replacement. Hyponatraemia and hypernatraemia require controlled correction because rapid osmotic shifts injure brain.

Fever risk changes sharply with age and immune status. Young infants may have serious bacterial infection with only poor feeding, hypothermia, irritability, or reduced tone. Non-blanching rash, neck stiffness, focal neurology, respiratory distress, impaired perfusion, or altered responsiveness mandates urgent assessment at any age. Antipyretics improve comfort and intake but do not distinguish viral from bacterial disease or prevent ordinary febrile seizures. Drug dosing must use measured weight and correct concentration; caregivers should receive milligrams and millilitres clearly rather than ambiguous spoon measures.

Paediatric prescribing requires an independent calculation chain: indication, current weight, milligrams per kilogram per dose or per day, interval, maximum adult dose, formulation concentration, volume, route, kidney and liver maturation, and interaction. Decimal errors are prevented by leading zeros and avoiding trailing zeros. Infants have higher total-body water and lower protein binding; hepatic enzymes and glomerular filtration mature at

different rates. A dose appropriate at one neonatal week may be wrong later. Excipients such as alcohol, propylene glycol, sodium, or benzyl alcohol can also matter.

Safeguarding is part of physiological care. Injury must fit developmental capability, mechanism, timing, and examination. Bruising in a non-mobile infant, patterned burns, unexplained fractures, repeated poisoning, delayed presentation, or inconsistent accounts require a structured multidisciplinary response while considering bleeding, bone, connective-tissue, and metabolic mimics. Speak with children at their developmental level, involve caregivers without silencing the child, and offer adolescents confidential discussion within clear safety limits. The core paediatric skill is recognising deviation from that individual child's normal before compensation fails.

Retrieval prompts

One. Which circulatory changes occur at birth?

Two. Why are neonates vulnerable to hypothermia and hypoglycaemia?

Three. What makes neonatal jaundice pathological?

Four. Why can children deteriorate before becoming hypotensive?

Five. Which observations reveal serious respiratory distress?

Six. What safeguards paediatric medication dosing?

Concise answers

One. Lung expansion lowers pulmonary resistance, cord clamping raises systemic resistance, and fetal shunts begin closing.

Two. High heat loss and metabolic demand combine with limited insulation, glucose stores, and immature regulation.

Three. First-day onset, rapid or prolonged rise, conjugated bilirubin, illness, haemolysis, dark urine, or pale stool.

Four. Strong vasoconstriction preserves pressure until cardiovascular compensation is nearly exhausted.

Five. Tachypnoea, recession, flaring, grunting, stridor, apnoea, poor feeding, altered interaction, and exhaustion.

Six. Current weight, correct concentration, age and organ adjustment, maximum dose, clear units, and independent calculation check.

Source map

Original synthesis informed by Guyton and Hall, neonatal transition, growth, and paediatric physiology; Robbins, congenital, neonatal, and childhood pathology; Katzung and OpenStax Pharmacology, developmental pharmacology and vaccines; OpenStax Anatomy and Physiology, Biology, Microbiology, and Medical-Surgical Nursing; and Talley and O'Connor, paediatric and developmental assessment.

Chapter 43: Ageing, Frailty, Polypharmacy, and Geriatric Assessment

Orientation

Ageing is heterogeneous. Chronological age does not define physiological reserve, cognition, function, or goals. Normal ageing narrows homeostatic range; disease, inactivity, nutrition, environment, and social conditions determine how much reserve is lost. Frailty is increased vulnerability to stress across systems, not simply old age or disability. Good geriatric medicine integrates diagnoses with medicines, mobility, cognition, continence, nutrition, support, and what matters to the person.

Physiological ageing

Cardiovascular ageing increases arterial stiffness, systolic pressure, afterload, and reliance on atrial filling. Adrenergic responsiveness, maximum heart rate, and baroreflex sensitivity decline, reducing exercise reserve and increasing orthostatic hypotension. Resting function may remain adequate until stress exposes limitation.

Lungs lose recoil, chest-wall compliance, clearance, cough strength, and gas-exchange reserve, increasing aspiration and infection risk. Kidney mass, blood flow, filtration, concentration, and acid-potassium reserve decline variably. Normal creatinine can conceal low filtration when muscle mass is reduced.

Total-body water and lean mass decrease while fat proportion rises. Water-soluble drugs may concentrate; lipid-soluble drugs may persist. Liver mass and phase-one metabolism may decline while conjugation is better preserved. Thermoregulation, thirst, skin, bone, immunity, and glucose control become less resilient.

Processing speed, sensation, reaction, sleep, motor units, muscle power, and anabolic response change, but dementia is not normal ageing and resistance training remains effective. Reserve varies greatly.

Frailty and resilience

Frailty can be conceptualised as a physical phenotype involving weakness, slow walking, low activity, exhaustion, and weight loss, or as accumulated deficits across disease, function, cognition, and social domains. Both predict falls, delirium, disability, prolonged admission, institutionalisation, treatment toxicity, and death.

Frailty is dynamic. Acute illness can cause abrupt decline, while nutrition, rehabilitation, medication review, social support, and treatment of reversible disease can improve function. A frailty score supports communication and risk estimation but must not be used to deny treatment automatically. The relevant question is whether an intervention's likely benefit, burden, and recovery path fit this individual.

Sarcopenia is loss of muscle strength and quantity or quality. Strength and performance matter more than mass alone. Causes include inactivity, inadequate protein or energy, inflammation, endocrine disease, neurological illness, and medicines. Progressive resistance exercise, adequate nutrition, and treatment of causes are central.

Comprehensive geriatric assessment

Comprehensive geriatric assessment evaluates disease, medicines, function, mobility, cognition, mood, nutrition, continence, senses, pain, oral health, skin, sleep, support, environment, finances, transport, caregiver burden, capacity, and wishes, producing a coordinated plan with accountable follow-up.

Assess basic activities such as feeding, dressing, toileting, bathing, transfers, and mobility, then instrumental activities such as shopping, cooking, medicines, finances, communication, and transport. Baseline decline may be the most sensitive illness sign.

Observe transfers and gait, inspect footwear and aids, test vision and hearing, and review the home environment. Include caregivers with permission but speak directly to the patient. Caregiver reports can clarify change, while private conversation protects autonomy and detects abuse or coercion.

Atypical illness presentation

Older adults may present with delirium, falls, immobility, incontinence, anorexia, or functional decline rather than organ-specific symptoms. Fever and leukocytosis can be absent during infection. Myocardial infarction may cause breathlessness, weakness, or confusion rather than pain. Drug toxicity may mimic dementia or frailty.

Avoid attributing new symptoms to age. Construct a timeline, compare baseline, review medicines, assess hydration and pain, and examine all systems. Several modest insults often interact: infection, constipation, sleep loss, unfamiliar environment, sedatives, and sensory impairment can together cause severe delirium.

Delirium, dementia, and depression

Delirium is acute fluctuating inattention and altered cognition caused by physiological disturbance. Hypoactive delirium is quiet and frequently missed. Predisposition includes dementia, frailty, sensory loss, and severe illness; precipitants include infection, drugs, withdrawal, retention, constipation, pain, hypoxia, metabolic disturbance, and surgery.

Treat causes while protecting hydration, oxygenation, sleep, mobility, hearing, vision, orientation, nutrition, and family contact. Avoid restraints and sedatives where possible. Antipsychotics do not cure delirium and are reserved for selected severe distress or danger after non-drug measures, with attention to cardiac, motor, stroke, and mortality risk.

Dementia is chronic acquired cognitive decline interfering with independence. Establish functional impact, progression, affected domains, neurological features, medicines, mood, sleep, and reversible contributors. Depression can mimic or worsen cognitive impairment and may coexist. Cognitive tests are influenced by education, culture, language, hearing, vision, pain, and fatigue.

Falls and mobility

Falls usually have multiple causes: weakness, impaired balance, neuropathy, vestibular disease, vision loss, cognition, urgency, foot problems, postural hypotension, arrhythmia, sedatives, environmental hazards, and risky tasks. Ask about prodrome, loss of consciousness, direction, activity, injury, time on floor, prior falls, fear, and ability to rise.

Examine lying and standing pressure, heart rhythm, gait, feet, joints, power, balance, sensation, vision, and cognition. Review bone health and fracture risk. Effective prevention combines strength and balance exercise, medication reduction, vision and footwear care, home modification, continence management, and treatment of cardiovascular or neurological causes. Fear-driven inactivity worsens risk.

After a fall, assess head injury, fracture, rhabdomyolysis, pressure injury, dehydration, and why the person could not recover. Anticoagulation lowers the threshold for head imaging according to clinical guidance. A fall is a syndrome to investigate, not a diagnosis.

Polypharmacy and prescribing

Polypharmacy means multiple medicines, but appropriateness matters more than a fixed count. Multimorbidity guidelines can accumulate drugs whose combined burden exceeds benefit. Prescribing cascades occur when a drug adverse effect is mistaken for a new disease and treated with another drug.

Ageing alters absorption modestly, distribution substantially, metabolism variably, and renal elimination commonly. Pharmacodynamic sensitivity to sedatives, anticholinergics, opioids, anticoagulants, glucose-lowering drugs, and pressure-lowering agents may increase. Interactions, duplicate classes, complex schedules, swallowing difficulty, cost, cognition, and dexterity affect safety.

At transitions, reconcile all products actually taken. For every drug state indication, benefit and time to benefit, dose, organ suitability, adverse effects, interactions, monitoring, adherence, and goal relevance. High-risk medicines require sick-day, bleeding, hypoglycaemia, sedation, and falls advice.

Deprescribing

Deprescribing is supervised dose reduction or cessation when harm or burden exceeds likely benefit. Prioritise drugs without indication, duplicates, those causing current harm, preventive therapies whose time to benefit exceeds prognosis, and medicines inconsistent with goals. Some require tapering to avoid withdrawal or rebound.

Shared decisions should explain uncertainty and distinguish stopping preventive therapy from abandoning care. Change a manageable number at once, document the plan, monitor symptoms and withdrawal, and provide restart criteria. Deprescribing can reduce burden while preserving treatments that materially support comfort or function.

Nutrition, swallowing, and continence

Weight loss may reflect inadequate access, isolation, dental disease, dysphagia, depression, dementia, medication, malabsorption, inflammation, endocrine disease, or cancer. Assess weight trajectory, intake, muscle, function, oral health, and feeding support. Albumin is altered by inflammation and hydration and is not a standalone nutrition measure.

Dysphagia can cause aspiration, dehydration, and malnutrition. Ask about cough, wet voice, prolonged meals, food sticking, recurrent chest infection, and weight loss. Speech pathology assessment and texture modification may improve safety but can reduce enjoyment or intake; decisions should reflect goals and evidence.

Incontinence may be transient from delirium, infection, drugs, mobility, stool impaction, or excess urine, or chronic from urge, stress, overflow, functional barriers, or mixed mechanisms. Bladder diaries, residual measurement, examination, and targeted testing guide treatment. Indwelling catheters rarely solve chronic incontinence without creating infection and trauma.

Capacity, goals, and advance care

Capacity is decision-specific and time-specific. A person must understand relevant information, retain it long enough, use or weigh it, and communicate a choice. Support communication, treat delirium, optimise hearing and vision, use interpreters, and simplify information before concluding incapacity. An unwise decision alone does not prove incapacity.

When capacity is absent, follow valid advance directives, appointed decision-makers, and local law, using the person's values and best interests. Goals-of-care discussions connect prognosis with outcomes the person values: living independently, cognition, symptom relief, longevity, avoiding hospital, or family events. Document clearly across settings.

Rehabilitation and transitions

Hospitalisation causes deconditioning through bed rest, sleep disruption, poor intake, unfamiliar routines, and devices. Restore mobility, clothing, toileting, meals, senses, and sleep early. Rehabilitation uses task-specific strength, balance, endurance, cognitive, and environmental goals.

Discharge safety depends on function. Reconcile medicines, follow-up, equipment, transport, food, caregiver capacity, wound or continence needs, warning signs, and coordination. Readmission often arises from gaps between services.

TTS module 2: Physiological reserve, medication burden, function, and goal-concordant ageing care

Ageing changes the distance between ordinary demand and physiological failure. Resting measurements can remain normal while cardiac chronotropy, baroreflexes, ventilatory reserve, kidney concentration, immune response, thermoregulation, muscle power, and cognitive flexibility narrow. Acute illness then crosses several thresholds at once. The older person may present with a fall, confusion, immobility, poor intake, or new dependence rather than pain, fever, or a textbook organ syndrome. Baseline function and the speed of change are therefore diagnostic vital signs.

Frailty describes reduced multidimensional reserve and impaired recovery after stress. It is related to, but distinct from, age, comorbidity, and disability. A

person can have several stable diseases without frailty, or substantial frailty without a long diagnosis list. Phenotype models emphasise weakness, low activity, exhaustion, slow gait, and weight loss; deficit-accumulation models integrate diseases, symptoms, function, cognition, and social vulnerability. Scores estimate population risk but cannot determine the value of surgery, intensive care, or rehabilitation for an individual without considering reversibility and goals.

Frailty can improve because its contributors are not all fixed. Bed rest, sedatives, dehydration, infection, pain, depression, sensory loss, inadequate protein, and environmental barriers are modifiable. Conversely, a minor procedure or urinary infection can initiate a cascade of delirium, immobility, deconditioning, incontinence, pressure injury, and institutional dependence. Prevention requires early mobilisation, ordinary clothing, meals out of bed, access to glasses and hearing aids, sleep protection, removal of unnecessary lines, and explicit rehabilitation from the first day.

Sarcopenia is diagnosed through low strength with reduced muscle quantity or quality, and severity is reflected in performance. Grip strength, chair rise, gait speed, stair ability, and falls reveal more than visual muscle bulk. Inflammation, endocrine disease, denervation, inactivity, and low energy or protein intake interact. Resistance exercise remains effective in advanced age and should be progressively loaded, not reduced to token movement. Protein and energy support training but do not rebuild function without mechanical stimulus. Vitamin or hormone treatment helps only when a relevant deficiency or indication exists.

Cardiovascular ageing increases arterial stiffness and pulse pressure, left-ventricular afterload, and dependence on diastolic filling. Reduced beta-adrenergic responsiveness limits maximal heart rate and rapid compensation. Blunted baroreflexes, venous pooling, dehydration, meals, heat, autonomic disease, and vasoactive drugs combine to cause orthostatic hypotension. Lying and standing pressure should be paired with symptoms and heart-rate response. Treatment targets falls and cerebral perfusion while avoiding supine hypertension, heart failure, and kidney stress.

Kidney function is easily overestimated when low muscle mass produces little creatinine. Estimated filtration equations are imperfect in extremes of frailty, amputation, oedema, or rapidly changing function. Drug dosing may require

trends, cystatin-based estimates, measured clearance, and clinical toxicity. Ageing kidneys conserve water and sodium less effectively, excrete potassium and acid with less reserve, and recover more slowly after haemodynamic or nephrotoxic injury. "Normal" laboratory values can therefore hide vulnerability during fasting, diarrhoea, diuretic use, contrast exposure, or infection.

Drug distribution changes as total-body water and lean mass decline and fat proportion rises. A water-soluble loading dose may create a higher concentration, while a lipid-soluble sedative can have a longer terminal effect. Albumin, liver blood flow, enzyme activity, kidney clearance, receptor sensitivity, and blood–brain barrier function vary. Pharmacodynamic vulnerability is often more important than kinetics: benzodiazepines, opioids, anticholinergics, antipsychotics, insulin, sulfonylureas, anticoagulants, and antihypertensives can produce disproportionate delirium, falls, constipation, retention, hypoglycaemia, bleeding, or syncope.

Medication reconciliation asks what is actually taken, not what appears on the list. Include non-prescription analgesics, antihistamines, sleep products, supplements, patches, inhalers, eye drops, and medicines borrowed or intentionally skipped. For each drug, identify indication, current benefit, time to benefit, dose, organ suitability, interaction, monitoring, practical administration, and alignment with goals. A prescribing cascade is suspected when a new medicine treats a plausible adverse effect of an earlier one, such as a diuretic for drug-related ankle oedema or a bladder drug for diuretic-associated urgency.

Deprescribing is a therapeutic trial with a withdrawal and monitoring plan. Immediate priorities include duplicate therapy, absent indication, current toxicity, dangerous interaction, and drugs made inappropriate by organ decline. Preventive medicines are judged by absolute benefit, time to benefit, treatment burden, and life expectancy rather than age alone. Proton-pump inhibitors, gabapentinoids, sedatives, anticholinergics, antihypertensives, glucose-lowering drugs, and supplements may warrant review, but abrupt cessation of benzodiazepines, opioids, corticosteroids, beta blockers, antidepressants, or anticonvulsants can cause withdrawal or rebound.

Falls are events with mechanisms. Ask what happened before, during, and after: posture, turning, trip, dizziness, palpitation, loss of consciousness, focal

weakness, urgency, footwear, lighting, injury, and time on the floor. A backward fall differs from slipping over an obstacle. Syncope, seizure, vestibular disease, neuropathy, parkinsonism, visual loss, painful joints, cognitive misjudgement, sedatives, and environmental hazards can coexist. Multifactorial intervention is effective when it treats identified contributors, particularly progressive balance and strength exercise, medication reduction, vision and foot care, home modification, and cardiovascular causes.

An inability to rise after a fall predicts further harm. Prolonged time on the floor causes pressure injury, hypothermia, dehydration, rhabdomyolysis, kidney injury, and fear. Assess head injury and occult hip, pelvic, wrist, rib, or vertebral fracture; pain may be muted and walking does not exclude fracture. Anticoagulation changes concern for intracranial bleeding. A personal alarm is useful only if worn, reachable, understood, and connected to a response system.

Delirium reflects acute brain-network dysfunction in a vulnerable host. Attention fluctuates, arousal may be high or low, and sleep, perception, thought, and motor activity change. Dementia strongly predisposes but does not explain an acute decline. Common interacting precipitants include infection, hypoxia, pain, medications or withdrawal, dehydration, electrolyte disturbance, urinary retention, constipation, unfamiliar surroundings, and sensory deprivation. A quiet sleeping patient may have hypoactive delirium, not benign fatigue. Treatment removes causes and reconstructs orientation, mobility, hydration, senses, and circadian cues.

Comprehensive geriatric assessment converts parallel problems into one prioritised plan. Medical diagnoses are considered alongside basic and instrumental activities, cognition, mood, mobility, nutrition, continence, pain, vision, hearing, oral health, sleep, social connection, housing, finances, transport, caregiver capacity, and medication management. The assessment is effective when recommendations are implemented, ownership is assigned, and follow-up crosses settings. Discovering fifteen problems without coordination merely increases burden.

Weight loss is investigated as both disease and function. Difficulty shopping, opening packaging, cooking, chewing, swallowing, seeing food, remembering meals, or affording nutrition may be more important than gastrointestinal absorption. Inflammatory illness lowers appetite and drives catabolism.

Texture modification can reduce aspiration in selected dysphagia yet also reduce intake and pleasure; decisions require observed swallowing, goals, hydration, and review. Serum albumin mainly reflects inflammation, distribution, and illness and cannot diagnose malnutrition alone.

Continence depends on bladder and bowel physiology, mobility, cognition, clothing, environment, and assistance. Acute incontinence can be triggered by delirium, excess urine, infection, stool impaction, medication, or loss of access to the toilet. Post-void residual helps detect retention or overflow. Anticholinergic treatment for urgency may worsen dry mouth, constipation, cognition, and falls. Indwelling catheters create infection, trauma, immobility, and dependence and require a defined indication and removal plan.

Capacity is assessed for a specific decision at a specific time. The person must understand relevant information, retain it long enough, use or weigh it in light of values, and communicate a choice. Hearing loss, aphasia, unfamiliar language, low literacy, delirium, pain, and rushed explanation can falsely appear as incapacity. Provide interpreters, communication aids, quiet, repetition, and treatment of reversible barriers. Disagreement or an unwise choice does not itself establish incapacity.

Goal-concordant care connects medical options to outcomes the person values. A treatment may prolong life but threaten cognition, mobility, independence, or the ability to remain at home; another may carry short-term burden with a realistic recovery path. Discuss best, worst, and most likely functional outcomes, including what rehabilitation would require. Advance directives and substitute decision-makers matter when capacity is lost, but prior values remain the central evidence. Good geriatric care does not offer less because a person is old; it offers proportionate care informed by reserve, reversibility, burden, and what makes the outcome worthwhile.

Retrieval prompts

One. Why can normal creatinine conceal kidney impairment in older adults?

Two. What distinguishes frailty from chronological age?

Three. Which domains belong in comprehensive geriatric assessment?

Four. Why are falls considered a syndrome?

Five. What makes deprescribing safe?

Six. How should decision-making capacity be assessed?

Concise answers

One. Reduced muscle mass lowers creatinine production even when filtration is poor.

Two. Frailty measures multidimensional vulnerability and reduced reserve, which vary widely at the same age.

Three. Medical, medication, functional, mobility, cognitive, mood, nutrition, continence, sensory, social, environmental, caregiver, and goal domains.

Four. Multiple interacting cardiovascular, neurological, sensory, medication, musculoskeletal, and environmental causes usually contribute.

Five. Shared prioritisation, gradual withdrawal when needed, documentation, monitoring, and clear restart criteria.

Six. For the specific decision, test understanding, retention, weighing, and communication after providing necessary support.

Source map

Original synthesis informed by Guyton and Hall, ageing physiology and reserve; Robbins, age-associated pathology; Katzung and OpenStax Pharmacology, geriatric pharmacokinetics and polypharmacy; OpenStax Anatomy and Physiology and Medical-Surgical Nursing; and Talley and O'Connor, geriatric, functional, cognitive, falls, and capacity assessment.

Chapter 44: Trauma, Resuscitation, Altered Consciousness, and Perioperative Assessment

Orientation

Acute care prioritises reversible threats before diagnosis. Trauma uses repeated primary survey, intervention, and secondary survey. Altered consciousness requires simultaneous stabilisation and cause-finding. Perioperative assessment estimates whether surgery, anaesthesia, disease, and reserve can be managed safely. Trajectory and treatment response matter more than one normal observation.

Preparation and primary survey

Before arrival, assemble personnel, protective equipment, warming, monitoring, blood access, airway tools, imaging, and surgical support. Establish mechanism, time, prehospital physiology, treatment, anticoagulants, pregnancy possibility, allergies, and major comorbidity. Transfer information while continuing care.

Use a structured sequence: catastrophic external haemorrhage, airway with cervical protection, breathing, circulation, disability, and exposure with temperature control. Treat each life threat when found rather than completing the survey first. Reassess after every intervention and whenever condition changes.

Direct pressure, wound packing, tourniquet, pelvic binder, splinting, and urgent surgery or embolisation control haemorrhage according to site. A tourniquet is placed proximal to severe limb bleeding, tightened until bleeding stops, and time documented. Do not periodically loosen it outside a controlled plan.

Airway and cervical spine

Look for obstruction, blood, vomit, facial or neck injury, burns, hoarseness, stridor, expanding haematoma, and reduced consciousness. Open the airway with jaw thrust when cervical injury is possible, suction, remove visible obstruction, provide oxygen, and use adjuncts appropriate to reflexes and anatomy.

Definitive airway is indicated when protection, ventilation, oxygenation, anticipated swelling, transport, or clinical course cannot be secured. Rapid-sequence intubation can precipitate hypotension and arrest in severe hypovolaemia, right-heart obstruction, or metabolic acidosis. Resuscitate physiology, choose drugs carefully, prepare failed-airway plans, and confirm tracheal placement with continuous waveform carbon dioxide.

Maintain spinal motion restriction when mechanism and findings justify it, but do not allow collars or imaging to delay airway or haemorrhage control. Prolonged rigid immobilisation causes pressure injury, aspiration risk, pain, and raised intracranial pressure. Clear the spine clinically or radiologically through validated pathways.

Breathing and chest injury

Inspect effort, symmetry, wounds, trachea, neck veins, and cyanosis; palpate chest and assess breath sounds, oxygenation, and ventilation. Immediately treat tension pneumothorax when shock or severe respiratory compromise accompanies suspected pleural pressure; do not wait for imaging. Open pneumothorax requires an appropriate vented seal and definitive pleural drainage.

Massive haemothorax causes blood loss and lung compression and requires drainage, transfusion, and surgical evaluation. Flail segments and pulmonary contusion cause pain, impaired ventilation, and delayed hypoxaemia; analgesia, physiotherapy, and selective ventilatory support are central. Cardiac tamponade causes obstructive shock and requires urgent decompression and definitive repair.

Circulation and haemorrhagic shock

Trauma shock is haemorrhagic until proved otherwise, but obstructive, cardiogenic, neurogenic, and distributive mechanisms may coexist. Early signs include tachycardia, cool skin, narrow pulse pressure, anxiety, delayed capillary refill, and reduced urine; young patients can maintain pressure until late. Beta blockade, pacing, pregnancy, and age alter signs.

Control bleeding while restoring perfusion with warmed blood components in major haemorrhage. Excess crystalloid dilutes clotting factors, worsens oedema, and causes hypothermia and acidosis. Balanced component therapy or whole blood follows local protocols. Give tranexamic acid early to eligible bleeding trauma patients within the evidence-based window.

Damage-control resuscitation limits permissive hypotension until haemostasis in selected patients, minimises crystalloid, corrects calcium and temperature, and moves rapidly to surgery or embolisation. This strategy is inappropriate when cerebral perfusion requires higher pressure, including severe traumatic brain injury. Monitor coagulation dynamically where available.

The lethal triad of hypothermia, acidosis, and coagulopathy is self-reinforcing. Exposure, cold fluid, shock, and long procedures worsen it. Warm the patient, room, fluids, and blood; control haemorrhage; restore perfusion; and shorten definitive procedures when physiology is exhausted.

Disability and altered consciousness

Assess level of consciousness, pupils, limb movement, lateralising signs, glucose, seizures, and temperature. A simple coma scale documents eye, verbal, and motor responses but is affected by intoxication, sedation, language, hearing, paralysis, and intubation. Record components, not score alone.

Causes include traumatic brain injury, hypoxia, hypercapnia, hypoglycaemia, seizure, stroke, infection, toxins, endocrine crisis, organ failure, electrolyte disturbance, and temperature extremes. Give glucose for documented or strongly suspected hypoglycaemia, naloxone for likely opioid toxicity with respiratory depression, and anticonvulsant treatment for ongoing seizure. Empirical therapy should not delay ventilation or imaging.

Prevent secondary brain injury by avoiding hypoxia, hypotension, fever, severe carbon-dioxide disturbance, and glucose extremes. Head elevation and neutral venous drainage help. An expanding pupil, declining consciousness, or new posturing suggests herniation and requires urgent hyperosmolar therapy, controlled rescue ventilation when imminent, imaging, and neurosurgical intervention.

Exposure, secondary survey, and imaging

Expose to inspect back, perineum, axillae, scalp, and skin while preventing heat loss and preserving dignity. Log-roll with adequate personnel when findings change care. Remove contamination and irrigate chemical exposure.

After immediate threats, obtain history and head-to-toe examination. Inspect surfaces, palpate bones and compartments, and assess wounds, joints, pulses, nerves, abdomen, pelvis, and spine. Repeat because pain, distraction, shock, and evolution conceal findings.

Focused ultrasound rapidly detects selected pericardial, pleural, and intraperitoneal fluid but cannot exclude all injury. Computed tomography is valuable in stable or stabilised patients and must not delay operative control in obvious ongoing haemorrhage. Pregnancy changes imaging discussion but does not justify withholding necessary diagnosis.

Specific injury patterns

Traumatic brain injury includes concussion, contusion, diffuse axonal injury, and extra-axial or intracerebral bleeding. Anticoagulation increases risk and may require urgent reversal. Serial neurological examination and repeat imaging are guided by trajectory. Discharge advice must address deterioration, activity, cognition, driving, and supervision.

Spinal-cord injury produces motor, sensory, autonomic, and respiratory dysfunction according to level. Neurogenic shock causes hypotension from lost sympathetic tone, often with relative bradycardia, after haemorrhage is excluded. Prevent pressure injury, maintain perfusion, and obtain early spinal specialist care.

Abdominal injury may bleed or leak without early peritonism. Pelvic fractures can cause major retroperitoneal haemorrhage; binders belong over the

greater trochanters. Limb injury threatens tissue through arterial disruption, compartment syndrome, open fracture, or crush. Assess and document neurovascular status before and after intervention.

Cardiac arrest and post-resuscitation care

Confirm unresponsiveness and absent normal breathing, summon help, begin high-quality chest compressions, attach a defibrillator, and minimise interruptions. Shock ventricular fibrillation or pulseless ventricular tachycardia promptly. For non-shockable rhythms, continue compressions, give appropriate adrenaline, and identify reversible causes: hypoxia, hypovolaemia, hydrogen-ion excess, potassium or metabolic disturbance, hypothermia, thrombosis, tamponade, tension pneumothorax, and toxins.

Airway interventions must not degrade compressions. Waveform carbon dioxide confirms tube placement and can indicate compression quality or return of circulation. After return of spontaneous circulation, optimise oxygen without prolonged hyperoxia, ventilation, pressure, coronary or other cause treatment, temperature control, glucose, seizures, and neurological prognostication. Delay definitive prognosis until confounders and appropriate time have been addressed.

Preoperative assessment

Preoperative care identifies modifiable risk and plans perioperative management; it is not clearance or a guarantee. Define procedure urgency, magnitude, blood loss, physiological stress, and postoperative destination. Assess functional capacity, frailty, airway, heart, lungs, kidney, liver, diabetes, anaemia, nutrition, infection, bleeding, thrombosis, cognition, pain, substance use, and prior anaesthetic complications.

Order tests only when results can change management. A normal test does not neutralise poor functional status, and routine indiscriminate testing creates false positives. Cardiac investigation follows symptoms, active disease, surgical risk, and functional capacity. Optimisation may include treating anaemia or infection, improving respiratory control, nutrition and exercise, smoking cessation, and planning critical care.

Medicines and fasting

Reconcile all medicines and supplements. Create explicit plans for anticoagulants, antiplatelets, diabetes therapies, steroids, pressure medicines, diuretics, immune suppression, seizure medicines, opioids, and herbal products. Balance bleeding against thrombosis rather than applying one stop interval. Patients with adrenal suppression may require perioperative glucocorticoid coverage.

Fasting reduces aspiration risk but excessive fasting causes dehydration and discomfort. Follow procedure- and patient-specific guidance for clear fluids and solids. Diabetes plans must prevent both hypoglycaemia and ketosis. Sodium-glucose cotransporter inhibitors require preoperative withholding because euglycaemic ketoacidosis can occur.

Postoperative physiology and complications

Anaesthesia, pain, opioids, inflammation, immobility, fluid shifts, blood loss, and surgery alter every organ. Monitor airway, breathing, circulation, consciousness, pain, nausea, urine, bleeding, temperature, and the surgical site. New deterioration is not simply expected postoperative change.

Prevent complications through multimodal analgesia, lung expansion, early mobilisation, thrombosis prophylaxis, nutrition, glucose control, device removal, sleep and orientation, and medication reconciliation. Consider haemorrhage, myocardial injury, pulmonary embolism, atelectasis, pneumonia, aspiration, kidney injury, retention, ileus, infection, delirium, and withdrawal according to timing and phenotype.

TTS module 2: Resuscitation physiology, occult injury, and perioperative risk control

Resuscitation is repeated hypothesis testing under time pressure. Identify the physiological threat, intervene, and immediately ask whether airway patency, ventilation, oxygen delivery, perfusion, consciousness, or temperature improved. A normal initial pressure or scan cannot end assessment because compensation hides injury and bleeding evolves. Mechanism predicts what may be concealed, while response to treatment reveals which physiological model is plausible. Definitive haemorrhage control, decompression, antidote, or surgery must proceed alongside—not after—diagnostic refinement.

Catastrophic external haemorrhage is controlled before a conventional airway sequence because exsanguination can kill within minutes. Direct pressure, wound packing, tourniquet, pelvic stabilisation, and rapid operative or interventional control are anatomical treatments. A pelvic binder belongs at the greater trochanters, reducing pelvic volume and motion; placement over the iliac crests is ineffective. Long-bone splinting limits bleeding and pain. Every intervention is followed by repeat distal perfusion and neurological documentation.

Airway assessment predicts both current obstruction and likely deterioration. Blood, vomit, loose teeth, facial fractures, burns, expanding neck haematoma, laryngeal injury, and declining consciousness can convert a manageable airway into an impossible one. Preoxygenation builds an oxygen reservoir, but shock, agitation, pulmonary injury, obesity, and pregnancy shorten safe apnoea. Rapid-sequence induction removes sympathetic tone and positive-pressure ventilation reduces venous return, potentially precipitating arrest in severe hypovolaemia, tamponade, tension pneumothorax, or right-heart failure. Resuscitation, haemodynamic drug choice, and a failed-airway plan precede induction.

Waveform carbon dioxide is the most reliable continuous confirmation of tracheal placement when circulation is present. Chest rise, mist, and breath sounds can mislead, particularly during noisy resuscitation or oesophageal ventilation. A falling carbon-dioxide trace may indicate disconnection, tube displacement, severe bronchospasm, reduced pulmonary blood flow, or arrest. Cervical protection should minimise harmful motion without preventing oxygenation. A rigid collar can impede mouth opening, raise venous pressure,

and cause pressure injury; manual stabilisation and definitive clearance are adapted to risk.

Tension pneumothorax is obstructive shock from rising pleural pressure that collapses lung and restricts venous return. Severe respiratory compromise or shock with compatible unilateral findings is treated immediately without imaging. Needle decompression can fail through inadequate catheter length, kinking, clot, or wrong position; finger thoracostomy or tube drainage provides more definitive relief according to setting and expertise. Cardiac tamponade similarly impairs filling, but trauma may not produce classic distended veins or muffled sounds when hypovolaemia coexists.

Haemorrhagic shock reduces oxygen delivery through falling circulating volume and haemoglobin. Tachycardia and vasoconstriction preserve pressure initially, so capillary refill, skin temperature, mental state, pulse quality, urine, acid-base change, and mechanism reveal earlier compromise. Older adults, athletes, pregnant patients, beta-blocked patients, and paced rhythms alter expected signs. Permissive hypotension can limit bleeding before haemostasis in selected torso trauma, but it is inappropriate when severe brain injury requires adequate cerebral perfusion or when pregnancy and other contexts demand different targets.

Damage-control resuscitation addresses the lethal interaction of bleeding, dilution, hypothermia, acidosis, and coagulation failure. Warmed blood components or whole blood restore oxygen carriage and clotting more effectively than large crystalloid volumes. Fibrinogen may fall early, platelets and clotting factors dilute, and shock activates endothelial and fibrinolytic pathways. Citrate in transfused blood binds calcium, reducing contractility and coagulation; ionised calcium is monitored and replaced. Tranexamic acid benefits selected bleeding trauma when given early within a validated window, not indiscriminately after thrombosis has stabilised.

The first survey finds immediate killers; the secondary survey finds injuries hidden by distraction, shock, intoxication, or altered consciousness. Scalp, posterior surfaces, axillae, perineum, joints, and compartments require inspection while heat is preserved. Abdominal bleeding may remain painless and without early rigidity. A seat-belt mark, lower-rib injury, distracting fracture, or falling haemoglobin increases concern. Serial examination and repeated focused ultrasound can detect evolution, but a negative focused

scan does not exclude retroperitoneal, hollow-viscus, diaphragmatic, or early intraperitoneal injury.

Computed tomography is a destination for a stable or sufficiently stabilised patient, not a treatment room for uncontrolled haemorrhage. Transfer risk includes disconnection, airway loss, deterioration beyond immediate surgical access, and delay to haemostasis. Contrast risk is usually secondary to identifying life-threatening injury. In pregnancy, maternal oxygenation and perfusion are the best initial fetal resuscitation, and necessary imaging should not be withheld because of exaggerated radiation fear. Position and gestational physiology alter venous return and resuscitation.

Traumatic brain injury is worsened by preventable secondary insults. Hypoxaemia and hypotension sharply reduce outcome; fever, seizures, severe hypo- or hypercapnia, anaemia, glucose extremes, and impaired cerebral venous drainage add injury. Sedation and paralysis obscure examination but may be required for airway and pressure control, so pre-treatment findings are documented. A unilateral enlarging pupil, falling motor response, new posturing, or bradycardic hypertension suggests herniation. Hypertonic saline or mannitol and brief controlled hyperventilation create time for neurosurgical treatment but do not remove the mass.

Altered consciousness is approached with simultaneous support and reversible-cause testing. Oxygenation, ventilation, pressure, glucose, temperature, pupils, motor asymmetry, seizure activity, trauma, and toxins are checked immediately. Naloxone is titrated to restore ventilation in suspected opioid toxicity rather than necessarily provoking full arousal and withdrawal. Thiamine is provided to at-risk patients but should not delay glucose for hypoglycaemia. Persistent unresponsiveness after convulsion may be postictal, structural, toxic, metabolic, or non-convulsive status, requiring imaging and electroencephalography according to context.

Cardiac arrest treatment maximises coronary and cerebral perfusion through high-quality compressions, minimal interruption, and prompt defibrillation of shockable rhythm. Airway procedures should not interrupt compressions unnecessarily. Reversible causes are searched by mechanism: hypoxia, hypovolaemia, acidosis, potassium and metabolic disturbance, hypothermia, thrombosis, tamponade, tension pneumothorax, and toxins. Point-of-care ultrasound can identify selected causes during planned rhythm checks but

prolonged image acquisition lowers perfusion. A sudden rise in expired carbon dioxide may indicate return of spontaneous circulation.

Post-arrest care treats the cause and prevents a second wave of brain, heart, and organ injury. Avoid both hypoxaemia and prolonged hyperoxia, control ventilation, restore pressure and coronary flow, identify seizure, manage temperature, and correct glucose extremes. Sedatives, paralysis, hypothermia, shock, and metabolic failure confound neurological assessment. Prognosis is multimodal and delayed until timing and confounders permit; an early poor motor response alone is insufficient to withdraw treatment.

Preoperative assessment is risk modification and planning rather than ritual clearance. Procedure urgency, expected blood loss, anatomical stress, anaesthetic technique, postoperative pain, and destination are matched to functional capacity and organ reserve. Active cardiac instability, uncontrolled respiratory disease, infection, anaemia, malnutrition, kidney dysfunction, frailty, and delirium risk may change timing or support. Routine tests in low-risk asymptomatic people create incidental abnormalities without improving outcomes. Each test should answer a question that could alter surgery, anaesthesia, treatment, or consent.

Functional capacity integrates cardiac, respiratory, musculoskeletal, neurological, and motivational limits. Inability to climb stairs may reflect knee pain rather than myocardial reserve, so the limiting symptom is identified. Frailty predicts postoperative delirium, complications, institutional discharge, and slow recovery beyond conventional organ scores. Shared decision-making should describe likely functional trajectory and rehabilitation burden, not only mortality. Prehabilitation may include exercise, nutrition, anaemia treatment, smoking cessation, respiratory optimisation, and medication rationalisation when time allows.

Perioperative medication plans balance competing hazards. Stopping anticoagulation reduces bleeding but may expose mechanical valves, recent thrombosis, or high-risk atrial fibrillation; bridging itself causes bleeding and is not universal. Antiplatelet decisions depend on coronary stents, indication, procedure, and urgency. Sodium–glucose cotransporter inhibitors are withheld sufficiently before many procedures because fasting and stress can produce euglycaemic ketoacidosis. Long-term corticosteroid exposure may suppress adrenal response. Insulin plans prevent both ketosis and fasting

hypoglycaemia, and essential antiseizure or Parkinson medicines should not be omitted inadvertently.

Postoperative deterioration is never dismissed as “just surgery.” Timing and phenotype narrow the cause. Early hypotension suggests bleeding, vasodilation, cardiac dysfunction, or obstruction. Hypoxaemia may reflect atelectasis, aspiration, pulmonary oedema, embolism, pneumonia, opioid depression, or residual neuromuscular block. Oliguria can follow low perfusion, congestion, obstruction, or kidney injury. Delirium may be the first sign of sepsis, hypoxia, retention, withdrawal, pain, or medication toxicity. Multimodal analgesia, lung expansion, mobility, thrombosis prevention, nutrition, orientation, sleep, and early device removal preserve reserve while diagnostic reassessment continues.

Retrieval prompts

One. Why is the trauma primary survey repeated?

Two. What constitutes damage-control resuscitation?

Three. Which factors cause secondary brain injury?

Four. What are the reversible causes of cardiac arrest?

Five. What is the purpose of preoperative assessment?

Six. Which postoperative measures prevent common complications?

Concise answers

One. Injury and response evolve, and treatment can reveal or create new threats.

Two. Rapid haemorrhage control, restrained crystalloid, early blood, temperature and calcium management, and abbreviated definitive procedures when physiology is failing.

Three. Hypoxia, hypotension, fever, carbon-dioxide extremes, glucose disturbance, seizures, and impaired venous drainage.

Four. Hypoxia, hypovolaemia, acidosis, potassium or metabolic disturbance, hypothermia, thrombosis, tamponade, tension pneumothorax, and toxins.

Five. To define and modify risk, coordinate medicines and physiology, and plan anaesthesia, surgery, and recovery.

Six. Analgesia, pulmonary care, mobilisation, thrombosis prevention, nutrition, glucose control, device removal, orientation, and reconciliation.

Source map

Original synthesis informed by Guyton and Hall, shock, circulation, ventilation, brain perfusion, and stress physiology; Robbins, trauma and organ injury; Katzung and OpenStax Pharmacology, emergency, anaesthetic, anticoagulant, and perioperative medicines; OpenStax Medical-Surgical Nursing; and Talley and O'Connor, resuscitation, trauma, coma, and preoperative assessment.

Chapter 45: Mental State Examination, Capacity, Risk, and Common Syndromes

Orientation

Psychiatric assessment combines the person's story, observed mental state, physical and neurological assessment, collateral information, longitudinal course, and social context. Symptoms can arise from primary mental illness, drugs, withdrawal, infection, neurological disease, endocrine disturbance, trauma, sleep loss, or severe stress. Safety and therapeutic alliance are developed together. Distress does not remove autonomy, and a diagnostic label does not determine capacity or risk.

Setting and history

Create privacy, introduce roles, explain confidentiality limits, and use an interpreter when needed. Address medical instability, intoxication, withdrawal, agitation, self-harm, violence, abuse, neglect, and dependent safety. Avoid crowding or confrontation.

Begin with what happened, why now, and what the person wants. Establish onset, triggers, course, function, sleep, appetite, energy, concentration, mood, anxiety, unusual experiences, substances, medicines, physical symptoms, and previous episodes. Ask about treatment, trauma, admissions, housing, relationships, culture, and supports.

Collateral information may be essential when cognition, psychosis, intoxication, communication, or risk limits history. Seek consent where possible. Clinicians may receive information without disclosing confidential details. Document source and discrepancies neutrally.

Mental state examination

The mental state examination describes the current encounter. Appearance includes dress, hygiene, nutrition, injury, and self-care. Behaviour includes engagement, agitation, slowing, movements, impulsivity, hostility, withdrawal, and response to unseen stimuli. Describe rapport without moral judgement.

Describe speech rate, volume, amount, latency, rhythm, articulation, and prosody. Mood is sustained subjective emotion; affect is observed range, intensity, stability, reactivity, and congruence. Reported and observed emotion can differ.

Thought form describes connections: linear, circumstantial, tangential, disorganised, racing, blocked, perseverative, or sparse. Content includes guilt, hopelessness, grandiosity, obsessions, delusions, self-harm, violence, and reference. Delusions are fixed false beliefs outside cultural context; explore certainty respectfully.

Perception includes hallucinations, illusions, depersonalisation, derealisation, and dissociation. Visual hallucinations raise neurological, substance, sleep, sensory, and delirium causes but also occur in psychiatric illness. Ask meaning and whether commands occur.

Cognition includes arousal, attention, orientation, memory, language, visuospatial and executive function. Inattention and fluctuation suggest delirium. Insight concerns awareness and interpretation; judgement is demonstrated decision-making. Neither means agreement with the clinician.

Formulation and differential diagnosis

Summarise predisposing, precipitating, perpetuating, and protective factors across biological, psychological, and social domains. State syndrome, severity, function, risks, strengths, likely diagnoses, and alternatives. A formulation explains this person's presentation and guides care; it is not a list of labels.

Exclude medical and substance causes according to phenotype. Sudden onset, late first presentation, fluctuating attention, abnormal observations, focal neurology, catatonia, new seizures, head injury, fever, endocrine

features, medication change, or atypical hallucinations increase concern. Tests are targeted; routine broad screening can produce incidental findings.

Decision-making capacity

Capacity is specific to the decision and time. The person must understand relevant information, retain it long enough, use or weigh it, and communicate a choice. Provide interpreters, communication aids, pain relief, sensory support, simple explanations, and treatment of reversible impairment. Capacity can fluctuate and may differ between simple and complex decisions.

An apparently unwise choice does not prove incapacity. Delusion affects capacity only if it prevents understanding or weighing the specific decision. If capacity is absent, follow emergency necessity, advance directives, substitute decision-makers, best-interest standards, and local law. Use the least restrictive alternative and document evidence, support offered, reasoning, and review plan.

Suicide and self-harm risk

Ask directly and calmly about wishing to die, suicidal thoughts, plans, intent, preparation, access to means, rehearsal, previous attempts, recent self-harm, substance use, agitation, hopelessness, psychosis, pain, loss, and reasons for living. Asking does not implant the idea. Explore what stopped action and whether risk is changing.

Risk cannot be reduced to a score or low-medium-high label. Formulate dynamic and enduring factors, foreseeable scenarios, protective resources, access to lethal means, and ability to seek help. Recent discharge, escalating intent, command hallucinations, severe agitation, intoxication, and high-lethality preparation increase urgency.

Safety planning identifies personal warning signs, internal coping, social distraction, trusted contacts, professional services, and means restriction. It is collaborative and specific, not a promise or no-suicide contract. Arrange the level of observation and care required, communicate across transitions, and involve supports with consent or necessary safety disclosure.

Violence, vulnerability, and neglect

Assess recent threats or acts, intent, targets, access to weapons, persecutory beliefs, command hallucinations, intoxication, impulsivity, previous violence, legal restrictions, and escalating stress. Most people with mental illness are not violent; substance use, past behaviour, situational factors, and access often predict more strongly than diagnosis.

Also assess vulnerability to exploitation, domestic violence, trafficking, homelessness, impaired self-care, wandering, financial abuse, and risk to children or dependent adults. Duty to protect or report follows local law and imminent risk. De-escalation uses calm voice, space, choices, reduced stimulation, clear limits, and attention to pain, fear, intoxication, and communication needs.

Depressive syndromes

Depression involves persistent low mood or loss of interest with changes in sleep, appetite, energy, movement, concentration, guilt, hopelessness, and suicidal thinking. Grief and adversity can cause intense distress without a disorder, but context does not exclude major depression. Bipolar history, psychosis, substances, thyroid disease, anaemia, sleep disorder, pain, and medicines must be considered.

Treatment combines psychological, social, physical, and medication care according to severity, preference, prior response, pregnancy, and risk. Antidepressants take time and can cause gastrointestinal, sleep, sexual, or activation effects. Severe psychotic, catatonic, life-threatening, or resistant depression may require electroconvulsive therapy.

Anxiety, obsessive-compulsive, and trauma syndromes

Anxiety disorders produce excessive fear, worry, avoidance, arousal, or panic. Panic can mimic arrhythmia, asthma, seizure, endocrine disease, or intoxication. Cognitive behavioural and exposure therapies reduce avoidance; antidepressants support persistent disorders. Long-term benzodiazepines cause tolerance, dependence, falls, and cognitive impairment.

Obsessions are intrusive unwanted thoughts, images, or urges; compulsions are repeated behaviours or mental acts intended to reduce distress.

Post-traumatic stress involves re-experiencing, avoidance, altered mood and cognition, and hyperarousal after trauma. Trauma-informed care emphasises choice, control, collaboration, and avoidance of unnecessary re-enactment. Detailed trauma retelling is not required during every acute assessment.

Mania and bipolar disorder

Mania is a sustained abnormal elevation or irritability with increased energy plus reduced need for sleep, pressured speech, racing thought, distractibility, increased activity, grandiosity, and risky behaviour, causing marked impairment or psychosis. Hypomania is less severe and does not cause marked impairment or psychosis. Antidepressant exposure, stimulants, steroids, thyroid disease, neurological illness, and substances can mimic or precipitate symptoms.

Acute mania may require antipsychotic or mood-stabilising treatment, reduced stimulation, sleep, and admission. Long-term therapy considers lithium, anticonvulsants, antipsychotics, relapse prevention, pregnancy, organ function, metabolism, and interactions. Dehydration and interacting drugs promote gastrointestinal, neurological, and cardiac lithium toxicity.

Psychosis and catatonia

Psychosis includes delusions, hallucinations, disorganisation, and impaired reality testing. Schizophrenia-spectrum illness also involves negative symptoms and cognitive dysfunction, but first-episode diagnosis requires longitudinal exclusion of mood, substance, neurological, autoimmune, endocrine, infectious, and developmental causes.

Antipsychotics reduce positive symptoms but can cause sedation, metabolic disease, prolactin elevation, QT prolongation, movement disorders, and malignant syndrome. Use shared choice, monitoring, and the lowest effective dose. Clozapine benefits resistant illness and suicide risk but requires blood and systemic monitoring.

Catatonia causes combinations of immobility, mutism, posturing, negativism, excitement, echophenomena, and autonomic disturbance. It occurs in mood, psychotic, medical, neurological, and toxic conditions. Recognise it urgently because dehydration, thrombosis, pressure injury, malnutrition, and malignant

deterioration occur. Benzodiazepine challenge and electroconvulsive therapy are key treatments; antipsychotics can worsen some cases.

Substance-related syndromes

Ask nonjudgmentally about substances, prescribed medicines, route, amount, frequency, last use, tolerance, withdrawal, overdose, and goals. Toxicology has detection windows; a positive result does not prove current impairment.

Alcohol and sedative withdrawal can cause seizure, delirium, and death. Opioid overdose causes respiratory depression and responds to ventilation and naloxone. Stimulants can cause agitation, psychosis, hyperthermia, ischaemia, and rhabdomyolysis. Treatment combines safety, harm reduction, withdrawal care, relapse prevention, and medication.

Eating and personality-related syndromes

Eating disorders involve restriction, bingeing, compensation, body-image disturbance, or avoidant intake at any body size. Assess weight trajectory, cardiovascular stability, electrolytes, glucose, blood count, liver, bone, hormones, purging, exercise, and suicide. Refeeding causes dangerous phosphate and fluid shifts; instability needs specialist care.

Enduring personality patterns affect emotion, relationships, impulse control, and self-concept, but diagnosis requires longitudinal context and should not become pejorative shorthand. During crisis, validate distress, maintain consistent boundaries, formulate triggers and skills, and avoid splitting care teams. Evidence-based psychological therapies can reduce self-harm and improve function.

TTS module 2: Phenomenology, dynamic risk, capacity, and psychiatric medical safety

Psychiatric assessment begins by describing experiences and observable functions before assigning a syndrome. "Hearing a voice" may represent an auditory hallucination, intrusive thought, dissociative experience, misinterpreted environmental sound, sleep-transition phenomenon, trauma memory, substance effect, sensory impairment, delirium, or psychosis. Clarify location, sensory quality, controllability, frequency, context, meaning, distress, commands, and insight. Precise phenomenology improves both diagnosis and rapport because the person is not forced into the clinician's vocabulary.

The interview is itself an examination. Notice whether arousal fluctuates, attention can be sustained, questions are understood, answers remain goal-directed, and behaviour changes with topics or environmental stimuli. Appearance is described without equating unconventional dress with illness. Psychomotor slowing, agitation, tremor, rigidity, stereotypy, akathisia, intoxication, and response to unseen stimuli are observations, not interpretations. Speech rate, latency, volume, spontaneity, prosody, and articulation can reveal mania, depression, thought disorder, anxiety, catatonia, neurological disease, or medication effects.

Mood is the person's sustained internal emotional state; affect is the observed expression during the encounter. Affect is described by range, intensity, stability, reactivity, and congruence with thought content. Tearfulness can occur in grief, depression, anxiety, neurological disease, pain, and relief. Apparent emotional flatness may reflect negative symptoms, severe depression, parkinsonism, medication, cultural communication, or guardedness. A single examination cannot establish a longstanding personality or functional baseline.

Thought form concerns how ideas connect. Circumstantial speech eventually reaches the point through excessive detail; tangential speech departs and does not return. Flight of ideas contains rapid understandable links, often in mania; loosening of associations makes links difficult to follow. Thought blocking is sudden interruption, while perseveration repeats a response despite a changed question. Thought content includes beliefs, preoccupations, obsessions, guilt, hopelessness, reference, grandiosity,

somatic concern, self-harm, and harm to others. Explore evidence, alternatives, conviction, effect, and cultural context without arguing.

An obsession is an intrusive unwanted thought, image, or urge that causes distress and is usually recognised as originating in one's own mind. A compulsion is a behaviour or mental act performed to neutralise fear or follow a rigid rule. Delusions are held with pathological conviction despite counterevidence and outside shared cultural context, but the boundary requires humility. Overvalued ideas occupy an intermediate space. Insight is dimensional: a person may recognise that others disagree, accept possible illness, and still retain strong conviction.

Cognition is screened whenever presentation is acute, atypical, late in onset, medically complicated, or fluctuating. Test arousal and attention before memory. Months backward, digit span, or following a multistep command may reveal inattention that orientation misses. Delirium is suggested by acute change, fluctuation, and impaired attention, but psychosis and delirium can coexist. Visual hallucinations, abnormal vital signs, focal neurology, recent medication change, withdrawal, infection, organ failure, or altered sleep–wake cycle increase concern for a medical cause.

Capacity is a functional legal-clinical assessment, not a global cognitive label. Define the exact decision and information required. The person must understand the nature, purpose, material benefits, risks, and alternatives; retain them long enough; use or weigh them in relation to values; and communicate a stable choice. A person can have capacity to accept food but not manage a complex high-risk discharge. Capacity can return after intoxication, delirium, pain, fear, or communication barriers are treated.

The reasoning process, not agreement with clinicians, is decisive. A patient may rationally refuse a recommended operation because preserving a particular function matters more than survival probability. A delusion impairs capacity only when it materially prevents understanding or weighing the decision. Support includes interpreter, hearing aid, visual material, simplified chunks, repetition, trusted support, and a quieter time. When capacity is absent, emergency necessity, advance instruction, appointed decision-maker, or best-interest law guides the least restrictive option, with explicit review.

Suicide assessment is a formulation of foreseeable pathways, not a score. Ask directly about passive death wishes, active thoughts, method, access, preparation, rehearsal, intent, timeframe, recent self-harm, previous attempts, rescue expectations, and what interrupted action. High-lethality planning, concealment, escalating intent, intoxication, agitation, psychotic commands, severe insomnia, unbearable pain, recent discharge, and sudden losses can raise acute danger. Prior attempts and enduring illness influence baseline vulnerability. Protective factors are tested for reliability rather than merely listed.

A useful risk formulation states the current risk state relative to the person's baseline, likely scenarios, triggers, means, warning signs, capacity to collaborate, supports, and interventions that change probability. "Low risk" can falsely imply certainty and does not specify action. Safety planning identifies personal warning signs, coping actions, people and places that provide distraction, direct support contacts, professional crisis routes, and practical restriction of lethal means. A no-suicide promise does not predict or prevent action. Handover must include what changed and what to do if it recurs.

Violence risk is similarly dynamic and contextual. Prior violence, current intent, target, access to weapons, intoxication, escalating persecutory belief, command phenomena, impulsivity, acute humiliation, and breakdown of support are more informative than diagnosis alone. Most people with mental illness are not violent and are more often victims. De-escalation preserves distance and exit routes, reduces stimulation, uses one calm speaker, offers choices, addresses pain and fear, and sets clear respectful limits. Restraint or emergency medication is a last-resort safety intervention requiring monitoring and review.

Depressive syndromes require persistent mood or anhedonia plus cognitive, somatic, and behavioural change with impairment. Grief can include intense sadness and preserved waves of connection; major depression can coexist with bereavement. Bipolar depression is important because antidepressant monotherapy may destabilise some patients. Thyroid disease, anaemia, sleep apnoea, pain, neurodegeneration, corticosteroids, alcohol, and other drugs can mimic or worsen depression. Treatment combines psychological therapy,

social intervention, exercise and sleep care, medication, and for severe psychotic, catatonic, or life-threatening illness, electroconvulsive therapy.

Mania is defined by abnormally elevated or irritable mood and increased energy with reduced need for sleep, pressured speech, accelerated thought, distractibility, increased goal-directed activity, grandiosity, and risky behaviour. Reduced need for sleep differs from insomnia because the person feels rested. Marked impairment, psychosis, or need for admission makes the episode manic rather than hypomanic. Stimulants, corticosteroids, antidepressants, thyroid excess, neurological disease, and substances are sought. Acute care restores sleep and safety while managing financial, sexual, driving, and interpersonal consequences.

Psychosis is a syndrome, not synonymous with schizophrenia. First-episode assessment examines substance use, mood episodes, developmental history, trauma, cognition, seizures, movement, autoimmune or infectious features, endocrine disease, and medication. Negative symptoms such as reduced motivation, speech, pleasure, and social engagement overlap with depression, sedation, poverty, stigma, and cognitive impairment. Antipsychotic choice balances prior response with metabolic, movement, prolactin, cardiac, sexual, sedation, and pregnancy effects. Baseline and longitudinal physical monitoring are part of psychiatric treatment.

Catatonia is a motor-behavioural syndrome of stupor, mutism, posturing, waxy flexibility, negativism, echophenomena, stereotypy, agitation, grimacing, or autonomic instability. It occurs with mood disorders, psychosis, neurological disease, infection, metabolic illness, and medication or withdrawal. Complications include dehydration, malnutrition, aspiration, thrombosis, pressure injury, contracture, infection, and malignant hyperthermia-like deterioration. A benzodiazepine challenge can support diagnosis, and electroconvulsive therapy is highly effective. Antipsychotic escalation can worsen malignant forms.

Eating disorders can be medically unstable at any body size. Assess weight trajectory, restriction, bingeing, vomiting, laxatives, diuretics, exercise, fluid manipulation, fainting, chest symptoms, menstruation or endocrine change, and suicidality. Bradycardia, hypotension, hypothermia, electrolyte disturbance, hypoglycaemia, corrected-QT change, organ injury, and severe malnutrition require specialist care. Refeeding increases insulin, driving

phosphate, potassium, and magnesium into cells while sodium and water are retained; thiamine deficiency and cardiac failure can follow. Planned nutrition and close biochemical monitoring prevent avoidable harm.

Substance assessment uses nonjudgmental specificity: substance, formulation, dose, route, frequency, last use, tolerance, withdrawal, overdose, combinations, and desired change. Alcohol and sedative withdrawal can cause seizure and delirium; opioid overdose is primarily ventilatory failure; stimulants can produce hyperthermia, ischaemia, psychosis, and rhabdomyolysis. Toxicology detects exposure within a window and does not prove impairment or cause. The final formulation integrates psychiatric syndrome, medical and substance alternatives, function, capacity, dynamic risk, strengths, culture, and a collaborative plan proportionate to the current threat.

Retrieval prompts

One. Which domains form the mental state examination?

Two. How is decision-making capacity tested?

Three. What belongs in a suicide-risk formulation?

Four. How do mania and hypomania differ?

Five. Why is catatonia an emergency?

Six. Which physical risks accompany eating disorders?

Concise answers

One. Appearance, behaviour, speech, mood, affect, thought form and content, perception, cognition, insight, and judgement.

Two. Assess understanding, retention, weighing, and communication for the specific decision after providing support.

Three. Current thoughts, plan, intent, preparation, means, history, dynamic and enduring factors, scenarios, protections, and help-seeking ability.

Four. Mania causes marked impairment, admission, or psychosis; hypomania does not.

Five. Immobility or excitement can rapidly cause dehydration, thrombosis, malnutrition, autonomic instability, and malignant deterioration.

Six. Arrhythmia, hypotension, electrolyte and glucose disturbance, organ injury, bone loss, refeeding syndrome, and suicide.

Source map

Original synthesis informed by Talley and O'Connor, mental state, cognition, capacity, and risk assessment; Katzung and OpenStax Pharmacology, psychotropic and substance-related therapy; Guyton and Hall, neural, sleep, stress, and autonomic physiology; Robbins, neurological and systemic mimics; and OpenStax Medical-Surgical Nursing, crisis, safety, and therapeutic communication.

Chapter 46: Integrated Ward Presentation, Handover, and Longitudinal Care

Orientation

Clinical care is the repeated conversion of incomplete information into safe action. A useful clinician identifies physiological threats, represents the problem concisely, ranks plausible diagnoses, chooses discriminating tests, begins proportionate treatment, measures response, and revises the model. Ward care adds coordination across people, shifts, disciplines, and settings. Good handover preserves reasoning and future tasks, not merely historical facts. Longitudinal care links today's episode to prevention, function, and patient goals.

First contact and immediate priorities

Before full history, treat threats to airway, breathing, circulation, consciousness, glucose, temperature, bleeding, seizure, infection, or pain. Use patterns and trends; normal pressure does not exclude shock. Confirm identity, language, allergies, precautions, pregnancy possibility, and decision support. Explain roles and seek consent while emergency necessity permits stabilisation. Record pre-intervention state because treatment response helps distinguish mechanisms.

Problem representation

A problem representation compresses context, tempo, syndrome, severity, and discriminating findings into one sentence. Use qualifiers such as acute or chronic, focal or diffuse, inflammatory or non-inflammatory, and stable or progressive. Include negatives only when reliable and discriminating. Avoid premature labels and update the representation whenever data or treatment response changes the model.

Problem list and differential diagnosis

Build an active problem list separating syndromes, confirmed diagnoses, complications, chronic conditions affecting care, functional or social barriers, and preventive needs. Merge duplicates while preserving causal relationships. Rank problems by physiological threat, reversibility, time sensitivity, and patient priority.

For each unresolved syndrome, generate a focused differential using anatomy, mechanism, and tempo. Include common likely causes, dangerous alternatives, and diagnoses made plausible by this patient's exposures or comorbidity. A long unranked list is not safer than a short reasoned one.

State evidence for and against leading possibilities and what would change the ranking. Diagnostic time can be therapeutic when the patient is stable; immediate empirical action is needed when delay risks irreversible harm. Revisit alternatives after treatment failure rather than simply escalating the original therapy.

Investigation strategy

Every test should confirm a syndrome, distinguish alternatives, measure severity, identify cause, establish baseline, monitor therapy, or screen for complication. Predict how each result changes action; if none would, reconsider testing. Collect good specimens before treatment when safe, match imaging to anatomy, and interpret against baseline and context. Verify unexpected results. Communicate critical findings through a closed loop confirming recipient, meaning, action, and escalation.

Treatment plan and therapeutic trial

Write treatment around goals and mechanisms. Specify drug, indication, dose, route, interval, duration or review date, monitoring, and stop conditions. Include non-drug measures, source control, nutrition, mobility, devices, prophylaxis, rehabilitation, and communication. A plan should allow another clinician to continue safely.

Therapeutic trials require a defined target, time frame, and alternative explanation. A diuretic trial may target work of breathing, weight, urine, and congestion; a bronchodilator trial targets airflow and symptoms. Lack of

response may reflect wrong diagnosis, inadequate delivery, advanced irreversible disease, or insufficient time.

Balance benefits over different horizons. Intensive glucose control may not help an acute admission, while preventing hypoglycaemia does. Long-term preventive drugs may be temporarily held during shock but need explicit restart review. Patient preferences influence whether burdens are acceptable.

Medication reconciliation

Reconciliation compares medicines the patient actually used before admission with current orders and the intended discharge regimen. Sources include patient, caregiver, pharmacy, primary care, dispensing record, and previous notes. Resolve name, formulation, dose, timing, indication, adherence, recent changes, non-prescription products, and allergies or intolerances.

Classify each difference as intentional or unintentional. Common errors include omissions, duplication, wrong concentration, failure to adjust kidney function, continuation of temporary therapy, and accidental restart of a previously harmful drug. High-risk transitions involve anticoagulants, insulin, opioids, steroids, anticonvulsants, immune suppression, and medicines causing withdrawal.

Document held medicines with reason and restart criteria. A discharge list without explanation transfers ambiguity to the patient and next clinician. Simplify schedules where possible and confirm access, affordability, dexterity, cognition, and monitoring.

Daily ward review

Review overnight events, concerns, observation trends, fluid and intake, elimination, mobility, pain, wounds, devices, results, medicines, and response. Examine active risks and ask bedside staff about change. For each problem, state status, interpretation, action, monitoring, and escalation. Identify discharge barriers and remove unnecessary devices or restrictions. Specify observation frequency, treatment limits, whom to call, and trigger thresholds; concern overrides a reassuring score.

Escalation and uncertainty

Escalate when physiology worsens, diagnosis remains unstable, required expertise or procedure exceeds the setting, or staffing and monitoring cannot safely deliver the plan. Communicate urgency first, then concise context, findings, actions, response, and specific request. Persist through hierarchy if the patient remains unsafe.

Uncertainty should be calibrated and documented: what is known, likely, possible but dangerous, and currently unsupported. Avoid false reassurance and defensive overtesting. Safety-net uncertainty through observation, repeat examination, trigger thresholds, and planned follow-up.

Diagnostic error often arises from anchoring, premature closure, framing, availability, and failure to integrate discordant data. Deliberate pauses ask: What else could this be? What does not fit? What would I expect next? Has treatment changed the probability? Team diversity and respectful challenge improve detection.

Structured handover

Handover states identity, severity, active syndrome, relevant background, recent change, findings, treatment response, tasks, risks, escalation, and goals. Lead with action-critical information. Each task needs what, why, when, owner, expected result, and abnormal response. Use closed-loop questions and restatement. Update confidential lists and directly communicate unresolved danger.

Documentation

Notes should be timely, factual, attributable, and useful, recording perspective, findings, interpretation, differential, decisions, consent, capacity, advice, and follow-up. Distinguish observation, reports, and inference. Review copied text, correct errors transparently, describe behaviour without judgement, and document uncertainty respectfully.

Discharge planning

Discharge planning begins at admission with baseline function, environment, supports, transport, medicines, cognition, equipment, and follow-up capacity. Stability means safe management outside current monitoring. Communicate diagnoses, unresolved questions, results, treatment, medication changes, pending tests with owners, warning symptoms, care instructions, restrictions, appointments, and contacts. Use teach-back, accessible written information, confirmed supply, monitoring, and early review when high risk.

Longitudinal and preventive care

After acute recovery, address the cause and the conditions that made harm more likely. Review vaccination, smoking, alcohol, nutrition, exercise, sleep, mental health, dental care, screening, contraception or pregnancy plans, falls, bone health, cardiovascular risk, and treatment adherence according to relevance.

Multimorbidity creates competing recommendations. Prioritise treatments with meaningful benefit, acceptable burden, and alignment with goals. Coordinate specialists so one organ's plan does not destabilise another. A named generalist or primary clinician often provides continuity and prevents fragmentation.

Track function and quality of life alongside biomarkers. Recovery from sepsis, trauma, cancer, surgery, and critical illness can involve weakness, cognition, pain, anxiety, and financial strain. Rehabilitation and social support are disease treatment, not optional extras.

Shared decisions and goals of care

Shared decision-making presents options, likely outcomes, uncertainty, and burdens, then elicits values and preferences. Decision aids support but do not replace conversation. Capacity, urgency, and emotional distress determine how information is paced.

Goals-of-care planning clarifies desired outcomes and limits when deterioration occurs. A resuscitation decision addresses cardiopulmonary arrest, not all treatment. Document which interventions remain appropriate,

who participated, the patient's reasoning and values, and when the decision should be reviewed.

Palliative care treats symptoms, communication, family needs, and future planning alongside disease-directed treatment. Recognising that cure is unlikely should intensify comfort and coordination rather than signal abandonment.

Quality, safety, and learning

After harm or near harm, stabilise, disclose appropriately, document facts, support those affected, and preserve evidence. Analyse workload, design, communication, staffing, policies, equipment, and individual actions. Create specific owned changes; standardisation and better design are stronger than education alone. Audit effects. A just culture distinguishes human error, risky shortcuts, and reckless conduct while preserving accountability.

TTS module 2: Clinical synthesis, closed-loop handover, transitions, and continuity

Safe ward care depends on making the patient's current state reconstructable by another clinician. A record can contain every historical fact and still fail if it does not identify what is dangerous now, what changed, what remains uncertain, what response is expected, and who owns the next action. Clinical synthesis compresses information without discarding causal structure. The goal is not brevity alone; it is the smallest representation that preserves decisions.

A problem representation includes patient context, tempo, dominant syndrome, severity, discriminating positives and reliable negatives, and modifiers such as immune state or recent procedure. "Older frail patient with two days of progressive hypoxaemic respiratory failure, focal consolidation, and shock after aspiration" directs action better than "chest infection." The representation remains provisional. New rash, absent response to antibiotics, falling haemoglobin, or pulmonary oedema should force revision rather than being appended to an unchanged label.

The problem list separates confirmed diagnoses from unresolved syndromes, treatment complications, chronic conditions affecting immediate choices,

functional barriers, and preventive issues. Each item should have status, interpretation, action, monitoring, and escalation threshold. Causal links reduce duplication: acute kidney injury may result from sepsis and nephrotoxins, then alter antibiotic dose and fluid tolerance. Ranking follows threat, reversibility, time sensitivity, and patient priority, not the order problems were discovered.

Differential diagnosis is a ranked prediction of mechanisms. Include the most likely explanation, dangerous alternatives that require exclusion, and patient-specific possibilities created by exposure, medication, anatomy, or immune state. State what evidence supports and contradicts each leading hypothesis. A diagnosis with no discordant data may reflect inadequate testing rather than certainty. Ask what should happen next if the model is correct; an unexpected trajectory is evidence and should prompt a diagnostic pause.

Tests are chosen for their expected decision impact. Before ordering, predict the result that would start, stop, narrow, escalate, or postpone an action. A test with no plausible management consequence may create false positives and downstream harm. Timing and specimen quality belong to the order: cultures after antibiotics, a superficial swab for deep infection, or a troponin without a time relation can answer a different question. Unexpected results are checked against identity, sampling, assay limitations, baseline, and clinical coherence before treatment.

A therapeutic trial is interpretable only when its target and timeframe are defined. A fluid challenge targets stroke volume or tissue perfusion and is reassessed for congestion. Diuresis targets work of breathing, weight, venous pressure, oedema, and oxygenation while kidney and pressure are monitored. Analgesia should improve pain and function without unacceptable sedation. If the target does not change, consider insufficient delivery, wrong mechanism, irreversible disease, or inadequate observation time. Repeating the same intervention without a response model converts treatment into ritual.

Ward review begins with trajectory: overnight events, patient and nursing concerns, vital-sign trends, oxygen and fluid support, intake and output, mobility, cognition, pain, wounds, bowel and bladder, sleep, and new results. Examine the systems that could explain change rather than performing a generic ritual. Devices are reviewed daily because lines, catheters, drains,

oxygen, and restrictions create infection and immobility. Discharge barriers are treated as active problems from admission, not discovered after medical stability.

Escalation communication leads with urgency and the requested action. State identity, current physiological threat, relevant context, key findings and trends, interventions already given, response, and what help is needed. “I need you to review now for possible ongoing haemorrhage” is safer than a long chronology with the request buried. Closed-loop communication requires the receiver to acknowledge the message and clarify the plan. If deterioration persists and the first route fails, escalate through the hierarchy; politeness does not require passive delay.

Handover transfers prospective responsibility. For every outstanding task specify what must happen, why it matters, when it is due, who owns it, the expected result, and the action for an abnormal result. “Check potassium” is incomplete; “repeat potassium at eight because replacement finished, and call if outside the stated range” is actionable. High-risk pending cultures, imaging, anticoagulation decisions, ceilings of care, or deteriorating trends require direct verbal transfer in addition to an electronic list.

Handover lists should distinguish stable background from active change and be maintained as clinical tools, not copied historical archives. Remove completed tasks, reconcile contradictory entries, and date uncertainty. Confidentiality requires secure systems and minimum necessary information, but fear of privacy breach should not prevent essential safety communication within the care team. The patient can often identify errors during bedside handover, provided sensitive information and privacy are handled deliberately.

Medication reconciliation follows medicines across transitions. Build the pre-admission list from several sources because patient memory, dispensing records, primary-care lists, and previous discharge summaries each contain gaps. Determine formulation, actual dose, timing, indication, adherence, last dose, and non-prescription products. Compare this with inpatient orders and classify every difference as intentional, temporary, or erroneous. Particular danger surrounds anticoagulants, insulin, opioids, corticosteroids, anticonvulsants, transplant drugs, Parkinson therapy, and agents causing withdrawal.

Every intentionally held medicine needs a reason, monitoring plan, and restart or permanent-stop decision. Kidney or liver recovery may require dose re-escalation. A diuretic stopped during shock may be needed when congestion returns. An anticoagulant withheld for bleeding needs a documented reassessment balancing recurrent bleeding and thrombosis. At discharge, the patient and next clinician need a reconciled final list that explicitly says what started, changed, stopped, and why, not merely a new set of prescriptions.

Documentation separates observation, reported history, and inference. "Patient shouted and left the room" is an observation; "aggressive and noncompliant" is a judgement that loses context. Record decision reasoning, uncertainty, consent, capacity, alternatives discussed, and safety-net advice. Copy-forward text must be reverified because stale examinations and resolved diagnoses create false authority. Correct errors transparently rather than deleting the history of a consequential change.

Discharge is safe when necessary care can be delivered in the destination, not when hospital tests are finished. Confirm baseline and current mobility, cognition, continence, eating, wound or device care, housing, caregiver capacity, transport, medicine access, monitoring, and follow-up. A person who cannot climb home stairs or understand a new insulin regimen is not safely discharged by a normal blood test. Rehabilitation, equipment, community nursing, and social services are therapeutic components of the plan.

The discharge communication should state the presenting syndrome, confirmed diagnoses, significant exclusions, treatment and response, current function, medication changes, pending results with named ownership, warning symptoms, restrictions, and appointments. Teach-back tests whether the explanation was understood: ask the patient to describe what they will do, not whether they understand. Written material should match language, literacy, vision, and cognition. Provide enough supply to bridge to follow-up and verify that services accepted the referral.

Pending results are a common ownership failure. Ordering clinicians or services remain responsible until a documented transfer is accepted. Every pending pathology, culture, imaging report, biopsy, or drug level needs a named reviewer, expected availability, communication method, and action threshold. Automated inbox delivery is not ownership if nobody is scheduled

to check it. Abnormal results require closed-loop contact and documentation of attempts, advice, and escalation.

Longitudinal care integrates disease-specific plans that may conflict. Tight pressure control can worsen falls; anti-inflammatory therapy can destabilise diabetes or infection; fluid restriction can impair nutrition; anticoagulation can reduce stroke while increasing bleeding. A coordinating generalist identifies which outcomes matter most and simplifies monitoring and treatment burden. Function, symptom load, caregiver strain, financial cost, and quality of life are tracked alongside biomarkers.

Shared decision-making presents reasonable options, likely outcomes, uncertainty, and burden, then elicits the patient's values. It is not asking the patient to choose without guidance. Use absolute rather than only relative benefits where possible and distinguish immediate from delayed effects. A cardiopulmonary-resuscitation decision concerns treatment after arrest and does not mean withholding antibiotics, analgesia, or ward care. Goals and treatment limits should identify which interventions remain appropriate, the reasoning, participants, and review trigger.

Safety systems learn from discrepancies and near misses as well as harm. First stabilise the patient and disclose according to ethical and legal duties. Preserve facts, timelines, equipment, and relevant records. Analysis examines workload, staffing, interface design, medication packaging, policy conflict, supervision, communication, and individual decisions. "Be more careful" is weak prevention. Standardisation, forcing functions, simpler processes, independent checks for high-risk steps, and feedback on outcomes are stronger when matched to the failure mechanism.

Continuity is achieved when the next person can see not only what happened but what the team believed, why it acted, how the patient responded, and what remains to be done. This chain extends from bedside to shift, ward to procedure, hospital to community, and acute episode to prevention. Reliable care is therefore a property of information design and accountable relationships as much as individual diagnostic knowledge.

Retrieval prompts

One. What belongs in a problem representation?

Two. How should a diagnostic differential be ranked?

Three. What makes a therapeutic trial interpretable?

Four. Which elements make handover actionable?

Five. What information must accompany discharge?

Six. How should teams learn from clinical error?

Concise answers

One. Patient context, tempo, dominant syndrome, severity, and discriminating findings.

Two. By likelihood, danger if missed, patient-specific plausibility, and time sensitivity.

Three. A defined intervention, physiological target, time frame, measurement, and alternative interpretation.

Four. Current severity, active problems, changes, treatment response, owned timed tasks, risks, escalation, and goals.

Five. Diagnoses, unresolved questions, key results, medication changes, pending tests with ownership, warning signs, care instructions, restrictions, and follow-up.

Six. Stabilise and disclose, analyse system and human contributors, implement owned design changes, and audit outcomes.

Source map

Original synthesis informed by Talley and O'Connor, clinical presentation, examination, reasoning, and communication; Guyton and Hall, integrative physiology and response measurement; Katzung and OpenStax Pharmacology, medication reconciliation and monitoring; Robbins, disease mechanisms and diagnostic pathology; OpenStax Medical-Surgical Nursing, ward care, handover, discharge, safety, and longitudinal support; and all preceding textbook chapters.

Chapter 47: Oesophageal, Gastric, Intestinal, and Anorectal Disease

Orientation

The gastrointestinal tract propels contents, protects its lining, digests food, absorbs nutrients and water, and hosts a dense microbiome. Disease appears through dysphagia, reflux, pain, vomiting, bleeding, altered stool, malabsorption, obstruction, inflammation, and nutritional change. Reasoning should localise the process, identify mechanism and tempo, estimate consequence, and recognise haemorrhage, perforation, ischaemia, obstruction, or cancer early.

Oesophageal disease

Dysphagia means impaired swallowing. Oropharyngeal dysphagia causes difficulty initiating a swallow, coughing, nasal regurgitation, wet voice, or aspiration. Oesophageal dysphagia produces a sensation of food sticking after initiation. Difficulty first with solids suggests mechanical narrowing; difficulty with solids and liquids from the outset suggests disordered motility, although patterns overlap. Painful swallowing suggests ulceration or inflammation.

Gastro-oesophageal reflux occurs when gastric contents repeatedly cross an incompetent antireflux barrier. Transient lower sphincter relaxation, hiatus hernia, obesity, pregnancy, delayed emptying, and impaired clearance contribute. Symptoms include heartburn and regurgitation, but cough, laryngeal symptoms, and chest pain are less specific. Persistent acid exposure can cause erosive oesophagitis, peptic stricture, and metaplastic Barrett oesophagus, which increases adenocarcinoma risk.

Management uses weight reduction when relevant, individual meal and positional measures, and proton-pump inhibition. Long-term therapy needs a clear indication and the lowest effective regimen. Progressive dysphagia, bleeding, anaemia, weight loss, persistent vomiting, or late-onset symptoms require endoscopy.

Achalasia results from loss of inhibitory myenteric neurons, causing impaired lower sphincter relaxation and absent organised peristalsis. Regurgitation, weight loss, and aspiration may follow. Manometry establishes the motor pattern; endoscopy excludes obstructing cancer. Pneumatic dilation, surgical myotomy, or endoscopic myotomy reduces outflow resistance. Oesophageal cancer commonly causes progressive dysphagia and weight loss; staging determines endoscopic, surgical, systemic, radiation, or palliative treatment.

Gastritis and peptic ulcer disease

The stomach is protected by mucus, bicarbonate, epithelial restitution, mucosal blood flow, and prostaglandins. Injury follows imbalance between these defences and acid, pepsin, *Helicobacter pylori*, non-steroidal anti-inflammatory drugs, bile, alcohol, severe physiological stress, or immune attack. Gastritis is histological inflammation; dyspepsia is a symptom syndrome and does not prove mucosal injury.

Helicobacter pylori colonises gastric mucus and produces chronic inflammation. Distribution and host response influence duodenal ulcer, gastric ulcer, atrophy, adenocarcinoma, and mucosa-associated lymphoid lymphoma. Diagnose with urea breath testing, stool antigen, or biopsy according to context. Acid suppression, antibiotics, and sometimes bismuth are combined in locally appropriate eradication regimens. Confirm eradication after treatment, allowing adequate time off agents that suppress the organism and produce false-negative results.

Non-steroidal drugs reduce protective prostaglandins and can cause ulceration without warning pain. Risk rises with age, previous ulcer, anticoagulation, corticosteroids, antiplatelet therapy, higher dose, and severe comorbidity. Avoid unnecessary exposure, use gastroprotection when indicated, and address *Helicobacter* infection. Ulcer complications include haemorrhage, perforation, penetration, and gastric outlet obstruction. Sudden severe pain with peritonism suggests perforation and requires resuscitation, antibiotics, imaging, and urgent surgical review.

Gastrointestinal bleeding

Haematemesis or melaena usually indicates upper bleeding, while haematochezia often indicates lower bleeding but can accompany brisk upper haemorrhage. Occult loss may present as iron deficiency. Initial care prioritises airway risk, perfusion, venous access, crossmatch, coagulopathy correction, and targeted transfusion. Pulse, posture, skin perfusion, cognition, urine, lactate, and serial haemoglobin matter more than one apparently normal observation.

After stabilisation, endoscopy localises and treats many upper lesions. Proton-pump inhibition supports ulcer haemostasis. Suspected variceal bleeding requires vasoactive therapy, antibiotics, and urgent endoscopic control. Lower bleeding may require colonoscopy, computed tomographic angiography, radiology, or surgery. Antithrombotic reversal and restart balance bleeding against thrombosis.

Coeliac disease and malabsorption

Coeliac disease is an immune-mediated enteropathy triggered by gluten in genetically susceptible people. Villous injury reduces absorptive surface and may cause diarrhoea, bloating, weight loss, iron deficiency, folate deficiency, low bone density, neuropathy, infertility, or minimal gastrointestinal symptoms. Test immunoglobulin A tissue-transglutaminase with total immunoglobulin A while gluten is still being eaten; use alternative serology when immunoglobulin A deficient. Small-bowel biopsy commonly confirms diagnosis. Treatment is lifelong gluten exclusion with dietetic support and correction of deficiencies.

Malabsorption can result from pancreatic enzyme deficiency, reduced bile, mucosal disease, bacterial overgrowth, infection, lymphatic obstruction, or short bowel. Distinguish global malabsorption from a specific deficit. Stool, weight, blood count, iron, vitamins, coagulation, and disease-specific tests guide evaluation. Albumin also reflects inflammation, liver disease, renal loss, and dilution.

Diarrhoea and infectious enteritis

Diarrhoea may be osmotic, secretory, inflammatory, fatty, or caused by altered motility. Acute disease is often infectious or drug related. Blood, fever, severe pain, dehydration, recent antibiotics, immune suppression, travel, outbreaks, or prolonged symptoms change testing and treatment. Replace water and electrolytes; oral rehydration works because sodium-glucose cotransport remains effective in many secretory illnesses. Antibiotics are selective because some infections are self-limited and treatment can worsen toxin-mediated disease or resistance.

Clostridioides difficile ranges from diarrhoea to toxic megacolon after microbiome disruption, commonly from antibiotics or healthcare exposure. Diagnose only compatible symptomatic patients, stop unnecessary precipitating drugs, use recommended enteral therapy, and apply infection-control precautions. Recurrence may require staged antibiotics, microbiota-based therapy, or specialist management.

Chronic diarrhoea requires review of medicines, diet, travel, endocrine disease, inflammation, malabsorption, and functional syndromes. Nocturnal symptoms, blood, weight loss, anaemia, fever, or late onset are warning features. Faecal inflammatory markers distinguish inflammation from functional disease but not its cause.

Inflammatory bowel disease

Ulcerative colitis causes continuous mucosal inflammation beginning in the rectum and extending proximally through colon. Crohn disease causes patchy transmural inflammation anywhere from mouth to anus, with strictures, fistulas, abscesses, and perianal disease. Both can produce diarrhoea, bleeding, pain, fatigue, weight loss, anaemia, and extraintestinal inflammation of joints, skin, eyes, or hepatobiliary tract.

Diagnosis integrates endoscopy, histology, imaging, stool testing, and exclusion of infection. Treatment matches location and severity, using aminosalicylates mainly in ulcerative colitis, corticosteroids for induction rather than maintenance, immune modulators, biologic or targeted small-molecule therapy, nutrition, and surgery. Before immune suppression, assess infection and vaccination. Acute severe colitis requires admission, thrombosis

prevention, infection testing, intensive monitoring, intravenous corticosteroid, and early rescue-therapy and surgical planning. Toxic megacolon, perforation, uncontrolled bleeding, obstruction, abscess, or dysplasia can require urgent surgery.

Diverticular and functional bowel disease

Diverticula are mucosal outpouchings through weak points in the colonic wall. Diverticulosis is often asymptomatic. Diverticulitis causes focal inflammation, usually left-lower pain, fever, and bowel change; complications include abscess, perforation, fistula, and obstruction. Computed tomography defines complicated disease. Stable uncomplicated cases may be managed selectively without antibiotics, while systemic illness or complications require antibiotics, drainage, or surgery.

Irritable bowel syndrome causes recurrent abdominal pain related to defecation or altered stool frequency or form without structural explanation. It is a disorder of gut-brain interaction, involving sensation, motility, microbiota, immune signalling, and context. Make a positive diagnosis after appropriate warning-feature assessment. Explanation, dietetic guidance, soluble fibre, symptom-targeted drugs, and psychological therapies can help. Avoid repetitive low-yield investigations that reinforce fear.

Constipation may reflect low fibre or fluid intake, immobility, medicines, metabolic disease, neurological dysfunction, pelvic-floor dyssynergia, slow transit, pain, or obstruction. Examine for impaction and warning features. Treatment combines cause correction, activity, appropriate fibre, laxatives, and bowel routine. Fibre can worsen pain with obstruction or severe loading.

Obstruction, ischaemia, and anorectal disease

Mechanical obstruction causes colicky pain, vomiting, distension, and failure to pass stool or flatus. Adhesions and hernias commonly obstruct small bowel; cancer, volvulus, and diverticular stricture commonly obstruct colon. Strangulation compromises blood flow and may produce continuous pain, fever, tachycardia, acidosis, local tenderness, or peritonism. Resuscitate, decompress when indicated, correct electrolytes, image promptly, and seek surgical review.

Mesenteric ischaemia causes pain that may initially exceed examination findings. Embolism, thrombosis, low-flow states, or venous thrombosis can progress rapidly to necrosis. Lactate may be normal early. Computed tomographic angiography and urgent vascular or surgical treatment are required when suspected.

Haemorrhoids are vascular cushions that become symptomatic through bleeding or prolapse. Anal fissure causes severe pain with defecation, usually from a linear tear. Perianal abscess causes constant pain and requires drainage; fistula may follow. New rectal bleeding should not automatically be attributed to benign disease, especially with bowel change, anaemia, weight loss, or cancer risk.

TTS module 2: Gastrointestinal localisation, mucosal injury, obstruction, bleeding, and inflammatory control

Gastrointestinal symptoms are interpreted by following the path of luminal contents. Difficulty initiating a swallow localises to coordinated mouth, pharynx, upper sphincter, cranial nerves, and airway protection. Retrosternal sticking after an effective swallow localises to oesophageal transit. Vomiting suggests proximal propulsion failure or central stimulation; distension and obstipation suggest distal obstruction; diarrhoea reflects excess secretion, impaired absorption, mucosal inflammation, osmotic solute, or rapid transit. The same symptom can arise at several levels, so anatomy, timing, content, and physiological consequence are combined.

Oropharyngeal dysphagia is dangerous because aspiration can be silent. Wet voice, repeated throat clearing, cough with meals, prolonged chewing, nasal regurgitation, recurrent pneumonia, weight loss, and inability to handle secretions indicate impaired safety or efficiency. Stroke, neurodegeneration, neuromuscular disease, structural tumour, frailty, and medication are common mechanisms. Bedside assessment identifies immediate risk, while videofluoroscopic or endoscopic swallowing assessment characterises phases and tests strategies. Texture modification may reduce aspiration yet also reduce hydration and pleasure, so it requires review rather than indefinite prescription.

Oesophageal dysphagia is classified by solids, liquids, progression, intermittency, pain, regurgitation, and weight. Progressive solids-first difficulty

suggests a narrowing lumen from cancer, peptic stricture, ring, or eosinophilic inflammation. Solids and liquids from onset suggest impaired motility, but advanced mechanical disease eventually affects both. Achalasia retains food and saliva above a non-relaxing lower sphincter and can cause nocturnal regurgitation and aspiration. Endoscopy excludes obstructing or mucosal disease, contrast studies show anatomy and emptying, and manometry defines motor physiology.

The antireflux barrier consists of lower sphincter tone, diaphragmatic crura, intra-abdominal oesophageal length, angle, clearance, and gastric emptying. Reflux symptoms do not always correlate with acid exposure; functional heartburn and hypersensitivity can persist despite normal mucosa. Proton-pump inhibitors work best when timed before meals that activate pumps and are highly effective for acid-mediated injury, but indefinite use should have a documented indication. Persistent dysphagia, anaemia, bleeding, vomiting, weight loss, or progressive symptoms shifts the task from empirical therapy to endoscopic diagnosis.

Gastric mucosal defence depends on a hydrophobic mucus–bicarbonate layer, tight epithelium, rapid restitution, blood flow, and prostaglandins. Non-steroidal drugs injure both systemically through cyclooxygenase inhibition and topically, while *Helicobacter pylori* creates chronic active inflammation adapted to the mucus niche. Their combination markedly increases ulcer risk. Dyspepsia alone cannot identify an ulcer. Testing and treating *Helicobacter* requires attention to proton-pump inhibitors, antibiotics, and bismuth, which suppress organism burden and cause false negatives; eradication is confirmed after an appropriate interval.

Upper gastrointestinal haemorrhage is a circulation and airway problem before an endoscopic diagnosis. Haematemesis, coffee-ground vomit, melaena, or brisk haematochezia indicate possible bleeding, but initial haemoglobin may remain normal because whole blood has been lost before equilibration. Assess shock, orthostasis where safe, cognition, perfusion, urine, ongoing output, liver disease, antithrombotics, and comorbidity. Transfusion is individualised rather than driven by one threshold, because excessive transfusion can raise portal pressure while inadequate oxygen delivery harms heart and brain.

Ulcer haemostasis combines injection, thermal, mechanical, or topical endoscopic methods according to lesion. Acid suppression stabilises clot by raising gastric pH. Rebleeding prompts repeat endoscopy, radiological embolisation, or surgery based on source and physiology. Anticoagulant reversal considers drug, timing, kidney function, bleeding severity, and thrombosis risk; restart is an active decision once haemostasis is secure. In suspected variceal bleeding, vasoactive splanchnic therapy and antibiotics begin before endoscopy because portal physiology and infection both influence outcome.

Diarrhoea is mechanistically classified. Osmotic diarrhoea falls with fasting because poorly absorbed solute retains water. Secretory diarrhoea continues despite fasting through active ion secretion or impaired absorption. Inflammatory diarrhoea includes blood, leukocytes, protein leakage, fever, and mucosal injury. Fatty diarrhoea reflects maldigestion or malabsorption and produces bulky difficult-to-flush stool, weight loss, and vitamin deficiency. Rapid transit can blend these patterns. Medication review includes laxatives, metformin, magnesium, antibiotics, acid suppressants, immune therapies, and many less obvious agents.

Oral rehydration succeeds because sodium and glucose cotransport remains functional in many secretory illnesses. Water alone does not replace sodium loss and very concentrated sweet drinks can worsen osmotic diarrhoea. Stool testing is selected for blood, fever, severe illness, immunosuppression, outbreak, travel, recent antibiotics, duration, and public-health consequences. Molecular panels detect more organisms but can identify colonisation or prolonged shedding. Antibiotics benefit selected invasive, severe, or high-risk infections and can harm in toxin-producing syndromes where bacterial lysis increases complication risk.

Clostridioides difficile disease follows loss of colonisation resistance and toxin-mediated colitis. Testing only unformed stool from a compatible syndrome prevents treating asymptomatic carriage. Nucleic-acid detection is sensitive for toxigenic organisms but does not prove active toxin injury; multistep algorithms and clinical probability improve interpretation. Stop unnecessary inciting antibiotics and acid suppression, treat with recommended enteral therapy, and avoid antimotility drugs in severe colitis.

Fulminant disease with shock, ileus, megacolon, or rising organ failure needs early surgical and multidisciplinary review.

Celiac disease is an antigen-driven small-bowel enteropathy with manifestations extending beyond diarrhoea. Iron deficiency may be the only sign; osteoporosis, infertility, neuropathy, dermatitis, and abnormal liver tests can occur. Tissue-transglutaminase immunoglobulin A is interpreted with total immunoglobulin A while gluten remains in the diet. Starting exclusion before testing can heal mucosa and obscure diagnosis. Histology, serology, genetics in selected uncertainty, dietetic review, correction of deficiency, and response establish a durable diagnosis that justifies lifelong dietary burden.

Inflammatory bowel disease is defined by pattern and objective inflammation after infection and mimics are excluded. Ulcerative colitis remains mucosal and continuous from rectum; Crohn disease is discontinuous and transmural, creating stricture, fistula, abscess, and perianal disease. Faecal calprotectin reflects neutrophilic intestinal inflammation but cannot distinguish infection from immune disease. Endoscopic appearance and histology define mucosal activity, while cross-sectional imaging examines small bowel and penetrating complications. Symptoms can persist despite healed inflammation through bile-acid diarrhoea, irritable bowel physiology, dysmotility, or medication.

Acute severe ulcerative colitis is a systemic emergency. Frequent bloody stool, tachycardia, fever, anaemia, inflammatory burden, dilatation, and nutritional decline require admission, infection testing, thrombosis prophylaxis, intravenous corticosteroid, fluid and electrolyte care, and early surgical involvement. Response is assessed within days, not weeks. Failure leads to rescue biological or calcineurin therapy or colectomy. Delaying surgery in toxic megacolon, perforation, uncontrolled bleeding, or worsening systemic toxicity increases mortality.

Mechanical obstruction produces upstream fluid and gas accumulation, vomiting, third-space loss, electrolyte disturbance, bacterial translocation, and aspiration risk. The pain of simple obstruction is often colicky as peristalsis works against a block. Continuous pain, local tenderness, fever, tachycardia, acidosis, closed-loop anatomy, reduced wall enhancement, pneumatosis, or portal venous gas suggests strangulation and ischaemia. Nasogastric decompression relieves selected proximal distension but does not remove the lesion. Surgery is urgent when perfusion or perforation is threatened.

Mesenteric ischaemia should be suspected when severe pain exceeds early examination, particularly with atrial fibrillation, atherosclerosis, low-flow shock, vasoconstrictive drugs, thrombophilia, or portal disease. Arterial embolism, arterial thrombosis, venous thrombosis, and non-occlusive low flow require different interventions. Lactate can be normal before bowel necrosis and must not reassure. Computed-tomographic angiography provides rapid vascular and bowel assessment, followed by revascularisation, anticoagulation where appropriate, resuscitation, antibiotics, and resection of non-viable bowel.

Constipation is more than stool frequency. Assess straining, hard stool, incomplete evacuation, blockage sensation, manual manoeuvres, medication, mobility, hydration, neurological disease, endocrine disturbance, and pelvic-floor function. Overflow diarrhoea can occur around impaction and can be mistaken for a primary secretory illness. New bleeding, anaemia, weight loss, severe pain, vomiting, or late change requires structural evaluation. Anorectal bleeding should be visualised and examined because haemorrhoids, fissure, inflammation, cancer, prolapse, and proximal disease coexist. The final gastrointestinal habit is to distinguish a symptom-control problem from a threatened bowel, blood volume, airway, or nutritional state.

Retrieval prompts

One. How does the pattern of dysphagia help localisation?

Two. Which mechanisms most commonly cause peptic ulcer disease?

Three. What are the immediate priorities in major gastrointestinal bleeding?

Four. How do ulcerative colitis and Crohn disease differ structurally?

Five. Which findings suggest strangulated obstruction or mesenteric ischaemia?

Six. Why should rectal bleeding not automatically be labelled haemorrhoidal?

Concise answers

One. Initiation difficulty and aspiration suggest oropharyngeal disease; solids-first suggests narrowing; solids and liquids together suggest dysmotility.

Two. *Helicobacter pylori* and non-steroidal anti-inflammatory drug injury, modified by acid and impaired mucosal defence.

Three. Protect the airway when needed, restore perfusion, obtain access and blood, correct major coagulopathy, and arrange definitive localisation and haemostasis.

Four. Ulcerative colitis is continuous and mucosal in colon; Crohn disease is patchy, transmural, and can affect the entire tract.

Five. Continuous severe pain, systemic toxicity, peritonism, acidosis, organ hypoperfusion, or pain disproportionate to early examination.

Six. Benign anorectal disease can coexist with colorectal cancer, inflammation, or proximal bleeding, particularly when warning features are present.

Source map

Original synthesis informed by Guyton and Hall, gastrointestinal motility, secretion, digestion, and absorption; Robbins, oesophageal, gastric, intestinal, inflammatory, vascular, and neoplastic pathology; Katzung and OpenStax Pharmacology, acid suppression, antimicrobial therapy, anti-inflammatory treatment, laxatives, and supportive drugs; Talley and O'Connor, gastrointestinal history and examination; OpenStax Medical-Surgical Nursing, bleeding, obstruction, inflammatory bowel disease, nutrition, ostomy, and perioperative care; and OpenStax Microbiology, enteric pathogens and host-microbiome interactions.

Chapter 48: Liver, Biliary Tract, and Pancreatic Disease

Orientation

The liver receives portal blood carrying nutrients, toxins, and microbial products, while arterial blood supplies oxygen. Hepatocytes synthesise proteins, regulate fuels, process drugs and hormones, convert ammonia to urea, and form bile. The biliary tract delivers bile for fat absorption; the pancreas supplies digestive enzymes and bicarbonate while its islets regulate metabolism. Disease can present through hepatocyte injury, impaired bile flow, synthetic failure, portal hypertension, infection, obstruction, inflammation, or malignancy. Symptoms may remain subtle until reserve is exhausted.

Interpreting liver tests

Alanine and aspartate aminotransferases rise with hepatocyte membrane injury but do not directly measure function. Alkaline phosphatase rises with cholestasis and also originates from bone; gamma-glutamyl transferase can support hepatic origin but lacks specificity. Bilirubin rises from overproduction, impaired uptake or conjugation, hepatocellular dysfunction, or biliary obstruction. Albumin reflects synthesis over weeks but is also altered by inflammation, nutrition, renal loss, and fluid state. Prothrombin time changes rapidly when hepatic clotting-factor synthesis fails, although vitamin K deficiency and anticoagulants are alternatives.

Interpret the pattern, magnitude, tempo, symptoms, medicines, alcohol, metabolic risk, infection exposure, haemodynamics, and previous results. A hepatocellular pattern emphasises aminotransferases; a cholestatic pattern emphasises alkaline phosphatase and bilirubin. Very high aminotransferases suggest ischaemia, toxins such as paracetamol, or acute viral or immune hepatitis, but degree does not reliably predict outcome. Ultrasound evaluates ducts, gallbladder, liver texture, vessels, and ascites; further serology, cross-sectional imaging, elastography, or biopsy follows the question.

Jaundice

Unconjugated hyperbilirubinaemia commonly follows haemolysis, ineffective erythropoiesis, or reduced conjugation. Conjugated bilirubin is water soluble and darkens urine; pale stool and pruritus suggest reduced bile reaching intestine. Examine for anaemia, fever, confusion, chronic liver signs, mass, tenderness, and sepsis. Painful jaundice with fever may be ascending cholangitis, while painless progressive jaundice raises malignant obstruction. Jaundice with coagulopathy, altered consciousness, hypoglycaemia, acidosis, or kidney injury requires urgent specialist assessment.

Acute hepatitis and liver failure

Acute hepatitis can be viral, drug-induced, toxic, immune, ischaemic, metabolic, or pregnancy related. Take a precise timeline of prescribed, non-prescription, recreational, and herbal exposures; dose and timing are crucial in paracetamol poisoning. Viral risks include travel, food and water, blood, sexual exposure, injection, tattoos, occupation, birth country, and immune status. Autoimmune hepatitis may occur without classic autoimmune symptoms.

Acute liver failure means new hepatic injury with coagulopathy and encephalopathy in a person without established cirrhosis. Cerebral oedema, hypoglycaemia, lactic acidosis, infection, bleeding, kidney failure, and circulatory collapse can follow. Stabilise, check glucose repeatedly, stop potential toxins, administer cause-specific antidotes when indicated, avoid sedating diagnostic confusion, and discuss early with a transplant centre. N-acetylcysteine replenishes glutathione in paracetamol toxicity and may benefit selected non-paracetamol failure.

Chronic liver disease and cirrhosis

Repeated injury from alcohol, metabolic dysfunction-associated steatotic liver disease, chronic viral hepatitis, immune disease, cholestatic disorders, iron or copper accumulation, vascular disease, or drugs can produce fibrosis. Cirrhosis reorganises liver into regenerative nodules separated by scar, increasing intrahepatic resistance and reducing function. Compensated disease may be silent; decompensation appears as ascites, variceal bleeding, encephalopathy, jaundice, infection, or kidney dysfunction.

Alcohol-associated disease ranges from steatosis through inflammatory hepatitis to cirrhosis. Assess amount, pattern, dependence, withdrawal risk, nutrition, and psychosocial context without judgement. Abstinence support is central. Metabolic steatotic disease is associated with central adiposity, insulin resistance, dyslipidaemia, and cardiovascular risk. Weight reduction and metabolic treatment can improve disease, but fibrosis stage predicts liver outcomes.

Chronic hepatitis B can integrate into hepatocytes and cause cancer even without cirrhosis; serology distinguishes infection phase, immunity, and previous exposure. Antiviral suppression is used according to viral activity, fibrosis, immune suppression, pregnancy, and transmission risk. Direct-acting antiviral combinations cure most hepatitis C, but advanced fibrosis still requires surveillance and complication care. Vaccination prevents hepatitis A and B but not hepatitis C.

Portal hypertension and ascites

Portal hypertension diverts blood through collateral vessels, enlarges the spleen, and promotes ascites. Oesophageal or gastric varices may bleed catastrophically. Screening and primary prevention use endoscopic assessment and non-selective beta blockade or banding according to risk. Acute bleeding requires resuscitation that avoids excessive pressure, vasoactive splanchnic therapy, antibiotics, endoscopic control, and consideration of a transjugular intrahepatic portosystemic shunt when bleeding persists or high-risk criteria are met.

Ascites results from portal pressure, splanchnic vasodilation, renal sodium retention, and reduced effective arterial volume. Diagnostic paracentesis is required for new, worsening, or hospitalised ascites to identify infection and alternative mechanisms. Spontaneous bacterial peritonitis may produce pain, fever, encephalopathy, kidney injury, or little local tenderness. Treat promptly with appropriate antibiotics and albumin in selected patients. Sodium moderation, diuretics, therapeutic drainage with albumin replacement when large volumes are removed, and transplant evaluation address recurrent ascites. Avoid non-steroidal drugs and other renal insults.

Encephalopathy and hepatorenal dysfunction

Hepatic encephalopathy causes altered attention, sleep, behaviour, movement, and consciousness through impaired toxin handling, inflammation, and altered neurotransmission. Search for bleeding, infection, constipation, dehydration, electrolyte disturbance, kidney failure, sedatives, and portosystemic shunting. Lactulose promotes colonic ammonia disposal and is titrated to regular soft stool; rifaximin reduces recurrence. An ammonia concentration neither confirms nor excludes the syndrome by itself.

Kidney dysfunction in cirrhosis may result from hypovolaemia, sepsis, nephrotoxins, intrinsic renal disease, abdominal pressure, or hepatorenal physiology. Evaluate and treat reversible causes. Hepatorenal syndrome reflects profound renal vasoconstriction amid systemic and splanchnic vasodilation; albumin plus vasoconstrictor therapy can bridge selected patients to transplantation.

Liver cancer and transplantation

Cirrhosis and chronic hepatitis B increase hepatocellular carcinoma risk. Surveillance aims to find tumours suitable for ablation, resection, or transplantation. Larger or advanced disease may receive arterial, radiation, targeted, immune, or palliative therapy. Alpha-fetoprotein can support surveillance or assessment but is not independently diagnostic. Transplantation replaces function and removes portal hypertension but requires lifelong immune suppression, infection prevention, malignancy surveillance, and careful drug-interaction management.

Gallstone and biliary disease

Gallstones form when bile becomes supersaturated and crystals grow amid gallbladder stasis. Transient cystic-duct obstruction causes episodic biliary pain, typically steady upper abdominal pain after meals without persistent inflammation. Acute cholecystitis causes prolonged pain, fever, local tenderness, and inflammatory change; ultrasound is first-line. Treatment combines analgesia, selected antibiotics, and early cholecystectomy when suitable.

A common-duct stone can obstruct bile and cause jaundice or pancreatitis. Cholangitis is infected obstruction, classically producing fever, jaundice, and pain, with hypotension or confusion in severe disease. Give fluids, antibiotics, and urgent biliary drainage, usually by endoscopic retrograde cholangiopancreatography. This procedure is therapeutic when obstruction is likely but can cause pancreatitis, bleeding, perforation, or infection.

Acute and chronic pancreatitis

Acute pancreatitis is premature enzyme activation causing pancreatic and systemic inflammation. Gallstones and alcohol are common causes; triglycerides, drugs, procedures, trauma, calcium disorders, and genetic factors also contribute. Diagnose with characteristic pain plus raised pancreatic enzymes or imaging evidence. Severity depends on persistent organ failure and complications rather than enzyme height.

Provide early analgesia, balanced fluid resuscitation adjusted to response, oxygen when needed, and enteral nutrition as tolerated. Routine prophylactic antibiotics do not treat sterile necrosis. Identify cholangitis or persistent biliary obstruction requiring urgent intervention. Complications include necrosis, infected collections, pseudocyst, bleeding, thrombosis, respiratory failure, kidney injury, and shock; intervention is delayed and minimally invasive when clinically possible.

Chronic pancreatitis causes irreversible fibrosis, chronic pain, duct distortion, exocrine failure, diabetes, and increased cancer risk. Alcohol and smoking accelerate disease, but genetic, obstructive, immune, and idiopathic causes occur. Management includes abstinence support, nutrition, pancreatic enzyme replacement with meals, fat-soluble vitamin review, diabetes care, analgesia, and endoscopic or surgical treatment of selected obstruction and pain.

Pancreatic cancer often presents late with weight loss, back or epigastric pain, new diabetes, thrombosis, or painless obstructive jaundice. Pancreas-protocol imaging establishes resectability; surgery offers the main chance of cure, combined with systemic therapy. Biliary drainage, enzyme replacement, glucose management, analgesia, coeliac-plexus intervention, nutrition, and palliative care relieve burden.

TTS module 2: Hepatic reserve, portal failure, biliary sepsis, and pancreatic systemic injury

Liver disease is interpreted along three axes: hepatocyte injury, bile-flow impairment, and loss of functional reserve. Aminotransferases leak from injured cells and can fall because injury resolves or because few functioning hepatocytes remain. Alkaline phosphatase rises with cholestasis but also bone turnover. Bilirubin reflects production, conjugation, transport, and excretion. Prothrombin time changes quickly when short-lived clotting factors are not synthesised, while albumin changes slowly and is strongly influenced by inflammation, renal loss, distribution, and nutrition. No single “liver test” measures the whole organ.

Pattern is quantified by comparing aminotransferase elevation with alkaline-phosphatase elevation relative to their upper reference limits. A hepatocellular pattern directs attention to viral, toxic, immune, ischaemic, and metabolic injury. A cholestatic pattern suggests duct obstruction, infiltrative disease, drug injury, or small-duct disorders. Mixed injury broadens the differential. Magnitude and trajectory matter: very high aminotransferases favour ischaemia, paracetamol toxicity, or acute viral and immune hepatitis, whereas modest values can accompany advanced cirrhosis. Previous normal results do not exclude abrupt failure.

Jaundice is localised before imaging. Predominantly unconjugated bilirubin suggests overproduction, ineffective erythropoiesis, impaired uptake, or conjugation. Conjugated bilirubin enters urine and reflects hepatocyte excretory failure or obstruction. Pale stool means little pigment reaches intestine; pruritus supports retained bile constituents. Painful jaundice with fever and systemic illness suggests infected obstruction, while progressive painless jaundice raises pancreatic, biliary, or hepatic malignancy. Ultrasound rapidly tests duct dilatation, gallbladder, masses, portal flow, and ascites, after which cross-sectional or duct imaging is selected.

Drug-induced liver injury requires a complete exposure timeline. Prescription, non-prescription, herbal, bodybuilding, weight-loss, recreational, and intermittent medicines are included with start, stop, dose, and re-exposure. Injury may be predictable and dose-related, as with paracetamol, or idiosyncratic and delayed. A causality assessment considers known phenotype, latency, dechallenge, competing viral or immune disease, alcohol,

shock, and biliary obstruction. Rechallenge can provoke more severe injury and is generally avoided. Apparent “natural” products may contain several undeclared hepatotoxins.

Acute liver failure combines recent hepatic injury, coagulopathy, and encephalopathy without established cirrhosis. The danger extends beyond bleeding: cerebral oedema, hypoglycaemia, lactic acidosis, infection, vasodilatory shock, kidney injury, and respiratory failure can progress rapidly. Repeated glucose, neurological status, acid-base state, kidney function, coagulation, and cause-specific tests are followed in intensive care. Sedatives cloud encephalopathy assessment. N-acetylcysteine is given promptly for suspected paracetamol exposure and can benefit selected non-paracetamol failure. Early transplant-centre discussion occurs before irreversible multiorgan collapse.

Cirrhosis changes both structure and circulation. Fibrous septa and regenerative nodules increase sinusoidal resistance, while splanchnic vasodilation lowers effective arterial volume. Neurohormonal sodium retention expands total extracellular fluid even as kidneys perceive underfilling. Portosystemic collaterals bypass hepatic clearance and create varices. Splenic congestion lowers platelets, and reduced synthesis alters albumin and haemostasis. Conventional coagulation tests capture loss of procoagulant factors but not parallel loss of anticoagulant factors, so cirrhosis is a fragile rebalanced state with both bleeding and thrombosis risk.

Ascites should be sampled when new, worsening, or associated with hospital admission. Cell count and differential identify spontaneous bacterial peritonitis; culture yield improves when fluid is inoculated directly into blood-culture bottles. Albumin gradient estimates whether portal hypertension drives fluid formation, while total protein and clinical context refine cardiac, malignant, infectious, and pancreatic causes. A low peripheral white count or absent fever does not exclude peritonitis. New encephalopathy, kidney injury, pain, hypothermia, or hypotension may be the only manifestation.

Ascites treatment balances sodium retention against effective volume. Dietary sodium moderation and aldosterone-antagonist with loop-diuretic therapy mobilise fluid while weight, pressure, sodium, potassium, and kidney function are monitored. Overdiuresis causes kidney injury, hyponatraemia, encephalopathy, and cramps. Large-volume paracentesis relieves pressure

and breathing but can worsen circulatory dysfunction unless albumin is replaced according to volume and protocol. Refractory ascites prompts assessment for shunt, transplant, repeated drainage, or goal-directed palliation, with cardiac and encephalopathy risks considered.

Variceal bleeding is managed differently from ordinary ulcer bleeding. Restore perfusion without excessive volume that raises portal pressure, protect the airway when bleeding and consciousness threaten aspiration, begin splanchnic vasoactive therapy and antibiotics, and perform urgent endoscopic ligation or other control. Balloon or covered stent tamponade is a temporary bridge in uncontrolled haemorrhage, not definitive therapy. Early transjugular intrahepatic portosystemic shunt benefits selected high-risk bleeding but can worsen encephalopathy by increasing portal bypass.

Hepatic encephalopathy is a clinical syndrome of attention, sleep, cognition, motor control, and consciousness. Ammonia contributes but its blood concentration correlates poorly enough that it cannot confirm or stage the condition alone. Search for gastrointestinal bleeding, infection, constipation, dehydration, excessive diuresis, kidney failure, hypokalaemia, sedatives, and new shunting. Lactulose traps nitrogen in the colon and accelerates elimination, titrated to regular soft stool without dehydration. Rifaximin reduces recurrent episodes. Focal signs or atypical course require neurological alternatives.

Kidney injury in cirrhosis is not automatically hepatorenal syndrome. Volume loss, sepsis, nephrotoxins, obstruction, glomerular disease, acute tubular injury, abdominal pressure, and cardiac congestion are assessed first. Hepatorenal physiology reflects extreme splanchnic vasodilation and renal vasoconstriction without primary structural kidney disease. Albumin challenge and withdrawal of contributors support diagnosis, followed by vasoconstrictor plus albumin therapy in appropriate patients. Improvement may bridge to transplantation; untreated liver failure remains the underlying cause.

Metabolic dysfunction-associated steatotic liver disease is part of systemic insulin resistance and cardiovascular risk. Steatosis alone does not predict outcome as strongly as fibrosis stage. Weight reduction through sustainable nutrition and activity improves liver and metabolic physiology, while diabetes, lipids, pressure, and sleep apnoea require treatment. Alcohol and metabolic injury can coexist and amplify risk. Non-invasive fibrosis scores and

elastography identify people needing specialist assessment, surveillance, or complication care without biopsying everyone.

Biliary pain results from transient cystic-duct obstruction and is typically steady rather than truly colicky despite the traditional name. Persistent obstruction and inflammation cause cholecystitis, with local tenderness, fever, and imaging change. Common-duct stones produce cholestasis, pancreatitis, or cholangitis. Cholangitis is sepsis behind an obstructed duct; antibiotics cannot reliably sterilise it without drainage. Resuscitation and antimicrobials are followed by urgent endoscopic decompression, while surgical or radiological routes are alternatives when endoscopy fails or anatomy is altered.

Acute pancreatitis begins with premature enzyme activation and acinar injury, but systemic severity comes from inflammatory vascular leak and organ dysfunction. Diagnosis requires a compatible combination of characteristic pain, enzyme elevation, and imaging, not all three. Enzyme height does not grade severity and can be modest with delayed presentation or hypertriglyceridaemia. Persistent respiratory, cardiovascular, or renal failure predicts severe disease. Early balanced crystalloid is titrated to perfusion and congestion because indiscriminate aggressive fluid worsens lung and abdominal oedema.

Enteral nutrition is introduced early as tolerated because it preserves gut integrity and reduces infectious complications compared with prolonged parenteral feeding. Routine antibiotics do not prevent infection of sterile pancreatic necrosis. New deterioration after several days, gas within a collection, bacteraemia, or persistent sepsis raises infected necrosis. When possible, intervention is delayed until collections become organised and follows a step-up approach from drainage toward minimally invasive or surgical necrosectomy. Immediate intervention is reserved for specific emergencies such as uncontrolled infection, bleeding, obstruction, or compartment physiology.

Gallstone pancreatitis requires definitive prevention of recurrence. Urgent endoscopic retrograde cholangiopancreatography is indicated for cholangitis or persistent biliary obstruction, not for every pancreatitis with gallstones. Cholecystectomy during the same admission is preferred after mild disease when safe. Alcohol use, triglycerides, calcium, medicines, trauma,

procedures, anatomic variants, autoimmune disease, and genetics are assessed when stones are absent. Smoking independently worsens recurrent and chronic pancreatic disease.

Chronic pancreatitis destroys exocrine and endocrine tissue, producing pain, maldigestion, weight loss, fat-soluble vitamin deficiency, and diabetes prone to both hyper- and hypoglycaemia because glucagon reserve also falls. Pancreatic enzymes are taken with meals and snacks and adjusted to nutritional response; acid suppression can assist selected inadequate response. Persistent pain may arise from duct obstruction, pseudocyst, neuropathic change, ongoing exposure, or pancreatic cancer. Cross-sectional and endoscopic assessment directs drainage, stone treatment, resection, or pain care.

Pancreatic and hepatobiliary cancers often present through physiological obstruction rather than a palpable mass. Painless jaundice, dark urine, pale stool, pruritus, weight loss, back pain, new diabetes, thrombosis, or recurrent pancreatitis are warning patterns. Tissue diagnosis, resectability imaging, biliary drainage when clinically required, and systemic staging are coordinated so a procedure does not delay curative surgery or compromise tissue. Across liver, biliary, and pancreatic disease, the central task is to recognise when apparently modest biochemical change conceals exhausted reserve, infected obstruction, vascular decompensation, or systemic organ failure.

Retrieval prompts

One. Which tests reflect hepatocyte injury, cholestasis, and liver function?

Two. What defines acute liver failure?

Three. Which events mark decompensated cirrhosis?

Four. Why is diagnostic paracentesis important in hospitalised ascites?

Five. What is the immediate treatment logic for cholangitis?

Six. What determines severity in acute pancreatitis?

Concise answers

One. Aminotransferases indicate injury, alkaline phosphatase and bilirubin suggest cholestasis, and prothrombin time and contextualised albumin reflect synthesis.

Two. New hepatic injury with coagulopathy and encephalopathy without established cirrhosis.

Three. Ascites, variceal bleeding, encephalopathy, jaundice, infection, or hepatorenal dysfunction.

Four. It detects spontaneous bacterial peritonitis and distinguishes portal hypertensive from alternative causes.

Five. Restore perfusion, give antibiotics, and urgently decompress the infected obstructed biliary system.

Six. Persistent organ failure and local or systemic complications, not the height of pancreatic enzymes.

Source map

Original synthesis informed by Guyton and Hall, hepatic blood flow, bile, metabolism, pancreatic secretion, and portal physiology; Robbins, hepatitis, cirrhosis, biliary pathology, pancreatitis, and malignancy; Katzung and OpenStax Pharmacology, antiviral, antidote, portal-pressure, encephalopathy, diuretic, antibiotic, and immune-suppressive therapy; Talley and O'Connor, jaundice, chronic liver signs, abdominal examination, and clinical pattern recognition; OpenStax Medical-Surgical Nursing, liver failure, ascites, bleeding, biliary care, pancreatitis, nutrition, procedures, and transplantation; and OpenStax Microbiology, viral hepatitis and biliary infection.

Chapter 49: Gastrointestinal History, Examination, Investigations, and Acute Presentations

Orientation

Gastrointestinal assessment converts symptoms into an anatomical and physiological model while simultaneously judging urgency at the bedside. Pain, vomiting, bleeding, altered stool, jaundice, distension, weight change, and dysphagia can each arise from benign, inflammatory, obstructive, vascular, infectious, metabolic, toxic, or malignant disease. The safest approach defines onset and trajectory, localises the likely compartment, assesses hydration and perfusion, searches for peritonitis or obstruction, and uses investigations to answer explicit questions.

Symptom history

For pain, establish onset, first and current site, radiation, character, intensity, constancy or colic, progression, movement, meals, defecation, urination, menstruation, pregnancy possibility, posture, medicines, and prior episodes. Visceral pain is diffuse and midline because organs share autonomic afferents; parietal inflammation produces sharper local pain and guarding. Migration can be informative, but textbook patterns are imperfect. Sudden maximal pain suggests perforation, vascular catastrophe, torsion, rupture, or obstruction and deserves urgent evaluation.

Clarify vomiting frequency, volume, colour, blood, bile, faeculent quality, relation to meals, and ability to retain fluid. Early bilious vomiting suggests obstruction beyond the pylorus; repeated vomiting causes chloride and hydrogen loss, contraction alkalosis, potassium depletion, and kidney injury. Faeculent vomiting suggests distal obstruction or fistula. Distinguish vomiting from effortless regurgitation and from rumination.

Characterise dysphagia by initiation, solids, liquids, progression, pain, regurgitation, aspiration, neurological symptoms, and weight loss. Ask about

heartburn, early satiety, postprandial fullness, appetite, and previous ulcer or reflux treatment. Dyspepsia is a symptom cluster, not a final diagnosis.

For altered bowel habit, define the patient's baseline, frequency, consistency, urgency, nocturnal symptoms, incontinence, straining, incomplete evacuation, mucus, blood, black stool, steatorrhea, and relationship to diet or travel. Bright blood coating stool differs from blood mixed throughout it, but neither reliably excludes proximal disease. Melaena is black, sticky, and offensive, although iron and bismuth darken stool without bleeding. Document unintentional weight change rather than relying on appearance.

Jaundice history includes dark urine, pale stool, pruritus, pain, fever, confusion, bleeding, alcohol, medicines, supplements, travel, food and water, sexual and blood exposure, tattoos, procedures, occupation, family disease, and previous abnormal tests. Ask about abdominal swelling, ankle oedema, sleep reversal, concentration, bruising, and gastrointestinal haemorrhage when chronic liver disease is possible.

Context and risk

Review surgery, hernias, endoscopy, inflammatory bowel disease, cancer, radiation, liver or pancreatic disease, diabetes, vascular disease, immune suppression, and pregnancy. Previous abdominal surgery increases adhesion risk but does not prove adhesional obstruction. Medicines can cause ulceration, bleeding, dysmotility, diarrhoea, constipation, pancreatitis, hepatitis, and microbiome disruption. Specifically review non-steroidal anti-inflammatory drugs, antithrombotics, opioids, antibiotics, acid suppressants, immune therapies, metformin, glucagon-like peptide one agonists, laxatives, iron, and herbal products.

Dietary questioning should be neutral and specific. Include food access, restriction, supplements, alcohol, fluid, fibre, lactose or gluten avoidance, cultural practice, and eating-disorder behaviour when relevant. Family history may indicate colorectal, gastric, pancreatic, hepatic, inflammatory, or polyposis syndromes. Social history identifies sanitation, travel, infectious contacts, smoking, substance use, housing, caregiving, and capacity to manage nutrition or stomas.

Immediate physiological assessment

Before a detailed examination, assess airway risk from haematemesis or reduced consciousness, respiratory effort, pulse, blood pressure, posture, capillary refill, skin temperature, mental state, urine, pain, fever, and glucose. Look for shock, sepsis, major bleeding, dehydration, perforation, obstruction, acute liver failure, and metabolic disturbance. A normal haemoglobin early in acute haemorrhage does not exclude major loss because plasma and red cells are initially lost together.

General and abdominal examination

Observe distress, stillness or restlessness, nutrition, muscle wasting, pallor, jaundice, pigmentation, bruising, scratch marks, oedema, dehydration, odour, and cognition. Hands may show clubbing, palmar erythema, contracture, tremor, or asterixis. Examine eyes and mouth for pallor, jaundice, ulcers, glossitis, dehydration, and dental disease. Chronic liver signs support context but neither establish cause nor reliably measure severity.

Position the patient supine with exposure from lower chest to groin while preserving dignity. Inspect contour, scars, striae, veins, hernias, stomas, masses, pulsation, and movement with respiration. Auscultation before repeated palpation is traditional, but bowel-sound interpretation has limited accuracy; absent sounds alone do not diagnose ileus, and high-pitched sounds do not prove obstruction.

Palpate gently away from pain, watching the face. Distinguish voluntary guarding from involuntary rigidity. Rebound is not required when percussion, cough, movement, or light palpation already demonstrates peritoneal irritation. Define masses by site, size, surface, edge, consistency, mobility, tenderness, pulsatility, and movement with respiration. Assess liver edge and span, spleen enlargement, kidneys, bladder, aorta, and focal signs in clinical context. Forceful palpation of a suspected aneurysm is unsafe.

Percussion estimates liver, spleen, bladder, gas, mass, and ascites. Shifting dullness supports moderate free fluid but ultrasound is more sensitive. Examine groins and hernial orifices when obstruction or pain is unexplained. Digital rectal examination is selective: explain, obtain consent, use a chaperone according to policy, inspect first, and assess tenderness, mass,

stool, blood, and sphincter only when it will affect care. Pelvic, testicular, cardiovascular, respiratory, skin, joint, or neurological examination may be essential because abdominal symptoms often arise outside the abdomen.

Baseline investigations

Tests follow the syndrome. Blood count assesses anaemia, leukocytosis, thrombocytopenia, and marrow response. Electrolytes, urea, creatinine, glucose, magnesium, phosphate, and venous blood gas reveal volume and metabolic consequences. Liver tests define injury pattern; lipase supports pancreatitis. C-reactive protein tracks inflammation imperfectly. Coagulation, crossmatch, lactate, cultures, pregnancy testing, haemolysis studies, iron, vitamins, coeliac serology, viral serology, or immune markers are selected by context.

Interpret trends and pretest probability. A normal lactate does not exclude early mesenteric ischaemia. Mild lipase elevation occurs outside pancreatitis. Positive faecal occult blood does not localise bleeding. Stool testing is most useful with inflammatory features, outbreaks, travel, immune suppression, prolonged illness, or healthcare exposure. Avoid testing formed stool for infectious diarrhoea syndromes.

Imaging and endoscopy

Plain abdominal radiography has limited indications but can demonstrate some obstruction, perforation, toxic megacolon, or foreign bodies. An erect chest film may show free subdiaphragmatic gas, yet a negative film does not exclude perforation. Ultrasound is preferred for gallstones, biliary dilation, ascites, portal flow, and many pelvic questions. Computed tomography evaluates obstruction, inflammation, perforation, abscess, cancer, and trauma. Computed tomographic angiography is urgent for suspected active bleeding or mesenteric ischaemia.

Endoscopy directly evaluates mucosa, obtains biopsy, and treats bleeding, strictures, polyps, stones, or obstruction. Upper endoscopy addresses dysphagia, warning-feature dyspepsia, upper bleeding, and selected anaemia. Colonoscopy evaluates lower bleeding, bowel change, inflammation, surveillance, and screening after appropriate preparation. Capsule endoscopy and enterography assess small bowel but capsule

retention is possible in strictures. Endoscopic retrograde cholangiopancreatography is mainly therapeutic; magnetic resonance cholangiopancreatography is non-invasive diagnosis.

Acute pain and surgical consultation

Resuscitation and diagnosis proceed together. Keep oral intake appropriate to likely procedures, establish access, provide analgesia and antiemetic treatment, correct fluid and electrolytes, obtain cultures and give antibiotics when infection or perforation is likely, and involve surgery early when time-sensitive disease is possible. Analgesia does not meaningfully conceal important signs when examination is repeated.

Peritonism, free gas, strangulated hernia, closed-loop obstruction, bowel ischaemia, uncontrolled bleeding, ruptured aneurysm, ectopic pregnancy, testicular or ovarian torsion, and severe sepsis require immediate escalation. Frail, pregnant, immune-suppressed, very young, and older patients may show muted signs. Reassessment after treatment is part of diagnosis: record pain, observations, examination, urine, laboratory trends, and response.

TTS module 2: Bedside abdominal reasoning, physiological consequences, and time-critical investigation

The abdominal history is a physiological map. Pain onset and movement identify the first affected structure and later spread of inflammation. Vomit content estimates the level and duration of impaired transit. Stool colour, consistency, blood distribution, and nocturnal passage distinguish bleeding, inflammation, malabsorption, and altered motility. Jaundice, distension, weight change, and dysphagia add hepatobiliary, mechanical, nutritional, and upper-tract information. Urgency is judged simultaneously through perfusion, sepsis, peritonism, obstruction, and organ failure.

Visceral pain arises from stretch, contraction, ischaemia, and inflammation of internal organs and is poorly localised because afferents converge across spinal segments. Foregut structures tend to produce epigastric pain, midgut structures periumbilical pain, and hindgut structures lower abdominal pain. When inflammation reaches parietal peritoneum, pain becomes sharper and localised, worsens with movement or cough, and may produce involuntary

guarding. Migration from central to focal pain can reflect this transition, but older, immunosuppressed, pregnant, or neuropathic patients may have muted findings.

Sudden maximal abdominal pain raises perforation, arterial occlusion, aneurysm rupture, torsion, ectopic pregnancy, haemorrhage, or abrupt obstruction. Colicky waves imply contraction against intermittent or fixed resistance in bowel, ureter, or biliary tract, although biliary pain is often steady after onset. Pain out of proportion to early palpation is a warning for mesenteric ischaemia because mucosa and muscle become hypoxic before peritoneum is inflamed. Analgesia should be given promptly; repeated examination, not untreated suffering, preserves diagnostic information.

Vomiting causes consequences according to site and duration. Gastric hydrogen and chloride loss produces metabolic alkalosis, while volume contraction activates sodium retention and distal potassium and hydrogen loss. Severe potassium depletion impairs gut motility and cardiac conduction, worsening obstruction and vomiting. Later or distal bowel loss may include bicarbonate and create acidosis. Aspiration risk rises with reduced consciousness, obstruction, pregnancy, reflux, and ineffective airway reflexes. Nasogastric decompression is therapeutic in selected obstruction but is not a substitute for correcting strangulation or perforation.

Bleeding severity cannot be inferred from visible colour alone. Coffee-ground material indicates blood altered by gastric acid but can coexist with active bleeding. Melaena commonly follows upper bleeding yet can arise from small bowel or right colon; brisk upper haemorrhage can cause bright rectal blood. The first haemoglobin may be normal because red cells and plasma are lost together. Tachycardia, postural change, narrow pulse pressure, cool peripheries, altered cognition, urine decline, ongoing loss, and comorbidity guide resuscitation. Anticoagulants and liver disease modify both bleeding and reversal choices.

Examination starts before touch. A motionless patient avoiding cough suggests peritoneal irritation; restless movement is more typical of colic but not diagnostic. Observe breathing, colour, sweating, nutrition, hydration, jaundice, scars, distension, hernias, stomas, and pulsation. Expose adequately while preserving warmth and dignity. Severe abdominal pain

accompanied by shock, respiratory effort, or altered consciousness demands resuscitation before a complete regional examination.

Gentle palpation away from pain evaluates guarding, rigidity, tenderness, masses, and organ enlargement. Voluntary guarding can lessen with reassurance and slow expiration; involuntary rigidity persists. Rebound testing is unnecessary when light percussion, coughing, or movement already demonstrates peritoneal irritation and can inflict pain. A pulsatile mass with shock suggests aneurysm and should not be repeatedly pressed. Hernial orifices, testes, pelvis, back, chest, heart, skin, and peripheral vessels are included when they could explain the syndrome.

Bowel sounds have low discriminatory accuracy. High-pitched or frequent sounds may occur early in obstruction; reduced sounds occur with ileus, peritonitis, opioids, electrolyte disturbance, or even normal variation. Their presence cannot exclude obstruction and their absence cannot diagnose ileus. More useful evidence comes from distension, vomiting, obstipation, prior surgery, hernia, tenderness, imaging, and trajectory. Rectal examination is performed only when blood, mass, impaction, perianal disease, sphincter function, or operative planning makes the result relevant.

Laboratory tests measure consequences and competing mechanisms. Blood count evaluates anaemia, inflammation, and platelets, but early haemorrhage and neutropenic infection may be hidden. Electrolytes, kidney function, magnesium, phosphate, glucose, and blood gas reveal dehydration and acid-base effects. Lipase supports pancreatitis only in a compatible syndrome and may rise with kidney failure, bowel disease, and other illness. Lactate reflects hypoperfusion and metabolism but can remain normal during early mesenteric ischaemia. A reassuring biomarker never overrides a high-risk anatomical pattern.

Pregnancy testing is part of abdominal diagnosis whenever biologically possible because ectopic pregnancy, miscarriage, torsion, and pregnancy-related physiology change imaging and treatment. It should be asked and tested respectfully without relying on relationship status or reported contraception alone. In pregnancy, the enlarging uterus displaces organs, baseline leukocytes may be higher, and peritoneal signs can differ. Necessary ultrasound, magnetic resonance, or computed tomography is chosen by

diagnostic value; fear of imaging should not delay maternal life-saving diagnosis.

Imaging follows the question and stability. Ultrasound is rapid for gallstones, biliary dilatation, free fluid, pelvic organs, aorta, and selected bowel disease but depends on operator, body habitus, and gas. Computed tomography defines obstruction transition points, perforation, abscess, diverticulitis, malignancy, trauma, and many inflammatory patterns.

Computed-tomographic angiography is urgent for mesenteric ischaemia or active bleeding because ordinary portal-phase imaging may miss vascular anatomy. A patient with uncontrolled haemorrhage or clear surgical catastrophe may need theatre rather than a prolonged imaging pathway.

Endoscopy is both diagnostic and therapeutic but requires physiological preparation. Upper endoscopy can secure ulcer or variceal bleeding, biopsy dysphagia, and dilate selected strictures. Colonoscopy evaluates mucosa and removes lesions but bowel preparation can worsen dehydration, obstruction, kidney injury, and electrolyte imbalance. In severe colitis, full colonoscopy can increase perforation risk and limited endoscopy may suffice. Capsule endoscopy examines small bowel but can lodge at a stricture; patency assessment or cross-sectional enterography reduces this risk.

Obstruction is classified as mechanical or functional, partial or complete, small or large bowel, simple or strangulated, and open-loop or closed-loop. A closed loop is obstructed at two points, trapping a segment whose distension rapidly compromises venous then arterial flow. Adhesions, hernias, tumours, volvulus, inflammatory strictures, and intussusception are important causes. Resuscitation corrects fluid and electrolyte loss, aspiration is prevented, and surgery is involved early when closed-loop anatomy, ischaemia, perforation, incarcerated hernia, or failure of conservative care is possible.

Perforation releases gas, fluid, organisms, and digestive contents into peritoneum. Sudden pain, rigidity, sepsis, free gas, or focal inflammatory source requires antibiotics, resuscitation, and urgent source control. Free subdiaphragmatic gas may be absent, especially with contained, retroperitoneal, or small perforations. Immune suppression and corticosteroids can blunt pain and fever. The operative question is not merely whether a hole exists, but whether contamination, sepsis, ongoing leak, and tissue viability can be controlled non-operatively or require intervention.

Extra-abdominal disease remains in the differential. Inferior myocardial ischaemia, lower-lobe pneumonia, pulmonary embolism, diabetic ketoacidosis, adrenal crisis, porphyria, radiculopathy, and abdominal-wall pain can all present predominantly as abdominal discomfort. Chest, cardiac, metabolic, neurological, and skin findings prevent false localisation.

Acute gastrointestinal bleeding, obstruction, and inflammation require serial reassessment. Record pain and analgesia, observations, perfusion, mental state, urine, abdominal findings, haemoglobin, electrolytes, acid-base state, and ongoing output. A response to fluid does not prove the problem is solved; transient improvement can mask continued bleeding. New constant pain during colicky obstruction suggests ischaemia. Falling temperature or alertness may signal sepsis even when tenderness changes little.

Safe discharge after undifferentiated abdominal symptoms requires more than pain improvement. The patient must tolerate appropriate intake, maintain hydration, mobilise, understand medication, and have no evolving physiological or surgical warning. Explain explicit return triggers such as worsening or localising pain, repeated vomiting, blood, black stool, fever, fainting, abdominal swelling, jaundice, reduced urine, or inability to eat. Assign ownership for pending imaging, cultures, biopsies, or endoscopy. The acute abdomen is ultimately a time-series diagnosis built from anatomy, physiology, and repeated observation.

Retrieval prompts

One. Which features of abdominal pain suggest urgent vascular or perforating disease?

Two. What complications arise from persistent vomiting?

Three. Why can early haemoglobin and lactate be falsely reassuring?

Four. What is the correct role of bowel sounds?

Five. Which modality is preferred for suspected mesenteric ischaemia?

Six. Why is repeated examination essential in acute abdominal illness?

Concise answers

One. Sudden maximal pain, pain disproportionate to examination, peritonism, shock, acidosis, or rapid progression.

Two. Dehydration, chloride and hydrogen loss, metabolic alkalosis, potassium depletion, aspiration, and kidney injury.

Three. Haemoglobin may not fall before redistribution, and lactate may remain normal before advanced ischaemia.

Four. They are contextual observations with limited diagnostic accuracy and should not determine obstruction or ileus alone.

Five. Urgent computed tomographic angiography.

Six. Disease evolves, treatment changes physiology, and serial findings often discriminate benign from time-critical mechanisms.

Source map

Original synthesis informed by Talley and O'Connor, gastrointestinal history, general inspection, abdominal, rectal, hernia, and systemic examination; Guyton and Hall, pain referral, vomiting, motility, fluid, and acid-base consequences; Robbins, inflammatory, obstructive, vascular, hepatic, pancreatic, and malignant patterns; Katzung and OpenStax Pharmacology, medicine adverse effects, analgesia, antiemetics, fluids, and acute treatment; OpenStax Medical-Surgical Nursing, bedside assessment, stomas, procedures, bleeding, abdominal emergencies, and reassessment; and OpenStax Chemistry, Biology, and Microbiology, laboratory principles, metabolism, and infectious diagnostics.

Chapter 50: Malnutrition, Micronutrients, Enteral and Parenteral Support, and Obesity Care

Orientation

Nutrition supplies energy, amino acids, essential fatty acids, vitamins, minerals, water, and electrolytes while food also carries cultural, social, and psychological meaning. Malnutrition can mean deficient intake, inflammatory catabolism, impaired absorption, altered utilisation, or excess adiposity with poor nutrient quality. Body size alone is misleading: a person with obesity can lose muscle and become severely malnourished. Assessment should identify mechanism, severity, functional consequence, refeeding risk, and the safest route of support.

Energy, protein, and adaptation

Basal metabolism supports ion gradients, protein turnover, circulation, breathing, thermoregulation, and cellular repair. Total expenditure adds physical activity, digestion, growth, pregnancy, lactation, fever, and illness. Carbohydrate supplies readily available glucose; fat provides concentrated energy and essential fatty acids; protein supplies nitrogen and indispensable amino acids. The body stores glycogen and triglyceride but has no inert protein reserve, so prolonged negative balance consumes muscle and organ protein.

During fasting, insulin falls and glucagon and stress hormones promote glycogenolysis, lipolysis, gluconeogenesis, and ketone production. Brain use of ketones limits but does not prevent protein loss. Injury, sepsis, burns, and cancer can drive inflammatory catabolism despite feeding. Nutritional support cannot fully reverse this response until the underlying disease is controlled, yet adequate protein and mobilisation help preserve function.

Recognising malnutrition

Risk rises with reduced intake, dysphagia, dental disease, nausea, pain, depression, cognitive impairment, poverty, isolation, substance use, restrictive diets, malabsorption, chronic organ disease, cancer, infection, wounds, surgery, and high losses through diarrhoea, fistulae, drains, or dialysis. Ask about usual and current intake, duration, weight trajectory, food access, chewing and swallowing, gastrointestinal symptoms, function, and prior support. Verify whether weight change reflects tissue, fluid, or both.

Examine muscle bulk at temples, shoulders, hands, thighs, and calves; subcutaneous fat; oedema; mouth and skin; wounds; grip or mobility; and signs of specific deficiency. Ascites and oedema conceal weight loss. Body mass index neither excludes malnutrition nor directly measures body composition. Screening tools identify risk; diagnosis integrates phenotypic loss of weight, muscle, or low body mass with reduced intake, malabsorption, or inflammatory disease.

Laboratory tests identify causes and consequences but no single blood test diagnoses malnutrition. Albumin falls mainly with inflammation, capillary leak, dilution, liver dysfunction, and renal loss. Review blood count, electrolytes, kidney and liver function, glucose, magnesium, phosphate, calcium, iron studies, folate, vitamin B twelve, vitamin D, coagulation, and disease-specific tests according to risk.

Protein-energy malnutrition

Insufficient energy and protein cause weight loss, weakness, immune dysfunction, poor wound healing, falls, pressure injury, impaired breathing, hypothermia, and reduced tolerance of treatment. Severe acute illness can produce rapid muscle loss even when weight appears stable. Sarcopenia is low muscle strength and quantity associated with ageing or disease; frailty is broader vulnerability across physiological systems. Resistance exercise, adequate protein, disease treatment, and reduction of unnecessary fasting are central.

Set goals collaboratively. Improve the eating environment, symptom control, assistance, texture, timing, fortification, and culturally suitable food before escalating route. Oral supplements add energy and protein but should not

displace preferred meals. Dietitians calculate individual requirements and monitor whether delivery actually occurs. Avoid rigid calorie targets that ignore fluid limits, kidney or liver disease, metabolic tolerance, and goals of care.

Micronutrient deficiencies

Iron deficiency causes microcytic or initially normocytic anaemia, fatigue, epithelial changes, and sometimes pica. Search for bleeding, menstrual loss, low intake, malabsorption, and increased demand rather than replacing indefinitely without explanation. Ferritin reflects stores but rises with inflammation; transferrin saturation and context help interpretation. Oral iron is effective when absorbed and tolerated; intravenous iron is used when rapid replacement, severe intolerance, ongoing losses, or poor absorption justify it.

Folate and vitamin B twelve deficiencies impair DNA synthesis and cause megaloblastic change. Vitamin B twelve deficiency can also cause neuropathy, posterior-column dysfunction, cognitive change, and optic injury, sometimes without anaemia. Causes include autoimmune loss of intrinsic factor, gastric or ileal disease, surgery, diet, and medicines. Treat suspected neurological deficiency promptly. Giving folate alone may improve anaemia while neurological injury progresses.

Thiamine is required for oxidative carbohydrate metabolism. Deficiency in alcohol dependence, starvation, vomiting, malignancy, or bariatric disease can cause heart failure, neuropathy, or Wernicke encephalopathy with confusion, eye-movement abnormality, and ataxia. Give thiamine before or with carbohydrate in high-risk patients; do not delay emergency glucose in hypoglycaemia. Vitamin C deficiency impairs collagen, causing bleeding and poor healing. Vitamin D deficiency contributes to osteomalacia and muscle weakness. Vitamin K deficiency impairs coagulation. Deficiencies of vitamins A and E, zinc, copper, selenium, and others occur with severe restriction, malabsorption, or prolonged artificial nutrition and require risk-directed testing.

Refeeding syndrome

After prolonged undernutrition, insulin-driven refeeding moves phosphate, potassium, and magnesium into cells, increases thiamine demand, retains sodium and water, and expands metabolism. Severe depletion can cause arrhythmia, heart failure, respiratory weakness, delirium, seizure, rhabdomyolysis, and death. Risk depends on low or rapidly falling weight, minimal intake, alcohol use, malabsorption, electrolyte depletion, and catabolic illness.

Identify risk before feeding. Give thiamine and multivitamin support, correct and closely monitor electrolytes, begin energy cautiously in high-risk patients, manage fluid and sodium, and increase delivery progressively with clinical review. Monitoring includes observations, fluid balance, oedema, cardiac or respiratory symptoms, glucose, phosphate, potassium, magnesium, and kidney function. Refeeding can occur by oral, enteral, parenteral, or intravenous glucose routes.

Enteral nutrition

If the gastrointestinal tract functions but oral intake is unsafe or inadequate, enteral feeding is generally preferred because it supports mucosa and carries fewer bloodstream and metabolic complications than parenteral nutrition. Nasogastric or nasojejunal tubes suit shorter use; gastrostomy or jejunostomy may suit longer support. Route depends on swallowing, gastric emptying, aspiration risk, anatomy, expected duration, and patient goals.

Confirm initial tube position by approved methods before use and recheck after displacement risk. Elevate the head when appropriate, provide mouth care, flush tubes, review medicines for safe formulations, and monitor delivery. Complications include aspiration, blockage, displacement, nasal or skin injury, diarrhoea, constipation, electrolyte change, and intolerance. Gastric residual volume alone is an imperfect aspiration marker. Diarrhoea often results from antibiotics, infection, medicines containing sorbitol, or excessive delivery rather than the formula itself.

Parenteral nutrition

Parenteral nutrition delivers glucose, amino acids, lipid, electrolytes, vitamins, trace elements, and water intravenously when the gut cannot safely absorb enough, as in prolonged obstruction, severe short bowel, high-output fistula, or selected critical illness. Central access permits concentrated solutions; peripheral support is limited by vein tolerance and duration.

Use a dedicated aseptic line and daily review because catheter infection and thrombosis can be fatal. Metabolic complications include hyperglycaemia, electrolyte disturbance, refeeding, hypertriglyceridaemia, liver dysfunction, gallbladder stasis, and micronutrient imbalance. Monitor clinical state, intake and output, weight, glucose, electrolytes, liver tests, triglycerides, and line site. Transition toward enteral or oral intake as gut function returns rather than stopping essential support abruptly without a plan.

Obesity as chronic disease

Obesity reflects interacting biology, food environment, sleep, stress, medicines, socioeconomic conditions, and genetics, not a simple failure of will. Adipose dysfunction promotes insulin resistance, fatty liver, sleep apnoea, osteoarthritis, reflux, cardiovascular disease, several cancers, reproductive dysfunction, and stigma. Assess waist and weight trajectory alongside blood pressure, glucose, lipids, liver disease, sleep, function, mental health, eating disorder, fertility, medicines, and readiness. Use respectful language and ask permission to discuss weight.

Treatment aims for health and function, not a cosmetic number. Individualised nutrition, resistance and aerobic activity, sleep, psychological support, and removal of weight-promoting medicines form the base. Anti-obesity medicines alter appetite, absorption, or metabolic signalling and require contraindication review, adverse-effect monitoring, contraception or pregnancy planning, and a long-term maintenance strategy. Metabolic surgery produces substantial durable loss and improves diabetes and mortality in selected patients, but requires lifelong nutrition, micronutrient monitoring, and psychosocial follow-up. Rapid loss can increase gallstones and expose sarcopenia.

TTS module 2: Nutritional diagnosis, refeeding physiology, artificial support, and metabolic treatment

Nutrition assessment distinguishes intake failure, malabsorption, inflammatory catabolism, and altered body composition. Weight is a mixture of fat, muscle, bone, organs, water, and luminal contents. Oedema, ascites, resuscitation, tumour, and obesity can conceal severe muscle loss. A useful diagnosis therefore combines percentage and rate of weight change, current intake, disease burden, muscle and fat examination, strength or performance, and the mechanism preventing recovery.

During uncomplicated fasting, falling insulin permits glycogen use, lipolysis, ketone production, and gluconeogenesis. As the brain adapts to ketones, amino-acid demand falls but never disappears because red cells and selected tissues require glucose. Critical illness differs: catecholamines, cortisol, glucagon, and inflammatory cytokines accelerate proteolysis and hepatic glucose production despite exogenous feeding. Nutrition cannot fully suppress this adaptive catabolism. Treating infection or injury, providing adequate protein, and loading muscle when safe are needed to preserve lean tissue.

Energy requirement is estimated rather than assumed. Predictive equations can be inaccurate during obesity, oedema, fever, ventilation, burns, or changing illness. Indirect calorimetry measures oxygen consumption and carbon-dioxide production and is useful when precision matters and conditions permit. Overfeeding increases carbon-dioxide production, hyperglycaemia, hepatic fat, and ventilatory burden; underfeeding prolongs negative balance. Protein targets depend on kidney and liver function, dialysis or losses, wounds, activity, and catabolic severity. Delivery is measured because prescribed feed is often interrupted.

Malnutrition screening identifies risk; diagnostic assessment establishes severity and cause. Ask about ordinary meals, recent reduction, fasting for procedures, taste, nausea, pain, early satiety, swallowing, teeth, bowel loss, food access, culture, mood, cognition, alcohol, restrictive behaviour, and who prepares food. Examine temples, clavicles, shoulders, interossei, thighs, and calves for muscle, and orbital, rib, and limb regions for fat. Grip strength, chair rise, gait, cough, and wound healing reveal functional consequence.

Albumin should not be used as a nutritional scoreboard. Inflammation reduces hepatic albumin production and increases capillary escape; fluid expansion dilutes it; kidney and gut can lose it; liver failure impairs synthesis. A low value predicts illness severity but does not prove inadequate protein intake, and infused albumin does not rebuild muscle. C-reactive protein and clinical trajectory help interpret biochemical change. Nitrogen balance is also approximate because urinary and non-urinary losses and changing body pools are difficult to measure.

Micronutrient deficiency is predicted from diet, anatomy, losses, drugs, and phenotype. Iron deficiency may precede microcytosis and requires investigation for bleeding, menstrual loss, malabsorption, or demand. Ferritin rises during inflammation, so transferrin saturation, soluble receptors in selected settings, and response add context. Vitamin B twelve deficiency produces neuropathy, posterior-column dysfunction, cognitive or optic injury with or without anaemia. Folate can correct megaloblastic anaemia while unrecognised B-twelve neurological damage progresses.

Thiamine is a cofactor for oxidative carbohydrate metabolism. Deficiency in alcohol dependence, prolonged vomiting, starvation, cancer, bariatric surgery, or dialysis can cause Wernicke encephalopathy, neuropathy, lactic acidosis, or high-output heart failure. The classic triad of confusion, eye-movement abnormality, and ataxia is often incomplete. Treat suspected deficiency parenterally without waiting for a low blood result when consequences are high. Give before or alongside carbohydrate where possible, but never delay emergency glucose in true hypoglycaemia.

Other deficiencies cluster by mechanism. Fat malabsorption causes deficiencies of vitamins A, D, E, and K, impairing vision, bone, nerve, and coagulation. Zinc loss impairs taste, skin, immunity, and wounds; copper deficiency causes anaemia, neutropenia, and myeloneuropathy and can follow excess zinc. Vitamin C deficiency produces perifollicular bleeding, gum disease, and defective collagen. Selenium deficiency can impair heart and thyroid pathways. Broad empirical replacement may itself cause toxicity, so testing and dosing follow the risk and urgency.

Refeeding syndrome is an intracellular redistribution emergency. During starvation, total-body phosphate, potassium, magnesium, and thiamine become depleted even when serum values appear normal. Carbohydrate

reintroduction raises insulin, drives these electrolytes into cells, increases phosphorylated intermediates and adenosine triphosphate demand, and retains sodium and water. Consequences include arrhythmia, respiratory muscle failure, heart failure, weakness, rhabdomyolysis, delirium, seizure, and death. Oral food, tube feed, parenteral nutrition, and glucose-containing intravenous fluid can all trigger it.

Risk is identified before feeding from very low or rapidly falling weight, minimal intake, alcohol use, malabsorption, cancer treatment, bariatric disease, uncontrolled diabetes, electrolyte loss, and prolonged catabolism. Give thiamine and multivitamins, correct electrolytes, begin energy cautiously according to risk, and increase progressively. Monitor phosphate, potassium, magnesium, glucose, fluid balance, oedema, pulse, breathing, and neurological state frequently. Feeding is not stopped reflexively for every fall; replacement, rate adjustment, and clinical severity guide action.

Oral nutrition is preferred when swallowing is safe and intake can meet goals. Treat pain, nausea, constipation, dry mouth, depression, and feeding dependence; remove unnecessary fasting; improve meal timing and environment; fortify familiar food; and provide culturally acceptable supplements. Texture modification requires swallowing evidence and review because thickened fluids or purees may reduce intake. Oral supplements are additions, not replacements for meals the patient enjoys. Mealtime assistance can be more effective than another prescription.

Enteral feeding is used when the gut can digest and absorb but oral intake is unsafe or insufficient. A nasogastric tube suits many short-term needs; post-pyloric feeding helps selected gastric-emptying or aspiration problems; gastrostomy or jejunostomy supports longer courses when consistent with goals. Tube position is confirmed by approved methods before first use and after possible displacement. Auscultating injected air is unreliable. Head elevation, mouth care, secure fixation, flushing, medication review, and delivery monitoring prevent common harm.

Aspiration risk is not eliminated by a feeding tube because saliva and reflux can still enter the airway. Reduced consciousness, poor cough, supine position, reflux, delayed gastric emptying, and excessive sedation matter. Gastric residual volumes alone poorly predict aspiration and can lead to avoidable feed interruption. Diarrhoea during enteral support often reflects

antibiotics, infection, sorbitol-containing medicine, rapid delivery, or underlying disease rather than formula intolerance. Constipation, tube blockage, nasal injury, electrolyte disturbance, and refeeding also require surveillance.

Parenteral nutrition is indicated when the gastrointestinal tract cannot absorb sufficient nutrition safely, such as prolonged obstruction, severe short bowel, high-output fistula, intestinal ischaemia, or selected postoperative complications. It is not automatically superior in critical illness. The prescription includes glucose, amino acids, lipid, electrolytes, vitamins, trace elements, and water adjusted daily. Central venous access allows concentrated long-term delivery but creates bloodstream infection, thrombosis, occlusion, and mechanical risks. A dedicated aseptic lumen and daily line necessity review are essential.

Metabolic monitoring of parenteral nutrition includes glucose, sodium, potassium, phosphate, magnesium, kidney and liver function, triglycerides, fluid, weight, and clinical examination. Excess glucose causes hyperglycaemia, lipogenesis, and carbon-dioxide burden. Excess or inappropriate lipid contributes to hypertriglyceridaemia. Long-term therapy can produce cholestasis, steatosis, gallbladder stasis, and trace-element accumulation or deficiency. Abrupt interruption can cause hypoglycaemia in a person receiving substantial insulin. Transition toward enteral intake as gut function returns while confirming that actual oral delivery is adequate.

Obesity is a chronic relapsing biological condition shaped by genetics, neuroendocrine appetite control, sleep, medications, stress, food availability, socioeconomic conditions, and built environment. Weight stigma delays care and produces avoidance. Ask permission to discuss weight and assess complications rather than assuming behaviour. Waist, weight trajectory, pressure, glucose, lipids, liver fibrosis, sleep apnoea, joints, reflux, fertility, mental health, eating disorder, function, and medications define treatment priorities.

Weight loss activates biological defence through increased hunger and reduced energy expenditure, so regain is not simply failure of will. Sustainable nutrition, resistance and aerobic activity, sleep treatment, psychological support, and reduction of weight-promoting medicines build the foundation. Pharmacotherapy alters appetite, absorption, or metabolic signalling and generally requires long-term use to maintain effect. Gastrointestinal adverse

effects, gallbladder disease, pancreatitis concerns, pregnancy, interactions, mood, and loss of lean mass are reviewed according to class.

Metabolic surgery produces large durable weight loss and can improve diabetes, sleep apnoea, cardiovascular risk, and survival in selected patients. It also changes anatomy and absorption, creating risks of leak, thrombosis, dumping, hypoglycaemia, gallstones, alcohol effects, and deficiencies of iron, B twelve, folate, thiamine, calcium, vitamin D, and others. Lifelong supplementation and monitoring are mandatory. Vomiting or neurological symptoms after surgery can signal urgent thiamine deficiency. Rapid loss without resistance exercise and protein support can expose severe sarcopenia.

Nutrition goals follow prognosis and values. Artificial feeding can bridge reversible illness, support treatment, or sustain life, but it may not prevent aspiration, restore function, or improve comfort in advanced irreversible disease. Discuss expected benefit, burdens, duration, review criteria, and what happens if goals are not met. Nutritional care is most effective when it treats food as metabolism, function, culture, and relationship simultaneously rather than reducing the patient to calories and body mass.

Retrieval prompts

- One. Why can obesity coexist with malnutrition?
- Two. Why is albumin not a direct nutrition marker?
- Three. Which neurological risk distinguishes vitamin B twelve deficiency?
- Four. What causes the dangerous shifts of refeeding syndrome?
- Five. When is enteral feeding preferred over parenteral feeding?
- Six. What must accompany long-term obesity treatment?

Concise answers

One. Excess fat does not guarantee adequate protein, muscle, micronutrients, absorption, or recent intake.

Two. It changes substantially with inflammation, fluid distribution, liver synthesis, and renal loss.

Three. Potentially irreversible neuropathy, posterior-column, cognitive, and optic injury.

Four. Insulin-driven intracellular uptake of phosphate, potassium, and magnesium with thiamine demand and sodium-water retention.

Five. When oral intake is unsafe or inadequate but the gastrointestinal tract can digest and absorb.

Six. Long-term nutrition, activity, adverse-effect and micronutrient monitoring, mental-health support, and prevention of weight regain and muscle loss.

Source map

Original synthesis informed by Guyton and Hall, fuel metabolism, fasting adaptation, appetite, energy balance, and gastrointestinal absorption; OpenStax Anatomy and Physiology, Biology, and Chemistry, macronutrients, vitamins, minerals, and cellular energy; Robbins, malnutrition, obesity, vitamin disease, and inflammatory catabolism; Katzung and OpenStax Pharmacology, replacement therapy, appetite and weight medicines, glucose management, and interactions; Talley and O'Connor, nutritional history and physical signs; and OpenStax Medical-Surgical Nursing, screening, feeding tubes, parenteral nutrition, bariatric care, wounds, and patient education.

Chapter 51: Urinary Tract Disease, Obstruction, Stones, Continence, and Urological Assessment

Orientation

The urinary tract transports urine from renal collecting systems through ureters to a low-pressure reservoir, then empties through coordinated bladder contraction and outlet relaxation. Disease may impair drainage, storage, voiding, continence, or protection from infection. Presentations include flank or pelvic pain, dysuria, frequency, urgency, haematuria, altered stream, retention, incontinence, fever, and kidney dysfunction. Immediate priorities are infected obstruction, acute retention, anuria, severe bleeding, testicular emergencies, and rapidly deteriorating renal function.

History and localisation

Clarify pain onset, site, radiation, colic or constancy, severity, movement, urinary relation, fever, nausea, trauma, and pregnancy possibility. Ureteric colic is severe fluctuating flank pain radiating toward groin as peristalsis works against obstruction; patients often move restlessly. Peritoneal inflammation more often produces stillness and guarding. Constant flank pain with fever suggests upper-tract infection, although older or immune-suppressed patients may lack classic features.

Lower urinary symptoms divide into storage and voiding domains. Storage symptoms include frequency, urgency, nocturia, and urgency incontinence. Voiding symptoms include hesitancy, weak or intermittent stream, straining, prolonged emptying, and terminal dribbling. Post-void symptoms include incomplete emptying and after-dribble. Quantify fluid, caffeine, alcohol, urine volume, nocturnal volume, bowel function, mobility, cognition, sleep, and impact rather than assuming prostate disease.

For dysuria and haematuria, ask visible colour, clots, timing through the stream, menstruation, discharge, sexual exposure, instrumentation, stone history, anticoagulants, smoking, occupation, radiation, chemotherapy, travel,

and family disease. Anticoagulation can reveal bleeding but should not be accepted as its sole cause. Painless visible haematuria requires malignancy assessment until adequately explained.

Review diabetes, neurological disease, pelvic surgery, childbirth, prolapse, cancer, kidney disease, infection, and medicines. Anticholinergic drugs, opioids, sympathomimetics, anaesthetics, and constipation can precipitate retention; diuretics and sedatives can worsen incontinence. Establish baseline renal function and whether the patient has one functioning kidney.

Examination

Assess observations, hydration, perfusion, sepsis, confusion, and pain. Examine abdomen for scars, masses, tenderness, hernias, enlarged bladder, or peritonism. Costovertebral tenderness supports upper-tract inflammation but is not specific. Inspect external genitalia when symptoms indicate, using consent, privacy, and a chaperone according to policy. Testicular pain requires assessment of lie, swelling, tenderness, cremasteric response, groin, and abdomen without delaying urgent surgical review for suspected torsion.

Digital rectal examination may assess prostate size, consistency, tenderness, and nodularity, but size correlates poorly with obstruction and a normal examination does not exclude cancer. Avoid vigorous prostatic massage in acute bacterial prostatitis because it is painful and may promote bacteraemia. Pelvic examination assesses prolapse, atrophy, masses, discharge, fistula, and stress leakage when relevant. Neurological examination includes gait, leg function, reflexes, perineal sensation, anal tone, and sacral function when retention or incontinence may reflect cord or cauda equina disease.

Urinalysis and microbiology

Dipstick detects blood, leukocyte esterase, nitrite, protein, glucose, ketones, concentration, and pH, but each requires context. Haem pigment may represent red cells, haemoglobin, or myoglobin. Nitrite depends on bacterial nitrate reduction and bladder dwell time, so a negative result does not exclude infection. Pyuria indicates inflammation and can occur with stones, sexually transmitted infection, tuberculosis, malignancy, or instrumentation.

Collect midstream urine after instruction, or sample aseptically from a catheter port rather than bag. Culture is particularly important in pregnancy, children, men with complicated presentations, pyelonephritis, recurrence, immune suppression, catheter-associated disease, resistant risk, or treatment failure. Asymptomatic bacteriuria usually should not be treated except in defined situations such as pregnancy or before selected invasive urological procedures. Treating colonisation causes adverse effects and resistance.

Urinary tract infection

Uncomplicated cystitis causes dysuria, frequency, urgency, and suprapubic discomfort without systemic illness. Vaginal symptoms lower the probability and suggest alternative diagnoses. Pyelonephritis produces fever, flank pain, nausea, and systemic features, with bacteraemia possible. Choose antibiotics from syndrome, severity, allergies, pregnancy, kidney function, previous cultures, resistance, tissue penetration, and local guidance. Reassess failure for obstruction, abscess, resistant organism, wrong diagnosis, or inadequate adherence.

Catheters bypass defence and develop biofilm. Diagnose catheter-associated infection from compatible symptoms, not cloudy or odorous urine alone. Remove or replace an unnecessary longstanding catheter when culturing and treating according to protocol. Prevention means avoiding insertion, aseptic placement, a closed unobstructed system, bag below bladder, daily indication review, and prompt removal.

Obstruction and retention

Obstruction raises upstream pressure and can reduce filtration, impair concentration, cause infection, and eventually atrophy renal tissue. Causes include stones, prostate enlargement or cancer, urethral stricture, pelvic mass, clot, neurogenic dysfunction, constipation, and retroperitoneal fibrosis. Bilateral obstruction or obstruction of a solitary kidney may cause anuria and acute kidney injury without dramatic pain.

Acute retention causes painful inability to void and a distended bladder, but neurological or chronic retention can be painless. Measure post-void residual with ultrasound. Drain the bladder when indicated while monitoring discomfort, haematuria, blood pressure, urine volume, and post-obstructive

diuresis. Difficult or traumatic catheterisation requires experienced help; repeated blind attempts can create false passages. Suprapubic drainage may be needed. Treat precipitating factors and arrange evaluation of outlet and bladder function.

Post-obstructive diuresis may represent excretion of retained salt and water or impaired tubular concentration. Excessive loss causes hypovolaemia, sodium disturbance, potassium loss, and shock. Record urine closely and replace a considered proportion according to volume status and electrolytes rather than automatically matching every millilitre.

Urinary stones

Stones form when urine is supersaturated with calcium salts, uric acid, cystine, or infection-related minerals, modified by volume, pH, inhibitors, anatomy, and microbes. Non-contrast computed tomography is highly sensitive; ultrasound avoids radiation in pregnancy and often detects hydronephrosis. Urinalysis, culture, kidney function, blood count, calcium, urate, and stone analysis identify complications and cause.

Provide analgesia, antiemetic therapy, hydration to correct deficit, and selected expulsive therapy. Forced excessive fluid does not push an obstructed stone through. Infection plus obstruction is an emergency requiring antibiotics and urgent decompression by ureteric stent or nephrostomy; definitive stone treatment follows stabilisation. Also intervene for threatened renal function, uncontrolled symptoms, solitary kidney, or unlikely passage. Prevention uses increased urine volume and cause-specific sodium, calcium, oxalate, citrate, purine, infection, or drug management. Dietary calcium is not routinely eliminated because low intake can increase intestinal oxalate absorption.

Haematuria and urological cancer

Confirm true haematuria and assess glomerular features such as protein, casts, dysmorphic cells, oedema, and hypertension. Urological bleeding may arise from infection, stone, prostate, trauma, kidney tumour, urothelial cancer, or instrumentation. Clots suggest non-glomerular bleeding and can obstruct outflow. Persistent microscopic or any unexplained visible haematuria is evaluated according to age and risk with upper-tract imaging and often cystoscopy.

Bladder cancer is strongly associated with smoking and some occupational exposures. Renal cancer may present with haematuria, incidental mass, systemic symptoms, or paraneoplastic effects. Prostate cancer ranges from indolent local disease to metastatic illness; prostate-specific antigen is organ-specific rather than cancer-specific and requires shared interpretation. Management spans surveillance, endoscopic treatment, surgery, radiation, systemic therapy, and palliation according to biology, stage, health, and preferences.

Incontinence and neurogenic bladder

Stress incontinence occurs when abdominal pressure exceeds urethral closure, producing leakage with cough or exertion. Urgency incontinence accompanies involuntary urgency; overflow results from retention; functional incontinence reflects mobility, cognition, environment, or assistance. Mixed mechanisms are common. Use bladder diary, urine testing, residual volume, medicine review, examination, and selective urodynamics.

Treat reversible contributors such as infection, constipation, excess urine production, atrophy, immobility, and drugs. Pelvic-floor training is first-line for stress and mixed disease; bladder training supports urgency. Antimuscarinic drugs can cause dry mouth, constipation, confusion, and retention; beta-three agonists can affect blood pressure. Procedures are selected after mechanism and goals are clear. Neurogenic bladder management protects kidney function through reliable low-pressure storage and emptying, sometimes using intermittent catheterisation and medicines.

TTS module 2: Urinary drainage physiology, infected obstruction, haematuria, and bladder function

The urinary tract protects kidney function by maintaining low-pressure drainage and coordinated storage and emptying. Obstruction can occur anywhere from renal pelvis to urethra and can be fixed, intermittent, unilateral, bilateral, anatomical, or functional. Symptoms do not measure severity: slowly progressive bilateral obstruction may be painless, while a small moving ureteric stone causes extreme colic. The urgent questions are whether infection is trapped behind obstruction, whether both kidneys or a solitary kidney are threatened, whether bladder pressure is damaging the upper tract, and whether tissue perfusion is falling.

Ureteric colic reflects peristaltic contraction and rising pressure above a mobile obstruction. Pain often begins in flank and migrates toward groin as the stone advances, accompanied by nausea and restlessness. Microscopic haematuria supports but is not required. Constant focal tenderness, fever, rigors, hypotension, confusion, anuria, or acute kidney injury changes the syndrome from uncomplicated colic to possible infection, infarction, rupture, or threatened renal function. Analgesia and antiemetics are given promptly while pregnancy, one-kidney status, and sepsis are established.

Infected obstruction is a source-control emergency. Pressure and impaired flow reduce antibiotic delivery and bacterial clearance, allowing rapid bacteraemia and septic shock. Urine from the bladder can be deceptively sterile when the infected system is completely blocked. Resuscitation and broad antibiotics are followed by urgent decompression with ureteric stent or percutaneous nephrostomy; definitive stone removal waits until infection is controlled. Forcing oral or intravenous fluid does not push a stone through and can worsen pain or overload.

Stone formation depends on supersaturation, nucleation, crystal growth, urinary inhibitors, pH, volume, anatomy, and infection. Low urine volume concentrates every lithogenic solute. Sodium intake increases urinary calcium; low dietary calcium can paradoxically increase oxalate absorption by leaving intestinal oxalate unbound. Citrate complexes calcium and inhibits crystallisation. Uric acid precipitates in acidic urine; cystine stones reflect inherited transport; urease-producing infection creates alkaline struvite stones

that can branch through the collecting system. Stone analysis and targeted blood and urine studies guide prevention.

Non-contrast computed tomography identifies most stones, obstruction, and alternative diagnoses, but radiation and context matter. Ultrasound detects hydronephrosis and many renal stones and is preferred initially in pregnancy, yet early obstruction or small ureteric stones may be missed. Passage probability depends on size, location, anatomy, oedema, and duration. Medical expulsive therapy has selected use. Persistent obstruction, uncontrolled pain or vomiting, infection, worsening function, solitary kidney, or low likelihood of passage requires urological intervention.

Bladder emptying requires detrusor contraction with relaxation of bladder neck, urethral sphincter, and pelvic floor. During storage, sympathetic and somatic pathways maintain outlet closure while parasympathetic activity is restrained. Pontine and cortical networks coordinate socially appropriate voiding. Outlet obstruction, weak detrusor, sensory impairment, spinal disease, autonomic neuropathy, drugs, pain, and pelvic-floor dyssynergia can all leave residual urine. Prostate size alone cannot define the mechanism.

Acute urinary retention may cause severe suprapubic pain, agitation, and a palpable bladder, but chronic high-pressure retention can be silent until kidney failure or overflow leakage. Bedside ultrasound estimates volume and avoids unnecessary catheterisation when the diagnosis is uncertain. Immediate drainage relieves pressure, but difficult male catheterisation, urethral trauma, blood at the meatus, or altered anatomy requires experienced technique and possible suprapubic access. Repeated blind attempts create false passages, bleeding, and later stricture.

After relief of chronic bilateral obstruction, post-obstructive diuresis can produce litres of urine as retained sodium and water are excreted and damaged tubules fail to concentrate. Monitor pressure, weight, fluid balance, sodium, potassium, magnesium, and kidney function. Replace a clinically judged portion of losses while allowing appropriate excess volume to clear; matching every millilitre can perpetuate diuresis, whereas inadequate replacement causes hypovolaemia and shock. Mild haematuria can follow decompression, but clot retention or ongoing bleeding requires reassessment.

Lower urinary symptoms are divided into storage, voiding, and post-void domains because treatment differs. Frequency can reflect small functional capacity or high urine production from diabetes, diuretics, polydipsia, or nocturnal fluid redistribution. Nocturia may arise from sleep apnoea, heart failure, oedema mobilisation, insomnia, or reduced concentrating ability rather than bladder disease. A frequency–volume chart records intake, time, voided volume, urgency, and leakage and often reveals mechanism more clearly than a symptom score.

Urinalysis is a screening tool with imperfect specificity. Dipstick blood detects haem pigment, so microscopy distinguishes red cells from haemoglobin or myoglobin. Leukocyte esterase indicates white cells, not necessarily bacterial infection. Nitrite requires nitrate-reducing organisms and sufficient bladder dwell time; frequent voiding and some organisms produce false negatives. Protein, glucose, ketones, concentration, and pH add kidney, metabolic, and stone clues. Menstrual, genital, catheter, and collection contamination must be considered.

Urinary infection is diagnosed from a compatible syndrome plus appropriate testing, not culture alone. Dysuria and frequency without vaginal symptoms strongly support cystitis in selected patients. Fever, flank pain, vomiting, or systemic illness suggests pyelonephritis and requires tissue-penetrating therapy. Prostatitis, urethritis, vaginitis, stones, interstitial bladder pain, and malignancy mimic infection. Treatment failure prompts review of adherence, resistance, wrong diagnosis, abscess, foreign material, and obstruction. Repeated empirical antibiotics without culture can obscure the cause and select resistance.

Catheter bacteriuria is expected because biofilm develops along internal and external surfaces. Cloudiness, sediment, odour, and pyuria do not establish symptomatic infection. New fever, rigors, pelvic or flank pain, haemodynamic change, or delirium without a better cause can support infection after assessment. Culture is obtained from a sampling port, often after replacing a longstanding catheter when treatment is indicated, never from the drainage bag. Prevention is chiefly avoidance, aseptic insertion, closed unobstructed drainage, and removal at the earliest safe time.

Haematuria is first confirmed, then localised as glomerular or urological. Proteinuria, dysmorphic red cells, casts, hypertension, and oedema support

glomerular injury. Clots, irritative symptoms, stone pain, and visible lesions support urinary-tract bleeding, but overlap occurs. Anticoagulation can amplify bleeding from cancer or stone and should not terminate investigation. Painless visible haematuria is a malignancy warning even if transient. Smoking, occupational aromatic amines, pelvic radiation, cyclophosphamide, age, and family disease increase urothelial risk.

Evaluation combines upper-tract imaging with cystoscopy according to age, risk, and haematuria type because imaging alone can miss flat or small bladder lesions. Severe bleeding can form clots that obstruct the outlet and distend the bladder; large-bore irrigation and endoscopic control may be needed. Kidney tumours can cause haematuria, mass, anaemia or erythrocytosis, hypercalcaemia, fever, and incidental imaging findings. A negative initial evaluation does not justify ignoring recurrent visible blood.

Incontinence is a functional diagnosis with several mechanisms. Stress leakage occurs when urethral closure cannot oppose a pressure rise. Urgency leakage follows an involuntary compelling need and often detrusor overactivity. Overflow leakage accompanies retention and high residual volume. Functional incontinence occurs when mobility, cognition, clothing, environment, or assistance prevents timely toileting. Fistula creates continuous leakage. Mixed disease is common, so treatment based on age or sex stereotypes often fails.

Pelvic-floor muscle training requires correct contraction, sufficient dose, and months of practice; simply advising “do exercises” is inadequate. Bladder training progressively lengthens intervals and uses urgency-suppression strategies. Antimuscarinic drugs may improve urgency but worsen dry mouth, constipation, cognition, glaucoma risk, and retention; beta-three agonists can raise pressure. Procedures, botulinum toxin, neuromodulation, sling, sphincter, or prolapse treatment follow confirmed mechanism and patient priorities.

Neurogenic lower-tract dysfunction is managed to protect kidneys, not only continence. Spinal cord disease, multiple sclerosis, diabetes, Parkinson disease, stroke, and peripheral injury can produce dangerous high-pressure storage, detrusor–sphincter dyssynergia, or ineffective emptying. Symptoms may underestimate pressure. Residual measurement, kidney imaging, renal function, and urodynamics are selected by risk. Clean intermittent

catheterisation, bladder-relaxing medicine, outlet procedures, and surveillance aim for reliable low-pressure storage and complete enough emptying while preserving independence and dignity.

Retrieval prompts

One. Which urinary presentations require immediate escalation?

Two. Why does a negative nitrite test not exclude infection?

Three. Why is bacteriuria in a catheter not automatically treated?

Four. What makes an infected obstructed kidney an emergency?

Five. What dangers follow relief of chronic obstruction?

Six. How do stress, urgency, overflow, and functional incontinence differ?

Concise answers

One. Infected obstruction, anuria, acute retention, severe bleeding, testicular torsion, and rapidly worsening renal function.

Two. Some organisms do not reduce nitrate and frequent voiding limits conversion time.

Three. Biofilm commonly causes colonisation; antibiotics without compatible illness add harm and resistance.

Four. Antibiotics cannot reliably sterilise a pressurised obstructed system, so urgent drainage is required.

Five. High-output diuresis, hypovolaemia, electrolyte loss, haematuria, and hypotension.

Six. Stress leaks with pressure, urgency with compelling need, overflow from retention, and functional disease from access or capability barriers.

Source map

Original synthesis informed by Guyton and Hall, urine transport, bladder storage, micturition, and obstruction physiology; Robbins, urinary infection, stones, obstruction, and renal, urothelial, and prostate pathology; Katzung and OpenStax Pharmacology, antibiotics, outlet drugs, bladder medicines, analgesia, and cancer therapy; Talley and O'Connor, urinary history, abdominal, genital, rectal, pelvic, and neurological examination; OpenStax Medical-Surgical Nursing, catheter care, retention, diversion, continence, infection, stones, and urological procedures; and OpenStax Microbiology, urinary pathogens, biofilm, culture, and resistance.

Chapter 52: Male Reproductive and Sexual Health

Orientation

Male reproductive health integrates hypothalamic-pituitary-gonadal signalling, testicular sperm and testosterone production, genital anatomy, sexual function, fertility, infection, and malignancy. Symptoms are often delayed by embarrassment or stigma, so clinicians should use inclusive language, clarify anatomy and goals, explain confidentiality, and ask permission before sensitive questions or examination. Testicular torsion, Fournier gangrene, priapism, severe trauma, and incarcerated hernia are time-critical diagnoses.

Hormonal control and testicular function

Pulsatile hypothalamic gonadotropin-releasing hormone stimulates pituitary luteinising and follicle-stimulating hormones. Luteinising hormone drives Leydig-cell testosterone production; follicle-stimulating hormone acts with intratesticular testosterone on Sertoli cells and spermatogenesis.

Testosterone and inhibin provide negative feedback. Peripheral conversion produces dihydrotestosterone and oestradiol, influencing genital tissue, hair, muscle, bone, erythropoiesis, libido, and metabolism.

Primary hypogonadism is testicular failure with low testosterone and elevated gonadotropins; causes include chromosomal disorders, orchitis, torsion, chemotherapy, radiation, trauma, and ageing-related disease. Secondary hypogonadism involves hypothalamus or pituitary and may follow tumours, hyperprolactinaemia, opioids, glucocorticoids, obesity, severe illness, undernutrition, or anabolic-steroid suppression. Symptoms include reduced libido and erections, infertility, fatigue, loss of shaving or body hair, low muscle, anaemia, hot flushes, and reduced bone density.

Testosterone varies diurnally and falls transiently in acute illness. Confirm compatible symptoms with properly timed repeated testing and interpret binding proteins, luteinising hormone, follicle-stimulating hormone, prolactin, iron, pituitary features, medicines, fertility goals, and comorbidity.

Testosterone replacement may improve symptoms in proven deficiency but

suppresses spermatogenesis and can cause erythrocytosis, acne, oedema, infertility, and prostate-related monitoring needs. It is not a general treatment for fatigue or ageing.

Sexual history and examination

Ask what concerns the person, partners and practices, desire, arousal, erection, ejaculation, orgasm, pain, curvature, contraception, fertility intentions, sexually transmitted infection exposure, coercion, medicines, substances, cardiovascular symptoms, mood, and relationship context. Do not infer identity from partners or anatomy. Determine onset, consistency, spontaneous or morning erections, situational variation, and whether masturbation differs from partnered activity.

With consent and an offered chaperone according to policy, inspect skin, pubic region, penis, meatus, scrotum, and groins. Palpate each testis systematically, then epididymis and spermatic cord. A normal adult testis is smooth and firm, not hard. Transillumination may support a fluid collection but ultrasound is more reliable. Examine abdomen, secondary sexual characteristics, vascular and neurological systems, and prostate when indicated.

Erectile and ejaculatory dysfunction

Erection requires parasympathetic nitric-oxide signalling, arterial inflow, smooth-muscle relaxation, venous occlusion, intact nerves, and psychological engagement. Erectile dysfunction can be vascular, neurological, endocrine, structural, medication-related, or psychogenic and often has mixed causes. It may precede symptomatic cardiovascular disease because penile arteries are small. Assess blood pressure, diabetes, lipids, smoking, sleep apnoea, exercise, obesity, medicines, and cardiovascular fitness for sexual activity.

Treatment includes explanation, risk-factor control, exercise, relationship or psychological therapy, and removal of contributing drugs when safe. Phosphodiesterase type five inhibitors amplify nitric-oxide signalling and require sexual stimulation. They must not be combined with nitrates because profound hypotension can occur; caution applies with unstable cardiovascular disease and interacting vasodilators. Vacuum devices, intraurethral or intracavernosal drugs, and prostheses are alternatives.

Premature ejaculation is persistent ejaculation sooner than desired with distress and poor control; behavioural, topical anaesthetic, and serotonergic approaches may help. Delayed or absent ejaculation can follow serotonergic drugs, neuropathy, pelvic surgery, endocrine disease, or psychological factors. Retrograde ejaculation sends semen into bladder because the outlet fails to close, often after surgery or from autonomic dysfunction.

Priapism and penile disease

Priapism is an erection lasting more than four hours without sexual purpose. Ischaemic priapism is painful, rigid, and a compartment syndrome that threatens smooth muscle; causes include blood disorders, intracavernosal drugs, psychotropics, and idiopathic events. It requires urgent aspiration, sympathomimetic treatment, and specialist care. Non-ischaemic priapism is often less painful after arterial trauma and is managed differently.

Phimosis prevents foreskin retraction; paraphimosis traps a retracted foreskin behind the glans and requires urgent reduction because oedema compromises blood flow. Balanitis reflects inflammatory, infectious, irritant, or metabolic causes. Peyronie disease is fibrous plaque causing curvature or pain. Penile cancer is uncommon but persistent ulcer, mass, bleeding, or skin change needs biopsy; smoking, phimosis, and oncogenic human papillomavirus increase risk.

Acute scrotal pain

Testicular torsion twists the spermatic cord, obstructing venous then arterial flow. Sudden unilateral pain, nausea, a high or horizontal testis, and absent cremasteric reflex support diagnosis, but no sign safely excludes it. Urgent surgical exploration and bilateral fixation should not be delayed for imaging when clinical suspicion is high; salvage falls rapidly with time.

Epididymitis typically develops more gradually with posterior tenderness, urinary or urethral symptoms, and sometimes fever. Causes vary with sexual exposure, urinary pathology, instrumentation, and age. Test urine and sexually transmitted infections as appropriate, treat likely pathogens, address partners, and reconsider torsion if uncertainty remains. Orchitis can accompany viral or bacterial infection. Fournier gangrene causes rapidly progressive genital or perineal necrosis with severe pain, oedema, skin

change, crepitus, toxicity, or pain beyond visible findings; resuscitation, broad antibiotics, and immediate debridement are essential.

Hydrocele is fluid around the testis; varicocele is dilated pampiniform veins, commonly left-sided and described as a bag of worms. A new right-sided, non-reducing, or supine-persistent varicocele raises secondary obstruction. Hernias may extend into scrotum. Any solid intratesticular mass is malignant until evaluated by urgent ultrasound and tumour markers; do not perform trans-scrotal biopsy when testicular cancer is suspected.

Testicular cancer

Germ-cell tumours often affect younger adults and present as painless lump, heaviness, ache, or metastatic symptoms. Risk rises with undescended testis, previous tumour, and some developmental disorders. Ultrasound distinguishes intratesticular lesions; alpha-fetoprotein, beta human chorionic gonadotropin, and lactate dehydrogenase aid classification and follow-up but normal markers do not exclude cancer. Radical inguinal orchidectomy provides diagnosis and local control. Germ-cell cancers are highly curable with stage-adapted surveillance, chemotherapy, radiation for selected seminoma, or surgery. Discuss sperm banking before gonadotoxic treatment when possible.

Prostate disorders

Benign prostatic hyperplasia enlarges transition-zone tissue through androgen-dependent stromal and epithelial growth. Symptoms depend on dynamic smooth-muscle tone, anatomical narrowing, bladder compensation, and neurological function rather than size alone. Watchful waiting suits mild symptoms. Alpha-one blockers relax outlet muscle quickly but can cause postural hypotension and ejaculatory change. Five-alpha-reductase inhibitors shrink larger glands over months and can affect libido and prostate-specific antigen interpretation. Retention, recurrent infection, stones, bleeding, or renal compromise may require surgery.

Acute bacterial prostatitis causes fever, pelvic or urinary pain, voiding symptoms, and a tender prostate. Obtain cultures and use antibiotics with prostate penetration; manage retention carefully and avoid forceful massage.

Chronic pelvic pain syndrome is multifactorial and should not trigger repeated antibiotics without evidence.

Prostate cancer may be screen-detected, incidental, locally symptomatic, or metastatic to bone. Prostate-specific antigen rises with cancer, benign enlargement, inflammation, retention, and manipulation. Screening is a preference-sensitive decision because early detection can prevent some deaths but creates false positives, biopsy harms, overdiagnosis, and treatment-related urinary, sexual, and bowel dysfunction. Magnetic resonance imaging and risk-adapted biopsy improve selection. Management includes active surveillance, surgery, radiation, androgen deprivation, androgen-pathway drugs, chemotherapy, radiopharmaceuticals, and palliation according to stage and biology.

Male infertility

Infertility evaluation involves both partners and begins after an appropriate duration of trying or earlier with risk factors. Causes include impaired sperm production, obstruction, ejaculation dysfunction, endocrine disease, toxins, heat, infection, varicocele, genetic disorders, and idiopathic factors. Obtain reproductive, developmental, surgery, infection, medicine, anabolic steroid, occupational, and sexual histories.

Semen analysis evaluates volume, concentration, motility, and morphology and is repeated because results vary. Hormonal tests, examination, ultrasound, genetic testing, or post-ejaculatory urine follow the pattern. Treat reversible endocrine, infectious, obstructive, or lifestyle factors; exogenous testosterone must be stopped when fertility is desired. Assisted reproduction ranges from intrauterine insemination to in-vitro fertilisation and intracytoplasmic sperm injection.

TTS module 2: Androgen physiology, sexual function, fertility, and male genital emergencies

Male reproductive and sexual symptoms arise from interacting endocrine, vascular, neural, anatomical, psychological, relational, and medication factors. Libido, erection, ejaculation, orgasm, and fertility are related but separable functions. A person can have normal erections with impaired spermatogenesis, or low desire with intact genital physiology. Assessment should establish the person's anatomy, concerns, partners, practices, fertility goals, and definition of satisfactory function without assuming identity or relationship structure.

Gonadotropin-releasing hormone is released in pulses; continuous exposure suppresses pituitary gonadotropins. Luteinising hormone stimulates Leydig-cell testosterone, while follicle-stimulating hormone and high intratesticular testosterone support Sertoli cells and sperm maturation. Circulating testosterone is mostly protein-bound, with sex-hormone-binding globulin altered by age, obesity, thyroid disease, liver disease, insulin, and medications. A total concentration can therefore misrepresent biologically available hormone when binding is abnormal.

Hypogonadism is diagnosed from compatible symptoms plus consistently low appropriately timed testosterone, not from fatigue or one result. Morning measurement is most useful in people with a preserved diurnal rhythm and is repeated when well because acute illness, sleep loss, energy deficit, and drugs suppress the axis. Raised luteinising and follicle-stimulating hormones indicate primary testicular failure. Low or inappropriately normal gonadotropins indicate hypothalamic or pituitary suppression and prompt assessment of prolactin, iron, pituitary symptoms, opioids, glucocorticoids, obesity, undernutrition, and systemic disease.

Testosterone replacement is appropriate only when deficiency is established and benefits outweigh risk. It can improve sexual symptoms, anaemia, muscle, and bone in selected patients but causes erythrocytosis, acne, oedema, and suppression of testicular volume and sperm production. Baseline and follow-up haematocrit, symptoms, dose exposure, and prostate-related assessment are individualised. Untreated severe sleep apnoea, active fertility desire, prostate or breast cancer context, thrombosis

risk, and cardiovascular instability require careful review. Exogenous anabolic steroids can cause prolonged secondary hypogonadism after cessation.

Erection begins with parasympathetic and endothelial nitric oxide, which raises cyclic guanosine monophosphate and relaxes cavernosal smooth muscle. Arterial inflow expands sinusoids and compresses subtunical veins, trapping blood. Sympathetic tone promotes detumescence and emission. Diabetes can impair endothelium, autonomic nerves, and sensation simultaneously. Pelvic surgery, spinal disease, vascular injury, smoking, hypertension, dyslipidaemia, sleep apnoea, depression, anxiety, alcohol, and medications commonly combine.

Erectile dysfunction can be an early vascular marker because smaller penile arteries may become clinically impaired before coronary or cerebral vessels. Assessment includes exercise symptoms and whether sexual activity is safe in unstable cardiac disease, but stable cardiovascular disease usually does not prohibit sex. Spontaneous morning or masturbation erections suggest preserved physiology but do not prove a purely psychological cause. Situational variability, performance fear, relationship stress, and pain are explored without dismissing physical contributors.

Phosphodiesterase type-five inhibitors preserve cyclic nucleotide signalling and require sexual stimulation. Timing, food effect, duration, dose, and repeated correct trials determine apparent response. Nitrates are absolutely incompatible because profound hypotension can occur; caution applies with alpha blockers, severe pressure instability, and selected eye or cardiac disease. Headache, flushing, dyspepsia, nasal symptoms, and visual changes are class effects. Failure should prompt technique, testosterone where indicated, vascular disease, neurological injury, and alternative devices rather than unsafe dose escalation.

Intracavernosal vasoactive injection and intraurethral therapy bypass some neural pathways but can cause pain, fibrosis, hypotension, and priapism. Vacuum devices draw blood into the penis and use a constriction ring, which must not remain indefinitely. Prostheses provide reliable rigidity when other treatments fail but introduce surgical and infection risk. Psychological and couples therapy can reduce performance anxiety, avoidance, and communication difficulty and should not be framed as evidence that symptoms are unreal.

Ejaculation involves sympathetic emission of sperm and glandular fluid into posterior urethra, closure of the bladder neck, then somatic rhythmic expulsion. Serotonergic drugs commonly delay ejaculation; alpha-blocking or pelvic surgery can impair emission or bladder-neck closure. Retrograde ejaculation produces low-volume or dry orgasm with sperm in post-ejaculatory urine and matters particularly for fertility. Premature ejaculation requires a persistent pattern, poor control, and distress rather than an arbitrary stopwatch alone; behavioural, topical, and serotonergic treatments are selected collaboratively.

Priapism is classified by perfusion. Ischaemic priapism causes painful rigid corpora with a relatively soft glans and represents stagnant hypoxic blood within a closed compartment. Smooth-muscle necrosis and later erectile fibrosis develop with time, so an erection lasting around four hours requires emergency aspiration, blood-gas confirmation where needed, intracavernosal alpha-agonist treatment, and surgical escalation. Sickle cell disease, haematological malignancy, psychotropics, erectile injections, and idiopathic episodes are important causes. Systemic treatment alone must not delay corporal decompression.

Non-ischaemic priapism usually follows perineal arterial injury, remains partially rigid and less painful, and does not carry the same immediate ischaemic threat. Observation or selective arterial embolisation may be appropriate. Recurrent brief ischaemic episodes, sometimes called stuttering priapism, require a prevention strategy before one becomes prolonged. Any priapism assessment includes medication, recreational drugs, blood disease, trauma, and neurological symptoms.

The acute scrotum is treated as torsion until a safer diagnosis is established. Cord twisting obstructs venous drainage first, then arterial inflow, producing infarction within hours. Sudden pain, nausea, high or horizontal lie, and absent cremasteric response support torsion, but age, urinary symptoms, partial relief, or one normal sign cannot exclude it. When clinical suspicion is high, surgical exploration should bypass imaging delay. Both testes are fixed because anatomical predisposition is commonly bilateral.

Epididymitis more often evolves gradually with posterior tenderness, urinary or urethral symptoms, and fever, but overlap with torsion is substantial. Pathogens depend on sexual exposure, urinary tract abnormality,

instrumentation, and age. Nucleic-acid testing, urine culture, partner treatment, and resistance-aware therapy follow the context. Mumps and other viruses can cause orchitis. Persistent swelling after infection requires reassessment because tumour can be mislabelled as inflammation.

Fournier gangrene is necrotising infection of perineal and genital fascia. Diabetes, immune suppression, obesity, malignancy, perianal or urinary disease, trauma, and procedures increase risk, but anyone can be affected. Pain beyond visible change, rapidly spreading oedema, dusky skin, bullae, anaesthesia, crepitus, fever, shock, or confusion demands immediate surgical exploration. Broad antibiotics and resuscitation accompany repeated debridement; computed tomography must not delay action in an obvious syndrome.

Any solid intratesticular mass is malignant until assessed. Ultrasound determines whether a lesion is intratesticular and solid; tumour markers support classification and monitoring but can be normal. Trans-scrotal biopsy disrupts lymphatic staging and is avoided. Radical inguinal orchidectomy provides diagnosis and treatment. Sperm cryopreservation is discussed before chemotherapy, radiation, retroperitoneal surgery, or other gonadotoxic treatment when time permits. Germ-cell cancers remain highly curable even when metastatic, making prompt referral important.

Prostate enlargement causes symptoms through fixed tissue bulk, dynamic smooth-muscle tone, and bladder adaptation. A small gland can obstruct and a large gland can be asymptomatic. Alpha blockers act quickly on tone but may cause postural hypotension and ejaculatory dysfunction. Five-alpha-reductase inhibitors shrink larger glands over months and lower prostate-specific antigen, which must be reinterpreted during cancer assessment. Recurrent retention, infection, stones, haematuria, high residual, or kidney injury strengthens the case for procedural treatment.

Prostate-specific antigen is produced by prostate tissue rather than cancer alone. Age, gland volume, inflammation, retention, ejaculation, manipulation, and drug therapy affect concentration. Screening is a preference-sensitive sequence, not a simple blood test: an abnormal result may lead to magnetic resonance imaging, biopsy, surveillance, surgery, or radiation with urinary, sexual, and bowel effects. The potential reduction in metastatic disease or

death is weighed against false positives and overdiagnosis. Active surveillance separates monitoring from neglect for suitable low-risk cancer.

Male infertility evaluation occurs in parallel with partner assessment. Semen analysis varies with abstinence interval, fever, collection completeness, laboratory methods, and biological fluctuation and is repeated when abnormal. Low volume suggests collection error, androgen deficiency, ejaculatory-duct obstruction, or retrograde ejaculation. Azoospermia may reflect absent production or obstruction; testicular size, gonadotropins, genetics, and examination distinguish them. Severe oligospermia or absent vas deferens can reveal chromosomal or cystic-fibrosis-related conditions with implications for offspring.

Lifestyle advice is targeted rather than moralising. Stop exogenous testosterone and anabolic steroids when fertility is desired; review heat exposure, smoking, cannabis, alcohol, toxins, obesity, severe undernutrition, and gonadotoxic medicines. Varicocele repair benefits selected clinical disease, not every ultrasound finding. Endocrine therapy can restore gonadotropin drive in secondary hypogonadism, while obstruction may be surgically corrected. Intrauterine insemination, in-vitro fertilisation, intracytoplasmic injection, donor sperm, and adoption are discussed according to sperm phenotype, partner factors, time, cost, values, and genetic risk.

Retrieval prompts

One. How are primary and secondary hypogonadism distinguished?

Two. Why can erectile dysfunction be a cardiovascular warning?

Three. Which drug combination with phosphodiesterase inhibitors is dangerous?

Four. Why must suspected testicular torsion bypass diagnostic delay?

Five. How should a solid intratesticular mass be approached?

Six. Why does testosterone therapy conflict with fertility treatment?

Concise answers

One. Primary disease has testicular failure with raised gonadotropins; secondary disease has inadequate hypothalamic-pituitary stimulation.

Two. Systemic endothelial and arterial disease may impair the smaller penile circulation before larger vessels become symptomatic.

Three. Nitrates, because combined vasodilation can cause profound hypotension.

Four. Ongoing cord ischaemia rapidly causes irreversible testicular loss.

Five. Arrange urgent ultrasound and specialist cancer evaluation without trans-scrotal biopsy.

Six. Exogenous testosterone suppresses gonadotropins and intratesticular testosterone, reducing or stopping spermatogenesis.

Source map

Original synthesis informed by Guyton and Hall, male reproductive endocrinology, spermatogenesis, erection, and ejaculation; Robbins, testicular, penile, and prostate pathology; Katzung and OpenStax Pharmacology, androgen therapy, erectile medicines, prostate drugs, antibiotics, and cancer therapy; Talley and O'Connor, sexual history and genital, groin, rectal, endocrine, vascular, and neurological examination; OpenStax Medical-Surgical Nursing, urological procedures, sexual counselling, cancer, infection, and postoperative care; and OpenStax Anatomy and Physiology and Microbiology, reproductive anatomy and sexually transmitted infection.

Chapter 53: Gynaecological Disease, Menstruation, Fertility, and Contraception

Orientation

Gynaecological care integrates endocrine physiology with bleeding, pain, infection, fertility, sexual health, pelvic-floor function, and cancer prevention. Symptoms are shaped by age, pregnancy possibility, anatomy, medicines, trauma, and reproductive goals. Explain each examination, obtain consent, offer a chaperone according to policy, and allow stopping. Instability, ectopic pregnancy, adnexal torsion, severe infection, haemorrhage, and genital trauma require urgent action.

Menstrual physiology and history

Follicular growth produces oestradiol, which proliferates endometrium. Sustained high oestradiol triggers the luteinising-hormone surge and ovulation. The corpus luteum then produces progesterone, converting endometrium into a secretory state. Without implantation, hormone withdrawal causes menstruation. Cycles vary across life, and ovulation may be inconsistent after menarche and before menopause.

Ask menarche, cycle interval and regularity, bleeding duration and volume, flooding, clots, pain, intermenstrual or postcoital bleeding, last normal period, pregnancy possibility, contraception, sexual exposure, discharge, fertility goals, menopausal symptoms, medicines, bleeding tendency, endocrine symptoms, weight change, and functional impact. Product counts are imperfect; anaemia, leakage, restriction, and patient experience matter.

Abnormal uterine bleeding

Abnormal bleeding may result from polyps, adenomyosis, leiomyomas, malignancy or hyperplasia, coagulopathy, ovulatory dysfunction, endometrial dysfunction, medicines, or procedures. Pregnancy-related bleeding is assessed separately. Adolescents commonly have immature anovulatory cycles, but inherited bleeding disorders must be considered. Perimenopausal anovulation is common, yet age also increases endometrial cancer risk.

Assess stability and pregnancy first. Blood count and iron define consequence; thyroid, coagulation, infection, endocrine testing, ultrasound, and endometrial sampling are selected by pattern and risk. Endometrial biopsy is important with older age, persistent irregular bleeding, prolonged unopposed oestrogen, obesity, polycystic ovarian syndrome, tamoxifen exposure, or failed treatment.

Acute heavy bleeding may require resuscitation, blood products, tranexamic acid, hormonal therapy, uterine tamponade, hysteroscopic treatment, embolisation, or surgery. Chronic treatment addresses cause and goals. The levonorgestrel intrauterine system, combined hormonal contraception, progestogens, tranexamic acid, and non-steroidal drugs can reduce bleeding. Structural lesions may require hysteroscopy, myomectomy, ablation, embolisation, or hysterectomy. Endometrial ablation is not contraception and is unsuitable when future pregnancy is desired.

Pelvic pain and endometriosis

Acute pelvic pain differentials include ectopic pregnancy, miscarriage, torsion, ruptured cyst, pelvic inflammatory disease, appendicitis, urinary stone, bowel disease, and trauma. Establish onset, laterality, bleeding, discharge, fever, vomiting, urinary and bowel symptoms, sexual exposure, procedures, and pregnancy possibility. A pregnancy test is routine when biologically possible. Sudden severe unilateral pain with nausea suggests torsion; normal Doppler flow does not completely exclude intermittent or partial torsion, so urgent specialist judgement matters.

Endometriosis is endometrium-like tissue outside the uterine cavity producing inflammation, fibrosis, adhesions, pain, and infertility. Symptoms include dysmenorrhoea, deep dyspareunia, chronic pelvic pain, painful defecation or

urination, and cyclical organ symptoms, but burden does not correlate reliably with lesion extent. Make a clinical diagnosis supported by examination and imaging; laparoscopy can diagnose and treat but is not required before all therapy. Analgesia, combined hormonal suppression, progestogens, gonadotropin-releasing hormone modulation, surgery, and fertility treatment are individualised.

Adenomyosis is endometrial tissue within myometrium, often causing painful heavy bleeding and a tender enlarged uterus. Leiomyomas are benign smooth-muscle tumours causing bleeding, pressure, fertility effects, or no symptoms according to size and location. Treatment balances symptoms, anaemia, fertility, uterine preservation, and procedural risk.

Infection and discharge

Vaginal discharge may be physiological or caused by bacterial vaginosis, Candida, Trichomonas, cervicitis, foreign body, atrophy, fistula, or malignancy. History alone is unreliable. Assess colour, odour, irritation, pain, bleeding, sexual exposure, pregnancy, antibiotics, diabetes, immune status, and retained products. Examination and microscopy, pH, nucleic-acid testing, or culture are selected by syndrome.

Pelvic inflammatory disease is ascending infection of endometrium, tubes, ovaries, or peritoneum, commonly associated with sexually transmitted pathogens but often polymicrobial. Lower abdominal pain with cervical motion, uterine, or adnexal tenderness supports empirical treatment after pregnancy and emergencies are considered. Delay increases infertility, ectopic pregnancy, abscess, and chronic pain. Treat broadly, test for infections, manage partners, advise abstinence until treatment is complete, and reassess. A tubo-ovarian abscess may require admission and drainage.

Genital ulcers can be infectious, inflammatory, traumatic, or malignant. Human papillomavirus causes warts and oncogenic types drive cervical, vulval, vaginal, anal, penile, and oropharyngeal cancers. Vaccination and cervical screening prevent disease but do not eliminate all risk. Trauma-informed assessment is essential when assault, coercion, or female genital cutting may be relevant; immediate medical care, forensic options, pregnancy prevention, infection prophylaxis, safeguarding, and psychological support should follow the person's choices and local pathways.

Polycystic ovarian syndrome and amenorrhoea

Polycystic ovarian syndrome combines ovulatory dysfunction, clinical or biochemical androgen excess, and characteristic ovarian morphology after excluding alternatives. Insulin resistance commonly contributes. Features include irregular cycles, infertility, acne, hirsutism, alopecia, obesity, sleep apnoea, dyslipidaemia, diabetes risk, and endometrial hyperplasia from prolonged unopposed oestrogen.

Management targets the person's priorities: nutrition and activity, cycle protection with combined contraception or progestogen, cosmetic and antiandrogen treatment with reliable contraception, metabolic risk management, and ovulation induction when pregnancy is desired. The name does not require ovarian cysts, and ultrasound alone does not establish the syndrome.

Amenorrhoea requires pregnancy exclusion. Primary causes include chromosomal, anatomical, gonadal, pituitary, hypothalamic, and endocrine disorders. Secondary amenorrhoea may result from pregnancy, lactation, menopause, weight change, intense exercise, stress, chronic disease, hyperprolactinaemia, thyroid disease, polycystic ovarian syndrome, ovarian insufficiency, uterine scarring, or medicines. History, examination, pregnancy testing, gonadotropins, oestradiol, prolactin, thyroid testing, and selective imaging localise the axis. Low-energy hypothalamic suppression threatens bone and fertility and is treated by restoring energy availability.

Infertility

Infertility is failure to conceive after an appropriate period of regular unprotected intercourse, assessed earlier with advanced reproductive age, irregular cycles, known pelvic or testicular disease, previous gonadotoxic treatment, or other risk. Evaluate both partners concurrently. Female factors include ovulatory dysfunction, diminished ovarian reserve, tubal damage, endometriosis, uterine abnormalities, and age-related oocyte decline.

Confirm ovulation from cycle pattern or timed progesterone. Ovarian-reserve tests estimate stimulation response better than natural fertility. Tubal testing and ultrasound assess anatomy; hysteroscopy selectively evaluates the cavity. Treatment may include timing education, ovulation induction, surgery,

insemination, in-vitro fertilisation, donor gametes, or adoption. Discuss multiple pregnancy, ovarian hyperstimulation, emotional burden, cost, and realistic success.

Contraception

Contraceptive choice is preference-sensitive and should consider effectiveness, bleeding effects, privacy, reversibility, adherence, sexually transmitted infection protection, medical conditions, medicines, breastfeeding, and fertility timing. Condoms reduce infection transmission. Long-acting reversible methods include intrauterine devices and implants and have low user-failure rates.

Combined oestrogen-progestogen methods suppress ovulation and improve bleeding, pain, acne, and endometriosis symptoms, but oestrogen increases thrombosis risk and is unsuitable with selected migraine, smoking, vascular, hepatic, postpartum, and cancer contexts. Progestogen-only pills, injection, implant, and hormonal intrauterine systems avoid oestrogen but differ in bleeding, bone, weight, and return-to-fertility effects. Copper intrauterine contraception is hormone-free but may increase bleeding and pain.

Emergency contraception includes a copper intrauterine device or oral agents chosen according to time, weight, interactions, ovulation timing, and access. It does not disrupt an implanted pregnancy. Permanent contraception requires informed voluntary consent and discussion of alternatives and regret risk. Enzyme-inducing medicines reduce some hormonal methods; contraceptive planning must accompany teratogenic treatment.

Menopause and pelvic-floor health

Menopause is diagnosed retrospectively after sustained amenorrhoea at the expected age; earlier ovarian insufficiency requires evaluation. Vasomotor symptoms, sleep disruption, mood change, urogenital dryness, pain, urinary symptoms, and bone loss vary. Menopausal hormone therapy is the most effective vasomotor treatment and protects bone while used, but choice depends on uterus, thrombosis, cardiovascular, liver, bleeding, and cancer history. Systemic oestrogen requires endometrial protection when a uterus is present. Local vaginal oestrogen has low systemic exposure and treats urogenital symptoms for many patients.

Pelvic-organ prolapse and incontinence reflect childbirth, ageing, connective tissue, nerve, and muscle factors. Assess urinary, bowel, sexual, and pressure symptoms. Pelvic-floor physiotherapy, pessary, bowel management, topical therapy, and surgery follow burden and goals.

Gynaecological cancer

Postmenopausal bleeding requires endometrial assessment. Cervical cancer may cause postcoital or irregular bleeding; screening detects precancer before symptoms. Ovarian cancer often produces persistent bloating, early satiety, pelvic pressure, urinary change, or weight change rather than a single specific symptom. Vulval cancer can present as persistent itch, ulcer, mass, or skin change. Persistent warning features require examination, imaging, biopsy, and specialist referral; tumour markers are not general screening tests.

TTS module 2: Menstrual mechanisms, pelvic emergencies, reproductive planning, and contraceptive safety

Gynaecological assessment begins with three simultaneous questions: could the patient be pregnant, is haemorrhage or infection causing physiological instability, and is an ovary or other organ losing blood supply? Pregnancy testing is routine whenever biologically possible because ectopic pregnancy and miscarriage can resemble menstruation, urinary disease, gastroenteritis, or musculoskeletal pain. Privacy, consent, inclusive language, and trauma-informed explanation are essential because examination and reproductive history can otherwise reproduce harm.

The menstrual cycle is a coordinated hypothalamic, pituitary, ovarian, and endometrial sequence. Early follicular loss of progesterone and oestradiol releases gonadotropin drive. Follicle-stimulating hormone recruits follicles; a dominant follicle produces rising oestradiol and inhibin. Sustained high oestradiol switches from negative to positive feedback and triggers the luteinising-hormone surge. After ovulation, corpus-luteum progesterone stabilises secretory endometrium. Without implantation, luteal regression withdraws hormones, spiral arterioles constrict, inflammatory mediators activate, and the functional layer sheds.

Cycle history is quantified rather than labelled “heavy” or “irregular.” Record interval, predictability, duration, flooding, clots, overnight changes, double protection, leakage, iron deficiency, work or school restriction, and pain. Bleeding between periods, after sex, or after menopause has different implications. A normal-looking haemoglobin does not exclude depleted iron stores. Menstrual products, cultural expectations, anticoagulants, bleeding disorders, contraception, and access influence reporting.

Abnormal uterine bleeding is classified into structural and non-structural mechanisms. Polyps and submucosal fibroids distort the cavity; adenomyosis enlarges and inflames myometrium; hyperplasia and malignancy follow abnormal proliferation. Coagulopathy, ovulatory dysfunction, local endometrial haemostasis, medicines, and procedures can produce bleeding without a visible lesion. Anovulation creates prolonged unopposed oestrogen followed by irregular unstable shedding, common soon after menarche and in perimenopause but also associated with polycystic ovarian syndrome, thyroid disease, hyperprolactinaemia, energy deficit, and obesity.

Acute heavy bleeding is treated according to circulation and ongoing loss. Establish access, blood count, crossmatch, pregnancy status, coagulation context, and pelvic source. Tranexamic acid reduces fibrinolysis; hormonal regimens stabilise endometrium when appropriate; uterine tamponade, hysteroscopy, embolisation, or surgery controls refractory anatomical bleeding. Oestrogen-containing treatment is inappropriate in selected thrombotic or vascular risks. Once stable, investigate cause and restore iron rather than repeatedly treating episodes without a long-term plan.

Endometrial sampling is based on cancer probability rather than ultrasound thickness alone in premenopausal patients. Age, persistent irregular or intermenstrual bleeding, obesity, chronic anovulation, diabetes, tamoxifen, family syndromes, and failed treatment raise concern. Postmenopausal bleeding always requires assessment because even a small amount may signal endometrial or cervical malignancy. Transvaginal ultrasound evaluates endometrium and adnexa; hysteroscopy visualises focal cavity lesions that blind biopsy may miss.

Acute unilateral pelvic pain with nausea or vomiting raises adnexal torsion. An enlarged ovary twists its vascular pedicle, obstructing venous outflow before arterial inflow and causing oedema and infarction. Doppler flow can persist

through dual blood supply or intermittent torsion, so a reassuring scan cannot override a convincing syndrome. Urgent laparoscopy detorses and preserves ovarian tissue where possible. Ruptured haemorrhagic cyst can also cause intraperitoneal bleeding, requiring serial perfusion, haemoglobin, imaging, and operative assessment when unstable.

Pelvic inflammatory disease is a clinical diagnosis made with a low enough threshold to prevent tubal injury. Cervical-motion, uterine, or adnexal tenderness with lower pain supports treatment after ectopic pregnancy, torsion, appendicitis, and other emergencies are considered. Fever and raised markers may be absent. Nucleic-acid tests identify important sexually transmitted pathogens but negative results do not exclude polymicrobial upper-tract infection. Broad treatment, partner management, infection screening, abstinence until therapy is complete, and reassessment reduce infertility, ectopic risk, chronic pain, and tubo-ovarian abscess.

Endometriosis produces inflammatory and fibrotic disease outside the uterine cavity, commonly affecting pelvic peritoneum, ovaries, deep ligaments, bowel, bladder, and scars. Pain severity reflects lesion location, nerve involvement, central sensitisation, pelvic-floor response, inflammation, and psychosocial burden more than visible volume. Normal ultrasound does not exclude superficial disease. Empirical hormonal suppression is reasonable when pregnancy is not desired and warning features are absent; laparoscopy is used when diagnosis remains uncertain, fertility or organ involvement matters, or symptoms resist treatment.

Amenorrhoea is localised after pregnancy is excluded. High gonadotropins with low oestradiol indicate primary ovarian insufficiency. Low or normal gonadotropins with low oestradiol suggest hypothalamic or pituitary suppression. Hyperprolactinaemia inhibits gonadotropin-releasing hormone and can follow pituitary tumour, antipsychotics, hypothyroidism, chest stimulation, kidney failure, or pregnancy. Polycystic ovarian syndrome usually has evidence of androgen excess and preserved oestrogen. Outflow obstruction or intrauterine scarring is considered when hormones and secondary sexual development appear intact.

Functional hypothalamic amenorrhoea reflects inadequate energy availability, stress, illness, or excessive exercise and is not harmless contraception. Low oestradiol compromises bone, cardiovascular physiology, fertility, and genital

tissue. Weight may be normal and eating disorder can be concealed. Treatment restores energy intake relative to expenditure, reduces excessive training, addresses psychological drivers, and protects bone. Combined hormonal pills can create withdrawal bleeding without correcting the underlying energy deficit.

Polycystic ovarian syndrome is diagnosed from a compatible combination of ovulatory dysfunction, androgen excess, and ovarian morphology after excluding thyroid disease, hyperprolactinaemia, non-classic adrenal hyperplasia, androgen-secreting tumour, and Cushing syndrome where appropriate. Rapid virilisation or very high androgen is not typical and demands urgent investigation. Management follows goals: cycle protection, metabolic prevention, hirsutism or acne treatment, fertility, sleep apnoea, and psychological wellbeing. Regular progestogen exposure reduces endometrial hyperplasia risk.

Infertility is a couple or reproductive-system assessment, not a female diagnosis. Evaluate ovulation, ovarian reserve in its limited role, uterine cavity, tubal patency, semen, sexual timing, age, and prior pelvic or gonadotoxic disease in parallel. Ovarian-reserve tests predict response to stimulation better than spontaneous conception and should not be presented as an egg-quality score. Tubal damage from infection or endometriosis raises ectopic risk. Treatment ranges from timing and ovulation induction to surgery, insemination, in-vitro fertilisation, donor gametes, or non-biological parenting.

Contraceptive counselling separates perfect-use efficacy from typical-use failure and considers autonomy, privacy, bleeding preference, reversibility, infection prevention, adherence, medical risk, drug interaction, and reproductive coercion. Long-acting implants and intrauterine devices minimise user-dependent failure. Condoms are the principal contraceptive method that also reduces sexually transmitted infection transmission. Dual protection may combine a highly effective pregnancy method with barriers.

Combined oestrogen–progestogen contraception suppresses ovulation but increases venous and selected arterial thrombosis risk. Migraine with aura, smoking at older reproductive age, severe hypertension, active vascular disease, early postpartum state, selected liver disease, thrombophilia, and breast cancer context influence eligibility. Progestogen-only methods avoid oestrogen but differ in irregular bleeding, bone effect, return to fertility, and

interaction. Enzyme-inducing anticonvulsants and other drugs reduce several hormonal methods, while intrauterine contraception remains unaffected.

Emergency contraception works before implantation. A copper intrauterine device is highly effective and provides ongoing contraception; oral options delay or inhibit ovulation and vary with elapsed time, body weight, interacting medicines, and subsequent hormonal-method timing. They do not end an implanted pregnancy. Sexual-assault care offers emergency contraception, infection prophylaxis, forensic options, safeguarding, and psychological support without making access contingent on police reporting.

Menopause results from depletion of ovarian follicular function. Vasomotor instability, sleep disruption, urogenital atrophy, sexual pain, mood symptoms, and bone loss vary. Systemic menopausal hormone therapy is the most effective vasomotor treatment but requires individual assessment of age, timing, uterus, thrombosis, cardiovascular, liver, bleeding, and cancer history. Oestrogen without adequate progestogen exposes an intact endometrium to hyperplasia. Local vaginal oestrogen has low systemic exposure and can improve dryness, pain, recurrent urinary symptoms, and tissue resilience for many patients.

Persistent postcoital bleeding, vulval itch or ulcer, pelvic mass, early satiety, bloating, or unexplained weight change requires examination rather than repeated empirical infection treatment. Cervical screening prevents cancer through detection of high-risk viral change but is not a diagnostic test for symptoms. Ovarian tumour markers lack specificity for population screening. The guiding principle across gynaecology is to protect life and organ perfusion first, then align endocrine, fertility, sexual, and cancer care with the person's own reproductive goals.

Retrieval prompts

One. What must be excluded first in abnormal bleeding or pelvic pain?

Two. Which mechanisms cause abnormal uterine bleeding?

Three. Why can endometriosis severity not be judged from pain alone?

Four. What long-term risks accompany polycystic ovarian syndrome?

Five. Which contraception also protects against sexually transmitted infection?

Six. Why does systemic oestrogen require progestogen when a uterus is present?

Concise answers

One. Pregnancy and immediate threats such as ectopic pregnancy, torsion, haemorrhage, and severe infection.

Two. Structural lesions, malignancy, coagulopathy, ovulatory or endometrial dysfunction, medicines, and procedures.

Three. Pain reflects inflammation, neural sensitisation, location, and context and correlates poorly with lesion burden.

Four. Infertility, diabetes, cardiovascular risk factors, sleep apnoea, and endometrial hyperplasia or cancer.

Five. Barrier contraception, particularly condoms.

Six. Unopposed oestrogen stimulates endometrium and increases hyperplasia and cancer risk.

Source map

Original synthesis informed by Guyton and Hall, ovarian cycles, reproductive endocrinology, pregnancy, and menopause; Robbins, uterine, ovarian, cervical, vulval, inflammatory, and neoplastic pathology; Katzung and OpenStax Pharmacology, hormonal contraception, fertility, bleeding, pain, infection, and menopause therapy; Talley and O'Connor, menstrual, sexual, fertility, pelvic, abdominal, and endocrine assessment; OpenStax Medical-Surgical Nursing, gynaecological procedures, cancer, infection, fertility, trauma-informed care, and education; and OpenStax Anatomy and Physiology and Microbiology, reproductive anatomy and sexually transmitted infection.

Chapter 54: Antenatal Care, Pregnancy Complications, Labour, and the Puerperium

Orientation

Pregnancy alters every major system while two patients share placental circulation. Normal adaptations can resemble disease, and minor maternal deterioration may threaten placental perfusion. Care combines dating, maternal assessment, fetal growth, screening, prevention, birth planning, and recognition of haemorrhage, hypertension, sepsis, thromboembolism, diabetes, and fetal compromise. Management varies with gestation and requires current obstetric protocols.

Confirmation, dating, and initial assessment

Urine or serum human chorionic gonadotropin confirms trophoblastic activity but does not by itself locate or prove viability. Date pregnancy from the last normal menstrual period and early ultrasound, accounting for cycle uncertainty and assisted conception. Early ultrasound identifies location, number, cardiac activity, and crown-rump length. A pregnancy of unknown location requires serial clinical assessment, hormone trends, and imaging until intrauterine pregnancy, ectopic pregnancy, or loss is established.

At booking, review obstetric history, disease, medicines, allergies, vaccination, infection, genetics, mental health, substance use, nutrition, violence, housing, and support. Record blood pressure, weight context, urinalysis, blood group and antibodies, blood count, infection screening, and selected tests. Reconcile teratogenic and essential medicines: abrupt withdrawal of anticonvulsants, steroids, psychiatric therapy, or other critical treatment may exceed exposure risk.

Folic acid before conception and early pregnancy reduces neural-tube defects, with higher doses for selected risks. Other supplementation follows diet, deficiency, and policy. Avoid retinol excess. Advise food and infection

precautions, exercise, travel, occupational exposure, and warning symptoms without unnecessary restriction.

Maternal adaptation and monitoring

Plasma volume rises more than red-cell mass, producing physiological dilutional anaemia. Cardiac output and heart rate increase, systemic resistance falls, and blood pressure often dips mid-pregnancy. Ventilation rises, lowering carbon dioxide and bicarbonate. Renal blood flow and filtration increase, altering creatinine and drug clearance. Gastrointestinal motility slows and clotting shifts toward thrombosis. These adaptations reduce reserve for haemorrhage, embolism, infection, and cardiac disease.

Routine visits monitor blood pressure, symptoms, fetal growth, movement, and risk. Ultrasound evaluates anatomy and growth. Genetic screening is voluntary and requires explanation of screening versus diagnosis, false results, next tests, and choices. Chorionic-villous sampling and amniocentesis are invasive diagnostic tests.

Early pregnancy pain and bleeding

Miscarriage can present with bleeding and cramping, but symptoms overlap viable and ectopic pregnancy. Assess perfusion, pain, bleeding, peritonism, cervical findings when indicated, haemoglobin, blood group, hormone level, and ultrasound. Management of confirmed loss may be expectant, medical, or surgical according to stability, gestation, infection, bleeding, and preference. Provide clear follow-up and compassionate acknowledgement of grief.

Ectopic pregnancy most often implants in a fallopian tube. Risk rises with previous ectopic, tubal disease or surgery, pelvic infection, smoking, and assisted reproduction, but many have no risk factor. Unilateral pain, bleeding, shoulder-tip pain, syncope, or shock are warning features. Rupture can occur with low hormone levels. Unstable or ruptured ectopic pregnancy requires immediate surgery and resuscitation; stable cases can receive methotrexate or observation with reliable follow-up.

Severe vomiting with weight loss, dehydration, ketosis, or electrolyte disturbance suggests hyperemesis gravidarum after alternatives are

considered. Treat thiamine, fluids, electrolytes, antiemetics, and nutrition; give thiamine before substantial carbohydrate in prolonged vomiting.

Hypertensive disorders

Chronic hypertension predates pregnancy or appears early. Gestational hypertension develops later without defining organ injury. Pre-eclampsia is new hypertension after mid-pregnancy with proteinuria or maternal organ dysfunction involving kidney, liver, brain, blood, or placenta. It arises from abnormal placentation, endothelial dysfunction, vasoconstriction, inflammation, and capillary leak. Severe headache, visual symptoms, epigastric or right-upper pain, dyspnoea, reduced urine, confusion, bleeding, or reduced fetal movement require urgent assessment.

Confirm blood pressure correctly and assess blood count, platelets, kidney and liver tests, urine protein, symptoms, fetal growth, fluid, and wellbeing. Antihypertensives reduce severe-pressure complications; magnesium sulphate prevents and treats eclamptic seizure. Definitive treatment is delivery, balanced against prematurity when mother and fetus are stable. Corticosteroids accelerate fetal lung maturation when early birth is likely. Low-dose aspirin reduces pre-eclampsia risk in selected patients. Postpartum deterioration remains possible.

Diabetes in pregnancy

Placental hormones increase insulin resistance to direct nutrients toward the fetus. Gestational diabetes occurs when beta-cell compensation is inadequate; pre-existing diabetes carries risk from conception.

Hyperglycaemia increases congenital anomaly with early exposure, fetal overgrowth, birth injury, pre-eclampsia, polyhydramnios, neonatal hypoglycaemia, and later metabolic disease. Screen at the recommended gestation or earlier with risk factors.

Management uses nutrition, activity, glucose monitoring, and medicines when targets are not met. Insulin is established therapy; some settings use selected oral agents after counselling. Monitor fetal growth and plan timing and mode of birth from glycaemia, fetal size, comorbidity, and obstetric factors. Insulin requirements fall sharply after placental delivery. Reassess maternal glucose postpartum because future type two diabetes risk remains high.

Placental and fetal complications

Placenta praevia overlies or approaches the cervical opening and commonly causes painless bleeding. Avoid digital vaginal examination until placental location is known. Placental abruption is premature separation, often causing pain, uterine tenderness, contractions, bleeding, and fetal compromise; concealed haemorrhage means visible loss underestimates severity. Resuscitation, blood-product readiness, fetal assessment, and timely delivery are essential.

Rhesus D alloimmunisation occurs when a susceptible pregnant person forms antibodies against fetal red cells. Antenatal and postpartum anti-D immunoglobulin prevents sensitisation after routine indications and potentially sensitising events. Established antibodies require fetal anaemia surveillance and specialist treatment.

Reduced fetal movement warrants assessment. Growth restriction reflects fetal, placental, or maternal disease and is evaluated with serial biometry, fluid, and Doppler flow. Preterm membrane rupture risks infection, cord complications, and prematurity. Fever, uterine tenderness, tachycardia, offensive fluid, or deterioration suggests intra-amniotic infection and prompts antibiotics and delivery planning.

Labour and birth

Labour is regular uterine activity producing cervical change. First stage covers cervical dilation, second stage birth, and third stage placental delivery. Confirm presentation, position, fetal heart pattern, contraction frequency, membrane status, cervical progress when examination is appropriate, maternal observations, pain, hydration, and preferences. Continuous support and respectful communication improve experience and outcomes.

Analgesia options include non-drug support, inhaled agents, opioids, and neuraxial techniques, each with maternal and fetal effects. Fetal heart monitoring identifies patterns requiring correction of hypotension, excessive contractions, cord compression, or hypoxia and possible expedited birth. Delay can result from ineffective contractions, malposition, disproportion, or obstruction. Operative vaginal birth or caesarean section requires clear

indication, consent proportional to urgency, skilled staff, anaesthesia, neonatal support, antibiotics when indicated, and thrombosis prevention.

Shoulder dystocia is failure of shoulders to deliver after the head and requires an immediate rehearsed sequence of manoeuvres, help, timing, and neonatal assessment. Umbilical cord prolapse compromises fetal blood flow and requires pressure relief and emergency delivery. Uterine rupture causes pain, bleeding, fetal compromise, loss of station, or shock, particularly with a scarred uterus, and demands laparotomy.

Postpartum haemorrhage

Postpartum haemorrhage arises from uterine atony, retained tissue, genital trauma, or coagulopathy. Call for help, assess airway and circulation, massage the uterus, establish large-bore access, obtain blood, warm the patient, quantify loss, and identify cause. Use uterotonics while respecting contraindications, tranexamic acid early, repair trauma, remove retained tissue, correct coagulation, and escalate to balloon tamponade, embolisation, compression sutures, arterial control, or hysterectomy. Shock may precede a dramatic blood-pressure fall.

Puerperium and lactation

After birth, monitor bleeding, uterine tone, observations, bladder, perineum or wound, pain, mobility, thrombosis, infection, blood pressure, mood, feeding, and infant wellbeing. Endometritis, wound infection, mastitis, urinary infection, and sepsis require prompt recognition. Unilateral leg swelling, chest pain, dyspnoea, syncope, severe headache, neurological deficit, or hypertension may indicate thromboembolism, pre-eclampsia, stroke, or anaesthetic complication.

Prolactin supports milk production and oxytocin causes milk ejection. Early skin contact, effective attachment, frequent milk removal, and practical support establish supply. Assess nipple trauma, engorgement, infant transfer, hydration, weight, jaundice, and maternal medicines. Mastitis is inflammatory and may become bacterial; continue milk drainage when safe, treat pain and infection, and image a suspected abscess.

Postpartum blues are brief and common; persistent depression, anxiety, intrusive thoughts, mania, psychosis, suicidality, or inability to care require urgent assessment. Postpartum psychosis is an emergency. Discuss contraception, pelvic-floor recovery, chronic disease, vaccination, sleep, support, birth experience, and future pregnancy planning. A postpartum review should connect complications such as diabetes or pre-eclampsia to long-term cardiovascular and metabolic prevention.

TTS module 2: Maternal adaptation, placental physiology, obstetric emergencies, and postpartum transition

Pregnancy requires maternal organs to support increased metabolic demand, uteroplacental flow, fetal growth, and later haemorrhage at delivery. These adaptations can mimic disease: heart rate rises, plasma volume expands, blood pressure falls mid-pregnancy, ventilation increases, creatinine falls, and coagulation becomes more prothrombotic. Interpretation must use gestational physiology. A creatinine considered normal outside pregnancy may represent kidney impairment, and apparently modest blood loss can become dangerous as compensation fails.

Plasma volume increases more than red-cell mass, producing dilutional anaemia while total red-cell mass still rises. Cardiac output increases through heart rate and stroke volume; systemic vascular resistance falls under hormonal and placental influence. Supine uterine compression of the vena cava can reduce venous return later in pregnancy and improves with lateral displacement. Physiological flow murmurs and mild breathlessness occur, but syncope, chest pain, resting hypoxaemia, pulmonary oedema, or progressive limitation require cardiac and thromboembolic evaluation.

Progesterone increases ventilatory drive, lowering arterial carbon dioxide and bicarbonate through renal compensation. Oxygen consumption rises and functional residual capacity falls as the diaphragm elevates, reducing oxygen reserve during apnoea. Airway mucosa becomes oedematous and aspiration risk increases. These changes make respiratory failure and anaesthesia potentially rapid. Pregnancy-specific blood-gas interpretation matters: a “normal” non-pregnant carbon dioxide can signal hypoventilation.

Renal plasma flow and filtration increase early, lowering urea and creatinine and changing clearance of medicines. Glycosuria can occur because filtered glucose exceeds tubular reabsorption, but it does not diagnose gestational diabetes. Ureteric dilation and urinary stasis increase pyelonephritis risk. Pregnancy is hypercoagulable through increased clotting factors, reduced fibrinolysis, and venous stasis, protecting against delivery bleeding while raising venous thrombosis risk, especially postpartum and with surgery, immobility, obesity, or thrombophilia.

The placenta is an exchange and endocrine organ. Maternal and fetal blood normally remain separate while oxygen, carbon dioxide, nutrients, waste, antibodies, and drugs cross by several mechanisms. Uterine spiral arteries should remodel into low-resistance vessels. Inadequate invasion and remodelling contribute to placental ischaemia, release of antiangiogenic and inflammatory signals, maternal endothelial dysfunction, pre-eclampsia, fetal growth restriction, and abruption. Maternal pressure can be high while placental flow remains poor.

A positive human chorionic gonadotropin test proves trophoblastic activity, not location, viability, or normal development. Early pain or bleeding requires assessment of stability, gestational dating, ultrasound, and serial hormone trends where location is unknown. Discriminatory hormone thresholds are not absolute and should not trigger treatment of a desired potentially viable pregnancy without adequate evidence. Ectopic rupture can occur at low or falling concentrations. Shoulder-tip pain, syncope, peritonism, or shock demands immediate surgical care.

Hyperemesis gravidarum is more than ordinary nausea. Weight loss, dehydration, ketosis, electrolyte disturbance, kidney injury, and inability to function require treatment and exclusion of multiple pregnancy, molar disease, thyroid, liver, gastrointestinal, neurological, and medication causes. Prolonged vomiting depletes thiamine; giving substantial carbohydrate first can precipitate Wernicke encephalopathy. Replace thiamine, fluid, potassium and other deficits, provide pregnancy-compatible antiemetics, prevent thrombosis when admitted, and escalate nutrition support when oral intake remains impossible.

Pre-eclampsia is a multisystem placental endothelial disease, not simply hypertension plus protein. New hypertension after mid-pregnancy with kidney,

liver, neurological, haematological, pulmonary, or placental dysfunction meets the syndrome even without proteinuria. Severe headache, visual change, right-upper-quadrant or epigastric pain, dyspnoea, reduced urine, confusion, brisk reflexes, bleeding, or reduced fetal movement are warning signs. Confirm pressure with correct cuff and positioning, then assess platelets, haemolysis, liver, creatinine, urine protein, symptoms, fetal growth, fluid, and Doppler physiology.

Severe hypertension is treated urgently to prevent maternal intracranial haemorrhage and other injury. Magnesium sulphate prevents or treats eclamptic seizure and requires monitoring of reflexes, breathing, urine, and toxicity, with calcium available as antidote. Fluid is cautious because endothelial leak and low oncotic pressure predispose pulmonary oedema. Delivery removes the placenta and is definitive, but timing balances maternal deterioration, fetal compromise, and prematurity. Corticosteroids support fetal lung maturation when early birth is expected but must not delay delivery during uncontrolled threat.

HELLP syndrome combines haemolysis, elevated liver enzymes, and low platelets and may present with epigastric pain, nausea, malaise, or hypertension. Eclampsia can occur before, during, or after birth and may be the first presentation. Postpartum pre-eclampsia remains possible after discharge, so severe headache, visual symptoms, dyspnoea, swelling with illness, epigastric pain, seizure, or high pressure needs emergency assessment. People with pre-eclampsia have increased lifelong hypertension, stroke, and cardiovascular risk and need preventive follow-up.

Gestational diabetes reflects placental-hormone insulin resistance exceeding pancreatic beta-cell reserve. Maternal glucose crosses placenta and drives fetal insulin, promoting growth and fat deposition. After birth, placental glucose stops abruptly while neonatal insulin remains high, causing hypoglycaemia. Management combines nutrition, activity, glucose monitoring, and medication when targets are unmet. Fetal growth, fluid, pressure, and timing of delivery are monitored. Insulin requirement falls sharply after placental delivery, and postpartum testing identifies persistent diabetes and future type-two risk.

Antepartum haemorrhage is managed by physiology and placental anatomy. Placenta praevia classically causes painless bleeding from a placenta near or

covering the cervix; digital vaginal examination is avoided until location is known. Placental abruption produces painful bleeding, uterine tenderness or hypertonus, contractions, fetal compromise, and coagulopathy, but blood can remain concealed. Visible volume therefore underestimates loss. Vasa praevia causes fetal vessels to cross membranes near the cervix and can produce catastrophic fetal bleeding when membranes rupture.

Reduced fetal movement is a symptom of possible fetal hypoxia or placental dysfunction and deserves same-day assessment according to gestation and protocol. Growth restriction is distinguished from a constitutionally small fetus using growth trajectory, fluid, maternal disease, and umbilical and other Doppler flow. Abnormal placental resistance redistributes fetal circulation toward brain and heart before decompensation. Timing birth balances worsening intrauterine hypoxia against prematurity and requires serial rather than single measurements.

Labour is diagnosed by regular uterine activity producing cervical change. Progress depends on power, passenger, passage, position, and maternal–fetal tolerance. Fetal heart monitoring is an indirect measure of autonomic response to oxygenation, not a direct oxygen meter. Baseline, variability, accelerations, decelerations, contraction frequency, gestation, medicines, and clinical context are interpreted together. Correct maternal hypotension, excessive uterine activity, position-related compression, fever, and reversible causes while preparing expedited operative birth when compromise persists.

Shoulder dystocia is an unpredictable mechanical emergency after delivery of the head. Traction on the fetal head can injure the brachial plexus and does not release the shoulder. Call for help, document time, stop pushing where directed, use maternal positioning, suprapubic pressure, internal rotational manoeuvres, or delivery of the posterior arm in a rehearsed sequence. Fundal pressure is avoided. The newborn is assessed for hypoxia and injury, and the parent for haemorrhage and trauma.

Umbilical cord prolapse compresses fetal blood flow between presenting part and pelvis after membrane rupture. Elevate the presenting part manually or through positioning, minimise cord manipulation, treat fetal compromise, and proceed to emergency birth unless vaginal delivery is immediately achievable. Uterine rupture can present with sudden pain, fetal bradycardia, bleeding, loss

of station, cessation of contractions, or maternal shock, especially with a uterine scar. Immediate laparotomy and haemorrhage resuscitation are required.

Postpartum haemorrhage is organised as tone, tissue, trauma, and thrombin. Uterine atony is most common and is treated with uterine massage, emptying the bladder, and uterotonics chosen around asthma, hypertension, and other contraindications. Retained placenta or accreta requires removal or surgery; genital laceration requires visualisation and repair; coagulopathy requires targeted blood products. Warmed balanced transfusion, calcium, tranexamic acid given early, and rapid escalation to balloon, embolisation, compression sutures, vessel control, or hysterectomy prevent the lethal triad.

The postpartum period includes large haemodynamic and endocrine shifts. Autotransfusion from uterine contraction and mobilisation of extracellular fluid can expose heart failure. Venous thromboembolism risk is high. Endometritis presents with fever, uterine tenderness, offensive lochia, or sepsis; wound and urinary infections are common alternatives. Chest pain, dyspnoea, syncope, unilateral leg swelling, severe headache, focal neurology, heavy bleeding, fever, or abdominal pain are emergency warnings, not routine recovery.

Lactation depends on prolactin-driven production and oxytocin-mediated ejection, reinforced by frequent effective milk removal. Assess attachment, swallow, transfer, infant weight and hydration, nipple injury, maternal pain, and medicines rather than attributing difficulty to motivation. In mastitis, ongoing drainage is usually encouraged and bacterial disease treated when indicated; a persistent focal mass may require ultrasound and abscess drainage. Feeding plans should support parental goals without compromising neonatal nutrition.

Postpartum mental illness ranges from transient blues to depression, anxiety, obsessive intrusive thoughts, mania, and psychosis. Intrusive thoughts can be ego-dystonic and frightening without intent, but risk assessment is explicit. Postpartum psychosis, severe mania, suicidality, inability to care, or thoughts of harming the infant require emergency specialist care and protection. Follow-up after pregnancy should also address contraception, pelvic floor, anaemia, blood pressure, diabetes, vaccination, birth trauma, sleep, and how placental complications predict long-term maternal health.

Retrieval prompts

One. Why does a positive pregnancy test not prove a viable intrauterine pregnancy?

Two. Which features define pre-eclampsia beyond hypertension?

Three. Why can placental abruption cause underestimated blood loss?

Four. What are the four major mechanisms of postpartum haemorrhage?

Five. Which postpartum symptoms require urgent escalation?

Six. Why is follow-up after gestational diabetes or pre-eclampsia important?

Concise answers

One. Human chorionic gonadotropin indicates trophoblast but not implantation site, viability, or normal development.

Two. Proteinuria or maternal kidney, liver, neurological, haematological, pulmonary, or placental dysfunction after mid-pregnancy.

Three. Blood can remain concealed behind the placenta or within the uterus.

Four. Uterine atony, retained tissue, genital trauma, and coagulopathy.

Five. Heavy bleeding, severe headache, visual or neurological symptoms, chest pain, dyspnoea, syncope, fever, severe abdominal pain, or psychiatric crisis.

Six. Both predict increased future diabetes, hypertension, cardiovascular disease, and recurrence in later pregnancy.

Source map

Original synthesis informed by Guyton and Hall, maternal adaptation, placental exchange, fetal circulation, labour, and lactation; Robbins, placental pathology, hypertensive disease, haemorrhage, infection, and trophoblastic disease; Katzung and OpenStax Pharmacology, pregnancy-safe prescribing, antihypertensives, magnesium, diabetes therapy, uterotonics, analgesia, and lactation; Talley and O'Connor, obstetric history and maternal examination; OpenStax Medical-Surgical Nursing, antenatal monitoring, labour, obstetric emergencies, postpartum care, feeding, and education; and OpenStax Anatomy and Physiology, Biology, Chemistry, and Microbiology, embryology, inheritance, metabolism, and perinatal infection.

Chapter 55: The Eye: Visual Physiology, Examination, and Common Disease

Orientation

Vision depends on transparent optical media, accurate focus, retinal phototransduction, intact optic pathways, coordinated eye movement, and cortical interpretation. Eye symptoms can reveal local disease, neurological emergencies, vascular injury, infection, inflammation, toxicity, or systemic illness. Sudden visual loss, a painful red eye with reduced vision, chemical injury, penetrating trauma, acute angle closure, retinal arterial occlusion, endophthalmitis, and giant-cell arteritis require immediate action.

Optics and retinal signalling

The cornea supplies most refractive power; the lens fine-tunes focus. Ciliary-muscle contraction reduces zonular tension, allowing the lens to thicken for near vision. Myopia focuses distant images before retina, hyperopia behind it, astigmatism unevenly, and presbyopia reflects age-related loss of accommodation. Refraction corrects optical blur but cannot restore vision lost from retinal or neural disease.

Rods are highly light-sensitive and support dim peripheral vision. Cones provide colour and high acuity, concentrated in the fovea. Light activates photopigment and hyperpolarises photoreceptors, altering glutamate release to bipolar and ganglion networks. Ganglion axons form the optic nerve. Nasal retinal fibres cross at the chiasm, so pathway lesions create characteristic field defects. Pupillary constriction tests afferent retina and optic nerve with parasympathetic efferent function; a relative afferent defect indicates asymmetric retinal or optic-nerve dysfunction.

Aqueous humour flows from ciliary body through pupil and trabecular meshwork; resistance determines intraocular pressure. Vitreous fills the posterior globe. Retinal pigment epithelium supports photoreceptors, while

choroidal and retinal circulations supply different layers. Small disruptions in perfusion, transparency, or alignment can markedly reduce function.

History

Define monocular or binocular change, onset, duration, progression, field or central loss, blur, distortion, double vision, flashes, floaters, curtains, halos, colour desaturation, pain, photophobia, discharge, redness, trauma, chemical exposure, contact lenses, headache, jaw pain, scalp tenderness, neurological symptoms, and systemic illness. Determine whether diplopia persists when either eye is covered: monocular diplopia is usually optical; binocular diplopia reflects misalignment.

Review previous vision, surgery, glaucoma, retinal disease, diabetes, hypertension, immune disease, infection, migraine, cancer, medicines, smoking, occupation, family history, and eye protection. Steroids can promote cataract, glaucoma, and infection. Anticholinergic dilation may precipitate angle closure in susceptible eyes. Hydroxychloroquine and several other drugs require retinal monitoring.

Examination

Measure visual acuity in each eye separately with habitual correction before other manipulations. If reduced, use a pinhole: improvement suggests refractive blur. Test near vision when relevant. Inspect lids, lashes, conjunctiva, sclera, cornea, anterior chamber, iris, and pupils. Fluorescein with blue illumination reveals epithelial defects; avoid pressure and shield rather than patch if globe penetration is suspected.

Assess pupil size, shape, direct and consensual responses, and swinging-light response. Test visual fields by confrontation, then eye position and movements in all directions. Note ptosis, nystagmus, pain, limitation, and diplopia. Cover testing reveals misalignment. Fundoscopy examines red reflex, disc margin and colour, vessels, macula, and retina; pharmacological dilation improves view when safe but does not replace urgent specialist examination.

Intraocular pressure is interpreted with corneal and clinical context and should not be measured when globe rupture is possible. Slit-lamp examination

magnifies anterior structures. Optical coherence tomography maps retina and optic nerve; formal perimetry quantifies fields. Ultrasound helps when media are opaque but is avoided or used cautiously with suspected open globe.

Red eye

Conjunctivitis causes diffuse injection and discharge with preserved vision and little true pain. Viral disease is watery and contagious; bacterial discharge is purulent; allergy causes bilateral itch. Contact-lens wear increases risk of rapidly destructive bacterial keratitis. Corneal ulcer causes pain, photophobia, reduced vision, focal opacity, and fluorescein uptake and needs urgent cultures and antimicrobial therapy.

Anterior uveitis produces aching pain, photophobia, ciliary injection, and inflammatory cells in the anterior chamber. Causes include immune disease, infection, trauma, and drugs; treatment requires ophthalmic diagnosis because steroid drops can worsen herpetic keratitis. Scleritis causes deep severe pain and may reflect systemic vasculitis, whereas episcleritis is more superficial and self-limited.

Acute angle-closure glaucoma causes severe ocular pain, headache, blurred vision or halos, nausea, a poorly reactive mid-dilated pupil, corneal haze, and raised pressure. Rapid pressure lowering and definitive angle treatment protect the optic nerve. Chemical injury requires immediate copious irrigation before detailed history, with pH rechecking and removal of retained particles. Alkali penetrates especially deeply.

Sudden visual loss

Retinal arterial occlusion causes sudden painless monocular loss from retinal ischaemia. Treat as a vascular emergency, assess for giant-cell arteritis, embolic source, and stroke risk, and involve ophthalmology immediately. Retinal venous occlusion produces variable painless loss with retinal haemorrhage and oedema and requires vascular-risk treatment and retinal therapy when indicated.

Retinal detachment often causes flashes, new floaters, and a curtain or field defect. Posterior vitreous separation can produce similar symptoms and may tear retina, so urgent dilated examination is needed. Macula-on detachment is

particularly time-sensitive. Vitreous haemorrhage causes floaters, haze, or loss and may arise from diabetic neovascularisation, tear, trauma, or vascular disease.

Optic neuritis commonly causes subacute monocular loss, pain with movement, colour desaturation, and an afferent pupillary defect; demyelination, infection, and immune disease are causes. Ischaemic optic neuropathy causes sudden loss, often with disc swelling. In older patients, headache, scalp tenderness, jaw claudication, polymyalgia, fever, or high inflammatory markers suggest giant-cell arteritis. Give immediate high-dose corticosteroid treatment when strongly suspected to protect the other eye; arrange biopsy or vascular imaging without delaying treatment.

Occipital injury causes homonymous field loss and may spare pupillary responses. Transient monocular loss can be retinal ischaemia; transient binocular positive visual phenomena may be migraine but diagnosis requires context. Never dismiss new visual loss as migraine before vascular and ocular emergencies are considered.

Glaucoma

Glaucoma is progressive optic neuropathy with characteristic disc and field loss, often but not always associated with raised pressure. Open-angle disease is usually painless and peripheral loss is unnoticed until advanced. Risk includes age, family history, ancestry, pressure, thin cornea, steroid exposure, vascular factors, and prior injury. Screening or case detection uses pressure, disc assessment, optical coherence tomography, fields, and angle examination.

Treatment lowers pressure with topical agents, laser, or surgery. Prostaglandin analogues increase outflow; beta blockers reduce production but can worsen asthma, bradycardia, or heart block; alpha agonists and carbonic-anhydrase inhibitors have systemic effects. Adherence and drop technique matter. Angle closure differs mechanistically and requires anatomical management.

Cataract, macular disease, and diabetic eye disease

Cataract is lens opacity causing painless progressive blur, glare, poor contrast, colour change, or refractive shift. Ageing, diabetes, steroids, smoking, ultraviolet exposure, trauma, and congenital factors contribute. Surgery replaces the lens when impairment justifies procedural risk.

Age-related macular degeneration damages central vision. Drusen and pigment change characterise early disease; geographic atrophy causes gradual loss, while choroidal neovascularisation causes new distortion or rapid central loss. Anti-vascular endothelial growth factor injections preserve vision in neovascular disease. Smoking cessation and selected supplements reduce progression in defined stages, not as universal prevention.

Diabetic retinopathy progresses from microaneurysms and leakage to ischaemia and fragile neovascularisation. Macular oedema threatens central vision at any stage. Glycaemic and blood-pressure control reduce risk, while retinal laser, intraocular therapy, and surgery treat sight-threatening disease. Rapid glucose improvement can transiently worsen retinopathy, so established disease needs coordinated monitoring.

Paediatric and neuro-ophthalmic principles

Visual development requires clear images and alignment during a sensitive period. Amblyopia is reduced vision from unequal focus, deprivation, or strabismus and becomes less reversible with age. Abnormal red reflex, leukocoria, constant strabismus, developmental concern, or inability to fix and follow needs urgent assessment. Retinoblastoma is an important cause of leukocoria.

Cranial-nerve palsies, neuromuscular junction disease, thyroid orbitopathy, and orbital lesions cause diplopia. Painful ophthalmoplegia, proptosis, reduced acuity, colour loss, pupil abnormality, fever, or severe headache suggests orbital cellulitis, cavernous sinus disease, aneurysm, or raised intracranial pressure. Papilloedema is optic-disc swelling from raised intracranial pressure and requires urgent neurological evaluation.

TTS module 2: Visual localisation, sight-threatening emergencies, and systemic ocular disease

Visual assessment localises dysfunction by asking which essential step has failed: light must pass through tear film, cornea, aqueous, pupil, lens, and vitreous; photoreceptors must transduce it; retinal ganglion cells and optic nerve must transmit it; eye movements must align images; and cortex must interpret them. Monocular loss usually lies in one eye or anterior visual pathway. Homonymous field loss lies behind the chiasm. Binocular diplopia disappears when either eye is covered and reflects misalignment, whereas persistent monocular diplopia is usually optical.

Visual acuity is measured first in each eye separately with the person's usual correction because later drops, pressure, pain, or examination can change it. Pinhole blocks peripheral rays and improves refractive blur but not most retinal or neural loss. Near and distance acuity sample different focusing demands. When chart letters cannot be read, record progressively counting fingers, hand movements, light perception, and projection rather than "poor vision." Baseline and change determine urgency as much as the absolute value.

Phototransduction begins when light changes retinal chromophore configuration, activating an opsin cascade that lowers cyclic nucleotide and hyperpolarises the photoreceptor. Darkness therefore produces tonic glutamate release, while light reduces it. ON and OFF bipolar pathways encode increments and decrements; horizontal and amacrine interactions create contrast, temporal sensitivity, and receptive fields. Ganglion-cell output already contains processed information rather than a pixel copy. The fovea sacrifices convergence for high acuity; peripheral retina sacrifices detail for sensitivity and motion detection.

The pupillary light reflex tests an afferent signal through retina, optic nerve, chiasm, and midbrain and an efferent parasympathetic pathway through the third nerve and ciliary ganglion. A relative afferent defect appears when moving light to the worse eye causes both pupils to constrict less or appear to dilate. It indicates asymmetric severe retinal or optic-nerve dysfunction but does not localise between them alone. Anisocoria greater in bright light suggests failure of constriction in the larger pupil; greater in darkness suggests failure of sympathetic dilation in the smaller pupil.

Field patterns follow fibre anatomy. Optic-nerve disease produces monocular defects. Chiasmal compression classically impairs crossing nasal retinal fibres and temporal visual fields. Optic tract, radiations, and occipital cortex produce contralateral homonymous loss, becoming more congruous posteriorly as a general tendency. Temporal-lobe optic radiations carry superior visual fields, while parietal fibres carry inferior fields. Confrontation testing can miss subtle defects; formal perimetry quantifies threshold and progression.

A red eye is triaged by pain, photophobia, acuity, corneal clarity and staining, pupil, anterior-chamber inflammation, pressure, contact-lens use, trauma, and systemic disease. Simple conjunctivitis should preserve vision and cause irritation rather than deep pain. Corneal keratitis creates epithelial staining, focal opacity, severe pain, and photophobia and can progress rapidly in contact-lens wearers. Cultures and intensive organism-directed drops are urgent; eye patching or casual steroid use can accelerate destruction.

Anterior uveitis causes ciliary injection, consensual photophobia, pain, small or irregular pupil, and cells or protein in the anterior chamber. It may accompany spondyloarthritis, sarcoidosis, infection, trauma, or be idiopathic. Topical corticosteroid and cycloplegia can preserve function but require ophthalmic confirmation because herpetic epithelial keratitis worsens with steroid. Scleritis produces severe boring pain, often radiating and waking sleep, and can indicate systemic vasculitis; posterior disease may threaten vision without dramatic redness.

Acute angle closure occurs when aqueous cannot reach or drain through the trabecular meshwork, abruptly raising pressure. A shallow anterior chamber, lens anatomy, pupillary dilation, and susceptible angle contribute. Severe unilateral pain, headache, halos, nausea, corneal haze, reduced vision, and a fixed or sluggish mid-dilated pupil form the syndrome. Pressure-lowering drops and systemic therapy begin urgently, but definitive laser treatment relieves pupillary block and the fellow eye may also require prevention. Nausea should not misdirect the patient to a gastrointestinal diagnosis.

Chemical injury is treated before history. Irrigate immediately with abundant appropriate fluid, remove contact lenses and retained particles, and continue until ocular-surface pH remains neutral after stopping. Alkali penetrates and saponifies tissue particularly rapidly, while acids can also cause deep injury.

Do not delay irrigation to identify the product or measure initial acuity. Penetrating trauma is protected with a rigid shield without pressure; avoid tonometry, manipulation, ointment pressure, food, or magnetic resonance when a metallic foreign body is possible.

Sudden painless monocular loss is a vascular emergency until proved otherwise. Retinal arterial occlusion interrupts inner-retinal perfusion and may represent embolic carotid or cardiac disease, giant-cell arteritis, or systemic vascular risk. Ophthalmic action and stroke evaluation proceed urgently. Transient monocular dimming or curtain can be retinal transient ischaemia and requires the same seriousness even when examination normalises. Retinal venous occlusion produces haemorrhage and oedema and signals pressure, diabetes, glaucoma, and thrombosis risk.

Giant-cell arteritis causes granulomatous inflammation of medium and large arteries and can infarct the optic nerve. New headache, scalp tenderness, jaw or tongue claudication, polymyalgia, fever, anaemia, thrombocytosis, or transient visual symptoms in an older adult raises probability. Inflammatory markers can occasionally be normal. High-dose corticosteroid is started immediately when visual threat is suspected because the fellow eye can become blind within days. Ultrasound or biopsy confirms and long-term therapy is adjusted later.

Flashes arise when vitreous traction mechanically stimulates retina. New floaters may represent vitreous condensation, posterior vitreous separation, haemorrhage, or inflammatory cells. A sudden shower of floaters, flashes, or curtain-like field defect requires urgent dilated peripheral retinal examination for tear or detachment. Ultrasound assists when haemorrhage or cataract obscures the view, provided open globe is not suspected. A macula-on detachment is exceptionally time-sensitive because central acuity can still be preserved before progression.

Optic neuritis often causes subacute central loss, colour desaturation, pain with eye movement, and a relative afferent defect, with a normal-looking disc when inflammation is posterior. Demyelination is common but infection, systemic immune disease, and unusual antibodies alter prognosis and treatment. Ischaemic optic neuropathy is typically sudden and altitudinal. Compressive optic neuropathy may progress slowly with colour and field change before acuity falls. Optical coherence tomography measures

nerve-fibre and ganglion layers but shows structural consequence, not cause by itself.

Diplopia is localised through eye position, direction of maximal separation, lid, pupil, proptosis, restriction, pain, fatigability, and other neurology. A third-nerve palsy with pupil involvement can indicate a compressive aneurysm and demands emergency imaging. Sixth-nerve palsy can arise from the nerve, brainstem, cavernous sinus, orbit, or raised intracranial pressure. Internuclear ophthalmoplegia localises the medial longitudinal fasciculus. Variable fatigable ptosis and diplopia suggest neuromuscular-junction disease; proptosis and restrictive movement suggest orbital disease.

Glaucoma is loss of retinal ganglion cells and optic-nerve axons with characteristic field damage, not simply high pressure. Some patients deteriorate at statistically normal pressure, while ocular hypertension may never damage the nerve. Corneal thickness and biomechanics affect measured pressure. Treatment lowers the only modifiable factor through drops, laser, or surgery, and progression is judged from serial disc, nerve imaging, and fields. Drop technique includes punctal occlusion to reduce systemic absorption, important with beta blockers and alpha agonists.

Diabetes damages capillaries, causing leakage, closure, retinal ischaemia, and vascular endothelial growth factor release. Macular oedema threatens central sight at any retinopathy stage; neovascularisation causes vitreous bleeding, tractional detachment, and neovascular glaucoma. Glycaemic, pressure, kidney, lipid, pregnancy, and smoking context affect progression. Rapid improvement in longstanding severe hyperglycaemia can transiently worsen retinopathy, so retinal surveillance is coordinated rather than used as a reason to leave glucose uncontrolled.

Age-related macular degeneration impairs central detail while peripheral navigation may remain. New distortion on an Amsler-like grid, central blur, or scotoma can signal choroidal neovascularisation and requires rapid retinal therapy. Cataract causes gradual glare, reduced contrast, and blur but should not be accepted as the cause until retina and nerve are assessed. In children, abnormal red reflex, leukocoria, constant strabismus, or failure to fix and follow threatens visual development or life; urgent assessment prevents amblyopia and detects cataract or retinoblastoma.

The eye is also a window into systemic disease. Papilloedema suggests raised intracranial pressure; retinal emboli and vascular changes reveal cardiovascular disease; uveitis and scleritis reveal inflammation; infection and drug toxicity appear in retina and nerve. The safe ocular habit is to document acuity first, recognise the sight-threatening pattern, and treat time-dependent tissue before diagnostic completeness.

Retrieval prompts

One. Which visual symptoms demand same-day emergency assessment?

Two. What does improvement through a pinhole suggest?

Three. How is monocular diplopia distinguished from binocular diplopia?

Four. Which red-eye features suggest corneal or intraocular disease rather than conjunctivitis?

Five. What warning symptoms precede retinal detachment?

Six. Why is giant-cell arteritis treated before confirmation?

Concise answers

One. Sudden loss, painful red eye with reduced vision, chemical or penetrating injury, acute angle closure, retinal occlusion, and suspected arteritis.

Two. Refractive optical blur.

Three. Monocular persists with the other eye covered; binocular resolves when either eye is covered.

Four. True pain, photophobia, reduced acuity, corneal opacity or staining, ciliary injection, abnormal pupil, or raised pressure.

Five. New flashes, sudden floaters, and a curtain or field shadow.

Six. Delay can cause irreversible blindness in the affected or fellow eye.

Source map

Original synthesis informed by Guyton and Hall, optics, accommodation, retina, visual pathways, pupils, and eye movements; Robbins, corneal, lens, retinal, vascular, inflammatory, glaucomatous, and neoplastic pathology; Katzung and OpenStax Pharmacology, ocular autonomic drugs, pressure-lowering agents, antimicrobials, corticosteroids, and vascular therapy; Talley and O'Connor, visual history, acuity, fields, pupils, movements, fundoscopy, and neurological localisation; OpenStax Medical-Surgical Nursing, eye injury, surgery, glaucoma, cataract, retinal disease, and patient education; and OpenStax Anatomy and Physiology, phototransduction and special-sense anatomy.

Chapter 56: Ear, Nose, Throat, Hearing, Balance, Voice, and Swallowing

Orientation

The ear, nose, pharynx, larynx, and upper airway concentrate sensory, respiratory, swallowing, speech, and immune functions in a small anatomical region. Minor swelling can obstruct airflow; infection can spread to orbit, skull base, deep neck, or brain. Assessment must first recognise stridor, drooling, respiratory effort, rapidly expanding neck swelling, severe haemorrhage, deep-space infection, sudden sensorineural hearing loss, and central vertigo.

Hearing physiology and history

The pinna and canal deliver sound to the tympanic membrane. Ossicles transmit and amplify vibration to cochlear fluid. Movement of the basilar membrane bends hair-cell stereocilia, converting mechanical energy into auditory nerve signals. High frequencies map near the stiff cochlear base and low frequencies near the apex. Brainstem pathways receive bilateral input, supporting localisation, while cortex interprets pitch, language, and meaning.

Conductive loss impairs external or middle-ear transmission; sensorineural loss reflects cochlea, auditory nerve, or central pathways. Ask onset, one or both ears, fluctuation, tinnitus, pain, discharge, pressure, infection, trauma, noise, medicines, neurological symptoms, family history, and functional impact. Sudden sensorineural loss developing within hours to days is an otological emergency and should receive urgent audiometry, specialist assessment, and time-sensitive corticosteroid treatment after alternatives are considered.

Inspect pinna, mastoid, canal, and tympanic membrane. Cerumen, foreign body, oedema, perforation, fluid, retraction, cholesteatoma, or mass may be visible. Pneumatic otoscopy or tympanometry assesses mobility. Whispered voice provides screening; formal audiometry quantifies air and bone

thresholds and speech discrimination. Weber and Rinne tuning-fork tests help classify asymmetry but do not replace audiometry.

External and middle-ear disease

Otitis externa causes canal pain, itch, discharge, and tenderness with tragal movement, often after moisture or trauma. Clean the canal safely and use topical antimicrobial and anti-inflammatory therapy when the membrane and organism context permit. Diabetes or immune suppression with severe persistent pain, granulation, cranial neuropathy, or systemic illness suggests invasive skull-base infection requiring urgent imaging and systemic antipseudomonal therapy.

Acute otitis media follows Eustachian-tube dysfunction and upper respiratory infection, especially in children. A bulging inflamed membrane and middle-ear effusion support diagnosis. Analgesia is central; antibiotics depend on age, severity, certainty, complications, and follow-up. Mastoid tenderness, swelling, displaced pinna, neurological signs, or toxicity suggests mastoiditis and requires urgent care.

Persistent middle-ear effusion causes conductive loss and can affect speech development. Cholesteatoma is keratinising epithelium expanding in middle ear or mastoid, producing foul discharge, hearing loss, bone erosion, facial palsy, labyrinthine fistula, or intracranial infection; definitive management is usually surgical. Tympanic perforation may follow infection, pressure, or trauma. Keep contaminated water out and avoid ototoxic drops when middle ear is exposed.

Tinnitus and hearing rehabilitation

Tinnitus is sound perceived without an external source. Characterise laterality, pulsatility, hearing change, medicines, noise, sleep, distress, and neurological symptoms. Unilateral tinnitus, asymmetric hearing loss, pulsatile sound, or focal neurology warrants targeted investigation. Pulsatile tinnitus may reflect vascular flow or raised intracranial pressure. Most non-pulsatile tinnitus accompanies hearing loss; explanation, hearing aids, sound strategies, and cognitive behavioural approaches reduce burden. Avoid promising a universal cure.

Age-related and noise-induced cochlear loss particularly affects high frequencies and speech in background noise. Prevention uses hearing protection without eliminating safe sound exposure. Hearing aids amplify according to profile; cochlear implants directly stimulate the auditory nerve in severe loss when conventional aids are insufficient. Communication support includes facing the person, reducing noise, checking understanding, captions, interpreters, and respecting sign language and Deaf identity.

Vestibular physiology and vertigo

Semicircular canals detect angular acceleration; utricle and saccule detect linear acceleration and gravity. Vestibular nuclei integrate bilateral labyrinth, vision, and proprioception to stabilise gaze and posture. Vertigo is an illusion of movement, while dizziness may mean presyncope, disequilibrium, or nonspecific light-headedness. Define timing, triggers, duration, hearing symptoms, headache, neck pain, neurological signs, medicines, and cardiovascular context.

Brief positional attacks triggered by head movement suggest benign paroxysmal positional vertigo from displaced otoconia. Positional testing provokes characteristic delayed fatigable nystagmus; canalith repositioning is treatment. Vestibular neuritis causes prolonged acute vertigo without hearing loss, while labyrinthitis includes cochlear symptoms. Meniere disease produces episodic vertigo with fluctuating hearing loss, tinnitus, and aural pressure.

Stroke can mimic peripheral vestibular disease. In continuous acute vestibular syndrome, trained bedside testing of head impulse, nystagmus, and skew may outperform early imaging, but misuse is unsafe outside the correct syndrome and expertise. New severe headache, inability to stand, focal neurology, direction-changing or vertical nystagmus, central eye signs, or high vascular risk increases concern. Do not diagnose peripheral vertigo from symptom intensity alone.

Vestibular suppressants may relieve severe acute symptoms briefly but delay compensation if prolonged and increase falls. Early mobilisation and vestibular rehabilitation support recovery. Treat nausea, hydration, migraine, infection, or vascular disease according to cause.

Nose and sinuses

The nasal cavity warms, humidifies, filters, and directs air while olfactory neurons detect volatile molecules. Rhinitis may be allergic, infectious, irritant, hormonal, medication-related, or autonomic. Allergy causes itch, sneeze, watery discharge, and congestion; management includes trigger reduction, saline, intranasal corticosteroids, antihistamines, and selected immunotherapy. Topical decongestants used repeatedly cause rebound congestion; oral sympathomimetics can raise pressure and heart rate.

Acute rhinosinusitis is usually viral. Persistent, severe, or worsening-after-improvement patterns increase bacterial probability. Antibiotics are selective. Unilateral swelling, severe frontal headache, eye pain or movement restriction, reduced vision, proptosis, neurological change, or systemic toxicity suggests orbital or intracranial spread and needs emergency imaging and treatment. Chronic rhinosinusitis requires prolonged symptoms plus objective inflammation; polyps raise allergic, inflammatory, and medication-related associations.

Epistaxis usually arises anteriorly. Sit the patient forward, pinch the soft nose continuously, clear airway risk, and assess haemodynamics and antithrombotics. Topical vasoconstriction, cautery, or packing follows visualised source and severity. Posterior bleeding can be brisk and requires specialist control. Recurrent unilateral obstruction, bleeding, facial numbness, or ear effusion can indicate malignancy.

Throat, tonsils, and deep-neck infection

Pharyngitis is commonly viral; cough, coryza, ulcers, and conjunctivitis support viral disease, while streptococcal probability uses age and clinical features. Test and treat selectively to reduce unnecessary antibiotics. Infectious mononucleosis causes fever, pharyngitis, lymphadenopathy, fatigue, and sometimes splenic enlargement; avoid contact risk while spleen is vulnerable. An aminopenicillin-associated rash can occur without true penicillin allergy.

Peritonsillar abscess causes unilateral severe throat pain, muffled voice, trismus, palatal swelling, and uvular deviation. Retropharyngeal or parapharyngeal infection can cause neck stiffness, swelling, drooling, toxicity,

or airway compromise. Ludwig angina is spreading floor-of-mouth cellulitis with tongue elevation and threatened airway, often from dental infection. Secure the airway early with experienced help, give intravenous antibiotics, image when stable, and drain collections or remove source.

Voice, larynx, and airway

Phonation requires airflow vibrating vocal folds, modified by laryngeal tension and resonating cavities. Hoarseness may follow viral laryngitis, voice overuse, reflux, smoking, inhaled steroids, vocal-fold paralysis, neurological disease, or cancer. Persistent hoarseness, neck mass, haemoptysis, dysphagia, ear pain, weight loss, tobacco or alcohol exposure, or stridor requires laryngoscopy.

Stridor is turbulent upper-airway sound and an emergency sign. Inspiratory stridor suggests extrathoracic obstruction; biphasic sound suggests glottic or subglottic narrowing. Causes include anaphylaxis, foreign body, infection, tumour, trauma, oedema, and vocal-fold dysfunction. Keep the patient calm, provide oxygen, call airway expertise, and avoid distressing examination when obstruction may worsen. Treat cause with adrenaline for anaphylaxis, foreign-body removal, antimicrobials, corticosteroids in selected oedema, or surgical airway when necessary.

Swallowing

Swallowing coordinates oral preparation, voluntary propulsion, pharyngeal airway protection, upper sphincter opening, and oesophageal transport. Oropharyngeal dysfunction causes coughing, choking, wet voice, repeated pneumonia, nasal regurgitation, or difficulty initiating; neurological disease, muscle disease, structural lesions, and frailty are common causes.

Assess alertness, voice, oral motor function, secretion handling, cough, cranial nerves, nutrition, and respiratory status. A bedside screen estimates risk but silent aspiration requires instrumental assessment by fibreoptic endoscopic or videofluoroscopic study. Management may include posture, texture and fluid modification, swallowing strategies, rehabilitation, mouth care, and enteral support. Thickened fluid can reduce aspiration in selected physiology but may worsen hydration and enjoyment; use individualised evidence and review.

TTS module 2: Auditory localisation, vestibular syndromes, deep-neck danger, and airway protection

Ear, nose, throat, and swallowing assessment begins with airway and neurological urgency. Stridor, drooling with inability to swallow, rapidly enlarging neck or floor-of-mouth swelling, severe epistaxis with airway contamination, deep-neck sepsis, sudden sensorineural hearing loss, and continuous vertigo with central signs are time-dependent syndromes. Because the anatomy is compact, infection or oedema can move rapidly from a local symptom to airway obstruction, intracranial spread, vascular thrombosis, or aspiration.

Hearing requires external sound collection, tympanic and ossicular transmission, cochlear transduction, auditory-nerve firing, bilateral brainstem processing, and cortical interpretation. Conductive loss attenuates sound before the cochlea; sensorineural loss impairs hair cells, nerve, or central processing. Air-conduction thresholds include both pathways, while bone conduction bypasses external and middle ear. An air–bone gap supports conductive loss. Speech discrimination disproportionately poor for measured thresholds can suggest neural or central pathology but depends on language and test conditions.

Tuning forks are bedside probability modifiers. In conductive loss, Weber tends to lateralise toward the affected ear because ambient masking is reduced, while Rinne may show bone conduction exceeding air. In unilateral sensorineural loss, Weber tends away and Rinne usually remains air greater than bone. Incorrect frequency, poor technique, profound loss, and bilateral disease mislead. Formal audiometry, tympanometry, otoscopy, and speech testing remain necessary when decisions depend on precision.

Sudden sensorineural hearing loss develops over hours to a few days and is often noticed on waking or during a phone call. It may be mistaken for blocked wax or congestion. Bedside fork testing helps separate conductive disease, but urgent audiometry and otology are required because corticosteroid benefit is time-sensitive. Neurological signs, severe headache, trauma, infection, bilateral loss, or vascular context broaden imaging and treatment. Normal otoscopy does not make sudden unilateral loss benign.

Otitis externa is usually confined to canal skin and causes tragal tenderness, itch, oedema, and discharge. Topical therapy achieves high local concentrations, but wick placement or careful cleaning may be needed when the canal is closed. Aminoglycoside-containing drops can injure the inner ear when the tympanic membrane is perforated, so membrane status matters. Severe nocturnal pain, granulation, cranial neuropathy, diabetes, or immune suppression suggests invasive skull-base osteomyelitis and requires systemic antipseudomonal therapy and imaging.

Middle-ear pressure depends on Eustachian-tube ventilation. Viral swelling creates negative pressure and effusion; bacterial acute otitis media is supported by a bulging membrane, not redness alone. Persistent effusion causes conductive loss and may affect language, attention, and learning in children. A unilateral adult effusion requires examination of the nasopharynx for obstruction. Cholesteatoma traps keratinising epithelium, erodes ossicles and adjacent bone, and can produce facial palsy, vertigo, meningitis, or abscess despite deceptively modest pain.

Tinnitus is characterised as pulsatile or non-pulsatile, unilateral or bilateral, tonal or complex, and associated with hearing or neurological change. Pulse-synchronous tinnitus raises vascular flow, dural fistula, arterial stenosis, venous abnormality, glomus tumour, anaemia, thyroid excess, or intracranial-pressure mechanisms. Unilateral tinnitus with asymmetric sensorineural loss warrants retrocochlear evaluation. For chronic non-pulsatile tinnitus, hearing rehabilitation, education, sleep care, sound enrichment, and cognitive behavioural therapy reduce distress even when the percept persists.

Vestibular hair cells encode head acceleration while paired labyrinths generate a balanced resting signal. A sudden unilateral peripheral loss creates an asymmetry interpreted as rotation, producing vertigo and nystagmus. The vestibulo-ocular reflex moves eyes opposite the head to stabilise vision. A catch-up saccade during rapid head impulse indicates deficient peripheral reflex on that side in the correct clinical setting. Central lesions can preserve the reflex yet disrupt gaze holding, skew, smooth pursuit, or cerebellar coordination.

Vertigo is diagnosed through timing and triggers rather than adjectives. Seconds-long episodes provoked by a particular head position suggest

benign positional vertigo. Spontaneous episodes lasting minutes to hours suggest vestibular migraine, Meniere disease, transient ischaemia, arrhythmia, or panic according to associated features. Continuous acute vestibular syndrome lasting days suggests vestibular neuritis or stroke. Light-headedness on standing points toward haemodynamic physiology rather than inner ear. "Worse with movement" is nonspecific because nearly every vestibular syndrome worsens when the head moves.

Benign paroxysmal positional vertigo results from otoconia entering a semicircular canal. Positional testing identifies a canal-specific delayed, transient nystagmus, and a repositioning manoeuvre moves particles back toward the vestibule. Persistent non-fatigable vertical or direction-changing nystagmus suggests central disease. Vestibular suppressants do not correct particles and prolonged use impairs central compensation. Early safe movement and vestibular rehabilitation promote recalibration after neuritis.

The head-impulse, nystagmus, and skew battery is validated only in continuous acute vestibular syndrome with spontaneous nystagmus and when performed by trained clinicians. It is not a general dizziness screen and can falsely reassure in episodic disease or inexperienced hands. New inability to stand, severe occipital headache, focal weakness or numbness, dysarthria, diplopia, vertical or direction-changing nystagmus, skew, or high vascular risk requires stroke assessment. Early posterior-fossa magnetic resonance can occasionally be negative, so trajectory and examination remain important.

The nasal mucosa conditions air and clears particles through cilia. Allergic rhinitis is immunoglobulin-E-mediated and causes itch, sneeze, watery discharge, and congestion, while non-allergic forms respond to irritants, temperature, hormones, drugs, or autonomic change. Intranasal corticosteroids require regular use and correct direction away from septum. Repeated topical decongestant causes rebound vasodilation and dependence. Unilateral obstruction, blood-stained discharge, facial numbness, cranial neuropathy, or persistent adult middle-ear effusion raises tumour concern.

Most acute rhinosinusitis is viral. Severe early fever and purulence, persistence beyond an expected viral course, or clear worsening after initial improvement increases bacterial probability. Eye pain, proptosis, reduced acuity or colour, restricted movement, severe frontal swelling, neurological

change, or toxicity indicates orbital or intracranial complication. The thin orbital walls and valveless veins permit spread. Emergency imaging, intravenous antibiotics, ophthalmic and surgical review, and drainage where needed protect vision and brain.

Epistaxis is managed by sitting forward, clearing the airway, and compressing the soft cartilaginous nose continuously for an adequate interval. Pinching the nasal bones is ineffective. Topical vasoconstrictor and cautery treat a visible anterior point; bilateral opposing septal cautery increases perforation risk. Posterior haemorrhage may flow into pharynx with little anterior blood and threatens airway and circulation. Anticoagulant reversal depends on severity and thrombosis risk, while packing requires monitoring for continued concealed bleeding and pressure injury.

Deep-neck infections exploit fascial planes. Peritonsillar abscess causes unilateral pain, trismus, muffled voice, and palatal bulge. Retropharyngeal infection can produce neck stiffness, drooling, and airway compromise, especially in children. Ludwig angina elevates the tongue through rapidly spreading submandibular cellulitis, often dental in origin. Airway loss can occur when sedation removes compensatory tone; experienced awake or surgical plans may be required. Antibiotics accompany drainage and dental source control.

Stridor is a sound of critical upper-airway narrowing. Inspiratory sound suggests extrathoracic obstruction, expiratory sound intrathoracic large-airway obstruction, and biphasic sound a fixed glottic or subglottic lesion. Severity is judged from work, air movement, voice, drooling, fatigue, oxygenation, and consciousness, not loudness; a quieter stridor can mean exhaustion. Keep the patient upright and calm, call airway expertise, avoid unnecessary throat manipulation, and treat anaphylaxis, croup, infection, foreign body, tumour, or oedema according to mechanism.

Persistent hoarseness requires visualisation of vocal folds, particularly with smoking, alcohol exposure, neck mass, dysphagia, haemoptysis, referred ear pain, weight loss, or stridor. Unilateral vocal-fold paralysis may follow recurrent-laryngeal nerve injury anywhere from skull base through neck into chest and can signal thyroid, lung, aortic, mediastinal, neurological, or surgical disease. Bilateral paralysis threatens airway. Voice therapy, injection,

framework surgery, or cause treatment follows closure, aspiration, and communication needs.

Swallowing safety depends on alertness, oral preparation, timely pharyngeal response, laryngeal closure, upper-sphincter opening, sensation, and cough. Aspiration can be silent when sensation is impaired. Bedside screening identifies obvious risk but cannot view bolus flow. Fiberoptic endoscopic evaluation directly assesses pharynx, secretions, penetration, and residue, while videofluoroscopy displays oral, pharyngeal, and upper-oesophageal mechanics across strategies. Texture, posture, pacing, rehabilitation, mouth care, and feeding route are individualised and reviewed because safety measures can reduce hydration, nutrition, autonomy, and pleasure.

Retrieval prompts

One. How do conductive and sensorineural hearing loss differ anatomically?

Two. Which hearing presentation is time-critical?

Three. What symptom pattern suggests benign positional vertigo?

Four. Which findings make sinus disease an emergency?

Five. What does stridor signify?

Six. Why can a normal bedside swallow screen miss risk?

Concise answers

One. Conductive loss blocks external or middle-ear transmission; sensorineural loss affects cochlea, nerve, or central pathways.

Two. Sudden sensorineural hearing loss over hours to days.

Three. Brief attacks triggered by particular head positions with characteristic positional nystagmus.

Four. Eye signs, reduced vision, proptosis, severe frontal symptoms, neurology, or toxicity suggesting orbital or intracranial spread.

Five. Turbulent upper-airway narrowing with potential impending obstruction.

Six. Aspiration may be silent and requires instrumental visualisation when suspicion remains.

Source map

Original synthesis informed by Guyton and Hall, cochlear transduction, auditory pathways, vestibular control, olfaction, phonation, and swallowing; Robbins, inflammatory, infectious, degenerative, and neoplastic ear, nose, and throat pathology; Katzung and OpenStax Pharmacology, antimicrobial, anti-inflammatory, allergy, vertigo, and airway therapy; Talley and O'Connor, hearing, vestibular, cranial-nerve, oral, neck, voice, and swallowing examination; OpenStax Medical-Surgical Nursing, airway, tracheostomy, hearing rehabilitation, sinus, laryngeal, and swallowing care; and OpenStax Anatomy and Physiology and Microbiology, special-sense anatomy and upper-respiratory infection.

Chapter 57: Oral and Dental Disease, Salivary Function, and Maxillofacial Warning Signs

Orientation

Oral health affects nutrition, communication, social participation, infection risk, and chronic disease. Dental disorders can progress into deep-neck infection, sepsis, airway compromise, or tooth loss. Oral lesions can reveal haematological, immune, endocrine, nutritional, infectious, gastrointestinal, or malignant disease. Assessment should respect pain, anxiety, trauma, disability, culture, and barriers to care.

Oral structure, saliva, and microbiome

Teeth have mineralised enamel over dentine surrounding vascular pulp and root canals. Periodontal ligament anchors cementum-covered roots to alveolar bone, while gingiva forms a protective seal. Mastication reduces particle size and mixes food with saliva. Parotid, submandibular, sublingual, and minor glands produce water, electrolytes, mucus, antimicrobial proteins, bicarbonate, calcium, phosphate, and amylase.

Parasympathetic stimulation drives abundant watery saliva; sympathetic activity changes protein composition. Saliva lubricates speech and swallowing, buffers acid, supports remineralisation, clears food, and restrains microbes. Reduced flow causes thirst, difficulty eating dry food, altered taste, fissures, candidiasis, caries, periodontal disease, and poor denture tolerance. Causes include dehydration, anticholinergic and psychotropic drugs, opioids, radiation, autoimmune disease, diabetes, obstruction, and ageing-associated polypharmacy.

Dental plaque is an organised biofilm rather than loose contamination. Frequent fermentable carbohydrate permits bacterial acid to demineralise enamel. Salivary buffering and mineral exchange can reverse early injury, but repeated low pH produces cavitation. Fluoride strengthens resistance and promotes remineralisation. Caries risk therefore reflects exposure frequency,

biofilm, saliva, fluoride, tooth structure, diet, and access—not sugar quantity alone.

History and examination

Ask about pain onset, exact tooth or region, triggers from heat, cold, sweetness, biting or posture, spontaneous night pain, swelling, discharge, trauma, bleeding, ulcers, dryness, taste, swallowing, trismus, fever, weight change, dental care, hygiene, fluoride, diet, tobacco, alcohol, substances, medicines, immune suppression, anticoagulation, diabetes, radiation, and previous procedures. Determine airway symptoms, drooling, voice change, neck swelling, eye symptoms, neurological change, and systemic toxicity immediately.

Inspect face for asymmetry, swelling, erythema, scars, fistulae, and eye involvement. Palpate temporomandibular joints, masticatory muscles, lymph nodes, salivary glands, and neck. Measure mouth opening. With light, gloves, tongue depressor, and dental mirror when available, inspect lips, labial and buccal mucosa, gingiva, teeth, palate, tongue surfaces, floor of mouth, tonsillar region, and posterior pharynx. Palpate suspicious lesions bimanually where appropriate. Note colour, ulceration, induration, fixation, bleeding, sensation, tooth mobility, percussion tenderness, fluctuance, and purulence.

Caries, pulpitis, and dental abscess

Early caries may be painless. Once bacteria and inflammation reach pulp, reversible pulpitis causes brief stimulus-provoked pain, whereas irreversible pulpitis causes spontaneous or lingering pain and needs root-canal treatment or extraction. Pulp necrosis permits apical infection. A periapical abscess causes local tenderness and swelling and may drain through a sinus.

Analgesia and definitive dental source control are central. Antibiotics do not cure contained pulp disease and are reserved for spreading infection, systemic involvement, immune risk, or situations defined by dental guidance. Repeated antibiotic courses without drainage or extraction delay cure and select resistance. Non-steroidal anti-inflammatory drugs, paracetamol, or their appropriate combination often control dental pain better than opioids, while contraindications and total dose must be checked.

Spreading odontogenic infection may enter facial spaces. Mandibular molar infection can produce Ludwig angina: bilateral submandibular and sublingual cellulitis with firm floor-of-mouth swelling, tongue elevation, dysphagia, drooling, and threatened airway. Pain, swelling, fever, trismus, voice change, inability to swallow, floor elevation, neck extension, or toxicity requires emergency airway, maxillofacial, antibiotic, imaging, and surgical assessment. Airway planning precedes supine imaging in a deteriorating patient.

Periodontal disease

Gingivitis is plaque-induced superficial inflammation with redness, swelling, and bleeding but no attachment loss and is reversible with plaque control. Periodontitis involves dysregulated host inflammation, periodontal-pocket formation, attachment loss, and alveolar-bone destruction. Smoking, diabetes, immune dysfunction, poor access, and genetic susceptibility increase risk. Advanced disease causes recession, sensitivity, halitosis, tooth migration, mobility, and loss.

Treatment uses effective daily plaque removal, professional debridement, risk-factor control, and surgery or antimicrobials selectively. Glycaemic control and periodontal health influence one another. Bleeding during brushing usually signals inflammation rather than a reason to stop cleaning, although unexplained or disproportionate bleeding warrants review for platelet, coagulation, liver, or marrow disease.

Oral mucosal disease

Aphthous ulcers are painful recurrent shallow ulcers without preceding vesicles. Frequent, severe, or atypical disease can accompany iron, folate, or vitamin B twelve deficiency, coeliac disease, inflammatory bowel disease, immune disorders, or medicines. Herpes simplex commonly causes clustered vesicles and ulcers on keratinised mucosa in recurrence. Oral candidiasis appears as removable white plaques, erythema, angular cheilitis, or denture-related inflammation and is promoted by antibiotics, inhaled steroids, diabetes, dry mouth, dentures, and immune suppression. Treat predisposing factors as well as infection.

White lesions include frictional keratosis, candidiasis, lichen planus, leukoplakia, and cancer. Red lesions can carry greater dysplasia risk. An

ulcer, red or white patch, lump, induration, unexplained bleeding, numbness, loose tooth, unilateral ear pain, or swallowing change persisting beyond a short healing interval requires urgent dental or specialist review and often biopsy. Oral squamous-cell carcinoma is associated with tobacco, alcohol, oncogenic human papillomavirus at oropharyngeal sites, ultraviolet exposure at lip, and immune suppression, but can occur without recognised risks.

Mucosal blistering or erosion may reflect immune disease, severe drug reaction, infection, or trauma. Widespread lesions with eye, genital, skin, fever, or systemic involvement demand urgent assessment. Pigmentation varies normally with ancestry but new focal pigmentation may require evaluation for melanoma, medicines, endocrine disease, or systemic syndromes.

Salivary-gland disease

Acute bacterial sialadenitis causes painful swollen gland, fever, and purulence from duct, often with dehydration or obstruction. Restore hydration, stimulate flow when safe, provide gland massage, antibiotics, and drain abscess.

Stones most often obstruct the submandibular duct and cause meal-related pain and swelling. Ultrasound or computed tomography identifies stone or abscess; persistent obstruction may need endoscopic or surgical removal.

Viral parotitis can be caused by mumps and other viruses; vaccination reduces mumps. Bilateral persistent enlargement occurs with autoimmune disease, human immunodeficiency virus, diabetes, alcohol-related disease, eating disorders, and medicines. Sjogren syndrome causes immune destruction of salivary and lacrimal glands with dry mouth and eyes, systemic disease, caries, and lymphoma risk. Management combines hydration, saliva stimulation or substitutes, fluoride, frequent dental prevention, eye care, and systemic evaluation.

Salivary tumours often present as painless slow-growing masses. Most parotid tumours are benign, but pain, rapid growth, fixation, skin involvement, lymph nodes, or facial weakness suggests malignancy. Do not casually biopsy a parotid mass through inappropriate routes; imaging and specialist tissue sampling preserve surgical planning.

Dental trauma and bleeding

Facial trauma assessment begins with airway, cervical spine, haemorrhage, vision, occlusion, sensation, and mouth opening. Hold an avulsed permanent tooth by its crown, rinse without scrubbing, and replant when safe or store appropriately for emergency dental care; do not replant primary teeth. Malocclusion, mobility, lower-lip numbness, sublingual bruising, double vision, cerebrospinal fluid leak, or airway bleeding suggests facial fracture. Avoid blind nasal instrumentation with possible skull-base injury.

Treat post-extraction bleeding with local pressure, visualisation, haemostatic material, suturing, and correction of systemic factors. Check the antithrombotic indication before reversal. Dry socket is painful clot loss days after extraction without spreading infection; use local care and analgesia rather than routine antibiotics.

Temporomandibular and facial pain

Temporomandibular disorders cause joint or muscle pain, clicking, restricted opening, or headache. Explanation, temporary activity change, physiotherapy, sleep care, and non-opioid analgesia usually help. Trauma, locking, inflammation, asymmetry, or persistent symptoms prompt imaging. Facial pain also arises from sinus disease, neuralgia, migraine, tumour, vascular inflammation, or cardiac ischaemia. Trigeminal neuralgia causes brief triggerable electric attacks; atypical features prompt imaging. Jaw claudication with visual symptoms suggests giant-cell arteritis.

Prevention and medically complex care

Prevention uses fluoride toothpaste, interdental cleaning, less frequent free sugar, tobacco cessation, saliva support, protection, and risk-based care. Treat active oral disease before head-and-neck radiation, transplantation, intensive cancer treatment, or potent antiresorptives where feasible. Jaw osteonecrosis is uncommon but risk rises with oncology-dose antiresorptive or antiangiogenic therapy and extraction. Endocarditis prophylaxis is limited to specified high-risk cardiac situations under current guidance. Diabetes, bleeding, immune suppression, pregnancy, organ disease, and frailty alter planning but rarely justify neglecting care. Coordination prevents unsafe medicine interruption.

TTS module 2: Oral ecology, odontogenic infection, mucosal diagnosis, and medically complex dental care

The mouth is a mechanically stressed, microbially dense interface between external environment and deep facial spaces. Teeth cannot repair cavitated enamel, pulp lies within a rigid chamber, and infection can track along roots and fascial planes toward orbit, neck, mediastinum, or airway. Oral pain is therefore localised by structure and urgency: distinguish enamel or dentine sensitivity, inflamed pulp, apical infection, periodontal disease, salivary obstruction, mucosal ulceration, temporomandibular pain, neuralgia, and referred cardiac or vascular disease.

Saliva is an active protective fluid. Parasympathetic drive produces abundant watery secretion, while sympathetic stimulation changes protein content and may leave the mouth subjectively dry. Bicarbonate buffers plaque acids; calcium and phosphate support remineralisation; mucins lubricate speech and swallowing; immunoglobulin, lysozyme, lactoferrin, peroxidases, and flow limit organisms. Salivary function varies with hydration, circadian rhythm, chewing, medications, gland injury, and autonomic state. Subjective xerostomia can occur before measured flow falls and deserves preventive treatment.

Dental caries is a biofilm-mediated ecological disease. Frequent fermentable carbohydrate selects acid-producing and acid-tolerant organisms and repeatedly lowers plaque pH. Enamel mineral dissolves when demineralisation exceeds saliva- and fluoride-supported repair. Frequency matters because each exposure creates another acidic interval; a slowly consumed sweet drink can be more damaging than the same amount taken with a meal. Once cavitation forms a sheltered niche, brushing and fluoride cannot restore lost architecture and dental restoration becomes necessary.

Pulpitis is painful because inflammation develops inside poorly compliant dentine. Brief stimulus pain that resolves suggests reversible irritation; spontaneous, nocturnal, or lingering thermal pain suggests irreversible pulp injury. Necrosis may temporarily reduce pain while infection extends through the apical foramen. Percussion tenderness reflects periodontal-ligament inflammation around the root. Antibiotics penetrate poorly into necrotic avascular pulp and cannot replace root-canal treatment, drainage, or extraction. They are reserved for spreading infection, systemic illness, selected immune risk, or guideline-defined indications.

Odontogenic infection spreads according to the root's relationship to muscle attachments and fascial spaces. Maxillary disease can extend to canine space, sinus, orbit, or infratemporal regions. Mandibular molars can enter sublingual, submandibular, or masticator spaces. Trismus suggests masticator or deep-space involvement. Fever, rapidly increasing swelling, floor-of-mouth elevation, tongue displacement, drooling, dysphagia, voice change, neck swelling, eye signs, cranial neuropathy, or toxicity requires emergency surgical and airway assessment rather than another outpatient antibiotic course.

Ludwig angina is rapidly spreading bilateral cellulitis of submandibular, sublingual, and submental spaces, often from a mandibular molar. The floor becomes firm and elevated, tongue is displaced upward and backward, and the patient may sit forward unable to swallow. Stridor and oxygen desaturation are late. Sedation or supine positioning can precipitate complete obstruction. Experienced airway planning precedes imaging, while intravenous broad antimicrobials, drainage where needed, extraction of the source, fluid, glucose control, and intensive monitoring proceed.

Periodontal disease begins when plaque at the gingival margin provokes inflammation. Gingivitis remains confined to soft tissue and is reversible. In susceptible hosts, dysregulated inflammation destroys periodontal ligament and alveolar bone, deepening pockets and loosening teeth. Smoking can mask bleeding while accelerating destruction; diabetes increases severity, and active inflammation can worsen glycaemic control. Treatment removes biofilm and calculus, improves daily interdental cleaning, controls risk, and uses surgery or adjunctive antimicrobials only when indicated.

Oral examination is systematic because lesions hide under tongue, behind molars, and beneath dentures. Inspect lips, buccal mucosa, gingiva, every tooth, hard and soft palate, dorsal, lateral and ventral tongue, floor of mouth, tonsillar region, and oropharynx. Palpate tongue and floor bimanually, nodes, salivary glands, jaw, and facial sensation. Describe site, size, border, surface, colour, ulceration, induration, fixation, bleeding, and sensory change. Remove dentures for examination.

An ulcer's duration and context narrow cause. Traumatic ulcers should heal after the irritant is removed. Recurrent aphthae occur on non-keratinised mucosa and may accompany iron, folate or B-twelve deficiency, coeliac

disease, inflammatory bowel disease, neutropenia, immune disease, or drugs. Recurrent herpes usually affects keratinised attached mucosa after grouped vesicles. A persistent ulcer, indurated margin, red or mixed red-white patch, unexplained numbness, loose tooth, unilateral referred ear pain, or neck node requires urgent biopsy assessment.

Oral squamous-cell carcinoma risk rises with tobacco, alcohol, immune suppression, and oncogenic human papillomavirus in oropharyngeal sites, but absence of exposure does not exclude it. Early lesions may be painless. Tumour can invade tongue muscle, jaw, nerves, skin, and lymphatics, impairing speech, swallowing, airway, and nutrition. Biopsy establishes diagnosis; imaging stages depth and nodes. Dental, speech, nutrition, reconstructive, pain, and psychosocial care should begin alongside surgery, radiation, or systemic treatment.

White oral lesions are tested for whether they wipe away. Candidal pseudomembranes can often be removed, leaving erythema, whereas keratosis and leukoplakia cannot. Erythematous candidiasis and angular cheilitis can occur without white plaques. Antibiotics, inhaled steroids, diabetes, dentures, xerostomia, and cellular immune deficiency predispose. Rinse after inhaled corticosteroids, disinfect and remove dentures overnight where appropriate, restore salivary function, and evaluate oesophageal involvement when painful swallowing or immune suppression is present.

Dry mouth demands cause review. Anticholinergics, antidepressants, antipsychotics, antihistamines, opioids, diuretics, dehydration, diabetes, Sjögren disease, head-and-neck radiation, and graft-versus-host disease are common. Management includes medication rationalisation, frequent water, sugar-free chewing stimulation when glands retain function, saliva substitutes, high-fluoride prevention, meticulous plaque control, and regular dental surveillance. Acidic sweets used to stimulate saliva can worsen erosion and caries. Persistent major-gland swelling or nodes in Sjögren disease raises lymphoma concern.

Salivary obstruction produces meal-related pain and swelling as secretion builds behind a stone or stricture. Submandibular stones are common because the duct is long and saliva more mucous and mineral-rich. Bacterial sialadenitis causes sustained tender swelling, fever, dehydration, and sometimes pus expressed from the duct. Hydration, warm compresses,

massage toward the opening, sialogogues when safe, antibiotics, and drainage treat infection; imaging identifies stone, abscess, or tumour. Facial weakness with a parotid mass is a malignancy warning.

Dental trauma is handled by tooth type and tissue. An avulsed permanent tooth loses periodontal-ligament viability as it dries. Hold only the crown, rinse gently without scraping, replant immediately if safe, or store in an appropriate medium while obtaining emergency care. Primary teeth are not replanted because they can damage the developing successor. Occlusal change, sublingual bruising, step deformity, numb lower lip, diplopia, restricted eye movement, or cerebrospinal fluid leakage indicates facial fracture requiring airway, vision, and specialist assessment.

Post-extraction bleeding is usually controlled locally through direct pressure, clot removal and visualisation, haemostatic packing, suturing, and treatment of an identifiable vessel. Anticoagulants are not stopped or reversed reflexively because thrombosis risk may exceed local bleeding risk. Determine drug, indication, last dose, kidney function, platelet and liver status, then coordinate. Dry socket produces severe pain several days after extraction from loss of clot and exposed bone; local irrigation, dressing, and analgesia help, while antibiotics add little without infection.

Temporomandibular pain is often muscular or joint-related and varies with chewing, clenching, sleep, stress, and range. Clicking alone does not require imaging or irreversible bite adjustment. Education, soft temporary diet, avoidance of extreme opening, physiotherapy, heat, and non-opioid analgesia are first-line. Persistent locking, trauma, inflammatory arthritis, asymmetry, neurological change, or mass requires targeted imaging. Electric triggerable facial pain suggests trigeminal neuralgia; jaw claudication with visual or constitutional symptoms suggests giant-cell arteritis.

Medical treatment and oral care interact bidirectionally. Before head-and-neck radiation, transplantation, intensive chemotherapy, or high-dose antiresorptive therapy, remove uncontrolled infection and plan prevention when time permits. Radiation reduces salivary flow and bone healing, increasing lifelong caries and osteoradionecrosis risk. Antiresorptive-related jaw osteonecrosis is uncommon but more likely with oncology doses, long exposure, dental infection, and extraction. Pregnancy generally does not preclude necessary dental treatment, local anaesthesia, or indicated imaging with appropriate

technique. Neglecting oral disease because a patient is medically complex often creates the infection that later interrupts essential care.

Retrieval prompts

One. Why does dry mouth accelerate dental disease?

Two. When are antibiotics appropriate for dental infection?

Three. Which oral infection threatens the airway?

Four. How do gingivitis and periodontitis differ?

Five. Which persistent oral features require biopsy assessment?

Six. How should an avulsed permanent tooth be handled?

Concise answers

One. Loss of lubrication, clearance, buffering, antimicrobial action, and remineralising calcium and phosphate promotes caries and infection.

Two. For spreading infection, systemic illness, selected immune risk, or other guideline-defined indications alongside definitive source control.

Three. Ludwig angina and other deep-space infections causing floor-of-mouth, tongue, neck, or laryngeal swelling.

Four. Gingivitis is reversible superficial inflammation; periodontitis destroys attachment and alveolar bone.

Five. Non-healing ulcer, red or white patch, induration, mass, bleeding, numbness, loose tooth, unilateral ear pain, or swallowing change.

Six. Hold the crown, rinse gently, replant promptly when safe or use appropriate storage, and obtain emergency dental care.

Source map

Original synthesis informed by Guyton and Hall, salivary secretion, mastication, swallowing, taste, and autonomic control; Robbins, dental, periodontal, salivary, immune, infectious, and oral neoplastic pathology; Katzung and OpenStax Pharmacology, analgesia, antimicrobials, saliva-modifying drugs, anticoagulation, and antiresorptive therapy; Talley and O'Connor, oral, facial, salivary, neck, cranial-nerve, and systemic examination; OpenStax Medical-Surgical Nursing, oral care, nutrition, cancer therapy, airway infection, trauma, and education; and OpenStax Anatomy and Physiology and Microbiology, tooth structure, salivary physiology, biofilm, and infection.

Chapter 58: Allergy, Anaphylaxis, Immunodeficiency, Transplantation, and Immune Therapy

Orientation

The immune system must eliminate threats without injuring self or harmless antigens. Failure produces allergy, autoimmunity, immunodeficiency, infection, graft rejection, or malignancy. Immune-modifying treatments can be transformative but create infection, cancer, metabolic, haematological, and organ toxicities. Care requires recognising threats, identifying the defective pathway, and preventing complications before suppression.

Hypersensitivity mechanisms

Immediate hypersensitivity occurs when allergen cross-links immunoglobulin E bound to mast cells and basophils, releasing histamine, tryptase, leukotrienes, prostaglandins, and cytokines. Vasodilation, capillary leak, smooth-muscle contraction, mucus, itch, and sensory activation cause urticaria, angioedema, bronchospasm, gastrointestinal symptoms, and shock. Sensitisation may precede a reaction, but detectable sensitisation does not always mean clinical allergy.

Antibody-mediated cytotoxic reactions target cells or matrix, activate complement, or alter receptors. Immune-complex deposition activates inflammation in vessels, glomeruli, joints, and skin. Delayed hypersensitivity is T-cell mediated and appears hours to days after exposure, as in contact dermatitis and many drug eruptions. These categories overlap in real disease and do not replace a precise clinical phenotype.

Allergy history and testing

Document trigger, route, dose, timing, symptoms, treatment, cofactors, prior tolerance, and later exposure. Exercise, alcohol, infection, non-steroidal drugs, and uncontrolled asthma can amplify reactions. Distinguish allergy from adverse effects, intolerance, toxicity, vasovagal episodes, and coincidence. An inaccurate label can deny first-line therapy and increase complications.

Skin-prick and specific immunoglobulin E tests detect sensitisation and should target plausible triggers. Results alter probability but not reaction severity. Supervised challenge is often the diagnostic standard. Patch testing assesses contact allergy. Broad food panels create false positives. Drug evaluation may include selected skin testing, graded challenge, or specialist desensitisation.

Anaphylaxis

Anaphylaxis is a rapidly evolving systemic hypersensitivity reaction threatening airway, breathing, or circulation, often with skin or mucosal features but sometimes without them. Hoarse voice, stridor, tongue swelling, wheeze, hypoxaemia, hypotension, collapse, severe gastrointestinal symptoms after a likely trigger, or rapidly progressive multisystem illness should prompt treatment. Do not wait for laboratory confirmation.

Give intramuscular adrenaline promptly into the anterolateral thigh and repeat according to response and protocol. Position supine with legs elevated when tolerated; sudden standing can precipitate collapse. Use a lateral position in pregnancy and recovery position if unconscious while protecting the airway. Call for help, remove an ongoing trigger when safe, provide high-flow oxygen, establish access, and give rapid isotonic crystalloid for shock. Nebulised bronchodilator treats persistent bronchospasm but does not replace adrenaline. Antihistamines relieve skin symptoms only; corticosteroids have no reliable immediate life-saving effect.

Refractory anaphylaxis requires experienced management with monitored intravenous adrenaline infusion, further fluid, airway planning, and evaluation of alternative or additional shock. Beta blockade may blunt response and glucagon can be considered. Measure acute and baseline tryptase when useful, without delaying care. Observe according to severity, treatment

required, comorbidity, trigger, and access to help because biphasic symptoms can recur.

Before discharge, document the event, refer for trigger evaluation, teach avoidance without unnecessary restriction, supply adrenaline autoinjectors when indicated, demonstrate technique, provide an action plan, and address asthma and cofactors. Adrenaline has no absolute contraindication in true anaphylaxis.

Urticaria, angioedema, and allergic disease

Urticaria consists of transient itchy wheals that migrate and resolve individually within a day. Acute episodes often follow infection, foods, drugs, or no identified trigger. Chronic spontaneous urticaria is commonly autoimmune rather than an undiscovered food allergy. Second-generation antihistamines are first-line and may be increased under guidance; resistant disease can receive biologic therapy.

Angioedema affects deeper skin and mucosa. Histamine-mediated disease usually accompanies itch or wheals and responds to standard allergy therapy. Bradykinin-mediated angioedema from angiotensin-converting enzyme inhibitors or hereditary complement-regulator defects lacks urticaria and responds poorly to adrenaline or antihistamine, though airway protection remains paramount and cause-specific inhibitors are used. Laryngeal symptoms require emergency assessment.

Allergic rhinitis, asthma, eczema, food allergy, and eosinophilic disease often cluster through epithelial-barrier and type-two immune pathways. Management combines trigger-specific avoidance, barrier care, topical anti-inflammatory treatment, and targeted systemic therapy. Food elimination should be nutritionally supervised, especially in children. Oral immunotherapy raises reaction threshold for selected foods but is not risk-free permission for unrestricted intake.

Primary immunodeficiency

Inborn errors can impair antibodies, T cells, phagocytes, complement, immune regulation, or combinations. Warning patterns include unusually severe, recurrent, persistent, opportunistic, or anatomically unusual infection; poor growth; chronic diarrhoea; deep abscesses; absent lymphoid tissue; delayed cord separation; autoimmunity; granulomas; family history; or adverse reactions to live vaccines. Common infections alone do not establish deficiency.

Start with blood count and differential, blood film, quantitative immunoglobulins, vaccine responses, lymphocyte subsets, human immunodeficiency virus testing, and complement or neutrophil tests according to phenotype. Timing, age, medicines, protein loss, and acute illness affect results. Genetic diagnosis can guide treatment and family counselling.

Management includes prompt cultures and antimicrobials, prophylaxis, immunoglobulin replacement for selected antibody failure, vaccination strategy, avoidance of live vaccines in defined cellular defects, and haematopoietic stem-cell transplantation or gene therapy for severe disorders. Irradiated or cytomegalovirus-safe blood products may be needed in particular syndromes.

Secondary immunodeficiency

Acquired immune impairment is more common and follows age extremes, malnutrition, diabetes, renal or liver failure, human immunodeficiency virus, malignancy, splenectomy, protein loss, burns, critical illness, and immune-suppressive treatment. Pattern matters: neutropenia favours bacterial and fungal invasion; impaired T-cell function permits viral, fungal, mycobacterial, and protozoal disease; antibody failure promotes encapsulated bacterial infection.

Human immunodeficiency virus depletes CD4 T cells and causes chronic immune activation. Diagnose with antigen-antibody testing and confirmatory algorithms, then measure viral load, CD4 count, resistance, coinfection, organ health, pregnancy, and medicines. Combination antiretroviral therapy should begin promptly and can achieve durable suppression, immune recovery, and

prevention of sexual transmission. Adherence, interactions, inflammatory recovery syndromes, and opportunistic infection timing require expertise.

Loss of splenic function increases fulminant infection from encapsulated bacteria. Prevention includes vaccination, education, rapid assessment of fever, and antibiotic strategies based on age and policy. Patients need a clear emergency plan because deterioration can be rapid.

Immune-suppressive and biologic therapy

Glucocorticoids broadly alter transcription and suppress inflammation but cause hyperglycaemia, infection, osteoporosis, myopathy, mood change, cataract, glaucoma, skin fragility, hypertension, and adrenal suppression. Use the lowest effective exposure and do not abruptly stop prolonged therapy without considering adrenal recovery.

Conventional agents inhibit lymphocyte proliferation or signalling and can injure marrow, liver, kidney, lung, fertility, or fetus. Biologic medicines target cytokines, receptors, cells, complement, or adhesion; kinase inhibitors alter intracellular signalling. Toxicity follows mechanism: tumour-necrosis-factor blockade can reactivate tuberculosis and hepatitis B; B-cell depletion impairs antibody responses; complement blockade increases meningococcal risk; checkpoint inhibitors can inflame almost any organ by releasing antitumour immunity.

Before treatment, screen relevant tuberculosis, hepatitis, infection, cancer, blood and organ function, pregnancy, vaccines, and interactions. Give indicated non-live vaccines beforehand when possible. Monitor response, infection, toxicity, and inflammatory syndromes. Fever may be muted and serious infection can occur with a normal leukocyte count.

Febrile neutropenia is fever with severe neutropenia and is an emergency requiring cultures and immediate broad antipseudomonal antibiotics without waiting for source. Avoid rectal procedures that disrupt mucosa. Growth factors are used according to chemotherapy risk and clinical setting.

Transplantation

Transplantation replaces failed organs or marrow but introduces alloimmunity, infection, malignancy, and drug toxicity. Hyperacute antibody rejection occurs within minutes to hours; acute cellular or antibody rejection develops over days to months; chronic rejection causes progressive vascular and interstitial injury. Obstruction, vascular compromise, recurrence, infection, dehydration, and drug toxicity also impair grafts, often requiring biopsy.

Maintenance often combines a calcineurin inhibitor, antiproliferative drug, and corticosteroid. Calcineurin inhibitors cause kidney injury, hypertension, neurological effects, diabetes, and metabolic interactions, requiring drug-level monitoring. Rejection treatment further increases infection risk.

Post-transplant infection reflects surgery, latent organisms, prophylaxis, exposures, and net suppression. Early problems are often surgical or nosocomial; later risks include cytomegalovirus, Pneumocystis, fungi, reactivation, and community infections. Prevention and rapid fever assessment are central. Long-term risks include skin cancer, lymphoproliferative disease, metabolic disease, kidney injury, and cardiovascular disease.

Stem-cell transplantation can cure marrow, immune, and malignant disease. Conditioning damages marrow and mucosa before engraftment. Allogeneic cells can cause graft-versus-host disease of skin, gut, liver, lung, and other organs while providing graft-versus-tumour benefit. Care includes infection prevention, blood support, immune suppression, nutrition, and revaccination.

TTS module 2: Systemic hypersensitivity, immune-defect localisation, and transplant complications

Immune disorders are localised by asking which defensive or regulatory function has failed. Allergy represents an inappropriate response to a usually harmless trigger. Immunodeficiency removes protection against characteristic organisms and sites. Autoimmunity redirects effectors toward self.

Transplantation deliberately suppresses alloimmune recognition but creates infection, malignancy, and drug toxicity. The clinical challenge is often to distinguish too much immunity from too little, because fever, rash, lung infiltrates, diarrhoea, cytopenia, and organ dysfunction can represent either.

Immunoglobulin-E-mediated allergy requires sensitisation. Allergen-specific B cells class-switch under T-helper signals, and immunoglobulin E binds high-affinity receptors on mast cells and basophils. Later cross-linking releases preformed histamine and tryptase plus newly generated leukotrienes, prostaglandins, and cytokines. Vasodilation and capillary leak lower effective volume; smooth muscle constricts in airway and gut; mucus, itch, urticaria, and oedema develop. The severity depends on dose, route, absorption, asthma, cardiovascular reserve, medication, and cofactors, not on the size of a skin-test wheal.

Anaphylaxis is diagnosed clinically when rapidly evolving airway, breathing, or circulation compromise follows a plausible trigger, often with skin, mucosal, or gastrointestinal features. Skin findings can be absent, particularly in profound shock. Differential diagnoses include vasovagal syncope, severe asthma, panic, vocal-fold obstruction, hereditary angioedema, sepsis, pulmonary embolism, and mast-cell disorders, but uncertainty must not delay adrenaline when the syndrome fits. A normal blood pressure does not exclude dangerous laryngeal or bronchial disease.

Intramuscular adrenaline into the anterolateral thigh is first-line because alpha-one vasoconstriction reverses vasodilation and oedema, beta-one activity supports the heart, and beta-two activity bronchodilates and reduces mediator release. Repeat promptly if airway, breathing, or circulation fails to improve. Position matters: sudden sitting or standing in low venous return can precipitate empty-ventricle arrest. Keep the patient supine with legs elevated

when tolerated, laterally positioned in pregnancy, or in a position necessary for breathing without allowing unsupported standing.

High-flow oxygen, monitoring, intravenous access, and rapid isotonic crystalloid support severe reactions, but none replaces adrenaline. Nebulised beta agonist treats residual bronchospasm and nebulised adrenaline may temporarily reduce upper-airway oedema. Antihistamines improve itch and wheals after stabilisation but do not treat shock or laryngeal obstruction. Corticosteroids act too slowly for initial rescue and do not reliably prevent biphasic reactions. Refractory disease requires expert intravenous adrenaline infusion, further haemodynamic evaluation, and an early advanced-airway plan before swelling makes access impossible.

Serum tryptase may support mast-cell activation when sampled at appropriate acute and baseline times, but sensitivity varies by trigger and a normal result does not exclude anaphylaxis. After recovery, reconstruct trigger, route, timing, cofactors, treatment, and prior tolerance. Provide an action plan, avoidance advice proportionate to evidence, autoinjectors when indicated, hands-on technique, and referral. Poorly defined “allergy” labels can cause inferior antibiotics, procedural delay, or unnecessary dietary restriction and should be clarified through targeted testing or supervised challenge.

Urticaria and angioedema separate histamine from bradykinin mechanisms. Histaminergic wheals are itchy, migratory, and individually last less than about a day. Histaminergic angioedema often accompanies wheals and responds to adrenaline in anaphylaxis and antihistamines for lesser disease. Bradykinin-mediated angioedema from angiotensin-converting-enzyme inhibitors or hereditary complement-regulator deficiency usually lacks itch and urticaria, develops more slowly, and responds poorly to antihistamine, steroid, or adrenaline. Airway protection remains paramount while a bradykinin-targeted inhibitor or replacement is arranged.

Food-allergy testing begins with a reproducible immediate phenotype. Specific immunoglobulin E and skin-prick tests detect sensitisation, not inevitable clinical reaction, and positive predictive value falls when broad panels are used in people without a compatible history. A supervised oral challenge is often the diagnostic standard. Food intolerance, coeliac disease, enzyme deficiency, toxin, and functional symptoms are not immunoglobulin-E

allergy. Removing staple foods from children without dietetic support can cause growth failure and reinforce fear.

Immunodeficiency is recognised through infection pattern rather than infection count alone. Antibody deficiency commonly produces recurrent sinopulmonary infection with encapsulated bacteria, chronic *Giardia*, and reduced vaccine response. T-cell defects permit viral, fungal, mycobacterial, and protozoal opportunists and may cause chronic diarrhoea, thrush, and poor growth. Neutrophil defects produce deep bacterial and fungal abscesses, impaired pus, poor wound healing, or unusual catalase-positive infection. Terminal-complement deficiency strongly predisposes to invasive *Neisseria*, while absent spleen impairs clearance of encapsulated organisms and parasites within red cells.

Initial tests are selected by phenotype: full blood count and film, quantitative immunoglobulins, vaccine-specific antibodies, lymphocyte subsets, human immunodeficiency virus testing, complement pathways, and neutrophil oxidative function. Age-specific ranges matter, particularly in infants. Acute infection, protein-losing gut or kidney disease, lymphoid malignancy, medications, and recent immunoglobulin therapy alter results. A normal screening panel does not exclude every signalling or functional defect when the clinical pattern remains strong.

Secondary immune impairment is far more common than primary genetic disease. Diabetes, malnutrition, renal and liver failure, malignancy, burns, age, pregnancy, critical illness, protein loss, splenectomy, human immunodeficiency virus, and immune-modifying drugs change different compartments. The net state reflects combinations: a patient receiving corticosteroid, B-cell depletion, and chemotherapy may simultaneously have T-cell dysfunction, poor antibody, neutropenia, mucosal injury, and a central line. Prophylaxis and diagnostic thresholds follow this integrated state rather than one drug name.

Human immunodeficiency virus is diagnosed through a validated antigen–antibody and confirmatory sequence, followed by viral load, CD-four count, resistance, organ function, hepatitis and other coinfection assessment. Combination antiretroviral treatment starts promptly and suppresses transmission when viral load remains undetectable. Early immune recovery can unmask or worsen inflammation against an existing opportunistic

infection, an immune-reconstitution syndrome. This does not usually mean treatment has failed, but timing and management vary by pathogen, especially central nervous system disease.

Glucocorticoids alter transcription across many immune and metabolic pathways. Infection risk depends on dose, duration, comorbidity, and combination therapy. They also cause hyperglycaemia, muscle loss, osteoporosis, avascular necrosis, mood or psychotic symptoms, cataract, glaucoma, skin fragility, pressure, and adrenal suppression. Abrupt withdrawal after prolonged exposure can precipitate adrenal crisis even while the treated inflammatory disease relapses. A plan should specify taper, infection and bone prophylaxis, glucose monitoring, and stress-dose considerations.

Biological therapy creates mechanism-specific vulnerabilities.

Tumour-necrosis-factor blockade disrupts granuloma maintenance and can reactivate tuberculosis or hepatitis B. B-cell depletion lowers new antibody responses and can cause prolonged hypogammaglobulinaemia or viral reactivation. Complement inhibition markedly increases meningococcal risk even after vaccination. Janus-kinase inhibition alters antiviral and broader immune signalling and can increase zoster, thrombosis, and other risks in selected patients. Checkpoint inhibitors do the opposite by releasing T-cell restraint and can inflame almost any organ.

Before immune therapy, document latent-infection risk, vaccination, blood counts, kidney and liver function, pregnancy and fertility plans, previous cancer, heart or neurological disease, and drug interactions according to mechanism. Give indicated vaccines before suppression when feasible; live vaccines may later be unsafe and responses to inactive vaccines may be weaker. Strongyloides exposure matters before corticosteroids. Pneumocystis, viral, fungal, or bacterial prophylaxis follows the cumulative regimen and host, not a universal threshold.

Solid-organ transplant dysfunction has several competing causes. Hyperacute antibody injury appears immediately and can destroy the graft. Acute cellular or antibody rejection develops over days to months but can occur later with reduced immunosuppression or non-adherence. Chronic rejection produces progressive vascular and fibrotic loss. Vascular thrombosis, obstruction, leak, recurrence, dehydration, calcineurin toxicity, and infection can look similar.

Drug levels, imaging, microbiology, donor-specific antibodies, and biopsy separate them; empirically increasing immunosuppression before excluding infection can be catastrophic.

Post-transplant infection changes over time. Early disease often reflects surgery, devices, donor or recipient organisms, and hospital exposure. During maximal suppression, cytomegalovirus, Pneumocystis, fungi, herpesviruses, and other opportunists emerge according to prophylaxis and geography. Later, community infections dominate unless rejection treatment intensifies suppression. Fever can be absent. Vaccination, food and environmental advice, skin surveillance, and rapid evaluation of subtle decline are long-term care.

Calcineurin inhibitors narrow the therapeutic window: inadequate exposure risks rejection, while excess causes kidney vasoconstriction, hypertension, hyperkalaemia, tremor, neurotoxicity, diabetes, and infection. Cytochrome interactions with azoles, macrolides, anticonvulsants, herbal products, and grapefruit can shift levels dramatically. Diarrhoea can increase or destabilise exposure. A trough concentration must be interpreted with timing, adherence, interacting changes, organ function, and clinical graft state.

Allogeneic stem-cell transplantation replaces marrow and immunity while donor lymphocytes can attack residual malignancy and recipient tissues. Graft-versus-host disease commonly affects skin, gut, and liver acutely and can later involve mouth, eye, lung, fascia, and other organs. Diarrhoea and rash may also be infection or drug toxicity, often requiring biopsy. Conditioning, neutropenia, mucosal barrier loss, delayed T- and B-cell recovery, and immunosuppression create sequential infection risks. Revaccination is required because previous immune memory may be lost.

The practical immune principle is to pair every therapeutic benefit with a surveillance map. Know which immune arm is altered, which organisms and toxicities follow, which vaccines and prophylaxis reduce risk, and which symptom demands urgent investigation. Immune-modifying treatment succeeds not merely when inflammation or rejection is suppressed, but when infection, malignancy, metabolic injury, and organ toxicity are anticipated early enough to preserve the benefit.

Retrieval prompts

One. What is the first-line treatment for anaphylaxis?

Two. Why should broad allergy panels be avoided?

Three. How does bradykinin-mediated angioedema differ clinically?

Four. Which infection patterns suggest immunodeficiency?

Five. What must be assessed before biologic immune therapy?

Six. Why is fever during severe neutropenia an emergency?

Concise answers

One. Prompt intramuscular adrenaline into the anterolateral thigh.

Two. They detect sensitisation without proving disease and generate false-positive restrictions.

Three. It usually lacks urticaria and itch and responds poorly to antihistamines, corticosteroids, and adrenaline.

Four. Severe, recurrent, persistent, opportunistic, unusual-site infections or infections accompanied by poor growth, autoimmunity, or family history.

Five. Infection and vaccine status, tuberculosis and hepatitis risks, blood and organ function, cancer, pregnancy, interactions, and monitoring plan.

Six. Infection can progress rapidly with little inflammatory response and requires immediate broad antibiotics.

Source map

Original synthesis informed by Guyton and Hall, immune regulation, mast-cell mediators, circulation, and shock; Robbins, hypersensitivity, immune deficiency, transplantation, rejection, and immune pathology; Katzung and OpenStax Pharmacology, adrenaline, antihistamines, corticosteroids, immunosuppressants, biologics, antiretrovirals, and prophylaxis; Talley and O'Connor, allergy, lymphatic, skin, respiratory, and systemic assessment; OpenStax Medical-Surgical Nursing, anaphylaxis, immune suppression, transplantation, infection prevention, and education; and OpenStax Biology and Microbiology, adaptive immunity, complement, human immunodeficiency virus, and opportunistic infection.

Chapter 59: Poisoning, Overdose, Envenomation, and Environmental Exposure

Orientation

Toxicology applies physiology under uncertainty. Substance, dose, route, formulation, time, patient, and prior treatment matter, but histories are incomplete and products mixed. Protect staff, stop exposure, stabilise airway, breathing, circulation, temperature, glucose, and seizures, then identify toxidromes, test selectively, contact a poisons service, and consider antidotes or enhanced elimination.

Immediate approach

Do not enter contamination without protection. Remove clothing and brush off dry chemicals before irrigation. Move inhalational exposures to fresh air without exposing rescuers. Irrigate skin and eyes generously. Protect staff from powders, pesticides, body fluids, and off-gassing substances.

Assess airway, ventilation, circulation, consciousness, pupils, muscle tone, skin moisture, bowel sounds, retention, temperature, and trauma. Check glucose and electrocardiogram early. Treat hypoxia, hypoventilation, shock, dysrhythmia, temperature disturbance, seizure, and agitation while diagnosis evolves. Normal early findings do not exclude delayed-release, metabolite, or hepatic toxicity.

Ask substance, maximum amount, time, route, co-ingestants, medicines, weight, pregnancy, illness, symptoms, intent, and treatment. Bring packaging but use tablet counts cautiously. Check for patches, body packing, and collateral evidence when appropriate. Deliberate self-poisoning requires psychosocial safety assessment after stabilisation; low intent does not reduce dose.

Toxidromes

Opioid toxicity produces depressed consciousness, slow breathing, and often small pupils. Sympathomimetic toxicity causes agitation, mydriasis, sweating, hypertension, tachycardia, hyperthermia, and sometimes chest pain, seizure, or psychosis. Antimuscarinic toxicity also causes agitation and mydriasis but with dry flushed skin, reduced bowel sounds, and urinary retention.

Cholinergic excess produces secretions, bronchospasm, bradycardia or tachycardia, diarrhoea, vomiting, small pupils, fasciculation, weakness, and seizure. Sedative-hypnotic toxicity generally depresses consciousness and breathing without a unique pupil pattern.

Serotonin toxicity causes mental change, autonomic activation, hyperreflexia, clonus, and heat. Neuroleptic malignant syndrome evolves more slowly with rigidity, fever, altered consciousness, and instability after dopamine blockade. Malignant hyperthermia is an anaesthetic-triggered crisis with rising carbon dioxide, rigidity, acidosis, hyperkalaemia, and heat. Mixed overdoses blur patterns.

Testing and decontamination

Select electrolytes, organ function, glucose, blood gas, lactate, creatine kinase, osmolality, pregnancy testing, and drug concentrations by syndrome. Measure paracetamol after possible overdose despite absent symptoms; interpret salicylate serially. QRS widening suggests sodium-channel blockade and QT prolongation increases polymorphic ventricular tachycardia risk.

Urine drug screens have false results, long detection windows, and poor correlation with impairment. Treat the patient, not the screen. Anion and osmolar gaps support toxic-alcohol assessment but vary with timing. Preserve samples when legally or epidemiologically relevant.

Activated charcoal adsorbs many substances if given early to a cooperative patient with a protected airway, but aspiration can be catastrophic. It poorly binds alcohols, metals, corrosives, and some small ions. Gastric lavage is rarely appropriate. Do not induce vomiting. Whole-bowel irrigation is selected for sustained-release drugs, substances not bound by charcoal, or body packing under specialist guidance. Corrosive ingestion should not be

neutralised or blindly lavaged; airway and endoscopic assessment are prioritised.

Paracetamol and salicylate

Paracetamol overdose saturates conjugation pathways, allowing a reactive metabolite to deplete glutathione and injure liver. Early symptoms may be mild. Interpret a correctly timed concentration on the treatment nomogram for a single immediate-release ingestion with known time; staggered, delayed, massive, or uncertain ingestion needs specialist protocols. N-acetylcysteine replenishes glutathione and is highly effective when early but remains useful after injury. Monitor liver, coagulation, kidney, glucose, acid-base status, and encephalopathy; acute liver failure requires transplant-centre discussion.

Salicylate uncouples oxidative phosphorylation and stimulates respiration, producing early respiratory alkalosis, later metabolic acidosis, heat, vomiting, tinnitus, confusion, pulmonary oedema, and glucose disturbance. Clinical toxicity can worsen while serum concentration falls as drug enters tissues. Avoid intubation if possible because loss of compensatory hyperventilation can cause sudden acidaemia; if essential, preserve high ventilation. Give glucose when needed, alkalinise serum and urine with bicarbonate, correct potassium, and use haemodialysis for severe neurological, pulmonary, acid-base, renal, or concentration criteria.

Opioids, sedatives, and antidepressants

Ventilation is the priority in opioid poisoning. Naloxone competitively reverses respiratory depression and is titrated to adequate breathing rather than full painful withdrawal. Its duration may be shorter than the opioid, so recurrent toxicity requires observation or infusion. Long-acting opioids, potent synthetic agents, patches, and mixed sedatives prolong risk. Withdrawal is distressing but usually less dangerous than untreated apnoea.

Benzodiazepines usually cause supportive-care sedation; flumazenil can precipitate seizure or withdrawal in dependent patients or mixed overdose and is reserved for carefully selected cases. Tricyclic antidepressants block fast sodium channels, causing hypotension, QRS widening, ventricular dysrhythmia, seizure, and antimuscarinic signs. Sodium bicarbonate treats cardiovascular toxicity. Selective serotonin-reuptake inhibitor overdose is

often milder but some agents cause seizures, QT prolongation, or serotonin toxicity.

Stimulants and hyperthermia

Cocaine, amphetamines, and related stimulants increase catecholamine effects, causing agitation, hypertension, tachycardia, coronary vasospasm, aortic injury, seizure, stroke, rhabdomyolysis, and hyperthermia.

Benzodiazepines reduce central activation; treat severe temperature elevation with rapid external cooling and sedation. Antipyretics do not correct toxin-driven muscle heat. Extreme agitation may require a team plan that minimises struggle, restraint time, acidosis, and staff injury.

Serotonin toxicity is treated by stopping agents, sedation, cooling, fluids, and airway or paralysis for extreme hyperthermia; selected antagonists may be used. Neuroleptic malignant syndrome requires drug cessation, supportive intensive care, and selected dopamine agonist or muscle-directed therapy. Malignant hyperthermia requires immediate dantrolene, anaesthetic cessation, cooling, correction of hyperkalaemia and acidosis, and monitoring for recurrence.

Alcohols and gases

Ethanol intoxication remains a diagnosis of exclusion when trauma, hypoglycaemia, infection, intracranial disease, or mixed ingestion is possible. Dependence risks withdrawal with tremor, autonomic activation, hallucinations, seizure, and delirium. Use protocol benzodiazepines, thiamine, fluid and electrolyte care, and assess injury and nutrition.

Methanol produces visual and basal-ganglia toxicity through formic acid; ethylene glycol causes acidosis, kidney injury, and calcium oxalate deposition. Block alcohol dehydrogenase with fomepizole, correct acidosis, give pathway cofactors, and dialyse severe cases. Do not wait for definitive levels when history and metabolic findings strongly support dangerous exposure.

Carbon monoxide binds haemoglobin and disrupts cellular respiration, causing headache, nausea, confusion, cardiac ischaemia, coma, and delayed neurological injury. Pulse oximetry can appear normal. Give

high-concentration oxygen and measure co-oximetry; consider hyperbaric consultation for severe neurological, cardiac, acidotic, pregnancy, or concentration features. Cyanide blocks mitochondrial electron transport, producing profound lactic acidosis and cardiovascular collapse after smoke or industrial exposure; give appropriate antidote promptly when suspected.

Pesticides, metals, and corrosives

Organophosphate pesticides inhibit acetylcholinesterase. Decontaminate safely, suction secretions, ventilate, give atropine until bronchial secretions and perfusion improve, and use an oxime early where indicated. Repeated atropine doses may be large; pupil size is not the treatment endpoint. Intermediate neuromuscular weakness can follow apparent recovery.

Lead causes abdominal, neurological, haematological, kidney, and developmental injury; children and pregnancy are particularly vulnerable. Exposure sources include old paint, contaminated soil or water, occupations, hobbies, spices, cosmetics, and traditional remedies. Remove exposure and use chelation according to concentration and symptoms. Other metals have distinct targets and chelators; specialist advice is essential.

Acids cause coagulative injury and alkalis penetrate deeply through liquefaction. Oropharyngeal appearance does not predict oesophageal damage. Avoid induced vomiting, neutralisation, and blind tube placement. Manage airway, pain, perforation risk, endoscopic timing, and surgical complications with toxicology and gastrointestinal teams.

Envenomation and bites

Envenomation varies geographically. Record time, place, description without pursuing the creature, progression, first aid, and systemic features. Keep the patient still, immobilise the limb, and follow regional pressure-immobilisation guidance. Do not cut, suck, freeze, or use an arterial tourniquet. Serial examination and targeted blood, neurological, and cardiac monitoring detect progression. Select antivenom by credible syndrome and severity with anaphylaxis readiness.

Bites cause crush, inoculation, tendon, joint, bone, and cosmetic injury. Irrigate, assess deep involvement, update tetanus, consider rabies by

geography and animal behaviour, and give antibiotics to high-risk wounds. Hand and clenched-fist injuries need urgent surgical review.

Heat, cold, and immersion

Heat exhaustion involves dehydration and cardiovascular strain without central neurological dysfunction. Heat stroke is hyperthermia with encephalopathy and organ injury and requires immediate rapid cooling, airway support, and complication care. Exertional heat stroke responds best to cold-water immersion when feasible; do not delay cooling for transport or tests.

Hypothermia causes bradycardia, dysrhythmia, coagulopathy, and reduced consciousness. Handle gently, remove wet clothing, insulate, and rewarm by severity; severe cases may need extracorporeal support. Drowning care prioritises ventilation and oxygenation, with selective spinal protection and observation rather than routine prophylactic antibiotics.

TTS module 2: Toxicokinetics, pattern recognition, antidotal logic, and environmental emergencies

Toxicology begins with exposure control and physiology because exact identification often arrives late. Protect rescuers, remove contaminated clothing, brush off dry powder before irrigation where appropriate, ventilate the area, and prevent secondary exposure from vomit, pesticide, or off-gassing chemicals. Then stabilise airway, ventilation, circulation, glucose, temperature, and seizures. A patient can look well while a delayed-release formulation is being absorbed or a toxic metabolite is accumulating, so observation follows expected kinetics rather than reassurance from one examination.

Toxic effect depends on dose reaching the target over time. Absorption varies with route, formulation, gastric emptying, pH, co-ingestion, skin integrity, and sustained-release technology. Distribution depends on protein binding, lipid solubility, tissue perfusion, and apparent volume. Metabolism can detoxify or create a more dangerous product, as with paracetamol, methanol, and ethylene glycol. Elimination changes with kidney and liver function. Saturation, enterohepatic cycling, bezoars, and ongoing release can turn apparently first-order decline into prolonged or nonlinear toxicity.

History seeks the maximum plausible dose, not an average estimate. Identify substance, concentration, formulation, time, route, co-ingestants, body weight, pregnancy, organ disease, regular medicines, dependence, and prehospital treatment. Packaging, dispensing records, witnesses, emergency personnel, and scene evidence may be more reliable than tablet count. Deliberate self-poisoning can involve concealed agents, and stated low suicidal intent does not reduce chemical dose. Psychosocial assessment follows medical stabilisation.

Toxidromes organise incomplete data. Opioids depress consciousness and ventilation, often with miosis. Sedative-hypnotics produce depression with less distinctive pupils. Sympathomimetics cause agitation, mydriasis, sweating, tachycardia, hypertension, hyperthermia, and bowel activity; antimuscarinics produce similar agitation and dilation but dry flushed skin, urinary retention, and ileus. Cholinergic poisoning floods muscarinic secretions and bronchial tone while nicotinic effects cause fasciculation then weakness. Mixed ingestion, hypoxia, trauma, and temperature blur every pattern.

The electrocardiogram is an early toxicology test because it reveals sodium-channel blockade, potassium-channel delay, bradyarrhythmia, ischaemia, and conduction disease. A widening QRS in tricyclic or other membrane-stabilising overdose predicts seizure and ventricular dysrhythmia and is treated with sodium bicarbonate to raise sodium and pH. QT prolongation requires correction of potassium and magnesium and removal of culprit drugs. Serial tracings matter because absorption and metabolites evolve.

Activated charcoal reduces absorption of many agents when given early enough, but benefit depends on substance, dose, timing, and airway safety. Aspiration can be fatal in a drowsy or vomiting patient. Charcoal binds poorly to alcohols, corrosives, metals, and several small ions. Whole-bowel irrigation has specialist use for some sustained-release products, metals, or body packets. Gastric lavage is rarely justified and induced vomiting is unsafe. Corrosives are not neutralised because exothermic reactions and re-exposure worsen injury.

Paracetamol toxicity is initially silent while a reactive metabolite consumes hepatic glutathione and binds cellular proteins. A treatment nomogram applies

only to a single acute immediate-release ingestion with a known time and correctly timed concentration. Staggered, repeated supratherapeutic, delayed-release, massive, or unknown-time exposure requires different interpretation. N-acetylcysteine restores glutathione capacity and improves outcomes even after liver injury. Rising aminotransferases, coagulopathy, hypoglycaemia, acidosis, kidney injury, encephalopathy, or shock requires intensive and transplant-centre involvement.

Salicylate stimulates respiratory drive and uncouples oxidative phosphorylation, generating heat and mixed respiratory alkalosis with metabolic acidosis. Tinnitus, vomiting, sweating, tachypnoea, agitation, confusion, pulmonary oedema, and low brain glucose can occur. Serial concentrations are essential because delayed absorption and tissue redistribution make one falling result misleading. Acidaemia increases non-ionised salicylate entry into brain. Sodium bicarbonate alkalinises serum and urine, but potassium must be corrected for urinary alkalinisation to work. Dialysis removes drug and corrects severe acid-base, renal, neurological, or pulmonary toxicity.

Intubating a salicylate-poisoned patient is hazardous because even brief apnoea or ordinary ventilator settings allow carbon dioxide and acidity to rise abruptly. If airway control is unavoidable, preserve pre-intubation minute ventilation, minimise apnoea, and begin aggressive ventilation immediately while dialysis proceeds. Sedation can also remove compensatory drive. This illustrates a general toxicology principle: a physiological adaptation that looks abnormal may be keeping the patient alive.

Opioid death is ventilatory failure. Open the airway and ventilate before or while naloxone is given. Titrate naloxone to adequate respiratory rate, depth, and airway protection rather than full alertness, reducing severe withdrawal, vomiting, agitation, and sympathetic stress. Long-acting opioids, patches, methadone, sustained release, and potent synthetics can outlast naloxone, requiring repeated doses or infusion. Normal pupils do not exclude mixed or atypical opioid toxicity.

Flumazenil can reverse benzodiazepine sedation but removes anticonvulsant protection and precipitates withdrawal in dependent patients or seizure in mixed tricyclic and other overdose. It is reserved for narrow low-risk scenarios. Serotonin toxicity is distinguished by rapid onset, clonus,

hyperreflexia, agitation, autonomic change, and heat after serotonergic exposure. Neuroleptic malignant syndrome is generally slower, with severe rigidity and dopamine blockade. Both require drug cessation, sedation and support, but extreme serotonin hyperthermia may require paralysis and ventilation because heat comes from muscle, not hypothalamic fever.

Methanol and ethylene glycol become toxic after alcohol-dehydrogenase metabolism. Methanol formate damages retina and basal ganglia; ethylene-glycol acids cause kidney injury and calcium oxalate deposition. The osmolar gap may be high early and fall as the anion gap rises, so a normal gap late is not reassuring. Fomepizole blocks metabolism, cofactors support alternative pathways, bicarbonate treats acidosis, and haemodialysis removes parent and metabolites in severe disease. Treatment begins on strong clinical evidence without waiting for delayed definitive levels.

Carbon monoxide impairs oxygen carriage and cellular use while ordinary pulse oximetry reads carboxyhaemoglobin as oxygenated. Headache, nausea, confusion, syncope, cardiac injury, fetal harm, and delayed neuropsychiatric sequelae can occur. Remove exposure, give high-concentration oxygen, measure co-oximetry, electrocardiogram and cardiac biomarkers where indicated, and discuss hyperbaric treatment for severe neurological, cardiac, acidotic, pregnancy, or exposure features. A low level after oxygen or delay underestimates peak burden.

Organophosphate toxicity combines muscarinic secretion with nicotinic weakness and central dysfunction. Decontamination protects staff. Oxygen, suction, and ventilation manage the airway. Atropine is rapidly escalated until bronchial secretions and perfusion improve; pupil size is not the endpoint and very large doses may be required. Oximes reactivate acetylcholinesterase before enzyme ageing for selected compounds. Weakness can emerge after cholinergic symptoms improve, requiring prolonged respiratory surveillance.

Envenomation is identified through geography, syndrome, and progression rather than risky pursuit of the animal. Immobilise and use regional first aid; avoid cutting, sucking, freezing, electric shock, and arterial tourniquets. Serial neurological, coagulation, kidney, muscle, and cardiac assessment detects systemic venom effects. Antivenom is given for credible significant envenomation when expected benefit exceeds reaction risk, with anaphylaxis

treatment immediately available. A bite without envenomation does not benefit from antivenom.

Heat stroke is central neurological dysfunction with dangerous hyperthermia and organ injury. Begin rapid cooling immediately; exertional disease is best treated with cold-water immersion when feasible, while airway and circulation are supported. Antipyretics do not help because the temperature is not prostaglandin-set fever. Monitor coagulation, liver, kidney, muscle, glucose, and electrolytes for delayed failure. Hypothermia causes reduced consciousness, bradycardia, coagulopathy, and irritable myocardium; handle gently, insulate, rewarm according to severity, and continue resuscitation because profound cold can mimic death.

Environmental and chemical injury frequently combines trauma, inhalation, and systemic poison. Smoke exposure may include carbon monoxide, cyanide, airway heat, and irritant lung injury. Cyanide blocks mitochondrial respiration and causes profound lactate elevation and cardiovascular collapse; give a suitable antidote promptly when the scenario fits. Drowning primarily causes hypoxaemia, so effective ventilation matters more than draining water. Across toxicology, the safest practice is to treat the threatened physiological process, consult specialist poison expertise early, and reassess through the full expected toxicity window.

Exact product formulation and regional antidote availability should be confirmed with the poisons service whenever uncertainty could change observation, decontamination, treatment, or disposition.

Retrieval prompts

One. Which priorities precede identifying a poison?

Two. How do sympathomimetic and antimuscarinic toxidromes differ?

Three. Why are routine urine drug screens limited?

Four. What is the goal of naloxone dosing?

Five. Why is salicylate intubation hazardous?

Six. What defines heat stroke?

Concise answers

One. Rescuer safety, decontamination, airway, ventilation, circulation, temperature, glucose, and seizure control.

Two. Both cause agitation and mydriasis, but sympathomimetics cause sweating while antimuscarinics cause dry skin, ileus, and retention.

Three. Detection windows, cross-reactivity, missing substances, and poor correlation with present toxicity.

Four. Restore adequate spontaneous ventilation while avoiding unnecessary precipitated withdrawal.

Five. Removing compensatory hyperventilation can abruptly worsen acidaemia and tissue salicylate entry.

Six. Hyperthermia with central neurological dysfunction and evolving organ injury.

Source map

Original synthesis informed by Guyton and Hall, ventilation, acid-base physiology, temperature control, neural function, and shock; Robbins, toxic liver, kidney, lung, blood, neurological, and thermal injury; Katzung and OpenStax Pharmacology, toxidromes, antidotes, drug interactions, elimination, and overdose treatment; Talley and O'Connor, exposure history, neurological, cardiovascular, respiratory, skin, and mental-state assessment; OpenStax Medical-Surgical Nursing, poisoning, substance withdrawal, environmental emergencies, bites, and decontamination; and OpenStax Chemistry, Biology, and Microbiology, chemical reactions, metabolism, venoms, and wound infection.

Chapter 60: Occupational, Travel, and Population Infection Risks

Orientation

Illness is shaped by where people live, work, travel, migrate, and gather. Exposure assessment links a patient to environments, vectors, animals, water, food, air, chemicals, tasks, and social conditions. The same symptom may have a different differential after healthcare work, mining, farming, displacement, a long flight, freshwater contact, or return from a malaria region. Protect the individual while recognising clusters that require workplace control or public-health notification.

Occupational history

Ask current and previous jobs, tasks, materials, dusts, fumes, organisms, radiation, noise, vibration, temperature, load, shifts, protection, spills, injuries, and affected colleagues. Job title is insufficient. Establish intensity, frequency, duration, latency, relation to work or leave, and exposures from contracting, military service, hobbies, renovation, caregiving, waste, and informal employment.

Consider inhalation, skin, ingestion, inoculation, noise, radiation, heat, and biomechanical load. Obtain product names and safety data without assuming listed limits prove safety. Work often worsens common disease; asthma, dermatitis, hearing loss, pain, infection, anxiety, and sleep disruption have mixed contributors.

Document objectively because findings may affect income, compensation, workplace relationships, and other workers. Explain confidentiality and seek consent before employer contact unless law requires notification. Removing a hazard is more effective than relying on individual behaviour: elimination and substitution outrank engineering controls, administrative controls, and personal protective equipment.

Occupational respiratory disease

Irritant fumes cause bronchospasm, pneumonitis, pulmonary oedema, or persistent airway dysfunction. Occupational asthma may follow sensitisation or irritation. Improvement away supports association. Serial peak flow at work and away, spirometry, bronchial testing, and exposure assessment establish linkage; early sensitiser removal improves prognosis.

Pneumoconioses follow inhalation and retention of mineral dust. Silica exposure in mining, cutting engineered stone, construction, and sandblasting causes nodular fibrosis, accelerated disease, autoimmune risk, cancer, and susceptibility to tuberculosis. Asbestos causes interstitial fibrosis, pleural plaques, lung cancer, and mesothelioma after long latency, with smoking strongly multiplying lung-cancer risk. Coal and other dusts have distinct patterns. Prevention requires exposure control and surveillance; fibrosis is generally irreversible.

Hypersensitivity pneumonitis is an immune response to inhaled antigens from moulds, birds, farming, water, or industry. Fever, cough, and breathlessness can evolve into fibrosis. Diagnosis integrates exposure, imaging, physiology, and sometimes lavage or biopsy. Antigen avoidance is central.

Skin, musculoskeletal, and physical hazards

Irritant contact dermatitis results from cumulative barrier injury; allergic contact dermatitis is delayed hypersensitivity to a specific agent. Distribution and timing guide patch testing. Gloves can protect but also occlude, irritate, or cause allergy. Restore the barrier, treat inflammation, and redesign exposure.

Work-related musculoskeletal disease reflects force, repetition, posture, vibration, fatigue, psychosocial demand, and limited recovery. Diagnosis should identify tissue and function rather than attributing every symptom to imaging changes. Early activity modification, ergonomic redesign, graded rehabilitation, and communication support sustained work. Prolonged complete rest often worsens recovery.

Noise causes irreversible cochlear injury, typically high-frequency loss and tinnitus. Control sound at source, measure exposure, maintain hearing protection, and conduct surveillance. Hand-arm vibration can produce vascular blanching, neuropathy, and musculoskeletal injury. Ionising radiation

damages DNA and tissues according to dose and timing; use time, distance, shielding, dosimetry, and pregnancy-aware protocols. Heat strain, ultraviolet exposure, and shift work add cancer, cardiovascular, sleep, and injury risks.

Needlestick and biological exposure

After sharps, splash, bite, or mucosal exposure, wash skin with soap and water, irrigate mucosa, do not squeeze or use caustics, and report urgently. Define fluid, route, depth, device, source status, and exposed person's vaccination, pregnancy, kidney health, and medicines. Human immunodeficiency virus post-exposure prophylaxis is most effective when started promptly and should not await all results when exposure is significant. Continue or stop after expert risk reassessment.

Hepatitis B risk depends on source and immunity; vaccination and immunoglobulin are used according to antibody status and exposure. There is no hepatitis C post-exposure prophylaxis, so baseline and follow-up testing seek early infection for curative treatment. Maintain source confidentiality and provide psychological support. Prevention uses safety-engineered devices, no recapping, immediate disposal, training, staffing, and hepatitis B vaccination.

Healthcare and laboratory workers may encounter tuberculosis, meningococcus, pertussis, measles, varicella, influenza, emerging respiratory pathogens, and specimen aerosols. Exposure management combines immediate infection-control advice, vaccination records, symptom monitoring, testing at biologically appropriate times, and prophylaxis when indicated.

Pre-travel assessment

Travel risk depends on destination, season, setting, duration, accommodation, activities, altitude, transport, care access, and traveller health. Start early for vaccine series and medicine trials. Review routine then destination vaccines and entry requirements. Live vaccines may be unsafe in pregnancy or immune suppression; inactivated vaccines may be less immunogenic.

Malaria prevention combines mosquito avoidance and chemoprophylaxis matched to region, resistance, pregnancy, age, comorbidity, interactions, and itinerary. Start and continue for the drug-specific interval. No regimen is

completely protective. Fever during or after travel to a malaria area is an emergency until malaria is excluded with appropriately repeated testing.

Prevent mosquito-borne disease with effective repellent, clothing, screened or air-conditioned rooms, bed nets, and reduction of breeding sites. Day-biting vectors transmit dengue, chikungunya, and Zika; night-biting vectors commonly transmit malaria. Dengue can progress from fever and pain to plasma leak, bleeding, and shock around defervescence. Avoid non-steroidal drugs until dengue is excluded because bleeding risk may rise.

Food and water precautions reduce but do not eliminate diarrhoea. Oral rehydration is primary treatment. Antibiotic self-treatment is selected by severity, destination, resistance, and patient risk; antimotility drugs are avoided with invasive dysentery or severe colitis. Hand hygiene, safe sex, animal avoidance, and freshwater precautions address hepatitis, enteric fever, rabies, sexually transmitted infection, and schistosomiasis.

Assess altitude, thrombosis, motion, sun, heat, cold, diving, injury, and medication supply. Gradual ascent prevents acute mountain illness. Headache with nausea or poor sleep suggests acute mountain sickness; ataxia or confusion indicates cerebral oedema, and breathlessness at rest may indicate pulmonary oedema. Descent and oxygen are definitive priorities, with preventive or therapeutic drugs used according to protocol.

The returned traveller

Ask itinerary and dates, setting, food, water, insects, animals, freshwater, caves, healthcare, sex, procedures, contacts, prophylaxis, vaccines, and symptom onset. Incubation narrows the differential. Reassess sepsis, bleeding, jaundice, respiratory distress, neurological change, and isolation needs.

Fever differentials include malaria, dengue, enteric fever, rickettsial disease, acute human immunodeficiency virus, viral hepatitis, influenza, coronavirus, leptospirosis, meningitis, and common local infections. Eosinophilia suggests some helminths but may be absent early or in severe disease. Blood cultures, malaria films or rapid tests, full blood count, liver and kidney tests, and pathogen-specific sampling follow exposure. Alert the laboratory before handling suspected high-consequence pathogens.

Diarrhoea lasting beyond two weeks raises protozoal and non-infectious causes. Skin lesions can reflect bites, larval migration, fungal infection, leishmaniasis, rickettsial eschar, or resistant bacteria. Respiratory illness after caves, birds, dust, healthcare, or mass gatherings changes fungal, zoonotic, and viral possibilities. A travel label should not distract from appendicitis, pulmonary embolism, drug reaction, or other universal disease.

Migration, displacement, and health inequity

Country of origin is not a biological diagnosis. Risk reflects prevalence, journey, living conditions, trauma, vaccination, nutrition, work, and healthcare exclusion. Use trained interpreters. Screen by individual risk for tuberculosis, hepatitis, human immunodeficiency virus, parasites, anaemia, vaccination gaps, pregnancy, dental disease, and mental health while preserving consent.

Displaced people may face interrupted medicines, overcrowding, unsafe water, violence, and exploitation. Trauma-informed care offers choice without forced disclosure. Treat urgent needs while building records, vaccination, continuity, and primary-care connection. Infection control must not become stigma.

Outbreak recognition and public health

An outbreak exceeds expected disease in a population, though one rare high-consequence case may trigger action. Confirm diagnosis and case definition, describe person, place, and time, identify source or transmission, sample appropriately, and control while investigating. Isolation separates infectious cases; quarantine restricts exposed people who may become infectious.

Notify public health promptly for locally reportable or urgent suspected disease. Explain why data and contacts are needed. Tracing, prophylaxis, vaccination, exclusion, remediation, recall, or workplace control follow the setting. Preserve privacy and use the least restrictive effective measure.

One Health links human, animal, and environmental health. Land use, farming, wildlife trade, antimicrobials, climate, travel, and urbanisation alter zoonoses and vectors. Prevention requires clinical, laboratory, veterinary, workplace, water, food, and community collaboration.

TTS module 2: Exposure reconstruction, travel syndromes, worker protection, and public-health action

Occupational and travel medicine begins by reconstructing exposure rather than naming a destination or job. A stone worker and an office manager may share an employer but not a hazard; two travellers to the same country may differ through season, altitude, rural contact, freshwater swimming, healthcare exposure, food, sexual activity, and prophylaxis. The useful history identifies agent, route, intensity, duration, timing, protection, latency, affected colleagues, and whether symptoms change away from the environment.

Hazard control follows a hierarchy because personal protective equipment is the least reliable barrier. Eliminate the task or agent where possible, substitute a safer process, isolate exposure through engineering such as enclosure or ventilation, then use administrative controls such as rotation and procedures. Respirators, gloves, hearing protection, and other personal equipment remain necessary for residual risk but depend on selection, fit, maintenance, training, and consistent use. A worker should not carry sole responsibility for an unsafe system.

Occupational asthma can arise through sensitisation to a specific high- or low-molecular-weight agent or after intense irritant exposure. Improvement on weekends or leave supports work association but may disappear once disease becomes persistent. Serial peak-flow records across work and non-work days, spirometry, bronchial responsiveness, immunological testing for selected agents, and industrial-hygiene assessment build causality. Early removal from a sensitiser improves prognosis, while delayed recognition can leave permanent hyperresponsiveness and major employment consequences.

Inhaled mineral and engineered dusts deposit according to particle size, airway geometry, and clearance. Crystalline silica activates macrophages and drives nodular fibrosis, raises tuberculosis and autoimmune risk, and is especially hazardous during dry cutting of engineered stone. Asbestos fibres cause pleural plaques, diffuse fibrosis, lung cancer, and mesothelioma after decades; smoking multiplies asbestos-related lung-cancer risk. Imaging and lung function assess disease, but primary prevention through dust suppression, enclosure, ventilation, and exposure prohibition is far more effective than surveillance.

Hypersensitivity pneumonitis is an immune response to inhaled antigens from birds, mould, farming, water aerosols, and industrial materials. Acute exposure may cause fever, cough, and breathlessness hours later; repeated exposure leads to chronic dyspnoea and fibrosis. Precipitating antibodies indicate exposure, not necessarily disease. Diagnosis integrates a credible source, high-resolution imaging, physiology, lavage or biopsy in selected uncertainty, and response to avoidance. Asking only about job title misses home humidifiers, pets, hobbies, and contaminated ventilation.

Occupational dermatitis is mapped against contact. Irritant disease follows cumulative wet work, solvents, friction, soaps, and glove occlusion; allergic disease is delayed T-cell hypersensitivity to a particular chemical and can spread beyond the contact edge. Patch testing identifies contact sensitisation, whereas skin-prick testing does not. Gloves can contain rubber accelerators or latex and may worsen sweating. Restore barrier, treat inflammation, substitute products, and redesign exposure rather than repeatedly assigning topical steroids while the cause continues.

Noise injury depends on intensity, duration, impulse peaks, frequency, and cumulative exposure. Cochlear hair-cell loss is irreversible and often first impairs speech in background noise. Engineering control at source and quieter equipment outrank hearing protectors. Audiometric surveillance detects threshold shift but is not prevention if exposure remains. Vibration can injure digital vessels, nerves, joints, and muscle, producing blanching, numbness, reduced dexterity, and pain; temperature and smoking modify vascular symptoms.

After needlestick or mucosal exposure, wash skin with soap and water, irrigate mucosa, avoid squeezing and caustics, report urgently, and characterise source fluid, route, depth, device, and timing. Human immunodeficiency virus post-exposure prophylaxis is most effective when started promptly and should not await every source result after a meaningful exposure. Hepatitis-B management depends on vaccination and antibody response; hepatitis-C has no prophylactic drug but early follow-up testing permits curative treatment. Confidentiality applies to source and worker alike.

The exposed worker's baseline human immunodeficiency virus, hepatitis, pregnancy, kidney function, drug interactions, and vaccination guide prophylaxis. A negative source test may not exclude very recent infection

without context. Follow-up timing follows marker biology rather than convenience. Adverse effects and anxiety can undermine completion, so active contact and counselling matter. Prevention includes safety-engineered needles, immediate disposal, no recapping, adequate staffing and lighting, vaccination, and a blame-free reporting pathway that learns from near misses.

Pre-travel assessment is a risk plan, not a vaccine transaction. Destination is specified down to region and elevation, with dates, season, duration, urban or rural setting, accommodation, transport, planned work, animal and freshwater contact, sex, medical care, and remoteness. Review routine immunisation before destination-specific vaccines because measles, influenza, and other familiar diseases often pose greater risk than exotic infections. Live vaccines require special consideration in pregnancy and immune suppression, and several series need advance time.

Malaria prevention combines avoiding mosquito bites with a drug chosen for exact region, resistance, pregnancy, age, psychiatric and kidney history, interactions, and itinerary. Starting before travel tests tolerance and establishes therapeutic levels; continuation after leaving covers parasites emerging from the liver according to the agent. No prophylaxis is fully protective. Fever during or after travel in a malaria area is malaria until urgently excluded through appropriate rapid test and repeated thick and thin blood films when initial testing is negative but suspicion remains.

Dengue commonly causes fever, severe aches, headache, and cytopenias, but dangerous capillary leak and bleeding often emerge around defervescence as the person appears to be leaving the febrile phase. Warning signs include abdominal pain, persistent vomiting, mucosal bleeding, lethargy, hepatomegaly, rising haematocrit with falling platelets, and fluid accumulation. Fluid is carefully titrated because both shock and later overload are dangerous. Avoid aspirin and non-steroidal anti-inflammatory drugs until dengue is excluded.

Traveller's diarrhoea is managed first with oral rehydration. Antibiotic self-treatment is reserved for functionally severe or high-risk disease according to destination resistance and patient factors. Loperamide can reduce symptoms in non-invasive disease but is avoided alone with dysentery, high fever, or severe colitis. Persistent diarrhoea beyond roughly two weeks shifts attention toward *Giardia* and other protozoa, post-infectious

bowel syndromes, inflammatory disease, and medication. A negative single ova-and-parasite examination may miss intermittent shedding.

Rabies prevention depends on animal species, behaviour, geography, exposure site, and access to observation or testing. Immediately wash wounds thoroughly with soap and water and virucidal agent where available. Post-exposure vaccination and immunoglobulin for previously unvaccinated people must not be delayed after a credible exposure because symptomatic rabies is almost uniformly fatal. Pre-exposure vaccination simplifies but does not eliminate post-exposure treatment. Avoid handling bats, dogs, monkeys, and other mammals even when apparently tame.

Altitude illness results from hypobaric hypoxia. Gradual ascent permits ventilation, renal bicarbonate loss, erythropoietic and vascular adaptation. Acute mountain sickness causes headache plus nausea, dizziness, fatigue, or sleep disturbance after ascent. Ataxia, confusion, or altered consciousness indicates cerebral oedema; breathlessness at rest, cough, reduced exercise, and crackles suggest high-altitude pulmonary oedema. Stop ascent and descend; oxygen and portable pressure support are bridges. Sedatives and continued ascent are dangerous.

The returned traveller is assessed first for immediate physiological threat and isolation needs. Exact dates allow incubation-based reasoning. Fever with thrombocytopenia may be malaria, dengue, rickettsial disease, sepsis, acute human immunodeficiency virus, or a common local virus. Jaundice may reflect viral hepatitis, malaria haemolysis, leptospirosis, drug injury, or obstruction. Eosinophilia suggests selected tissue-invasive helminths but may be absent early or during overwhelming infection. Inform the laboratory before samples from suspected high-consequence pathogens.

Migration history is not a proxy for biology. Risk reflects prevalence in places lived, route and duration of journey, camp or detention conditions, vaccination interruption, occupational exposure, nutrition, trauma, and barriers to care. Use trained interpreters and explain screening. Tuberculosis, hepatitis, human immunodeficiency virus, parasites, anaemia, dental disease, pregnancy, mental health, and chronic medicines are assessed by individual evidence. Infection-control measures should protect without stigmatising a community.

Cluster recognition turns one consultation into prevention for many. Ask whether coworkers, household members, travellers, diners, patients, or animals are ill. A rare high-consequence disease may require action after one suspected case. Notify public health according to local law and urgency before complete confirmation when delay risks transmission. A case definition, line list, person-place-time pattern, specimens, and exposure map guide control. Isolation separates infectious cases; quarantine restricts exposed but currently well people during possible incubation.

Public-health interventions should be effective, proportionate, and least restrictive. Contact tracing, prophylaxis, vaccination, exclusion, food recall, water remediation, ventilation, workplace closure, or vector control are selected by transmission. Genomic similarity supports but does not prove an epidemiological link. One Health connects human infection with animals, food production, antimicrobial use, land change, climate, and water systems. Clinical care is complete only when preventable exposure has been addressed for the patient and others still at risk.

Retrieval prompts

- One. Which details make an occupational history useful?
- Two. What hierarchy best controls workplace hazards?
- Three. What are the priorities after a needlestick exposure?
- Four. What must be assumed about fever after malaria-region travel?
- Five. Which travel details narrow an infectious differential?
- Six. What is the difference between isolation and quarantine?

Concise answers

- One. Exact tasks, agents, routes, intensity, duration, latency, protective measures, symptom timing, and affected coworkers.
- Two. Elimination, substitution, engineering, administrative control, then personal protective equipment.

Three. Wash or irrigate, report urgently, characterise risk, test appropriately, and begin indicated human immunodeficiency virus or hepatitis B prophylaxis promptly.

Four. Malaria is possible and potentially fatal until appropriately excluded.

Five. Exact locations and dates, season, vectors, animals, water, food, sex, healthcare, activities, vaccines, prophylaxis, and incubation.

Six. Isolation separates infectious cases; quarantine restricts exposed people during the period they may become infectious.

Source map

Original synthesis informed by Guyton and Hall, respiratory, temperature, altitude, and environmental adaptation; Robbins, pneumoconiosis, radiation, infection, toxic, and occupational pathology; Katzung and OpenStax Pharmacology, vaccines, prophylaxis, post-exposure therapy, travel drugs, and interactions; Talley and O'Connor, occupational, travel, exposure, respiratory, skin, neurological, and systemic assessment; OpenStax Medical-Surgical Nursing, worker safety, infection control, needlestick response, travel illness, and community care; and OpenStax Biology, Chemistry, and Microbiology, ecology, vectors, zoonoses, outbreaks, antimicrobial resistance, and environmental exposure.

Chapter 61: Pain Medicine, Analgesia, Anaesthesia, Sedation, and Procedural Safety

Orientation

Pain is protective but causes disability when persistent or poorly controlled. Analgesia should relieve suffering while preserving breathing, circulation, cognition, mobility, and reassessment. Anaesthesia and sedation alter consciousness, sensation, reflexes, or movement and can rapidly create airway and haemodynamic emergencies. Safe care combines mechanism-based treatment, consent, preparation, monitoring, rescue, and recovery planning.

Pain mechanisms and assessment

Nociceptors detect mechanical, thermal, and chemical threat. Primary afferents enter spinal cord, where signals are amplified or inhibited before ascending through thalamic and cortical networks. Descending pathways, attention, expectation, fear, memory, sleep, mood, and social context modulate experience. Pain is real even when tissue damage is not visible, but description alone does not identify mechanism.

Nociceptive pain arises from tissue injury and may be somatic or visceral. Neuropathic pain results from lesion or disease of somatosensory pathways and is often burning, electric, shooting, numb, or allodynic. Nociplastic pain reflects altered processing without sufficient ongoing tissue injury or identifiable neural lesion. Mixed mechanisms are common in cancer, back pain, diabetes, surgery, and trauma.

Ask onset, site, radiation, character, severity, timing, triggers, function, sleep, mood, beliefs, prior treatment, substance use, and goals. Use a validated scale suited to communication, not as a substitute for observation and function. In dementia, critical illness, or communication impairment, combine behavioural signs, physiological context, caregiver knowledge, and

therapeutic trials. Reassess both benefit and adverse effects after intervention.

Multimodal analgesia

Treat the cause when possible and combine non-drug and drug methods that act through different pathways. Explanation, positioning, ice or heat when suitable, splinting, physiotherapy, graded activity, sleep care, psychological therapy, and procedural source control reduce pain and disability. Regional anaesthesia can provide powerful opioid-sparing relief.

Paracetamol provides analgesia and fever reduction with little anti-inflammatory action. Check total dose across combination products and reduce exposure with low body weight, malnutrition, alcohol-related risk, or liver disease. Overdose can cause silent early hepatic injury and needs timely antidote assessment.

Non-steroidal anti-inflammatory drugs inhibit cyclooxygenase and prostaglandin synthesis. They benefit inflammatory and acute musculoskeletal pain but can cause ulceration, bleeding, kidney injury, sodium retention, hypertension, bronchospasm, and cardiovascular events. Risk rises with age, dehydration, kidney disease, anticoagulation, steroids, heart failure, and prior ulcer. Use the lowest effective dose for the shortest suitable period; topical preparations reduce systemic exposure for local pain.

Opioids

Opioid receptors reduce neurotransmitter release and neuronal excitability. Opioids relieve severe acute and cancer pain but cause sedation, respiratory depression, nausea, constipation, itch, urinary retention, endocrine effects, tolerance, dependence, and overdose. Risk increases with sleep apnoea, frailty, kidney or liver dysfunction, opioid naivety, high dose, and concurrent alcohol, benzodiazepines, gabapentinoids, or other sedatives.

Select agent, route, and dose from urgency, prior exposure, organ function, and duration. Titrate small intravenous doses with monitoring when rapid effect is needed; oral treatment is safer for stable pain. Equianalgesic tables are estimates and cross-tolerance is incomplete, so reduce calculated dose when rotating and reassess. Some metabolites accumulate in kidney disease.

Long-acting preparations are inappropriate for rapid titration in opioid-naïve acute pain.

Respiratory depression presents with reduced rate, depth, arousal, and ventilation; oxygen saturation may remain deceptively normal with supplemental oxygen. Stimulate, open airway, ventilate, stop opioid, and titrate naloxone to adequate breathing. Naloxone may wear off first. Co-prescribe bowel prevention for continuing therapy and explain driving, storage, and disposal.

For chronic non-cancer pain, define functional goals and review whether benefit exceeds harm. Higher doses do not reliably restore function and increase overdose. Monitor sedation, sleep breathing, mood, substance use, endocrine effects, and aberrant supply without stigmatising. Dependence is expected adaptation; addiction is impaired control despite harm. Avoid abrupt discontinuation unless immediate danger, and offer evidence-based opioid-use-disorder treatment when indicated.

Neuropathic and specialised analgesia

Neuropathic pain often responds incompletely to conventional analgesics. Tricyclic or serotonin-noradrenaline antidepressants, gabapentinoids, topical lidocaine, capsaicin, or condition-specific therapy can help. Start low and review dizziness, falls, anticholinergic effects, cardiac risk, oedema, sedation, misuse, and organ function. Combination sedatives increase respiratory risk.

Ketamine antagonises N-methyl-D-aspartate receptors and provides analgesia with relative preservation of breathing at subanaesthetic doses, although dissociation, hypertension, nausea, salivation, and emergence phenomena occur. Local anaesthetics block voltage-gated sodium channels; systemic toxicity causes circumoral symptoms, tinnitus, seizure, conduction delay, ventricular dysrhythmia, and collapse. Stop injection, call for help, manage airway and seizure, use modified resuscitation, and give lipid emulsion according to protocol.

Cancer pain management combines disease-directed treatment, opioids, non-opioids, adjuvants, radiation, nerve blocks, and psychosocial care. Incident pain may need pre-emptive rescue dosing. Bone pain, neuropathy, obstruction, raised pressure, and muscle spasm require mechanism-specific

treatment. Palliative intent does not remove the need for careful titration and adverse-effect prevention.

Pre-anaesthetic assessment

Define procedure, urgency, goals, prior anaesthetic problems, allergies, medicines, airway, reflux, fasting, exercise tolerance, cardiorespiratory and neurological disease, sleep apnoea, pregnancy, substances, bleeding, infection, dentition, frailty, and discharge support. Retrieve records for difficult airway, malignant hyperthermia, awareness, or family reactions.

Examine mouth opening, dentition, jaw, pharyngeal view, neck mobility, pathology, and ability to lie flat. No single test predicts difficulty; prepare for ventilation, device, intubation, and front-of-neck rescue. Test only when results change care. Optimise anaemia, glucose, pressure, infection, volume, medicines, and thrombosis risk without delaying emergency surgery.

Fasting reduces aspiration risk but excessive fasting causes dehydration and discomfort. Follow current timing for clear fluids and solids and recognise delayed gastric emptying from obstruction, pregnancy, diabetes, severe pain, opioids, and selected metabolic medicines. A fasting patient can still aspirate.

General and regional anaesthesia

General anaesthesia combines hypnosis, amnesia, analgesia, immobility, and reflex suppression using intravenous and inhaled agents, opioids, and neuromuscular blockers. These reduce sympathetic tone, vascular resistance, myocardial function, and breathing to different degrees. Continuous observation, oxygenation, ventilation, circulation, temperature, and neuromuscular monitoring guide dosing and detect equipment or physiological failure.

Airway management ranges from natural airway with support to supraglottic device or tracheal tube. Preoxygenation extends safe apnoea time. Confirm tracheal placement with sustained exhaled carbon dioxide, not chest movement alone. Failed oxygenation requires a rehearsed algorithm prioritising oxygen over repeated traumatic attempts.

Spinal and epidural anaesthesia block nerve roots and sympathetic outflow. Benefits include excellent analgesia and reduced systemic opioid, while

complications include hypotension, urinary retention, motor block, post-dural-puncture headache, infection, bleeding, nerve injury, and rare high block with respiratory and cardiovascular collapse. Antithrombotic timing is critical to prevent spinal haematoma. Peripheral nerve blocks require ultrasound or anatomical skill, dose calculation, monitoring, and protection of the numb limb.

Procedural sedation

Sedation is a continuum from anxiolysis to unresponsiveness; patients can unexpectedly enter deeper levels. The team must rescue one level deeper than intended. Assess airway, fasting context, comorbidity, medicines, consent, escort, and procedure stimulation. Allocate a trained clinician whose primary role is monitoring rather than performing the procedure when moderate or deep sedation is used.

Prepare oxygen, suction, bag-mask ventilation, airway devices, reversal drugs, intravenous access, and emergency equipment. Monitor responsiveness, respiratory rate and effort, oxygen saturation, circulation, and exhaled carbon dioxide for ventilation when appropriate. Capnography detects hypoventilation before desaturation, especially with supplemental oxygen. Titrate slowly and allow effect-site time before repeating doses.

Common combinations of benzodiazepine and opioid synergistically depress breathing. Propofol can rapidly cause apnoea and hypotension and has no reversal agent. Ketamine generally preserves reflexes but can still cause airway obstruction, laryngospasm, apnoea, or emesis. Nitrous oxide provides rapid anxiolysis and analgesia but expands closed gas spaces and requires scavenging and contraindication review.

Perioperative emergencies and recovery

Perioperative anaphylaxis may cause sudden hypotension, bronchospasm, or difficult ventilation without rash. Give adrenaline, fluid, oxygen, stop triggers, and investigate later. Malignant hyperthermia produces rising carbon dioxide, rigidity, acidosis, hyperkalaemia, and heat; stop triggers, give dantrolene, cool, and treat complications.

Awareness is uncommon but distressing, especially with emergency surgery, haemodynamic limitation, paralysis, or equipment failure. Take reports seriously, document, investigate, explain, and support. Hypothermia increases bleeding, infection, discomfort, and delayed drug clearance; use active warming.

Recovery requires stable airway, breathing, circulation, consciousness, pain, nausea, temperature, bleeding, hydration, and motor function. Sedation discharge also requires mobility, escort when indicated, written advice, and driving, alcohol, decision, and activity restrictions. Escalation instructions address bleeding, infection, respiratory depression, or block complications.

TTS module 2: Mechanism-based analgesia, sedation rescue, and anaesthetic physiological safety

Pain care has two simultaneous goals: relieve suffering and preserve the ability to breathe, think, move, and reveal changing disease. A pain score is one measurement, not the endpoint. Assess function, sleep, cough, mobilisation, anxiety, adverse effects, and whether the trajectory fits the presumed lesion. Escalating pain despite increasing analgesia can signal compartment syndrome, ischaemia, perforation, infection, or treatment complication. Analgesia should enable repeated examination rather than be withheld in the belief that pain itself is diagnostically useful.

Nociceptive pain begins with tissue-threat transduction and is divided broadly into somatic and visceral patterns. Neuropathic pain follows disease or lesion of the somatosensory system and may include allodynia, hyperalgesia, burning, electric shocks, numbness, and sensory change. Nociplastic pain reflects altered processing and network amplification without enough ongoing tissue or nerve injury to explain severity. These mechanisms overlap and can evolve; persistent nociceptive input can sensitise spinal and supraspinal circuits, while fear, sleep loss, depression, memory, and social threat alter descending control.

Multimodal analgesia combines treatments with different targets so no one toxicity dominates. Positioning, splinting, drainage, reduction, heat or cold, physiotherapy, psychological strategies, regional blocks, paracetamol, anti-inflammatory drugs, opioids, and neuropathic agents are selected by mechanism. "Opioid-sparing" is useful only if the replacement is itself safe;

stacking gabapentinoids, sedatives, and muscle relaxants may worsen falls and ventilation. Every addition needs a defined benefit and review point.

Paracetamol has a ceiling effect and hidden duplication across combination products is a major overdose route. Dose should reflect low body weight, malnutrition, alcohol-related risk, and liver disease. Non-steroidal drugs reduce prostaglandin-mediated inflammation but also remove gastric, renal, platelet, and vascular protection. Dehydration, heart failure, kidney disease, anticoagulation, corticosteroids, previous ulcer, and older age amplify harm. Topical treatment can provide local musculoskeletal benefit with lower systemic exposure, though absorption is not zero.

Opioid effect depends on receptor, route, active metabolites, prior tolerance, organ clearance, and concurrent sedatives. Intravenous boluses reach the effect site rapidly and are titrated in monitored severe pain. Long-acting preparations are unsuitable for initial opioid-naïve titration because accumulation outlasts reassessment. Equianalgesic conversions are population estimates; incomplete cross-tolerance means the calculated new opioid dose is reduced and retitrated. Morphine metabolites accumulate in kidney failure, while every alternative carries its own interaction and organ constraints.

Respiratory rate alone can miss opioid ventilatory impairment. A sedated patient may take apparently regular shallow breaths with rising carbon dioxide, especially on supplemental oxygen that preserves saturation. Monitor arousal, airway obstruction, rate, depth, capnography where appropriate, and risk factors including sleep apnoea, obesity, frailty, chest disease, kidney failure, and benzodiazepines. Initial management is stimulation, airway opening, oxygen, and effective ventilation. Naloxone is titrated to adequate breathing rather than full painful withdrawal and may require infusion when the opioid lasts longer.

Long-term opioid therapy is judged by sustained functional benefit, not dose stability alone. Tolerance and physical dependence are expected adaptations and differ from opioid-use disorder, which involves impaired control despite harm. Constipation usually persists despite analgesic tolerance and needs prevention. Endocrine suppression, sleep-disordered breathing, falls, cognitive effects, hyperalgesia, and overdose risk are reviewed. Abrupt involuntary discontinuation can trigger withdrawal, illicit substitution, and

suicide risk; taper collaboratively unless immediate toxicity demands faster action, and offer agonist therapy for opioid-use disorder.

Neuropathic medicines work slowly and often partially. Tricyclics add anticholinergic, postural, cardiac, and overdose risk. Serotonin–noradrenaline reuptake inhibitors affect pressure, nausea, withdrawal, and interactions. Gabapentinoids cause dizziness, oedema, sedation, and respiratory synergy and require kidney adjustment. Topical lidocaine or capsaicin suits selected focal disease. Treatment success may mean better sleep and movement rather than zero pain. Escalation without benefit should stop because polypharmacy itself worsens function.

Local anaesthetics block voltage-gated sodium channels and preferentially affect rapidly firing and smaller fibres according to drug, concentration, and anatomy. Adding vasoconstrictor can prolong local exposure and reduce systemic absorption in suitable sites. Total dose includes every infiltration and block, adjusted for weight, pregnancy, low cardiac output, liver dysfunction, acidosis, and protein binding. Accidental intravascular injection or rapid absorption causes circumoral numbness, metallic taste, tinnitus, agitation, seizure, then conduction delay, ventricular dysrhythmia, and collapse.

Local-anaesthetic systemic toxicity requires immediate cessation, help, airway oxygenation, seizure control, and a modified resuscitation strategy with intravenous lipid emulsion. Large propofol doses and selected antiarrhythmics can worsen physiology; protocol and specialist guidance matter. Ultrasound reduces but does not eliminate intravascular or neural injury. Fractionated injection with repeated aspiration and observation detects early symptoms. A numb limb is protected from pressure, heat, and falls until sensation and power return.

Pre-anaesthetic assessment predicts difficulty in oxygenation, ventilation, circulation, aspiration prevention, positioning, drug handling, and recovery. Mouth opening, dentition, mandibular movement, neck mobility, pharyngeal view, obesity, facial hair, airway pathology, and previous records contribute, but no single score excludes a difficult airway. Prepare primary and backup devices, skilled assistance, suction, and front-of-neck rescue. The first priority in difficulty is oxygen delivery, not completing tracheal intubation at any cost.

Preoxygenation replaces nitrogen in functional residual capacity with oxygen, extending tolerable apnoea. Its duration is shorter in pregnancy, obesity, children, lung disease, agitation, and low cardiac output. Head-up positioning and continuous oxygen during attempts can help. Induction removes airway tone and sympathetic support; positive-pressure ventilation reduces venous return. Severe hypovolaemia, tamponade, tension pneumothorax, pulmonary hypertension, or metabolic acidosis can therefore arrest immediately after an anatomically successful intubation unless physiology is prepared.

General anaesthesia combines hypnosis, amnesia, analgesia, immobility, and reflex suppression, but no single agent provides all components safely. Neuromuscular paralysis creates no unconsciousness or analgesia and can conceal awareness. Depth and drug delivery are cross-checked, especially during low-dose emergency anaesthesia. Tracheal placement is confirmed through sustained waveform carbon dioxide. Hypotension, bronchospasm, anaphylaxis, equipment failure, circuit disconnection, and malignant hyperthermia are distinguished by timing, airway pressures, carbon dioxide, skin, and haemodynamics.

Neuraxial anaesthesia blocks sympathetic fibres as well as sensation and movement, producing vasodilation and reduced venous return. A high block can impair intercostal function, arms, consciousness, and cardiac accelerator fibres, causing profound bradycardia and shock. Treat airway and circulation immediately. Anticoagulant timing is crucial because spinal haematoma can cause irreversible cord compression; new severe back pain, weakness, sensory change, or sphincter dysfunction after neuraxial procedure is an emergency.

Procedural sedation is a continuum because individual response varies and painful stimulation changes over time. The sedation clinician must be able to rescue at least one level deeper than intended and should not be simultaneously absorbed in the procedure during moderate or deep sedation. Prepare oxygen, suction, bag-mask ventilation, airway adjuncts, reversal agents, access, defibrillation, and a plan for difficult ventilation. Supplemental oxygen delays desaturation but can mask hypoventilation; capnography detects absent or obstructed ventilation earlier.

Benzodiazepines and opioids synergistically depress ventilation. Propofol can abruptly cause apnoea, airway collapse, and vasodilation and has no

pharmacological reversal. Ketamine tends to preserve respiratory drive and sympathetic tone but can still cause laryngospasm, obstruction, apnoea, vomiting, or cardiovascular stress. Repeating doses before peak effect is a common route to unintended deep sedation. Dose is reduced and titrated slowly in frailty, shock, organ failure, and concurrent medication.

Recovery is a distinct high-risk phase because procedure stimulation ends while sedative absorption and active metabolites continue. Confirm stable airway, ventilation, oxygenation, circulation, consciousness, pain, nausea, temperature, bleeding, hydration, and motor or sensory block. Discharge requires responsible support when indicated, safe mobility and intake, and written restrictions on driving, alcohol, legal decisions, machinery, and additional sedatives. Clear return instructions address respiratory depression, bleeding, fever, severe pain, weakness, retention, or block complications. Procedural success includes safe recovery, not merely completion of the intervention.

Retrieval prompts

One. Which three broad pain mechanisms guide treatment?

Two. What increases opioid respiratory risk?

Three. What are the warning features of local-anaesthetic systemic toxicity?

Four. What is the central principle in a failed airway?

Five. Why is procedural sedation a continuum rather than a fixed state?

Six. What must be stable before recovery discharge?

Concise answers

One. Nociceptive, neuropathic, and nociplastic mechanisms, often mixed.

Two. Opioid naivety, high dose, sleep apnoea, frailty, organ failure, and concurrent sedatives or alcohol.

Three. Circumoral or auditory symptoms, neurological excitation or seizure, conduction delay, dysrhythmia, and collapse.

Four. Prioritise oxygenation and stop repeated attempts that cause trauma and swelling.

Five. Individual responses vary and intended sedation can deepen unpredictably, requiring rescue capability.

Six. Airway, ventilation, circulation, consciousness, pain, nausea, temperature, bleeding, hydration, mobility, and support plan.

Source map

Original synthesis informed by Guyton and Hall, nociception, central modulation, consciousness, neuromuscular function, ventilation, and circulation; Robbins, tissue injury, inflammation, neuropathy, and anaesthetic complications; Katzung and OpenStax Pharmacology, analgesics, opioids, local and general anaesthetics, sedatives, reversal agents, and interactions; Talley and O'Connor, pain, airway, cardiorespiratory, neurological, and preoperative assessment; OpenStax Medical-Surgical Nursing, perioperative care, pain measurement, sedation monitoring, recovery, and education; and OpenStax Anatomy and Physiology and Chemistry, neural signalling and drug physicochemistry.

Chapter 62: Critical Care Physiology, Oxygen Support, Ventilation, and Organ Support

Orientation

Critical care supports failing physiology while treating reversible disease and preventing harm. Organ failures interact: ventilation alters circulation, shock injures kidney and brain, inflammation changes drug handling, and immobility weakens respiratory muscles. Good care uses explicit targets, frequent reassessment, minimal invasiveness, multidisciplinary coordination, and discussion of prognosis and goals.

Recognition and initial stabilisation

Critical illness may appear as airway obstruction, increased work of breathing, hypoxaemia, hypercapnia, hypotension, poor perfusion, altered consciousness, oliguria, acidosis, severe temperature disturbance, or rapidly changing observations. Trends and clinician concern matter more than a single score. Begin airway, breathing, circulation, disability, and exposure assessment while calling appropriate help and treating immediately reversible causes.

Match monitoring to instability: electrocardiography, saturation, pressure, temperature, urine, glucose, consciousness, fluid balance, and selected invasive haemodynamics. Monitoring should answer a question. Lines, catheters, alarms, and blood draws also cause infection, thrombosis, anaemia, sleep disruption, and false intervention.

Oxygen physiology and delivery

Oxygen delivery equals cardiac output multiplied by arterial oxygen content. Haemoglobin-bound oxygen dominates content, so normal saturation does not compensate for profound anaemia or low flow. Tissue extraction rises when delivery falls, but sustained imbalance causes lactate, organ dysfunction, and cell death. Lactate also rises from adrenergic glycolysis, impaired clearance, drugs, seizure, and mitochondrial dysfunction and is not a direct oxygen meter.

Give oxygen for hypoxaemia and emergencies, then titrate. Excess causes absorption atelectasis, oxidative injury, vasoconstriction, and hypercapnia in susceptible patients. Nasal cannulae provide low flow, masks higher concentration, and Venturi systems controlled concentration. Heated high-flow nasal oxygen reduces dead-space rebreathing, provides modest distending pressure, and decreases work.

Escalate when oxygenation, ventilation, work, consciousness, haemodynamics, or airway protection worsens. Pulse oximetry is affected by perfusion, motion, skin pigmentation, dyshemoglobins, nail products, and device limitations. Arterial blood gas measures oxygen and carbon dioxide tensions and acid-base status but should be interpreted with clinical trajectory.

Non-invasive respiratory support

Continuous positive airway pressure recruits alveoli, improves oxygenation, and reduces left-ventricular afterload. Bilevel support adds inspiratory assistance to improve ventilation. Strong uses include acute cardiogenic pulmonary oedema and selected hypercapnic exacerbations of obstructive lung disease. Hypoxaemic respiratory failure can benefit in selected cooperative patients with close monitoring.

Non-invasive support is unsafe with inability to protect airway, severe vomiting, facial barrier, untreated pneumothorax, profound instability, or immediate intubation need. Reassess within minutes to hours. Worsening acidosis, respiratory rate, oxygenation, effort, consciousness, or circulation signals failure. Prolonged unsuccessful trials delay definitive airway and increase harm.

Intubation and invasive ventilation

Intubation is a physiological as well as anatomical procedure. Preoxygenate, optimise position, prepare suction, drugs, devices, backup airway, and skilled assistance. Correct hypotension and anticipate reduced venous return after positive pressure. Rapid-sequence induction uses an induction agent and neuromuscular blocker to facilitate prompt intubation while limiting aspiration time. Confirm tracheal placement with sustained exhaled carbon dioxide.

Ventilation supplies tidal volume and respiratory rate to remove carbon dioxide, while inspired oxygen and positive end-expiratory pressure support oxygenation. Pressure and volume modes differ in what is set and what varies, but patient outcome depends more on appropriate targets, synchrony, lung protection, and reassessment than mode names.

Ventilator-induced injury arises from excessive stretch, pressure, repeated alveolar opening, oxygen toxicity, and inflammatory amplification. Use predicted body weight, not actual weight, to set lung-protective tidal volume because lung size follows height and sex-related anatomy. Limit plateau and driving pressures and use sufficient positive end-expiratory pressure to prevent collapse without unacceptable overdistension or haemodynamic compromise.

Acute respiratory distress syndrome is acute inflammatory lung injury with bilateral opacities and hypoxaemia not fully explained by cardiac failure. Treat cause, use low tidal volume, appropriate pressure, conservative fluid after shock resolves, and prone positioning for severe disease. Neuromuscular blockade and extracorporeal support are selective. Permissive hypercapnia can reduce ventilator stress but is unsuitable in some raised intracranial pressure, severe pulmonary vascular, or acid-base conditions.

Ventilator interaction and liberation

Agitation or poor synchrony may reflect pain, fear, delirium, tube discomfort, secretions, bronchospasm, acidosis, inappropriate settings, or neurological injury. Treat cause before escalating sedation. Analgesia-first and light sedation reduce ventilation duration and delirium for many patients. Daily goals should include wakefulness, mobility, and readiness to breathe independently.

Liberation requires improving cause, adequate oxygenation on modest support, haemodynamic stability, respiratory drive, manageable secretions, cough, and airway protection. A spontaneous-breathing trial tests integrated reserve but successful breathing does not guarantee safe extubation if upper-airway obstruction or poor secretion handling remains. After extubation, selected high-risk patients benefit from high-flow or non-invasive support.

Shock and cardiovascular support

Shock is inadequate cellular perfusion from low volume, pump failure, vasodilation, obstruction, or combinations. Assess pulse pressure, skin, mentation, urine, lactate trend, bedside ultrasound, passive leg response, and disease context. Static central venous pressure poorly predicts fluid responsiveness. A fluid challenge is a therapeutic test with defined dose, response, and stopping rule; unnecessary fluid causes oedema, impaired gas exchange, abdominal pressure, and delayed recovery.

Balanced crystalloids suit many resuscitations; blood products treat haemorrhage. Norepinephrine is common first-line vasopressor for distributive shock, raising vascular tone with less tachycardia than some alternatives. Add agents according to physiology. Inotropes increase contractility when low output persists despite adequate pressure and volume but can provoke arrhythmia and oxygen demand. Source control, antibiotics, reperfusion, drainage, anticoagulation, or surgery treats the mechanism.

Kidney and metabolic support

Acute kidney injury requires treatment of perfusion, infection, obstruction, toxins, and drug dosing. Diuretics manage fluid overload but do not restore injured nephrons. Renal replacement therapy treats refractory hyperkalaemia, acidosis, fluid overload, selected toxins, or uraemic complications rather than a creatinine threshold alone. Intermittent haemodialysis removes solute rapidly; continuous therapy permits slower control in unstable patients. Anticoagulation, catheter infection, bleeding, hypotension, electrolyte shift, and nutrient loss require monitoring.

Control glucose while avoiding hypoglycaemia; severe tight control is harmful. Feed enterally when the gut functions, generally starting early after stabilisation. Monitor phosphate, magnesium, potassium, and refeeding risk.

Stress-dose corticosteroids are used for defined adrenal insufficiency and selected refractory septic shock, not as universal critical-care therapy.

Neurological critical care

Protect airway, glucose, oxygenation, perfusion, temperature, and seizure control. Serial consciousness, pupils, motor responses, and sedation interruptions identify change. Raised intracranial pressure is worsened by hypoxia, hypotension, fever, seizure, venous obstruction, and extreme carbon dioxide. Elevate head, align neck, treat causes, use osmotic therapy selectively, and seek neurosurgical input. Brief hyperventilation is a rescue bridge for impending herniation, not routine therapy.

Status epilepticus requires rapid benzodiazepine followed by longer-acting antiseizure treatment and investigation of glucose, electrolytes, infection, toxins, withdrawal, and structural disease. Persistent coma may require electroencephalography for non-convulsive seizure. Brain death determination is a formal legal and clinical process requiring prerequisites, exclusion of confounders, and jurisdiction-specific testing.

Infection prevention and devices

Obtain cultures promptly when useful but do not delay necessary antibiotics. Reassess daily for narrowing, duration, source control, and non-infectious mimics. Use aseptic line insertion, suitable site and dressing, closed urinary drainage, ventilator care, oral hygiene, hand hygiene, and daily device removal review. Fever may result from drugs, thrombosis, transfusion, inflammation, or central injury as well as infection.

Pressure care, eye care, thrombosis prevention, stress-ulcer prophylaxis only when indicated, bowel management, sleep support, and early mobilisation prevent secondary harm. Physical, occupational, speech, pharmacy, nutrition, nursing, and medical expertise should converge on shared daily goals.

Delirium, communication, and long-term outcomes

Delirium is acute fluctuating inattention and cognition promoted by illness, drugs, sleep loss, immobilisation, pain, sensory deprivation, and unfamiliarity. Treat causes, orient, support hearing and vision, normalise day-night cues, mobilise, involve family, and minimise benzodiazepines and anticholinergics. Antipsychotics do not routinely shorten delirium but may be used briefly for dangerous distress after reversible causes are addressed.

Communicate prognosis with uncertainty and distinguish survival from likely function. Review advance directives, substitute decision-makers, and treatment goals early. Time-limited trials specify intervention, hoped-for outcome, review point, and response if goals are not met. Palliative care can accompany organ support.

Survivors may develop weakness, neuropathy, cognitive impairment, distress, pain, sleep disorder, and family strain. Minimise deep sedation, immobility, hypoglycaemia, and delirium; provide rehabilitation, medication reconciliation, follow-up, and explanation.

TTS module 2: Oxygen transport, ventilator mechanics, haemodynamic support, and intensive-care recovery

Critical care temporarily substitutes for failing physiology while definitive treatment and recovery occur. Every support has a cost: oxygen can injure lung, positive pressure can reduce cardiac output, vasopressors can impair peripheral flow, dialysis can destabilise circulation, sedation can prolong ventilation, and invasive devices cause infection and thrombosis. The central discipline is to define the physiological target, choose the least harmful effective support, measure response, and remove support as soon as reserve returns.

Systemic oxygen delivery equals cardiac output multiplied by arterial oxygen content. Arterial content is dominated by haemoglobin concentration times saturation; dissolved oxygen contributes little at ordinary pressures. A saturation of one hundred percent cannot compensate for profound anaemia or very low cardiac output. Tissue oxygen consumption depends on delivery and extraction until a critical threshold is crossed. Venous saturation and

lactate provide indirect clues but are altered by shunting, adrenergic metabolism, liver clearance, seizures, and mitochondrial dysfunction.

Oxygen is prescribed to a target appropriate to the patient. Pulse oximetry estimates saturation but is affected by poor perfusion, motion, dyshemoglobins, pigmentation-related device bias, and probe placement. Supplemental oxygen can conceal hypoventilation because saturation remains high while carbon dioxide rises. Arterial blood gas measures tensions and acid-base state, while co-oximetry detects carboxyhaemoglobin and methaemoglobin. Persistent hypoxaemia prompts analysis of low inspired oxygen, hypoventilation, diffusion limitation, ventilation–perfusion mismatch, or shunt.

High-flow nasal oxygen delivers heated humidified gas at flows that meet inspiratory demand, washes upper-airway dead space, reduces work, and creates modest variable distending pressure. Continuous positive airway pressure primarily recruits alveoli and reduces left-ventricular afterload; bilevel support adds inspiratory assistance and improves ventilation. Non-invasive ventilation is particularly effective in cardiogenic pulmonary oedema and selected hypercapnic obstructive exacerbations, but success requires cooperation, secretion control, mask fit, and rapid improvement.

Failure of non-invasive support is recognised early. Rising respiratory rate, worsening acidosis, falling consciousness, poor secretion handling, shock, persistent hypoxaemia, or exhaustion should trigger intubation rather than prolonged mask escalation. Vomiting, facial trauma, upper-airway obstruction, untreated pneumothorax, and immediate arrest risk are major limitations. The apparent comfort of avoiding intubation can become harmful if delayed invasive support occurs after physiological collapse.

Intubation changes cardiovascular physiology abruptly. Induction reduces sympathetic tone and vascular resistance; neuromuscular blockade removes respiratory muscle pump; positive intrathoracic pressure reduces right-heart venous return and can increase pulmonary vascular resistance.

Hypovolaemia, right-ventricular failure, tamponade, tension pneumothorax, severe acidosis, and pulmonary hypertension increase peri-intubation arrest risk. Preoxygenate, optimise pressure and volume appropriately, prepare vasopressor, select drugs, minimise apnoea, and confirm tracheal placement with sustained waveform carbon dioxide.

Mechanical ventilation controls pressure, volume, flow, time, and oxygen. Tidal volume and respiratory rate determine minute ventilation, but effective alveolar ventilation subtracts dead space. Carbon dioxide rises when alveolar ventilation falls or production increases. Oxygenation responds to inspired oxygen, end-expiratory pressure, recruitment, perfusion, and shunt. A mode name does not ensure safety; inspect delivered tidal volume, pressures, waveforms, synchrony, gas exchange, and haemodynamics after every change.

Airway pressure has components. Peak pressure reflects resistive plus elastic load, while plateau pressure approximates alveolar pressure during no flow. A rising peak with stable plateau suggests increased airway resistance from secretions, kink, biting, bronchospasm, or flow. Rising peak and plateau suggest reduced compliance from oedema, acute respiratory distress syndrome, pneumothorax, atelectasis, abdominal pressure, or position. Driving pressure is plateau minus total end-expiratory pressure and reflects tidal strain relative to available aerated lung.

Ventilator-induced lung injury results from overdistension, excessive pressure, repeated opening and closure, high oxygen, and inflammatory amplification. Low tidal volume is based on predicted body weight from height because aerated lung size does not scale with adipose mass. Limit plateau and driving pressures, use adequate end-expiratory pressure without overdistending lung or obstructing venous return, and accept selected carbon-dioxide elevation when safer. Sudden deterioration requires immediate disconnection and manual assessment when equipment or tension pneumothorax is possible.

Acute respiratory distress syndrome is inflammatory permeability oedema with bilateral opacities and hypoxaemia not explained solely by heart failure. Lung becomes heterogeneous: dependent regions collapse while small remaining regions receive most tidal volume. Prone positioning redistributes stress, improves dorsal recruitment and perfusion matching, and reduces mortality in severe disease when applied early and for sufficient duration. Conservative fluid after shock resolves reduces lung water. Neuromuscular blockade and extracorporeal membrane oxygenation are reserved for selected refractory cases.

Patient–ventilator dyssynchrony is a sign, not a sedation indication by itself. Pain, anxiety, delirium, acidosis, fever, bronchospasm, secretions,

auto-positive end-expiratory pressure, weak effort, and poorly matched trigger or cycling can cause conflict. Waveforms reveal missed triggers, double triggering, flow starvation, and ineffective exhalation. Correct the physiological or setting problem before deepening sedation. Analgesia-first light sedation, daily wakefulness goals, sleep, communication, and early mobility shorten many courses and reduce delirium.

Liberation tests integrated respiratory and cardiovascular reserve. Improving cause, manageable oxygen requirement, haemodynamic stability, spontaneous drive, strength, cough, secretions, and airway protection are required. A spontaneous-breathing trial can fail through respiratory load, diaphragm weakness, cardiac dysfunction, fluid overload, anxiety, or metabolic demand. Passing it does not guarantee extubation success if laryngeal oedema, poor consciousness, or secretion burden remains. High-risk patients may transition to high-flow or non-invasive support.

Shock is inadequate tissue perfusion from reduced volume, pump failure, vasodilation, obstruction, or combinations. Blood pressure is a means, not the final target. Assess mentation, skin gradient, capillary refill, urine, lactate trajectory, venous congestion, ultrasound, and response. A fluid challenge is a measured intervention with a defined stroke-volume or perfusion endpoint and stopping rule for oedema. Fluid responsiveness does not automatically mean fluid is beneficial when the circulation is adequate or lungs and right heart cannot tolerate it.

Norepinephrine restores vascular tone in distributive shock and may be started early through a suitable monitored peripheral vein while definitive access is considered. Vasopressin can reduce catecholamine requirement; epinephrine adds inotropy but raises arrhythmia and lactate. Inotropes are used when low cardiac output persists despite adequate preload and pressure. In right-heart failure, excessive fluid dilates the ventricle, shifts the septum, reduces left filling, and worsens coronary perfusion; treat afterload, rhythm, oxygenation, and cause.

Acute kidney injury alters volume, acid, potassium, uraemic toxin, and drug homeostasis. Creatinine lags injury and is diluted by fluid; urine output provides earlier but nonspecific evidence. Correct perfusion, sepsis, obstruction, nephrotoxins, and dose. Diuretics treat congestion but do not regenerate nephrons. Renal replacement begins for refractory hyperkalaemia,

acidosis, fluid overload, selected poison, or uraemic complication rather than a number alone. Continuous therapy permits slower shifts in unstable or brain-injured patients but requires anticoagulation and sustained access.

Nutrition begins enterally after initial stabilisation when the gut functions, with refeeding-risk assessment and measured protein and energy delivery. Overfeeding raises carbon dioxide and liver fat; underfeeding accelerates muscle loss. Glucose is controlled while avoiding hypoglycaemia. Phosphate, magnesium, and potassium affect diaphragm, heart, and weaning. Critical illness rapidly weakens muscle through inflammation, denervation, immobility, corticosteroid and neuromuscular exposure, and inadequate loading. Early mobilisation is organ support.

Neurological critical care protects brain from secondary injury. Maintain oxygenation and pressure, avoid fever and seizure, control carbon dioxide within the relevant intracranial physiology, and preserve venous drainage. Sedation interruption enables examination when safe. A declining pupil or motor response requires immediate herniation assessment. Hyperosmolar therapy and brief hyperventilation are bridges to definitive drainage or surgery. Persistent coma after convulsion may be non-convulsive status and requires electroencephalography.

Intensive-care devices must justify their continued risk daily. Central lines cause bloodstream infection, thrombosis, and mechanical injury; urinary catheters cause infection and immobility; arterial lines cause ischaemia and blood loss; endotracheal tubes cause pneumonia and laryngeal injury. Aseptic insertion, closed systems, hub and mouth care, pressure prevention, thrombosis prophylaxis, eye protection, bowel care, and prompt removal prevent secondary disease. Routine blood draws themselves contribute to anaemia.

Critical-care outcomes include survival, cognition, muscle, sleep, mood, pain, employment, and family wellbeing. Delirium is reduced through light sedation, orientation, hearing and vision support, normal day–night cues, mobilisation, analgesia, and removal of anticholinergic and benzodiazepine burden. Time-limited trials of organ support specify the intervention, hoped-for physiological and functional outcome, review date, and action if recovery does not occur. Palliative care can coexist with ventilation or dialysis and improves symptom and decision quality.

Recovery planning starts before discharge from intensive care. Explain the illness and frightening memories, reconcile medicines, document new organ impairment, screen swallowing and mobility, and arrange rehabilitation and psychological support. Families may have traumatic stress and caregiver burden. The goal of critical care is not merely to normalise monitored numbers; it is to return enough integrated reserve for the patient to live an outcome they consider worthwhile.

Retrieval prompts

One. What determines systemic oxygen delivery?

Two. When should non-invasive respiratory support be abandoned?

Three. Which principles reduce ventilator-induced lung injury?

Four. What makes a fluid challenge interpretable?

Five. Which indications justify renal replacement therapy?

Six. How can intensive-care delirium be reduced?

Concise answers

One. Cardiac output multiplied by arterial oxygen content, which depends mainly on haemoglobin and saturation.

Two. With worsening oxygenation, ventilation, effort, acidosis, consciousness, circulation, or airway protection.

Three. Low tidal volume based on predicted body weight, limited plateau and driving pressure, appropriate end-expiratory pressure, and restrained oxygen.

Four. A defined volume, physiological target, observation period, and stopping rule for harm.

Five. Refractory hyperkalaemia, acidosis, fluid overload, selected toxins, or symptomatic uraemia.

Six. Treat causes, use light sedation, support sleep and senses, orient, mobilise, manage pain, and remove unnecessary drugs and devices.

Source map

Original synthesis informed by Guyton and Hall, oxygen delivery, ventilation, circulation, renal regulation, cerebral perfusion, and shock; Robbins, acute lung injury, sepsis, organ injury, and critical-illness pathology; Katzung and OpenStax Pharmacology, sedatives, paralytics, vasoactive drugs, antimicrobials, diuretics, and anticoagulation; Talley and O'Connor, airway, respiratory, cardiovascular, neurological, perfusion, and consciousness assessment; OpenStax Medical-Surgical Nursing, ventilation, intensive monitoring, devices, renal support, delirium, mobility, and family care; and OpenStax Anatomy and Physiology, Chemistry, Biology, and Microbiology, gas exchange, acid-base chemistry, cellular injury, and infection.

Chapter 63: Fever, Collapse, Dizziness, Weakness, and Undifferentiated Acute Illness

Orientation

Undifferentiated symptoms are not diagnoses. Fever, collapse, dizziness, and weakness can arise from benign self-limited illness or time-critical failure of circulation, respiration, brain, metabolism, or host defence. The clinician must stabilise first, define the symptom precisely, identify dangerous patterns, build a mechanism-based differential, and use repeated examination and response to treatment to revise probability.

Universal first assessment

Observe before questioning. Assess airway sound and protection, respiratory rate and effort, oxygenation, pulse, pressure, perfusion, temperature, consciousness, pupils, movement, glucose, bleeding, rash, pain, and environmental context. Obtain an electrocardiogram early with collapse, dizziness, chest symptoms, palpitations, older age, cardiac disease, or electrolyte risk. Pregnancy testing can alter almost every acute differential.

Treat immediate hypoglycaemia, hypoxia, anaphylaxis, seizure, haemorrhage, shock, dangerous dysrhythmia, opioid respiratory depression, severe temperature disturbance, and suspected sepsis while investigation proceeds. A normal observation set after transient illness does not prove safety; establish what happened before arrival, recovery completeness, baseline function, and recurrence risk.

Fever and temperature

Fever is a regulated rise in hypothalamic set point driven by inflammatory mediators; hyperthermia is unregulated heat accumulation and does not respond to antipyretics. Confirm measurement site and method. Chills suggest rapid temperature rise but not necessarily bacteraemia. Older, frail, pregnant, immune-suppressed, or medication-treated patients may have serious infection without high fever.

Ask onset, pattern, rigors, localising symptoms, contacts, travel, occupation, animals, devices, procedures, medicines, immune status, vaccination, substance use, and prior antimicrobials. Examine skin, mouth, sinuses, neck, lungs, heart, abdomen, flanks, joints, spine, wounds, lines, genital or pelvic sites when indicated, and neurological function. Avoid indiscriminate cultures from colonised sites.

Sepsis is life-threatening organ dysfunction from dysregulated response to infection. Suspect it with altered consciousness, hypotension, tachypnoea, hypoxaemia, oliguria, mottling, lactate rise, coagulopathy, or organ injury. Obtain suitable cultures without delaying antibiotics, measure severity, restore perfusion, give oxygen when needed, and pursue source control. Antimicrobial choice follows likely site, organisms, resistance, allergy, organ function, and local protocol.

Non-infectious fever includes inflammation, malignancy, thrombosis, tissue infarction, transfusion, endocrine crisis, drug reaction, and central neurological injury. Drug fever may lack rash or eosinophilia. Hyperthermia syndromes include heat stroke, serotonin toxicity, neuroleptic malignant syndrome, antimuscarinic poisoning, stimulant toxicity, thyroid storm, and malignant hyperthermia. Rapid cooling and mechanism-specific treatment are urgent.

Persistent fever without diagnosis requires reassessment of history, exposures, hidden infection, inflammatory disease, cancer, and medication rather than automatic broadening of antibiotics. Time course and repeated targeted testing outperform enormous one-time panels.

Collapse and transient loss of consciousness

Clarify whether collapse meant loss of consciousness, loss of posture, weakness, fall, seizure, dizziness, or sleep. Syncope is transient global cerebral hypoperfusion with rapid onset, brief duration, and spontaneous complete recovery. Presyncope has similar warning without full loss. Obtain patient and witness accounts: posture, activity, prodrome, colour, breathing, movement, duration, injury, tongue bite, continence, confusion, chest pain, palpitations, exertion, and family sudden death.

Reflex syncope follows pain, fear, heat, prolonged standing, or visceral triggers and often has nausea, warmth, sweating, and visual dimming. Orthostatic syncope follows inadequate vascular compensation to standing from volume loss, medicines, autonomic failure, or deconditioning. Measure pressure and pulse after supine rest and standing while ensuring safety. A pressure drop can be asymptomatic and an initially normal test does not exclude intermittent disease.

Cardiac syncope arises from tachyarrhythmia, bradyarrhythmia, outflow obstruction, pulmonary embolism, ischaemia, or pump failure. Exertional or supine collapse, sudden event without prodrome, palpitations, abnormal electrocardiogram, structural disease, family sudden death, or injury raises risk. Monitoring duration should match event frequency; a normal short tracing does not exclude rare dysrhythmia. Echocardiography follows suspected structural disease.

Seizure is suggested by stereotyped aura, tonic then clonic movement, lateral tongue injury, prolonged post-event confusion, or focal deficit, but brief jerks occur in syncope. Psychogenic non-epileptic events are genuine functional episodes and require positive neurological diagnosis, not dismissal. Hypoglycaemia, intoxication, withdrawal, head injury, stroke, infection, and metabolic disturbance can mimic or provoke seizure.

Collapse without loss of consciousness may reflect fall, cataplexy, vestibular disease, transient ischaemia, weakness, or mechanical instability. Older adults often have multiple contributors including vision, feet, cognition, urgency, medicines, pressure, and environment. Assess injury and safeguarding as well as cause.

Dizziness and vertigo

Words such as dizzy, light-headed, spinning, floating, and off-balance are inconsistent. Classify by timing and triggers: episodic triggered, spontaneous episodic, or continuous acute. Determine whether symptoms reflect vertigo, presyncope, disequilibrium, oscillopsia, or nonspecific distress. Ask hearing change, tinnitus, headache, neurological symptoms, neck pain, chest symptoms, medicines, fluid loss, anxiety, and migraine.

Brief position-triggered vertigo suggests benign paroxysmal positional vertigo and is assessed with positional testing. Spontaneous episodes suggest vestibular migraine, Meniere disease, dysrhythmia, or panic according to associated features. Continuous vertigo with nausea and gait difficulty may be vestibular neuritis or posterior-circulation stroke.

In continuous acute vestibular syndrome with spontaneous nystagmus, trained head-impulse, nystagmus, and skew testing can identify central patterns, but it is invalid for brief episodic symptoms and unsafe when performed without expertise. New severe headache, focal neurology, direction-changing or vertical nystagmus, inability to sit or stand, new deafness, vascular risk, or neck trauma increases stroke concern. Early computed tomography poorly excludes posterior fossa ischaemia; magnetic resonance can also be falsely negative early.

Generalised weakness and fatigue

Weakness means reduced force; fatigue means difficulty sustaining activity or low energy; sleepiness means propensity to sleep. Ask onset, progression, proximal or distal distribution, symmetry, fluctuation, sensory symptoms, pain, swallowing, voice, breathing, autonomic symptoms, fever, weight, mood, sleep, medicines, toxins, and function. Sudden focal weakness is stroke until assessed. Ascending weakness, respiratory symptoms, bulbar dysfunction, or rapidly progressive paralysis requires emergency monitoring.

True weakness can arise from upper motor neuron, anterior horn cell, peripheral nerve, neuromuscular junction, muscle, electrolyte, endocrine, inflammatory, infectious, toxic, or critical-illness disease. Examination localises using bulk, tone, power, reflexes, plantar response, sensation,

coordination, cranial nerves, gait, and fatigability. Pain and poor comprehension can mimic weakness.

Guillain-Barre syndrome commonly causes progressive symmetrical weakness and reduced reflexes after infection, with autonomic instability and respiratory failure. Measure forced vital capacity serially; oxygen saturation falls late in pure ventilatory weakness. Myasthenia gravis causes fluctuating ocular, bulbar, neck, or limb weakness with preserved sensation; infection and certain drugs can precipitate crisis. Botulism causes descending paralysis with autonomic and pupil features. Hypokalaemia, hyperkalaemia, hypophosphataemia, and magnesium disorders can cause paralysis and dysrhythmia.

Fatigue without objective weakness is common in anaemia, infection, endocrine disease, heart or lung disease, kidney or liver failure, cancer, sleep disorder, medicines, pain, depression, anxiety, undernutrition, and post-infectious syndromes. Select tests from history rather than ordering every possible marker. Validate disability while avoiding false certainty about a single unproven mechanism.

Targeted investigations

Baseline testing may include blood count, electrolytes, kidney and liver function, glucose, calcium, magnesium, phosphate, inflammatory markers, electrocardiogram, urinalysis, and pregnancy testing according to syndrome. Add troponin, blood gas, lactate, cultures, creatine kinase, thyroid tests, cortisol, toxicology, imaging, lumbar puncture, electroencephalography, or cardiac monitoring only when they answer a clinical question.

Interpret results using pretest probability. Mild abnormalities are common and may distract from the cause. A normal troponin at the wrong time, a negative early scan, or one normal observation cannot close a dangerous differential. Conversely, indiscriminate testing creates incidental findings, radiation, false alarms, and cascades. Document why the patient is safe for observation, admission, or discharge.

Treatment, observation, and safety net

Treat the physiological threat and likely cause while preserving diagnostic information when safe. Replace fluid when depleted but avoid reflex boluses in heart or kidney failure. Review and hold contributing medicines deliberately, with restart criteria. Mobilise under supervision when testing posture or gait. Provide analgesia and antiemetics without obscuring serial neurological or abdominal assessment.

Observation is an active diagnostic strategy with specified duration, repeat observations, examination, testing, oral intake, mobility, and escalation thresholds. Before discharge, confirm recovery toward baseline, safe function, supervision and transport, medicine plan, follow-up ownership, and access to emergency care. Explain uncertainty and give specific return triggers such as recurrent collapse, chest pain, palpitations, breathlessness, persistent fever, confusion, severe headache, new weakness, bleeding, or inability to drink.

TTS module 2: Syndrome clarification, high-risk collapse, dynamic observation, and safe uncertainty

Undifferentiated illness becomes tractable when vague labels are translated into failed functions and a timeline. “Collapse” may mean syncope, seizure, fall, loss of tone, sudden weakness, intoxication, or sleep. “Dizziness” may mean motion illusion, impending faint, imbalance, visual instability, or dissociation. “Weakness” may mean reduced muscle force, fatigue, pain-limited movement, dyspnoea, or reduced initiative. Clarification should occur alongside stabilisation because airway, breathing, circulation, glucose, temperature, bleeding, seizure, anaphylaxis, and focal neurological deficits cannot wait.

The first observations include appearance, speech, posture, work of breathing, skin colour and moisture, spontaneous movement, interaction, and environmental clues before instruments are attached. Then measure pulse, pressure, oxygen saturation, respiratory rate, temperature, glucose, consciousness, pupils, perfusion, and rhythm. Supplemental oxygen, prehospital fluid, antipyretic, naloxone, or spontaneous recovery may normalise numbers without removing the cause. Record the lowest or most abnormal pre-intervention state whenever reliable.

Fever is a cytokine-driven increase in hypothalamic set point. Vasoconstriction and shivering generate the subjective chill while core temperature rises; vasodilation and sweating occur when the set point falls. Hyperthermia is uncontrolled heat production or impaired dissipation without a raised set point and will not respond to antipyretics. Heat stroke, serotonin toxicity, neuroleptic malignant syndrome, malignant hyperthermia, stimulant poisoning, antimuscarinic toxicity, and thyroid storm require rapid cooling and mechanism-specific treatment rather than waiting for infection tests.

Serious infection may present without fever in older, frail, neutropenic, pregnant, immunosuppressed, or antipyretic-treated patients. Conversely, fever does not prove infection. Examine devices, skin folds, mouth, lungs, heart, abdomen, flanks, joints, spine, wounds, and neurological state according to risk. Blood cultures are useful in sepsis, endocarditis, and selected focal syndromes but contaminated or low-volume cultures create harm. Empirical antibiotics are proportional to infection probability and consequence of delay, with daily diagnostic reassessment.

Syncope is abrupt transient loss of consciousness and postural tone from global cerebral hypoperfusion with rapid spontaneous recovery. Establish whether consciousness was truly lost and obtain witness detail: posture, trigger, prodrome, pallor or cyanosis, breathing, eye position, movements, duration, injury, and recovery. Nausea, warmth, sweating, and visual dimming before prolonged standing support reflex syncope. Sudden unheralded collapse during exertion or while supine, palpitations, chest pain, abnormal electrocardiogram, structural heart disease, or family sudden death raises cardiac risk.

Brief tonic or myoclonic movements can accompany syncope and do not prove epilepsy. Seizure is more likely with stereotyped aura, sustained tonic then clonic activity, lateral tongue injury, prolonged confusion, focal postictal weakness, or events from sleep. Urinary incontinence is nonspecific. Hypoglycaemia, arrhythmia, convulsive syncope, psychogenic non-epileptic seizures, sleep disorders, and intoxication are common mimics. A normal interictal electroencephalogram does not exclude epilepsy, and a positive functional diagnosis should be based on characteristic signs rather than exclusion alone.

Orthostatic physiology is measured after sufficient supine rest and during standing with symptoms. A fall in pressure accompanied by an appropriate tachycardia suggests volume loss or venous pooling; a disproportionately small heart-rate response suggests autonomic failure or rate-limiting medication. An asymptomatic numerical fall may not explain collapse, while a normal brief measurement can miss delayed hypotension. Meals, heat, alcohol, deconditioning, neuropathy, Parkinsonian disease, vasodilators, diuretics, and impaired thirst modify risk.

The electrocardiogram after collapse searches for conduction disease, pre-excitation, repolarisation syndromes, long or short corrected QT, ischaemia, arrhythmia, hypertrophy, and pulmonary strain. A normal tracing cannot exclude intermittent dysrhythmia. Monitoring duration must match event frequency: telemetry suits frequent inpatient risk, ambulatory patches suit days to weeks, and implantable loop recorders can capture rare events. Echocardiography answers suspected structural or functional heart disease rather than being routine for every faint.

Dizziness is classified by timing and triggers. Recurrent seconds-long episodes triggered by rolling or looking up suggest benign positional vertigo. Spontaneous episodes lasting minutes or hours can reflect vestibular migraine, Meniere disease, panic, arrhythmia, or transient ischaemia. Continuous vertigo with nausea, nystagmus, and gait difficulty is an acute vestibular syndrome caused by neuritis or posterior-circulation stroke. "Triggered by movement" must mean movement initiates an otherwise absent episode; symptoms merely worsened by movement occur in both central and peripheral disease.

Central vestibular concern rises with inability to sit or stand, severe new headache or neck pain, diplopia, dysarthria, weakness, numbness, new deafness, direction-changing or vertical nystagmus, skew, abnormal pursuit, or high vascular risk. The head-impulse, nystagmus, and skew examination is validated only in continuous acute vestibular syndrome and trained hands. Computed tomography poorly excludes posterior-fossa ischaemia and early magnetic resonance can be negative. Repeated neurological examination and gait assessment remain essential.

True weakness is demonstrated by reduced force after accounting for pain, comprehension, effort, fatigue, and mechanical limitation. Localise through

distribution, tone, reflexes, sensation, cranial nerves, coordination, and fatigability. Upper motor-neuron disease combines pyramidal pattern, brisk reflexes, and pathological plantar response. Lower motor-neuron disease causes reduced tone, atrophy, fasciculation, and depressed reflexes. Neuromuscular-junction disease fluctuates without sensory loss, while muscle disease tends to be proximal with preserved sensation.

Rapid ascending weakness, bulbar dysfunction, weak cough, neck flexion loss, autonomic instability, or breathlessness requires respiratory monitoring. Pulse oximetry stays normal in pure ventilatory weakness until late; serial forced vital capacity, inspiratory force, carbon dioxide, voice, swallowing, and secretion clearance provide earlier warning. Guillain–Barré syndrome can deteriorate suddenly and causes pressure and rhythm instability. Myasthenic crisis often follows infection or medication and may present with ocular, bulbar, or respiratory failure before dramatic limb weakness.

Electrolyte and endocrine disorders can imitate neurological disease. Severe hypo- or hyperkalaemia produces paralysis and dysrhythmia; low phosphate weakens diaphragm; magnesium disorders alter reflexes, conduction, and potassium correction. Adrenal crisis causes weakness, vomiting, hypotension, sodium and potassium change, and hypoglycaemia and is treated before definitive cortisol results. Thyroid disease, hypercalcaemia, ketoacidosis, uraemia, liver failure, and toxic exposure produce fatigue or altered consciousness. Medication review includes sedatives, antihypertensives, diuretics, insulin, opioids, anticholinergics, anticonvulsants, and recent withdrawal.

Fatigue is real but not synonymous with muscle disease. Clarify exertional intolerance, sleepiness, post-exertional worsening, breathlessness, mood, cognition, pain, sleep quality, weight, fever, and functional pattern. Anaemia, heart or lung disease, sleep apnoea, infection, cancer, endocrine disease, kidney or liver failure, malnutrition, depression, and post-infectious syndromes are selected by context. Exhaustive untargeted panels generate incidental abnormalities and can entrench an unsupported explanation. A negative investigation does not invalidate disability.

Observation is a planned diagnostic intervention. Specify duration, repeat vital signs, rhythm monitoring, neurological or abdominal examinations, orthostatic or walking challenge, oral intake, urine, laboratory trend, and escalation

thresholds. A fluid trial tests a volume hypothesis; symptom improvement alone may be nonspecific. Fever settling after antipyretic does not distinguish bacterial from viral illness. No recurrent event during a short stay does not exclude rare arrhythmia. Interpretation requires prediction in advance.

Disposition integrates residual probability, physiological reserve, function, supervision, distance from help, and reliability of follow-up. A normal test cannot make an unsafe home environment safe. Before discharge, confirm recovery toward baseline, ability to eat, drink, walk, void, and manage medication, plus injury review after collapse. Assign pending-result ownership and provide specific return triggers: recurrent faint, chest pain, palpitations, breathlessness, persistent fever, confusion, severe headache, new weakness, bleeding, reduced urine, or inability to hydrate.

The final synthesis should state what happened, the most likely mechanism, dangerous alternatives still possible, evidence changing their probability, and why the chosen monitoring or discharge plan is proportionate. Diagnostic uncertainty is not failure when it is calibrated and safety-netted. The unsafe alternative is false precision: converting fever, dizziness, collapse, or weakness into a premature label that stops observation before physiology declares itself.

Driving, bathing, swimming, heights, machinery, and solitary activity may need temporary restriction until recurrent collapse or seizure risk has been adequately assessed.

Retrieval prompts

One. How does fever differ from hyperthermia?

Two. What defines syncope?

Three. Which collapse features suggest cardiac risk?

Four. How should dizziness be classified?

Five. Which weakness patterns require emergency respiratory monitoring?

Six. What makes observation an active diagnostic process?

Concise answers

One. Fever raises the regulated hypothalamic set point; hyperthermia is uncontrolled heat accumulation.

Two. Brief transient loss of consciousness from global cerebral hypoperfusion with rapid onset and spontaneous complete recovery.

Three. Exertional or supine onset, absent prodrome, palpitations, abnormal electrocardiogram, structural disease, or family sudden death.

Four. By timing and triggers into episodic triggered, spontaneous episodic, or continuous acute syndromes.

Five. Rapidly progressive, ascending, bulbar, neck, or breathing weakness and autonomic instability.

Six. Defined repeat observations, examinations, tests, functional challenges, time frame, and escalation thresholds.

Source map

Original synthesis informed by Guyton and Hall, thermoregulation, cerebral perfusion, posture, vestibular function, neuromuscular transmission, and fatigue; Robbins, infection, inflammation, infarction, neurological and muscle pathology; Katzung and OpenStax Pharmacology, antimicrobials, vasoactive drugs, vestibular medicines, toxins, and medicine adverse effects; Talley and O'Connor, fever, collapse, dizziness, weakness, neurological, cardiovascular, and gait assessment; OpenStax Medical-Surgical Nursing, rapid assessment, sepsis, falls, observation, mobility, and discharge safety; and OpenStax Anatomy and Physiology, Biology, Chemistry, and Microbiology, homeostasis, metabolism, and infection.

Chapter 64: Common Laboratory Tests, Diagnostic Imaging, and Test Stewardship

Orientation

A test changes uncertainty but does not replace reasoning. Results reflect probability, variation, specimen quality, error, timing, and treatment. Testing may diagnose, stage, monitor, or exclude but also causes pain, anaemia, radiation, contrast injury, incidental findings, cascades, and delay. Order when results change action.

Probability and thresholds

State the target condition and pretest probability. Sensitivity is the proportion with disease testing positive; specificity is the proportion without disease testing negative. Predictive values vary with prevalence. Likelihood ratios convert pretest into post-test odds, though performance changes with population and threshold.

A sensitive negative test helps exclude disease in its validated context; a specific positive helps confirm. Shortcuts fail when timing, technique, spectrum, or cut-off differs. Continuous values carry more information than binary labels. Thresholds reflect consequences: dangerous treatable disease justifies action at lower probability.

Reference intervals commonly contain ninety-five per cent of a selected healthy population, so healthy people fall outside. Multiple testing increases chance abnormality. Distinguish statistical from clinically important change using baseline, trend, precision, variation, and management impact.

Specimen quality and pre-analytical error

Confirm identity, timing, preparation, tube, site, and transport. Fasting, posture, exercise, circadian rhythm, pregnancy, medicines, infusion, and tourniquet alter results. Haemolysis raises potassium and enzymes; delay changes glucose and gases; underfilled citrate tubes distort coagulation; contaminated cultures drive antibiotics.

For unexpected results, assess the patient and check units, previous values, specimen comments, and plausibility. Repeat urgently when error could harm, but do not dismiss critical results. Closed-loop communication identifies recipient, meaning, action, and owner.

Full blood count and film

Haemoglobin, haematocrit, red-cell indices, leukocyte count and differential, and platelets provide a map rather than a diagnosis. Mean cell volume classifies anaemia broadly, but mixed deficiencies can appear normal. Reticulocytes indicate marrow response after accounting for anaemia severity. High neutrophils may reflect infection, inflammation, steroids, stress, or marrow disease; low neutrophils raise infection risk according to depth and duration. Lymphocyte and eosinophil changes require age, medicines, infection, allergy, parasites, and malignancy context.

Platelet number does not measure function. Bleeding risk also depends on vessel, coagulation, drugs, kidney or liver disease, and procedure. The blood film can reveal abnormal shape, fragmentation, blasts, parasites, platelet clumping, and marrow stress. Severe anaemia, blasts, schistocytes with organ injury, or profound cytopenia demands prompt assessment.

Electrolytes, kidney function, and osmolality

Sodium mainly reflects water balance, not body sodium content. Correct interpretation requires volume state, glucose, osmolality, urine osmolality, and urine sodium in context. Correct slowly when chronic disturbance risks osmotic brain injury. Potassium is influenced by sampling, acid-base state, insulin, cell injury, kidney excretion, and drugs. Confirm unexpected hyperkalaemia but obtain an electrocardiogram and treat immediately when severe or accompanied by electrical change.

Creatinine reflects filtration imperfectly and lags acute injury; it varies with muscle, diet, secretion, age, and drugs. Estimated glomerular filtration is less reliable during rapidly changing function, pregnancy, extremes of muscle, and dialysis. Urea reflects filtration, protein breakdown, gastrointestinal bleeding, steroids, and volume. Urinalysis and urine microscopy often identify glomerular, tubular, infectious, or obstructive patterns that serum values cannot.

Measured serum osmolality compared with calculated osmolality produces an osmolar gap, which can suggest unmeasured alcohols but varies with timing and other solutes. The anion gap identifies unmeasured anions in metabolic acidosis but should be interpreted with albumin and baseline. Neither gap can independently exclude dangerous poisoning.

Liver, cardiac, inflammatory, and endocrine markers

Aminotransferases indicate hepatocyte injury; alkaline phosphatase and gamma-glutamyl transferase support cholestatic pattern; bilirubin reflects production, processing, and excretion; prothrombin time and contextualised albumin assess synthesis. Large enzyme elevation does not necessarily mean worse function than modest elevation in advanced failure.

Troponin indicates myocardial injury, not automatically infarction. Diagnose myocardial infarction from a rise or fall plus clinical evidence of ischaemia. Sepsis, tachyarrhythmia, heart failure, pulmonary embolism, kidney disease, and myocarditis also elevate troponin. Natriuretic peptides rise with myocardial wall stress and help evaluate heart failure but vary with age, rhythm, kidney function, body size, and treatment.

C-reactive protein and erythrocyte sedimentation rate indicate inflammation nonspecifically. Trends can support response but should not override deterioration or improvement. Procalcitonin may support selected antimicrobial decisions but is not a universal infection test. Autoantibodies require compatible phenotype because low-level positivity is common.

Thyroid-stimulating hormone and free thyroid hormone are interpreted together; pituitary disease, acute illness, pregnancy, and medicines alter patterns. Cortisol is strongly time and context dependent; suspected adrenal crisis is treated before definitive testing. Glycated haemoglobin estimates

prior glycaemia but is distorted by altered red-cell survival, haemoglobin variants, transfusion, pregnancy, and kidney disease.

Coagulation and transfusion testing

Prothrombin time reflects selected clotting pathways and is used for vitamin-K antagonist monitoring; activated partial thromboplastin time assesses other pathways and some heparin effects. Normal tests do not exclude platelet dysfunction, von Willebrand disease, factor thirteen deficiency, or direct anticoagulant effect. Abnormal tests can result from deficiency, inhibitor, liver disease, anticoagulant, consumption, or specimen error. Mixing studies and factor assays follow the clinical question.

Blood group, antibody screen, and crossmatch reduce incompatible transfusion risk. Emergency uncrossmatched blood may be life-saving when haemorrhage outweighs compatibility delay. Transfusion decisions use symptoms, bleeding, physiology, comorbidity, and alternatives rather than one threshold. Monitor for acute haemolysis, allergic reaction, lung injury, circulatory overload, sepsis, and delayed antibodies.

Microbiology and molecular testing

Collect specimens from the actual infected site before antimicrobials when safe. Blood cultures need adequate volume and separate collections; contamination is reduced by skin antisepsis. A positive culture may represent pathogen, colonisation, or contaminant. Interpret organism, number of sets, time to positivity, host, device, and syndrome. Susceptibility results guide narrowing but laboratory sensitivity does not guarantee penetration or source control.

Polymerase chain reaction detects nucleic acid and may remain positive after viable organisms disappear. Antigen tests are often faster but less sensitive. Serology reflects immune response and may be negative early or confounded by vaccination, prior infection, or immune suppression. Cycle thresholds and multiplex detections require cautious clinical interpretation. Do not test asymptomatic colonised sites unless the result has a defined purpose.

Plain imaging, ultrasound, computed tomography, and magnetic resonance

Plain radiography evaluates chest patterns, fractures, alignment, foreign bodies, and selected obstruction, but overlap hides disease. A normal chest film does not exclude early infection, embolism, or small pneumothorax. Compare priors and know view and position.

Ultrasound has no ionising radiation and dynamically assesses gallbladder, pelvis, vessels, heart, pleura, kidneys, soft tissue, and procedures. It is operator dependent and limited by gas, depth, and windows. Point-of-care scanning should answer focused questions.

Computed tomography rapidly supplies cross-sectional anatomy for trauma, stroke, chest, abdomen, vessels, and cancer. Radiation accumulates, especially in children and pregnancy, but should not delay necessary life-saving imaging. Iodinated contrast improves diagnosis; assess benefit, kidney context, hydration, prior reaction, and alternatives.

Magnetic resonance provides neural, soft-tissue, marrow, cardiac, pelvic, and biliary contrast without ionising radiation but is slower and constrained by movement, monitoring, claustrophobia, and implants. Verify device compatibility and use gadolinium cautiously in severe kidney failure or pregnancy. Do not withhold necessary ionising imaging in pregnancy.

Nuclear and interventional imaging

Radiotracers map perfusion, metabolism, bone, thyroid, kidney, biliary, infection, and cancer physiology; inflammation can mimic cancer and low-metabolic tumours may be missed. Interventional imaging enables biopsy, drainage, embolisation, thrombectomy, ablation, and stenting but still requires consent, coagulation, infection, contrast, sedation, and recovery planning. Biopsy routes must preserve later cancer surgery.

Stewardship and incidental findings

For every test, ask: what diagnosis or severity question is being addressed, how will each plausible result change care, what is the right timing, and what harm could follow? Avoid repetitive daily panels without indication. Stop monitoring when it no longer guides treatment. Choose the least harmful test that answers the question, but do not substitute an inadequate low-risk test for a necessary definitive one.

Screening differs from diagnosis: it tests asymptomatic populations where prevalence is low, so false positives and overdiagnosis matter greatly. Incidental findings should be described by risk, patient context, and recommended follow-up rather than ignored or catastrophised. Assign ownership for pending and incidental results at every transition. A test without a responsible reviewer is unfinished care.

TTS module 2: Measurement uncertainty, longitudinal interpretation, imaging choice, and result ownership

A diagnostic test is an intervention in a probability model. It has value when its result changes treatment, monitoring, prognosis, or the need for another test. It has harm through pain, blood loss, radiation, contrast, false reassurance, false alarms, incidental findings, cost, and delay. Before ordering, state the target condition, pretest probability, relevant threshold, expected timing, and what each plausible result will make the team do differently.

Sensitivity and specificity are properties measured in a defined population at a defined threshold. Positive and negative predictive values change with prevalence and individual pretest probability. A positive result in a low-risk population may be mostly false positives even when specificity appears high. Likelihood ratios formally update odds, but published performance may not transport to early disease, older adults, pregnancy, immunosuppression, or a different assay. No slogan such as “sensitive tests rule out” overrides validation context.

Continuous values contain more information than binary flags. A potassium of six point eight is not equivalent to five point two despite both being high; a troponin just above a limit differs from a large dynamic rise. Reference intervals usually contain the central ninety-five percent of selected healthy

people, placing about one in twenty healthy results outside by design. Ordering twenty independent tests makes at least one flagged result likely. Clinical significance depends on effect size, biological and analytical variation, baseline, trajectory, and consequence.

Pre-analytical error occurs before the analyser sees the specimen. Wrong identity, wrong tube, underfilling, prolonged tourniquet, exercise, posture, fasting, circadian timing, contamination from an infusion, delayed transport, temperature, and haemolysis all alter results. Haemolysis releases potassium and intracellular enzymes. An underfilled citrate tube contains excessive anticoagulant and prolongs clotting times. Blood gas changes when air enters or cellular metabolism continues. Unexpected critical values require immediate patient assessment and rapid confirmation when plausible error exists, not automatic dismissal.

Analytical variation reflects instrument precision, calibration, interference, and method. Post-analytical error occurs when units, reference ranges, transcription, inbox routing, or interpretation fails. A result is unfinished care until a named clinician reviews it, communicates it when necessary, and acts. Electronic delivery alone is not closed loop. At discharge and transfer, every pending culture, biopsy, image, send-away test, and incidental follow-up needs explicit ownership, expected availability, and action threshold.

The full blood count is interpreted across lineages and indices. Mean cell volume can appear normal when iron deficiency and B-twelve deficiency coexist. Reticulocytes show marrow response only after correction for anaemia severity and maturation time. Neutrophilia can reflect infection, inflammation, corticosteroid demargination, stress, smoking, or neoplasia; neutropenia risk depends on absolute count, duration, barrier injury, and immune state. Platelet number does not measure adhesion or aggregation. A blood film can reveal schistocytes, blasts, parasites, dysplasia, spherocytes, target cells, platelet clumping, and marrow stress that automated counts cannot explain.

Sodium concentration primarily expresses water relative to exchangeable sodium and potassium, not total body sodium. Interpret measured or effective osmolality, glucose, volume state, urine osmolality, urine sodium, kidney function, endocrine context, and drugs. Acute severe hyponatraemia threatens cerebral oedema and may require hypertonic saline; chronic

correction that is too rapid risks osmotic demyelination. Hypernatraemia reflects water deficit or sodium gain and requires calculation and controlled replacement while ongoing losses are addressed.

Potassium is governed by total stores, intracellular distribution, kidney excretion, acid-base state, insulin, beta signalling, tissue breakdown, and medications. Pseudohyperkalaemia from haemolysis, fist clenching, thrombocytosis, or leukocytosis should be recognised, but electrical instability is treated immediately while confirmation proceeds. Electrocardiographic progression is variable and a normal tracing does not make severe hyperkalaemia safe. Magnesium deficiency makes potassium refractory to replacement and increases dysrhythmia risk.

Creatinine is a delayed imperfect filtration marker. Production varies with muscle mass, diet, illness, and amputation; tubular secretion and drugs alter it; fluid expansion dilutes it. Estimated filtration assumes steady state and becomes unreliable during evolving acute kidney injury, pregnancy, dialysis, or extremes of body composition. Cystatin C has different confounders and may help in selected cases. Urinalysis, sediment, protein quantification, urine output, ultrasound, and drug exposure often explain mechanism more effectively than serum creatinine alone.

Liver chemistry separates injury, cholestasis, and function. Aminotransferases indicate cell injury, alkaline phosphatase with corroborating hepatic markers indicates cholestatic tendency, bilirubin reflects production and excretion, and prothrombin time changes with synthesis, vitamin K, anticoagulants, and consumption. Albumin is slow and nonspecific. Falling aminotransferases during worsening acute liver failure may mean loss of viable cells, not recovery. Pattern and trajectory are more useful than the phrase “abnormal liver function tests.”

Troponin proves myocardial injury. Myocardial infarction requires a rise or fall with evidence of ischaemia from symptoms, electrocardiogram, imaging, or coronary findings. Sepsis, pulmonary embolism, rapid rhythm, myocarditis, heart failure, kidney disease, and critical illness also injure myocardium. High-sensitivity assays detect smaller injury and require time-specific algorithms. Natriuretic peptides reflect wall stress but rise with age, atrial fibrillation, kidney failure, pulmonary pressure, and sepsis and can be lower in obesity.

Endocrine tests are feedback-system measurements. Thyroid-stimulating hormone and free hormone are interpreted together, but acute illness, pregnancy, pituitary disease, assay interference, biotin, and drugs distort patterns. Cortisol depends on circadian time, stress, binding proteins, and exogenous steroid; suspected adrenal crisis is treated before testing. Glycated haemoglobin reflects glycation over red-cell lifespan and becomes misleading after bleeding, haemolysis, transfusion, iron deficiency, pregnancy, kidney disease, or haemoglobin variants. No endocrine number is independent of physiological context.

Coagulation screens sample limited pathways. Prothrombin time and activated partial thromboplastin time can be normal in platelet dysfunction, von Willebrand disease, factor-thirteen deficiency, and some direct-anticoagulant effects. They can be prolonged by factor deficiency, inhibitor, heparin, vitamin-K antagonism, liver failure, consumption, or specimen error. A mixing study asks whether normal plasma corrects deficiency or an inhibitor persists, but time-dependent inhibitors complicate interpretation. Bleeding history and procedure matter more than indiscriminate screening.

Microbiology results require site and viability reasoning. A high-quality deep specimen collected before antibiotics can establish cause and susceptibility. A superficial swab may report colonisers. Blood-culture significance depends on organism, number of bottles and sets, time to positivity, devices, host, and syndrome. Nucleic-acid amplification can detect dead organisms, carriage, or contamination as well as disease. Serology measures host response and may be negative early or after immune suppression. Susceptibility in vitro assumes a validated dose and does not guarantee penetration or source control.

Plain radiography is a projection with overlapping structures and limited sensitivity. Its value depends on view, position, inspiration, exposure, and comparison. Ultrasound provides dynamic, radiation-free bedside information but is operator- and window-dependent; a focused point-of-care study should not be misrepresented as a comprehensive formal examination. Computed tomography offers rapid high-resolution cross-sectional anatomy and angiography but uses ionising radiation and often iodinated contrast. Radiation risk is cumulative yet should not delay a necessary life-saving scan.

Iodinated contrast can cause immediate hypersensitivity-like reactions and is associated with kidney risk that depends strongly on baseline function, acute illness, volume, dose, and alternative diagnosis. A previous unrelated allergy does not automatically prohibit contrast. Premedication does not eliminate reactions. Gadolinium-enhanced magnetic resonance provides tissue and vascular information but requires device compatibility and kidney-context review. Magnetic resonance is slower, sensitive to movement, difficult during instability, and not inherently the best test simply because it lacks radiation.

Imaging choice is anatomical and temporal. Non-contrast computed tomography rapidly detects acute intracranial blood; vascular imaging detects occlusion; magnetic resonance diffusion detects early infarction. Ultrasound is first-line for gallstones and many pelvic questions. Computed-tomographic angiography is urgent for mesenteric ischaemia and selected bleeding. Endoscopy sees mucosa and treats lesions that cross-sectional imaging may miss. Choose the least harmful test that adequately answers the question, not a safer test that cannot.

Incidental findings create a second diagnostic problem. Estimate whether the abnormality is common, whether morphology and patient risk make it dangerous, and what validated follow-up is recommended. Avoid catastrophic language for low-risk lesions, but never bury action-critical findings. Overdiagnosis occurs when screening detects disease that would never cause symptoms or death, exposing people to labels and treatment without benefit. Screening therefore demands stronger evidence than testing a symptomatic high-risk patient.

Test stewardship is longitudinal. Stop daily blood panels when results no longer guide care, reduce duplicate imaging, batch necessary phlebotomy, and monitor only drug levels with an interpretable sampling time. Repeating a test can be valuable when trajectory is the signal, as with troponin, haemoglobin, sodium, cultures, or tumour response. The final discipline is to document why the test was ordered, how it altered probability, what action followed, and who owns the next result. Measurement becomes medicine only when it closes that loop.

Retrieval prompts

One. Why does predictive value change between populations?

- Two. What does a reference interval represent?
- Three. Why can creatinine underestimate early acute kidney injury?
- Four. What does an elevated troponin prove?
- Five. Which limitations affect molecular pathogen testing?
- Six. What questions define good test stewardship?

Concise answers

- One. Predictive value depends on pretest disease prevalence as well as test performance.
- Two. Usually the central ninety-five per cent of results in a selected healthy reference group.
- Three. It accumulates after filtration falls and is influenced by muscle, volume, secretion, and drugs.
- Four. Myocardial injury; infarction additionally requires dynamic change and clinical evidence of ischaemia.
- Five. Timing, residual nucleic acid, colonisation, sampling, immune suppression, multiplex false discovery, and imperfect correlation with viable pathogen.
- Six. Define the question, expected decisions for each result, timing, accuracy, harms, alternatives, and ownership.

Source map

Original synthesis informed by Guyton and Hall, physiological measurement, blood, kidney, liver, endocrine, cardiac, and gas principles; Robbins, laboratory-pathology correlations and tissue diagnosis; Katzung and OpenStax Pharmacology, therapeutic monitoring, interactions, contrast, coagulation, and toxicology testing; Talley and O'Connor, test selection and interpretation across clinical systems; OpenStax Medical-Surgical Nursing, specimen collection, imaging preparation, transfusion, procedures, and result follow-up; and OpenStax Chemistry, Biology, and Microbiology, analytical chemistry, probability, molecular assays, and culture.

Chapter 65: Epidemiology, Screening, Vaccination, Prevention, and Population Health

Orientation

Population health asks why disease occurs unevenly and how to prevent suffering before clinical care. Treatment remains essential, but housing, income, education, food, work, discrimination, environment, commercial influence, and access determine illness and recovery. Prevention combines evidence with fairness, feasibility, trust, and attention to harm.

Measures of disease

Incidence counts new events among people at risk over time and reflects transition into disease. Prevalence counts existing disease at a point or period and depends on both incidence and duration. A chronic survivable condition may have high prevalence despite modest incidence. Mortality counts death; case fatality is the proportion of diagnosed cases who die. Survival after diagnosis can improve through earlier detection without changing time of death, creating lead-time bias.

Risk is probability over a defined period. Rate incorporates person-time and permits unequal follow-up. Absolute risk difference expresses additional or prevented events; relative risk expresses a ratio. A large relative effect on a rare outcome may produce a small absolute benefit. Number needed to treat is the reciprocal of absolute risk reduction over the study period, while number needed to harm uses absolute risk increase. Always state time horizon and baseline risk.

Age, sex, and other population structures distort crude comparisons. Stratification or standardisation permits fairer comparison but depends on chosen reference. Ecological data describe groups and cannot establish that group-level relationships apply to individuals. Surveillance data also reflect who is tested, diagnosed, reported, and able to access care.

Prevention levels

Primordial prevention prevents emergence of risk through safe housing, clean air, education, food policy, active transport, and regulation. Primary prevention prevents disease before onset through vaccination, tobacco control, contraception, sun protection, blood-pressure management, and occupational controls. Secondary prevention detects early disease through screening and prompt treatment. Tertiary prevention limits complication and disability after disease. Quaternary prevention protects people from unnecessary medical intervention.

Upstream interventions often reach more people than counselling. Safe water prevents infection without repeated individual choice. Taxes, labelling, engineering, and marketing restrictions change exposure but should avoid regressive burdens. Clinical prevention uses shared decisions and support rather than blame.

Screening principles

Screening tests asymptomatic people to reduce future morbidity or mortality, not simply to find abnormalities. A worthwhile programme addresses an important condition with a detectable preclinical phase, acceptable accurate testing, effective earlier treatment, defined pathway, quality assurance, equitable access, and favourable balance of benefit, harm, and cost.

Low disease prevalence means false positives can outnumber true positives even with a good test. Harms include anxiety, invasive follow-up, complications, radiation, false reassurance, overdiagnosis of disease that would never cause symptoms, and overtreatment. Length bias preferentially detects slow disease; lead-time bias makes survival appear longer merely because diagnosis starts earlier. Randomised mortality or meaningful morbidity outcomes are stronger evidence than stage shift alone.

Sensitivity and specificity may change with threshold. Screening often chooses sensitivity to avoid missed disease, then uses a specific confirmatory test. Interval and stopping age depend on disease natural history, prior results, life expectancy, treatment fitness, and preferences. A programme that improves average outcomes can widen inequity if access is hardest for those at greatest risk.

Explain purpose, outcomes, false results, downstream procedures, overdiagnosis, and choice to decline. Programmes need registers, recalls, standards, result ownership, follow-up, audit, and data protection. Opportunistic testing without follow-up is not a programme.

Vaccination and immune protection

Vaccines expose antigen without full disease, generating memory B and T cells. Platforms include live, inactivated, protein, polysaccharide, conjugate, vector, and nucleic-acid vaccines. Conjugation recruits T-cell help against polysaccharides; adjuvants enhance innate signalling; boosters restore or broaden waning protection.

Effectiveness depends on age, comorbidity, immune suppression, pathogen variation, cold chain, timing, and outcome. Vaccines may prevent severe disease better than infection. Herd effects protect others by interrupting spread, but thresholds vary with transmissibility, clustering, waning, and performance.

Common reactions include local pain, fever, and fatigue. Serious events are rare and require surveillance. Events after vaccination are not necessarily caused by it; compare background rates, timing, mechanism, and controlled evidence. Distinguish contraindications from precautions. Mild fever, family allergy, or antibiotic use rarely justifies permanent avoidance.

Live vaccines can replicate and are generally avoided in severe immune suppression and pregnancy, with exceptions governed by specialist guidance. Inactivated vaccines cannot cause the target infection but may produce weaker responses during immune suppression. Give indicated vaccines before immune-suppressive treatment when feasible. Premature infants are usually vaccinated by chronological age; pregnancy vaccination protects parent and infant against selected diseases.

Maintain routine schedules across life, including influenza, coronavirus, tetanus, pertussis, pneumococcus, shingles, human papillomavirus, and travel or occupational vaccines according to current local guidance. Missed doses usually continue rather than restart a series. Record product, batch, date, site, consent, and adverse events, and report defined events to surveillance systems.

Cardiovascular and metabolic prevention

Risk factors act together, so estimate absolute cardiovascular risk rather than treating each number in isolation. Smoking, pressure, lipids, diabetes, kidney disease, age, family history, activity, diet, sleep, and social conditions matter. Lifestyle support should be specific and feasible: tobacco treatment, affordable dietary change, physical activity including resistance work, sleep evaluation, and reduced harmful alcohol.

Blood-pressure measurement requires correct cuff, rest, repeated readings, and often home or ambulatory confirmation. Lipid-lowering benefit relates to baseline risk and magnitude of low-density lipoprotein reduction. Statins reduce vascular events but can cause muscle symptoms, liver-enzyme change, interactions, and a small diabetes risk; apparent symptoms should be evaluated without prematurely abandoning prevention. Antiplatelet therapy is valuable after established atherosclerotic disease but bleeding often outweighs benefit in low-risk primary prevention.

Diabetes prevention combines weight and activity support, sleep, medicines for selected high-risk people, and treatment of cardiovascular risk. Obesity care should avoid stigma and preserve muscle, nutrition, and mental health. Prevention goals should be adjusted for frailty and limited life expectancy when treatment burden exceeds delayed benefit.

Cancer prevention and early detection

Tobacco control prevents multiple cancers and cardiorespiratory disease. Human papillomavirus and hepatitis B vaccination prevent virus-driven cancer. Sun protection reduces ultraviolet injury. Alcohol reduction, healthy weight, activity, occupational control, and treatment of chronic infections reduce risk. Chemoprevention is appropriate only when individual risk justifies adverse effects.

Screening may include cervical, breast, colorectal, and lung programmes in defined populations, while prostate testing remains preference-sensitive. Screening is not appropriate merely because a test exists; thyroid, ovarian, pancreatic, and whole-body imaging can generate substantial overdiagnosis in average-risk asymptomatic people. High-risk genetic syndromes require different surveillance and counselling.

Infectious-disease prevention

Transmission is interrupted through clean water, sanitation, ventilation, vaccination, hand hygiene, respiratory precautions, safer sex, sterile injection, food safety, vector control, isolation, and treatment that reduces infectiousness. Match precautions to contact, droplet, airborne, blood, vector, food, or environmental route. Personal protective equipment is one layer and fails when systems, training, ventilation, staffing, or supply are poor.

Antimicrobial resistance is driven by exposure in human health, animals, agriculture, and environment. Prevention includes vaccination, infection control, rapid diagnostics, narrow effective therapy, correct dose and duration, source control, and avoidance of antibiotics for viral illness or colonisation. One Health coordination recognises shared ecosystems.

Tobacco, alcohol, and substance prevention

Tobacco dependence responds to behavioural support plus nicotine replacement, varenicline, or bupropion according to contraindications and preference. Combined long-acting and short-acting nicotine improves control. Relapse reflects chronic dependence and prompts renewed support, not blame.

Alcohol risk varies by dose, pattern, health, pregnancy, medicines, and activity. Brief intervention helps risky use; dependence may require supervised withdrawal and relapse-prevention medicines. Naloxone, sterile equipment, safer-use education, and opioid agonist treatment reduce death and infection when abstinence is not chosen.

Health equity and implementation

Equal care gives the same resource; equitable care responds to barriers and need. Analyse outcomes by social groups without treating identity as biology. Racism, ableism, stigma, colonisation, rurality, language, violence, and digital exclusion shape exposure and access. Community co-design improves relevance and trust.

Implementation defines target population, intervention, workforce, supply, financing, data, harms, and who may be missed. Process measures matter when linked to outcomes. Evaluate reach, uptake, fidelity, benefit, harm, cost,

and distribution. Prevention succeeds when healthier choices become practical and ordinary.

TTS module 2: Population measurement, screening trade-offs, vaccine strategy, and equitable prevention

Population health moves the unit of attention from one patient to distributions of exposure, disease, access, and outcome. Individual treatment remains essential, but clean air, safe work, housing, water, food systems, education, transport, income, and regulation determine how many people become patients. Prevention therefore asks not only whether an intervention works, but who can reach it, who bears its burden, whether benefits are durable, and whether the programme narrows or widens avoidable inequality.

Incidence is the occurrence of new events among people at risk. Cumulative incidence is a probability over a defined interval, whereas incidence rate divides events by person-time and accommodates different follow-up. Prevalence is the proportion living with a condition and rises with either higher incidence or longer duration. Successful treatment can increase prevalence by extending survival without increasing new disease. Mortality rate describes deaths in a population; case fatality describes death among identified cases and changes when diagnosis captures milder disease.

Comparisons require compatible denominators and time. Crude rates can differ because populations have different age or sex structures rather than different underlying risk. Direct standardisation applies stratum-specific rates to a common reference population; indirect methods compare observed events with those expected from reference rates. Standardised estimates improve fairness but are not the actual local burden and vary with the chosen standard. Always present counts and population context alongside adjusted measures where practical.

Relative risk compares probabilities and can look impressive when events are rare. Absolute risk difference states how many events are actually prevented or caused. Number needed to treat is the reciprocal of absolute reduction and inherits its population, comparator, adherence, and time horizon. If baseline risk changes, number needed changes even when relative effect is stable. Prevention discussions should use natural frequencies with a common denominator and include adverse outcomes, not only a relative percentage.

Screening is offered to people without symptoms to improve future patient-important outcomes. It is not justified merely because disease can be found earlier. The condition needs a detectable phase in which earlier action improves mortality, morbidity, function, or quality of life; the test and confirmation pathway must be acceptable and accurate; treatment must be available; and the entire programme must deliver net benefit. An excellent assay attached to poor follow-up is not an effective screening programme.

Lead-time bias makes survival from diagnosis appear longer when detection moves earlier but death is unchanged. Length bias preferentially finds slower lesions because they remain detectable longer. Overdiagnosis identifies genuine pathology that would never cause symptoms during the person's lifetime; it cannot be corrected by better confirmatory accuracy because the abnormality is real. Overtreatment follows when such disease is treated. Randomised reduction in disease-specific or all-cause harm is stronger evidence than improved stage or five-year survival alone.

Low prevalence magnifies false positives. Even high specificity applied to millions of low-risk people can generate more false than true alarms, leading to imaging, biopsy, procedure complications, anxiety, and labels. Raising the threshold improves specificity but misses disease. The optimal threshold depends on consequences of delayed diagnosis versus unnecessary follow-up, not mathematics alone. Screening interval and stopping age reflect natural history, prior results, competing mortality, fitness for treatment, and preferences.

Screening can widen inequity when uptake is easiest among people already well served while high-risk groups face transport, language, cost, disability, mistrust, or leave barriers. Equity requires outreach, interpreters, accessible facilities, culturally safe co-design, reminder and navigation systems, and affordable confirmatory care. Uptake alone is not success if abnormal results remain unresolved. Programmes track invitation, participation, positivity, confirmation, treatment, interval cancers, complications, overdiagnosis, and outcomes by population group.

Vaccination creates memory without the full disease burden. Live attenuated organisms replicate and generate broad responses but may be unsafe in severe cellular immunodeficiency and pregnancy. Inactivated, subunit, conjugate, vector, and nucleic-acid platforms cannot cause the target infection

through replication but differ in adjuvant need, dose schedule, and immune profile. Conjugating a polysaccharide to protein recruits T-cell help, enabling affinity maturation and memory in young children who respond poorly to plain polysaccharide.

Vaccine efficacy is measured under study conditions; effectiveness includes delivery, cold chain, uptake, host variation, pathogen evolution, and real-world exposure. Protection may be stronger against severe disease than infection or transmission. Waning is outcome-specific and can be restored by boosters. Indirect protection depends on transmissibility, network clustering, immune escape, duration, and how well vaccination reduces infectiousness. A single universal herd threshold is rarely exact in a heterogeneous population.

An event after vaccination is temporally associated, not necessarily caused. Background illness, coincidental timing, stimulated reporting, and diagnostic awareness can create apparent clusters. Safety assessment compares observed with expected rates, examines biologically plausible windows and mechanisms, uses controlled and self-controlled designs, and investigates product or batch signals. Rare genuine harms matter and should be communicated transparently in absolute terms beside the much larger disease risk when that is the evidence.

Contraindications are narrower than common myths. Mild illness, antibiotic use, breastfeeding, or family history usually does not prevent routine vaccination. Severe immediate allergy to a component and selected live-vaccine immune or pregnancy contexts require specialist guidance. Immunosuppressed patients may respond weakly and should receive indicated non-live vaccines before treatment when possible. Household vaccination can reduce exposure, while live products with shedding implications are selected carefully. Catch-up schedules usually continue rather than restart previous valid doses.

Cardiovascular prevention is based on absolute risk because age, smoking, pressure, lipids, diabetes, kidney disease, family history, and social determinants interact. The same lipid reduction prevents more events in a high-risk person than a low-risk person. Blood pressure must be measured accurately and confirmed outside clinic when appropriate before long-term labels. Treatment combines tobacco therapy, affordable dietary change, activity and resistance work, sleep assessment, weight-neutral or

weight-directed metabolic care, and medication whose expected absolute benefit exceeds burden.

Primary prevention can become overtreatment when delayed benefit is unlikely within life expectancy or adverse effects threaten current function. Frailty, falls, polypharmacy, kidney function, bleeding, and patient goals modify pressure, glucose, lipid, and antiplatelet strategies. Secondary prevention after established vascular disease usually carries greater absolute benefit. A therapy should not be stopped solely because of chronological age, nor continued automatically when its original evidence no longer matches the person's prognosis.

Cancer prevention includes tobacco and alcohol reduction, healthy weight and activity, ultraviolet protection, occupational control, vaccination against oncogenic infections, and treatment of selected chronic infections. Screening differs by organ and risk. Cervical, colorectal, breast, and selected lung programmes have evidence in defined groups, while broad ovarian tumour markers, pancreatic screening, thyroid ultrasound, and whole-body imaging cause substantial false positives and overdiagnosis in average-risk asymptomatic populations. Genetic high-risk syndromes require separate surveillance and counselling.

Infectious prevention interrupts the actual route: water and sanitation for faecal–oral disease, ventilation and respiratory protection for airborne exposure, sterile injection for blood transmission, condoms and treatment for sexual transmission, vector control for mosquito-borne disease, and vaccination where available. Antimicrobial resistance prevention begins with fewer infections through vaccination, device removal, hand hygiene, source control, and safer systems, then accurate diagnosis and the narrowest reliably effective regimen. Human, animal, food, and environmental antimicrobial use share one ecology.

Tobacco dependence responds to combined behavioural support and pharmacotherapy. Nicotine replacement separates nicotine from combustion; combining long-acting baseline delivery with short-acting rescue often improves control. Varenicline and bupropion suit selected patients after contraindication and interaction review. Relapse is expected in chronic dependence and should prompt another treatment attempt rather than blame. Harm reduction for opioids similarly includes naloxone, sterile equipment,

drug checking where available, and agonist therapy while preserving access to abstinence-based goals.

Social categories should not be treated as fixed biology. Racism, colonisation, poverty, disability exclusion, rurality, unsafe work, gendered violence, language, and commercial marketing shape exposure and access through mechanisms that can be changed. Equality gives the same intervention; equity modifies resources and design so people with greater barriers can achieve comparable benefit. Population averages can conceal worsening outcomes in a minority, so evaluation must examine distribution.

Implementation specifies target population, intervention, workforce, supply chain, funding, data, consent, harms, accountability, and exit strategy. Process measures such as invitations or doses matter only when linked to reach and outcomes. Evaluate adoption, fidelity, effectiveness, unintended consequences, cost, and sustainability. The best prevention often makes the safer option ordinary through engineering and policy while still supporting informed individual choice. Population medicine succeeds when fewer people develop disease and that benefit reaches those carrying the greatest preventable burden.

Retrieval prompts

One. How do incidence and prevalence differ?

Two. Why should relative effects be paired with absolute effects?

Three. Which biases make screening appear more effective than it is?

Four. How do live and inactivated vaccines differ clinically?

Five. Why use absolute cardiovascular risk?

Six. What distinguishes equality from equity?

Concise answers

One. Incidence counts new events among those at risk; prevalence counts existing disease and depends on incidence and duration.

Two. Baseline risk determines how many actual events a relative reduction prevents.

Three. Lead-time, length, selection, overdiagnosis, and healthy-participant effects.

Four. Live vaccines replicate and may be unsafe in pregnancy or severe immune suppression; inactivated vaccines cannot cause target infection but may be less immunogenic.

Five. It integrates interacting factors and estimates the likely absolute benefit of intervention.

Six. Equality supplies the same resources; equity adjusts resources and design to overcome unequal needs and barriers.

Source map

Original synthesis informed by Guyton and Hall, physiological risk and adaptation; Robbins, environmental, infectious, neoplastic, metabolic, and vascular disease causation; Katzung and OpenStax Pharmacology, vaccines, tobacco and alcohol therapy, cardiometabolic prevention, and chemoprevention; Talley and O'Connor, preventive, family, occupational, and social history; OpenStax Medical-Surgical Nursing, screening, vaccination, community health, education, and chronic-disease prevention; and OpenStax Biology, Chemistry, and Microbiology, evolution, immunity, ecology, transmission, and population measurement.

Chapter 66: Evidence-Based Medicine, Causation, Bias, and Interpreting Clinical Research

Orientation

Evidence-based medicine integrates the best available research with clinical expertise, patient values, biology, resources, and context. It does not mean obeying a hierarchy without judgement. Research estimates effects under uncertainty; clinicians decide whether those estimates are valid, important, applicable, and aligned with a particular person's goals. Good interpretation begins with a focused question, examines methods before results, and distinguishes absence of evidence from evidence of no effect.

Framing the question

For treatment, define population, intervention, comparator, and outcomes. For diagnosis, define target condition, test, reference standard, and clinical setting. For prognosis, specify starting point, follow-up, and outcome. For harm, define exposure, latency, and competing explanations. A vague question encourages selective evidence and irrelevant endpoints.

Prioritise outcomes that patients experience: survival, symptoms, function, quality of life, treatment burden, and adverse effects. Biomarkers and composite outcomes can increase statistical efficiency but may mislead if weakly linked to benefit or driven by less important components. Surrogate validation requires evidence that changing the surrogate reliably changes the clinical outcome across relevant mechanisms.

Study designs

Randomised controlled trials allocate intervention by chance, balancing known and unknown prognostic factors on average. Concealed allocation prevents prediction before enrolment; blinding reduces differential care, reporting, and assessment. Intention-to-treat analysis preserves randomisation by analysing assigned groups, estimating the effect of a treatment strategy under adherence patterns. Per-protocol analysis estimates effects among adherent participants but loses randomisation and is vulnerable to selection.

Crossover trials suit stable conditions without carryover. Cluster trials randomise groups and must account for within-cluster similarity. Factorial trials test multiple interventions. Pragmatic trials emphasise routine-care effectiveness; explanatory trials test efficacy under controlled conditions.

Cohorts follow exposed and unexposed people. Case-control studies begin with outcome and compare prior exposure, efficiently studying rare disease but vulnerable to selection and recall. Cross-sectional studies cannot easily establish temporal order. Ecological group data risk individual inference. Case reports generate signals, not frequency or causation.

Systematic reviews explicitly search, select, appraise, and synthesise. Meta-analysis combines compatible studies, increasing precision but not correcting biased inputs. Heterogeneity reflects populations, interventions, outcomes, methods, or chance. Pooling incomparable studies can precisely estimate the wrong construct.

Random error and estimation

A point estimate is the best single estimate from a sample; its confidence interval expresses precision under assumptions. A ninety-five per cent confidence interval from repeated identical studies would contain the true parameter in ninety-five per cent of intervals. It is not the probability that this one interval contains truth, though that shorthand is common.

A p value is the probability of results at least as extreme as observed if the null model and assumptions were true. It is not the probability that the null hypothesis is true, the chance results occurred randomly, or the size and

importance of effect. Statistical significance depends on sample size. Large studies detect trivial differences; small studies can miss important effects.

Examine absolute effect, relative effect, confidence interval, baseline risk, outcome importance, harms, and duration. Number needed to treat is not a fixed property of a drug because it varies with baseline risk and follow-up. Non-significance does not establish equivalence. Equivalence and non-inferiority trials require prespecified margins, adequate adherence, and careful analysis because bias toward no difference can falsely support success.

Bias within studies

Selection bias occurs when inclusion or retention creates groups that differ in prognosis. Allocation concealment, representative recruitment, high follow-up, and accounting for missing data reduce it. Performance bias arises when groups receive different co-interventions or attention. Detection bias occurs when outcome measurement differs. Attrition bias follows missing outcomes related to treatment and prognosis. Selective-reporting bias hides outcomes or analyses based on results.

Measurement error can be random, reducing precision, or systematic, shifting estimates. Misclassification may affect groups equally or differently. Recall bias particularly affects retrospective exposure histories. Interviewer expectations can shape questioning. Standard definitions, validated instruments, blinding, training, and repeated measurement improve reliability.

Confounding occurs when a third factor causes or predicts exposure and independently affects outcome, creating a non-causal association. Restriction, matching, stratification, regression, propensity methods, and instrumental variables address measured confounding but cannot guarantee removal of unmeasured confounding. Adjusting for a variable caused by exposure, a mediator, or a collider can introduce bias.

Causation

Association is more plausibly causal when exposure precedes outcome, findings are consistent, dose-response and coherent mechanism exist, alternatives are reduced, and intervention changes outcome. No checklist proves causation. Strong associations can be confounded and plausibility evolves.

Randomisation best supports intervention causality, but trials may be impossible, unethical, short, or underpowered for rare harm. Triangulation across designs with different biases strengthens inference. Natural experiments and time series help when assumptions are explicit.

Correlation between biomarker change and outcome does not prove targeting that marker helps. Reverse causation occurs when early disease changes apparent risk. Immortal-time bias grants exposed patients a period when outcome was impossible. Time-varying exposure requires appropriate methods.

Diagnostic evidence

Accuracy studies need representative patients, independent index and reference tests, and complete verification. Partial verification, incorporation, and review bias exaggerate performance. Sensitivity and specificity vary with spectrum and thresholds; predictive values vary with prevalence.

Receiver-operating-characteristic curves compare sensitivity and false-positive rate across thresholds. Area under the curve measures discrimination, not calibration, utility, or outcomes. A discriminating test adds little if management does not change.

Prediction models require internal and external validation. Discrimination separates risk; calibration compares predicted with observed risk. Performance degrades in new populations or after practice changes. Complexity does not guarantee accuracy or fairness.

Prognosis and survival analysis

Prognostic cohorts need defined stage, representative inception, adequate follow-up, objective outcomes, and competing-risk adjustment. Kaplan-Meier curves assume censored patients have comparable future risk. Hazard ratios compare instantaneous rates under proportional-hazard assumptions; they are not risk ratios.

Competing events prevent outcomes, such as death before fracture; ignoring them overestimates incidence. Median survival conceals tails and heterogeneity. Group prognosis should not become deterministic individual prediction.

Harms and pharmacovigilance

Trials commonly lack size or duration for rare, delayed, pregnancy-related, or interaction harms. Observational databases, spontaneous reports, registries, and case-control studies add evidence. Spontaneous reports detect signals but cannot determine incidence because reporting and denominator are unknown. Disproportionality is hypothesis-generating.

Assess temporality, dechallenge and rechallenge, dose, known mechanism, alternative causes, and consistency. Channeling bias occurs when high-risk patients preferentially receive one treatment. New-user designs and active comparators improve fairness. A safety signal requires investigation, not automatic dismissal or proof.

External validity and applicability

Ask whether the patient resembles participants in biology, severity, comorbidity, age, pregnancy, adherence, and baseline risk. Compare dose, expertise, monitoring, comparator, and system. Trial exclusion increases uncertainty rather than proving ineffectiveness. Absolute effects depend strongly on baseline risk.

Consider feasibility, cost, access, burden, environmental impact, and opportunity cost. Guidelines combine evidence with values and resource assumptions and may lag new research. Local protocols improve reliability but require justified exceptions for individual circumstances.

Subgroup claims are credible when prespecified, few, biologically plausible, supported by a statistically tested interaction, consistent, and independently replicated. Significance in one subgroup and non-significance in another does not itself prove a difference. Data-driven subgroup discovery commonly produces chance findings.

Publication and research integrity

Positive novel studies are more likely to appear. Registration, protocols, statistical plans, regulatory reports, and unpublished-data searches reduce selective visibility. Funnel asymmetry does not prove publication bias.

Many outcomes, time points, subgroups, and analyses increase false positives. Prespecification, correction, replication, and exploratory labels matter. Fabrication, manipulation, plagiarism, inappropriate authorship, and hidden conflicts undermine trust, but disclosed conflict does not automatically invalidate sound methods.

From evidence to decision

Summarise certainty across risk of bias, consistency, directness, precision, and publication bias. High certainty does not mean a large benefit; low certainty does not mean no benefit. Present expected outcomes in absolute natural frequencies using the same time frame and denominator. Include harms, inconvenience, uncertainty, and the option of no intervention.

Shared decision-making identifies what matters to the patient and whether likely benefit justifies burden. When evidence is weak but urgency is high, use physiological rationale, expert consensus, close monitoring, and a time-limited therapeutic trial. Document assumptions and stopping rules. Evidence-based practice is disciplined humility: act, measure, learn, and revise.

TTS module 2: Causal inference, effect interpretation, diagnostic evidence, and patient-level application

Evidence-based practice starts by separating four questions: is the estimate valid, what is the magnitude and uncertainty, does it apply here, and is the outcome worth the burden to this patient? A prestigious journal, randomised label, small p value, or guideline recommendation answers none alone. Methods determine whether groups are comparable and outcomes trustworthy; effect measures determine clinical size; external validity connects the study to context; shared decision-making connects it to values.

A focused treatment question defines population, intervention, comparator, and patient-important outcomes with a time horizon. Diagnostic questions define the index test, target condition, reference standard, and point in the care pathway. Prognostic questions require a clear starting state and follow-up. Harm questions need exposure timing, dose, latency, and competing causes. Vague questions invite selective searching and surrogate answers that do not resolve the actual decision.

Random allocation supports causal inference by making prognostic factors comparable on average, but sequence generation must be concealed before enrolment so recruiters cannot manipulate assignment. Blinding reduces differences in co-intervention, reporting, and outcome assessment, though objective outcomes are not automatically immune to bias. Loss to follow-up can destroy balance when related to prognosis and treatment. Intention-to-treat analysis preserves the effect of assignment; per-protocol analysis answers a different adherence-conditioned question and requires stronger assumptions.

Randomisation does not guarantee balance in a small trial, and a baseline significance test is not the solution. Inspect clinically important imbalances and whether adjusted analysis was prespecified. Cluster trials must account for correlation among people treated in the same clinic or community; failure produces confidence intervals that are too narrow. Crossover designs require stable disease, reversible effect, and negligible carryover. Factorial trials estimate interventions efficiently only when interaction and adherence are handled appropriately.

Observational studies are indispensable for prognosis, rare harm, long latency, and exposures that cannot be randomised. Confounding occurs when a pre-exposure variable influences both exposure and outcome. Restriction, matching, stratification, regression, propensity scores, and weighting adjust measured confounders but do not erase unmeasured ones. Channeling bias occurs when clinicians preferentially give one drug to sicker patients. New-user active-comparator designs improve fairness by aligning treatment initiation and clinical indication.

Causal diagrams help distinguish confounders from mediators and colliders. A mediator lies on the causal path; adjusting for it can remove part of the effect of interest. A collider is caused by two variables; conditioning on it can create a false association between them. Selection into hospital, testing, or study participation often acts as a collider. More covariates do not always mean less bias. Adjustment must follow a causal question, not automatic statistical availability.

Selection bias arises when inclusion, retention, or analysis depends jointly on exposure and prognosis. Recall bias occurs when cases reconstruct prior exposure differently from controls. Detection bias follows differential outcome measurement. Performance bias reflects unequal co-interventions. Misclassification can dilute or exaggerate effects depending on whether error differs by group. Missing data require reasons and sensitivity analysis; simply analysing complete cases assumes missingness unrelated to unobserved outcome after conditioning, often implausibly.

A relative risk expresses a ratio of event probabilities; an odds ratio approximates it only when events are uncommon and otherwise appears farther from one. A hazard ratio compares instantaneous event rates under a model and is neither a risk ratio nor a simple statement that events happen a percentage sooner. Absolute risk difference translates relative effect through baseline risk. Number needed to treat is one divided by absolute reduction, rounded appropriately, and must retain population, comparator, and time.

Confidence intervals express a range of estimates compatible with data and assumptions under repeated-sampling logic. Width reflects information and variability. A narrow interval around a trivial effect can be precise but unimportant; a wide interval crossing no effect may include meaningful benefit and harm. Failure to reject a null is not proof of equivalence. Non-inferiority

trials need a clinically justified margin, reliable active control, preserved conduct, and both intention-to-treat and per-protocol scrutiny because non-adherence biases toward apparent similarity.

A p value asks how surprising the observed or more extreme data would be under a specified null model. It does not give the probability the null is true, probability the result is “due to chance,” size of effect, reproducibility, or clinical importance. Multiple outcomes, repeated looks, flexible stopping, subgroups, and analytic choices inflate false positives. Prespecification, correction, replication, and transparent exploratory labelling limit this problem. A threshold of zero point zero five does not divide truth from falsehood.

Composite outcomes increase event count but can mislead when components differ in importance, frequency, or treatment effect. A composite of death, hospitalisation, and a laboratory change may be driven entirely by the least important component. Competing risks matter when one event prevents another; death prevents later fracture or dialysis, for example. Standard Kaplan–Meier methods can overestimate cumulative incidence of a nonfatal event when competing death is treated as ordinary censoring.

Surrogate outcomes are useful only when treatment effects on the surrogate reliably predict effects on how patients feel, function, or survive across relevant mechanisms. Lowering a biomarker may fail because the marker is not causal, because off-target harms offset benefit, or because timing differs. Correlation between biomarker and prognosis is insufficient validation. Disease-oriented improvement can coexist with worse mortality. Patient-important outcomes and adverse effects deserve priority even when they require larger, longer studies.

Diagnostic accuracy studies need a representative clinical spectrum, blinded interpretation, an acceptable reference standard, and verification independent of index-test result. Partial verification inflates accuracy when only positive patients receive the reference. Incorporation bias occurs when the index test forms part of the reference diagnosis. Sensitivity and specificity change with disease severity, threshold, and population; predictive values change with prevalence. A result useful in a referral centre may perform differently in primary care.

Receiver-operating-characteristic area measures discrimination across thresholds but not calibration, net benefit, or whether testing improves outcomes. A prediction model can rank high-risk above low-risk people yet systematically overpredict everyone. Calibration compares predicted with observed risk. Internal validation estimates optimism; external validation tests transport to another setting and time. Updating may be needed after treatment patterns or disease prevalence change. Algorithmic complexity and machine learning do not guarantee fairness or clinical utility.

Treatment-effect heterogeneity is credible when subgroups are prespecified, few, biologically plausible, measured reliably, supported by a formal interaction test, consistent across outcomes, and replicated. A significant effect in one subgroup and non-significant effect in another does not prove groups differ. Continuous modifiers should not be cut into arbitrary categories without justification. Data-driven subgroup discovery is exploratory and often fails replication because many comparisons produce chance extremes.

Systematic reviews reduce selective citation through explicit search, eligibility, appraisal, and synthesis, but meta-analysis cannot repair biased component studies. Heterogeneity may reflect population, dose, comparator, outcome definition, follow-up, or methods. A random-effects model acknowledges variation but does not make incompatible questions identical. Prediction intervals can show the range expected in a new setting. Publication bias, selective outcomes, duplicate reports, and inaccessible negative studies can make pooled precision illusory.

Evidence for harms often comes from several imperfect sources. Randomised trials may be too small or short for rare, delayed, pregnancy-related, or interaction effects. Registries and databases provide denominators but suffer confounding and coding error. Spontaneous reports identify unexpected signals but cannot estimate incidence because reporting and exposure counts are unknown. Causality is strengthened by temporality, dose response, dechallenge, coherent mechanism, recurrence on inadvertent rechallenge, and triangulation across methods, while deliberate rechallenge may be unethical.

External validity asks whether biology, severity, comorbidity, age, pregnancy, baseline risk, adherence, dose, expertise, monitoring, comparator, and health system resemble the patient's context. Exclusion from trials creates

uncertainty, not automatic contraindication. Relative effects may transport better than absolute effects, but not always. Feasibility, cost, travel, caregiver burden, monitoring, environmental impact, and opportunity cost are part of effectiveness. Guidelines encode evidence plus values and resource assumptions and may require justified individual deviation.

Certainty assessment separates confidence in evidence from magnitude of benefit. High-certainty evidence can show no meaningful effect; low certainty can surround a large possible benefit. Consider risk of bias, inconsistency, indirectness, imprecision, and publication bias, with upgrading logic for compelling observational evidence. Communicate effects as natural frequencies with the same denominator and period, showing outcomes with and without intervention, harms, burden, and uncertainty. Relative framing alone can manipulate perception.

When evidence is weak and urgency high, clinicians still act. Use biological rationale, indirect evidence, expert consensus, and the patient's risk tolerance, then design a time-limited therapeutic trial with target, timeframe, monitoring, and stopping rule. Record assumptions so response can update the model. Evidence-based medicine is not passive obedience to published averages; it is disciplined causal reasoning that exposes uncertainty, measures what matters, and adapts population estimates to one person without pretending certainty the research cannot provide.

Retrieval prompts

One. What four elements frame a treatment question?

Two. Why is intention-to-treat analysis important?

Three. What does a p value not tell us?

Four. How does confounding differ from mediation?

Five. Which features make subgroup findings credible?

Six. How should effects be communicated to patients?

Concise answers

One. Population, intervention, comparator, and patient-important outcomes.

Two. It preserves the prognostic balance created by random assignment and estimates the assigned strategy.

Three. It does not give the probability that the hypothesis is true, effect size, importance, or probability of random causation.

Four. A confounder precedes and influences exposure and outcome; a mediator lies on the causal path from exposure to outcome.

Five. Prespecification, few tests, plausible mechanism, significant interaction, consistency, and independent replication.

Six. Use absolute natural frequencies, a common time frame and denominator, and include harms, burden, uncertainty, and alternatives.

Source map

Original synthesis informed by OpenStax Biology and Chemistry, experimental design, probability, measurement, causation, and biological mechanisms; OpenStax Microbiology, diagnostic accuracy, surveillance, trials, and outbreak inference; Guyton and Hall and Robbins, mechanistic plausibility and clinicopathological outcomes; Katzung and OpenStax Pharmacology, clinical trials, dose-response, adverse effects, and pharmacovigilance; Talley and O'Connor, diagnostic probability and clinical application; and OpenStax Medical-Surgical Nursing, patient-important outcomes, implementation, shared decisions, and quality improvement.

Chapter 67: Ethics, Consent, Confidentiality, Safeguarding, and Medical Law Principles

Orientation

Clinical ethics addresses interactions among values, duties, evidence, rights, and consequences. Law sets jurisdiction-specific minimum standards; ethics often asks more. Practice protects autonomy, welfare, fairness, trust, and integrity amid uncertainty and unequal power. Use current local law, policy, indemnity, and senior advice for specific legal questions.

Ethical reasoning

Respect for autonomy supports informed self-determination. Beneficence promotes welfare; non-maleficence avoids disproportionate harm; justice concerns fair distribution, equal respect, and protection from discrimination. These principles can conflict. Refusing beneficial treatment may respect autonomy; allocating one scarce organ to one patient excludes another. Principles structure reasoning but do not mechanically produce an answer.

Clarify facts, uncertainty, urgency, options, outcomes, burdens, values, capacity, relationships, resources, and law. Identify affected parties and conflicting duties. Seek the least restrictive effective option, document reasoning, and revisit changes. Ethics consultation supports but does not replace accountable decisions.

Informed consent

Valid consent is voluntary, informed, given by a person with decision-making capacity, and specific enough for the proposed intervention. Consent is a process, not a signature. Explain diagnosis and uncertainty, purpose, material benefits and risks, reasonable alternatives including no treatment, recovery, costs or burdens, and who will perform key parts. Material risk includes what a reasonable person in this position would consider significant and what this particular patient is likely to value.

Use plain language, interpreters, diagrams or communication aids when helpful, and teach-back. Allow questions and time when possible. Do not ask unconsented family to translate sensitive or complex information. A patient may decline details, but verify that this choice is voluntary and explain essential safety information. Consent can be withdrawn at any time, though an irreversible step cannot be undone.

Emergency necessity permits immediately required treatment when a person lacks capacity, no valid refusal applies, delay risks serious harm, and intervention is proportionate to presumed wishes and interests. This does not authorise convenient non-urgent care. As soon as possible, restore normal consent, involve authorised decision-makers, and document why action was necessary.

Therapeutic privilege—avoiding disclosure because it may distress—is extremely narrow. Unexpected findings, trainee participation, photography, intimate examination, blood products, sterilisation, research, and special consequences may require explicit discussion. Disclose conflicts that could influence choice.

Decision-making capacity

Capacity is decision-specific and time-specific. The person must understand relevant information, retain it long enough, use or weigh it in reaching a choice, and communicate the decision. A mental illness, cognitive diagnosis, intoxication, communication disability, detention, or unwise choice does not automatically remove capacity. Values can differ from the clinician's without being irrational.

Support capacity by treating pain, delirium, hypoxia, withdrawal, fear, and sensory barriers; use interpreters, accessible formats, trusted supporters, and optimal timing. Ask the person to explain the decision in their own words and explore how they weigh consequences. Capacity may fluctuate and may be present for a simple decision but absent for a complex high-risk choice.

When capacity is absent, follow valid advance directives, legally appointed decision-makers, hierarchy of substitutes, or best-interest standards under local law. Substituted judgement asks what the patient would choose; best interests integrate welfare, values, relationships, least restriction, and

previously expressed wishes. Family members are not automatically legal decision-makers and may have conflicts. Disagreement requires careful communication, second opinion, mediation, ethics support, and legal review when necessary.

Children and adolescents

Parents or guardians generally authorise care for young children, but their authority is limited by the child's welfare. Include children in explanations and seek assent according to development. Some adolescents can consent independently when they understand the decision and consequences; exact rules depend on jurisdiction and treatment type.

Confidential adolescent care encourages disclosure, but explain limits for serious safety risk and safeguarding. Consider maturity, coercion, family environment, sexual health, substance use, mental health, and access to follow-up. If parental refusal exposes a child to serious preventable harm, escalate through safeguarding, institutional, and legal pathways rather than abandoning care.

Confidentiality and privacy

Confidentiality enables trust and candid disclosure. Collect only necessary information, access records for legitimate care, discuss patients privately, secure devices, verify recipients, and avoid identifiable details in casual teaching or messaging. De-identification is not guaranteed when rare details permit re-identification. Patients may request access and correction according to law; records should remain accurate and transparent rather than silently altered.

Share information with consent and on a need-to-know basis for direct care. Disclosure without consent may be justified or required for serious imminent harm, child or vulnerable-person protection, notifiable disease, court order, fitness to drive, firearm risk, or other statutory duties. Use the minimum necessary information, tell the patient when safe and lawful, consult senior or legal support, and document authority and reasoning.

After death, confidentiality generally continues, though access and disclosure rules differ. Genetic information creates familial implications: encourage

patient-led sharing and specialist counselling. Exceptional disclosure to relatives may be considered when serious preventable harm is likely and legal and ethical criteria are met.

Documentation and professional boundaries

The clinical record is a care tool and legal document. Record contemporaneous facts, sources, examination, assessment, uncertainty, options, consent, capacity, decisions, advice, follow-up, and changes. Distinguish observation from allegation and inference. Correct errors with dated, traceable amendments. Copying inaccurate text propagates harm.

Professional boundaries prevent exploitation. Sexual or financial relationships, coercive gifts, preferential access, social media, and dual relationships impair judgement; power persists after care ends. Chaperones support but do not replace consent. Address abusive behaviour while preserving emergency care and staff safety.

Safeguarding and violence

Safeguarding concerns abuse, neglect, exploitation, coercive control, trafficking, self-neglect, and preventable harm affecting children or adults with vulnerability. Warning signs include inconsistent injury, delay in care, fearful behaviour, controlling companions, malnutrition, poor hygiene, repeated emergencies, financial dependence, genital injury, unsafe housing, or caregiver strain. None proves abuse alone.

Speak privately using a trained interpreter. Ask direct, nonjudgmental questions, validate disclosure, assess immediate danger, dependent children, weapons, strangulation, sexual assault, and safe communication. Do not confront a suspected perpetrator or promise absolute secrecy. Document the patient's words, injuries, body maps or photographs with consent and policy, and actions. Follow mandatory reporting and multidisciplinary pathways.

Intimate-partner violence can affect anyone and often escalates during separation or pregnancy. Strangulation may cause delayed airway, vascular, or neurological injury with few external marks. Offer medical and forensic care, emergency shelter, advocacy, police options, and safety planning

without forcing a competent adult to leave unless law requires action. Abrupt intervention without planning can increase danger.

Error, disclosure, and duty of candour

After harm or near harm, stabilise, preserve evidence, notify, and disclose known facts compassionately. Explain uncertainty, consequences, treatment, investigation, prevention, and contact. Apology is compatible with analysis. Do not speculate, blame prematurely, or alter records.

Report through safety systems and analyse human, technical, environmental, organisational, and communication factors. A just culture distinguishes human error, at-risk shortcuts, and reckless conduct while maintaining accountability. Clinicians also need support after adverse events.

Negligence and professional standards

Negligence generally requires duty, breach of standard, causation, and compensable harm. Poor outcome alone is not negligence. Standards reflect reasonable practice, circumstances, resources, urgency, and law. Work within competence, seek help, communicate referrals and results, and ensure handover.

Abandonment can occur when care ends without transition. Conscientious objection does not justify discrimination, misinformation, emergency abandonment, or obstruction; duties vary locally. Impairment, boundary breach, dishonesty, or unsafe colleagues may trigger reporting duties.

Resource allocation and public health ethics

Scarcity requires transparent, consistent clinical criteria rather than social worth, wealth, influence, disability stereotypes, or inequitable first-come rules. Separate bedside advocacy from system allocation when possible, use prognosis carefully, and allow review. Disability is not lower life value.

Restrictive public-health measures require legality, necessity, effectiveness, proportionality, least restriction, reciprocity, equity, time limits, transparency, and review. Coercion without accessible services or support damages trust.

Research ethics

Research requires validity, favourable risk-benefit balance, fair selection, independent review, consent, privacy, and continuing respect. Equipoise means genuine expert uncertainty. Consent distinguishes research from therapy and explains randomisation, placebo, alternatives, data use, withdrawal, and injury arrangements. Placebo must not expose participants to serious avoidable harm. Genomics, artificial intelligence, and biobanks add consent, bias, ownership, security, and governance concerns.

TTS module 2: Resolving ethical conflict, supported decisions, digital privacy, and safeguarding practice

Ethically difficult cases usually become clearer when the team separates facts from values and process. First define the clinical state, realistic options, urgency, reversibility, and uncertainty. Then identify whose interests are affected, which values conflict, who has legitimate decision-making authority, and whether structural disadvantage or unequal power is distorting the discussion. Finally choose a fair process: support the patient, seek missing evidence, obtain senior or independent review, document the reasoning, and set a review point. A morally uncomfortable outcome is not necessarily an unethical one; some conflicts reflect unavoidable loss rather than misconduct.

Autonomy is relational as well as individual. People make choices through language, family, culture, disability supports, finances, and prior experience of healthcare. Supported decision-making aims to strengthen the person's own agency before substituting another decision-maker. Useful adjustments include treating pain or delirium, offering a professional interpreter, presenting information in short stages, providing accessible written material, allowing extra time, choosing the patient's best time of day, and inviting a trusted supporter selected by the patient. The supporter may help communication but must not quietly replace the patient's voice.

Capacity assessment should track the actual decision. State what choice is required, what consequences matter, and why capacity is questioned. Ask the patient to explain the condition, options, principal benefits and harms, and likely consequences of accepting or declining. Explore the reasoning without demanding agreement with medical values. A choice based on unusual religious or personal beliefs may still be capacitous; a familiar-sounding

answer may not be if delirium prevents genuine weighing. Repeat assessment after reversible impairment is treated, and record both abilities and supports used rather than merely writing “has” or “lacks” capacity.

When capacity is absent, decision-making remains centred on the person. A valid applicable advance refusal may control. Otherwise the authorised substitute should use the patient’s known values and previously expressed wishes, not the substitute’s preference. If those values are unknown, best-interest reasoning considers benefit, burden, dignity, relationships, prognosis, least restriction, and the possibility of recovery. The patient should still participate as far as possible. Disagreement among relatives does not become a vote, and forceful relatives do not acquire extra authority. Clarify the legal role, seek consensus, and escalate unresolved high-stakes conflict.

Consent discussions should be proportional but not perfunctory. Material information includes common harms, rare severe harms, expected recovery, uncertainty, alternatives, and the option of no intervention. The clinician should connect probabilities to consequences: a small risk of permanent voice loss may matter greatly to a singer, while another patient may prioritise avoiding repeated hospital visits. Teach-back tests the explanation, not the patient. A signed form cannot repair coercion, inadequate disclosure, or incapacity. Conversely, excellent consent may be documented in the clinical record even when a particular form is not legally required.

Voluntariness can be undermined by family pressure, institutional dependence, threats, financial incentives, immigration concerns, or the clinician’s framing. Recommendations are appropriate, but manipulation is not. Presenting only the favoured option, exaggerating certainty, or describing refusal as abandonment constrains choice. When coercion is suspected, speak privately, identify the source of pressure, assess immediate safety, and offer time and independent support. A competent refusal should prompt exploration and harm-reduction planning, not punishment or withdrawal of unrelated care.

Treatment over objection requires especially careful justification. In an emergency, temporary restraint or treatment may be necessary to prevent serious immediate harm when capacity is absent, but the least restrictive effective measure should be used for the shortest period. Outside emergency conditions, legal authority, necessity, proportionality, alternatives, trauma

history, and foreseeable harms require explicit review. Chemical restraint is medication used primarily to control behaviour rather than treat an underlying syndrome; calling it sedation does not remove the need for safeguards, monitoring, and review.

Confidentiality decisions should use a structured threshold. Identify the information, proposed recipient, purpose, patient preference, legal basis, seriousness and likelihood of harm, and whether a less intrusive route exists. Share the minimum necessary content and limit onward disclosure. A worried relative can provide information without automatically being entitled to receive information. The clinician may listen, explain general principles, encourage family communication, and preserve the patient's confidentiality. When disclosure is required, tell the patient beforehand when safe and lawful and document why the threshold was met.

Digital systems create privacy hazards beyond the traditional bedside. Accessing a record out of curiosity, photographing a screen, copying identifiable information into an unapproved application, or discussing a rare case online may breach confidentiality even without malicious intent. Metadata, dates, location, images, and combinations of unusual facts can re-identify a supposedly anonymous patient. Use approved secure channels, verify recipients, restrict downloads, lock devices, and promptly report misdirected messages or suspected breaches so that harm can be contained.

Clinical documentation should make the decision reproducible. Record who was present, the patient's communication needs, information provided, questions, expressed values, material risks, alternatives, recommendation, decision, and follow-up. For capacity, document the functional assessment and reversible factors. For safeguarding, distinguish exact quotations, direct observations, collateral reports, and clinical inference. Never alter the original record invisibly. A dated correction that explains the change protects both care and integrity better than retrospective reconstruction.

Safeguarding begins with recognition but depends on a safe response. Injuries inconsistent with the history, repeated missed care, controlling accompaniment, poor access to money or medicines, fearfulness, sexual health concerns, neglect, exploitation, and caregiver exhaustion are signals rather than proof. Arrange privacy without increasing danger, use a qualified interpreter rather than a potentially involved relative, and ask simple

normalising questions. Immediate priorities are urgent treatment, danger assessment, safe communication, dependent children or adults, access to weapons, threats to kill, strangulation, stalking, and escalation around separation.

A disclosure should be met with belief, validation, and explanation of choices and confidentiality limits. Avoid interrogating, promising a particular outcome, or confronting an alleged perpetrator. Preserve forensic options without making support conditional on police involvement. Document the patient's words and clinically relevant findings precisely. Mandatory reporting duties vary by age, vulnerability, injury, profession, and jurisdiction, so consult current local pathways. If reporting against a competent adult's wishes is not required, safety planning and continued access to care may be safer than a dramatic intervention.

Children require both welfare protection and developmentally appropriate participation. Explain care in language they can understand, seek assent, and attend to behaviour as communication. Parental decisions usually deserve respect, but authority narrows when refusal creates a substantial risk of serious preventable harm. Urgency, effectiveness, burden, prognosis, and alternatives determine whether escalation is justified. Adolescents may have independent capacity for some decisions; privacy promotes honest care, but limits should be explained before sensitive questions when serious harm or abuse may require action.

Professional boundaries are tested in small increments. Repeated personal messaging, special access, gifts with obligation, financial dealings, social-media contact, self-disclosure that burdens the patient, and care for close associates can shift attention from patient welfare. The appropriate response may be declining, documenting, seeking supervision, transferring care safely, or establishing a clear communication channel. Boundary protection is not emotional coldness; warm, compassionate care is compatible with limits that prevent exploitation and preserve clinical judgement.

After an adverse event, ethical duties include immediate care, honest communication, preservation of evidence, reporting, and learning. Explain what is known, what remains uncertain, the consequences, current treatment, and how further information will be provided. A sincere apology acknowledges suffering and responsibility without inventing facts. Just-culture analysis asks

how design, workload, communication, equipment, supervision, and normalised shortcuts contributed while still addressing reckless conduct. Supporting involved clinicians must never displace accountability to the patient and family.

Scarcity decisions should rely on relevant clinical criteria applied consistently and transparently. Prognosis estimates can encode disability, racial, socioeconomic, and access bias, so teams should examine the data and permit review. Social worth, fame, wealth, perceived moral deservingness, or unsupported assumptions about quality of life are inappropriate. When equally suitable patients cannot all receive treatment, an agreed fair tie-breaker may be needed. Separating allocation policy from bedside advocacy helps clinicians remain loyal to individual patients within legitimate system constraints.

Ethical practice ends with communication and review. State the recommendation, acknowledge burdens, summarise the reasons, identify disagreement, and specify what could change the decision. Seek ethics, legal, cultural, disability, safeguarding, or spiritual expertise when it adds relevant perspective. Law defines enforceable boundaries but cannot substitute for compassion, fairness, or careful reasoning. The durable habit is to make power visible, support the patient's voice, use the least restrictive option, and leave a transparent record that another clinician can understand and reassess.

Retrieval prompts

One. What elements make consent valid?

Two. How is decision-making capacity assessed?

Three. When can confidential information be disclosed without consent?

Four. What are the first duties after a clinical error?

Five. Which elements generally establish negligence?

Six. What principles justify restrictive public-health measures?

Concise answers

One. Voluntariness, adequate information, capacity, and specificity for the intervention.

Two. Assess understanding, retention, use or weighing, and communication for that decision at that time after support.

Three. When law requires it or proportionate disclosure is necessary to prevent serious harm, using the minimum information.

Four. Stabilise, preserve facts, notify, disclose compassionately, treat consequences, investigate, and prevent recurrence.

Five. Duty, breach of standard, factual and legal causation, and compensable harm.

Six. Legality, necessity, effectiveness, proportionality, least restriction, reciprocity, equity, transparency, time limits, and review.

Source map

Original synthesis informed by Talley and O'Connor, consent, communication, capacity, sensitive examination, documentation, and professionalism; OpenStax Medical-Surgical Nursing, advocacy, safeguarding, privacy, error disclosure, end-of-life decisions, and culturally safe care; Katzung and OpenStax Pharmacology, research ethics, conflicts, adverse events, and access; Robbins and Guyton and Hall, uncertainty, prognosis, brain function, and treatment limitation; and OpenStax Biology, Microbiology, and Chemistry, research integrity, genetics, public-health ethics, and environmental justice.

Chapter 68: Palliative Medicine, Symptom Control, Dying, Grief, and Bereavement

Orientation

Palliative medicine improves life with illness by treating symptoms, supporting decisions, coordinating care, and addressing psychological, social, cultural, and spiritual needs. It begins alongside disease treatment and is not limited to cancer or final days. Recognising dying shifts effort toward comfort and communication, not abandonment. Complex symptoms require assessment and specialist advice.

Identifying need and setting goals

Palliative need is indicated by progressive illness, repeated admissions, refractory symptoms, declining function, frailty, caregiver strain, or uncertainty about burdensome treatment. Ask whether the team would be surprised if the patient died within months; this prompt identifies unmet planning but is not a prognosis. Disease trajectories differ: cancer may decline predictably, organ failure fluctuates with crises, and frailty or dementia declines slowly with acute interruptions.

Clarify what the person understands, how much detail they want, what matters now, unacceptable outcomes, hopes, fears, cultural or spiritual commitments, and who should participate. Goals may include living longer, maintaining cognition, returning home, attending an event, avoiding hospital, relieving pain, or supporting family. Hope can shift from cure toward achievable priorities without becoming false reassurance.

Shared decisions compare likely benefit, burden, uncertainty, and alternatives in light of these goals. A time-limited trial specifies intervention, objective markers of benefit, duration, and what happens if benefit is absent. Revisit after deterioration and transitions. Record substitute decision-makers, advance directives, resuscitation decisions, preferred place of care, and emergency plans where legally applicable.

Communication and prognosis

Use privacy, chosen supporters, and an interpreter. Assess understanding, warn that difficult news is coming, speak plainly in small portions, pause, acknowledge emotion, and check meaning. Avoid euphemisms that obscure death. Ask which questions matter.

Prognosis is a range, not a date. Describe best, worst, and likely scenarios and function using broad time frames. State uncertainty. If a patient declines details, identify a recipient. Explore family requests to conceal information while respecting the patient's preferences.

Conflict often reflects different facts, values, grief, or trust. Name shared goals, summarise disagreement, obtain second opinion, and involve palliative, ethics, cultural, spiritual, or mediation support. Offer a recommendation when resuscitation or other treatment is unlikely to achieve goals.

Holistic symptom assessment

Symptoms are multidimensional. Assess onset, mechanism, severity, pattern, triggers, function, sleep, mood, meaning, previous treatment, medicine burden, and family observations. Examine and investigate only when findings will change management. Consider reversible contributors while avoiding burdensome tests that cannot improve outcomes. Reassess after every intervention.

Non-drug care includes explanation, positioning, pacing, fan or cool air, mouth care, massage, heat or cold, relaxation, physiotherapy, equipment, environmental change, and caregiver education. Simplify medication schedules and stop preventive drugs whose delayed benefit no longer matches prognosis, while avoiding withdrawal from corticosteroids, benzodiazepines, anticonvulsants, opioids, and other dependence-producing therapy.

Pain

Identify nociceptive, neuropathic, inflammatory, visceral, colic, bone, or incident pain and treat the cause when proportionate. Paracetamol, anti-inflammatory drugs, corticosteroids, antidepressants, anticonvulsants, bisphosphonates, radiotherapy, nerve blocks, surgery, or disease-directed therapy may reduce specific pain. Opioids are central for moderate to severe cancer and end-of-life pain but are not the only treatment.

For existing opioid use, assess regular and breakthrough exposure, adherence, effect, toxicity, organ function, and route. Titrate from response. When swallowing fails, use an appropriate alternative; transdermal systems adjust slowly and do not suit rapidly escalating pain. Conversion tables are estimates, so reduce for incomplete cross-tolerance and seek specialist advice.

Prevent constipation unless inappropriate. Treat nausea, itch, sedation, retention, confusion, and dry mouth. Respiratory depression can follow rapid escalation, interaction, kidney failure, or error. Distinguish tolerance and dependence from addiction; fear of addiction should not leave terminal pain untreated.

Breathlessness

Breathlessness reflects respiratory, cardiac, haematological, metabolic, neuromuscular, anxiety, and perceptual mechanisms. Treat reversible burdens such as pulmonary oedema, bronchospasm, pleural effusion, infection, anaemia, embolism, or airway obstruction when consistent with goals. Oxygen helps hypoxaemia but often adds little for a non-hypoxaemic patient.

Position upright, use a handheld fan across the face, pace activity, open the environment, and provide calm coaching. Low-dose systemic opioid reduces refractory breathlessness by altering central perception and ventilatory response; titrate carefully. Benzodiazepines do not directly treat breathlessness but may help severe associated anxiety after other measures. Non-invasive ventilation can relieve symptoms or prolong life, so its purpose and stopping plan must be explicit.

Nausea, bowel, and secretions

Nausea may follow drugs, stasis, obstruction, constipation, vestibular or metabolic disease, raised pressure, or anxiety. Choose antiemetic by mechanism and adverse effects. Bowel obstruction may require corticosteroid, antisecretory therapy, antiemetic, analgesia, decompression, stent, or surgery according to goals; avoid prokinetics in complete obstruction.

Constipation causes pain, nausea, overflow, retention, and delirium. Assess loading and obstruction. Combine stimulant and osmotic agents for opioid constipation; avoid rectal measures with neutropenia, thrombocytopenia, mucosal injury, or refusal. Review diarrhoea for overflow, infection, malabsorption, and drugs.

Noisy respiratory secretions near death are usually more distressing to family than to an unconscious patient. Reposition, reduce non-beneficial fluids, provide mouth care, and explain the sound. Antimuscarinic drugs may reduce new secretion but do not remove existing fluid and can cause dry mouth or delirium. Deep suction often causes distress without durable benefit.

Delirium, agitation, and neurological symptoms

Delirium may follow infection, retention, constipation, pain, drugs, withdrawal, organ failure, electrolytes, or brain disease. Reverse causes without disproportionate burden. Provide familiarity, orientation, sensory support, sleep cues, and safety. Antipsychotics may help dangerous agitation but can worsen some syndromes.

Terminal agitation may signal pain, bladder distension, fear, withdrawal, or spiritual distress. Treat causes and use proportionate sedation. Seizures require rescue and ongoing anticonvulsant. Catastrophic haemorrhage or airway obstruction needs dark towels, rapid sedation, calm presence, and family preparation.

Depression, anxiety, and existential distress

Sadness and grief are expected but do not exclude treatable depression. Ask about persistent loss of pleasure, hopelessness, guilt, worthlessness, suicide, panic, trauma, and desire for hastened death. Such a desire may express uncontrolled symptoms, loss of autonomy, family burden, depression, spiritual suffering, or a stable request under local law. Respond with curiosity and safety assessment rather than judgement.

Psychological, dignity, meaning-centred, spiritual, social, practical, and medication support can help. Antidepressant onset may exceed prognosis; specialists may consider faster options. Family meetings address caregiving, finances, children, conflict, and anticipatory grief.

Nutrition, hydration, and weakness

Reduced appetite accompanies advanced illness. Treat mouth pain, nausea, constipation, depression, and early satiety without forcing intake. Preferred foods and social eating may matter more than calorie targets. Inflammatory cancer cachexia is not reversed by calories alone.

Artificial nutrition may help selected reversible conditions but often does not improve survival, aspiration, function, or comfort in advanced dementia or active dying. Assisted hydration may worsen oedema, effusions, secretions, and line burden; mouth care often relieves dryness better. Use goals and time-limited trials.

Last days of life

Signs of dying include bedbound state, profound weakness, minimal intake, reduced consciousness, altered breathing, mottled extremities, low urine, and inability to swallow. Treat reversible causes only when aligned with goals. Explain expectations and contacts. Stop observations, tests, preventive drugs, and devices that do not improve comfort.

Prescribe anticipatory medicines for pain, breathlessness, agitation, nausea, and secretions by a usable route. Continue essential anticonvulsants and withdrawal-preventing drugs. Reposition, protect skin, provide mouth and eye care, manage elimination, and avoid disturbance. Presence, ritual, music, silence, and privacy follow preferences.

Resuscitation decisions concern arrest treatment, not abandonment. Withholding and withdrawing are ethically equivalent when based on benefit and goals. Proportionate symptom medication differs from intending death. Palliative sedation for refractory intolerable suffering uses minimum required sedation, consent or substitute process, multidisciplinary review, and continued care.

Grief and bereavement

Verify death and follow requirements for certification, official referral, donation, and family notification. Explain next steps and offer time with the body according to culture and safety. Debrief staff after difficult deaths.

Grief is not linear. Yearning, disbelief, anger, guilt, relief, sleep change, and poor concentration can be normal. Risk rises with traumatic death, conflict, prior illness, isolation, multiple losses, financial strain, and low support. Follow-up identifies suicide risk, severe impairment, persistent disabling grief, depression, trauma, and practical needs without medicalising normal sorrow.

TTS module 2: Advanced symptom reasoning, opioid safety, treatment withdrawal, and care in the final phase

High-quality palliative care combines meticulous medicine with attention to personhood. Symptoms should not be dismissed as inevitable merely because disease is advanced. At the same time, investigation and treatment are useful only when they can improve comfort, function, understanding, or a goal valued by the patient. For every new problem, ask what mechanism is likely, what reversible contributors exist, how burdensome clarification would be, what time is available for benefit, and how the result would change management. This approach prevents both therapeutic neglect and burdensome reflex intervention.

Total suffering can include physical discomfort, fear, depression, disrupted identity, family conflict, financial strain, spiritual distress, and uncertainty. These dimensions interact: breathlessness can cause panic, panic intensifies breathlessness, and a caregiver's alarm may further amplify distress. Assessment therefore includes symptom intensity and mechanism, but also function, meaning, coping, relationships, practical needs, and what the person

fears will happen next. A single numerical score is useful for tracking but cannot replace narrative assessment.

Pain evaluation distinguishes continuous background pain from incident and breakthrough pain. Somatic pain is often localised and aching, visceral pain may be diffuse or referred, colic fluctuates, and neuropathic pain may burn, shoot, or follow a nerve distribution. Bone instability, cord compression, pathological fracture, raised intracranial pressure, infection, retention, and ischaemia require prompt recognition because targeted treatment may rapidly improve comfort or prevent catastrophe. Treatment can combine disease-directed therapy, physical support, psychological techniques, non-opioid analgesia, adjuvants, procedures, and opioids.

Opioid prescribing begins with the patient's current exposure, route, renal and hepatic function, frailty, interacting sedatives, and previous response. Opioid-naïve patients need lower starting doses than tolerant patients. Breakthrough doses should be related to the regular regimen and reassessed according to onset and duration. Repeated rescue use suggests uncontrolled background pain, predictable incident pain needing anticipatory dosing, or a new cause. Escalation without reassessment can worsen opioid-induced hyperalgesia, delirium, myoclonus, nausea, constipation, and sedation while failing to treat the true mechanism.

Equianalgesic conversions are estimates derived from populations, not exact exchange rates. When changing drug or route, calculate the approximate equivalent, reduce for incomplete cross-tolerance, then titrate to effect. Kidney failure permits accumulation of active metabolites from some opioids and increases neurotoxicity; specialist advice helps select and dose alternatives. Transdermal delivery has delayed onset and offset, so fever, cachexia, adhesion, and rapidly changing requirements complicate use. A syringe infusion provides continuous subcutaneous medication when swallowing fails but does not automatically imply imminent death.

Constipation prophylaxis should accompany regular opioids unless diarrhoea or obstruction makes it inappropriate. Ask about last comfortable bowel action, stool consistency, flatus, colic, vomiting, rectal symptoms, drugs, hydration, and mobility. Overflow diarrhoea may conceal impaction. In malignant obstruction, distinguish partial from complete obstruction and consider inflammation, secretion, motility, pain, and vomiting. A prokinetic

may help functional stasis or partial obstruction but can worsen colic in complete mechanical obstruction. Rectal examination or intervention is chosen according to likely benefit, consent, blood counts, mucosal risk, and burden.

Breathlessness severity correlates imperfectly with oxygen saturation. Facial airflow, upright positioning, activity pacing, reassurance, and a calm plan can reduce neural threat perception. Oxygen is indicated for symptomatic hypoxaemia when it helps, but tubing and equipment may burden a non-hypoxaemic patient without greater relief than air. Low-dose systemic opioid can reduce refractory breathlessness; careful titration does not equate to euthanasia. Acute change still demands consideration of pulmonary oedema, embolism, effusion, bronchospasm, infection, anaemia, pneumothorax, metabolic acidosis, and airway obstruction when treatment fits goals.

Nausea treatment is mechanistic. Gastric stasis, biochemical disturbance, drug toxicity, vestibular stimulation, raised intracranial pressure, constipation, bowel obstruction, and anticipatory anxiety favour different strategies. Review the route because vomiting or poor absorption can make an apparently adequate oral regimen ineffective. Combining agents with complementary actions may help refractory nausea, while stacking drugs with similar receptor effects increases sedation, hypotension, extrapyramidal reactions, anticholinergic burden, or cardiac risk. Hydration may help a reversible biochemical cause but should be judged by symptom response, not routine habit.

Delirium is common, fluctuating, and often missed when hypoactive. Establish the patient's baseline, identify inattention and altered arousal, review medicines, and look for pain, retention, constipation, infection, hypoxia, withdrawal, organ failure, and metabolic disturbance. Correct causes whose treatment is proportionate. Familiar people, hearing aids, glasses, daylight, reduced noise, and simple explanations support orientation. Medication is reserved mainly for severe distress or dangerous agitation, with syndrome-specific caution. Families should be told that confused speech or picking behaviour can be part of brain failure and is not deliberate rejection.

When agitation persists near death, reassess for untreated pain, urinary retention, faecal loading, fear, akathisia, medication toxicity, nicotine or

alcohol withdrawal, and spiritual distress. Proportionate palliative sedation is considered only for intolerable refractory suffering after reasonable options have failed or are unavailable within the clinical time frame. Define the target symptom, discuss expected reduction in consciousness, obtain consent or an authorised decision, choose the minimum effective depth, monitor comfort, and continue nursing and family support. The intention is relief, and dosing is titrated to that endpoint rather than to hasten death.

Deprescribing is an active clinical intervention. Medicines for distant prevention may offer no realistic benefit while creating swallowing burden, bleeding, hypotension, hypoglycaemia, monitoring, cost, and interactions. Continue drugs that prevent near-term suffering or dangerous withdrawal, and taper corticosteroids, benzodiazepines, anticonvulsants, some antidepressants, and opioids when abrupt cessation would cause harm. Decisions should consider indication, time to benefit, prognosis, current route, patient priorities, and rebound effects. Explaining why a medicine is stopped prevents patients and families from interpreting simplification as abandonment.

Nutrition conversations require sensitivity because feeding symbolises care. In advanced disease, reduced intake often reflects metabolic change and diminishing physiological demand rather than neglect. Offer desired foods, small portions, texture modification, assistance, mouth care, and social company without coercion. Treat reversible thrush, pain, nausea, constipation, and depression. Tube feeding or parenteral nutrition may help selected reversible conditions, but in advanced dementia or active dying it may add aspiration, fluid overload, restraint, line infection, and transfer burden without restoring function or prolonging meaningful life.

Hydration decisions are similarly individual. Thirst and dry mouth often respond better to frequent mouth care, ice, sips, saliva substitutes, and lip care than to intravenous fluid. A time-limited trial may be reasonable for suspected reversible dehydration, medication toxicity, or delirium, with explicit endpoints and a stopping rule. Conversely, fluid can worsen oedema, ascites, pleural effusions, respiratory secretions, urine burden, and cannula discomfort. Explain that withholding non-beneficial artificial hydration does not mean withholding attention or basic comfort.

Recognising the final days depends on trajectory and repeated assessment rather than one sign. Increasing sleep, bedbound state, minimal intake, weak circulation, reduced urine, altered breathing, mottling, and loss of swallowing collectively raise probability, but reversible sedation, infection, hypercalcaemia, or medication toxicity may mimic dying. Communicate uncertainty honestly: the person may be dying, recovery is unlikely, and the team will review. Families need guidance about expected breathing patterns, cool extremities, reduced responsiveness, and whom to call, including what they can do at the bedside.

Anticipatory prescribing should match likely symptoms, organ function, previous exposure, allergies, and an available route. Orders need clear indications, dose ranges, intervals, maximums, monitoring, and escalation instructions. Medicines alone are insufficient: stop painful observations, silence unnecessary alarms, remove non-beneficial devices, reposition gently, protect pressure areas, manage bladder and bowel needs, and provide mouth and eye care. Ensure that nursing staff can obtain and administer drugs promptly, especially at home or after hours.

Withdrawal of life-sustaining treatment is ethically based on the same question as withholding: does the intervention provide benefit consistent with the patient's goals and values? Before withdrawal, confirm authority, address disagreement, explain the expected course, plan symptom control, and clarify which care continues. Ventilation, vasopressors, dialysis, artificial nutrition, or monitoring may be reduced or stopped in a staged way appropriate to the situation. Analgesia and sedation are titrated pre-emptively to distress, not withheld from fear that correctly used medication might shorten life.

After death, care continues through verification, culturally respectful handling, clear explanation of certification or coroner processes, return of belongings, donation pathways where relevant, and practical guidance. Invite questions and correct misconceptions, especially guilt about medication, food, hydration, or treatment withdrawal. Bereavement follow-up should be proportionate: normal grief needs compassion and community, while suicidal thinking, severe persistent impairment, traumatic intrusions, substance misuse, profound isolation, or prolonged disabling yearning warrants active assessment and treatment.

The practical standard is anticipatory, revisable care. Make a plan before crisis, prescribe before swallowing is lost, identify who can decide, define emergency contacts, and document what matters most. Reassess because symptom mechanisms, prognosis, and family capacity change. Palliative medicine is not less medicine; it is disciplined selection of treatments most likely to preserve comfort, agency, connection, and dignity when time and physiological reserve are limited.

Retrieval prompts

One. When should palliative care begin?

Two. What belongs in a time-limited treatment trial?

Three. Which measures relieve refractory breathlessness?

Four. Why may artificial hydration worsen comfort near death?

Five. How does palliative sedation differ ethically from euthanasia?

Six. Which bereavement features warrant further assessment?

Concise answers

One. Alongside disease treatment whenever serious illness creates symptoms, decision needs, functional decline, or family burden.

Two. The intervention, intended outcome, measurable markers, review time, and action if benefit is absent.

Three. Treat reversible causes, position and pace, use facial airflow, calm support, and carefully titrated systemic opioid; oxygen helps hypoxaemia.

Four. It can increase oedema, effusions, respiratory secretions, urination, and line burden without relieving dry mouth.

Five. It proportionately reduces consciousness to relieve refractory suffering without the intention of causing death.

Six. Suicide risk, severe or persistent functional impairment, traumatic symptoms, depression, isolation, conflict, or prolonged disabling grief.

Source map

Original synthesis informed by Guyton and Hall, pain, breathlessness, consciousness, thirst, appetite, and dying physiology; Robbins, advanced disease, cachexia, organ failure, and causes of death; Katzung and OpenStax Pharmacology, opioid conversion, antiemetics, sedatives, antisecretory therapy, and deprescribing; Talley and O'Connor, symptom assessment, communication, cognition, and functional examination; OpenStax Medical-Surgical Nursing, palliative care, last-days care, family support, grief, and bereavement; and all preceding chapters on ethics, cancer, organ failure, pharmacology, and communication.